Viscosity

The Evolution of Intelligence in Bits, Atoms, and Life

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The world does not run at inference speed.

First edition
2026

Preface

AI can produce ten proposals capable of setting another meeting in motion before the first meeting ends. Proposals keep getting cheaper; meetings, construction work, experiments, and the people who bear the consequences have not multiplied at the same rate. Debates about artificial intelligence often leap across this gap. Technological optimists say models will become more capable and costs will fall further; skeptics say deployment always lags behind demonstrations. Both can find evidence, yet both share one premise: once intelligence is abundant and cheap enough, the other obstacles will give way in turn. This book begins with that premise.

Generation has already become cheap. Over the past few years, models, data, training methods, and inference computation have together driven gains in capability, bringing the generation of text, images, and code into everyday work. But those cost reductions have not spread evenly along the whole chain. A fully developed proposal can be written in tens of seconds, while a production line still takes months or years to bring into operation: equipment must be purchased, staff trained, procedures approved, and responsibilities reassigned. The bottleneck has moved away from generation. Treating that shift as evidence that the entire world is accelerating is where the judgment starts to go wrong.

The first premise to examine is that intelligence can be measured in isolation and grow continuously along a single axis. This premise has an engineering history: model size, data volume, and compute once rose alongside performance, and extrapolation became a habit. But intelligence is not a scalar. Recognition, reasoning, memory, planning, practical action, empathy, and bearing responsibility cannot substitute for one another, and each requires different conditions; machines already surpass humans in some of these abilities and remain poor at others, and a single extrapolated scale cannot establish when those differences will disappear. A system trained only on text will not automatically extend its capabilities to physical operations, long-term commitments, and responsibility for consequences simply because it has more text. Treating intelligence as a single scale leads to the inference that "a little more capability will solve everything," an inference that has yet to be borne out.

The second premise is that the low cost of copying can extend to execution. Documents, images, and programs can be copied repeatedly, and digitization has indeed replaced much of the work once performed with maps, film, records, and over a counter. But copying a blueprint

and building a bridge consume different resources. A bridge's blueprint can be distributed ten thousand times, yet each bridge still has to be built separately; concrete's strength development depends on its mix, temperature, humidity, and curing conditions, and the 28-day age commonly used in engineering is a testing benchmark; drug candidates can be generated rapidly, but efficacy and long-term safety still require trials and observation. Tools for design, screening, and validation can change the efficiency of these processes, while the materials, exposure time, and real-world feedback each process requires must still be supplied, item by item.

The third premise is that the better proposal will automatically prevail. Engineering history is full of counterexamples. A better technical approach losing to a more mature supply chain, a larger installed base, and lower switching costs is the rule, not the exception. To reach implementation, a proposal must pass through the three constraints of resources, time, and existing structures, while optimization affects only a small part of that passage. A related set of misconceptions concerns people. The prevailing narrative is replacement: machines will take people's jobs, or machines will set people free. What actually happens is more fragmented and harder to deal with. What gets displaced is often the evidentiary link between people and the work they produce, rather than the people themselves. A person can write, calculate, and summarize, abilities that make that person needed and make their place self-evident; when a system can generate the same output in tens of seconds, the abilities remain, but their power to establish that person's value is gone. Chapters 5 and 6 address this problem through the mechanisms of collapse and the migration of value, respectively.

These premises persist because they are all internally consistent at the same level. At the level of information, marginal copying costs can be very low, optimization can work against a given target, and some capabilities can be extrapolated over limited ranges; each proposition holds in its own context, and the error comes when it is extended beyond that context. To judge where they fail, we must go one level deeper and ask more basic questions: what is intelligence, where does it come from, why has it taken its present form, and what does it encounter when it enters the physical world and the world of life? Physics, biology, and information science already offer mechanisms and evidence for these questions, though those answers rarely appear in the same context. This book connects them and uses them to examine the premises above.

The central concept connecting these answers is viscosity. Viscosity is not slowness. Slowness is a judgment about speed; viscosity is a judgment about cost. Viscosity is what a system has to pass through to move from state A to state B; it asks not how long the step took, but what it consumed, what it left behind, and whether it can be reversed if something goes wrong. Viscosity is not conservatism or fear of technology, either. Digital payments, navigation, industrial simulation, automated experiments, and code completion have all reduced viscosity in certain parts of a process. The reality is more uneven: viscosity differs greatly between stages, and the same technology settles into different stages at different rates. The question to keep asking is the same: which kind of viscosity has this technology reduced, and where has

it moved the viscosity?

Any real-world state transition requires us to examine at least three kinds of constraint. The materials, energy, equipment, and capital needed to drive change form resource barriers (R); the process time occupied by reactions, growth, testing, and coordination forms temporal rigidity (T); and the difficulties of synchronization, isolation, and rollback encountered as change propagates through a dependency network form topological coupling (G). Together, they describe where the state transition sits. Engineering improvements may trade additional resources for time, or use new processes to reduce several costs at once; a local reduction in resistance may also transfer the burden to maintenance, verification, or other parties who must bear it. Assessing a technology's effects requires checking both the change in total cost and the distribution of that cost.

The same constraints behave very differently in different media, so this book divides the world that intelligence encounters on entering reality into three media. **Bits** are the medium of information, where copying, transmission, modification, and rollback all cost little: they are the first to be transformed by generative systems and the most likely to create the impression that everything can be reconfigured; yet their low viscosity is concentrated at the level of generation, not trustworthiness, and as generation constraints ease, verification can become a new priority. **Atoms** are the medium of matter, where manufacturing, construction, transportation, energy, logistics, chips, and infrastructure all confront mass, energy, friction, inertia, material strength, and spatial constraints; a linguistic error can be revised before execution, but a robot's error may break an object, injure a person, or damage the robot itself. **Life** is the third medium, responding to interventions rather than instructions: a change entering a living system may be recognized, compensated for, adapted to, or even met with a rebound; DNA can be written as a sequence, but gene function is not textual meaning. The difference between the three media lies in how they absorb intelligence: bits translate intelligence into diffusion, atoms into physical work, and life into feedback. Applying one logic of acceleration to all three overlooks their different conditions.

The book has seven chapters and an appendix. Chapter 1 examines the gap in our sense of time: why the speed of generation gets mistaken for the speed of getting things done, and why viscosity acts as both resistance and a stabilizer. Chapter 2 establishes the physical boundaries—where intelligence comes from, and what reconfiguring matter costs in resource barriers, temporal rigidity, and topological coupling—with the central conclusion that computing power can optimize a path but cannot abolish a potential difference, compress search and improve processes while the chosen physical processes must still be completed, and generate proposals without generation itself becoming a state transition. Chapter 3 places these constraints at the interface, examining how each of the three media absorbs intelligence as an input, whether real systems reroute, accumulate backlogs, or become unstable under overload, and under what conditions a generated output counts as confirmed.

Chapter 4 turns to life and bodies, carrying the same constraints through to mobility systems. The first three sections examine why living systems respond to interventions rather than instructions, and why machine bodies cannot simply inherit the capabilities of biological bodies; Section 4 explains why exploration, control, and verification impose different requirements, whose relationships must be arranged even within one system; Section 5 brings this differentiation back to people, discussing their place in terms of task energy consumption, feedback stability, and admission to action, and distinguishing present engineering advantages from the standing to exercise judgment that institutions must establish; Sections 6 and 7 apply the same accounting to ground transportation and low-altitude systems, tracing how costs move between propulsion platforms, existing infrastructure, and governance; Section 8 explains how to use this analysis: in managing systems coupled across multiple rates, the objective is to maintain a feasible range of flow regimes, rather than seek an extremum.

Chapter 5 is the book's center of gravity: what counts as existing, what kinds of judgment can be relied on, when a representation can cross a transition edge into action, where the consequences of action return, and what kind of boundary qualifies as a bearer of consequences. The first two sections establish these criteria, and the remaining four follow them through, examining potentially interacting collapses at the three levels of functionality, authenticity, and subjecthood, and how those collapses combine into a downward spiral. From here, the question is no longer only what can be generated, but what can be borne. Chapter 6 addresses the migration of value: when technology reduces an old viscosity, the way people establish their value can shift—writing, printing, clocks, steam engines, and computers offer different examples; as generation costs fall, verification, care, long-term commitments, and bearing consequences may become new points of scarcity; whether those activities receive payment and social recognition depends on the institutions of distribution. The latter half of the chapter gathers the direction of this migration into four principles—damping ontology, the incompleteness principle, the symbiosis principle, and the recursive optimization principle—which address, respectively, how costs enter a boundary's formation and maintenance, what is lost when value is compressed into an objective function, how a division of labor creates complementarities and addresses competition, and why adjustments must proceed level by level, and explains why they resemble articles of a constitution more closely than a technical proposal. Chapter 7 brings the argument back to practice: how individuals can retain the roughness of reality, how organizations can rebuild trust and verification, why civilization still needs frontiers of high resistance, and why self-replicating systems become a stress test for the entire theory. The appendix proposes an initial route toward formalizing generalized viscosity, places it in the context of existing theories, and outlines how to develop testable propositions. Measuring the variables, modeling their coupling, and quantitative validation remain open work; readers are invited to carry it forward.

The order of these chapters is not arbitrary. The first three establish the constraints; Chapter

4 examines how those constraints become concrete costs in life and bodies; Chapters 5 and 6 address the gaps left when constraints are bypassed; Chapter 7 returns to action. Each chapter picks up where the previous one leaves off, so the book is best read in sequence; a chapter can also be read on its own, but each chapter's conclusions depend on what has already been established, and readers who skip ahead will need to return to the relevant premises.

This book connects engineering questions with questions of value through an analysis of constraints. Unproductive waiting and redundancy should be eliminated, and the costs of execution and verification reduced, while the functions of safety, trust, and responsibility must be retained or taken over by other reliable mechanisms. Physical models identify the relationships at work; decisions about specific deployments must also take account of task data, organizational conditions, and the people who bear the consequences.

If AI can replace an ever larger share of what people produce, what grounds remain for people to have a place? The question concerns ability, but also the standards by which society judges people. The human condition encompasses bodies, time, relationships, judgment, and responsibility for consequences; output describes only part of it. Recognizing the value of these dimensions requires corresponding arrangements for rights, distribution, and responsibility. Freedom is not the absence of viscosity, but the ability to choose it, distribute it, and bear it.

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Chapter 1

The Evolutionary Paradox: The Gap Between Technological Acceleration and Physical Constraints

Open your phone and the screen's contents change by the day; open the window and the street below has barely changed in a decade. The aging drains still flood after a downpour, the elevated roads still clog at rush hour, and a food delivery still takes a rider half an hour of cycling through the rain. A generative model can produce a street video convincing enough to pass for the real thing in sixty seconds, yet it cannot go downstairs to collect a parcel for you. Both speeds are running; the problem is that they are so often treated as the same speed.

1.1 Technological Acceleration: The Illusion of Collapsed Timescales and the Viscous State

1.1.1 Misaligned Perception

The faster a screen's contents update, the easier it becomes to mistake the pace of updates for the pace of change. Judgments about employment, education, industry, and policy therefore often conflate two processes: information being generated in a space of representations, and materials, energy, and living systems being rearranged in reality. Copying and retrying allow the first to expand rapidly; the second must still pass through execution, verification, and coordination. Force the two clocks into alignment, and resistance disappears from the judgment.

Two Speeds

The shock of generative AI comes from a change in the speed of writing, drawing, programming, and planning. A proposal appears in seconds, and people readily extrapolate that speed to the world the proposal describes. The speed of bits and the speed of reality become

conflated here. The former shows how quickly candidates appear; the latter also depends on whether those candidates can pass through specific systems.

AI has indeed sharply reduced the costs of search, generation, simulation, and expression. It can rapidly produce architectural designs, traffic models, drug candidates, business plans, code, and policy documents. Yet reality is neither a text to be rewritten directly nor a picture to be redrawn at will. It consists of materials, energy, bodies, time, equipment, institutions, responsibility, and complex feedback. Generating a proposal does not mean implementing it; a model's answer does not mean a real system has completed its state transition. AI can make finding possible solutions cheap without automatically making their realization equally cheap. Its essential capability, in other words, is to discover, generate, and screen paths faster, rather than abolish the real costs of executing and verifying them and bearing their consequences.

There is an everyday way to gauge this mismatch. The clock of public perception has been set to the 22nd century, while the physical world still moves along the tracks of Newtonian mechanics, and some infrastructure, including aging power grids and drainage systems, even keeps a 19th-century rhythm. Advances in algorithms have widened the rift between expectations and physical reality rather than closed it. The gap itself creates no danger; the danger lies in losing sight of it, and that loss of perception begins when experience on a screen is taken directly as experience of the world. To see how the misreading arises, we need to examine the two sides separately.

Digital representations are cheap to copy and transmit, and their working states are easy to restore. Once a model has been trained, each additional generation mainly consumes the computing resources needed for inference; drafts and test versions can be replaced, and code can be rolled back to a saved state. Users can therefore retry quickly. This convenience depends on the operation having triggered no external consequences: information already leaked, transactions already executed, and errors already introduced into production systems each carry their own recovery costs.

Concrete examples make the point more clearly than a list of properties. A city-street video with correct lighting and apparently plausible physics can be computed in sixty seconds, through nothing more than microscopic transitions of electrons on silicon chips. A translation, a summary, and a code completion rest on the same transitions, billed in thousandths of a yuan. In a remarkably short time, writing, translation, summarization, search, customer service, programming, and image generation have been recast from skills requiring training into interfaces requiring a call. The speedup people experience is real, but what has accelerated is only the first part of the chain.

The difference between the two domains can be written as growth laws with different exponential terms,

$$u_{\text{bits}}(t) = u_{\text{bits}}(t_0)\, e^{a(t-t_0)}, \qquad u_{\text{atoms}}(t) = u_{\text{atoms}}(t_0)\, e^{b(t-t_0)} \tag{1.1}$$

Here u_{bits} and u_{atoms} denote the amount of work performed on a selected task per unit cost, t_0 is the starting point of the comparison, and a and b are growth rates fitted over a given interval. If $a > b$, the logarithm of their normalized ratio grows at rate $a - b$. This is a working form for describing the separation of speeds. Computing power, energy conversion, and material throughput each have their own statistical definitions, and their growth rates must be established from the corresponding data; Figure 1.1 illustrates the difference with schematic curves. A particular system may exhibit exponential, linear, stepwise, or saturating growth; technology and process jointly determine the shape of its curve.

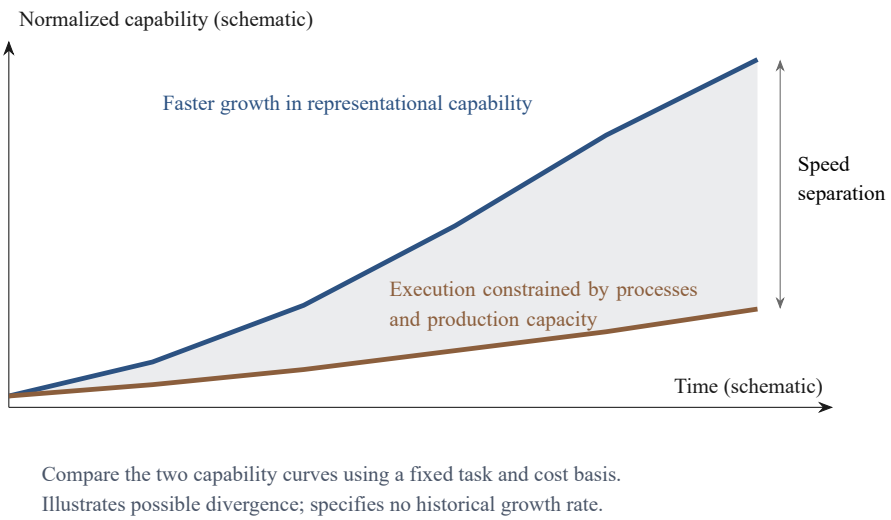


Figure 1.1 A schematic of two speeds. Both axes use illustrative scales; the curves show how representational capability and execution capability may improve at different rates. Actual growth rates must be measured for a fixed task and cost basis. The distance between the curves is not a generalized viscosity coefficient.

The Twentieth Century Outside the Window: Locked in by Mass and Entropy

When this capability enters material processes, execution imposes its costs anew. Traffic in a video is computed, while real vehicles need energy and must undergo braking, steering, and contact control. Drawings for a sea-crossing bridge can be revised quickly, while construction still requires materials, equipment, and people to be organized. Steel strength, the development of concrete strength, and foundation conditions all enter the same engineering account. Concrete's 28-day strength is a common testing benchmark, and its mix, temperature, and curing conditions can change how quickly strength develops; faster drawing generation changes only the design stage of this chain.

The second constraint is the installed base. Cities rest on road networks planned in the past, and underground utilities pass through neighborhoods where people already live. How much

infrastructure autonomous driving and intelligent urban systems need to modify depends on the technical approach; some reuse existing roads as far as possible, while others add sensing, communications, and power facilities. Every approach must connect vehicles, road users, maintenance systems, and arrangements for responsibility. An old street cannot be deleted like code, because real people live there and real goods move through it. An upgrade in the real world is, first of all, a transition that must keep the system running.

The third constraint is entropy. A simulator's designer specifies its rules and boundaries; the real environment also contains unmodeled disturbances, changing states, and other participants. AI may drive one hundred million accident-free kilometers in a simulator, yet a pedestrian darting across a real road, a sudden downpour, or an obscured sign can bring the system to a halt. The disorder of the physical world extends far beyond what digital logic covers. A road crew deals with more than laying pavement: ground subsidence, conflicting utility lines, residents' complaints, and administrative coordination all enter the work. Algorithms can simulate order; they cannot instantly erase entropy increase in reality.

The Category Error Between Simulation and Reality

Of the three properties, reversibility is the easiest to misuse. Time in a digital twin has a degree of reversibility, and rolling back requires only a command, encouraging aggressive iteration because the penalty for failure approaches zero. The world of atoms offers no such retreat. A demolished bridge pier cannot be rolled back, a drug taken by mistake cannot be untaken, and a scrapped battery cannot be regenerated. Reversibility and irreversibility differ in kind, rather than degree: they determine the shape of the trial-and-error cost function. Transplant a methodology developed in reversible systems directly into irreversible ones, and the result is often greater expense rather than greater speed.

A subtler move is to equate predictive capability directly with the capability to intervene. A model can learn from data the relationship between a cup's falling and breaking, but to catch the cup, a robot must also estimate its instantaneous position, produce the right torque, establish contact, and correct its movements through feedback. Learned models can contribute to every step, but success at one step cannot stand in for success at the next. The smoother a generated video becomes, the clumsier the construction crew outside the window appears; that comparison omits most of the work the crew actually has to do.

Public arguments about a technological takeover often skip this distance: once simulation looks real, execution is only a matter of time. What must actually be crossed involves sensing, actuators, energy, contact, and feedback. Pixels may have converged, but torque still has to be controlled anew; drawings may have passed review, but components still have to withstand their loads. The crucial issue lies in the blank space between those two records of achievement.

The central judgment of this section can therefore be condensed to one sentence: **Speed in the space of representations must be accounted for anew in the real world.** The physical world has high viscosity, made up of energy costs, material properties, logistical reach, and the

complexity of social cooperation. This viscosity absorbs the shock wave of AI's exponential growth and forms a first line of defense against technological fervor, slowing the rate at which algorithms sweep through the real economy and giving people time to breathe and adapt. That is no bad thing, but it rests on an easily overlooked premise: viscosity serves as a buffer only when it is acknowledged; if it is treated as nonexistent noise, people will leave no margin for it.

We can now answer the opening question. A screen changes by the day while a street stays the same for a decade, and the difference between these speeds is itself a fact that needs no further explanation. What does need explaining is why public judgment keeps leaning toward the faster side despite so obvious a gap. The answer lies not in the physical world, but in how information is filtered, amplified, and priced. This mechanism has had a name for many years.

1.1.2 Survivorship Bias in the Media

How Attention Is Filtered

This mismatch has a source. The lightest, fastest technologies, the ones easiest to demonstrate, attract the most attention. Video generation can be shown on the spot, while the substation powering the servers rarely appears on camera. Chip manufacturing, power transmission, cooling, mining, and construction each advance at their own pace, which cannot be inferred from how often a chat window updates. The interface is shown; the world supporting it is left out.

The filtering mechanism is straightforward. Media and capital markets reward things that can be demonstrated, replicated, and valued quickly; pouring concrete for a data center has no publicity value, while a model that generates video does. As a result, the demonstrable part gets a full narrative, while the part that cannot be shown does not even get a name. This asymmetry reinforces itself: the more something is talked about, the more readily it attracts resources and attention, and the more readily it attracts resources, the more likely it is to remain in the story. After decades of this process, the public receives a sample that has already been filtered, then uses it to infer the properties of the whole.

The bias has a psychological version too. The human mind has a natural preference for exponential growth curves and little patience for linear ones. Software tells precisely the exponential story: user numbers increase tenfold in months, and valuations multiply dozens of times over in two years. The physical world tells a linear story: base stations are built one at a time, fiber is laid meter by meter, and yields improve a percentage point at a time. The former lends itself to publicity; the latter only to execution. The resulting fallacy of induction is to believe that the whole world has become weightless.

Once accepted, that fallacy supports far-reaching inferences. Every industry, whether healthcare, education, energy, or manufacturing, is expected to leap forward along a Moore's-

law curve; any industry's lag is attributed to insufficient digitization or an AI that is not yet smart enough. This leads to a specific prediction: make a model's brain powerful enough, and it can take over every assembly line and every heavy truck as readily as it takes over a chat window. People remember the lightness of code and forget the rigidity of gravity. The hardest part of correcting this error is that it keeps confirming itself until it is disproved: the samples filtered out never speak.

The Silent Foundation: An Atomic World Governed by Thermodynamics

What this narrative folds out of view is far larger. Energy extraction, heavy freight, infrastructure construction, precision manufacturing, and agriculture—the industries absent from the headlines—form the greater part of the global real economy and obey the laws of thermodynamics rather than Moore's law. Here, efficiency gains come from confronting and transforming the physical world, rather than copying and pasting; there is no miracle of zero marginal cost, only the iron rule that every act of physical work requires energy. The misjudgment persists because feedback from this side arrives too slowly to interrupt the story told on the other.

Telecommunications provides the clearest example. A phone's signal bars suggest instantaneous coverage, yet behind them lie conduits, fiber, equipment rooms, towers, and maintenance. Laying fiber requires coordination with roads and underground utilities, while upgrading equipment requires arranging power, transport, installation, and testing. Updating a protocol and updating a physical network follow different critical paths. The difference comes from the operations required, rather than a universal multiplier by which "atoms are inherently slower than electrons."

Energy makes this distance greater still. Dispatch algorithms can improve the use of existing capacity, while adding capacity still requires equipment, land, financing, construction, and a grid connection. Berkeley Lab's 2026 edition of *Queued Up* reports approximately 2,060 gigawatts of generation and storage capacity in US interconnection queues at the end of 2025; in regions with available data, projects completed in 2025 had a median interval of more than five years between requesting interconnection and entering commercial operation (Lawrence Berkeley National Laboratory and GridTracker 2026). The queues include projects that will eventually withdraw, so queued capacity cannot simply be counted as future supply. Designing a generating facility, submitting an interconnection request, and actually delivering electricity to a load are three different things.

The contrast is sharper in precision manufacturing. Lithography equipment embodies highly complex knowledge, yet manufacturing, assembly, transport, installation, and calibration must each move beyond knowledge. Yield ramps emerge from engineers' continual encounters with material tolerances and process variation. Algorithms can shorten the feedback cycle and make each round of experimentation more valuable; the equipment must still pass through that round of feedback.

The Truth About the Cloud: The Heaviest Structures on Earth

Survivorship bias reaches its most extreme expression in the collective misreading of the word "cloud." In the public imagination, the cloud is light and immaterial: uploaded data seems to escape Earth's gravity and enter a pure paradise of bits, as though a life in the cloud were freed from physical-resource constraints. Physically, the cloud may be among Earth's heaviest and most resource-dependent structures. The cloud is not in the sky but anchored in physical facilities: data centers in deserts, deep in the mountains, or beneath the sea all require extensive engineering for equipment, power, cooling, and protection.

The physical foundations of computing have now forced their way into grid planning. The International Energy Agency's *Powering Artificial Intelligence*, published on 14 October 2026, identifies available network capacity, interconnection queues, and permitting times as constraints on data-center construction (International Energy Agency 2026b). Chips, racks, electrical equipment, copper cables, and fiber each have their own supply chains; cooling methods, climate, and load jointly determine water and electricity demand. Each increment of capability in the cloud requires answers on the ground: where will it go, how will it be powered, and where will its heat be discharged?

Seen in terms of how information spreads, this bias needs no deliberate author. What can be shown automatically gains attention and what cannot be shown automatically loses it, together producing a picture of the world in which weight is systematically underestimated. This also explains a peculiar phenomenon: the more heated the discussion of AI becomes, the less attention electricity, cooling, minerals, and supply chains receive, because they remain off camera. Correcting this bias requires a concept that can hold both sides in view, rather than more reporting.

Naming the Resistance: Technological Viscosity

This chapter uses **technological viscosity** to describe the difference. Technological viscosity is the generalized damping encountered when intelligence, algorithms, or technical proposals move a real system from state A to state B. The point goes beyond saying that reality changes slowly: real-world change depends on whether a solution can be executed, verified, maintained, and coordinated, and whether someone can bear the consequences of failure, as well as on whether a solution exists. AI's core advantage is to reduce search costs, but a real state transition must also pay for execution, verification, maintenance, coordination, and responsibility. The key word is "generalized." In physics, viscosity is the property within a fluid that resists relative motion, and this book borrows precisely that meaning: resistance arises from the medium's internal structure rather than an external obstacle. What intelligence actually encounters on entering a real system is the character of its medium.

A counterintuitive but consequential inference follows: **The cheaper generation becomes, the scarcer trustworthy results become; the more abundant the proposals, the more reality's capacity to absorb them becomes the bottleneck.** Where the structure of scarcity

changes, the structure of competition changes too. In many fields, generation used to be scarce: the ability to write text, draw images, find molecules, propose solutions, or produce code. That capability is now expanding rapidly, candidate proposals are multiplying, and the information space is beginning to experience inflation. The problem is no longer an absence of proposals but too many of them, no longer an absence of answers but answers that are too cheap, and no longer a shortage of expression but a shortage of trustworthy expression. Future bottlenecks will increasingly move from generation to verification, from creation to confirmation, and from possibility to what can be borne.

Every real state transition must pass through at least three constraints. It needs sufficient resources to drive it, sufficient time to unfold, and a way to propagate through existing structures. A minimal model for analyzing this damping is RTG: **resource barriers** (R), **temporal rigidity** (T), and **topological coupling** (G). Resource barriers answer whether a change can be driven, temporal rigidity whether it can happen in time, and topological coupling whether local change can propagate into systemic change. Together they determine the degree of viscosity encountered as intelligence enters a real system, and these three constraints also provide the minimal framework for understanding the book.

1.1.3 The Material Dependence of Virtual Escapism

Collapsing Inward

Since the real world is heavy and slow, the easiest response is to go around it. An impulse to escape follows: confronted by a burdensome physical world, technological imagination keeps moving toward dematerialization, seeking freedom at the level of pure symbols. This escape contains an internal paradox: the more the virtual world flourishes, the more deeply it depends on an energy-hungry physical foundation. Trying to escape gravity demands more fuel.

Over the past decade, this tendency has taken a more complete form that we might call virtual escapism. It begins with real frustrations: housing in skyscrapers is unaffordable, urban congestion is entrenched, and the routes to changing one's circumstances through physical relocation are narrowing. A civilization can command only a finite amount of power, making the reconstruction of the physical world both slow and costly. If securing a physical coordinate in the world of atoms costs so much, why not move life's center of gravity into the world of bits? At a deeper level, this echoes an ancient impulse: to escape material shackles through pure knowledge and treat the body as a burden to be shed. Silicon Valley becomes the new preacher, promising code as a path to a liberated form of existence.

The central change is a reordering of values. To those seeking escape, the physical world is losing its appeal, even becoming a burden to transcend. Clothing, food, shelter, and transport are demoted to the minimum requirements for sustaining vital signs, and a strategy of physical minimalism and spiritual abundance begins to spread. If a headset can put you on a beach, why

take on a thirty-year mortgage to secure a place downtown; if work can be remote, social life virtual, and parties held in digital settings, big-engine cars and rush-hour congestion lose their purpose. This is a deliberate spatial retreat: compressing the space of a life from an expanse of geographical coordinates into a fiber-optic connection.

A political vision accompanies this way of life. Silicon Valley thinkers, including Balaji Srinivasan, have proposed the "network state": begin in the cloud and establish a physical presence later, using software's flexibility to circumvent the rigidity of traditional states. From physical first principles, these projects share a flaw: they depend parasitically on the physical foundations of existing states. Digital nomads scattered across Bali or Lisbon neither generate electricity nor lay submarine cables, and cannot secure shipping lanes or handle large-scale epidemics. They remain attached to the parent civilization of steel and fossil energy that they seek to escape. A digital utopia unable to establish independent energy supplies and closed material cycles can only be a club dependent on the real economy, rather than an independent form of civilization.

The Thermodynamic Bill

This vision contains a physical paradox concealed by technology's halo. The virtual world has no independent existence, but is an arrangement of the physical world: at the microscopic level, controlled electrons in silicon chips, precise reversals of magnetic states on hard-drive platters, and optical signals traveling by total internal reflection through fiber. Keeping virtual systems running requires energy for physical equipment and the release of waste heat from actual irreversible processes. That is the bill presented by the second law of thermodynamics, and there is no way to skip it.

Every inference, rendering operation, and act of communication is performed by specific circuits and equipment. Where the electricity comes from, and what produces it, depends on location, time, and procurement arrangements. Moving an activity online may save transport and space while increasing demand for computation, communications, and equipment. Its net effect must account for both the activity displaced and the infrastructure added. The virtual world can change the bill; it cannot detach itself from where the bill is incurred.

A life freed from the constraints of physical space depends on spatial infrastructure of immense cost. Someone can retreat to the mountains as a digital nomad only if those mountains have stable electricity, high-bandwidth communications stations, and a logistics network capable of delivering emergency medicine within an hour. The cloud is no cloud in the sky, but enormous data centers lying on the ground, enclosed in tens of thousands of tons of reinforced concrete; these buildings need protection against earthquakes, fire, and moisture, along with vast cooling systems to carry away computing's waste heat. Servers, chips, fiber, and copper cables are physical objects, and their extraction, smelting, manufacture, and transport follow the traditional logic of heavy industry. Every moment of lightness in the virtual world imposes a heavier burden on the world of atoms.

The continuous operation of digital systems depends on conditions that can fail. Submarine cables, power supplies, underlying dependencies, and maintenance teams connect to different paths of failure. Redundancy, recovery at another site, and isolation can improve reliability, at an additional cost in resources. Maintenance costs and robustness must be compared for the particular architecture: thermodynamics does not dictate that greater order means greater fragility. The physical requirements of virtual life must be accounted for in these facilities and this work.

Another part of the bill has been transferred geographically. The interface keeps the result in view while sending power supply, cooling, repairs, and waste disposal elsewhere. Tidiness on a screen is not thermodynamic entropy; establishing the physical costs requires examining the equipment performing these operations and its environment. For a system that forms an isolated whole together with its environment, the second law is written as

$$\Delta S_{\text{local}} + \Delta S_{\text{environment}} = \Delta S_{\text{total}} \geq 0 \tag{1.2}$$

All three entropy changes are measured in joules per kelvin. If local entropy decreases, environmental entropy must increase by at least the same amount, with equality corresponding to an ideal reversible process. Electricity conversion, cooling, manufacturing, and transport involve irreversible processes, and actual costs also depend on the particular route. This overall account constrains physical processes; it cannot assign a fixed entropy price to every image, item of knowledge, or inch of a tidy screen. Water use, pollution, and electronic waste must each be measured separately.

Life’s Anchor: The Body That Cannot Be Uploaded

Beyond energy and infrastructure lies a more fundamental constraint: life itself. As long as consciousness remains anchored in neural networks and cannot be uploaded, and as long as birth, aging, illness, and death remain biological processes beyond our mastery, people cannot leave this viscous world altogether. The body depends on a narrow range of physiological conditions: core temperature must remain around 37 degrees Celsius, and its functions depend on precise biochemical feedback; substantial temperature deviations or uncontrolled infections can threaten life. Evolution has written this fragility into our biology. Software can be upgraded in an instant, but wounds cannot heal in an instant, and brain development cannot have its timescale compressed. This temporal incompressibility constitutes life’s most fundamental viscosity.

The supposedly lightweight life is, in substance, an exchange between order and disorder. When someone is immersed in a frictionless virtual experience, its smoothness does not come from nothing: a complex social division of labor transfers its physical burden to the underlying systems of atoms. The lightness of digital life must still be settled at the physical level: the electrician maintaining servers, the rider out in wind and rain, the cooling tower releasing waste

heat. A smooth virtual experience ultimately still depends on equipment, labor, and dissipative processes in the physical world.

There is therefore no escape. Virtual escapism is a cognitive retreat in the face of physical barriers: it seeks cheap shortcuts in the world of bits and becomes more deeply dependent on its substrate. This reveals an inverse relationship: **Every inch of order on the screen is paid for in dissipation off it.** The higher one hopes to ascend in the world of bits, the deeper one's roots must reach into the world of atoms. Without a thermodynamic foundation, there is no virtual dream; without a healthy biological medium, consciousness has nowhere to dwell. The capital-intensive industries that look obsolete—road building, mining, electricity generation, and grain production—become some of civilization's hardest parts to dislodge precisely because they sustain the digital world's operation.

This brings the problem to a more fundamental point. Physical constraints have so far appeared as obstacles: they keep proposals from being implemented, pull escapees back, and discount promises of acceleration. But widen the scale of observation, and those same constraints show another face. They do more than slow change; they also keep systems from falling apart. Seeing that side requires reading viscosity as damping, rather than as resistance.

1.2 The Dual Nature of Viscosity: Damping as a System Stabilizer

1.2.1 The Physical Anchor: Stability in Dissipative Structures

Consider an anomalous fact. If intelligence were expanding in the exponentially explosive tsunami described by the most radical forecasts, the material world should already have been engulfed once computing power passed a critical point. What actually happens is that capabilities in the domain of bits expand rapidly while civilization's physical and social structures remain remarkably viscous. This calm cannot be attributed to a special human privilege or to technological stagnation; the reason is simpler: civilization has never evolved in a frictionless vacuum. Two forms of damping stand between exponential growth in the world of bits and linear renewal in the world of atoms.

We need to distinguish execution from confirmation here. Execution is constrained by resources, process time, and existing structures, while confirmation depends on rules, evidence about the object in question, and arrangements for responsibility. Both affect the pace of deployment, and engineering improvements can alter both. The length of the resulting buffer must be judged for the particular system. A damped oscillator provides an initial illustration: under the same external drive, differences in inertia, damping, and restoring force produce different responses.

$$m\frac{d^2x}{dt^2} + \gamma\frac{dx}{dt} + kx = F(t), \qquad \zeta = \frac{\gamma}{2\sqrt{mk}}, \qquad \tau \approx \frac{\gamma}{k} \quad (\zeta \gg 1) \tag{1.3}$$

x represents how far a real system departs from its original state, and m, γ, and k are equivalent parameters: if x is taken as a length, m is the equivalent inertia, corresponding to established physical mass and sunk capital, in kilograms; γ is the equivalent damping coefficient, corresponding to the book's three forms of viscosity—resources, time, and structure—in kilograms per second; k is the equivalent restoring stiffness, representing the system's ability to maintain itself, in kilograms per second squared; $F(t)$ is the external driving force, or technological shock, in newtons; ζ is the dimensionless damping ratio; and τ is the relaxation time in the overdamped regime, in seconds. For a constant step input of amplitude F_0, starting from equilibrium at rest, when $\zeta \gg 1$ the response approaches the new equilibrium monotonically, with a slow-mode time constant of approximately $\tau \approx \gamma/k$: greater damping means slower relaxation. The equilibrium displacement is determined by F_0/k and the slow timescale of the transition by γ/k, so they can be examined separately: an abrupt change is stretched into a transition because damping is greater, rather than because the driving amplitude is smaller. The model's scope must be explicit: it is a linear second-order model with constant coefficients, used here only to illustrate the relationship between damping and response time through a constant step response starting from rest; the three real forms of viscosity are not constants and redistribute with technological inputs, so τ provides only an order-of-magnitude estimate, not a prediction of a particular year or rate of progress.

Keeping things running requires continual payment. Bodies rely on metabolism, servers on power and cooling, and transport networks on inspection and repairs; buildings can endure unattended for a considerable time while gradually losing performance as their materials age. These structures have different maintenance mechanisms and timescales. An old road is often harder to work on than an empty site because it must still carry traffic while being repaired. Maintenance is the price of keeping a system in existence and in use.

Resource Barriers: The Physical Bill That Must Be Paid

The first expenditure occurs along the path of material transformation. Ironmaking requires the reduction of iron oxides, while cement production requires reactions to be driven and hot materials to be handled. Fuel, equipment, reactions, and heat flows all belong in the account. Activation barriers, reaction free-energy differences, and actual factory energy consumption are three different quantities; catalysis can change the path and its barriers, processes and energy routes can change actual expenditure, and a catalyst does not change the free-energy difference between given initial and final states. Intelligence finds better paths within these conditions, and every path must still actually be traveled.

A change needs sufficient resources to drive it: these are resource barriers (R). Resource barriers comprise the energy, materials, equipment, capital, space, supply chains, and main-

tenance costs that real state transitions must pay. Information can describe a bridge, but the bridge must be built from steel, cement, machinery, land, and labor; an algorithm can optimize grid dispatch, but electricity still needs generation, transmission, storage, and peak balancing; AI can generate a robot's action policy, but the robot must rely on batteries, motors, sensors, joints, materials, and a maintenance system to move. AI can reduce design costs but cannot cancel the physical bill. The cloud is not light; it merely hides its atomic burden behind the user interface. Intelligence at larger scales depends ever more on chips, electricity, data centers, cooling systems, manufacturing capacity, and infrastructure.

The human body belongs on this bill too. A daily intake of 2,500 kilocalories corresponds to an average metabolic input of about 120 watts over the full day, sustaining temperature regulation, cognition, movement, and repair simultaneously. Comparing a robot with a human requires selecting the same task, load, quality of work, and duty cycle, and measuring each one's average input. An adaptable human may be economical in a cluttered setting, while a specialized machine may win by a wide margin in a standardized operation. The difference must be measured, rather than read directly from the labels "carbon-based" and "silicon-based."

The conversion gives the arithmetic behind that bill,

$$P_{\text{human}} = \frac{E_{\text{food}}}{\Delta t} = \frac{2500 \times 4.18 \times 10^3}{86400} \approx 1.2 \times 10^2 \text{ W}, \qquad \frac{P_{\text{machine}}}{P_{\text{human}}} = \frac{(1\text{–}3) \times 10^3}{1.2 \times 10^2} \approx 8\text{–}25 \tag{1.4}$$

P_{human} is the average metabolic power corresponding to dietary energy intake, in watts, E_{food} is one day's dietary energy, and Δt is the duration of a day. At 2,500 kilocalories, average power is about 120 watts, while mechanical output must be calculated separately for the activity and efficiency involved. P_{machine} should be the measured average input power for the same task; the 1 to 3 kilowatts shown in the equation are only illustrative comparison values, yielding ratios of about 10 to 25. Energy prices, equipment costs, and quality of work enter the economic account separately.

This scissors gap has another easily overlooked property: it sets a price, not a direction. It protects labor's relative cost advantage, rather than any particular skill. Once a task has been sufficiently structured and the human actions broken into an enumerable set of steps, the energy advantage ceases to be decisive, because the comparison is then between two specialized devices rather than general and specialized intelligence. The thickness of resource barriers differs greatly across settings, which is why it must be calculated case by case rather than cited wholesale.

Temporal Rigidity: Processes That Cannot Be Compressed

A change needs sufficient time to unfold: this is temporal rigidity (T). Temporal rigidity is the duration of real processes that cannot be compressed at will. AI can generate building plans instantly but cannot make concrete cure instantly; it can design molecules rapidly but cannot

make long-term side effects appear in a human body instantly; it can simulate traffic but cannot change roads, vehicles, regulations, insurance, and driving habits all at once. Physical systems involve heat transfer, diffusion, fatigue, and wear; living systems involve growth, metabolism, immunity, and long-term feedback; engineering systems involve testing, commissioning, certification, and the accumulation of reliability; institutional systems involve negotiation, legitimation, and behavioral adaptation. Time is not simply inefficiency: some delay is redundant, while some marks the boundaries of safety, trust, and responsibility. Compressing time often transfers risk into the future rather than providing free acceleration.

Temporal rigidity belongs to specific processes. Concrete's strength development depends on its mix and curing conditions; crop growth on variety, light, temperature, and physiological processes; nuclear plants and high-speed railways must also pass through coordination of land, environmental conditions, supply chains, and construction. Infrastructure service lives set the pace of renewal too: housing and bridges are designed on the scale of decades (Miatto et al. 2017), and modifications must accommodate existing use. Parallel construction, prefabrication, and better experimental design can shorten parts of a cycle, while material maturation, the emergence of damage, and the reconciliation of interests each have necessary conditions. Temporal rigidity must be identified stage by stage.

The disconnect has a direct consequence. AI might compress design from a year to a day, yet struggle to compress construction from three years to three months. For a considerable time, people's minds may inhabit a future woven by AI while their bodies remain lodged in an old shell that changes slowly. Institutions face a similar situation: a technology gains legitimacy through negotiation and trial and error, and the time negotiation takes is often determined by how many people need persuading rather than by technological efficiency. Mistaking temporal rigidity for inefficiency produces a recurrent error: assuming that replacing tools faster will make the whole process faster. What actually accelerates is the production of proposals, rather than their implementation.

Topological Coupling: Embedded in Existing Structures

A change must propagate through existing structures: this is topological coupling (G). Topological coupling comprises the connections, interdependencies, and feedback loops among system components, together with the limits that past choices impose on present options. Real systems are complex networks of equipment, people, institutions, habits, standards, interests, and environments, rather than collections of isolated variables. A smarter autonomous-driving algorithm does not automatically optimize the transport system; a better medical model does not simultaneously adjust hospital procedures, doctors' responsibilities, patient trust, health-insurance payments, and the legal system; a better energy device does not instantly align the grid, storage, land, capital, and policy structures. Many technologies fail because they cannot fit into the structure of an existing system, rather than because their capabilities at an individual point are insufficient.

Moravec's paradox illuminates this difficulty. A game with explicit rules can have an enormous search space while still specifying its states, rules, and winning conditions clearly. A robot handling a pile of clothes must contend with occlusion, deformation, contact, and friction at once. Actions humans perform easily rest on bodies adapted to their environments and sensorimotor capacities developed over time. The difficulty comes from the structure of the environment and the task; calling it "high entropy" cannot substitute for analysis.

The exponential growth here has a specific source: the dimension of configuration space,

$$N_{\text{configuration}} = m^n, \qquad \ln N_{\text{configuration}} = n \ln m \tag{1.5}$$

N_{config} is the number of configurations obtained by discretizing n degrees of freedom into m states each, and n, m, and N_{config} are all counts. The number of configurations grows exponentially with the number of degrees of freedom, so exhaustive point-by-point search soon becomes infeasible. Practical control can exploit geometry, learning, feedback, and the body's passive dynamics to narrow the search, with computational cost depending on the method. Grasping also faces uncertainty in contact positions, friction, and object deformation. State-space size supplies one source of difficulty; explaining Moravec's paradox also requires evolutionary experience and the mechanisms of perception and control.

The disorder of the physical world does protect people, but it protects tolerable ambiguity rather than all labor. Judgment in the field is costly because decisions must be made with incomplete information and their consequences borne, rather than because hands are involved. Once a high-entropy setting is sufficiently structured—a warehouse converted to standardized shelving, for example—that protection diminishes. This also shows that R, T, and G are not three independent walls: tidying a setting reduces topological coupling while also moving that work out from under the protection of resource barriers.

Three Media: Bits, Atoms, and Life

Different media absorb AI differently. Bits, atoms, and life are three typical media intelligence encounters on entering the real world. They form a progressively heavier gradient rather than a static classification: bits are easier to generate and roll back, atoms must bear materials, energy, and maintenance, and life adds growth, compensation, long-term consequences, and interventions that cannot easily be undone. To see how this gradient grows heavier, we need to examine each medium's mechanism of absorption.

The domain of bits includes text, images, code, data, models, and virtual spaces. Copying, transmission, modification, and rollback all cost little, so this domain was the first transformed by generative AI. Writing, translation, summarization, search, customer service, programming, and image generation have expanded rapidly because they operate in low-RTG environments. AI's search and generation capabilities translate almost directly into visible output. Yet viscosity in this domain is not zero. Its low viscosity lies mainly in generation, rather than trustworthiness. AI can produce text, images, video, and code cheaply, but society needs reliable

code, authentic images, verifiable knowledge, and judgments for which someone can be held accountable. Once generation ceases to be the problem, verification becomes the problem; once expression ceases to be the problem, judgment becomes the problem; once information ceases to be the problem, trust becomes the problem.

The domain of atoms includes manufacturing, construction, transport, energy, logistics, robots, chips, cities, and infrastructure. It faces mass, energy, friction, inertia, material strength, spatial constraints, and maintenance costs. The atomic world cannot be copied, deleted, rewritten, or rolled back like text. AI can optimize design, dispatch, and control, but cannot abolish construction, transport, assembly, commissioning, losses, and responsibility. Embodied intelligence concentrates the viscosity of this domain. Ordinary generative AI works mainly in a space of representations, where an erroneous output can be regenerated; a robot must turn representations into actions, predictions into control, and objectives into forces, contact, motion, and feedback. An error in text can be regenerated, while an error cast into a component must be chiseled out and the work done again. Chatbots can therefore enter screens quickly, while general-purpose robots, autonomous driving, and intelligent infrastructure enter the real world much more slowly than public expectations suggest.

The domain of life includes bodies, cells, drugs, heredity, ecology, and biological feedback. It is more complex than ordinary atomic systems because life is a complex system that responds, compensates, adapts, rebounds, and even evolves, rather than a static material. AI can accelerate target discovery, protein prediction, molecular design, and experimental screening, but cannot compress toxicology, pharmacokinetics, clinical trials, long-term safety studies, and observations of individual differences at the same rate. DNA can be written as a sequence, but gene function is not textual meaning; it must be realized within cellular environments, developmental stages, regulatory networks, and the surrounding ecology. Life is not software and cannot be rolled back as easily as software. Discovery may accelerate, while verification encounters stronger temporal rigidity and nonlinear risks.

The basic picture of the AI era is therefore uneven absorption rather than uniform acceleration. Bits flow, atoms resist, life responds. AI spreads rapidly through low-viscosity domains, embeds itself slowly in high-viscosity ones, and encounters stronger constraints of verification, responsibility, and trust in living and social systems. AI changes the world through intelligence being absorbed, impeded, amplified, distorted, and redistributed to different degrees in different media, rather than penetrating them all at the same speed.

The relative levels are not fixed; a medium's properties change with conditions. Text acquires responsibility when it is used for decisions, code acquires rollback costs when it enters production, and a drug candidate acquires temporal rigidity when it enters clinical trials. The medium sets an initial level of viscosity, rather than its final numerical value. This is why the same model capability can cause disturbances of very different magnitudes in different settings.

R, T, and G are not isolated; they trade off against one another. Compressing time often

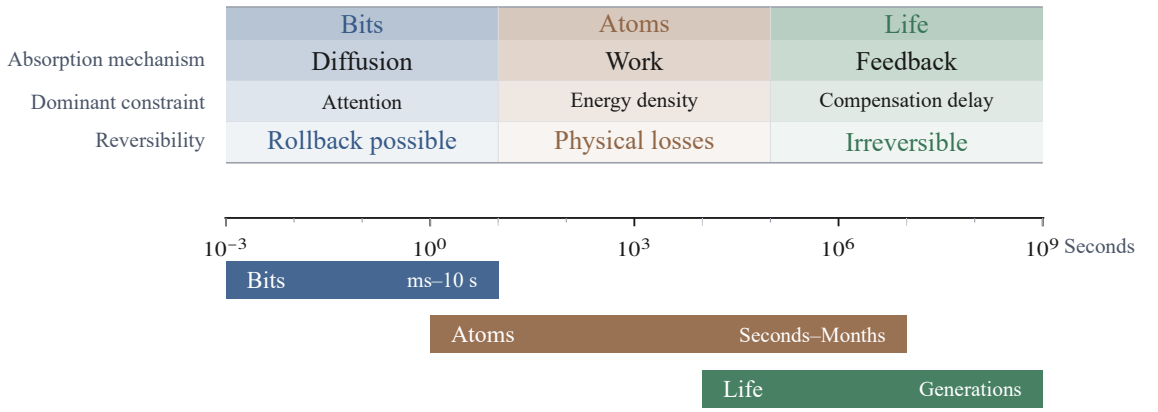


Figure 1.2 Bits: diffusion; atoms: work; life: feedback. In the domain of bits, low viscosity is concentrated in generation; trustworthiness remains costly; in the domain of atoms, every stage consumes matter and energy; in the domain of life, consequences unfold across time and generations.

requires more resources; saving resources often takes more time; reducing local friction often means adding interfaces, standards, redundancy, monitoring, audits, and compliance structures. This relationship deserves a pause. It means that technological progress often redistributes resistance among resources, time, and structure rather than eliminating it. A system that feels lighter to the user may have become heavier behind the scenes; an apparently automated process may have become harder to maintain as a whole. Thus, **The real question is not whether a technology is accelerating, but which kind of viscosity it reduces and where it transfers that viscosity.** Higher energy consumption buys time, longer time buys safety, complexity buys local efficiency, and redundancy buys reliability; every exchange has a cost, and the cost has simply moved to another place.

Resources, time, and structure jointly establish the conditions of real change. Technology can reduce total costs or transfer a stage's burden elsewhere; the two outcomes need separate accounts. Physical constraints set the lower bounds of processes, while engineering and institutions determine how actual costs arise and how they are shared. Once the conditions for execution have been met, another question remains: on what grounds can a generated judgment be relied upon?

1.2.2 The Semantic Anchor: Incompleteness and the Need for Openness

Conditions of execution explain how a proposal enters reality; conditions of confirmation explain why it can be relied upon. Complete formatting, fluent reasoning, and a real object each require different checks. Models can participate in those checks, but the rules, materials, experiments, and arrangements for responsibility on which the checks depend must themselves be examined. These are the points at which the costs of verification separate.

The Verification Pyramid

Verification becomes a bottleneck because generation takes place in a space of representations, while verification must return to reality. Reality contains materials, energy, time, noise, complex feedback, and consequences that cannot be rolled back. AI can generate a thousand drug candidates, but experiments, clinical trials, and long-term observation cannot expand at the same rate; AI can propose countless driving strategies, but a single simulation cannot exhaust real roads' weather, pedestrians, construction work, responsibility, and long-tail risks; AI can generate many policy proposals, but institutional effects must unfold through organizations, behavior, and public trust. Verification is a concentration of RTG rather than merely a cognitive review: it needs resources, time, and coverage of complex structures.

Verification is a continuous climb from formal correctness to consequences that can be borne over the long term, rather than a single act. We can understand it as a pyramid. At the base lies formal verification: checking format, syntax, rules, and local logic, mainly in the domain of bits, where viscosity is relatively low. The second layer is semantic verification: checking meaning, facts, references, and relationships within knowledge, beginning to involve knowledge systems and source tracing. The third is causal verification: establishing whether a proposal actually produces the intended result, rather than merely appearing correlated with it. The fourth is engineering verification: determining whether a system can operate stably amid real noise, failures, maintenance, costs, and scaling. The fifth is long-term verification in living systems: determining whether the long-term consequences of drugs, gene therapies, ecological, educational, or psychological interventions are acceptable. At the top lies verification of social responsibility: whether the system can be explained, audited, held accountable, and backed by compensation, and whether it can earn public trust. The pyramid's shape can be expressed numerically,

$$N_k = N_0 \prod_{i=1}^{k} p_i, \qquad 0 \le p_i \le 1 \tag{1.6}$$

Here N_0 is the initial number of candidates, p_i is the fraction of candidates that have passed the first $i-1$ layers and then pass layer i, and N_k is the number that pass the first k layers. The product follows from conditional counting and requires no assumption of independence; if a layer filters out every candidate, the count remains zero thereafter. Beyond the counts, each layer incurs testing costs. Those costs, together with actual capacity, determine how many candidates the upper layers can handle.

Formal checks, source verification, experiments, and long-term observation can improve at different speeds. Automated tools reduce some verification costs, while clinical exposure, field operation, and the handling of responsibility still take time and resources. When generation outpaces these stages, candidates awaiting verification accumulate. Each layer's throughput determines where the bottleneck lies, and the verification pyramid identifies stages that require

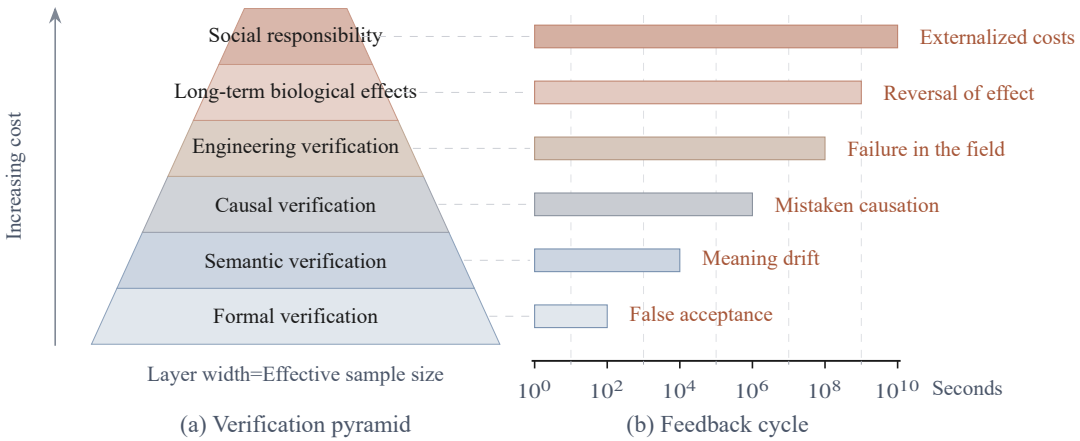


Figure 1.3 The verification pyramid: from formal, semantic, causal, and engineering verification to long-term biological effects and social responsibility, each higher level is harder to automate.

separate investment.

Another force reshapes the pyramid. The higher the layer, the more its criteria depend on context and on who bears the consequences, and the harder they are to express as an executable rule. Formal verification can be defined as a set of syntax checks and semantic verification as a set of source comparisons, but verification of social responsibility cannot be defined by any static rule set because its content changes with institutions, affected parties, and the distribution of consequences. Recasting upper-level verification as a lower-level computable problem is common engineering practice, and a common mode of failure: once a metric replaces an objective, the measured parties reorganize their behavior around the metric. Measurement itself enters the system, a problem we will examine separately below.

Formal Proof and the Grounds for Confirmation

What a formal system can prove depends on its axioms and inference rules. Gödel's incompleteness theorems show that a consistent, effectively axiomatizable system strong enough to express basic arithmetic contains propositions it cannot decide; under the usual conditions on provability, it also cannot prove its own consistency from within (Gödel 1931). The theorems describe the boundaries of formal proof. Learning and empirical judgment in neural networks also involve data, training, and feedback from the object of inquiry, and must be examined through the corresponding mechanisms.

Engineering verification offers a more direct distinction: proving that a program meets its specification still leaves the specification to be checked against actual requirements. Machines can perform formal checks, and other tools can go on checking specifications, sensors, and test settings; the entire chain of confirmation ultimately depends on the evidence and criteria selected. Judging whether that chain is sufficient requires identifying the errors it covers and

the consequences of omissions.

The human role thus enters the institutional domain. Standards involving interests and risks require channels through which those bearing the consequences can speak, authorize, and appeal. People may use machines to form judgments, but the legitimacy of those judgments still needs evidence, procedures, and responsibility. Formal capability and social authorization answer different questions, a distinction that will run through the later discussion of agency and accountability.

The Anchor of Trust: Guarantees Backed by Consequences

Trust also depends on guarantees. Nassim Taleb's "skin in the game" emphasizes that a judgment's credibility depends on how decision-makers' interests relate to those of the people who bear its consequences. Individuals can pledge property, careers, and reputations, while organizations can provide guarantees through assets, identity, and arrangements for responsibility. The question is whether losses from a wrong decision can return to the boundary that made it, and whether correction and compensation can actually occur.

Medical, legal, and other consequential decisions require reliability and accountability mechanisms together. Professional review, records, the power to suspend action, and arrangements for responsibility make decisions inspectable and ensure that someone handles errors. Human involvement needs real authority and the necessary information, while automated execution needs a deployment and guarantee structure against which claims can be pursued. Accuracy and accountability each have their own role, and neither substitutes for the other.

Trust also grows from shared experience and continuing relationships. Understanding in counseling, care, and companionship involves memory, judgment about circumstances, and promises kept. A model can generate a response, but whether the relationship endures, who handles a crisis, and how errors are corrected depend on the concrete arrangements behind the service. The value of human involvement must be explained through these identifiable relationships and actions.

Confirmation and trust meet here. A judgment needs a traceable basis, and a commitment needs a bearer who can be held accountable. Verification can be automated and guarantees institutionalized; both require stable identities, records, and paths along which consequences return. Human bodies, experiences, and finitude supply concrete conditions for bearing consequences, while institutions determine how those conditions enter judgment and responsibility.

Together, the two forms of damping stretch a potentially instantaneous rupture into a period of transition. That period is no safe haven. When two timetables run at once, the first thing to fail is people's understanding of their own place: labor markets reorganize at one speed while self-worth collapses at another. This split is not a psychological problem: it has a definite structure, which we can begin to trace from where replacement occurs.

1.3 The Evolutionary Threshold: The Crisis of Instrumental Reason

1.3.1 The Uneven Timing of Replacement

Combining the analysis of media with the structure of verification yields a direct inference about employment, opposite to the prevailing narrative. Physical viscosity delays AI's replacement of physical labor. Moving bricks, assembly, repair, nursing, driving, and inspection belong to the domain of atoms, where mass, energy, friction, and unpredictability in the field protect these jobs against cheap replacement in the near term. Low viscosity at the logical level accelerates AI's replacement of cognitive labor. Writing, translation, organization, basic programming, routine analysis, and standard document processing belong to the domain of bits, precisely where generative AI is strongest. This uneven timing creates a social separation: **The body remains in the old world while professional dignity and intellectual self-worth have already been rewritten.**

The Physical Defense: The Energy-Efficiency Gap and Error Tolerance

This line of defense begins with the cost curve. Robot deployment incurs procurement, integration, depreciation, maintenance, electricity, supervision, and downtime costs. Hiring people also means wages, benefits, training, and safety expenditure; some costs of raising and sustaining a person are borne by families, society, and the individual, and never appear on the employer's books. Low wages may reflect a transfer of costs. A job's survival for now means that the comparison between deployment options has not yet reversed; it does not establish that people are inherently cheap.

The last meter often concentrates the least regular work. A parcel's route may be planned, yet the entrance can be mistaken, access control can fail, or the recipient can be absent; warehouse containers deform, and objects depart from their intended poses. Automation is handling more of these situations, but every addition must satisfy both engineering reliability and the operating accounts. Explaining the present with a cartoon of a robot that always fails is the same kind of shortcut as declaring total replacement after a single demonstration.

Recent developments place the two clocks inside one company. On 27 October 2026, Waymo announced plans to expand its service areas by a combined total of nearly 250 square miles across five US markets over the following week; its Madrid plan, announced on 20 October, still began with human driving and local technical validation, targeting its first passengers in 2027; the Toronto plan, announced on 16 October, targeted its first passengers in 2028 (Waymo 2026a, 2026c, 2026b). Expanding an existing service, conducting local validation, and setting a future launch target cannot be collapsed into one capability metric. Figure 1.4 places these different states alongside one another.

Plot the two sets of costs as curves, and the condition for replacement is their intersection,

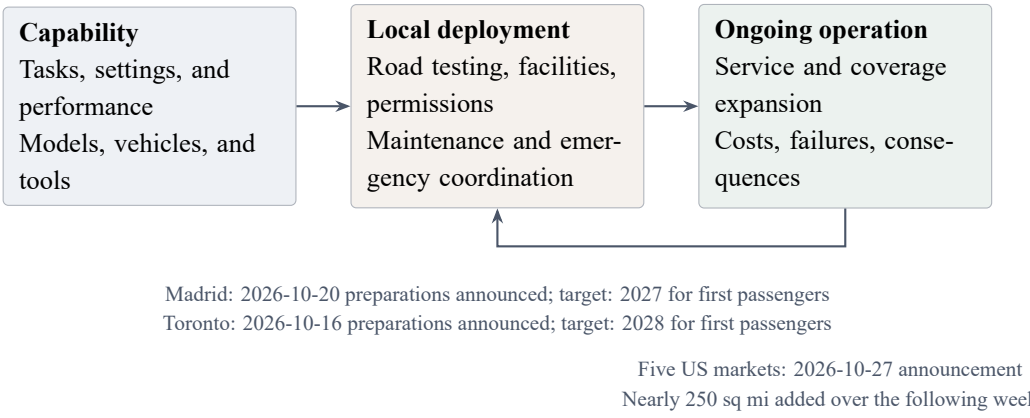


Figure 1.4 Capability, deployment, and operation represent three different kinds of progress. The Waymo examples come from October 2026 announcements. Coverage expansions and launch years retain their announced or planned status; arrows show the work that must connect the stages.

$$C_{\text{human}} = w, \qquad C_{\text{machine}} = \frac{K}{L} + m \tag{1.7}$$

Here w is the annualized cost assigned to the human option, K is the investment in machine purchase and installation, L is the depreciation period, and m is annual operating and maintenance expenditure. Given comparable output, quality, availability, and risk, the lower annualized cost drives replacement. Financing, utilization, switching costs, and the allocation of responsibility must be accounted for separately. Once the numbers favor the machine, procurement, installation, certification, and staff transfers still have to happen one by one.

This defense has clear limits. The scissors gap protects labor's relative price, rather than labor itself. An order-of-magnitude fall in electricity costs, through commercially realized controlled fusion, for example, would change the comparison between 100 and 3,000 watts; higher robot production would spread capital costs further; sufficiently structuring a setting would also eliminate the premium for error tolerance. The relative positions of two cost curves protect human jobs, rather than an inviolable physical law. This distinction must be clear, or physical protection will be mistaken for permanent exemption.

Psychological Collapse: From the Pinnacle of Creation to a Cheap Interface

Physical security has brought no peace of mind, instead creating a new problem. This explains a common perplexity. Many people still have jobs, yet have begun questioning the meaning of their work. Their positions survive because the high viscosity of their systems protects them; the way their skills are priced, however, has already been rewritten by generative capabilities in low-viscosity domains. The danger lies in the premature failure of the interface through which people establish their value, rather than in immediate replacement.

A distinction is easily lost here. Uneven timing means that cognitive labor's way of demon-

strating its value fails first, rather than that cognitive labor disappears first. The work remains, but its pricing changes; the position remains, but the training needed to enter it no longer automatically secures a place. Separating these two processes explains why unemployment data can miss a transformation that workers experience intensely. The statistics count whether jobs exist; the experience concerns whether one's place is still recognized.

Understanding this failure requires returning to the old equation of value. Since the Enlightenment and the Industrial Revolution, modern civilization's values have rested on an equation: a person's value equals their social function, which equals the skills they command. Education emphasizes mathematics, physics, and chemistry, workplaces encourage greater professional expertise, and social standing is allocated according to productive efficiency. This hierarchy assumes that work demanding complex thought ranks higher, while purely physical labor ranks lower. Large models strike at the foundation of this hierarchy. Logical reasoning, programming, writing, painting, and legal advice, abilities once treated as the province of intelligence, become cheap commodities purchased by volume. A generative model can supply in a second the lighting effects an artist has spent ten years learning; a completion tool can produce in an instant what once rested on a programmer's five years of architectural experience.

The consequences of skill depreciation are independent of income. The fear this depreciation brings is not "I cannot feed myself," because one can still make a living delivering for DoorDash, but "I am no use." For middle-class and knowledge workers, being stripped of intellectual self-respect is harder to bear than a fall in income. In the past, being able to write, calculate, and summarize showed that a person had been trained, had engaged with the subject, and occupied a place in a community; now, these abilities are copied, amplified, and automated by external systems, and ability no longer automatically proves value. What technology externalizes is not some isolated capability but the interfaces through which the old society recognized training, experience, qualifications, and responsibility; once those interfaces are taken over, people must migrate to other interfaces to establish their value.

A new form of work has emerged in the gap between physical employment and psychological disempowerment. Platforms can join algorithmic planning and dispatch to actions still needing human hands. DoorDash couriers and Uber drivers offer early examples: platforms organize tasks through dispatch, prices, route suggestions, timing, rewards, and penalties, while people drive, climb stairs, deliver, and absorb customers' emotions. People handle the final stretch of friction in the physical world. The work can provide an income, sometimes a good one; the question is how much control workers retain over objectives, conditions, and assessment. As decisions move to the platform, operators can become components called by an algorithm, temporarily cheaper than robotic arms; their actual labor still creates value, while their standing depends on usable refusal, negotiation, organization, and accountability.

Yuval Harari has proposed the idea of a "useless class." Within this framework, the idea needs one amendment: people remain economically useful, indeed highly useful; it is the nar-

rative that has no use for them. Civilization's central story has always been one of human-led exploration, creation, and conquest; when systems make frontier scientific discoveries and decide the most complex schedules, people become spectators. If the system has read every book I have read and remembers it better, can imitate every poem I write with surer rhythm, and can optimize every decision I make with greater stability, then who am I, and why do I need to exist? This fear is less acute than hunger, but persistent and penetrating. Its practical consequence is often an entire generation withdrawing from the climb up the skill tree, rather than unemployment.

The picture this section presents has two faces. On the physical side, entropy increase and high viscosity keep people temporarily safe in the high-entropy labor of moving atoms, fighting friction, and tending bodies. On the psychological side, the bit world's light-speed iterations remove the old explanation of their place. Two phrases sum it up: you keep your job, but lose your soul. The body remains fixed at the old world's worksite while the mind drifts through territory taken over by algorithms. This is harder to address than a problem of livelihood alone, because no directly corresponding compensation exists and because it takes place within a larger systemic change: efficiency gains themselves are producing new scarcities.

1.3.2 Efficiency Inflation and Structural Bottlenecks in Development

New demand can soon fill the space that greater efficiency releases. Cheaper generation brings a flood of proposals, code, and drafts; if review and maintenance cannot keep up, people move from drafting to processing candidates in a review queue. Work has no conservation law that requires it to return; the arrival rate has exceeded the capacity to receive it. The real question is how much extra work a saving at one step creates at the next.

Technological progress often begins with a local reduction in resistance. A route becomes shorter, a device more efficient, a model cheaper, a process automated, a metric clearer, or a risk reduced by a safety device. Intuition says that lower local resistance should improve the whole system. Costs fall, efficiency rises, choices expand, risks shrink, and control strengthens. Yet complex systems often deliver the opposite. An extra road may worsen congestion; greater efficiency may raise total consumption; stronger safety may encourage riskier behavior; clearer metrics may corrode the actual objective; greater faith in a model may change what it predicts; automation may reduce routine work while leaving people with rarer, harder exceptions carrying greater responsibility. These phenomena have different names.

Local Resistance Reduction and Actor Response

These phenomena have different names and are scattered across six unrelated fields: energy, transport, management, prediction, safety, and automation. Set them alongside one another, and their shared structure becomes visible.

Table 1.1 lists six different entries in its second column. Its fourth column points to the same

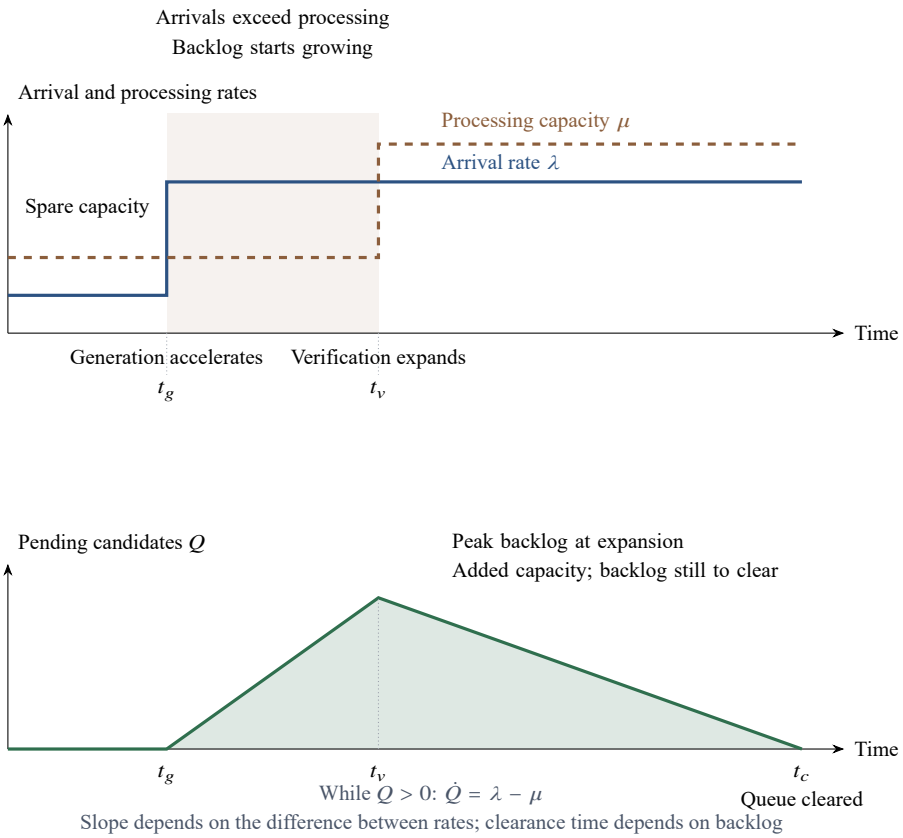


Figure 1.5 How bottlenecks migrate. Here λ is the arrival rate of candidates into the specified verification queue, and, μ is processing capacity measured on the same basis. Faster generation gives $\lambda > \mu$, and candidates accumulate; verification expansion gives $\mu > \lambda$, and the queue gradually clears. Curves are schematic, not measured values. Queue size follows the balance of inflows and outflows; this equation does not determine society's total amount of labor.

class of consequence. This is no coincidence. Every row follows three steps: a local condition changes, actors within the system choose again according to their interests, and aggregate quantities reorganize at a new equilibrium. We will examine all six rows below; first, remember the third column: technology changes actors' field of choice, and even if their behavioral rules remain unchanged, their choices may change with it. This also explains why reports of these paradoxes find local indicators improving while overall conditions deteriorate: they measure quantities at different levels, and conflating those levels leaves only a contradiction.

Technology does more than subtract a patch of resistance from a system. It is more like a change in the system's cost landscape. Lower a local barrier, and actors choose new paths rather than stand still. Once actors choose again, the fields of flows, risks, and metrics, and the distributions of tasks and responsibility, all reorganize. Instead of staying in its original state and enjoying a local benefit, the system reaches a new equilibrium under changed boundary

Table 1.1 A shared structure across six paradoxes: local resistance reduction, actor response, global reorganization

Paradox	Local resistance reduction	Actor response	Global outcome
Jevons	Cost per use falls	More applications and more uses	Total consumption rises instead of falling
Braess	A new route becomes available	Travelers reroute according to local optima	A worse equilibrium increases total network travel time
Goodhart and Campbell	A clear, measurable metric	Measured parties reorganize behavior around the metric	The metric detaches from its original objective; measurement fails
Lucas and reflexivity	A model predicts more accurately	Those being predicted adjust in response to the model	The old relationship fails; the system changes trajectory
Peltzman	Failure probability per exposure falls	Actors increase speed, frequency, or leverage	Total risk exposure expands, offsetting net benefits
Automation paradox	Routine operations are automated	People are left with rare exceptions	Remaining tasks grow harder; responsibility concentrates

conditions. The paradox arises here: a local barrier falls, yet total viscosity is higher at the new equilibrium.

Technological effects therefore cannot be judged by local parameters alone. Faster, cheaper, more accurate, or safer performance at one point does not automatically make the whole faster, cheaper, more accurate, or safer. The global cost of a complex system is generated jointly by flows, bottlenecks, risk, verification, and responsibility after actors respond, rather than by a linear sum of local gains. The common framework can be stated as follows: local resistance reduction changes the cost landscape, actor response redistributes the system, a new equilibrium forms, and hidden bottlenecks emerge at new locations.

Following this framework, the six kinds of paradox fall into three groups. The first type is Jevons's paradox: expansion through lower resistance. Jevons's paradox originally concerned energy efficiency. Equipment that uses less coal, oil, or electricity should intuitively reduce total consumption. Yet when efficiency lowers the cost per use, more people use it, more applications become viable, and more demand is released, so total consumption may end up rising rather than falling (Jevons 1865; Allwood and Cullen 2012). This is a throughput rebound, rather than an isolated counterintuitive feature of energy history. Lowering resistance in a channel attracts more flow from the system. Activities once too expensive become common;

capabilities once used sparingly because they were slow become routine calls; resources once constrained by scarcity find their way into more applications.

Efficiency lowers resistance per unit; total throughput may expand. In the AI era, Jevons's paradox becomes more conspicuous. Cheaper generation produces more content; easier code production increases software complexity and maintenance burdens; easier proposal generation raises verification pressure; easier access to computation drives further growth in total demand for it. Technology lowers resistance per operation while sending more operations through the system. Efficiency therefore does not automatically mean lower total cost. Whether a system is truly relieved depends on whether energy, cooling, verification, maintenance, responsibility, and absorptive capacity can still bear the expanded throughput, rather than simply on cheaper individual operations. In the framework of technological viscosity, Jevons's paradox shows how lower local viscosity may induce greater global throughput. The relationship between unit cost and total use can be expressed through elasticity,

$$E = q\,N(p), \qquad p = \alpha q, \qquad \varepsilon \equiv -\frac{d\ln N}{d\ln p}, \qquad \frac{d\ln E}{d\ln q} = 1 - \varepsilon \tag{1.8}$$

Here q is physical consumption per operation, p is unit price, and $N(p)$ is the number of operations per unit time; a fixed α links the two, and E is total consumption per unit time. When elasticity $\varepsilon < 1$, efficiency improvements reduce total consumption; when $\varepsilon > 1$, additional demand outweighs per-operation savings. In April 2026, the International Energy Agency recorded substantial gains in energy efficiency for individual AI tasks alongside a 17% increase in total data-center electricity use in 2025 (International Energy Agency 2026a). Savings per operation and rising totals can coexist; the size of the rebound must still be identified from demand data.

The second type is Braess's paradox: a new route, a worse equilibrium. Braess's paradox shows that, in some traffic networks, adding a road increases everyone's travel time. The problem is not that the road itself is harmful, but that it changes the choices available to each driver. Everyone selects a new route according to a local optimum, traffic redistributes, and the system settles into a worse equilibrium (Braess 1968). This is a general problem of topological coupling rather than something peculiar to traffic. A path's cost is a property produced by the traffic loaded onto it, rather than a static attribute of the path. A road may offer low resistance to a few users and become a new bottleneck when everyone enters. We can place each traveler's minimization alongside minimization for the whole system,

$$f_i > 0 \implies c_i(\mathbf{f}) = \min_{j\in\mathcal{P}} c_j(\mathbf{f}); \qquad \min_{\mathbf{f}} \sum_{e\in E} f_e\, c_e(f_e) \tag{1.9}$$

Here f_e is flow on a road segment, $c_e(f_e)$ is the corresponding travel time, $\mathcal{P}$ is the set of routes between the same origin and destination, with route costs obtained by adding the costs of their segments. With fixed demand, user equilibrium requires every used route to have

the current minimum cost, while the system optimum minimizes total network travel time. Adding a road expands the feasible set, so the system optimum cannot worsen when the new road need not be used; the new equilibrium reached through independent choices, however, may be worse. Braess's paradox occurs in this gap: more options do not themselves deliver coordination.

A tool can confer an advantage on a few users and become a bottleneck once everyone copies its use. A new road enlarges the set of physically reachable options, while the flow that users collectively create is a separate matter. Technology supplies paths, actors choose them, and pressure redistributes after those choices. The same question deserves asking in software and organizational systems, where the outcome depends on the particular mechanisms.

Measurement and Models

Flows and paths reorganize at the level of material movement, but for actors to perceive, compare, and choose among those changes, they must pass through an information layer, which the same reorganization also rewrites. The third type comprises Goodhart's law and Campbell's law: metric corruption. Goodhart's law is commonly summarized as follows: once a measure becomes a target, it ceases to be a good measure (Goodhart 1975). Campbell's law likewise observes that the more a social indicator is used for decision-making, the more vulnerable it becomes to corruption and the more it distorts the process it was meant to monitor (Campbell 1979). The root of these paradoxes is measurement's entry into the system. A metric begins as an observable, approximating a complex reality. Once it enters systems of rewards, penalties, resource allocation, and organizational control, however, it ceases to be merely an observable and becomes a control variable. Those being measured learn the metric, cater to it, game it, and may even sacrifice the real objective to optimize it. Behavioral feedback thus rewrites the relationship between metric and objective. Schools adapt teaching to rankings, hospitals adapt services to performance measures, platform creators adapt content to recommendation algorithms, and businesses adapt their actions to KPIs. The more consequential a metric becomes, the more behavior reorganizes around it; the more behavior reorganizes around it, the further the metric may drift from the real objective it was meant to measure.

In terms of physical mechanism, a metric is no disturbance-free measurement. Once connected to an incentive structure, it changes the distribution of the system's degrees of freedom. It compresses a complex reality into a few optimizable variables while inducing actors to concentrate their actions on those variables. The stronger the metric, the more likely reality is to be reshaped by it.

Verification systems therefore generate viscosity of their own. Mature verification must improve measurement accuracy while also preventing behavior from taking over the measurement channel in reverse. Otherwise, the clearer the metric, the more likely the system is to develop new paths of low resistance around it, while obscuring the actual objective. Goodhart's law and Campbell's law arise on this side of measurement: clarity in local measurement

can induce erosion of the global objective.

The fourth type is the Lucas critique and reflexivity: a model changes its object on entering the system. The Lucas critique observes that once a model built from past behavioral relationships is used in policy, actors adjust their behavior and the old relationships may fail (Lucas 1976). Reflexivity emphasizes that, in markets, organizations, and social systems, participants' perceptions and actions change the very object they perceive (Soros 1987). The root of this problem is the model's entry into the action loop. A model initially describes a system, but once it is used for decisions, actors within the system change their behavior in response to it. Market forecasts influence traders' behavior, risk models influence financial institutions' strategies, policy models influence the expectations of firms and the public, and organizational diagnoses influence members' responses. Once a model enters the system, it ceases to be merely an external representation and becomes an internal variable.

This differs from a physical object. A cup does not deliberately change itself because you have modeled it; markets, organizations, public opinion, and negotiating counterparts do. They understand, adapt to, counteract, and exploit models. The more a model is believed, the more likely it is to change what it predicts; the more widely predictions enter action, the less the system follows its original trajectory. The difficulty in these systems is that model output changes boundary conditions, rather than that the model is insufficiently accurate. The model initially faced a system in a particular state; after the model is used, the system is no longer the same system.

A prediction becomes a new source of disturbance once it enters the action loop. It may fulfill itself or defeat itself. A model in a highly coupled system therefore needs more than static accuracy. It must also ask: how will its use change the system itself? The model enters the flow and changes the next round of movement, beyond drawing the system from outside. The Lucas critique and reflexivity carry the issue into prediction: improved local predictive capability may induce reconstruction of the object being predicted.

Risk and Task Distribution

Once measurements and models are rewritten, actors' judgments about the world change, and with changed judgments come changed ways of acting. The fifth type is the Peltzman effect: risk compensation. The Peltzman effect describes how, after safety technology lowers a local risk, actors may take greater risks and offset part of the safety benefit. Safer roads may bring faster driving; stronger driving assistance may invite more distraction; finer financial risk controls may encourage more aggressive institutions; more automated platform moderation may prompt users to test boundaries more frequently (Peltzman 1975). This means that risk is no static substance, rather than that safety is useless. Risk is a distribution jointly produced by technology, actors, and the environment. A safety device lowers a local failure probability, but actors adjust their exposure, frequency, and intensity of risk-taking to the new safety boundary. Lower risk per exposure does not guarantee a proportional fall in total risk.

The key is expansion of the boundary. After a system gains a greater safety margin, actors may turn that margin into higher speeds, greater frequency, more leverage, less attention, or more aggressive behavior. Safety lowers a local barrier while also changing the space of action. Whether total risk ultimately falls depends on whether the local risk reduction exceeds the additional exposure created by expanding behavioral boundaries. Risk governance must therefore count total exposure after actors respond, as well as failure probability per exposure. Otherwise, safety systems will overestimate their net benefits. Reducing risk does not eliminate it; it can change how risk is used. The Peltzman effect is the risk-domain version of this logic: greater local safety may induce greater global exposure. The comparison can be written directly,

$$R = p\,n\,d, \qquad \Delta R \approx n\,d\,\Delta p + p\,d\,\Delta n, \qquad \Delta R < 0 \iff \frac{|\Delta p|}{p} > \frac{\Delta n}{n} \tag{1.10}$$

R is expected loss per unit time, measured in loss per unit time, such as accidents per year; p is the probability of failure per exposure, a dimensionless quantity; n is the number of exposures per unit time, measured in exposures per year; and d is the mean loss per failure, measured in loss per failure. A safety device lowers p ($\Delta p < 0$), while actors respond by increasing exposure ($\Delta n > 0$); to first order, the two effects on R have opposite signs, and total risk falls if the relative decrease exceeds the relative increase. The "magnitude of the local risk reduction" in the text corresponds to $|\Delta p|/p$, and the "additional exposure created by expanding behavioral boundaries" corresponds to $\Delta n/n$: safety produces a net benefit only if the reduction exceeds the expansion, which is also why risk governance cannot count failure probability per exposure alone. The limits of applicability are these: the first-order approximation holds only when both relative changes are small; if actors respond discretely, by switching the safety device off altogether or withdrawing from the activity, or if mean loss d itself varies with exposure intensity, this linear decomposition fails. The equation also addresses only whether total risk rises or falls, rather than how risk is distributed across a population. A safety device often protects only part of the population, while another part bears the expanded exposure.

The sixth type is the automation paradox: the concentration of residual burdens. Also known as the ironies of automation, it describes how automation reduces routine operations yet may make it harder for people to take over when something goes wrong. Machines handle routine processes while people lose contact with their details; when an exception occurs, people lack continuous involvement yet must judge quickly. Routine burdens fall, but what remains becomes more concentrated, harder to formalize, and harder to assign responsibility for (Bainbridge 1983). The burden changes distribution rather than disappearing. Automation usually absorbs the frequent, regular, repeatable parts first, leaving people with the rare, marginal, ambiguous, and consequential ones. People therefore cease to bear the burden of continuous operation and instead bear the burden of residual tasks in the tail. Task numbers fall while task density rises; everyday burdens decline while responsibility for exceptions grows. The system

looks more automated, yet the human role moves into harder territory.

Stronger AI makes this problem more pronounced. Models can handle many routine judgments, but the questions left behind are often harder to verify, explain, and take responsibility for. Automation reduces viscosity in front-line operations while concentrating residual burdens in exceptions, verification, and accountability. It rearranges people's place in a system rather than freeing them from it altogether. In the framework of technological viscosity, the automation paradox shows how automating local tasks can concentrate residual burdens in the tail.

Redistributing the Field of Resistance

These paradoxes differ in name and setting but share an underlying structure. Jevons's paradox concerns throughput response. Braess's paradox concerns route response. Goodhart's and Campbell's laws concern metric response. The Lucas critique and reflexivity concern model response. The Peltzman effect concerns risk response. The automation paradox concerns the response of task distributions.

Together they form a paradigm of technological paradox: technology first lowers a local barrier; actors act again in response to the new barriers; their actions change the fields of flows, risks, and measurements, and the distributions of tasks and responsibility; the system forms a new equilibrium under changed boundary conditions. The paradox arises when the total cost at the new equilibrium exceeds the benefit of the local reduction in resistance. This places existing paradoxes within one mechanistic framework, rather than inventing another name for a paradox.

To judge whether a technology has truly reduced system viscosity, we cannot look only at local efficiency, cost, speed, safety, or accuracy. We must examine whether, after entering the system, it changes total throughput, equilibrium, behavior around metrics, model reflexivity, risk exposure, verification pressure, and structures of responsibility. Technological paradox is no argument against technology. It simply reminds us that **Local optimization is not system optimization.** An open system responds to benefits, amplifies them, exploits them, and turns them into new pressures, rather than receiving them passively. The more effective a technology, the more likely it is to redistribute behavior; the better a path of low resistance, the more flow it attracts; the more useful a metric, the more it invites optimization around itself; the more a model is trusted, the more the object of its predictions changes through being predicted. Here, too, lies the deeper significance of technological viscosity.

Viscosity is a field of resistance that redistributes in response to technological inputs, rather than a fixed wall. When technology lowers resistance in one place, the system may develop new resistance elsewhere. Resistance can move from energy to cooling, generation to verification, paths to equilibrium, safety to behavior, automation to residual burdens, and models to responsibility.

Mature technological judgment asks not "Is this local improvement faster, cheaper, more

accurate, or safer?" but: after local resistance falls, how does the cost landscape change? How will actors act again? Where will flows be directed? Where will bottlenecks reappear? Who will bear the risks and responsibility? What new equilibrium will the system enter? Has total viscosity actually fallen, or merely changed location?

Energy and infrastructure continue to set conditions for this reorganization. Obtaining energy requires an investment of energy, and the net surplus depends on the technological route, location, and accounting boundary; materials age, and maintenance expenditure depends on asset condition and service requirements. No broad historical curve of decline can determine the future of every energy technology. Computing's promised expansion must still be realized through particular power sources, equipment, and maintenance capacity.

The common form of technological paradox is a global reckoning after local resistance falls. Local resistance reduction is not the endpoint. The new equilibrium is the answer.

This step, however, remains qualitative. The six kinds of paradox have their own terminology, settings, and research traditions, and can be placed in one framework because they share an underlying structure, not because a common set of variables already describes them. Whether the costs of lower local resistance emerge in throughput, equilibrium, measurement, risk, or task distribution must still be answered case by case in the current formulation. If this common form holds, the transfer of resistance, the response of throughput, and the formation of new equilibria should admit a reusable set of quantities, beyond a classification by names. That formulation remains to be written.

1.3.3 The Ultimate Question: The Paradox of the Post-Tool Age

The previous section treated the redistribution of the field of resistance as a purely structural matter, yet resistance never falls evenly on people. The resistance that falls is removed from some jobs, while the new resistance is imposed on others. Who bears the transferred viscosity is a question of distribution rather than efficiency; before discussing distribution, we must establish the limits of these judgments, or technological viscosity can easily be read as just another expression of sentiment.

The Full Account of Technological Effects

Digital payments, navigation, industrial simulation, automated experiments, and code tools have reduced transition costs at different stages. Evaluating those improvements requires examining output, execution, verification, maintenance, and the handling of failure together. Costs reduced, burdens transferred, and risks not yet revealed should be recorded separately within the appropriate system boundaries.

Technological viscosity is no universal explanation, either. Politics, culture, capital cycles, business models, organizational capabilities, structures of interest, and social psychology also shape technological deployment. Technological viscosity offers a foundational perspective

rather than reducing every social issue to resources, time, and topology: when intelligence tries to move from representational space into real systems, it must encounter resistance from its medium. That resistance is a structural property of the real system itself, rather than incidental noise. The decisive capability in the AI era is therefore not merely stronger generation, but the ability to turn generated results into real changes that can be executed, verified, maintained, and answered for. Future competition will concern verification systems, engineering systems, organizational systems, infrastructure, and institutions of responsibility, as well as model capabilities. Whoever manages viscosity better can turn intelligence into real productivity more effectively.

Three Interlocking Paradoxes

With those limits clear, the chapter's argument can be drawn together into three interlocking paradoxes. The first is the efficiency paradox. People once established their value by becoming faster, more accurate, and stronger, yet the end of that road is the disappearance of the person defined as a tool: the more people optimize themselves in logic, computation, and execution, the closer they come to the biological limits of carbon-based brains, while silicon-based systems advance exponentially on the same track. On this track, the finish line is the line of elimination. The issue is the choice of track, rather than a gap in ability.

The second is the escape paradox. People try to evade the material world's weight through virtualization, only to find that more freedom in the virtual world demands deeper dependence on the physical one. The more people desire a frictionless experience, the more energy-intensive, entropy-producing infrastructure they need in the world of atoms to support it. Trying to sever dependence on physical reality only makes that dependence harder. The desire to ascend intensifies the fall.

The third is the paradox of meaning, arising from the first two together. Modern people's sense of meaning rests on being needed: instrumental value. When the efficiency paradox declares people unnecessary as intelligent tools and the escape paradox declares escape impossible, meaning has a vacancy. People lose the pride of being producers, yet cannot rest content as pure experiencers, like ancient aristocrats, because someone must still maintain the physical foundation. They are caught between useful machines and free agents, unable to enter either position.

From Doing to Being

All three paradoxes share a premise: the problem is one of capability, and running faster will solve it. If that premise fails, technical reskilling and economic compensation are only pain relief, not a solution. Once the path of becoming a better tool reaches its end, and escape into a weightless virtual world is physically impossible, the question must change. The old question was: how can we do better than machines? We cannot. The new question is: when the capacity for *doing* is outsourced to machines, what does *being* consist of?

A turn is needed here, and it does not occur at the level of technology. The historical period that took efficiency as its sole coordinate defined a person's place through function; once function can be copied cheaply, that place is left suspended. The new place lies outside function: in the ability to bear, to judge, and to receive the consequences of action. These capacities cannot be bought centrally like computing power; they develop only in particular people over particular stretches of time. The era in which human civilization secured its right to survive by imitating machines, refining skills, and pursuing efficiency has ended, and the next question is to redefine the exam, rather than score higher on the old one.

A temporal limit must be made clear here. The physical world's lag is not a consequence of people failing to try hard enough; it is a window of time. During that window, people must both accept being surpassed in efficiency and preserve the ground on which machines do not define them. The window will not remain open indefinitely, because the three accounts discussed earlier keep moving: energy costs may fall, robot capital costs may be spread further, and high-entropy settings may be structured piece by piece. The buffer's value lies in giving time for a turn, not in guaranteeing its outcome.

History reveals the direction of that turn. Whenever technology externalizes an ability, it breaks an old bond between capability and value. Writing lightens memory, shifting value from recitation to understanding; printing lightens copying, shifting value from possessing texts to interpreting them and organizing trust; steam engines lighten muscular work, shifting value from doing work to control, maintenance, and dispatch; computers lighten rule execution, shifting value from execution to problem definition; large models lighten symbolic generation, requiring value to move toward verification, authorization for action, and bearing consequences. Each time, the structure repeats: external systems take over an ability, and people move to positions harder to counterfeit. Yet this trajectory also shows that migration never happens automatically. Writing did not naturally turn every memorizer into an interpreter; printing did not naturally bring every scribe into publishing institutions; steam engines did not naturally make every artisan an engineer. Migration requires new interfaces, new institutions, and sustained struggles over those interfaces.

This trajectory has a further qualification, which Chapter 6 develops. Migration is a contest, rather than a gentle upward curve, and high-viscosity positions do not inherently belong to people: the interfaces treated today as evidence of human value can also be absorbed by platforms, priced by capital, divided by contracts, and redistributed between expert systems and institutional machinery. If a few institutions monopolize these interfaces, people gain no automatic promotion, but become objects to be classified, scored, and tuned instead of producers. Migration cannot be achieved by individual effort alone; it requires new channels through which more people can express real experience, reject mistaken objectives, and participate in the distribution of consequences.

Technological progress will not turn the world into a frictionless information space. It light-

ens some stages while burdening others, accelerates some processes while shifting bottlenecks to verification, maintenance, and responsibility, and rapidly expands representational space while forcing real systems to redistribute resources, time, and structure. Understanding the AI era requires examining how intelligence passes through reality, as well as how capable it is. Durable, effective technological paths belong to people and institutions able to identify and distribute viscosity, reduce what is unnecessary, retain what is necessary, and achieve reliable state transitions under viscous conditions. The future does not belong to fantasies of a world without resistance.

Why intelligence itself encounters these forms of resistance remains unanswered. To answer that question, we must first ask an earlier one: in a world driven by energy gradients and sustained by dissipation, why does intelligence arise at all? Without an answer, all the preceding accounts amount to no more than a checklist.

Further reading

Kurt Gödel, *Über formal unentscheidbare Sätze der Principia Mathematica und verwandter Systeme I*
Hans Moravec, *Mind Children*
W. Stanley Jevons, *The Coal Question*
Dietrich Braess, "Über ein Paradoxon aus der Verkehrsplanung"
Donald T. Campbell, "Assessing the Impact of Planned Social Change"
Sam Peltzman, "The Effects of Automobile Safety Regulation"
Lisanne Bainbridge, "Ironies of Automation"
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Chapter 2

The Physical Boundaries of Reconfiguring Matter: Fundamental Forces and a Generalized Viscosity Model

Chapter 1 ended with a question that comes before all the others: why does something like intelligence emerge in a world driven by energy gradients and sustained by dissipation? Without an answer, all the constraints of resources, time, and structure that follow can only be read as external obstacles, as though intelligence could otherwise pass lightly through the material world and merely happened to meet resistance. The reality is the reverse: resistance is not an obstacle external to intelligence; it comes from the way intelligence originated.

Intelligence is neither a miracle arriving from nothing nor a decorative faculty that life acquired by chance; it is more like a mechanism for selecting paths that the thermodynamic world produced over the course of evolution. To understand why intelligence emerged, and why it encounters constraints of resources, time, and structure, we cannot start with brains, language, or computers alone, but must begin deeper: with energy gradients, dissipative structures, autocatalytic networks, physical trial and error, and virtual simulation.

2.1 Intelligence and the Reconfiguration of Matter: The Physical Limits of Optimization

2.1.1 The Principle of Least Action and the Path Selector

Dissipative Structures in Energy Gradients

The universe is not still. Wherever differences in temperature, potential, concentration, and chemical energy exist, matter flows, collides, reorganizes, and dissipates. Isolated systems tend toward increasing entropy, but open systems can form locally ordered structures through sustained flows of energy and matter. Storms, cells, organisms, cities, and machines are not the opposite of increasing entropy; they are local channels formed within an increase in entropy on a larger scale. They maintain order not by escaping thermodynamics but by taking in free energy, reorganizing matter, releasing waste heat, and using their own structures to participate in a larger process of energy release.

What sustains this local order requires a clearer account of the mechanism. In *What Is Life?*, Schrödinger posed a question with lasting influence: how do living organisms avoid rapidly sliding into thermal equilibrium? His answer was that life maintains its own order by taking in "negative entropy" from its environment (Schrödinger 1944). In more precise contemporary terms, life does not violate the second law of thermodynamics: an open system maintains local order by continuously taking in low-entropy resources and releasing high-entropy waste and waste heat. Life does not acquire its order for free; it must keep paying for it.

Prigogine's theory of **dissipative structures** took the question further. Far from equilibrium, open systems subject to sustained flows and nonlinear interactions can form new ordered structures (Prigogine 1977; Nicolis and Prigogine 1977). Their order is dynamic, dependent on inputs and outputs, rather than the closed order of a static crystal. Once the flows of energy and matter stop, they disintegrate, cool, die, or return to a state closer to equilibrium. Life is one of the extreme forms of this dynamic order: a complex, persistent, replicable, and evolvable dissipative structure within thermodynamics, rather than an exception outside it. The maximum entropy production principle offers a heuristic picture: in some open systems far from equilibrium, structures that can stably organize flows may be more readily sustained and amplified (Martyushev and Seleznev 2006). Storms organize flows of heat and momentum, organisms capture and convert chemical or solar energy, and ecosystems and cities reorganize flows of energy, matter, and information on larger scales.

A distinction matters here: the maximum entropy production principle is not an uncontested universal law that can be applied directly to every aspect of the evolution of life and intelligence (Bruers 2007). This chapter uses it as a picture, not as proof. It helps us see a possibility: **Order does not necessarily resist entropy increase; often, order itself is a more complex, stable, and enduring channel of dissipation.**

Autocatalysis and the Coupling of Conditions

Once local order forms, the next problem is persistence. Several critical transitions separate ordinary chemical reactions from life, and one of the most important is autocatalysis. In primordial oceans, hot springs, mineral surfaces, or small chemical environments, thermal motion, random fluctuations, and local differences in concentration can produce transient pockets of molecular order. Most soon disperse into a disordered distribution again. Some structures, however, are unusual: they do more than exist; they promote their own formation. A and B react to produce C, and C promotes further reactions between A and B: this is autocatalysis. Autocatalysis amplifies small pockets of order that arise by chance, giving transient structures the ability to reinforce themselves.

Autocatalytic networks matter because they turn a structure that occurs by chance into one that can persist. They are positive feedback networks, rather than single chemical reactions. Given sustained inputs of energy and raw materials, such networks can expand, replicate, compete, and differentiate. They absorb free energy and maintain local order while releasing heat and waste. More complex structures can process flows of matter and energy more effectively and are more likely to survive competition. Kauffman's work on self-organization and autocatalytic sets sought to explain precisely this: understanding the origin of life requires an account of how chemical networks spontaneously form structures that can persist and expand, as well as an account of external selection (Kauffman 1993). This account also has a limit. Autocatalytic networks are not a complete answer to the origin of life. In their discussion of autocatalytic sets, Hordijk and colleagues emphasize that these sets are important conditions for lifelike behavior but are insufficient on their own to constitute life in full (Hordijk and Steel 2018). A self-reinforcing reaction network also needs boundaries, information storage, variation, replication, selection, and internal regulation before it can approach life. Autocatalysis provides a route from transient pockets of order to structures that persist, rather than explaining the origin of life in a single stroke.

Each missing condition has its own possible origins, and information carriers have received the most attention. The RNA world hypothesis provides another important lead here. RNA is unusual because it can carry sequence information and, in some circumstances, also act as a catalyst. In 1986, Walter Gilbert used the phrase "RNA world" to describe a hypothetical stage before modern life came to be dominated by DNA and proteins, when RNA may have performed both information storage and catalysis (Gilbert 1986). This hypothesis does not mean that the origin of life has been solved, but it brings out a crucial point: the emergence of life requires information, reactions, and replication to become progressively linked. With energy flow but no information, structures struggle to accumulate a history; with information but no reactions or energy, structures cannot sustain themselves; with reactions but no boundaries, local order struggles to concentrate and be inherited.

Dissipative structures, autocatalytic networks, RNA, lipid membranes, and primitive

metabolism should therefore be understood as several kinds of conditions that must gradually couple, rather than as a single historical sequence already proved in full: energy flows, boundaries, information storage, replication, variation, selection, and internal regulation. Membranes distinguish a system from its environment, RNA or other information carriers preserve inheritable structures, and autocatalytic networks organize flows of energy and matter. Primitive life was not a soul that suddenly appeared; it arose as certain dissipative structures acquired boundaries and the capacities for replication and internal regulation through prolonged selection.

Geodesics in Phase Space

Dissipative structures and autocatalytic networks explain how local order can emerge and persist, but they do not explain where intelligence fits into this picture. To answer that question, we need a different description: treat a manufacturing activity as the movement of a system's state through a high-dimensional space.

Space in everyday language has just three dimensions—length, width, and height—while the state of a simple pendulum can be specified by two variables, position and momentum. Describing the state of a chemical plant or an assembly line, however, requires thousands of independent variables: reaction temperature, pressure, catalyst concentration, material strength, transport distance, inventory cycles, equipment depreciation rates, and the cost of capital, each constituting a dimension. Raw materials occupy one point in this space, the finished product another, and production is the trajectory traced by the state point between them. In this description, a process improvement is a new choice of trajectory.

In state space, engineering optimization compares paths under constraints: can the same objective be reached with less material, lower energy consumption, a shorter schedule, or less risk? The stationary action principle in physics provides another rigorous formulation in terms of paths. It derives equations of motion from a specified Lagrangian and boundary conditions, and cannot be directly equated with minimizing economic cost. Both formulations concern paths, but they use different quantities and criteria.

$$S[q] = \int_{t_1}^{t_2} (T - V)\, dt, \qquad \delta S = 0 \tag{2.1}$$

S is the action, measured in joule-seconds; q is a generalized coordinate; T and V are kinetic and potential energy, respectively; and $\delta S = 0$ means that the actual trajectory makes the action stationary under the specified endpoint conditions. A stationary value can be a minimum or have other properties. Nonconservative effects such as friction require the corresponding generalized forces or a description of dissipation. The principle supplies equations of motion; how intelligence develops choice, memory, and learning requires a separate account of the mechanisms.

Industrial waste has specific sources: inaccurate demand forecasts create inventories, inadequate control of operating conditions increases losses, and poorly planned routes cause

redundant transport. AI can reduce these expenditures through search, modeling, and control. The size of the improvement must be measured for the same task, quality, and system boundary; “extra action” cannot replace the separate accounts for energy consumption, money, and time.

Intelligence acts here as a **path selector**. It compares actions in light of objectives, available resources, and feedback, seeking routes with a higher probability of success and a more manageable cost. This choice can change materials, processes, and initial states, as well as actual losses; the specific lower bound must be recalculated for the task and its physical boundaries.

Path selection moves some expensive trial and error into a model before execution. Comparison itself has a cost, and the selected path must still be executed in reality. We therefore need to distinguish three accounts: a reaction’s net thermodynamic cost, the barrier governing its rate, and the operation and maintenance of equipment and structures. The three interact, but they are not the same expenditure of energy.

2.1.2 Two Unavoidable Expenditures: Activation Energy and the Entropy Tax

Change and maintenance face different constraints. Activation barriers primarily affect reaction rates, free-energy differences constrain net work exchange under given conditions, and maintenance concerns damage, disturbances, and the continuity of function. Stable structures such as crystals can persist passively, while life and continuously operating equipment generally require supporting flows. Keeping these separate makes it possible to see which part intelligence can improve.

The Thermodynamic Bottleneck of Physical Trial and Error

Once life emerges, however, it faces a harsher problem: how to find better paths through complex environments. Early living systems relied primarily on **physical trial and error**. Mutation, replication, competition, elimination, and death make up the oldest search mechanism. It works because reality selects; it is extraordinarily expensive because every failure is settled in the real world. A harmful mutation changes a body, rather than a line of code; a mistaken action consumes energy, exposes the organism to risk, costs opportunities to reproduce, and may cause death, rather than merely requiring a corrected answer. Natural selection computes in a real environment, and matter, time, and life pay the bill.

This is the thermodynamic bottleneck of physical trial and error. The more complex the environment and the greater the number of possible paths, the more expensive it becomes to select among them through collisions with reality alone. Evolution can search through vast populations and immense stretches of time, but for any particular organism, real-world trial and error is too slow, too costly, and too dangerous. An individual cannot use its body every time to test whether it can leap a cliff, whether it can approach a fire, whether an unfamiliar animal will attack, or whether spoiled food is safe to eat. Reality supplies an answer, but often

too late.

Intelligence emerged within this bottleneck. A nervous system does more than let life sense the world: it builds a low-cost model of the environment inside the body. An animal need not try every possible action in reality; it can first rehearse internally where food might be, where danger might lie, which route takes less effort, which movement is likelier to succeed, which object is worth approaching, and which signal must be avoided. Intelligence moves some external collisions inward, converting some physical trial and error into **virtual simulation**.

This is a profound substitution of costs. Real-world trial and error requires movement, injury, energy consumption, material losses, and sometimes death; internal simulation requires neural signals, memory, attention, and computation. Neither is free, but their costs differ by orders of magnitude. Intelligence lets a system choose before doing work, compare before acting, and rule out errors before exposure to risk. It does not cancel physical costs; it reduces the number of times they are paid blindly.

From the perspective of biological evolution and systems that act, **The primary function of intelligence is to avoid costly mistakes, rather than to think abstractly.** This judgment sounds less like the traditional philosophical celebration of intelligence, but it is closer to the predicament of life. An organism must act with limited energy, limited time, a limited body, and limited information. It need not know the truth of the universe to qualify as intelligent; first it needs to reduce risk before acting, assess objects before contact, choose paths before expending resources, and exclude obvious errors before failure. Intelligence is a mechanism through which a body selects paths that allow it to persist in a heavy world, rather than a luxury of the abstract mind.

Information Hubs and Two Kinds of Entropy

Once this mechanism operates, a finer question arises: what exactly does it reduce? Through internal models, intelligence reduces an agent's uncertainty about relevant environmental states and the consequences of action, thereby reducing the agent's internal information entropy. Two kinds of entropy must be distinguished here. Thermodynamic entropy describes energy dissipation and irreversible processes; information entropy describes uncertainty about states. In founding information theory, Shannon formalized uncertainty as information entropy (Shannon 1948). Intelligence reduces an agent's ignorance of its environment and the consequences of action, allowing clearer judgments about paths, rather than reducing the thermodynamic entropy of the external world itself; this reduction in information entropy must be carried by real physical processes. Brains need glucose and oxygen, neural signals require electrochemical processes, and memory and learning require changes in material structures. Intelligence is a highly organized dissipative process inside an open system, rather than mental activity without a cost.

The human brain provides a concrete factual anchor. In discussing the brain's energy budget, Raichle and Gusnard note that it accounts for about 20% of the body's oxygen and calorie

consumption despite comprising only a small fraction of its mass (Raichle and Gusnard 2002). That proportion alone shows that intelligence is not free. The brain is expensive because it compresses vast amounts of environmental information, bodily states, memory, prediction, and action selection into an energy-intensive system. Life's willingness to pay a high price for a brain shows how internal simulation can save even more expensive external trial and error.

Landauer's principle also reminds us that information processing cannot be separated from physical costs. In 1961, Landauer pointed out that erasing information in logically irreversible operations entails heat generation (Landauer 1961). The principle cannot simply be used to calculate the energy consumed by all thought or all AI training, but it establishes a basic fact: information is no ghost detached from the material world. Recording, erasing, transmitting, and computing information all require a physical substrate. Intelligence reduces uncertainty without escaping thermodynamics; it saves the cost of mistaken actions while still consuming energy itself.

Living organisms consequently develop a functional division of labor. A complex living system needs an energy system, an information system, and a system for doing work. The energy system takes in, converts, and supplies free energy; the work system moves, hunts, builds, escapes, and changes the environment; the information system senses, remembers, predicts, and selects paths. Cells have energy metabolism, genetic information, and protein machinery; multicellular organisms have digestive, circulatory, nervous, and motor systems; human civilization has energy infrastructure, production systems, bodies of knowledge, and computing systems. Intelligence is the information hub of a dissipative structure. It does not directly replace energy or work, but determines where they are directed. The brain is expensive because it compresses a large amount of environmental information; civilization developed language, writing, mathematics, and science because these systems compress scattered experience into a capacity for path selection that can be transmitted, used in simulation, and reused. The more complex the information and the more advanced its compression, the more the system can reduce blind trial and error.

Friston's free energy principle has something in common with this line of thought. It seeks to place perception, action, and learning within a unified framework: organisms maintain their states by reducing prediction error or surprise (Friston 2010). This chapter neither simply equates intelligence with free energy minimization nor applies one grand theory directly to every intelligent phenomenon. It borrows a deeper intuition: life continuously predicts, acts, and revises to keep itself within a viable range, rather than passively receiving the world. Intelligence continuously selects paths in a closed loop between body and environment, rather than observing from outside the world.

The Activation Barrier: The Cost of Breaking Chemical Bonds

Reconfiguring matter requires an actual reaction path. The free-energy difference between reactants and products determines the thermodynamic direction of net change under given con-

ditions, while the activation free energy affects the rate of crossing the transition state. Breaking old bonds requires energy, and forming new bonds releases it, so the net account cannot include bond-breaking costs alone. Manufacturing does not always push matter into a higher-energy state; the purpose of a process is to produce the target state at a viable rate, purity, and yield.

At constant temperature and pressure, under the corresponding reaction conditions, the change in Gibbs free energy can be written as $\Delta G = \Delta H - T\Delta S$. A negative reaction free-energy change indicates the spontaneous direction at the given state, but supplies no rate. Changes in reaction quotient, pressure, temperature, and composition also change the direction and the equilibrium position. A catalyst changes the path and the rate, but not the free-energy difference between the same initial and final states.

Ironmaking transfers oxygen from iron-bearing feedstocks to other substances. Carbon and carbon monoxide participate in reduction inside a blast furnace, while coke also supplies heat and supports the burden; energy consumption depends on the feedstock, reduction route, temperature, heat transfer, and equipment (Babich 2021). Algorithms can improve burden preparation, flow, and control, while alternative reducing agents and process routes can also change the entire cost structure. The energy of a single iron–oxygen bond cannot directly serve as the minimum energy bill per tonne of iron.

Producing cement clinker involves carbonate decomposition and the subsequent formation of minerals. Calcium carbonate decomposition is endothermic and its equilibrium also depends on the partial pressure of carbon dioxide; clinker burning requires specified temperature and residence-time conditions. The theoretical heat consumption of about 1.7 GJ/t reported in the literature is a process estimate for a particular clinker composition and system boundary, while actual heat consumption also includes heat-transfer, exhaust-gas, and equipment losses (Madivate 1999; Madlool et al. 2011). Changing the clinker proportion, feedstock, or binder route also changes the basis of comparison.

High temperatures affect equilibrium, reaction rates, and transport processes at the same time. Raising the temperature can accelerate some steps while also increasing losses or damaging the product; catalysis and route redesign can achieve the same objective at a lower temperature. The thermodynamic constraints of given initial and final states, the kinetic barriers along the selected path, and the losses in actual equipment must be distinguished. Intelligence can help change the latter two, and can also help find new combinations of initial states and objectives.

The Entropy Tax: Maintaining Order and Countering Randomness

"Entropy tax" here means the continuing expenditure needed to maintain function. Life depends on metabolism, repair, and regulation, while operating equipment needs energy, cooling, and maintenance. Buildings, crystals, and passive storage structures have different mechanisms of stability; not every form of order requires continuous external work. Maintenance costs should be estimated from the specific mechanisms of damage and disturbance.

Agriculture organizes light, water, nutrients, land, and biological processes into harvestable output. Breeding can change the harvest index and improve disease resistance, photosynthesis, root systems, and resource use. Yield and inputs do not simply redistribute a fixed total: they change together with variety, environment, and management. Solar energy flows, external inputs, land occupation, and harvest outcomes should be accounted for separately.

Structural safety requires dealing with uncertainty in loads, materials, construction, and models. Safety factors and partial factors are specified by codes, failure modes, and structural components; they cannot all be interpreted as "one tonne in an ideal case, two tonnes in reality." Redundant material can also carry loads and dissipate energy. Intelligent design aims to reduce unnecessary expenditure while meeting reliability requirements and retaining structures that withstand disturbances.

The Landauer limit supplies a precise microscopic account: under the corresponding thermal and information conditions, resetting one unbiased bit requires a minimum heat release of $k_{\mathrm{B}}T\ln 2$. At room temperature, this is approximately 2.9×10^{-21} joules. Raising a one-kilogram object by one meter requires about 9.8 joules, but these are entirely different units of operation, and their ratio cannot estimate the costs of complete reasoning and manufacturing tasks. Comparing complete tasks also requires accounting for the number of operations, actual devices, communications, equipment, and auxiliary systems.

2.1.3 The Conservation Equation for Reconfiguring Matter

Amplification Through Virtual Simulation

Human civilization continued and amplified this route. Language compresses individual experience into collective memory, maps turn spatial trial and error into representations of paths in advance, mathematics compresses relationships into forms that support deduction, scientific theories organize scattered experience into testable models, and engineering simulation moves expensive experiments into representational space before execution. Many of civilization's inventions essentially move physical trial and error into symbols, models, and computation in advance. People need not travel every road, because maps can exclude bad routes beforehand; engineers need not let every bridge collapse once, because mechanical models can exclude dangerous structures; doctors need not try every treatment directly on patients, because experiments, animal models, clinical trials, and statistical analysis jointly narrow the space of error.

Artificial intelligence further increases the density of virtual simulation. It can search enormous spaces of possibilities for candidate solutions, discover hidden structures, generate models, predict outcomes, and screen at scale before real action occurs. AI is a new stage of this evolutionary route, rather than a superphysical intelligence outside thermodynamics. Brains simulate the world with neural signals, science with equations and experimental systems, engi-

neering with computer simulations, and AI with high-dimensional statistical models and generative systems. Their common function is to narrow the scope for error, screen paths, and reduce blindness before physical costs are actually paid.

Virtual simulation separates selection from execution, and a classic thought experiment illustrates the distinction. Maxwell imagined a demon stationed at a vessel's valve, able to see the speed of every molecule, admit fast molecules, and block slow ones, thereby creating a temperature difference without consuming energy. From Szilard through Landauer to Bennett, the problem led researchers to include measurement, storage, and resetting in the account of a complete cycle; the preceding section discussed the thermodynamic cost of erasing information (Landauer 1961; Bennett 1982). Applying this thought experiment to industrial systems reveals another account: algorithms have no hands. They can decide when a valve should open or close, but cannot supply the actuator with the mechanical energy needed to move it. AI in industrial systems occupies the information control layer: it can reduce uncertainty about relevant states, resolve airflow vortices in blast furnaces, track vehicles through logistics networks, and detect frequency fluctuations in power grids; it does not directly produce the force of execution, and every change to the state of atoms must pass through actuators powered by electric current, hydraulics, or combustion. From a control perspective, industrial civilization depends on power amplification: in a schematic hierarchy running from neural electrical signals at the nanowatt scale through muscles at the watt scale, vehicles at the kilowatt scale, and factories at the megawatt scale, control signals can mobilize a capacity for work far greater than their own power. Intelligence occupies the information control layer, where signal power can be far below the power of the execution it directs. A control signal is not a power flow: a logic gate can determine whether a dam gate opens or closes, but cannot provide the mechanical energy to lift it; when energy is insufficient, the gate will not move, however correct and timely the command. For an actual amplification chain with an independent energy source at every stage, the power relationship can be written as a product,

$$P_k = P_0 \prod_{i=1}^{k} G_i, \qquad G_i = \frac{P_i}{P_{i-1}} \gg 1 \tag{2.2}$$

Here P_0 is the power of the control signal at the start of the chain, in watts; P_k is the final execution power, in watts; G_i is the dimensionless power gain at stage i; and k is the number of stages. The examples in the text—neural electrical signals at the nanowatt scale, muscles at the watt scale, vehicles at the kilowatt scale, and factories at the megawatt scale—show the range of power scales that control and execution can span. Each stage treats the preceding stage's signal as a control input, rather than an energy source, and draws power from its own energy system; the product accumulates the ratios of output to input power at successive stages, rather than creating energy from nothing. $G_i \gg 1$ also shows that the power carried by the signal layer can be small relative to execution power, so "the command is correct" and "the action

is possible" are two different conditions. A switch can determine whether passage is allowed, while the actuator still needs energy; the product above is the arithmetic expression of this distinction. The product holds when each gain is defined by actual input and output power and every stage has the power source it needs. Each stage must also meet the next stage's drive requirements; if an energy source or actuator is missing, the chain breaks there, and no amount of upstream computational accuracy can compensate.

Virtual Simulation and Its Limits

Virtual simulation, however, is not reality itself. Intelligence can select a path but cannot replace it, schedule energy but cannot cancel it, reduce wasteful dissipation but cannot make the reconfiguration of matter cease to dissipate. It is the arrangement of forces, rather than a force; a mechanism for choosing where energy is used, rather than energy itself. The essence of intelligence is to reduce the number of times the world is moved in the wrong way while it remains heavy, rather than to make the world light.

AI can design a bridge that uses less material but cannot abolish the mass of steel, gravitational potential energy, or the strength of connections; it can optimize a robot's movements but cannot supply a motor's torque, bear a joint's wear, or handle noise on a sensor's behalf; it can screen drug candidates but cannot perform metabolism, toxicology, clinical validation, or long-term feedback in place of a human body. **Intelligence can approach a given task's lower bound on physical cost, but computation cannot abolish that bound.** It can reduce mistaken paths but cannot eliminate the paths themselves; reduce wasted experiments but cannot eliminate verification; reduce waste but cannot eliminate work.

Reconfiguring matter requires connecting descriptions, inputs, processes, and outcomes; a preliminary accounting relationship can be written as follows:

$$\begin{aligned}&\text{Information (recipe)} + \text{Matter (medium)} + \text{Free energy (work)} + \text{Time (process)}\\&\longrightarrow \text{Ordered structure} + \text{Waste heat}\end{aligned} \tag{2.3}$$

Each of the four inputs represents a condition that cannot be replaced by the others. Information is the recipe, expressed dimensionally in bits: it specifies what the structure should become but does not itself move any atom; matter is the substrate, measured in kilograms, supplying the atoms the final structure needs; free energy is the work term, measured in joules, with the required net work determined by the initial and final states and the process conditions, rather than invariably pushing matter from a lower to a higher energy state; time is the process term, measured in seconds, because state transitions must follow a real trajectory through time, rather than jump only within a space of representations. The two terms on the right are outcomes: the ordered structure is the target organization formed by the process, and waste heat is an output of actual irreversible processes. For a continuously operating dissipative system, the boundaries for free-energy input and entropy output must be specified; a closed system can still exchange heat and work and must not be confused with an isolated system. Without entropy

export, continuing irreversible processes consume available free energy, and local order cannot accumulate indefinitely. This is a relationship among conditions for a process, rather than an accounting balance: with information but no matter, the result cannot appear; with matter but no available driving mechanism, the desired change does not happen automatically; with energy but no time, the process cannot unfold; an ordered system that continuously dissipates cannot sustain its function without the corresponding supporting flows. Intelligence can change selection, not the conditions of existence; it can optimize paths, not reality's physical accounts.

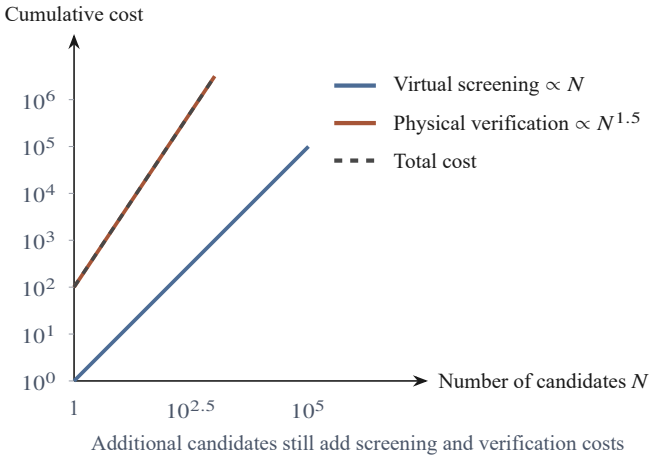
Candidate Proliferation and the Return of Costs

This is precisely what is most easily misunderstood in the AI era. AI reduces the cost of finding possible paths, rather than automatically reducing every cost of completing a state transition. It can generate a thousand building designs, but the buildings still have to be built; propose ten thousand molecular candidates, but experiments and clinical work cannot expand in step; simulate countless robot movements, but real robots still face friction, collisions, batteries, joints, and maintenance. The search space can expand rapidly, while verification and execution must still return to the physical world.

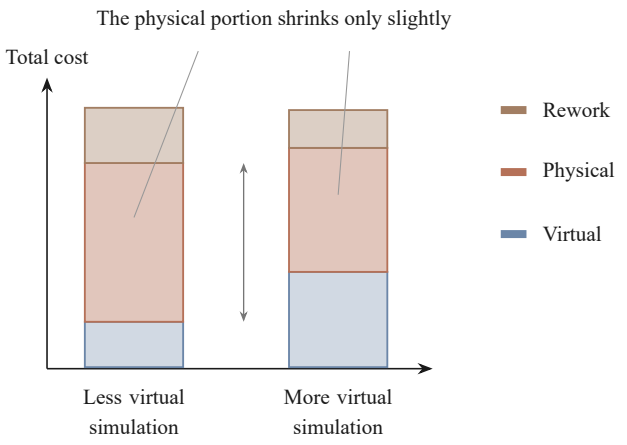
AI itself also has thermodynamic costs. Training and inference need chips, electricity, data centers, cooling systems, supply chains, and maintenance systems. The cloud is not weightless; it merely hides materials, energy consumption, and waste heat behind an interface. Internally, AI appears as information compression and path search; externally, it appears as a high-density dissipative structure. It converts some physical trial and error into computational trial and error, but computation still needs a physical substrate. Substituting bits for some trial and error with atoms does not eliminate atoms; substituting computation for some experiments does not eliminate energy consumption; screening paths with a model does not mean those paths have been completed.

Greater intelligence does not automatically make reality lighter. What intelligence expands first is the space of candidates. Where only ten solutions could once be tried, ten thousand can now be generated; where only a few conditions could once be simulated, vast parameter spaces can now be explored; where only a few experts' intuitions were available, models can now compress enormous bodies of experience. This is certainly progress. Yet more candidates also push pressure toward verification, execution, and those who bear the consequences. Intelligence lowers search costs but may increase the burden of screening; lowers the barrier to proposing solutions but raises the requirements for confirming them; lowers the cost of trial and error in bit space but brings the verification bottlenecks in the domains of atoms and life into view.

Intelligence forms internal representations, allowing some expensive trial and error to take place in models first. Technological viscosity examines how paths enter reality after screening. The two meet at the question of cost: computation and learning can reduce some execution



(a) Candidate expansion



(b) Cost reallocation

Figure 2.1 Virtual simulation changes the allocation of trial-and-error costs without automatically eliminating physical verification. (a) Assuming every candidate proceeds to physical verification, the illustrative costs $C_v = N$ and $C_p = 100N^{1.5}$ represent virtual screening and physical verification, with a total cost of $C_v + C_p$. The coefficients and exponents are schematic assumptions. Both terms increase with the number of candidates; determining an optimal number also requires benefits, success probabilities, or budget constraints. (b) The costs of the same task are allocated among virtual work, physical work, and rework; the figure illustrates a case in which total cost is nearly unchanged, while actual total costs may fall or rise depending on screening benefits and verification requirements.

costs while also adding expenditure on data, verification, and maintenance. Whether a change is worthwhile requires comparing the entire path, rather than counting only how quickly candidates appear.

The distinction becomes particularly clear across the three media of bits, atoms, and life. In bit space, path search can be extremely fast: text, code, images, proposals, and simulations can be generated, revised, copied, and withdrawn at high speed. In the domain of atoms, path selection must be realized in materials, energy, positions, friction, heat, forces, and maintenance; errors cannot be undone as easily as text. In the domain of life, path selection is heavier still, because living processes involve development, metabolism, immunity, reproduction, and long-term feedback; an apparently beneficial intervention may reveal its costs over a longer timescale. The deeper intelligence enters these media, the more it must acknowledge that paths can be screened in advance, while state transitions must still pay real-world costs.

Once a path has been selected, materials, energy, and structures for execution must still realize it. Computation can help reduce this expenditure, but cannot replace the actual process. The next step is to examine resource conditions in detail, distinguishing costs that arise from initial and final states from those that arise from processes and losses.

2.2 Resource Barriers: Power Flux, Binding Energy, and Material Abundance

Once intelligence has selected a path in representational space, the cost must be settled in the medium, and the medium's first account is resources. Of the three kinds of constraint, resource barriers are the easiest to misjudge. Discussions of intelligence's capabilities usually place the constraints on the information side: is the model strong enough, is there enough data, is the algorithm good enough; these conditions are indeed improving rapidly, and the improvements can be measured. For a proposal to become a fact rather than a description, however, it must pass another set of checks with entirely different criteria: can the materials be obtained, can energy be supplied at the required power, can the process finish within an acceptable time, and can the resulting structure be maintained over the long term? The material world does not change automatically because a proposal has been made. A drawing can be generated instantly, a route rapidly optimized, and countless possibilities simulated on a screen, but any proposal entering reality must answer older questions: where do the materials come from, how is energy supplied, how does the process unfold, how does the structure remain stable, how are losses handled, and how is order maintained? This is the resource barrier.

A resource barrier is neither a simple "lack of resources" nor a temporary shortage of supply, but the conditions for state transitions that every material reconfiguration must pass through. A real system's transition from state A to state B reconfigures material positions, energy distributions, structural relationships, and processes in time, rather than merely rearranging symbols. Information can provide the recipe, intelligence can select the path, and organizations can coordinate action, but matter does not rearrange itself automatically because an algorithm is better. Resources are not isolated "things" in nature either: they are neither names in the

periodic table nor total reserves in the crust, but matter in a state that permits extraction, separation, processing, transport, control, and maintenance. A resource is a combination of matter, available energy, location, structure, purity, interfaces, and usability, rather than atoms alone. Without a usable state, existence does not amount to a resource; without a route into a technical system, reserves do not amount to capability. The first question about a resource barrier is therefore whether matter can be brought into a usable state, beyond whether it exists.

Iron, silicon, nitrogen, lithium, copper, and aluminum all occur in nature, but natural occurrence does not automatically confer technological usability. To enter a technical system, matter must undergo extraction, separation, purification, transport, processing, stabilization, and maintenance. Every step requires available energy, incurs losses, and depends on equipment, organization, and time. Usability is therefore the result of a technical system's work rather than an intrinsic property of matter, which in turn explains why the same land supports vastly different populations and industries at different technological levels, and why resource statistics always lag changes in technical capability. What we call a resource is usability that a technical system creates in natural matter.

2.2.1 Power-Flux Limits and the Kardashev Scale

Levels of Available Energy

The "available-energy constraint" here does not mean applying one particular thermodynamic potential to every technical system; it refers more broadly to the energy conditions that can drive state transitions, sustain local order, and actually be called upon by technical systems. For an engineering system, the key is the energy's entry into the process in the appropriate form, at the required power and location, within the right time window and with sufficient control precision, as well as its total amount. Heat, fuel, electricity, mechanical work, chemical potential, and potential energy can all contribute to real-world change, but become capabilities only when technical systems can dispatch, convert, and constrain them. Available energy can also be stored in structures: cities and infrastructure lift, fix, and connect vast quantities of material, retaining some energy as gravitational potential energy and in other forms while dissipating the rest during construction; changing a structure requires rearranging these energy accounts, and concentration, location, and purity are prerequisites for state transitions alongside energy.

How much reconfiguring capability a form of available energy can support thus depends on its energy scale. In 1964, Kardashev proposed classifying civilizations by the power they could command (Kardashev 1964); the scale is commonly written as

$$K = \frac{\log_{10} P - 6}{10}, \tag{2.4}$$

Here K is the dimensionless civilization index and P the total power the civilization can command, in watts; the denominator 10 means that a 0.1 change in the index corresponds to one

order of magnitude in power, while each increase of 1 in K corresponds to ten orders of magnitude. On this scale, Type I corresponds to 10^{16} watts; humanity's present power consumption is on the order of 2×10^{13} watts, or 20 terawatts, giving $K \approx 0.73$, about 2.7 orders of magnitude short of 1.0 (Smil 2017). Another convention often cited in the literature places Type I at the total solar radiation received by Earth, approximately 1.7×10^{17} watts, corresponding to $K \approx 1.12$ in this formula; the two conventions differ only in their zero point, which does not affect the conclusion below. In this coordinate system, the gap between 0.73 and 1.0 is not 27%.

Chemical and nuclear reactions have different energy scales per event, nuclear fission already supplies electricity, and solar, hydroelectric, and wind power do not require humans to burn chemical fuels (Jaffe and Taylor 2018). The microscopic energy scale of a source cannot directly determine the macroscopic power available to a factory. Industrial processes are also constrained by energy conversion, transmission, equipment, materials, and control.

This scale, however, is easily misread as a one-dimensional ranking. Total power is only one coordinate of a civilization's capabilities: the same power can produce vastly different results depending on its form, location, and precision of release; conversely, a system with less total power but strong dispatch capabilities can achieve state transitions in particular domains that a more powerful system cannot. The Kardashev index addresses a ceiling rather than the whole problem, and treating K as a civilization score omits three variables: the form of available energy, its release location, and the control window.

From this perspective, technology's primary object is locally ordered structure sustained jointly by matter, energy, and constraints, rather than information. Bridges, chips, batteries, drugs, factories, cities, and data centers are more than "objects." They are functional structures formed by using energy to move matter away from dispersed, disordered, or unusable states. They must be manufactured, retain their functions amid disturbances, and continually resist wear, corrosion, heat, noise, aging, and failure. Technology involves more than turning ideas into reality: it converts available energy into controllable, stable, maintainable local order, and thermodynamics and energy history repeatedly show civilization to be an open process dependent on energy conversion, material organization, and waste-heat removal, rather than a symbolic system detached from energy (Smil 2017; Callen 1985).

The Step Beyond Chemical Energy

Nuclear fission supplies power through nuclear fuel, reactors, and thermodynamic cycles, while imposing requirements for fuel supply, materials, waste, and safety governance. Its scope for sustained expansion depends on resource pathways, technologies, costs, and institutions; neither the natural abundance of uranium-235 nor the existence of waste can decide the question alone. The respective constraints of chemical, nuclear, and other energy sources must be assessed across the complete system.

Each deuterium-tritium fusion event releases approximately 17.6 megaelectronvolts. Ob-

taining useful net energy also requires meeting conditions for plasma confinement, materials, the tritium cycle, heat removal, and equipment operation (Atzeni and Meyer-ter-Vehn 2004). If fusion develops into a mature energy system, it may broaden supply options; until then, industrial capability can continue to improve through existing energy sources, conversion, and processes, without being uniformly confined to a single valence-electron energy scale.

The rocket equation shows that, for a given effective exhaust velocity, larger velocity increments require a rapidly increasing mass ratio. Chemical propulsion has a practical performance range: about 4.5 kilometers per second is representative of high-performance chemical engines, rather than a strict ceiling shared by every design. Missions can also use staging, gravity assists, or other forms of propulsion. The force of these constraints lies in their item-by-item calculability, rather than in declaring a radius of activity directly from the energy of a single reaction.

$$\Delta v = v_e \ln \frac{m_0}{m_f}, \qquad \frac{m_0}{m_f} = \exp\left(\frac{\Delta v}{v_e}\right) \tag{2.5}$$

Here Δv is the velocity increment gained by the rocket, in meters per second; v_e is effective exhaust velocity, in meters per second, determined jointly by the propellant, reaction energy, expansion process, engine efficiency, and other factors, with the text's 4.5 kilometers per second representing the order of magnitude for high-performance chemical engines; m_0 is mass before ignition and m_f mass after propellant exhaustion, in kilograms; their ratio is the dimensionless mass ratio. The logarithm is the key: each additional v_e of velocity increment multiplies the mass ratio by e; for a given mode of propulsion, the larger the desired velocity increment, the greater the mass fraction that must be reserved for propellant, while the attainable range of v_e is constrained jointly by reaction energy and engineering design. This transport cost of chemical propulsion takes the form of a mass ratio rising exponentially with $\Delta v / v_e$; staging, gravity assists, and other modes of propulsion require their own mission-specific accounting. The limits of applicability are these: the equation gives an ideal velocity increment, neglecting drag and gravity losses, so a real mission requires more propellant with other conditions held equal; this book substitutes no target Δv for a particular mission, since that depends on the mission itself, and uses the equation here only to show how the order of magnitude of v_e constrains the cost of macroscopic transport.

The Mismatch Between Computing Speed and Power Delivery

Computation, energy storage, and actuators improve along different paths. A battery's specific energy, specific power, service life, safety, and cost must be balanced together; cell-level values, pack-level systems, and theoretical material values must also be distinguished. Comparisons of fuels and batteries must also account for engine and motor efficiencies and mission mass. Historical annual improvement rates are not eternal constants, and percentage-based compound growth is not linear growth (Ziegler and Trancik 2021; Viswanathan et al. 2022).

Theoretical limits such as those of Bremermann and Bekenstein concern information processing and information capacity, respectively, under particular assumptions (Bremermann 1982; Bekenstein 1981). They cannot predict how many more years commercial chips can improve at their present pace. Practical improvements in devices also face limits in materials, interconnects, memory, cooling, and manufacturing economics; robot motion requires a separate power account for actuators and energy storage.

Four Sets of Conditions

Once technology is understood as the generation of locally ordered structures, the conditions that material reconfiguration must pass through can be listed one by one. The first set concerns materials: real-world change needs a physical medium, bridges need steel and concrete, batteries need metals, separators, and electrolytes, chips need high-purity materials, and drugs need molecules that can be synthesized, remain stable, and enter the body. Without a medium, information remains a description; with the wrong medium, a proposal cannot enter the world. Many technologies fail for reasons of material form, purity, technological chains, service life, and long-term stability, rather than a lack of imagination. The second set concerns available energy: moving a material from one state to another generally requires crossing barriers, with ores to reduce, molecular bonds to break, crystal lattices to rearrange, objects to transport, structures to lift, and equipment to drive. Energy is a condition of state transitions rather than an optional addition; without enough available energy that can be called upon, controlled, and released in concentrated form, matter will not spontaneously acquire a new order merely because humans become more intelligent, and production, transformation, and disposal in economic processes are likewise bound to entropy increase, dissipation, and irreversibility (Georgescu-Roegen 1971). The third set concerns processes, whose distinctive feature is the sensitivity of their rates to temperature. Real-world change takes time: steel must be smelted, cement calcined, drugs synthesized and verified, chips put through many processing steps, and buildings constructed, cured, inspected, and delivered; processes require equipment, sequencing, environments, people, and stable conditions, and many physical processes cannot be copied, rolled back, or restarted like software, while greater precision imposes stricter requirements for temperature, pressure, purity, timing, and surroundings. This sensitivity has a definite physical form: the Arrhenius relation describes the relationship between reaction rate and temperature, with its expression and conditions of applicability given below in the discussion of strong covalent-bond cleavage. For now, retain its conclusion: the higher the activation energy, the more sharply temperature affects the rate; in high-temperature processing, temperature acts as a switch for crossing barriers rather than a linear amplifier.

The fourth set concerns maintenance, which turns the first three accounts from one-time investments into long-term expenditure. An ordered structure does not persist forever once built: machines wear, buildings age, pipes corrode, materials fatigue, data centers need cooling, infrastructure needs maintenance, and pharmaceutical systems need oversight; locally ordered

structure is a state of continually resisting dissipation rather than a one-time achievement, and maintaining order is itself part of the resource barrier. These four sets of conditions form material reconfiguration's underlying relationship: information provides the recipe, materials the medium, available energy performs the work, processes unfold the structure, and maintenance mechanisms resist dissipation. Information cannot fully replace any element of this relationship: a recipe without a medium produces no result; a medium without available energy undergoes no state change; energy without a process stabilizes no structure; and local order without continuous maintenance cannot sustain a system over the long term.

The four sets of conditions are easily read as an approval checklist, as though passing each item separately were enough to enter reality. The actual situation differs: all four must be present at once, and if any one is almost absent, investment in the other three cannot deliver; material shortages leave available energy idle, insufficient available energy halts a process midway, and loss of process control can make maintenance cost more than the structure itself is worth. Here lies the core meaning of the resource barrier: **Reality consists of executable states, rather than possibilities.** Possibilities can be generated without limit; executable states must satisfy material conditions. A viable proposal does not mean materials are already in place; available materials do not mean available energy has already been supplied; sufficient energy does not mean the process is stable; a completed process does not mean the structure can be maintained over the long term; real-world change must pass through the full chain from materials to available energy, processes to maintenance, and local states to lasting order.

These four sets of conditions do not apply with equal force at every stage of history. For a long time, technical systems bypassed the most expensive one: instead of manufacturing the low-entropy structures they needed, they took products that natural processes had already made. Wood had strength, fibers toughness, rivers a drop in elevation, and coal chemical energy; from that starting point, the accounts for materials, processes, and maintenance were much lighter. The cost of using nature's ready-made structures is dependence on their distribution, stocks, and rates of replenishment; once ready supply cannot meet demand, the costs of forming and processing materials, previously borne by nature, return to the accounts. How long this use can continue determines the form in which resource barriers later reappear.

2.2.2 Using Low-Activation-Energy Materials and Reconfiguring Physical Form

The history of technology begins with selecting matter rather than manufacturing it. Almost none of the materials and power sources humans first used were of their own synthesis; they were structures made in advance by photosynthesis, geological processes, and evolution. Understanding this use of ready-made structures explains both why early technology seemed light and why that lightness could not last.

The earliest technology used parts of the world that the world had already made, rather than

manufacturing the world. Wood, cotton, hides, bones, flowing water, wind, animal muscle, coal, and oil were not resources that humans created from scratch. They were structures already organized, or forms of energy already accumulated, by natural processes. Trees turn sunlight, carbon dioxide, and water into fibers, plants and animals organize environmental matter into bodies, rivers and wind fields give energy flows direction, and fossil fuels seal the solar energy captured by ancient life inside chemical bonds. Early technology seemed light because it drew on work nature had already completed, rather than escaping physics; this is the real meaning of low-hanging fruit.

Using Structures Prefabricated by Nature

Low-hanging means that nature has prepaid part of the bill, rather than that the technology is primitive. Humans need not synthesize wood, design cotton fibers, or produce the chemical energy in coal on demand: nature has prepaid energy, structure, and time for the technical system, which merely cuts, carries, weaves, burns, domesticates, and rearranges these ready-made structures. In other words, low-hanging fruit identifies places where nature has already generated low-entropy structures, rather than fruit that is easy to obtain. Wood already has strength, stone hardness, fiber toughness, muscle mechanical work, rivers a drop in elevation, and coal chemical energy; early technology follows the paths of low resistance nature provides, chiefly changing form rather than rewriting matter itself and drawing on existing order rather than creating order from disorder. This also explains why many early technologies seem especially "natural": a windmill intercepts wind rather than making it; a waterwheel uses a drop rather than creating it; a wooden house comes from cutting trees rather than manufacturing cellulose structures; domesticating animals organizes their existing metabolism and movement rather than manufacturing muscle. Technology has costs, but geological, biological, and climatic processes have already amortized a large share of them in advance.

Cotton brings naturally formed cellulose structures into textile processing. The plant has already synthesized the material; textile machinery chiefly organizes drafting, twisting, and winding (Klemm et al. 2005). This fit of material form to process was an important condition for mechanization, while the Industrial Revolution also involved energy, trade, capital, labor, and institutions; "the energy density for breaking strong chemical bonds was unavailable at the time" cannot explain it alone.

Early rough mechanical processing depended on a fit of microscopic form as well as the material's chemical properties. Cotton fibers have natural convolutions along their axes, related to cell-wall structure and collapse during drying: after the boll ripens and opens and water evaporates, the roughly cylindrical cells become twisted ribbons. Friction between fibers and cohesion produced by twisting allow loose staple fibers to form a continuous yarn. Scales on wool create directional friction and make fibers prone to felting and entanglement, requiring control of carding intensity and fiber alignment; fine flax yarn is often wet-spun to improve fiber slippage, while coarser yarn can also be dry-spun. Cotton requires neither resynthesizing

cellulose nor chemical bonding to form yarn, providing favorable conditions for early mechanical processing. Raw-material preparation is the other half of the explanation. Wool's diameter, length, and crimp vary with the animal, body region, and environment, and input variation adds to the control burden of fixed mechanical motions. Cotton, as a seed fiber, also varies in fineness, length, and maturity; grading, cleaning, and blending bring these differences within a range machines can handle. The water frame need not identify each fiber: it repeats drafting, twisting, and winding; biological form and raw-material preparation have already reduced input variation enough for fixed mechanical motions to run continuously (Cotton Incorporated, n.d.; Alliance for European Flax-Linen and Hemp, n.d.). Low-hanging fruit does not extend indefinitely. As demand expands, technical systems cease to be satisfied with the forms nature supplies: wood is not hard enough, requiring steel; natural fertility is not stable enough, requiring fertilizer; natural fibers are not diverse enough, requiring synthetic materials; natural crystals are not precise enough, requiring semiconductors; ordinary combustion is not controllable enough, requiring electricity systems; natural medicines are not precise enough, requiring synthetic molecules and targeted interventions. Technology shifts from drawing on existing structures to actively manufacturing them.

The Activation-Energy Staircase

Moving from existing forms to active manufacture usually makes process conditions more complex. This book calls that change the "activation-energy staircase," pointing to thresholds in purification, reactions, lattices, defects, and interface control. Individual processes have activation energies with precise physical meanings; a whole factory's capital and maintenance burdens have no single activation-energy value. The two must be measured separately.

Technological usability depends on whether a material has the required purity, form, concentration, and location. Sand is not a wafer, ore is not steel, and lithium in the ground is not a battery. Producing a usable state requires extraction, transport, processing, reactions, and quality control, with costs arising from thermodynamics and kinetics as well as equipment and organization.

Precision manufacturing often needs stricter cleanliness, defect control, measurement, and process windows. As chips shrink, some control tasks become harder, while process innovation may also lower cost per function. The account must cover the energy use, yield, capital, and maintenance of particular steps, rather than assigning every control difficulty to one progressively higher activation energy.

Startup and maintenance belong in the same life cycle. Finer structures may increase control and maintenance requirements, or perform the same function with less material and energy. Deeper penetration into matter does not guarantee that total cost rises in one direction; it requires accounting for the complete path from discovery through manufacture to long-term operation.

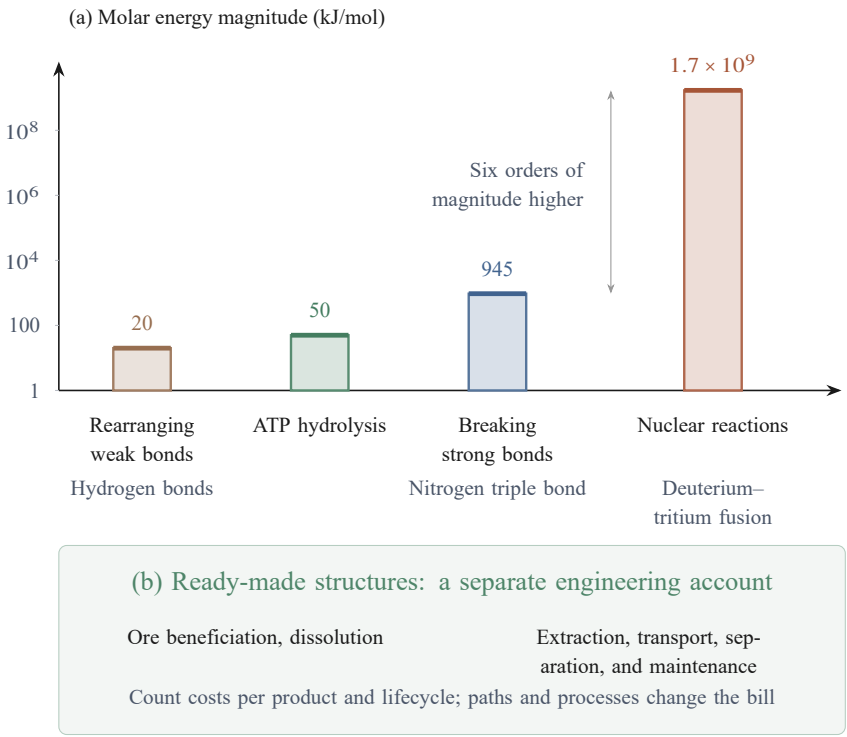


Figure 2.2 Microscopic energies and engineering costs require separate accounts. The upper panel gives typical energy magnitudes per mole of the indicated bonds or reaction events: about 20 kilojoules for hydrogen bonds, 50 for ATP hydrolysis free energy under cellular conditions, 945 for gas-phase nitrogen dissociation, and 1.7×10^9 for deuterium–tritium fusion, equivalent to 17.6 megaelectronvolts per reaction (Darwent 1970; ITER Organization, n.d.). These are bond dissociation energies, reaction free energies, and nuclear energy releases, rather than a common activation energy or ceilings on energy supply. The lower panel accounts for ready-made structures by product, process, and lifecycle; beneficiation, dissolution, and transport have no common "mole of task." Feasibility also depends on reaction paths, energy conversion, temperature, power, and control. Ammonia synthesis and ironmaking require full thermodynamic, kinetic, separation, and equipment accounts; catalysts and process improvements can change actual cost.

From Biological to Mechanical Power

Beyond these thresholds comes a change in the form of motive power, affecting how work is done rather than the material's interior. Before mechanization became widespread, skeletal-muscle contraction was a major source of production power, with humans and draft animals converting chemical energy into mechanical energy. Muscle can convert energy efficiently, but its sustained power, recovery needs, and consistency of output constrain its use as an industrial power source. For prolonged work, about 75 watts is a common order-of-magnitude estimate of sustained human mechanical output, varying with fitness, task, and duration; higher output

requires more metabolic energy and heat removal, sustained work still requires rest and recovery, and fatigue involves several cellular and neural mechanisms (Smil 2015; Westerblad et al. 2002). On a schedule of 8 to 12 working hours a day, a single biological power source has a daily utilization rate of roughly 30 to 50 percent. Output consistency is another problem: physiological tremor, attention, and fatigue introduce fluctuations, while hand-spun yarn's twist and strength also vary with the operator and raw material, making uniformity a matter of continuous control. A production system dependent on one biological power source must therefore manage low sustained power, intermittent operation, and variation in output together.

Early textile machinery's significance lay in decomposing this biomechanical process into rigid-body motions. Hand spinning coordinates the opening and closing of fingers, rotation of the wrist, and pull of the arm in a nonlinear movement with many degrees of freedom; rigid linkages, cams, and gears constrain those degrees of freedom, retaining only the necessary rotation and reciprocation. In the ideal kinematic relationship, geometric constraints determine the trajectory, and with a steady input speed the transmission ratio constrains roller and spindle motion; power source and control system are thereby decoupled, with the waterwheel providing torque and the transmission ratios embodying control logic in the structure. The machine need not "know" how to spin: its physical structure confines it to a trajectory that performs the spinning motions; this determinacy provides a physical condition for production at scale, while actual uniformity still depends on raw-material input, slippage, and mechanical condition.

One crucial source of power for the early textile industry was water, drawing on gravitational potential energy. A waterwheel uses a drop in elevation to deliver rotational torque, with power determined by flow, head, and efficiency; a small or medium-sized river with suitable flow and head can provide tens of kilowatts, equivalent to the sustained mechanical output of hundreds of people, concentrated on a drive shaft to run machines with many linked axes. A river needs no sleep, and when hydrological conditions permit it can supply power day and night, substantially extending a machine's available operating hours. This period contains another easily overlooked detail: Newcomen engines were already pumping water before Watt's improvements, repeatedly heating and cooling their cylinders each cycle and wasting large quantities of heat; Watt's separate condenser moved condensation out of the main cylinder so that it could remain hot (Science Museum Group, n.d.). Early steam power could also supply a waterwheel first: a steam pump returned water from downstream to an upstream pond, and the existing waterwheel and transmission then drove the machinery (Historic England, n.d.). This roundabout route shows how location, continuity of water supply, and existing mechanical interfaces can determine whether a power technology is adopted, alongside energy-conversion efficiency.

The Supply Ceiling for Ready-Made Structures

Nature's prepayment is limited by geometric conditions on the supply side. Biomass energy at Earth's surface depends on stellar radiation: the Sun emits about 3.8×10^{26} watts, but

after spreading across roughly 150 million kilometers of space, the irradiance reaching the top of Earth's atmosphere is only about 1,361 watts per square meter, the solar constant (Prša et al. 2016). Atmospheric absorption, scattering, cloud reflection, and the geometric reductions from day-night cycles and latitude make the annual mean surface irradiance available at middle latitudes much lower. Plant conversion efficiency compresses the allowance further: photosynthesis uses photons between 400 and 700 nanometers, with this photosynthetically active radiation accounting for about 43% of total radiation; the quantum requirements of photochemical reactions limit theoretical maximum conversion efficiencies to about 4.6% for C3 plants and 6.0% for C4 plants (Zhu et al. 2008). In actual agriculture and forestry, water, temperature, nutrients, and the growing season usually keep realized solar-energy conversion below 1%. This keeps average biomass power density low: at an estimated 0.4 watt per square meter, obtaining 1 watt of sustained biochemical power requires about 2.5 square meters of land, with the actual value varying by climate, species, and harvesting method (Smil 2015). Write these three factors as a product,

$$P = \Phi A \eta \tag{2.6}$$

Here P is the sustained biochemical power obtained, in watts; Φ is mean irradiance under the chosen reference-surface and time convention, in watts per square meter; A is the area receiving sunlight, in square meters; and η is the dimensionless overall efficiency of conversion from that reference radiation to usable biochemical power. The reference surface and averaging convention must be consistent: starting from approximately 1,361 watts per square meter at the top of the atmosphere requires η to include atmospheric, day-night, and geometric reductions; using annual mean surface irradiance instead incorporates those reductions into Φ, so they must not be applied twice. Written as a product, the relationship yields both judgments in the text: breeding, irrigation, and agronomy can improve the efficiency term η, subject to photochemical limits, with the theoretical maxima of about 4.6% for C3 plants and 6.0% for C4 plants using the incident-solar-radiation conventions of their respective studies; the area term A cannot increase incident sunlight without limit by stacking leaves within the same projected area, which gives the quantitative basis for capital accumulating in three dimensions while the photosynthetic energy flow powering it remains constrained by sunlit area. The limits of applicability are these: this is a steady-state mean-power relationship, excluding storage, transport, seasonal fluctuations, and other land demands involving water, fertility, and ecology; substituting the theoretical maximum for η overestimates real supply, and the average solar-energy conversion achieved in actual agriculture and forestry depends on field conditions.

This area constraint shaped the geography of pre-industrial society. Agriculture, forestry, and pastoralism supplied food, fiber, fuel, timber, and draft power, with many demands competing for the same land. Draft horses could sustain greater mechanical output than people, but also needed feed and pasture (Smil 2015). Energy flows and land use were therefore closely

linked, while trade, property rights, and the organization of labor determined how the constraints were distributed.

The calculation measures a flow, the sunlight received anew each year, rather than stocks already accumulated in nature; that is its boundary. Wood, fibers, and animal power depend on energy flows, and once stocks accumulated over geological time can replace them, the same land can support much greater power, which is why fossil energy later transformed the situation. Before those stocks were used at scale, however, the limits above applied to every system dependent on the current year's photosynthetic output; Malthusian stagnation was therefore the inevitable saturation of an energy system with low power density on a finite two-dimensional surface, rather than a failure of agricultural technology.

The early route beyond this ceiling shifted constraints onto other land rather than improving efficiency. Wool is a secondary product: sunlight is first captured by pasture and then transformed through sheep metabolism, with 1,000 pounds of wool requiring 50 to 100 acres of pasture; cotton is a primary product, converting sunlight directly into cellulose, and the same output requires only 1 to 2 acres at low latitudes with sufficient light and heat. Had Britain's textile industry around 1830 replaced imported cotton with domestic wool, the raw material alone would have required more than the country's entire arable area. Importing cotton, sugar, and timber was physically equivalent to renting overseas land for photosynthesis, transferring workers' food, industrial feedstocks, and part of the biological power supply to someone else's sunlight. Colonies played two roles at once, supplying low-entropy matter and receiving high-entropy waste, with environmental improvements at home paid for by ecological degradation elsewhere. The constraint remained, appearing in another place; this is an operation the book will return to repeatedly.

In technological history, this is a move from drawing on nature's ready-made structures to manufacturing low-entropy structures, rather than a simple progression from the "primitive" to the "advanced." Early civilizations mainly used energy and forms already concentrated by nature, industrial civilization began rearranging macroscopic matter at scale, modern manufacturing extended control to chemical, crystalline, and microscopic structures, and still deeper technologies began intervening in living networks, complex materials, and planetary cycles. The essential character of this path remains unchanged: **The further technology moves beyond forms supplied by nature, the more it must pay to generate those forms itself.** Civilizational capability involves injecting free energy into structures that are harder to control and keeping them operating stably, beyond commanding more energy. More total energy does not imply greater technological capability; whether energy can become real structure depends on releasing it at the right place, time, and scale, with sufficient power density and control precision, while handling waste heat, by-products, failures, and maintenance. Civilization is a more finely organized mechanism for free energy, rather than a larger fire.

Turning a laboratory result into a product requires raw materials, purity, synthesis con-

ditions, yield, stability, scale-up, and waste treatment to work together. A breakthrough in catalysts, batteries, or new materials may transform the entire process; whether it does must be measured at the same output and within the same boundaries. An isolated result provides an entry point; the full process determines usability.

Depleting Stocks of Low-Energy States

The ceiling on flow-based energy matters because it makes industry's rate of expansion dependent on the current year's sunlight and rainfall. When forest regeneration cannot match metallurgical consumption, the system reaches a throughput ceiling: surface ecosystems cannot capture enough solar energy in real time to feed industrial machinery. Two routes remain, retreating to a low-energy agricultural society or finding a denser energy source unconstrained by real-time sunlight. Coal provides the latter. Burning a ton of coal releases solar energy stored in plant remains and preserved through carbonization under geological pressure over vast periods, compressing a stock accumulated over tens of millions of years into minutes of power flow.

Replacing flows with stocks changes scale and continuity of supply first, rather than merely the temperature of an individual fire. Furnace temperatures with wood, charcoal, and coke all depend on fuel, air supply, and furnace structure, and charcoal ironmaking long predates large-scale coal use; coke provided blast furnaces with a more stable burden, reaction conditions, and fuel supply at a larger scale. High temperature changes reaction rates and equilibria, while oxygen in iron ore is transferred chiefly through reductants such as carbon monoxide, rather than carbon atoms using kinetic energy to tear iron-oxygen bonds apart directly. The transition from wood and charcoal to coal and coke therefore strengthened industry's ability to rearrange matter, rather than making chemical change possible for the first time. Drawing on geological stocks expanded the scale of processing humans could sustain, while adding the costs of mines, transport, furnaces, and emissions.

What Intelligence Can Change

Intelligence can search for new structures, predict properties, find catalytic pathways, reduce experimental blind spots, and improve maintenance. Better paths can reduce trial and error while also changing production itself. Assessing their contribution requires recording what is saved, what is added, and how long the resulting structures can be maintained.

One further point is needed: intelligence's improvements are uneven across the staircase. In representational space, the benefits of search, screening, and optimization grow rapidly with candidate numbers, almost unconstrained by physical rates; where physical execution is required, experimental apparatus, material batches, working hours, and environmental control constrain the benefits, making growth much slower. Extrapolating the first rate of improvement to the second is among the most common errors in these judgments, and a recurring source of disappointment when new proposals are scaled up.

As technology moves from using natural forms to actively constructing functions, it takes on more of the work of generating structure. Functions that were expensive can still become cheap through new principles, materials, and production at scale. Resource analysis identifies which condition currently controls cost and which improvements actually change it.

The staircase extends in more than the vertical direction. Technologies reaching deeper into matter encounter several distinct boundaries at once: some arise from immense masses already fixed in place, others from thermal noise and defects, and still others from how elements are distributed across the world. They demand different forms of payment and respond to different solutions, so treating any one as the whole produces false cost judgments. Only by separating them can we judge which resistance can be compressed and which must be paid in full.

2.2.3 Breaking Strong Covalent Bonds and Material Transformation with Fossil Energy

Expanding the supply of high-temperature processing, hydrogen, and high-pressure operation allowed industrial systems to fix nitrogen at rates far above natural replenishment. One decisive target here was the triple bond in the nitrogen molecule. Nitrogen is a structural element of proteins and nucleic acids, while atmospheric nitrogen exists chiefly as a stable diatomic molecule that must be converted into a usable form before entering crops. Life had long been fixing nitrogen at ambient temperature and pressure; industrial synthesis partly freed the scale of supply from the rates of land-based and biological processes. Turning coal and gas into food translates geological stocks into biological stocks, through processing conditions that combine high temperature, high pressure, catalysis, and hydrogen supply.

Between a Solution and a Usable Result

This complete chain explains why technologies that seem "solved" often struggle to change reality quickly. An energy device's efficiency in a model does not mean that the grid, storage, land, dispatch, maintenance, and replacement cycles are ready. A robot's feasible movements in simulation do not mean that motors, joints, sensors, friction, collisions, and safety boundaries have been resolved. Material selection, material efficiency, and life-cycle analysis matter precisely because material systems are constrained by how materials are obtained, processed, used, maintained, recycled, and replaced, beyond whether materials exist (Ashby 2012; Allwood and Cullen 2012). **Resource barriers are boundary conditions of the material world, rather than accidental consequences of technological backwardness.** Every real system must answer the same questions: can enough material be obtained, can states be changed with sufficient power, can processes finish within an acceptable time, can by-products and waste heat be handled, and can the resulting order be maintained over the long term? A proposal that cannot answer them remains a proposal in information space, short of capability in real space. Civilizational capability therefore cannot be measured only by information possessed,

or even energy commanded; what matters is whether available energy can enter the right material structures with sufficient power density, control precision, and low losses, while waste heat, losses, and maintenance are handled; total energy is only one condition, and usable form, release location, control windows, and capacity for maintenance determine whether it becomes real change. More concrete comparisons make the distinction clear: a heap of coal is not industrial civilization, but steadily converting chemical energy into mechanical work, electricity, transport, metallurgy, and factory systems is; sunlight on Earth is not a photovoltaic system, but collecting, converting, storing, dispatching, and connecting dispersed energy flows to a grid is energy technology; silicon's abundance is not information civilization, but bringing silicon into a high-purity, processable, operable technical chain is chip-making capability.

A misreading must be prevented here. Resource barriers are neither a doctrine of resource scarcity nor technological pessimism: they describe conditions rather than a ceiling's height. If a technology finds a shorter chain and achieves the same state transition with less material, lower temperatures, and fewer steps, the barrier moves, as it has repeatedly over the past two centuries. The argument objects to mistaking relocation for cancellation of conditions, rather than bypassing a particular obstacle: shifting consumption into deeper technological chains, more distant production sites, or later maintenance cycles leaves the condition in place, appearing elsewhere.

A "resource" is therefore a usable state produced by a technical system, rather than a static object given by nature. More advanced technologies make resources increasingly through high-purity materials, stable power, clean environments, precise interfaces, and maintainable structures, instead of drawing directly on natural resources. Resource barriers are conditions of generation that technology must continually bear, beyond a wall placed before it; studies of global material stocks and critical minerals show how more complex modern technical systems depend increasingly on large material flows, long-lived infrastructure stocks, and capacity to mine, refine, transport, and process particular minerals (Krausmann et al. 2017; International Energy Agency 2021).

Crossing the Barrier: Ammonia Synthesis

Whether a reaction can proceed and how fast it proceeds are different questions, the first addressed by free energy and the second by kinetics, whose expression was deferred in the earlier discussion. The temperature dependence of reaction rates is commonly described by the Arrhenius relation:

$$k = A \exp\left(-\frac{E_a}{RT}\right), \tag{2.7}$$

k is the reaction-rate constant, whose units depend on reaction order; A is the pre-exponential factor, related to collision frequency and orientation; E_a is activation energy, in joules per mole; R is the gas constant, approximately 8.314 joules per mole kelvin; and T is absolute temperature, in kelvins. The relationship assumes that a single dominant barrier controls the reaction and E_a is approximately constant over the temperature range. Because E_a appears in

the exponent, higher activation energy makes the rate more sensitive to temperature: doubling temperature does not double the rate, but can change its order of magnitude. Steelmaking and calcination usually require high temperatures, while conventional ammonia synthesis also uses high pressure; temperature, pressure, and catalysis jointly determine rates and equilibria, with the specific conditions set by the reaction and process. A catalyst takes another route, lowering E_a and increasing the rate at the same temperature; it changes the pathway's height rather than bond energy itself.

Ammonia synthesis brings this mechanism into sharp relief. Nitrogen makes up about 78% of dry air by volume, and the two atoms are held together by a triple bond comprising one σ bond and two π bonds, with a bond energy of approximately 945 kilojoules per mole and bond length of about 0.110 nanometer, giving the molecule high kinetic stability at room temperature. Before industrialization, new reactive nitrogen came chiefly through two pathways: lightning's high temperatures drove nitrogen-oxygen reactions whose products reached the surface by deposition; nitrogen-fixing microorganisms used nitrogenase to reduce nitrogen at ambient temperature and pressure, with the standard reaction consuming at least 16 ATP per nitrogen molecule fixed and the cell paying additional metabolic costs. Both pathways have limited throughput, and estimates of global natural fixation also depend on the accounting boundaries for land, oceans, and geological nitrogen; in 2013, Vitousek and colleagues estimated preindustrial terrestrial biological nitrogen fixation at 58 teragrams of nitrogen per year, within a range of 40 to 100 teragrams (Vitousek et al. 2013). On farmland where nitrogen is the limiting factor, more sunlight, water, or carbon dioxide cannot replace reactive nitrogen: Liebig's law of the minimum imposes a conditional constraint here, rather than one bottleneck shared by all farmland.

The Haber-Bosch process broke this lock by placing the reaction in an extreme physical environment. Ammonia synthesis is itself exothermic, yet proceeds at almost zero rate at room temperature because the activation barrier associated with the nitrogen triple bond must be crossed. Temperatures must be maintained at 400 to 500 degrees Celsius, a thermodynamic compromise: by Le Chatelier's principle, higher temperature favors the reverse reaction and lowers single-pass yield, but brings the rate into an industrially usable range. Pressure is also raised to 150 to 200 atmospheres to counter the equilibrium shift from high temperature and increase gas-molecule collision density. Hydrogen production itself consumes substantial energy, commonly through methane steam reforming, a strongly endothermic reaction driven by burning natural gas. Across the complete chain, modern agricultural output is more than nature's gift: fossil energy stored through geological history is used under high temperature and pressure to break some of the microscopic world's most resistant bonds, translating geological stocks into biological stocks. Catalysts can lower temperature and pressure requirements, and materials screening can find more efficient iron- or ruthenium-based catalysts, but converting inert nitrogen to reactive ammonia still requires bonds to be broken and rearranged (Erisman

et al. 2008).

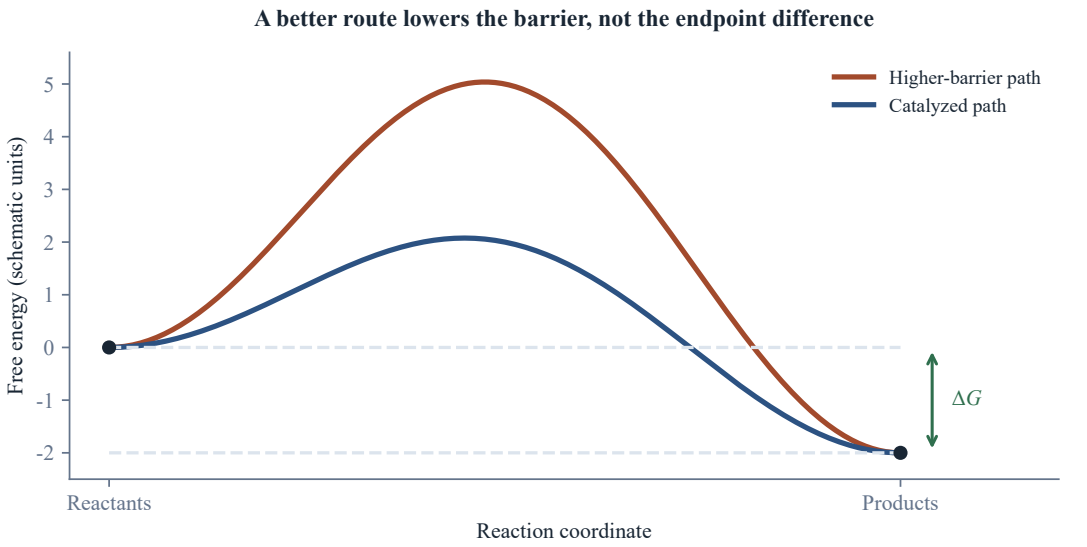


Figure 2.3 Catalysis changes the pathway and kinetic barrier. The two schematic routes have the same initial and final states and the same ΔG. Lowering activation free energy helps increase the reaction rate; the initial and final states determine equilibrium.

Where Intelligence Enters the Chain

Intelligence can screen catalysts, find new reaction routes, optimize process windows, dispatch energy, and predict failures. Catalysis changes reaction pathways and rates, while process integration can recover heat and reduce separation costs. Improvements should be measured through the whole process's energy use, yield, capacity, safety, and downtime.

Intelligent systems themselves also face resource barriers. Hardware, cooling, and utilization improvements can make data centers substantially more energy-efficient, but "greater efficiency" does not mean "the physical foundation disappears": large-scale computation still occupies equipment, land, electricity supplies, cooling systems, technological chains, and maintenance capacity (Masanet et al. 2020). Information processing is no ghost detached from physics either: Landauer's principle reminds us of thermodynamic lower bounds for information erasure and logically irreversible operations, although the principle cannot simply be converted into total energy consumption for every real computing system (Landauer 1961). Judging whether a technology actually lowers real-world cost therefore requires asking whether it reduces material consumption, energy losses, processing steps, maintenance burdens, and long-term irreversible consequences, beyond how many proposals it generates in information space. A proposal that merely shifts consumption into deeper technological chains, energy systems, computing infrastructure, or maintenance structures relocates the resource barrier rather than overcoming it. Reliable technological progress completes unavoidable state transitions

through better paths within atoms' hard constraints, rather than escaping atoms through bits; real-world change consists in matter entering a maintainable new order driven by available energy, beyond possibilities merely appearing.

2.2.4 Accumulating Macroscopic Mass and Building Gravitational Potential Energy Stores

As technical systems push deeper into matter, the form of the boundary they encounter depends on scale. Larger, smaller, and more dispersed objects meet different kinds of obstruction; the three boundaries just mentioned therefore take turns dominating at different scales, and the dominant one determines where that layer's costs must be recorded.

The most common misjudgment in material reconfiguration imagines the atomic world as software space, scalable, editable, and recyclable at will. At the macroscopic level, better planning seems sufficient to rewrite cities and infrastructure; at the microscopic level, continued progress seems to promise unlimited miniaturization and precision; at the resource level, sufficiently complete recycling seems to promise endless material circulation. Reality differs. The atomic world has scales, and changing scale changes the form of resistance: mass, inertia, and sunk structures at the macroscopic scale; thermal noise, defects, quantum effects, and cooling pressures at the microscopic scale; abundance, distribution, purification, and irreversible circulation at the elemental scale. Material reconfiguration crosses different physical boundaries at different scales, beyond moving symbols on a uniform plane; one sentence captures it: **The macroscopic world is too heavy, the microscopic world too noisy, and the world of elements too dispersed.**

Mass and Inertia

Start with the macroscopic scale, whose primary constraint is mass. Here mass carries more than weight: it embodies past work, and everything extracted, transported, lifted, fixed, connected, and maintained is more than an object, taking structural form through materials, energy, land, labor, institutions, and time together; buildings, roads, bridges, subways, grids, pipes, ports, and factories are physical memories of past investment in reality. Cities make this especially clear: a city is neither an area on a map nor an abstract container for economic activity, but a system of high potential energy formed by lifting, fixing, and connecting immense masses of steel reinforcement, cement, glass, asphalt, copper cables, pipes, and equipment. Buildings resist gravity, bridges cross space, subways penetrate strata, grids transmit power, pipes sustain flows, and roads organize the movement of people and things; a city is energy fixed in place and historical work solidified.

Material flows measure the scale of this accumulation. Cement, steel, sand, gravel, and other construction materials make up most modern societal material stocks; global socioeconomic material stocks grew more than twentyfold during the twentieth century, overwhelm-

ingly in buildings and infrastructure (Krausmann et al. 2017). People now move annual quantities of earth and sediment comparable to or greater than the natural sediment transport of the world's rivers, with human movement exceeding natural erosion in some basins (Syvitski et al. 2005). Cities are the final deposits of these flows, which is why the atomic world changes slowly: old buildings are not deletable files, old roads are not versions to roll back, and old pipes are not layers to redraw; they occupy space, bind lives, and connect property rights, finance, regulations, habits, and everyday routes. Reconfiguring a district requires discarding already fixed materials and potential energy, then paying for demolition, transport, pollution, replacement, construction, and social coordination, beyond generating a better proposal.

A reasonable objection can now be answered: if mass is the main resistance, why does the proposal with lower unit cost not win automatically? Mass constrains more than construction cost: through property rights, pipe-network interfaces, construction windows, habits, and existing routes, it spreads replacement costs over long periods, while an individual decision-maker often sees only one short stretch; a better proposal can therefore pay on a local account while failing to pay on the whole account. This property of mass has two faces, reliability and difficulty of change. A bridge works because it is sufficiently strong; a city resists change because it is sufficiently heavy. Infrastructure serves over long periods by fixing matter in stable structures; it resists rapid iteration because those structures are embedded in everyday life; macroscopic resistance is the inertia left by successful past construction, rather than accidental obstruction.

Many technologies therefore founder on accumulated mass, beyond conservatism in people's thinking: a superior transport plan cannot automatically move old roads, a better energy system cannot instantly replace a grid, and a smarter urban algorithm cannot rearrange buildings, pipes, parking lots, subway lines, and land patterns simultaneously. Macroscopic reconfiguration rewrites weight within weight already present. Digital technologies therefore often enter the atomic world as overlays: adding navigation to existing roads, sensors to factories, AI triage to hospitals, and monitoring platforms to cities costs far less than rebuilding their underlying structures. Overlay before rebuilding is a practical path under the inertia of macroscopic mass, rather than an accidental choice: the old world is too heavy, and new technology must first attach to its surface.

Gravitational Potential Energy Stores

Raising building materials increases gravitational potential energy. For a given mass and mean height of elevation, the ideal net work is

$$W = m\,g\,h \tag{2.8}$$

W is the increase in gravitational potential energy, m the mass, g local gravitational acceleration, and h mean elevation. With $m = 10^8$ kilograms and $h = 250$ meters, W is about 2.45×10^{11} joules. Design can change material quantities, structure, and height; the lifting ac-

count becomes fixed only once those quantities are given. Construction dissipation, material production, and other energy demands are additional. Potential energy released in demolition usually dissipates, and how much can be recovered depends on the actual equipment. Urban renewal also faces constraints in dismantling, transport, infrastructure, property rights, and continuity of service; stored potential energy is only one part.

Annual material flows reveal another scale. USGS and worldsteel publications issued in 2026 summarize 2025 cement and steel production, respectively, placing the former on the order of 4 billion metric tons and the latter near 2 billion (U.S. Geological Survey 2026; World Steel Association 2026). Cement differs from clinker, and theoretical reaction heat differs from actual whole-process energy use; accounting must use the corresponding production quantities and processes. Mining, processing, transport, and construction form a continuously operating material channel, whose long accumulation produces the city.

Once the reservoir of potential energy has been built, it constrains differently from flow-based energy. Wood and animal power are constrained by throughput, and insufficient current-year output halts expansion; fixed mass constrains through stocks, allowing further accumulation while raising the cost of change until it must be amortized over long periods. A city's center of gravity, pipe alignments, road pattern, and land ownership jointly constrain renovation proposals for decades afterward, forcing new proposals to find gaps in existing geometry. Macroscopic resistance thus has a counterintuitive property: it obstructs rebuilding rather than building.

2.2.5 Quantum Effects and Heat-Flux Limits in Miniaturization

Reducing the scale does not simplify the problem. Larger objects are harder to move at the macroscopic scale; smaller structures are harder to control at the microscopic scale, where the object of control shifts from position to probability. Physical miniaturization is more than proportional shrinkage: each order-of-magnitude reduction changes the set of physical mechanisms at work. Chips have taken this path further than almost anything else, exposing its two kinds of limit most clearly, one from quantum mechanics and the other from thermodynamics.

Noise and Heat

People often assume small means light and light means cheap, but as we approach atomic, electronic, and quantum scales, the world's stability and controllability cease to resemble those of macroscopic objects: thermal noise, surface effects, defect probabilities, quantum tunneling, material fluctuations, measurement disturbances, and heat-removal density become impossible to ignore. Microscopic resistance lies in unstable control rather than immovability, in excessive disturbances rather than excessive mass. Chips are the clearest example: bits seem light and software endlessly reproducible, yet the physical foundation that produces bits is immensely heavy. Making electrons move reliably at the nanoscale as intended requires high-purity ma-

terials, cleanrooms, vacuum, precision lithography, temperature control, chemical processing, defect detection, and global supply chains. Smaller chips require larger environmental-control systems, and shorter electronic paths require a greater supporting density of order; the microscopic world becomes more dependent on extensive supporting engineering as precision increases, beyond any promise of lightness and cheapness.

Miniaturization is more than proportional shrinkage. Macroscopic processing can tolerate roughness, while microscopic manufacturing must suppress impurities, vibration, thermal disturbances, and defects: an invisible particle can destroy a complex structure, a small process drift can affect many devices, and a local hot spot can limit the power density of an entire system. Smaller structures depend on larger environmental-control systems; smallness converts weight into the density of order rather than eliminating it. Engineering history records a turning point along this path: under ideal proportional scaling, transistor dimensions and operating voltages decrease together, increasing switching speed while keeping power per unit area roughly constant; voltage scaling slowed around 2005, after which mainstream processor clocks remained in the few-gigahertz range, and performance gains increasingly depended on parallelism and specialized architectures rather than higher single-core frequencies (Esmaeilzadeh et al. 2011). Dark silicon followed: power-density and cooling constraints prevent most transistors on a chip from running at maximum frequency simultaneously, requiring part of its area to remain in a low-power state at any given moment (Esmaeilzadeh et al. 2011). This path remains open, but its workarounds relocate constraints. Three-dimensional stacking, specialized accelerators, and advanced packaging shift some pressure away from size, at the cost of longer thermal paths, more complex interconnects, and more sensitive yields, transferring limits once borne by transistors to heat flux density and packaging processes. Here is the microscopic paradox: lighter products, heavier manufacturing conditions; smaller components, larger supporting systems; cheaper information, more expensive information infrastructure. Advanced manufacturing drives material requirements deeper into processes, equipment, energy, and environmental control, rather than making materiality disappear.

Quantum Effects: Tunneling and Statistical Fluctuations

Quantum effects enter manufacturing first through tunneling, then through statistical fluctuations. A very thin insulating layer separates a transistor's gate, which controls current, from the channel; earlier generations of silicon-oxide scaling brought that layer down to 1.2 to 1.5 nanometers, on the order of five or six atomic layers. At that thickness, the barrier no longer blocks electrons completely: they have a definite probability of passing directly through the insulator, producing tunneling leakage. Tunneling leakage rises sharply as thickness decreases, and further thinning ceases to pay when leakage power and charge-retention problems offset switching gains; high-permittivity materials offer another route, increasing the dielectric's physical thickness while retaining a small equivalent oxide thickness and suppressing tunneling. Here quantum mechanics directly enters process decisions: making a layer thinner is

possible, but the thinner device may no longer work as circuit design requires.

Statistical fluctuations arise from the same direction. Shrinking channel volumes reduce the number of dopant atoms involved in conduction; once only a few dozen remain, random atom-count fluctuations translate directly into differences between devices in parameters such as threshold voltage, and adjacent transistors in the same chip batch may behave noticeably differently. Better fabrication precision cannot eliminate a constraint arising from dopant discreteness and Poisson statistics rather than machine-positioning errors. Similar fluctuations appear upstream in manufacturing: EUV lithography uses 13.5-nanometer light, with photon energy of about 92 electronvolts, more than ten times that of deep-ultraviolet photons, leaving only roughly one photon for every dozen or so at the same exposure dose. Photon arrivals follow a Poisson distribution, so the relative fluctuation in the count within a small region decreases inversely with the square root of that count; shrinking feature sizes and regions sharply reduces photon numbers and raises relative fluctuations, turning nominally straight edges jagged. Halving the fluctuations requires quadrupling the exposure dose. The statistical relationship can be written directly,

$$\sigma_N = \sqrt{N}, \qquad \frac{\sigma_N}{N} = \frac{1}{\sqrt{N}} \tag{2.9}$$

Here N is the dimensionless count of photons falling within a small region, or of other independent, randomly arriving events; σ_N is its standard deviation, with the same dimension as N; and σ_N/N is the dimensionless relative fluctuation. The second equation makes relative fluctuations inversely proportional to the square root of the count: halving them requires increasing N fourfold. At the same dose per unit area, shrinking the feature area reduces the photon count, so maintaining the same statistical certainty requires a higher dose or improvements elsewhere. The constraint applies within these limits: the square-root relationship holds only for independent Poisson events with a stable mean arrival rate; if source intensity drifts or exposure in neighboring regions is correlated, actual fluctuations depart from $1/\sqrt{N}$, and the required dose must be measured (De Bisschop and Hendrickx 2018). The source itself is also improving: in October 2026, ASML Japan described an increase in source output from 500 to 600 watts, while ASML's 2025 annual report recorded a 1,000-watt EUV-source demonstration in April 2025 (Kuramoto 2026; ASML 2026). These are source-output figures: the light must still pass through collection and exposure optics, so they cannot be treated directly as power delivered to the wafer. Producing a definite line on silicon requires sustained upstream investment in lasers, plasma, optical control, and cooling; the issue is the cost of certainty itself, beyond simple roughness in processing.

The Heat Flux Density Wall

Heat is another microscopic boundary: computation, communication, manufacturing, and control consume energy that must be dissipated, and microscopic structure can increase den-

sity without eliminating heat flow. Denser devices concentrate power and make cooling more restrictive; often the obstacle to further progress is how heat is removed, noise suppressed, and defects controlled, rather than logical design. Cooling capacity depends jointly on thermal conduction, interface resistance, and the cooling medium, with narrower paths making it harder to increase heat removal per unit area; 100 watts per square centimeter is already difficult for conventional air cooling, and further increases usually require liquid cooling, phase change, or more complex cooling and packaging structures. Write this constraint as an equation,

$$q = \frac{P}{A} = -k \frac{\Delta T}{L} \tag{2.10}$$

q is heat flux density, in watts per square meter, with the text's 100 watts per square centimeter corresponding to the order of 10^6 watts per square meter; P is the power to be removed, in watts; A is cooling area, in square meters; k is the material's thermal conductivity, in watts per meter kelvin; ΔT is the final-minus-initial temperature difference along the chosen direction, in kelvins; L is path length, in meters; and the minus sign indicates flow toward decreasing temperature. The quantities on the right have their own conditions: ΔT is limited by device operating temperature and the external environment, k depends on the material and its state, and L on structure and the cooling path. A longer path reduces heat flux density with other conditions unchanged, while a smaller cross-sectional area also limits total cooling power; these are distinct changes. The limits of applicability are these: this is a linear relation for steady, approximately one-dimensional conduction; phase change, convection, and radiation require other relations, and interface resistance in a real package may constrain q more than the bulk material does. If demanded power density exceeds cooling capacity, the cost falls on lower clock rates, a smaller proportion of simultaneously active devices, or redesigned cooling structures.

At facility scale, this constraint has the same shape. Data centers concentrate computation in a few locations, concentrating cooling, electricity supply, and land use with it; the scale of the cooling system becomes a prerequisite for siting and expansion, while energy-efficiency gains reduce energy per unit of computation without eliminating heat flow itself. The deeper information systems go into the microscopic world, the more they depend upstream on vast macroscopic engineering, whose capacity to remove heat has limits. More advanced chips tighten this ceiling. **Every information system ultimately rests on a thermodynamic system.**

2.2.6 Cosmic Elemental Abundances and the Thermodynamic Limits of Material Cycles

If macroscopic resistance comes from mass and microscopic resistance from noise, elemental resistance comes from abundance and distribution. Elemental quantities are determined by

nucleosynthesis and planetary differentiation rather than demand; technology can improve extraction efficiency, but cannot change the distribution itself. This layer constrains differently from the preceding two: mass can be moved and noise isolated, while elemental distribution must be accepted and accommodated through larger processing volumes, longer processes, and greater energy use. Elemental constraints are therefore particularly hard to bypass through engineering and readily appear as prices and lead times when supply tightens.

Abundance and the Entropy Tax

Elements are no evenly stocked materials warehouse on Earth's surface: they come from cosmic nucleosynthesis, planet formation, geological differentiation, mineralization, and long geological history, and carbon, oxygen, silicon, iron, aluminum, copper, lithium, nickel, cobalt, rare earths, and platinum-group elements are not distributed uniformly to suit human needs. An element's existence in the universe does not guarantee sufficient usable reserves on Earth; its presence in the crust does not guarantee mineable deposits; its presence in products does not guarantee cheap recovery. Crustal distribution is itself highly uneven: oxygen and silicon together account for about three-quarters of crustal mass, aluminum and iron a few percent each, copper an average on the order of 60 parts per million, lithium about 20 parts per million, rare earths together around 100 to 200 parts per million, and platinum-group elements as little as parts per billion (Rudnick and Gao 2003). Stellar nucleosynthesis and planetary differentiation determine these proportions independently of demand; technology can improve extraction and find substitutes, but cannot make an element more common in the world.

The resulting judgment agrees with the preceding two sections: the key resource question is whether an element exists in a usable state at sufficient concentration, scale, stability, and low cost, beyond whether it exists at all. Low-grade ore contains resources too dispersed to use without greater energy and processing volumes; rare elements are likewise present, but at low concentrations, unevenly distributed, difficult to extract, closely associated with other substances, and often constrained by geography and industrial supply chains. Material circulation therefore cannot be thermodynamically free: recycling restores dispersed, mixed, contaminated, degraded materials embedded in complex products to a usable state, requiring collection, disassembly, separation, purification, reprocessing, and quality control, beyond merely taking matter back. Circulation can relieve pressure on primary extraction, but itself consumes energy, equipment, and time.

Recycling requires collection, disassembly, separation, and reprocessing. These steps consume resources and can also save substantial energy compared with primary extraction and smelting. The comparison must cover the whole process for the same material, equivalent quality, and the same boundaries: concentration in the product, contamination, design for disassembly, and transport distances can all change the result. Circulation's value comes from actual reductions in new extraction, energy use, and waste burdens.

This can be called the elemental entropy tax. Once matter is dispersed, mixed, contam-

inated, and degraded, concentrating, separating, and purifying it again requires new energy, equipment, and time; technology can lower this tax without reducing it to zero. Dispersed matter does not regain high concentration for free, mixtures do not separate into pure components for free, and circulation pays again to restore order after entropy increase, rather than escaping it. The elemental scale thus reveals a material world constrained jointly by abundance, location, concentration, chemical state, and circulation losses, beyond an infinitely adjustable inventory. Technology can improve extraction, substitutes, and recycling processes, but cannot reshuffle cosmic elemental abundance, concentrate dispersed matter without cost, separate mixtures without energy, or make circulation exempt from the entropy tax.

Where the Elements Come From

Element supply carries cosmic and geological history. Big Bang nucleosynthesis mainly formed hydrogen and helium, while reactions inside stars and their subsequent evolution produced many heavier elements. Production beyond the iron peak includes slow neutron capture in stars and rapid neutron capture in environments such as neutron-star mergers (Steigman 2007; Kobayashi et al. 2020). Usable deposits on Earth also require planetary differentiation and magmatic, hydrothermal, and weathering enrichment. Engineering faces the concentrations, locations, and chemical states those processes leave behind.

Planet formation and geological differentiation redistributed this inventory again. During Earth's differentiation, siderophile elements entered the core while lithophile elements remained in the mantle and crust, whose usable mineral deposits formed through billions of years of magmatism, hydrothermal transport, and weathering enrichment. Two things follow: cosmic and planetary processes determine elemental quantities, while later geological history determines whether those elements become concentrated into near-surface deposits. The fraction humans can mine consists of the few anomalously enriched areas those processes jointly leave behind. Technology can change the requirements for enrichment, lowering grades from a few percent to a few parts per million, but each order of magnitude down pushes processing volumes, energy use, and waste up another step; the entropy tax retains its slope. The relationship between processing volume and grade can be written directly,

$$M = \frac{m}{\theta} \tag{2.11}$$

Here M is the mass of raw ore that must be processed to obtain the specified target element, in kilograms or metric tons; m is the target element's mass within it, in the same units as M; and θ is grade, the dimensionless mass fraction of that element in the ore, commonly expressed as a percentage or parts per million. The inverse relationship captures the judgment: each order-of-magnitude decline in grade requires an order-of-magnitude increase in material mined, crushed, leached, and processed for the same target-element quantity, with energy, reagents, and waste rising alongside it; the entropy tax's slope is this inverse relation. There are two limits: first, this accounts for material quantity alone, excluding extraction efficiency, and since

actual mining and mineral-processing recovery rates are below 1, real throughput exceeds M; second, complex mineral phases and associated elements often make energy use rise faster than $1/\theta$ as grade falls, because finer grinding and more separation stages are needed. At a real mine, the equation is therefore an optimistic lower bound rather than a prediction.

What the Three Scales Have in Common

The three scales together show why the atomic world cannot be rewritten at will like software. At the macroscopic scale, mass leaves inertia; at the microscopic scale, noise limits control; at the elemental scale, abundance and distribution constrain usability. Each layer has its own boundaries and demands its own forms of payment; material reconfiguration creates and maintains structures across scales, beyond optimization in an abstract space. Its difficulty therefore extends beyond design: design occurs in representational space while scale boundaries exist in real space, so a better urban plan must cross macroscopic mass inertia, a better chip design microscopic manufacturing limits, and a better energy proposal elemental abundance, mineral distribution, and the thermodynamics of circulation. A proposal can be light; scale cannot.

The three scales present different boundaries pointing to the same fact: material reconfiguration must happen within scale. It can optimize urban renewal and reduce demolition waste, improve chip design and manufacturing control to reduce defects, find substitutes and improve resource circulation, and cut waste in complex supply chains. All this is possible, yet reducing waste in crossing scale boundaries does not remove those boundaries. Scale boundaries show how progress must occur within scale, beyond saying that progress is impossible; mature material reconfiguration understands resistance at different scales, respecting inertia macroscopically, managing noise microscopically, and recognizing scarce usability at the elemental level, rather than imagining the world flattened into an editable plane. The atomic world is more than slow software: it has weight, heat, noise, abundance, and history, and changes slowly because matter retains its own resistance at different scales, alongside inefficiencies in human organization. **The world can be changed, but cannot be rewritten at will.**

The three boundaries above concern space and matter: how much mass is fixed, how small a structure is maintained, and how dilute an element is concentrated. They explain how matter resists reconfiguration across scales, leaving open what happens when time is compressed. Time compression leaves these boundaries in place while reorganizing them: the same mass, heat, and defects must pass through the system in a shorter window. Once throughput becomes the main constraint, the three scale boundaries reappear in another form.

2.3 Temporal Rigidity: Processes That Resist Compression

2.3.1 Physical Processes Take Nonzero Time

The resource ledger answers where matter comes from; it has yet to answer how long matter takes to arrive. Acceleration removes none of the load; it changes the rate at which that load passes through the system. The same materials, energy, heat, and error must pass through the system in a shorter window. Once time becomes the constraint, the three spatial boundaries converge on a single quantity: load per unit time.

Treating time as a resource that can be compressed is the most widespread assumption of the age of intelligence. On the information side, this assumption does hold: waits can be eliminated, retries run in parallel, and searches pruned; a delay in one stage need not propagate automatically to the next. On the material side, however, the materials consumed, energy injected, heat generated, damage accumulated, and maintenance required by the same state transition must all be dealt with in a shorter window.

One of the easiest mistakes to make in the AI era is to equate speed with efficiency. Models produce answers quickly, systems respond quickly to signals, platforms distribute content quickly, and organizations obtain metrics quickly; it is natural to conclude that faster is better, and that compressing processes, eliminating waits, and increasing feedback frequency will make real systems more efficient. But time is not empty space that can simply be deleted. In a real system, a state transition passes through materials, energy, equipment, bodies, organizations, and the environment; compressing that transition into less time does more than "save time"—it raises the throughput passing through the system per unit time. The shorter the time, the more energy, matter, heat, error, and risk the system must handle per unit time; the substance of acceleration is this rise in flux density.

Reaction, diffusion, curing, growth, and service each have their own timescales. Computation can shorten some of these processes by helping change formulations, equipment, temperatures, sequences, and parallel arrangements; some long-term outcomes still require sufficient exposure before they can be established. The speeds of design, manufacture, and evidence acquisition need to be recorded separately.

Flux Density

A simple quantity describes flux density:

$$J = \frac{\Delta X}{A\,\Delta t}, \tag{2.12}$$

Here J is flux density: the load passing through a unit cross-section per unit time; ΔX is the total load that must pass through during this state transition, which may be energy, matter, or heat, with dimensions appropriate to the quantity; A is the effective cross-sectional area through which the load passes; and Δt is the time window allowed for the transition, measured in seconds. Applying the same form to errors or consequences requires a definition of how the

load is counted and through which channels it is handled; organizational capacity cannot simply be treated as physical area. The relationship gives a proportional comparison: with ΔX and A fixed, reducing Δt to one-tenth raises J tenfold. Whether the system can bear this load depends on its materials, structure, operating conditions, and permissible flux density, as well as the size of the cross-section. Completing the same task in ten days and in one day confronts the system with different processes: energy must be injected more intensively, waste heat removed faster, equipment subjected to higher loads, and people given less time for corrections; errors spread faster, and there are fewer opportunities to stop a failure early. Compressed time does not disappear; it returns as power, heat flux, stress, and pressure on inspection and repair.

Whether acceleration is feasible therefore depends on the available cross-section for throughput, rather than on whether a target date can be written into a plan. Here "cross-section for throughput" is a general name for a system's capacity to handle load per unit time; extending it from physical area to organizational capacity is a functional analogy, encompassing energy supply, heat removal, load bearing, inspection, repair, and responsibility. Insufficient energy supply leaves the process unable to run; insufficient heat removal raises its temperature; insufficient load-bearing capacity fatigues the structure; insufficient inspection lets defects through; insufficient repair turns small errors into major failures; and insufficient responsibility leaves the consequences of rapid action without anyone to bear them. This goes deeper than saying that speed has a price: speed compresses loads previously spread across time into a narrower window. Time provides part of the buffer: time for heat to diffuse, materials to release stress, people to inspect, organizations to coordinate, and errors to become visible. Removing these buffers through acceleration requires stronger structures, greater redundancy, faster inspection, and more maintenance capacity in their place.

A common reply is that, if technology can replace buffers, the limit to acceleration is simply a matter of investment rather than principle. This is often true of an individual stage; at the system level, the coupling among the six kinds of cross-section must be examined. Thickening a structure increases load-bearing capacity while adding mass and heat capacity; raising the refresh rate expands inspection capacity while increasing energy use and false alarms; adding redundancy improves repair capacity while increasing coordination costs; automating decisions speeds execution, but leaves responsibility less clear if arrangements for it fail to keep pace. Widening one cross-section often narrows another, so acceleration carries a structural cost that must be sustained, rather than an investment that can be paid off once. This also explains its nonlinearity: cutting ten days to five may require only less waiting, better scheduling, and less rework; cutting five days to one may demand more equipment, greater parallelism, more inventory, and denser inspection; cutting one day to an hour may encounter limits of heat removal, reliability, safety, and responsibility. Near the system's limits, each further reduction in time can demand a disproportionate increase in supporting capacity; marginal resistance to acceleration rises rapidly.

Concrete gains strength through hydration; 28 days is a common testing age, while actual curing requirements depend on the materials and engineering conditions. Mass concrete pours also require control of hydration heat and temperature differences; placing concrete in lifts, cooling it, and insulating it all change the feasible pace. High-rate charging of lithium-ion batteries can be limited by transport, intercalation, and lithium plating. Accelerating these processes requires changes to materials, temperature, structure, or operating methods, followed by checks on the resulting costs and service life.

Communication and computation provide two further boundaries. Signals in ordinary silica fiber travel at roughly two-thirds the speed of light in vacuum, putting one-way propagation delay across an ocean on the order of tens of milliseconds; adding wavelength channels, improving modulation and coding, and installing parallel fibers can increase bandwidth but cannot eliminate propagation time along a given route in a given medium. Queueing, processing, and detours can be improved; changing the propagation floor requires a shorter route or a different medium (Corning Optical Communications, n.d.). Growth in single-core clock frequency slowed sharply around 2005; frequencies have continued to develop within the several-gigahertz range, while power density and cooling constraints have shifted more of the growth in computing capacity toward multicore parallelism. Parallelism spreads out independent steps; it does not directly reduce the number of steps along a given dependency chain, a boundary that will recur in the discussions ahead.

2.3.2 The Power–Time Trade-off

A rise in flux density is still only a description; it does not specify which quantities bear the load. Compressing the same task into a shorter window displaces load into several places that can be examined separately; their order matters, because the preceding quantity sets the scale of the next.

Four Layers of Resistance

The first layer of resistance is power. Changing reality requires work: moving objects, driving equipment, compressing processes, raising output, and speeding transport all require more state transitions per unit time. Energy divided by time is power:

$$P = \frac{\Delta W}{\Delta t}, \qquad \Delta W \text{ held fixed } P \propto \frac{1}{t}, \tag{2.13}$$

Here P is power, measured in watts, or joules per second; ΔW is the work required for this state transition, measured in joules; and Δt is the time taken, measured in seconds. The inverse relationship assumes that total work remains fixed: the route, process, and materials stay the same, and the task is simply performed faster; if acceleration changes the process or makes ΔW depend on speed, the relationship must be recalculated. The conclusion follows directly: halving the time doubles the power; reducing time by an order of magnitude raises power by an

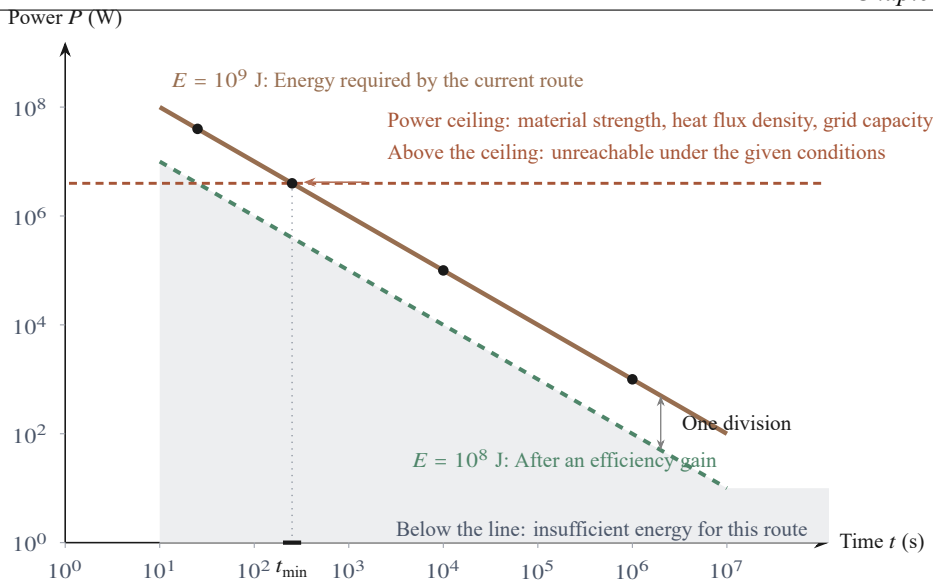


Figure 2.4 The power–time trade-off. On logarithmic axes, a fixed energy requirement gives a constant-energy line $P \cdot t = E$, along which power buys time; reducing the energy requirement shifts the whole line downward, with one grid interval here representing an order of magnitude and the slope unchanged. The horizontal dashed line illustrates a power ceiling for given material, heat-transfer, and supply conditions; its intersection with the energy line gives the shortest time t_{min} under those conditions. Changing the process or its boundary conditions can move either the energy line or the power ceiling.

order of magnitude. Compressing the same energy consumption into less time requires greater power input; without the corresponding supply, transmission, and control capacity, acceleration becomes overload. Industrial production makes this particularly clear: a faster line requires feeding, processing, inspection, packaging, logistics, and maintenance to sustain the faster pace together; a shortfall in power capacity at any stage becomes a bottleneck, and forcing one stage to accelerate creates a backlog at another. Speed is settled across the entire chain.

Power systems build different response speeds into their choice of equipment. Adjusting power faster requires more than pushing the same generating unit harder; it also requires equipment with different ramping capabilities: conventional coal units often adjust output by a few percent of rated capacity per minute, flexible gas units are generally faster, and battery storage can respond within milliseconds to seconds where its inverter and control system permit. Their costs must be compared separately in terms of power, storage duration, fuel, life, and operating conditions; a single "cost per unit of capacity" cannot place them in a fixed order. The wind and solar fluctuations a grid can accommodate depend on ramping capability, network capacity, reserves, storage, demand response, forecasting, and control (International Renewable Energy Agency 2017; Pacific Northwest National Laboratory 2019). The same principle

appears in manufacturing and transport: cutting delivery from six weeks to one usually means airfreighting some components, running some processes in parallel, and positioning some inventory in advance, as well as running equipment harder; the time saved is bought back at a higher unit cost. Time, power, and capacity can be exchanged in this way; near the limits of a given system, the margin required for further compression can become increasingly disproportionate.

Beyond power, the second layer of resistance is heat. Engineering must account for losses: computation generates heat, mechanical motion encounters friction, power transmission incurs losses, chemical reactions may produce by-products, and production may leave waste; if losses per operation stay fixed, higher speed means more waste heat to remove per unit time. Heat removal is a central boundary to acceleration: many systems could compute, transmit, or manufacture faster if they could remove heat, losses, and by-products at the same rate. Data centers, chips, engines, batteries, factories, and urban transport all face this constraint: energy can enter faster, but waste heat must leave faster too; if incoming flux rises beyond outgoing capacity, the system heats up, throttles, ages, or fails. Heat makes acceleration a question of whether unusable energy can leave as well as whether energy can enter.

Beyond power and heat, the third layer of resistance is stress. Physical objects are not ideal symbols: materials bear forces, structures bear vibration, equipment bears wear, connectors bear shocks, and supply chains bear fluctuations; driving a system faster exposes its weak points faster. Faster machinery increases friction and fatigue, faster construction puts more pressure on quality control, faster logistics reduces buffers, and faster organizational activity leaves less time for communication and repair. Stress is dangerous because it often accumulates as microscopic damage before causing failure; fatigue, wear, loosening, cracks, and quality fluctuations can result from prolonged high-frequency loading rather than a single large shock. Acceleration brings a system closer to its limits while accumulating damage faster.

The fourth layer of resistance, and the easiest to overlook, is inspection and repair. Faster systems need to find errors sooner: in a slow system, an error may have time to surface, be located, isolated, and repaired; in a fast one, a small error may be copied, spread, and amplified before it is found. A faster production line has a shorter inspection window; faster financial trading spreads erroneous orders faster; faster content generation concentrates review pressure; more scientific candidates deepen the verification bottleneck. Inspection and repair form a distinct cross-section because their costs do not fall linearly with speed: doubling inspection frequency often requires corresponding increases in equipment, staffing, standards, and organizational processes, while repair capacity remains constrained by the time needed to locate faults and complete rework.

Taken together, the four layers lead to a counterintuitive conclusion: speed need not mean efficiency. Efficiency means completing an effective state transition at lower total cost; speed measures how much transition is completed per unit time. A system can be fast yet consume

much energy, generate large heat loads and losses, spread errors quickly, and be difficult to maintain; it can also be slower, stable, economical, and repairable. The real question is whether going faster reduces the system's total bill.

Speed and Efficiency

Accelerated testing is engineering's formal method for trading away time; its boundaries show how far that trade can go. Materials and device life assessments commonly raise temperature to accelerate reactions, then use an Arrhenius relationship to extrapolate service life at ordinary temperatures; a test that would take years can thus finish in months. There is a strict condition: the extrapolation must stay within the same failure mechanism. At sufficiently high temperatures, new reaction pathways, stress states, or failure modes appear, and the life being extrapolated belongs to a different process. Testing time is compressed, while service still unfolds under its own conditions and over its own duration; conversion between them is valid only where the acceleration model applies and the failure mechanism remains the same (National Institute of Standards and Technology, n.d.). This condition recurs wherever acceleration is applied: directly reducing the time parameter first requires evidence that the shortened process still follows the same path.

Concrete curing illustrates a productive use of this exchange. Twenty-eight days is a common testing age, not the end of strength development; with a suitable mix and temperature–pressure schedule, high-pressure steam curing can produce a 24-hour strength close to the 28-day strength under ambient curing, with the autoclaving stage itself taking on the order of a dozen or more hours. Precast pipe piles and other high-strength components use processes of this kind (Pennsylvania State University, n.d.; Chen et al. 2017). The twenty-odd days saved become autoclaves, boilers, pressure vessels, and the energy needed for a continuous steam supply; component dimensions are also constrained by the sealed vessel. High temperature and pressure alter hydration products and long-term properties too, so the exchange involves mix design, microstructure, and performance targets as well as time and energy. The curing route chosen depends on which resources are scarcer on site and what properties the product requires.

AI is especially liable to create an illusion here. It greatly reduces the time spent on search, expression, generation, and preliminary screening; this speed is real and valuable, reducing pointless waits and blind trial and error, narrowing the candidate field, and improving early judgment. Once AI output enters reality, it encounters materials, equipment, bodies, processes, experiments, organizations, and responsibility, rather than an information space that can be compressed in the same way; the faster the generation stage, the more likely execution, verification, and maintenance are to become throughput bottlenecks. Reducing ten thousand candidates to one hundred can be quick; turning those hundred into reliable conclusions requires experiments, tests, replication, environmental exposure, and long-term feedback. Proposals can be generated at high frequency, but real verification cannot run at an unlimited frequency;

faster generation demands sufficient verification capacity, or speed merely transfers pressure from generation to verification.

Automation downstream can expand parallelism, reduce waits, and remove steps by improving the route. The remaining constraints must be specified: which steps genuinely have to occur in sequence, how long each requires under current conditions, and which consequences become observable only after exposure. Only with these conditions fixed does the chain have a corresponding lower bound on elapsed time.

Open the acceleration ledger, and benefits and costs appear in different accounts. Cutting a workshop's cycle time from 60 seconds to 45 raises output by one-third on paper, but feeding, changeovers, inspection, packaging, and logistics must be rescheduled to the new cycle; if one stage cannot keep up, extra work in progress accumulates between operations, tying up space, capital, and subsequent rework time. The equipment account is more direct: higher rotational speed means faster wear and shorter maintenance intervals, so some of the apparent increase in utilization must be offset against maintenance downtime. Speed gained at one stage is forfeited at another; acceleration moves costs from the time account to equipment and maintenance accounts.

Acceleration is therefore no one-way road to efficiency. Effective acceleration removes pointless waits, rework, and needless friction, completing the same state transition at lower total cost; ineffective acceleration merely raises flux density and pushes power, heat, stress, errors, and maintenance into the background. A mature system should seek a sustainable range of speeds rather than the shortest possible time; energy supply, heat removal, load bearing, inspection, repair, and responsibility jointly define that range: below it the system wastes time, above it the system consumes its own structure. Real efficiency means completing a state transition within the throughput the system can bear. Resistance to acceleration is the throughput ledger of compressed time. Time cannot simply be deleted: it is either wasted or used for buffering, cooling, inspection, repair, and revealing hidden effects. AI can help eliminate wasted time, but cannot remove for free the time needed to bear a process; information can arrive first, while reality must still pass through the available cross-section. The physical world accepts speed on the condition that the system can bear it.

The discussion so far concerns one kind of acceleration: compressing a process into a shorter window. Another kind increases how often updates are imposed on the same system, rather than shortening the duration of a particular process. The two face different constraints: the first depends on sufficient throughput capacity, the second on how long a system takes to complete an update. If inputs arrive more frequently than the system can absorb them, the system is being interrupted rather than accelerated.

2.3.3 The Bandwidth of Time: Update Frequency and Response Timescales

Temporal rigidity means more than an inability to compress a process at will. It has a subtler form: real systems cannot be updated at arbitrary frequencies. AI readily creates the illusion that faster generation, prediction, and scheduling should make real systems change faster too: models issue new judgments every second, platforms refresh data in real time, sensors collect signals continuously, organizations revise metrics whenever they like, and algorithms repeatedly recompute plans. Speed comes to be mistaken for intelligence itself, and real-time operation for efficiency itself. But a real system is not a screen that completes a state change whenever an input arrives; after accepting a change it still has to transmit, execute, obtain feedback, correct, and settle again. Only then has it completed an update. This is the bandwidth of time.

Bandwidth and Timescales

A system's response bandwidth describes how much of the variation in its input appears in its output. Materials, actuators, procurement, and organizations each have their own dynamics; high-frequency signals may be attenuated, or may elicit different responses through the control structure. A new input arriving before the previous round finishes does not automatically cause instability; the interactions among inputs, states, and the feedback loop must be examined.

$$\frac{\mathrm{d}x}{\mathrm{d}t} = \frac{u - x}{\tau}, \tag{2.14}$$

x and u have the same state units, and $\tau > 0$ is the time constant. With a fixed input, the error decays exponentially, leaving about 37% after one time constant. With a continuously varying input, the system acts as a stable first-order low-pass filter, attenuating amplitude and introducing phase lag. Closed-loop oscillation requires a separate analysis that includes a controller, delays, or other such structure.

$$f_c = \frac{1}{2\pi\tau}, \tag{2.15}$$

f_c is the cutoff frequency of a first-order system, measured in hertz. For a sinusoidal input at frequency f, the steady-state amplitude ratio is $[1 + (2\pi f\tau)^2]^{-1/2}$; at $f = f_c$ it falls to $1/\sqrt{2}$ of its low-frequency value. High-frequency response is attenuated, but the equation itself still describes a stable first-order tracking system. Greater tracking bandwidth requires changing the plant or the control structure.

Sampling frequency and a plant's ability to follow an input are distinct design questions. Faster sampling can provide more state information; the controller must also account for delay, noise, gain, and actuator limits in choosing how to update. A first-order low-pass system responds to high-frequency inputs with reduced amplitude; whether a closed loop oscillates depends on the loop formed jointly by controller and plant. Fast sampling alone neither guarantees stability nor necessarily causes instability.

Some processes can respond faster with faster actuators, shorter coordination chains, and greater sensor bandwidth; others require different materials or structures to change their timescales. Protective action also requires distinguishing measurement, judgment, and execution: fast measurement combined with filtering and thresholds can preserve both responsiveness and resistance to disturbances. Immune regulation, circuit protection, and quality inspection work through different mechanisms; their pace must be designed around the signals being identified and the consequences of errors.

Reality is therefore a system of many timescales, rather than a slow system. Near contact, protection, balance, fault isolation, and immediate safety, fast response is essential; near goals, meaning, institutions, trust, and long-term structure, stability is essential. Planning, scheduling, inventory, maintenance, coordination, and resource allocation lie between: they cannot be too slow, but cannot be driven directly by high-frequency signals either. Mature systems assign update times across levels, rather than letting a central intelligence control everything at one frequency.

The layers of a power system are a mature expression of this principle. In the first tens of milliseconds after a fault, protective relays begin identifying it and issuing trip commands, while actual clearance time also depends on circuit breakers and protection coordination; over the following seconds to minutes, frequency control, stability controls, dispatch, and related functions adjust output, shed load, or isolate areas; above these sit day-ahead and longer-term markets, maintenance, and planning, operating over days, months, and years. Each layer has different authority, action speeds, and grounds for judgment: protection and local control that cannot wait for human reaction rely on automatic action, while goals, rules, and long-term commitments require identifiable people and institutions to take responsibility; dispatch and planning can also use automated optimization (North American Electric Reliability Corporation 2024). Repeated accidents and engineering practice have shaped this layering, because no single timescale can handle both faults and planning.

Embodied intelligence makes the point especially clear. A robot cannot simply connect a large model directly to every motor: a body is not a screen, and movement is not text. Language understanding can be slower, vision must update faster, motion planning must turn goals into trajectories, and low-level control must handle balance, force, friction, slipping, and error at high speed. A system can take its time deciding "this is a cup," but when its fingers touch the cup it must correct pressure and posture quickly; semantic correctness cannot replace force feedback. High-level intelligence handles meaning and goals, low-level loops handle stability and safety, and the levels between must translate across timescales; otherwise greater intelligence can leave the body less controlled. Nor can a power grid be rewritten in real time at any frequency one chooses. Fault protection must be extremely fast, frequency regulation fast, load forecasting somewhat slower, market dispatch slower still, and the generation mix and grid plan slower again. AI can forecast load at high frequency, but cannot rewrite the

whole grid at that frequency; protection that is too slow causes accidents, while planning that changes too fast creates disorder. Grid stability depends on each level having an appropriate update time.

Forms of Frequency Mismatch

The same holds for supply chains. Demand signals can fluctuate in real time, while capacity, procurement, inventory, transport, and quality checks have their own cycles; passing high-frequency demand straight into production without filtering can turn short-term fluctuations into excess inventory, shortages, rework, and scheduling friction. A small fluctuation at the point of sale can be amplified through procurement, production, and logistics into oscillation across the whole chain; the problem is a slow system forced to respond to fast fluctuations it should not be following, rather than an excess of demand signals. Organizations provide an even clearer case: data can refresh in real time, metrics can change daily, and AI can offer continuous advice, but trust, capability, habits of cooperation, and relationships of responsibility cannot be rewritten every day. A team needs time to understand its goals, establish a rhythm, and stabilize who is responsible for what; if every day brings a new priority, every week a new objective, and every model update new advice, an apparently more agile organization may lose continuity. Attention fragments, boundaries of responsibility blur, executors keep changing direction, and the system spends its energy responding to change rather than completing work.

The real danger of high-frequency intelligence is updating before completion. The next change arrives before the system has absorbed the last; the next metric replaces the previous one before behavior has fed back; the next plan is issued before the consequences of the previous plan have appeared. The system then switches among unsettled changes instead of accumulating capability; it keeps updating without completing an update, and keeps responding without establishing stable control. Frequency mismatch usually takes several forms: first, noise, as inputs arrive too densely for the system to distinguish real change from short-term disturbance, producing more filtering, explanation, and coordination rather than more action; second, interruption, as high-frequency advice repeatedly cuts into slower processes, forcing people to reinterpret, realign, and reorder their work until time is fragmented into pieces too small to finish anything; third, oscillation, as delayed feedback meets overly frequent adjustment and the system keeps chasing past errors; fourth, drifting responsibility, as automatic advice outpaces confirmation, tracing, and accountability, leaving it hard after a failure to identify which output, interface, person, or organization actually changed the system's state.

Putting these four forms together can suggest that frequency mismatch only runs one way: fast disturbances striking a slow system. The reverse mismatch exists too, with more immediate consequences: slowing a layer that requires fast response produces accidents rather than mere inefficiency. Late protection burns out equipment, late obstacle avoidance strikes pedestrians, and late emergency treatment misses its window; response time here is part of the function

itself, rather than an exchangeable cost. The question is whether each layer's time constant matches what it must handle; mismatch causes losses in both directions, though the forms of loss differ.

Interfaces for Update Times

High-frequency intelligence therefore needs interfaces governing update times as well as faster output to enter reality. Such an interface requires at least four structures: rate limiting, buffering, holding, and layering. Rate limiting prevents high-frequency inputs from repeatedly rewriting slower structures; buffering keeps short-term fluctuations from striking slow systems directly; holding gives a decision enough time for execution, feedback, and correction; layering keeps fast feedback local and slow judgment at the structural level, so that different timescales do not damage one another. Responsibility also belongs in the time interface: a real system is more than a signal-processing system, and whether a high-frequency recommendation can execute automatically depends on who will bear a failure as well as whether the system can respond. Without an interface for responsibility, stronger high-frequency intelligence can push decisions into automatic processes whose consequences no one bears in full; the more a system operates in real time, the clearer it must be which changes can execute automatically, which need confirmation, which may only advise, and which must not become actions. Likewise, "end to end" cannot simply mean one model controlling everything: it can reduce hand-written rules and learn complex mappings, while real execution still requires layers, rate limits, and boundaries of responsibility. The danger extends beyond the black box to making the time interface opaque: if we cannot see which layers are fast or slow, which feedback is filtered, which commands are held, and who bears which failures, a system may lose its stabilizing structure beneath an appearance of integration.

One further point about interfaces is easily reversed: they are often imagined as brakes, as though lowering all update frequencies would make the system safe. What is required is matching across layers: fast layers must be fast enough, and slow layers must be allowed their time. AI's value does not depend on making reality match the model's speed. It can observe frequently without intervening frequently, predict continuously without changing things continuously, generate plans quickly while issuing them on a stable basis, and strengthen decisions while respecting the real system's own update times. Mature intelligence embeds model capability in the world's temporal structure. Real-time operation means being timely at the right level: fault protection needs it, strategic planning does not; contact feedback needs it, organizational culture does not; obstacle avoidance needs it, urban renewal does not. Making processes real time where they should not be exposes slow systems to high-frequency disturbances.

The bandwidth of time reminds us that reality's pace reflects a need shared by every enduring system: allocating time between change and stability. A change is complete only after transmission, absorption, correction, and stabilization; faster intelligence makes it still more essential to distinguish output frequency from the frequency of real updates. High-frequency

intelligence must know which layers can be fast, which must be slow, and which changes require rate limiting, buffering, holding, and layering. Intelligence can drive the world, but cannot drive all of it on the same timescale.

An interface can assign update frequencies to ranges each layer can sustain; it cannot change the fact that some results exist only after a process has run its course. A higher refresh rate does nothing for such problems, because what is awaited is the formation of an answer rather than the arrival of one already written. Beyond the boundary of frequency lies another boundary belonging to the process itself.

2.3.4 Sequence Dependence in Living Systems

Rate limiting and buffering address update frequency, preventing overly rapid change from interrupting a system; they do not explain why another kind of time can resist compression. Some changes cannot be skipped because their results do not exist before the process has run, rather than because the system cannot absorb them fast enough. Fatigue accumulates through cycles, side effects appear through metabolism, institutional effects emerge through implementation, and ecological consequences propagate through networks; these facts are products of the process, rather than answers stored at its starting point waiting to be read faster.

Materials, Engineering, and Life

Materials make this particularly clear. Good initial laboratory performance does not establish stability in long-term use: fatigue requires cycles, stress accumulation, corrosion exposure, and aging time. Many failures develop gradually through microscopic damage rather than appearing on day one; short tests screen out obvious errors but cannot fully replace service over time. That time is the process through which loads, temperature, moisture, oxidation, vibration, and maintenance conditions reveal hidden defects. Asbestos gives an extreme timescale: symptoms of mesothelioma may appear twenty to forty years or more after fibers are inhaled; an assessment comparing exposure and outcomes only within the same year will miss such long-latency consequences (MedlinePlus, n.d.). Engineering systems work the same way. A design may pass simulation, a device pass initial tests, and a system operate in ideal conditions, while reliability has to develop amid noise, faults, maintenance, scaling, edge cases, and user behavior. Many problems arise only through prolonged operation, use by different people, and different loads, even when designers have anticipated them; reliability is a property of the process as well as the drawing. A system's reliability depends on how it ages, is maintained, is misused, and degrades under faults, as well as how it was designed. The duration allocated to testing itself makes the point: some semiconductor reliability tests run continuously under high-temperature bias for over a thousand hours, full-scale structures are assessed through load spectra and cycle counts, and cumulative vehicle durability test distances can exceed a hundred thousand kilometers. The object and the standard determine the duration, load spectrum, and

equivalence calculation; these numbers matter because they provide enough of a process for hidden defects to emerge.

Life unfolds as material processes in a sequence: gene expression has an order; protein synthesis, folding, modification, transport, and degradation have rates; cell division has cycles; immune responses have initiation, amplification, regulation, and resolution; tissue repair has stages; development and aging are themselves long processes. Biological reality lies in the progressive interpretation of sequences by cells, tissues, environments, and time, as well as in the sequences themselves. That is why AI-accelerated molecule discovery cannot eliminate clinical time from drug development. AI can find candidate molecules, predict structures, and screen targets faster, but a candidate entering a body still faces absorption, distribution, metabolism, excretion, toxicity, immune responses, dose, individual differences, and long-term feedback. Short-term efficacy does not establish long-term safety, a correct local target does not establish that the whole-body network can bear the intervention, and efficacy in an animal model does not establish lasting efficacy in human populations; living systems respond, compensate, rebound, and evolve. Time here is part of biological reality, rather than merely regulatory delay: the body needs time to reveal side effects, disease to reveal its course, the immune system to reveal its response, and population differences to reveal their distribution. A drug truly enters reality by surviving the tests of living processes, rather than by looking plausible in a model.

Drug safety requires effects to become visible in appropriate populations, observation periods, and measurements. Rentosertib reached first-patient dosing in Phase III in September 2026, a new milestone in its discovery and development (Insilico Medicine 2026). It also shows that generating a candidate, entering clinical trials, and establishing evidence of long-term benefit are distinct milestones. Trial design and regulatory capacity can improve; follow-up results that have yet to occur still cannot be filled in ahead of time.

Not all biological time resists compression. Screening can occur earlier, biomarkers can reveal damage sooner, adaptive trial designs can reduce uninformative enrollment, and well-designed dose escalation can increase the information provided by early participants; these improvements do shorten the time needed to identify individual differences. They compress observation and selection, however, rather than exposure itself: metabolism in the body, accumulation in tissue under long-term load, and the distribution of effects across populations in real use must still unfold through their own steps. Distinguishing compressible observation from exposure that must be preserved lets us judge whether an acceleration shortens the process or moves observation ahead of its endpoint. Verification is how living systems turn candidates into facts.

Organizations and Society

Organizations and societies also require evidence through time. An institution may look reasonable on paper, a policy model effective in simulation, and a management plan logically

clear, but implementation encounters incentives, trust, habits, interpretations, resistance, and arbitrage; people cannot simply be assigned values like variables. Institutional effects emerge through implementation, trust through repeated interaction, responsibility through events, and organizational capability through sustained cooperation; social facts form through people's responses over time, rather than following statically from a proposal. Many reforms look successful at first and reveal costs later; platform mechanisms initially improve efficiency and later generate abuse, competitive overinvestment, and systemic fragility; educational, medical, and urban governance plans improve short-term metrics while producing more complex long-term consequences. Logical completeness does not establish an institution's success: what matters is how people understand, exploit, bypass, and counter it. Social systems learn and change shape, adapt and push back; time is the space in which social consequences form.

Ecosystems extend this structure across longer timescales. A chemical, an introduced species, a form of land use, or a water project can deliver clear short-term benefits while producing different long-term consequences through food chains, water cycles, soils, species relationships, and climate feedback. Ecology is a network of multiple feedbacks: local changes travel along relationships, short-term gains can become long-term system costs, and consequences are difficult to establish fully in advance because they form as interventions run through an open network. Such processes often span decades: a centimeter of topsoil can take hundreds of years to form, a logged forest commonly needs decades or more to recover its former structure and species composition, and lake sediments can continue releasing phosphorus after nutrient inputs stop, delaying water-quality recovery for years after emissions decline. Earlier conclusions in these systems require observation series, rather than faster reasoning alone.

Several more basic processes lie behind these examples. First, accumulation: microscopic damage, toxicity, erosion of trust, and ecological pressure can build through repeated cycles rather than appearing all at once. Second, thresholds: a system can look stable for a long time, then fail suddenly when a boundary is crossed. Third, adaptation: organisms, organizations, markets, and institutions compensate for, bypass, and counter interventions. Fourth, diffusion: local consequences spread through material cracks, metabolic networks, social relationships, and ecological chains. Evidence through time consists of these processes turning invisible states into observable facts, rather than waiting alone.

A crucial difference between social and material processes makes the former harder to converge. When observation does not significantly disturb a material, it does not adjust its strategy because it knows it is being observed; people and organizations do. Announcing a policy reshapes expectations; putting a metric into an evaluation reorganizes behavior around it; stronger supervision shifts the supervised activity toward areas left unsupervised. The effect of the same intervention can therefore vary across measurement dates, following a trajectory that changes with observation and rules. This is why "run it again and we will know" often

fails in social settings: the first run has already changed the initial conditions of the second. Social facts take time to form and, once formed, still cannot be treated as final conclusions established once for all.

2.3.5 Evidence Through Time: The Process as a Source of Information

In the age of intelligence, the time needed for some facts to form is easier to underestimate than the power to generate. A model can quickly generate answers, designs, molecules, strategies, and predictions, compressing search in a space of representations and presenting possibilities that once required prolonged trial. Yet real problems involve more than finding a plausible solution among possibilities: many outcomes gradually form, emerge, and become established as materials, organisms, organizations, and ecosystems operate. A single inference cannot fully replace them, and a static metric cannot directly establish them. This is evidence through time.

The Process as a Source of Information

Evidence through time is neither waiting alone nor a compensation for insufficient intelligence. It means putting candidates into processes, so that cycles, exposure, metabolism, use, diffusion, and responses turn hidden consequences into facts. A candidate may already be complete in representational space, but its validity in reality depends on how it bears cycles in a material, is metabolized in a body, is understood and countered in an organization, and spreads through an ecological network; time is a condition for some facts to form and appear. Computational irreducibility can offer an analogy, but not a blanket claim that complex systems are unpredictable. Many systems can be simulated, predicted at coarse resolution, or assessed early through proxy indicators; the crucial distinction is between what computation, inference, and simulation can compress in advance and what the running process must supply.

The latter are outcomes carried by the process itself: a material's fatigue history, a body's metabolic response, an organization's path of adaptation, and an ecosystem's long-term feedback are facts formed through interaction with the environment over time, rather than simple labels attached to a static candidate. Processes of this kind have a lower bound on duration:

$$T \geq N\,\delta t, \tag{2.16}$$

Here T is the total time required to complete the process along a given path, measured in seconds; N is the dimensionless number of evolutionary steps that must be completed in sequence along that path; and δt is a common lower bound on step duration, measured in seconds, such as the minimum duration specified for a fatigue cycle, a metabolic round, a reproductive generation, or an exposure period. Both sides have the dimensions of time; the bound requires dependent steps, each beginning from the result of its predecessor and each taking at least δt. Where a process contains independent branches, its critical dependency path constrains total duration; the steps of all branches cannot simply be added. This relationship describes the

execution time of a specified physical path; computational irreducibility additionally requires showing that a shorter computation cannot produce the output sought. Better algorithms, different paths, and faster individual physical steps are three distinct kinds of progress, whose boundaries cannot simply be transferred from one to another.

The bound is simple in form, but both quantities are difficult to establish; that is what makes it hard to judge whether an acceleration actually shortens a process. N is not an intrinsic property of the process: it depends on what counts as a step. Material fatigue can be counted by stress cycles or by stages of crack growth; different resolutions can change N by orders of magnitude, while coarse-graining cannot change T itself, only the steps we count. Nor can δt always be read directly from a parameter: the same metabolic round differs in speed across individuals, and the same exposure period differs in duration across climates and loads. Looking at the two quantities separately gives a practical criterion: technical progress that reduces δt changes the process rate; a claim to reduce N requires an account of what changed in the dependency structure, implementation path, or target. A different counting scale alone does not shorten the process.

This distinction also has a symmetrical misuse. The claim that some facts must come from processes can expand into a dismissal of all advance judgments and predictions, as though outcomes could only be awaited. That does not follow: a process that cannot be skipped can still be interrupted, recognized early, or redirected midway. Early signs of failure, intermediate metabolites, changing organizational attitudes, and ecological indicator species all provide information before completion; they do not establish the final outcome, but can turn blind waiting into directed observation. The distinction is between final facts and process signals: the former require the full course, while the latter can be collected along the way; confusing them leads either to premature conclusions when observation should continue or repeated demands for conclusions while the process still needs time.

Specific indicators show the value of process signals. Glycated hemoglobin reflects average blood glucose over roughly the previous two to three months, with greater weight on recent levels; a single glucose measurement captures one moment, while this measure captures accumulated exposure across a period, making it more useful for assessing long-term control but unable on its own to say what happened this morning (National Institute of Diabetes and Digestive and Kidney Diseases, n.d.). Cracks inside metal structures emit acoustic signals before reaching a critical size; monitoring those signals can indicate the direction of damage before failure, but cannot establish service life in the first loading cycle. Such signals have a common role: revealing part of an invisible process early and directing observation, while leaving the process itself to run.

What Can Be Compressed, and What Cannot

AI's role remains substantial. It can narrow the candidate space, find more promising paths, identify risks early, design better experiments, simulate more scenarios, detect early signals,

and reduce unproductive trial and error, making verification more intelligent, directed, and economical. AI still cannot eliminate real verification: it can reduce what we must wait for without abolishing the need to wait, improve prediction without allowing every step to be skipped, and find many errors early without relieving the world of having to run. Mature judgment distinguishes two kinds of time while recognizing AI's speed: search, candidate generation, and formal deduction can be compressed, while the time in which facts form cannot be canceled at will. Slowing what can be computed is inefficient; skipping what only the process can supply is dangerous. Real technical progress identifies which time is wasted waiting and which is necessary for reality to provide facts.

This is also why cheaper generation makes evidence through time more important. A model can generate a thousand drug candidates, engineering proposals, business strategies, or policy documents, while reality cannot expand clinical trials, long-term tests, service experience, organizational feedback, and ecological observation at the same rate; as information space expands, scarcity shifts toward the capacity to verify a few crucial candidates through real processes. AI reduces the cost of proposing possibilities; evidence through time bears the cost of turning possibilities into credible facts. Some processes resist complete compression into a simple shortcut because they generate new constraints, states, and feedback: a trajectory can be predicted but still has to be traveled, a risk simulated but still tested against reality, and search compressed without eliminating the information supplied by the process. Reality is a process that must unfold, rather than a table that can be filled out in advance.

Whether a result can be obtained quickly depends on problem structure, algorithms, required precision, and the sources of information. Complete initial conditions do not guarantee a shortcut; gradual unfolding does not establish that every approximation or redesign must fail. Only after the necessary sequential steps, conditions per step, and observation requirements are specified can $T \geq N\delta t$ be applied to a given process. Computational irreducibility does not supply a proof for every engineering wait.

This is not technological pessimism. Mature technology makes evidence through time more efficient: better sensors detect fatigue earlier, better biomarkers shorten clinical judgments, better digital twins reduce physical trial and error, better statistical design improves experimental efficiency, and better AI helps select the paths most worth verifying. Technology's task is to reduce wasted waiting and amplify useful feedback. The key lies in boundaries: which results computation can provide in advance and which only a process can provide; which risks a model can rule out and which must appear through exposure, use, and long-term feedback; which candidates are ready for the next step and which must remain in representational space. Distinguishing time that can be compressed in advance from time that must be preserved is a central form of judgment in the age of intelligence. Evidence through time reminds us that reality does not always preexist in the form of an answer: some reliable facts form through fatigue, metabolism, immune responses, organizational adaptation, ecological feedback, and

prolonged use. They have to be produced by the world running, rather than merely thought of by someone clever. AI can compress the search for an answer, but cannot eliminate the time in which the answer becomes fact; some facts can only come from the process.

Once a process has supplied facts, a further gate remains. Verification establishes a candidate at its own scale, without establishing that it can be integrated into an existing system. Real systems consist of interdependent equipment, people, institutions, habits, and interests; the actual cost of changing a node depends on its structural position and how many other parts connect to it. The resistance met by the same verified technology can differ by an order of magnitude depending on where it is installed.

2.4 Topological Rigidity: Coupling Density, Phase Transitions, and Structural Memory

2.4.1 Network Coupling Density and Virtual Mass

The time dimension has made one point clear: some facts can only come from processes; a candidate must enter materials, bodies, and organizations before hidden consequences gradually emerge. But a process has a prerequisite: the candidate must be able to enter it, and the system must be able to accommodate the target. Resource barriers concern the price of changing reality, and temporal rigidity concerns the limits to compressing real processes; topological coupling raises a deeper question: even with a target state identified and resources and time available, does the system have a path to that state? That is the question topological coupling addresses.

From Targets to Paths

Seeing the problem clearly requires a different description of the system. A real system is a network of relationships among variables, rather than a collection of isolated variables. Roads connect to roads, equipment to equipment, departments to departments, rules to rules, people's behavior to institutional expectations, and technical interfaces to legacy systems. An element derives its significance from its position as well as its own properties; changing it also changes its relationships with the surrounding structure. In a complex system, local improvement therefore has no guaranteed passage to system improvement. A node can be optimized, a process accelerated, and a model made more accurate; whether those improvements change the system's state depends on their position in the network, the connections, and the boundary conditions. Local optimization improves a point; system change moves along a path. Topology lies between the two.

The effect of connection density can first be seen through the accounts. A node's own size determines the work required to change the node itself; the relationships that must change with it add another component. As coupling density rises, this additional component grows; the

node behaves less like an object that can be moved alone and more like one carrying an extra load: the same push produces less displacement. Accounting for that additional cost separately gives what we can call the node's virtual mass in the network. This is not physical mass and has no fixed unit; it serves comparisons among positions within the same system, answering a concrete question: why does the same change take a day here and become a six-month project there?

Network science offers corresponding structures. Erdős–Rényi random graphs and regular lattices generally have relatively narrow degree distributions; many real technical and organizational networks have broader distributions, with a few nodes carrying many connections, while transport requires distinguishing structures such as road and airline networks. Stanley Milgram introduced the small-world letter-passing experiment in 1967; in the study published by Travers and Milgram in 1969, 64 of 296 initiated chains reached the target, and successful chains passed through an average of 5.2 intermediaries, near the familiar six-step scale. This does not prove that any two people are connected within six steps (Milgram 1967; Travers and Milgram 1969). In 1998, Duncan Watts and Steven Strogatz supplied a network model in which short paths coexist with high clustering; short paths alone do not require extremely high-degree hubs (Watts and Strogatz 1998). These studies suggest examining path length, clustering, and degree distribution separately: short paths make it possible for disturbances to reach distant locations quickly, while hubs can concentrate the cost of change. A position with large virtual mass may be only a field, an interface, or a table, yet connect an entire system; the number and type of connections, and the commitments crossing boundaries, all affect its weight.

Software engineering has repeatedly observed a correspondence between technical and organizational interfaces: the structure of the organization designing a system tends to be reproduced in the system's interfaces, as communication boundaries between teams become module boundaries (Conway 1968). In real systems this creates a double coupling, with technical dependencies and organizational coordination reinforcing each other: changing an interface is difficult both because of its downstream systems and because the change crosses departmental responsibilities. Assessing a node's virtual mass therefore requires asking whether its connections cross organizational boundaries, involve external commitments, or are embedded in contracts and performance evaluations, as well as counting them.

Target States and State Transitions

Whether a path exists first depends on the space in which the proposed state is valid. AI is good at generating target states: a more efficient urban transport plan, a better hospital triage process, a supply chain with less inventory, a more precise teaching evaluation model, or more automated factory scheduling. A target generated in a model establishes only that the state can be described in representational space; it does not establish that a real system can reach it cheaply, safely, and stably. A describable target does not guarantee an executable transition.

Many errors of technological optimism begin here. Knowing a better state is mistaken for being able to enter it. Real systems cannot be changed arbitrarily. Cities already have roads, subways, buildings, utility networks, property rights, residents' habits, and fiscal constraints; hospitals have departments, medical records, insurance payments, physicians' responsibilities, and patient trust; companies have legacy software and processes, departmental interests, performance systems, and data formats; education has examinations, admissions, teacher evaluation, family expectations, and social selection. A proposal entering these systems must find a channel through an existing topology, rather than draw lines on an empty plane. The first implication of topological coupling is therefore that a real system's state space is not uniformly open. Some imaginable states have no affordable path; some locally effective proposals cannot pass through system interfaces; some improvements valid in a model are absorbed, delayed, distorted, or amplified in the opposite direction by the real structure. Deployment requires both a good solution and a system that allows that solution to become reachable.

This explains many familiar outcomes. An AI diagnostic model can be highly accurate on a test set, yet in a hospital it encounters physicians' workflows, patient histories, equipment differences, insurance rules, liability for misdiagnosis, and clinical trust; its local performance cannot directly rewrite the healthcare system. A traffic algorithm can reduce congestion in simulation, yet in a real city drivers change routes, commercial districts change flows, construction interrupts plans, enforcement changes behavior, and public and private transport feed back on one another; the seemingly better traffic state may lack a stable path through the system.

Improvements Absorbed by the System

One common sign of an unstable path is that the system absorbs the improvement. An AI assistant may help every employee produce documents faster without making the company more efficient overall. Organizational efficiency also depends on judgment, communication chains, approval structures, the allocation of responsibility, and alignment of goals. If everyone produces material faster while review, coordination, and decision-making remain unchanged, the system may simply move from a generation bottleneck to a screening bottleneck. Greater local speed need not improve the overall state.

System change is not the simple sum of local capabilities. A complex system's overall state often depends on the structure of its relationships more than on the performance of an individual component (Boccaletti et al. 2006). In a net, strengthening one node does not mean the whole net can move into a new shape; force must pass along its strands to other nodes, and the connections may disperse, cancel, or lock that local force. Such resistance is often misread as conservatism or opposition to change. Topological coupling offers another explanation: many forms of resistance arise because relationships cannot be rearranged immediately. A technology entering a system often changes who is responsible, who benefits, who bears risk, who must learn, whose workflow is interrupted, and whose power is redistributed. What looks like a tool touches structure; the more complex the structure, the less likely the technology is

to be a neutral plug-in.

This absorption has an observable form: improvements farther downstream are more easily flattened by a fixed upstream rhythm. A workshop may cut processing time per piece by one-third, yet if assembly takt, material delivery cycles, and inspection batch rules stay fixed, the time saved becomes waiting between operations rather than a proportional rise in output. Conversely, even a small improvement at the stage setting the pace can propagate along the whole chain. To judge whether an improvement can change the system, first locate the source of its rhythm, then ask whether the improvement touches it.

Structural Position and the Distribution of Viscosity

The same technology can have entirely different fates in different positions. Many AI applications follow a familiar trajectory: easy demonstrations, difficult deployment; successful pilots, slow scaling. Demonstrations usually occur in low-coupling environments, with clear tasks, clean inputs, preset interfaces, responsibility suspended, and failures that can be disregarded; pilots also tend to select cooperative, standardized settings with the best prospects of success. Deployment at scale faces the heterogeneity of real systems: institutions differ in data structures, staff capabilities, processes, interests, compliance requirements, and boundaries of responsibility. Topological costs emerge when a showcase enters the system.

Assessing an AI technology's real potential therefore requires asking where it will be embedded, as well as whether it can do the task. Low-coupling positions are easier to automate; high-coupling positions face more constraints. Advertising copy generators, meeting-note tools, and first-pass customer service systems are generally easier to deploy than medical diagnosis, autonomous driving, educational assessment, judicial decisions, and energy dispatch, because their structural positions are lighter, their consequences more limited, and their failures easier to roll back—not necessarily because their models are stronger. The same capability encounters different viscosity in different structural positions. Text generation for a marketing draft connects primarily to communication effectiveness; for a medical report, to clinical responsibility; for a judicial decision, to rights, obligations, and coercive power. Similar task forms occupy different topological positions and face entirely different resistance.

Interface counts show the scale of this positional difference. An author bears the risk of changing a function called by a few internal scripts; changing an interface used in fixed formats by more than a hundred downstream systems distributes the risk among all callers, and any change to a field's meaning must pass through notification, compatibility work, parallel operation, and retirement of the old version. The first change takes hours, the second quarters. A few lines of code can differ by orders of magnitude in virtual mass because of how much they connect to.

Technologies That Change Connections

If resistance comes from position, the entry point for change lies beyond capability alone.

Technologies that change systems often do so by changing connections, rather than by offering the strongest capability at a single point. Smartphones changed the world by reorganizing mobile networks, app stores, payments, developer ecosystems, and user behavior as well as improving hardware; electric vehicles spread as battery supply chains, charging networks, policies, maintenance systems, and public understanding developed, as well as through efficient motors. A technology becomes a force within a system by finding or creating its niche. Deployment means giving a tool new interfaces within reality. Clear interfaces, explicit responsibility, closed feedback, and controllable failure ease entry; confused interfaces, unclear responsibility, delayed feedback, and irreversible failures leave technology in demonstrations and pilots. Topological coupling says that system change needs structural channels: the target state does not automatically supply a path; connections, history, institutional interfaces, and patterns of behavior jointly open or close it. A technology needs both a clear answer and a position in which the system can accommodate that answer.

We can now answer the opening question: a path exists because structural channels exist, rather than because the target is clear. Resource barriers say that changing reality requires paying a physical bill; temporal rigidity says it requires going through a process; topological coupling says it requires passing through structure. Without resources, the system cannot be moved; without time, the process cannot unfold; without a path, the target cannot be reached. Deployment gives an answer a structural channel into reality.

Opening a channel is only the first step. Changing one node along it often moves interfaces, data, permissions, training, responsibility, and expectations too; some must move together, and failure may leave no way back. These costs explain both the different speeds of the same improvement in different systems and whether a technology is absorbed as local efficiency or pulls the whole dependency network into adjustment.

2.4.2 Topological Phase Transition: Percolation Thresholds and Gelation

Beyond a certain range of coordinated movement costs, a system changes how it responds. The difficulty of complex systems often changes character abruptly rather than increasing gradually. A few added connections may improve coordination; a few more may merely increase coordination costs. But when connection density, feedback loops, and dependencies reach a critical range, the system takes on a new state: a network spanning the whole, rather than loose parts that can be adjusted locally. Local changes travel along connections to distant places; local errors become harder to isolate and may enter feedback loops across the system. This is the basic meaning of a "gelled system."

Gelation here is a structural analogy rather than a strict physical equivalence. In polymers, colloids, and material networks, gelation occurs when dispersed units form a spanning structure under particular conditions, giving a flowing system a supporting network. The image points to a shared mechanism across social, organizational, transport, and intelligent systems: **Beyond**

a range of connectivity, the system changes the phase of disturbance propagation rather than merely becoming "a little more complex." The system moves from locally absorbable changes to system-wide transmission, from local adjustment to whole-system entanglement, and from a flexible space for trial and error to a taut net.

The physical process itself offers useful comparisons; the analogy has a material source. As cross-linking increases in a polymer solution, viscosity initially rises slowly and then sharply near a particular cross-link density; a gel network spans the sample, while a residual sol fraction can remain. Materials science calls this the gel point: the first spanning connectivity across the sample, rather than a local region becoming hard. Functionality and reaction extent can be used to estimate the gel point; it belongs to the same class of mathematical structure as the percolation thresholds discussed next. Its defining property is worth remembering: before it, a local region can be stirred; after it, local forces can travel through the spanning network, with their range and timing still dependent on the material and loading.

Percolation Thresholds and Interdependence

Models can calculate the width of this region and the location of its threshold. Percolation theory gives a simple, powerful picture: with few connections, nodes form small disconnected clusters and disturbances cannot readily cross the system; near the critical threshold, clusters merge; beyond it, a giant component appears, making system-wide transmission possible (Broadbent and Hammersley 1957; Erdős and Rényi 1959). The picture matters because additional connections can produce a phase transition, rather than a linear change. The system shifts from one mechanism of disturbance propagation to another.

For a random graph with independent edges, the threshold can be expressed through mean degree. Let $\langle k \rangle$ be the network's mean degree, the average number of connections per node; the condition for a giant component to appear is

$$\langle k \rangle_c = 1. \tag{2.17}$$

For an Erdős–Rényi random graph, a giant component emerges asymptotically as mean degree crosses 1. This concerns the growth of random branches, rather than a guarantee of spanning connectivity whenever every node has one connection. The mean degree of $G(n,p)$ is $(n-1)p$, giving a threshold edge density of roughly $1/n$. Real dependency networks require examining direction, correlations, weights, and clustering.

In the corresponding infinite-system limit of ordinary percolation, the order parameter can grow continuously from zero above the threshold. Finite networks exhibit changes in spanning probability and cluster size. Let p be the occupation probability, p_c the threshold, and P_∞ the fraction in the giant component; near the threshold the following scaling form can arise:

$$P_\infty(p) \;\propto\; (p - p_c)^\beta, \qquad p \to p_c^+, \tag{2.18}$$

Here β is the critical exponent, and P_∞ lies between 0 and 1; the expression applies only in the infinite-system limit. The value of p_c depends on network type and dimension: site

percolation on a two-dimensional square lattice has a threshold of about $p_c \approx 0.5927$, so a spanning component appears when random occupation exceeds roughly 59%; bond percolation on the same lattice has a threshold of 0.5; a one-dimensional chain has $p_c = 1$, requiring full occupation. A finite system has a transition region that steepens near the threshold, rather than an exact switching point; smaller systems have broader transition regions.

The sensitivity of the threshold to network type matters more for real systems than the number 0.5927 itself. A two-dimensional square lattice has coordination number 4, while site percolation on a three-dimensional simple cubic lattice has a threshold near 0.31; higher coordination and dimension lower the occupation fraction needed to span. In some large-network models with a divergent second moment of the degree distribution, the retained-node fraction needed to sustain a giant component after random removal can approach zero; the corresponding removal fraction approaches one, indicating tolerance of substantial random losses of ordinary nodes (Albert et al. 2000). Tolerance of random disturbance and vulnerability to targeted attack can coexist in the same network; removing key hubs can disrupt connectivity more readily than removing the same number of nodes at random. Percolation thus gives a differentiated judgment: thresholds depend on structural form as well as total connection count.

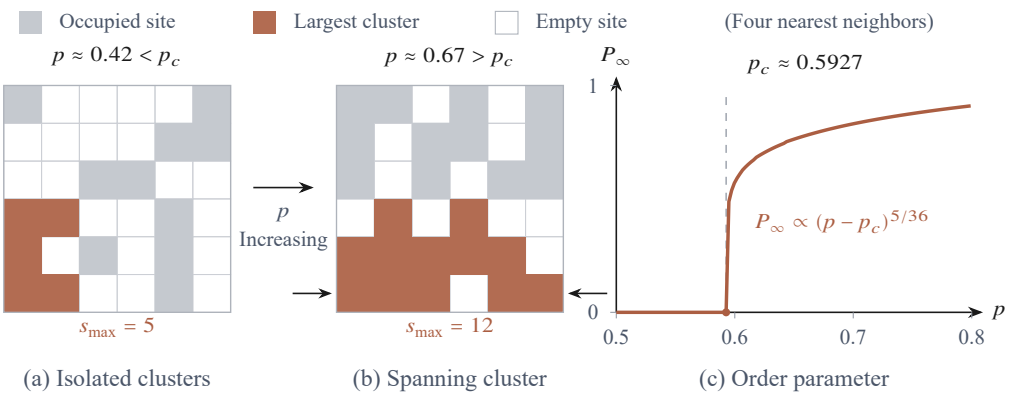


Figure 2.5 A phase transition near the percolation threshold. Below the occupation threshold, only separate small clusters exist and disturbances cannot span the system; above it, a spanning cluster appears and behavior changes. The order parameter P_∞ at lower right rises rapidly near the threshold; system size sets the width of the transition region.

What is cleanly separated in a mathematical model is mixed together in real systems. Organizations are not random graphs, city traffic is not a static network, supply chains contain more than nodes and edges, and finance contains more than trading relationships. Real systems also have delays, incentives, power distributions, standard interfaces, boundaries of responsibility, information asymmetries, and expectations. This makes real-world gelation harder to see: connections, standards, processes, platforms, metrics, permissions, and automation accu-

mulate until the system can no longer be changed locally, without a clearly dated crossing of a measurable threshold. Under low coupling, changes can be isolated, failures repaired locally, and technologies tried on a small scale; the system has gaps, buffers, and local degrees of freedom. Under high coupling, spanning connections turn local changes into global channels: one interface affects upstream and downstream systems, one metric affects multiple actors, and one model involves data, review, execution, compliance, and responsibility. The system hardens as a whole, with less freedom for local movement. This hardening comes from the spanning structure: a loose net permits a short section to be pulled locally; pulling one point in a taut net makes the whole net respond. Beyond a connection threshold, whole-system structure can take over local degrees of freedom: one department's speed requires absorption by others, one module's upgrade requires interfaces, data, permissions, training, and responsibility to update together, and one deployment requires simultaneous changes in processes, standards, incentives, and risk allocation.

More connections therefore do not necessarily make a system more flexible. Connections improve coordination while increasing entanglement, improve efficiency while increasing propagation risk, and reduce local friction while potentially increasing system friction. Seamless coordination can also mean seamless transmission of shocks. Fluidity and fragility often come from the same network. This is counterintuitive because modernization usually treats connection as progress: connect information systems, supply chains, cities, and finance; break down organizational data silos; connect AI to processes, interfaces, tools, and execution systems. Connections do bring efficiency: disconnected parts cannot coordinate, scale, or produce complex functions. But connections change failure: a sparsely connected system may fail locally, while a densely connected one may cascade, amplify, synchronize, or stop globally. Connection increases the ability to transmit both coordination and error.

Research on interdependent networks makes this clearer. Buldyrev and colleagues showed that a failed node in one network can cause a dependent node in another to fail, recursively propagating a cascade; under some conditions, removing a small fraction of nodes can fragment interdependent networks extensively (Buldyrev et al. 2010). The mechanism concerns systems supported by multiple layers—power, communications, finance, logistics, organizations, and information—whose fragility comes from dependencies between layers as well as within them. A shock tolerated by an independent network can become a cascade across interdependent layers. The strength, direction, distribution, and failure rules of those dependencies jointly affect cascade size; specifying them makes it possible to calculate propagation paths and locate weak links.

Cascades: Cases and Costs

Risk in complex systems is often underestimated. Assessments check whether nodes are reliable, modules stable, processes compliant, and models accurate, while overlooking the collective state formed by their connections. A module reliable in local tests may not be reliable

within the whole system; a model accurate offline may not be reliable in an organizational workflow; a strategy effective in a single-site experiment may cease to be effective once multiple actors respond, feedback is delayed, and incentives change. The risk in a gelled system lies in the way connections change error propagation, beyond the failure of individual parts.

Large organizations make gelation especially visible. Changing tools in a small team may take a few conversations; changing the same tool in a large organization involves permissions, data, security, budgets, training, audit, performance evaluation, and compliance. The tool has entered a different network, without itself becoming more complex. A tool choice in a low-coupling setting becomes a structural migration in a high-coupling organization. Many digital transformation projects fail because the software touches a gelled process network: every node favors greater efficiency, but none wants to bear alone the risks of changed interfaces, redistributed responsibility, and invalidated old processes.

City traffic works the same way. Optimizing a signal in a low-density area may directly improve traffic; in a dense urban network it may merely deliver vehicles faster to the next bottleneck, leaving overall speed unimproved. Braess's paradox, discussed in Chapter 1, is an extreme case: under certain network conditions, adding a road worsens traffic because route choices and network equilibrium redistribute flows. Traffic is a network of strategic actors: adding a connection changes many people's behavior. Connections change structure, structure changes behavior, and behavior changes the system's state in turn.

Supply-chain gelation is more direct. In a loosely coupled chain, a late component may create only a local delay; in a chain with lean operations, low inventories, global specialization, and dense standardized interfaces, a small component shortage may stop an entire production line. The system is highly efficient in normal operation because redundancy and inventory have been reduced and connections strengthened; the same structure lets local shocks propagate across the whole. Efficiency and fragility can be two sides of one structure.

The Suez Canal blockage provides a vivid example. The Ever Given grounding in 2021 blocked the canal for almost a week; the backlog was cleared only after 422 waiting ships had passed (Suez Canal Authority 2021; Japan Aerospace Exploration Agency 2021). The canal normally serves as an efficient corridor for global trade, shortening routes between Asia and Europe and saving time and fuel; a single ship's blockage exposed the other side of that tightly coupled network. A New Zealand Ministry of Foreign Affairs and Trade report put the canal's share at roughly 12% of global trade and 30% of global container traffic (New Zealand Ministry of Foreign Affairs and Trade 2021). The case illustrates a structural dependency: the same efficient corridor reduces friction in normal times and amplifies blockage in a crisis.

Power systems offer a more severe picture of cascading failure. The August 14, 2003 blackout in the northeastern United States and Ontario affected an estimated 50 million people and 61,800 megawatts of load, with outages lasting up to four days in some areas, according to the official report (U.S.-Canada Power System Outage Task Force 2004). It was a cascade

involving protection, dispatch, information displays, vegetation management, monitoring, and regional coordination, rather than a simple case of one failed device. The report also describes how, after a key line tripped, the cascade spread into multiple states and provinces within minutes, with most subsequent events concentrated in its final, extremely short window (U.S.-Canada Power System Outage Task Force 2004). Interconnection normally improves supply reliability and resource dispatch; in an unsafe system state, it also becomes a channel for spreading faults.

Finance works the same way. Financial connections speed capital flows, make asset prices more continuous, and allow more flexible risk transfer; when banks, funds, insurers, derivatives, collateral, ratings, algorithmic trading, and expectations are bound together, a local price fall can trigger margin calls, forced liquidation, shrinking liquidity, panic selling, and regulatory responses. In calm periods the connections let the system flow like a liquid; in a crisis the same network can make it contract like a gel. On May 6, 2010, U.S. stock prices plunged within minutes, with the Dow Jones Industrial Average at one point down almost a thousand points, then rebounded rapidly; the joint SEC–CFTC report described mutually reinforcing pressure from a large automated sell program, high-frequency traders reducing positions and rapidly passing contracts among themselves, and declining buy-side liquidity (U.S. Securities and Exchange Commission and Commodity Futures Trading Commission 2010). Prices moved away from fundamentals and back within minutes, while gains and losses had already settled in real accounts: a local instruction had become a system event.

Knight Capital's trading accident shows the speed of automation in tightly coupled finance: the SEC reported that, in the first 45 minutes after the market opened on August 1, 2012, Knight's automated router sent more than four million erroneous orders while attempting to fill just 212 customer orders, traded more than 397 million shares, acquired billions of dollars in unintended positions, and ultimately lost more than $460 million (U.S. Securities and Exchange Commission 2013). The case shows how a software defect can rapidly become market behavior, asset positions, and a threat to a company's survival when high frequency, automation, market access, and live execution are coupled together. Once a bit error enters a financial execution network, its consequences extend beyond bits.

Recent chip shortages again exposed supply-chain cascades. Automakers had long relied on lean inventories and just-in-time supply, with low-cost chips requiring long lead times distributed across dozens of control units; when wafer capacity was taken by other uses, the missing part was often a chip costing a few dollars inside a body-control module. Without that chip, the module could not be assembled and a whole vehicle might not leave the line. Capacity recovery takes time: expanding a wafer fab takes years, while demand can restart much faster; this mismatch stretched shortages across several quarters. The case shows two features of gelation together: the break occurs at the lowest-value point, and the slowest layer sets recovery time.

Education has the same structure, though its cascades reorganize behavior rather than stop operations. A changed assessment metric can affect teaching, student strategies, family investment, tutoring businesses, school rankings, and local policy as well as internal school management. Once a metric enters a tightly coupled network, it becomes a driver of an entire field of behavior as well as a measurement tool. Many reforms look reasonable locally: clearer metrics, more transparent rankings, finer assessment, and more frequent data feedback. Entering a gelled system of admissions, resource allocation, reputational competition, and family expectations, however, those metrics change its strategic structure: denser measurement organizes behavior around measurement, and more frequent assessment makes it easier for indicators to displace the real goal (Goodhart 1975; Campbell 1979).

Three Local Outcomes of AI

AI enters networks already approaching gelation, rather than isolated tasks. It feels light in low-coupling settings—copywriting, summaries, search assistance, code completion, and sketches—where failures can be isolated, outputs rolled back, and responsibility bounded. In healthcare, finance, transport, education, energy, and public administration, its outputs immediately connect to data, workflows, review, execution, responsibility, and behavior, making it a source of disturbance within the network. This explains the sudden slowdown as large models move from bit space into real systems: bit tasks often permit rapid trial and error, rewritten mistakes, withdrawn results, version updates, and parallel candidate generation; tightly coupled systems link medical advice to tests, treatment, insurance, and liability, traffic dispatch to vehicles, roads, passengers, accidents, and public order, financial decisions to assets, exposure, regulation, and expectations, and educational assessment to family investment, school behavior, and students' paths. Once connected, AI output becomes a structural disturbance rather than candidate generation alone. The bottleneck is then the system's ability to absorb the output.

Local AI optimization in a gelled system can produce three outcomes. First, absorption: the structure stays intact, and AI becomes a peripheral assistant that improves local efficiency without changing the overall state; meeting notes, email drafts, and preliminary document classification fit into existing workflows, improving local experience while leaving responsibility, power, and action chains untouched. Second, cancellation: AI speeds one stage and another becomes the bottleneck. Reports arrive faster while review slows; more proposals make decisions harder; predictions outpace execution; faster customer-service replies lead to more escalated complex complaints; more code candidates increase pressure on testing, integration, and security review. Overall speed stays fixed while the blockage moves; local acceleration may approximate a gain in a low-coupling task, but in a gelled system it often moves the queue. Third, amplification into new problems: automation improves local efficiency while weakening recovery from exceptions; recommendations raise clicks while amplifying emotional contagion; high-frequency algorithms speed trades while increasing turbulence; assessment

systems measure more often while amplifying teaching to the test; triage models process cases faster while potentially embedding bias in a larger workflow. AI can change how a system responds as well as replace an action, making a behavior cheaper and easier to scale and institutionalize.

The boundary among these outcomes depends on the room the system retains to absorb new inputs, beyond model capability alone. The same model can produce opposite results in two environments: where responsibility can be isolated and failure rolled back, it improves efficiency; where neither is possible, it transfers risk. The assessment must ask whether the change touches the source of the system's rhythm, crosses organizational boundaries, or drives a layer's inputs faster than its update time. These three checks provide a starting point for judging absorption or amplification; deployment must test the outcome.

Technology assessment in a gelled system must therefore examine how the system absorbs performance, beyond measuring that performance locally. A tool entering a low-coupling setting resembles water entering a channel; entering a gelled system resembles a force applied to a taut net. The same input produces different responses. Model accuracy, task completion, response speed, and user satisfaction are local metrics. At system level, the questions are: how do errors spread? How are exceptions stopped? How does responsibility return to those who act? Does the bottleneck move? Do people change strategy? Must existing workflows be rearranged? Does the system have enough buffering and isolation to keep trial and error from spanning the whole?

Scaling and Near Decomposability

Many technologies are easy to pilot and hard to scale. Pilots usually occur within local clusters: connections are deliberately cut, failure effects isolated, participants selected, resources specially assigned, and organizational attention concentrated. At full scale, connections return and hidden coupling emerges through different data standards, processes, responsibilities, acceptance, regulation, and legacy interfaces. Scaling means carrying a pilot through a larger network, rather than copying it many times. Many technologies harden when they span the network, rather than failing functionally.

Digital transformation, smart cities, intelligent healthcare, and AI public services repeatedly encounter this problem. A successful departmental pilot does not establish successful operation across departments; a result validated in one clinical department does not establish compatibility with the whole hospital's workflow; optimizing one city road does not establish better citywide traffic; automating one approval does not establish faster public service end to end. A pilot thins the system's connections; scaling reconnects it at full thickness. The resistance of a gelled system often becomes visible during that reconnection.

One engineering fantasy deserves particular scrutiny: that more data connections, interfaces, and models automatically make a system more intelligent. Reality can run the other way. More connections can improve sensing while making change harder, increase automatic

responsiveness while making stopping harder, and improve global coordination while making global resonance easier. A highly interconnected system has less isolation, which can accelerate both discovery and propagation of errors, both optimization and instability, without necessarily adding wisdom.

Gelled systems show why more connections are not always better. Too few prevent coordination; too many remove flexibility. A good structure holds connectivity and isolation in tension: information, resources, and action can travel, while local errors remain bounded, local experiments leave the whole undisturbed, and failures have buffers. This is the value of modularity, redundancy, sandboxes, layered interfaces, staged deployment, and circuit breakers: they govern how connections span the system, allowing necessary coordination while containing experiments, errors, and exceptions. These designs preserve plasticity in complex systems: a system without isolation may look smooth briefly and prove fragile over time; redundancy and module boundaries may appear inefficient briefly while improving the ability to absorb change.

This is close to Herbert Simon's idea of **nearly decomposable** systems. Complex systems can often evolve and be managed because they contain hierarchies, modules, and relatively independent substructures (Simon 1962). Subsystems remain connected without unrestricted spanning, allowing changes to be absorbed, tested, and corrected locally before affecting the larger system through interfaces. Without near decomposability, every local adjustment becomes a wholesale rearrangement, making evolution extremely expensive. AI entering reality must follow this structural logic: low-risk bit tasks can permit rapid trial and error, while tightly coupled systems require layered verification, local isolation, failure rollback, and buffers for responsibility. Healthcare, transport, finance, grids, education, and public administration require designs that prevent model output from propagating directly through the whole system, as well as stronger models; otherwise high-frequency bit inputs strike slower structures, creating backlogs, oscillation, and systemic risk.

The Media Gradient and Technological Viscosity

This connects directly to the book's central argument about technological viscosity. Viscosity arises as intelligent outputs enter reality and encounter damping from resources, time, and structural coupling. Gelled systems display nonlinearity in the structural dimension: once connections span the whole, local capability is settled in system consequences rather than local gains; a 20% improvement in a model's single-task performance does not imply a 20% improvement in system efficiency. The improvement may be absorbed, canceled, or generate new costs of coordination, verification, and responsibility.

The media of bits, atoms, and life also differ in their degree of gelation. Bit space connects fastest: interfaces, protocols, APIs, databases, and model calls can quickly form high-frequency networks, making information cascades, recommendation resonance, amplified public opinion, and software accidents particularly easy. Atomic connections form more slowly,

but errors in transport, power, manufacturing, logistics, and urban infrastructure are harder to reverse; recovery requires materials, energy, repairs, and time. Life is heavier still: development, metabolism, immunity, reproduction, and ecology couple its processes, so an intervention can change a long-term path rather than a single point. The heavier the medium, the harder it is to withdraw the costs of gelation.

Complex networks cannot always be changed smoothly. They may be flexible under low coupling and harden suddenly under high coupling. Technology entering such a system must ask whether a connection threshold has been crossed as well as whether its local capability is sufficient: can local changes still be absorbed locally? Can errors still be isolated? Can new proposals enter gradually? Can feedback return stably? Without answers, deployment merely pushes local capability into an already taut net. Resistance in a tightly coupled system can harden suddenly beyond a critical region; once connections span the system, local changes cease to stay local.

Technology entering a tightly coupled system must therefore address the structure of connections as well as the size of resistance. Spanning connectivity gives local changes system-wide consequences, shifting the system from local adjustability toward collective hardening. The most dangerous misjudgments of the intelligent era equate local capability with system capability, connection with flexibility, automation with control, and pilot success with success at scale. Gelled systems reverse the starting question: does the system retain enough room, buffering, and boundaries for the technology to enter in a form it can absorb, reverse, and take responsibility for?

Once a system crosses the threshold, its state depends on more than its current connections. Assets already committed, interfaces already adopted, habits already formed, and behavior already changed remain in the structure when the goal changes. Whether a tightly coupled system can recover depends on why these residues cannot simply be cleared away when a technology is withdrawn.

2.4.3 Hysteresis and Structural Memory

Hardening raises the cost of local adjustment without stopping change. Completed investments, adopted interfaces, and established habits continue to act within the system, regardless of its current goal. Real systems have memory. This is a mechanism of system evolution, not just a figure of speech. A system does not choose an optimum from scratch at every moment; its present form is the sediment of past investments, institutions, interfaces, errors, compromises, and habits. Roads, hospitals, schools, companies, software, supply chains, legal institutions, and urban spaces are built up through historical paths as well as abstract design.

Physical Residue

Structural memory is more than “historical baggage.” **structural memory** is the physical

residue the past leaves in a system: it changes path costs, interface forms, connection strengths, transmission across scales, directions of retreat, and the ways the system can subsequently be changed. The past persists because it has already altered the system. A road changes land values, residential patterns, commercial locations, commuting habits, and subsequent infrastructure; an adopted software interface changes data formats, development processes, compatibility requirements, and the design boundaries of later systems; a long-used assessment metric changes where people direct effort, how organizations allocate resources, and how later proposals are judged. Time turns choices into structure, and structure into inertia.

"Physical residue" is literal here. Once railway gauge is laid down, rolling stock, platforms, tunnel clearances, maintenance equipment, and dispatch rules take shape around it; subsequent change costs are set by this whole supporting system as well as the rails. The 1,435-millimeter standard gauge emerged from historical competition rather than geometric necessity; some railways in Commonwealth of Independent States countries retain 1,520 millimeters, India widely uses 1,676 millimeters, and Spain's traditional lines use 1,668 millimeters, with gauge-changing equipment or transshipment connecting different systems at borders. The gauges differ by only a few hundred millimeters, yet conversion involves entire fleets, lines, and supporting facilities. Grid voltages and frequencies, appliance ratings, screw-thread standards, and container dimensions have the same property: an initially contingent choice becomes part of the environment once adopted by mass production, certification, and training.

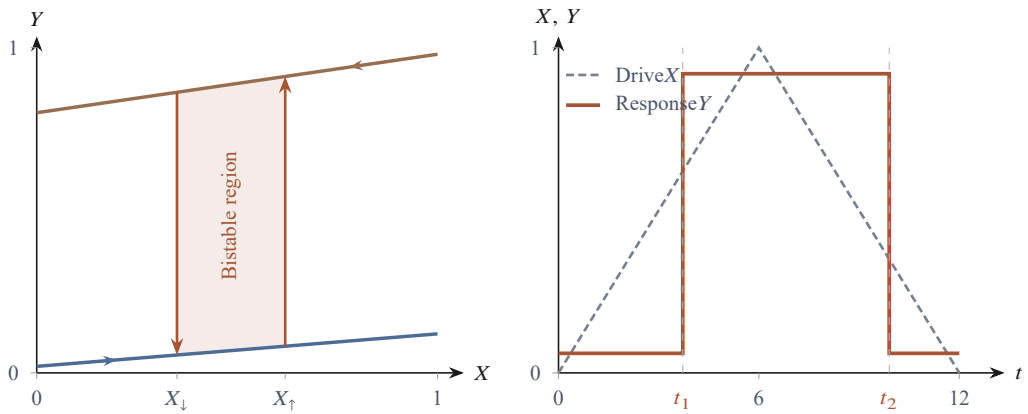
Path Dependence and Hysteresis

The first consequence of inertia is that the path already taken constrains the direction ahead. This is the basic meaning of **path dependence**. Real systems continue along paths already formed, with past choices changing future options (David 1985). Standards, interfaces, assets, and habits built for an earlier goal become boundaries on future choice, making some changes easier and others more expensive. Path dependence goes beyond a habit of taking the old road. The road taken changes the roads ahead. Moving from A to B changes the system itself: people are retrained, assets reallocated, interfaces redesigned, data reformatted, responsibilities redrawn, and interests redistributed. Once the system reaches B, the original A no longer exists; returning often requires a different route.

This is **hysteresis**. Hysteresis gives change a direction: the cost of moving from A to B differs from that of returning from B to A. A company migrating software may spend years cleaning data, rebuilding processes, training staff, and adapting interfaces; after migration, going back takes more than uninstalling the new system, because people, processes, data habits, and supplier relationships have changed. A city with an established road and housing pattern cannot return to an earlier state as easily as redrawing a plan, even after a better plan is found; buildings, land ownership, residents' lives, and infrastructure have fixed past choices in space. Irreversibility can mean that retreat has become a new engineering project, rather than that retreat is absolutely impossible. A path changes the system that travels it; returning means

changing the altered system again. The destination may be called A, but materials, interfaces, habits, trust, and responsibility make it structurally different from the original A.

Physical measurement gives hysteresis a precise form, helping distinguish it from mere delay. Magnetize a ferromagnet with an external field, then remove the field: some magnetization remains, called remanence; bringing magnetization to zero requires a reverse field, whose magnitude is the coercivity. Plotting magnetization against the applied field gives a closed loop; with the appropriate magnetic constant and unit conversion, its area gives hysteresis energy loss per unit volume per cycle, one source of heat in transformer and motor cores. The material's present state depends on the history of the applied field as well as its current value; reversing the field's path traces a different curve. In any system with such a loop, the present input alone does not determine the state, and removing the input does not imply a return to the starting point.



(a) Hysteresis loop (b) Time course

Figure 2.6 Hysteresis: state depends on the path, rather than instantaneous input alone. On the input-state loop at left, the same input value corresponds to two different states; at right, input and response share a time axis, showing how the state remains in a new position after the input is removed, with a reverse cost required to return to the starting point.

Frozen Degrees of Freedom

This gives structural memory a physical meaning. Repeated loading leaves fatigue and residual stress in materials; magnetic fields leave hysteresis in magnetic materials; cooling freezes part of a glass's disorder; prolonged use forms high-weight edges, critical interfaces, and low-resistance paths in complex networks. The materials differ, but the mechanisms resemble one another: the past reorganizes internal degrees of freedom, making arbitrary rearrangement harder than at the outset. **Structural memory is frozen degrees of freedom.** A system may begin with many possible connections, standards, and paths; repeated use of particular connections, widespread compatibility with interfaces, broad adoption of standards, and sustained investment in assets reduce its degrees of freedom. Directions are no longer equally

feasible, nor paths equally cheap; history turns some possibilities into low-resistance paths and pushes others behind high-resistance boundaries.

Glass cooling offers a particularly clear version of this mechanism. In a liquid, atoms can rearrange relatively freely among many configurations; as temperature falls, structural relaxation time rises sharply, with some glass formers exhibiting super-Arrhenius growth. Below a certain temperature, rearrangement can no longer finish within the observation time, freezing the configuration in its disordered state. Provided both cooling routes avoid crystallization and form glass, slower cooling generally lets the structure relax into a lower-energy state, while faster cooling freezes a higher-energy structure. The compositions can be identical and the properties different, solely because of the paths taken. The strength of structural memory thus depends on the degrees of freedom available at the time and the speed of change.

This is why legacy systems are so stubborn. Old software may be awkward, but data, permissions, reports, and staff training revolve around it; an old standard may be imperfect, but equipment, suppliers, certification, and customer relationships revolve around it; an old institution may be flawed, but resource allocation, promotion, and risk-bearing revolve around it. They have become low-resistance paths rather than optima; continued use attracts further investment, training, interfaces, and expectations, reinforcing those paths into structure. A local optimum is a property of the surrounding energy landscape, rather than the point alone. A system may stay because leaving requires too many simultaneous changes: interfaces, standards, data, processes, habits, and responsibility. Each edge adds a cost; together they form a stable basin. Knowing that a better place exists does not make climbing out of the present basin any lighter.

Scale Filtering and Structural Shielding

Structural memory extends across scales. Complex systems typically comprise individuals, processes, organizations, institutions, industries, and social environments; a lower-level change need not arrive unchanged at a higher level. Higher structures filter, absorb, rewrite, or shield lower-level changes, passing on what fits their logic; many real microscopic changes never become macroscopic variables. The renormalization group offers an intuition: moving from microscopic details to macroscopic behavior coarse-grains away many details, retaining a few variables decisive for the large-scale state. Not every microscopic perturbation changes a macroscopic phase; many local differences are averaged, filtered, or screened out in the change of scale (Wilson 1971). Its use in critical phenomena is worth examining. In the critical region of a continuous transition, fluctuations at different scales have similar structures; enlarging the scale preserves statistical features, allowing scale dependence to be expressed as power laws. A few critical exponents characterize the leading scaling behavior, while many microscopic details of a material drop out. Different physical systems in the same universality class can therefore share critical exponents: scale transformations erase their microscopic differences. For sociotechnical systems, this is an intuition rather than a rigorous model, pointing toward

macroscopic behavior governed by variables that remain effective across scales, which may be far from the easiest variables to observe.

Sociotechnical systems show similar patterns. AI-assisted lesson preparation can improve a teacher's efficiency without changing education's macroscopic state if curriculum standards, examinations, school evaluation, admissions competition, and resource allocation stay fixed; faster clinical judgment need not improve a hospital's efficiency if departments, insurance payments, records, liability, and patient flows stay fixed; local workshop automation need not upgrade a supply chain if orders, supplier capabilities, logistics, quality standards, and capital investment stay fixed. The renormalization group serves here as an intuition about scale: technical improvements must change structural variables that remain effective macroscopically to become system change; otherwise the system coarse-grains them into individual experience, local showcases, or short projects. This is **structural shielding**. The system filters microscopic innovation into forms compatible with its structure; "useful locally, invisible overall" often means a reform has not passed through cross-scale filtering into evaluation, resource allocation, responsibility, and institutional processes. It has not become a macroscopic variable.

AI is especially likely to create an illusion here. Its sudden expansion of individual abilities—faster writing, search, programming, design, synthesis, and analysis—makes it tempting to expect organizations, industries, and institutions to improve in step. Structural change depends on reconnecting those abilities to processes, evaluation, resources, and responsibility; without crossing these levels, enhanced abilities are absorbed as local experience. A stronger node does not imply a changed network. Helping someone complete an existing task faster improves node performance; changing process interfaces, the distribution of responsibility, resource allocation, and organizational boundaries changes structure. In tightly coupled systems, the bottleneck can lie in the organization of edges rather than the ability of nodes: unchanged interfaces, processes, authority, responsibility, and evaluation absorb new local capabilities into the old structure. This is a real boundary for many AI applications: more proposals, text, analysis, scripts, and local efficiencies generate microscopic perturbations, but structural change depends on whether effective coupling changes; technology that leaves interfaces, standards, processes, incentives, and responsibility intact often remains experience, a patch, a pilot, an add-on, or a report.

To judge whether an improvement has passed through scale filtering, examine three kinds of variable: what is evaluated, what is allocated, and who bears the consequences. They directly organize evaluation, resources, and responsibility, making them priorities for a cross-scale assessment. A technology that only speeds work without changing any of these variables can deliver real individual efficiency while being absorbed by existing system bottlenecks. Conversely, even a small change in evaluation criteria or responsibility can leave a higher-level trace and trigger further adjustments.

Withdrawal Is Not Restoration

Structural memory also means withdrawal is not restoration. New infrastructure changes land use, a new platform changes habits, a new standard changes supply chains, a new algorithm changes expectations, and a new institution changes the distribution of interests. Withdrawing the technology later does not fully restore the original system: people have learned, organizations adjusted, capital been committed, competitors followed, and users developed new expectations. Technology governance must therefore ask more than whether a deployment can be withdrawn if it fails. In tightly coupled systems with serious consequences, withdrawal is not restoration. A flawed educational assessment may already have changed students' paths; a flawed financial model, asset allocation; a flawed recommender, the structure of public discussion; a hasty medical AI deployment, trust between physicians and patients. Change leaves memory.

This irreversibility includes institutions, behavior, cognition, and trust, as well as physical processes. The losses already caused by a bridge collapse cannot be undone, and some drug side effects cannot be reversed; damaged trust, standards lock-in, path dependence, and organizational inertia also have irreversible or partly irreversible properties. Even where physical reversal remains possible, social restoration can be extremely difficult. Systems with serious consequences need more careful technology integration. Some trials change system memory: low-risk bit tasks permit rapid retries because errors can be cleared easily; high-risk institutional systems cannot experiment at unlimited frequency because errors accumulate as new structure. Healthcare, education, finance, law, transport, energy, and public administration all require boundaries between experimentation and irreversible consequences.

Electrifying infrastructure shows the scale of this irreversibility. Replacing a combustion-engine fleet with electric vehicles also changes parking-site power supply, distribution capacity, fire requirements, maintenance systems, battery-recycling chains, and insurance terms; after conversion, reversing course takes more than replacing the vehicles, because electrical capacity, site layout, and maintenance capability have been shaped for the new use. The pace is jointly constrained by construction capacity, social coordination, financing, permits, supply chains, and parallel construction; it cannot be inferred solely from progress in vehicle technology or accelerated wholesale by one decision.

Organizing History

Structural memory suggests that real technical progress includes dealing with the old system, rather than simply replacing it. A technology that cannot accommodate historical paths meets strong resistance; one that obeys them completely is domesticated by the old structure and produces little change. Mature transformation often lies between: understanding why the old structure exists while designing migration paths, retaining the possibility of reconstruction without imagining that it can happen at a click. This requires incremental interfaces, transitional structures, parallel systems, data migration, retraining, reassigned responsibility, and

mechanisms for retiring old institutions. Orderly retirement of the old is often harder than invention of the new. Without retirement, new technologies pile onto old systems and add layers; with it, technology can change structure.

Structural memory is also an asset. Without it, a system has no stability. Hospital workflows, legal institutions, engineering standards, traffic rules, financial credit, and scientific norms use memory to maintain predictability; institutionalizing experience reduces repeated trial and error. Memory preserves errors as well as experience, and constrains change as well as providing stability. It lets systems run while making fresh starts difficult. A mature system can organize its history: retain some old interfaces and retire some standards; distinguish safety experience from outdated habits; maintain some sunk structures while gradually loosening others. A system with no memory lacks stability; one unable to organize its memory lacks plasticity.

The AI era makes this awareness especially necessary. AI can generate future plans without automatically clearing historical liabilities: it may describe how a system should work while leaving old data, standards, interests, habits, responsibilities, and structures of trust to be dealt with. Every new system enters a reality already written by the past. Complex systems travel along paths shaped by historical investment, institutional interfaces, and behavioral habits, rather than abstract optima. AI can generate future states, but the future must pass through structures left by the past. Structural memory is the residue, hysteresis, and irreversibility the past leaves within the system.

Structural memory explains both why systems cannot start from zero and why retreat is itself an engineering project. It leaves a more basic question: which states can a system already written by history actually reach and sustain? Memory makes some directions cheaper and others dearer, narrowing the reachable range. Representational space alone cannot tell us which states remain reachable.

2.4.4 Reachable Space: Local Optima and Braess's paradox

AI expands what can be imagined; real systems need states they can reach. Connections, coupling density, thresholds, history, and filtering across levels all constrain change, leading to one conclusion: a real system's state space is not fully open; a target may be imagined, computed, simulated, and optimized without being stably reachable. We must distinguish imaginable states from reachable space.

Imaginable and Reachable States

An imaginable state belongs to representational space: a plan, an optimum, a model output, an institutional design, an engineering proposal, or a blueprint for the future; AI vastly expands this space by generating many routes, structures, molecules, processes, and organizational designs at once, making other possibilities easier to find. A reachable state is one a real

system can actually reach and sustain under given constraints; a valid target is necessary, but there must also be a path from the present state that is affordable in resources, possible in time, absorbable by the structure, controllable in risk, and manageable in failure. Imaginable states concern whether an endpoint can be described; reachable states concern whether we can get there from here.

The most common error of technological optimism is to mistake imaginable states for reachable ones. Seeing AI generate a better target encourages the belief that reality will naturally move toward it. But a real system occupies a terrain of slopes, ravines, old roads, bridges, cliffs, and sunk structures; marking a target on the map leaves the hard questions of whether a road exists, whether it is too steep, whether a bridge can bear the load, whether supplies are available along the way, and whether retreat is possible after failure. Generative capability determines the number of imaginable states; media, structure, and history jointly determine the number of reachable ones. Completeness in representational space does not establish reachability in reality.

Intermediate states are central to deployment. Many proposals remain suspended because they specify B without explaining how the system can move from A to B. Real change is usually a sequence of continuous or partly continuous transitions: old structures cannot vanish instantly and new ones cannot instantly take over every function, so the system needs parallel operation, test areas, compatibility layers, buffers, alternative paths, and exit mechanisms; without these intermediate states, a perfect target resembles an image hanging in midair. Several constraints narrow reachable space: resources exclude unaffordable states, time excludes paths that cannot finish soon enough, coupling excludes local changes that cannot occur alone, history rules out starting from zero, and risk boundaries exclude paths with unacceptable failure consequences. The choices available to a real system are the few paths jointly selected by these conditions.

Intermediate states have their own tests: the system must remain operational during transition, and the transition itself must have an exit. Running two systems in parallel doubles maintenance commitments, and a longer transition costs more; too short a transition leaves functional gaps before the old process has handed over. A workable design typically divides transition by function, allowing each segment to be accepted and rolled back independently, rather than treating the entire system as a switch. Treating reachability as a path problem produces concrete design questions; treating it only as a target problem produces ever more proposals.

Braess's paradox and Multi-Agent Equilibrium

Feasible paths and the equilibrium actually reached must be distinguished. If travelers may ignore a new road, adding it cannot shrink the original set of feasible paths; it can nevertheless change their choices and raise total travel time at user equilibrium. Braess's paradox concerns precisely this divergence. Similar effects from new tools, metrics, or models require specifying the actors and cost structures involved.

Braess's paradox matters beyond traffic because it changes how reachability is understood. A path's cost is formed through joint use, rather than fixed in the path itself; a path is a dynamic relationship among topology, cost functions, choices, and flow distribution. A new connection changes the system's cost landscape; behavior redistributes, and a seemingly shorter road can become a new source of congestion. In systems with autonomous choices of this kind, reachability must also account for the equilibrium among multiple actors.

User equilibrium gives this property a formal expression. Let $\mathcal{P}_w$ be the set of feasible paths for origin–destination pair w, x_f the flow on path f, $\mathbf{x}$ the distribution of all path flows, and $c_f(\mathbf{x})$ the travel cost of path f, determined by the full flow distribution and measured in time. At a user equilibrium $\mathbf{x}^*$, for every origin–destination pair w, each path with positive flow costs no more than any other path for that pair:

$$c_f(\mathbf{x}^*) \leq c_g(\mathbf{x}^*), \qquad \forall f \in \mathcal{P}_w \text{ and } x_f^* > 0,\ \forall g \in \mathcal{P}_w. \tag{2.19}$$

This condition captures the possible divergence between individual and system optima: each driver compares personal costs, while no one bears responsibility for total system cost, so equilibrium can leave everyone wishing for a better outcome that none can achieve alone. It assumes informed travelers who can choose paths freely and decide solely on their own travel time; tolls, restrictions, or information interventions can move the equilibrium. A path one person can traverse need not work for the system; low resistance for an individual need not mean low resistance for the whole; a locally better proposal need not remain better under shared use. A complex system's reachable space depends on where simultaneous use by all actors takes the system, as well as which paths exist (Wardrop 1952). The gap between individual and system optima can be quantified, with stronger consequences than intuition suggests. For a routing model with fixed demand, travelers treated as infinitesimal flows, and nonnegative affine edge-delay functions, total travel time at user equilibrium is at most four-thirds of the system optimum, about 1.333 times; some networks attain this bound (Roughgarden and Tardos 2002). Road supply offers further empirical evidence: a study of U.S. cities found that a roughly 10% increase in interstate highway lane-kilometers led over the long run to roughly 10% more vehicle-kilometers traveled, leaving far less improvement in travel time than a fixed-flow calculation would suggest (Duranton and Turner 2011). New capacity changes choices of route, travel time, and residence, generating new traffic rather than merely accommodating existing traffic.

More Paths and the Equilibrium Test

This matters especially in the AI era. AI continually adds "paths": recommendations, proposals, organizational tools, evaluation metrics, automated decisions, and low-cost entry points, turning choices once expensive, infrequent, and cautious into cheap, frequent, widespread ones. A low-resistance path used at scale attracts flow and imitation, creates congestion and arbitrage, encourages homogenization, and ultimately changes equilibrium.

Navigation can send everyone down the same small road; recommendations can make creators chase the same signals; research tools can crowd investigations into the same candidate directions; metrics can reshape departments around the same numbers. Each actor appears to choose a better path while the system may become narrower, more congested, and more fragile. Reachable space is the set of states sustainable under shared use, rather than a count of paths.

This also requires reconsidering redundancy, alternative routes, and buffers. Redundancy is more than adding connections, and more alternative paths are not automatically better. Braess's paradox shows how added connections can induce bad equilibria; valuable redundancy increases choice, isolation, recovery, and rollback capacity. It gives the system room to divert, isolate, wait, withdraw, and reorganize under disturbance, rather than sending everyone toward one new shortcut. Reachable space expands through **governable paths**. Governable paths avoid drawing all traffic to one bottleneck, share pressure under exceptions, isolate local failures, preserve rollback, and avoid worse equilibria. A seemingly easy shortcut that draws all actors in together can become a new entrance to congestion.

Power systems turn redundancy into explicit security assessments. N-1 examines specified single-component outages and N-2 specified double outages, requiring the system to meet stability, thermal-capacity, supply, and related criteria under the relevant contingencies and permitted measures; standards and system rules specify the combinations, corrective actions, and allowable loss of load (Morison et al. 2004; North American Electric Reliability Corporation, n.d.). These assessments require real resources: reserve capacity, transmission corridors, and dispatchable units can all raise normal operating costs. That margin buys the ability to maintain or restore a safe state after disturbance, avoiding losses that cannot be undone. Governable redundancy requires checking flows, controls, and equilibrium both in normal operation and after contingency switching, to establish that a new channel offers room for recovery without creating a new shared bottleneck.

Local Optima and Governable Paths

Governable paths are hard to open because systems often rest at local optima. A local optimum here is a nearby state that is easiest to sustain, rather than the best state overall. It may be inefficient, cumbersome, unfair, or outdated, yet stable. Stability is itself a force: a workflow accommodated by an organization, a standard adopted by an industry, an evaluation organized around by behavior, and a city pattern inhabited by daily life become parts of the environment. The defining feature of this local optimum is the difficulty of leaving it. Many systems know better proposals but cannot afford the departure. Awkward old software anchors data, permissions, processes, and training; a flawed assessment anchors schools, teachers, students, and resource allocation; an imperfect industry standard anchors suppliers, equipment, certification, and customers. They are sustainable local basins rather than global optima.

AI makes this problem more visible. It reveals more possible heights and makes the present

system's imperfections clearer. Seeing higher ground does not provide a path uphill; intelligence without path design can sharpen the anxiety of knowing where things are better without knowing how to get there. AI's higher-order value in complex systems therefore includes finding reachable paths: which can proceed in stages, which interfaces must change first, which old structures must remain temporarily, which nodes can be tested first, which risks require isolation, and which failures must be reversible. Path design must ask what equilibrium an opening will induce, as well as whether the individual path is shorter. A channel that lowers local cost while drawing everyone to one place can narrow the system's reachable space; a tool that makes everyone chase one metric can reinforce a local optimum; a model that homogenizes decisions can improve short-term efficiency while reducing adaptability. The central design question is whether use of a path pushes the system toward a narrower state.

Efficiency and Robustness

Reachable space also involves robustness. Beyond equilibrium, it depends on keeping paths available under disturbance. In pursuit of efficiency, systems often reduce redundancy, compress buffers, raise utilization, refine control, add connections, and eliminate alternatives. These changes can make a system faster, cheaper, and more precise in the short term while narrowing its reachable space over time. A system with no redundancy may perform superbly under normal conditions yet lack alternatives under disturbance: it can travel fast along a fixed track while struggling to change tracks (Carlson and Doyle 1999). Efficiency seeks full use of resources in a known environment; robustness seeks survival as the environment changes. Efficiency often removes idle capacity while robustness preserves margin; efficiency favors close connections while robustness needs isolation and buffering. A system optimized for efficiency can look excellent in normal times and expose its poverty of paths when external conditions change.

A hospital schedule optimized to the limit has high normal utilization but little buffer for emergencies; a grid minimizing only short-term cost may lack flexibility under extreme weather or load swings; a platform maximizing clicks may sacrifice long-term trust and public discussion; an organization maximizing immediate output may leave no time to learn, repair, or transform. Redundancy, buffering, and alternative paths have value beyond their apparent waste or inefficiency. They form part of reachable space: without redundancy there is only one route, without buffers no ability to wait, without alternatives no detour, and without rollback no confidence to experiment. A system compressed to extreme efficiency is often hardest to change, restore, or adapt to new technology. But redundancy must be governable, rather than a blind increase in connections or unlimited choice. Useful redundancy provides room that can open under exceptions without damaging normal equilibrium, keeping the system from being locked to one path at a crucial moment. Equilibrium remains the criterion: giving everyone the same new shortcut may create a congestion entrance rather than redundancy.

Plasticity

Plasticity is therefore closer than efficiency to long-term vitality. System plasticity is the ability to change structure without collapsing, retaining room for adjustment rather than becoming loose or chaotic. Too loose a system cannot function effectively; too tight a system cannot adapt. A mature structure can operate and be transformed, maintain order and room to move, and use local optimization without being locked into it.

From this perspective, technical progress expands reachable space as well as speeding arrival at a target: lower resource costs make more states affordable; shorter necessary processes make more paths timely; reduced unnecessary coupling pushes the reachable boundary outward; clearing historical liabilities opens blocked routes; sandboxes, buffers, and rollback mechanisms also expand the space. Each must face an equilibrium test: does it genuinely add sustainable paths, or create another low-resistance point of congestion? This brings the preceding sections together: system change needs paths beyond local optimization; coupling density adds weight by reducing degrees of freedom; gelation hardens a system as spanning connections carry local disturbances globally; structural memory prevents arbitrary resets because the past shapes the future; Braess's paradox shows how adding paths can change equilibrium and narrow reachable space.

AI can open immense possibilities in representational space. Without a larger reachable space in reality, many remain models, reports, pilots, plans, and visions rather than system states. Scarcity shifts from targets toward paths, from imagination toward reachability, and from pictures of global optima toward passages out of local ones. Managing viscosity ultimately means managing reachable space: putting targets back into real paths and asking which resistance can be reduced, which connections loosened, which new paths induce bad equilibria, which redundancy retained, which intermediate states designed, which experiments isolated, which failures rolled back, and which old structures retired in order. AI expands possibility; reality needs opened paths. Yet true reachable space counts the states a system can stably reach and sustain after all actors have chosen together, rather than the roads drawn on a map.

So far, reachability has been treated as structural: connections, historical residues, and equilibrium distributions determine which states can be reached. One layer remains uncounted. A path's viability depends on the relationships attached to its ends, as well as its terrain and how many actors use it at once. The same change applied to an isolated node is one alteration; applied to a node surrounded by dependencies, it becomes a structural movement.

2.4.5 Structural Weight: Dependency Surface, Synchronization Cost, and Effective Inertia

A viable path tells us where a system can go, but not where the cost of movement comes from. Replacing a component may be routine maintenance in one system and a redesign in another; deploying a model may be an experiment in one setting and a reorganization of responsibility in

another. Path length, connection count, and equilibrium position cannot individually explain this difference. Three separately measurable variables unpack the dependency: how many relationships a change involves, whether they must move together, and whether failure permits retreat. When all three rise, the system exhibits **effective inertia**.

Effective Inertia

The cost of movement depends on more than path length. An isolated component is easy to move; even a small component embedded in a network can be difficult to move because its relationships with neighboring nodes must change too. An independent script can be changed quickly, while a module linked to finance, customers, inventory, approvals, and compliance is harder to change; a laboratory model can be updated quickly, while a model in a hospital, bank, grid, or transport system is a node in a web of relationships and cannot be updated at will.

This is structural weight. The discussion of network coupling called the additional change cost attached to a node virtual mass, counting only the relationships a change involves; including synchronization and rollback expands the account to all three components. **Structural weight is the effective inertia relationships give a system, rather than physical mass.** With more connections, a node's changes belong less to itself: more interfaces, rules, upstream and downstream dependencies, and responsibilities must be satisfied together. What looks like one point pulls a whole structure. More precisely, structural weight comes from the cost of coordinated movement. Effective inertia depends on how many relationships must adjust, whether they must adjust simultaneously, and whether a failed adjustment can be rolled back locally. A component may be light in itself and heavy in its dependency surface, synchronization requirements, and difficulty of rollback.

These variables rarely act alone. When they rise together, their cost becomes one collective movement: dependency surface determines who must move, synchronization cost whether they can move separately, and rollback cost whether they can return. An internal processing script may need one changed line, go live the same day, and be restored after an error, leaving all three variables near zero; changing a hospital pricing rule may require billing, medical orders, insurance interfaces, reconciliation, and patient notifications to change together, with inconsistent accounts if one lags. The same technical capability is a tool in the first position and an engineering project in the second. The same team can therefore display very different efficiencies: the capability stays fixed while the structure it must move changes.

Dependency Surface

Dependency surface is the easiest of the three to measure. **dependency surface** describes how many relationships a change involves. With a small dependency surface, changing a personal note, an advertising image, or an independent script usually remains local; failure can be rolled back, costs isolated, and responsibility kept relatively simple. Such settings suit AI because outputs can be tried quickly and local effects translate almost directly into

practical effects. With a large dependency surface, a diagnostic model touches physicians' workflows, patient trust, insurance payments, responsibility, and regulation; a bank risk model affects credit, capital allocation, compliance, and exposure; a grid dispatch system connects generation, load, storage, safety, and accident consequences. These are nodes in dependency networks.

A technical object can be light while the structure it pulls is heavy. A model file can be replaced quickly, but a model embedded in business rules, staff roles, customer commitments, compliance, and accountability cannot be replaced as freely as a file; a button can change instantly, but acquires structural consequences when it changes behavior, transaction paths, and risk distribution. Dependency surface translates directly into replaceability: an independently replaceable object is light; one that requires associated replacements becomes heavy.

Engineering can count counterparts of dependency surface: interface documents touched, approvals required, responsible parties involved, and outside suppliers affected. Industry offers a physical version. Large blast-furnace campaigns can be planned over fifteen to twenty years or longer, with maintenance outages during the campaign; changes to the furnace body, blast supply, cooling, and raw-material systems must fit shutdown windows across the production chain (Dong et al. 2022). In October 2025, Třinecké železárny in the Czech Republic reported a 42-day shutdown to rebuild blast furnace 4 after a 20-year campaign, involving about 30 suppliers and changes to hot blast, gas piping, cooling, water systems, cables, and controls (Třinecké železárny 2025). The difficulty includes the dependency surface as well as the technology: a furnace connects the logistics, energy, and orders of an entire steelmaking process, rather than being an independently replaceable module. Likewise, the actual schedules of an urban arterial road, a substation, or a section of railway include the time for the existing network to adapt, as well as construction itself.

Synchronization and Rollback Costs

The second variable is whether the relationships must move at the same time. **synchronization cost** concerns this need for simultaneous movement. Systems that permit staged updates, local experiments, and gradual migration have low synchronization costs: one module can be replaced first, one interface made compatible first, and one workflow tried at the periphery; failure remains local. They absorb new technology more easily. Other systems require coordinated changes to standards, synchronized data, reassigned responsibilities, joint training, and compatibility between parallel old and new systems. Changing one stage without the others creates a break. Greater synchronization costs make the structure heavier. Many technologies are usable but cannot be moved into place across the system at once.

The third variable is the way back after failure. **rollback cost** concerns whether a failed adjustment can be reversed. Low-cost rollback permits bold trials: software features, recommendations, local processes, and low-risk assistants can use staged deployment, sandboxes, and rollback. Systems with high rollback costs cannot be tried so readily. In diagnosis, fi-

nancial risk control, traffic dispatch, power control, and public administration, a failure after processes and responsibility have changed takes more than restoring an old version. People have learned new methods, organizations reallocated resources, users formed expectations, and risk shifted. Dependency surface, synchronization cost, and rollback cost together form effective inertia.

Measurement units offer an old, clear example of synchronization cost. In 1999, Mars Climate Orbiter was lost during orbit insertion; investigators found that ground software supplied thruster impulse data in pound-force seconds while ground navigation software used the numbers as the newton-seconds required by the interface specification, a conversion error by a factor of 4.45. The resulting trajectory-estimation error brought the spacecraft too low over Mars (NASA 1999). A specification existed; the producers and users of the data failed to verify the same convention. Units, fields, timestamps, coordinate conventions, and state definitions form the smallest interfaces of a dependency surface; their number grows with complexity, and one mismatch can concentrate synchronization costs in integration or even operation, when rework is far more expensive than during design.

Rollback cost is easily underestimated when the old system no longer holds the full post-switch state. Core banking migration is a typical case. After TSB migrated to a new platform in 2018, major service failures affected digital channels, used actively by about 1.9 million customers, as well as branches and telephone banking; the following year the bank reported £330.2 million in post-migration costs, including compensation, extra resources, fraud, and lost income, partly offset by recovery from its IT provider (TSB Bank 2019b, 2019a; House of Commons Treasury Committee 2018). Once new transactions enter the new system, a return requires bringing the old system's transactions up to date, reconciling states, or making a second migration; restoring old files is insufficient. High-cost rollback systems therefore often begin alongside the main process, run in parallel, and then switch, specifying logs, state alignment, and compensation before cutover; switching changes the conditions for retreat rather than automatically making retreat impossible.

Coupling has a third consequence: one input may acquire authority to act on the system. The original Boeing 737 MAX Maneuvering Characteristics Augmentation System used one sensor for its angle-of-attack input; erroneous angle-of-attack data triggered repeated nose-down commands in the two crashes in 2018 and 2019, killing 346 people. The U.S. grounding lasted almost twenty months; return-to-service dates differed elsewhere (Joint Authorities Technical Review 2019). Key changes before return to service compared both angle-of-attack inputs, disabled relevant functions when disagreement exceeded a specified threshold, and limited commands and repeated action to preserve sufficient pilot control (Federal Aviation Administration 2020). Redundancy draws an authority boundary here: which inputs can directly authorize execution, and which require comparison and confirmation. Design can change dependency surface, synchronization cost, and rollback cost through structural isolation, parallel

operation, cross-checks, and reversibility.

Uneven Diffusion and Hidden Weight

A recurring pattern is that powerful-looking AI applications remain pilots while more ordinary ones spread rapidly. Document processing, initial customer-service screening, code completion, contract-clause extraction, local quality inspection, and image annotation are often easier to deploy. Their boundaries and inputs and outputs are clear, failures easier to isolate, and responsibility more manageable; their light structural positions help them spread. Entering core workflows is harder. Such AI can be more valuable while carrying a larger dependency surface: staff roles, interfaces, compliance, accountability, user trust, and long-term maintenance. Technical capability is only part of the problem; moving the entire dependency network is the harder part.

Enterprise choices about deployment positions show the same pattern. AI often gathers first at the edges of text and image work—clause extraction, ticket classification, code completion, inspection-image annotation—where inputs and outputs can be explicitly listed and downstream human review can catch errors. At positions such as credit decisions, diagnosis, dispatch, and pricing, the same model enters a chain of responsibility: its grounds, reviewers, and bearers of failure must be identified, potentially extending deployment from months to years. Position strongly affects speed and can be chosen before deployment, making it one of the few variables a technical team can decide for itself.

Structural weight often hides behind a smooth experience. A smoother user experience often rests on a more complex supporting system. One click may coordinate payments, authentication, risk models, logistics, customer service, data compliance, and platform rules. Users see a low-friction interface; the system maintains tight coupling. Technology can move friction out of sight and into the supporting structure without eliminating it (Star and Strauss 1999). A system's ease of transformation depends on how many hidden dependencies must hold together, beyond the interface shown to users.

Payment clearing and settlement give this a concrete form. In systems using end-of-day batch processing, reconciliation, error handling, and consolidation of funds occupy fixed windows, and the next business day depends on updated balances; this invisible window constrains parameter changes and upstream release schedules (Kahn and Roberds 2009). Other payment architectures change the schedule: FedNow's 2026 operating procedures specify continuous message processing, 24 hours a day throughout the year, illustrating how different settlement and liquidity arrangements can replace old windows (Federal Reserve Banks 2026). Smooth interfaces still coexist with timing constraints behind the scenes, while architecture changes where those constraints sit. The real cost of a change must include functions absent from the interface that nevertheless determine operating and release schedules.

The Dual Cost of Connection and Modularity

Connections have costs as well as benefits. They give a system coordination and inertia. Without them, loose parts cannot produce complex functions; too many make the system heavy and local adjustment difficult without disturbing the whole. Much modern efficiency rests on tight coupling: grids, financial networks, healthcare, supply chains, internet platforms, transport, and research all use dense connections to gain scale, speed, and coordination. Those connections also remove many local freedoms.

The system trades degrees of freedom for coordination. Good structure organizes connections appropriately. Critical parts need stable links while others remain replaceable; modules can be tightly coupled internally while retaining clear interfaces between them; efficiency gains must leave room for buffers, redundancy, and rollback. The location and strength of coupling matter more than making structure uniformly tighter or looser (Boccaletti et al. 2006). This is the purpose of modularity. It controls dependency surfaces, rather than merely breaking a system into pieces. It bounds the spread of local changes, isolates failures, and creates places to insert technology. Standard interfaces, clear inputs and outputs, responsibility boundaries, sandboxes, staged deployment, and rollback mechanisms all reduce effective inertia. They make structure manageable (Baldwin and Clark 2000). This also distinguishes structural weight from topological coupling: coupling explains why local change moves the whole, while weight explains why the affected whole is hard to move.

Industrial history offers a thorough demonstration of modularity. Before containers, cargo was handled piece by piece, loading and unloading a ship took weeks, and port charges formed a large share of transport costs; standardized steel boxes turned handling into crane lifts, with goods packed at the factory and left inside during transit, changing the structure of freight costs (Bernhofen et al. 2016). The steel box itself is no mystery; its system effect was to separate the dependency between cargo and ship, with the container handling adaptation. Standardization concentrates change in a separately replaceable layer: the box adapts, and ship and cargo evolve independently.

Modularity can also fail. Splitting a monolith into microservices adds interfaces as services multiply, and one business change may require several services to be released together; if data, transactions, and release rhythms remain coupled, splitting code moves coupling onto network interfaces and raises operating costs, producing the "distributed monolith." (Taibi et al. 2018) The test is whether dependency surfaces actually shrink: if several parts must still go live together, the split has moved coupling without changing structural weight.

Matching Intelligence to Structure

AI must confront this structural weight to enter real systems. Without clear inputs, outputs, applicability, failure handling, and accountability, a tool is hard to deploy at scale. It may occupy a position with large dependencies, high synchronization costs, and expensive rollback regardless of how intelligent its model is. A bounded setting with isolatable risks and reversible

failures can turn into real capability more readily, even without a grand ambition. Deployment tests the match between intelligence and structure as well as the level of intelligence.

The match can be specified more concretely. The same model in an assisting role has human-reviewed outputs and intercepted errors, giving it a small dependency surface; in a decision-making role, outputs enter execution directly and errors travel through dependencies, giving it a large surface. Success in the first position cannot simply be transferred to the second; a different structural position puts the same outputs into a different chain of consequences. Deployment choices require three questions: which actions will the output trigger, can those actions be isolated, and can the system continue operating after isolation?

AI can make a node smarter while leaving it slow to move if it pulls an entire network. Structural weight lies in the relationships that must move with the node. A point is easy to change; a point surrounded by relationships is hard. The heavier the structure, the more technology must ask about dependency surface, synchronization cost, and retreat after failure, as well as basic feasibility.

Dependency surface, synchronization cost, and rollback cost can each be measured, but their combination need not accumulate smoothly. When they rise together, locality is the first thing lost: one adjustment draws in a whole structure, and failure becomes harder to isolate where it begins. The question shifts from "how heavy?" to "what state?", which separate cost measurements cannot answer. The three variables also change how a medium responds: the same disturbance may be absorbed in a loosely coupled structure and amplified in a tightly coupled one. Describing that difference requires coordinates encompassing resources, time, and structure together.

2.5 A Generalized Viscosity Framework: The R–T–G Trade-off Model

Barriers in matter and energy, the time occupied by processes, and entanglement through connections jointly affect the cost of a state transition. Comparison requires fixing the task, outcome standard, system boundary, and period under study. This book uses (R, T, G) to describe these three constraints and a generalized viscosity coefficient η to record relative transition cost under a chosen accounting convention. The coordinates describe the sources of resistance; the coefficient describes the difficulty of transition.

From Three Observations to a Set of Parameters

The relationship between transition cost and the three constraint coordinates is written as:

$$\eta = f(R,\ T,\ G) \tag{2.20}$$

Here R is the resource barrier, T temporal rigidity, and G topological coupling. R compares task requirements with available resources; T compares process time with the available window; G

examines dependency surface, synchronization requirements, and difficulty of rollback. Each coordinate requires its own evidence. They may improve together or constrain one another; political, organizational, and distributive conditions affect which paths can be chosen.

Quantitative estimation requires a testable form for f. This book proposes an exponential working model that remains to be calibrated:

$$\eta = \eta_0 \exp(\alpha R + \beta T + \gamma G) \tag{2.21}$$

η is the dimensionless ratio of actual transition cost to reference cost for the same task under a fixed convention. Cost can mean money, total labor hours, or average passage time per qualifying result; a study must keep its convention consistent and report different costs separately where they cannot yet be converted. η_0 is the baseline coefficient at the reference values of the three coordinates. R, T, and G are normalized to an explicit reference state as zero; α, β, and γ describe changes in log cost per unit increase in their respective constraints, with nonnegative values constituting the working model's monotonicity assumption. The exponential form assumes approximately constant marginal effects within the range studied and does not separately include interactions; data and residual analysis determine its applicability.

The three coordinates, what they govern, and the costs of reducing them can first be set out in a table; later chapters will repeatedly return to these three rows.

Table 2.1 The three coordinates of η: the questions they address, sources of resistance, and where the costs of compression fall

Coordinate	Question	Source of resistance	Means of compression	Where costs fall
R Resource barriers	Can the change be driven?	Energy, materials, equipment, capital, land, supply chains, and maintenance	Increase investment and concentrate resources on production lines	Energy use, emissions, committed capital, and resource depletion
T Temporal rigidity	Can it happen in time?	Heat transfer, diffusion, hydration, growth, metabolism, certification, and negotiation	Parallelize, expedite work, and compress processes and public-notice periods	Risk shifts into the future; defects emerge after acceptance
G Topological coupling	Can change propagate?	Interfaces, standards, patterns of interests, path dependence, and allocation of responsibility	Add interfaces, redundancy, monitoring, and compliance structures	System complexity and maintenance burdens rise; local efficiency incurs system-wide costs

Table 2.1 lists ways to reduce constraints and the expenditures they may add. More equipment may shorten a schedule, more review may reduce failure risk, and redesigned interfaces

may reduce both coordination and maintenance costs. The outcome requires accounting for all relevant stages. Contracts, institutions, and participants' decision rights also determine who pays and who bears the risks.

Parameter Trade-offs

The three coordinates often move together in engineering. More equipment and parallel operations can ease time constraints while adding capital and maintenance costs; standardization can reduce both resource use and coordination difficulty. Comparing natural and artificial synthesis likewise requires placing sunlight, land, reactors, feedstocks, energy carriers, and control conditions within the same boundary. Changing the path changes the combination of resources, time, and structure; the particular process determines the net benefit.

The trade-off follows directly from (2.21): hold η fixed, take logarithms of both sides, then differentiate to obtain

$$\alpha \, \mathrm{d}R + \beta \, \mathrm{d}T + \gamma \, \mathrm{d}G = 0, \qquad \left.\frac{\mathrm{d}R}{\mathrm{d}T}\right|_{\eta, G} = -\frac{\beta}{\alpha} \tag{2.22}$$

The subscripts η, G denote fixed generalized viscosity and topology. Within the exponential model, with fixed coefficients and a nonzero resource weight, the slope along the constant-viscosity surface is $-\beta/\alpha$: reducing T by one unit increases R along that surface by β/α units. The ratio describes coordinate combinations at the same modeled cost. Whether engineering can move in that direction depends on paths allowed by the equipment, process, and budget.

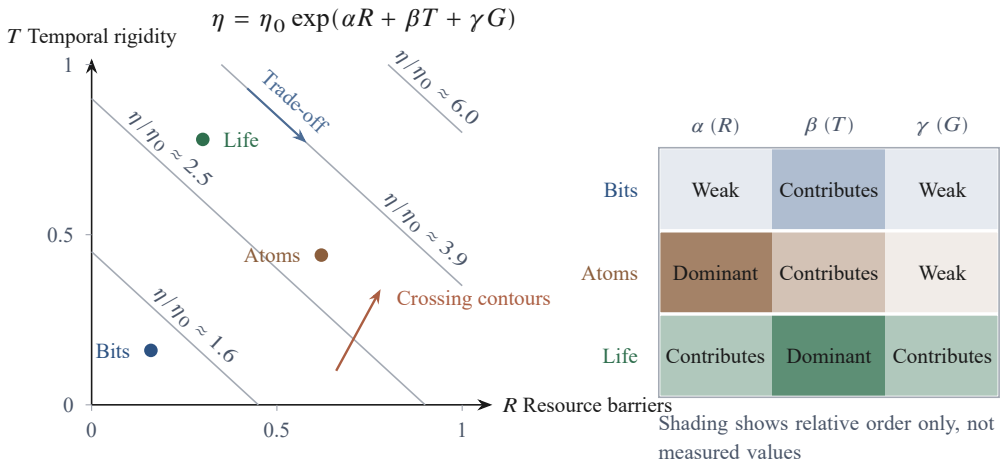
Coefficient values depend on the task, scales, and reference state. A different normalization changes the exchange ratio. Comparisons across systems require common reference conventions. Where marginal ratios vary with state, coefficients can be calibrated by region or interactions added to log cost, followed by testing the new form.

Taking more time to reduce risk is standard practice where consequences are serious. Drug development includes toxicology, Phase I, II, and III trials, and post-market observation; nuclear facilities undergo repeated reviews and redundant design; aircraft undergo full-life fatigue testing. These processes combine observation with verification in high-consequence systems, involving temporal rigidity as well as G: errors can propagate through connections and be difficult to isolate or reverse. Compressing T without replacement evidence or controls may merely move risk exposure into the future. A further trade-off is easy to overlook: adding structure to reduce local friction. Sensors, middleware, adapters, dispatch platforms, data interfaces, and audit modules can smooth the operating layer of a legacy system. If additions neither integrate nor replace old dependencies, G may rise as local friction falls and system entanglement grows.

Power transmission shows another configuration. Raising the voltage of a direct-current link can, under the relevant design conditions, increase capacity and reduce long-distance losses, at the cost of converter stations, insulation, and control systems. Once built, the corridor must operate in coordination with the grids at both ends. Capital expenditure, line losses,

and network dependencies are recombined; the net effect requires accounting across the whole transmission chain.

The three trade-offs share a property: they are strategic compromises rather than exact exchanges. What is given up and what is gained depend on the particular structure; there is no universal exchange rate. Benefits must therefore be calculated case by case rather than inferred from a general principle.



(a) Isoviscosity contours

(b) Relative ordering of the three weights (schematic)

Figure 2.7 The constraint space of resources, time, and topology. A single state transition can follow several paths, which may change the total cost as well as the combination of the three constraints. (a) A family of isoviscosity contours, with the same viscosity coefficient along each contour; the arrows show two kinds of movement, trading costs between two dimensions and crossing to a contour of higher viscosity. (b) The ordering of three weights across three media, setting out mechanistic expectations for testing; both the ordering and the magnitudes must be determined from task data.

Where Costs Come From

Cost transfers and physical lower bounds require separate accounts. Consider a closed reacting system exchanging heat and pressure–volume work with an environment at temperature T_0 and pressure p_0, with the initial reactants and final products at that temperature and pressure; the Gibbs free-energy difference gives the reversible minimum net input of non-expansion work. Entropy generation in the actual process adds to that requirement:

$$W_{\min} = \Delta G = G_{\mathrm{f}} - G_{\mathrm{i}}, \qquad W_{\mathrm{act}} = \Delta G + T_0\, S_{\mathrm{gen}}, \qquad S_{\mathrm{gen}} \geq 0 \tag{2.23}$$

G_{i} and G_{f} are the initial and final Gibbs free energies, while $W_{\min}$ and W_{act} are the reversible

and actual net non-expansion work inputs, all measured in joules; S_{gen} is entropy generation, and $T_0 S_{\text{gen}}$ the associated irreversible loss. For the same endpoints and environment, process improvements can reduce losses and bring actual expenditure closer to the lower bound. Total energy use must also include material handling, heat sources, and auxiliary equipment within the boundary.

Aluminum offers a direct comparison. Producing primary aluminum by electrolysis of alumina typically uses more than 13,000 kilowatt-hours per tonne; recycled aluminum bypasses reduction, and under the industry's common production-energy convention, recycling and remelting can save about 95% of primary-production energy. The scrap already contains metallic aluminum, leaving remelting and refining to be paid for, rather than reduction from the oxide (Quinkertz et al. 2001; International Aluminium Institute, n.d.). The routes begin in different states: one with oxide, the other with metal. The reference state is therefore set by the actual availability of materials, energy, and waste.

For given endpoints and environmental conditions, theoretical minimum work sets a process floor. Algorithms can screen processes, reduce trial and error, and help reduce irreversible losses. Changing feedstocks, products, or the environment requires recalculating the bound. **Intelligence can optimize a path without abolishing the physical conditions of a given state transition.**

Structural costs require separate measurement. Let C_{loc} be complexity at the operating stage and C_{rest} complexity elsewhere within the same boundary; with a consistent metric, the change in total complexity is

$$\Delta C_{\text{tot}} = \Delta C_{\text{loc}} + \Delta C_{\text{rest}} \tag{2.24}$$

All three quantities must use the same complexity metric, such as maintained interfaces, dependencies, or an explicitly defined weighted workload. Local simplification accompanied by increased background cost indicates transfer; a negative sum indicates lower total complexity. Modularity, standardization, and removal of redundant elements can produce the latter. Only after fixing the boundary and metric can transfer and net reduction be tested separately.

Uses and Limits of the Framework

Together, the parameters address a difficult question: why can a better technology still fail to move a system? The answer is often a mismatch of dimensions. A proposal improves its own performance—less energy, more speed, or greater accuracy—while entering a system already saturated in another dimension. A more fuel-efficient powertrain requiring a rebuilt fueling network meets G; a more precise therapy requiring ten years of follow-up meets T; a more efficient industry requiring a new grid meets R. The proposal is evaluated on a different scale from the system's ability to absorb it. In (2.21), adopting a technology changes all three coordinates:

$$\frac{\eta_{\text{new}}}{\eta_{\text{old}}} = \exp(\alpha\,\Delta R + \beta\,\Delta T + \gamma\,\Delta G), \qquad \Delta X = X_{\text{new}} - X_{\text{old}}, \quad X \in \{R, T, G\} \tag{2.25}$$

η_{old} and η_{new} use the same task, cost convention, reference, and coefficients. A negative weighted sum of the three increments gives lower modeled cost; a positive sum gives higher cost. Savings in one component's energy use may be offset by added infrastructure, switching, and verification. Accuracy and response gains must be assigned to the appropriate coordinates; savings and new expenditure within the same resource dimension must first be combined into a net change before entering the formula.

Two contrasting cases illustrate dimensional mismatch. Magnetic-confinement fusion binds R and T closely: confinement requires large magnets, powerful electrical supplies, and cryogenic systems; the Lawson condition connects density, temperature, and energy confinement time, with the idealized ignition threshold for deuterium–tritium fuel in a suitable temperature range taking the following order of magnitude:

$$n\,T\,\tau_E \;\gtrsim\; 3 \times 10^{21}\ \text{keV} \cdot \text{s} \cdot \text{m}^{-3} \tag{2.26}$$

Here n is plasma ion density in inverse cubic meters, T ion temperature in kiloelectronvolts, and τ_E energy confinement time in seconds. At fixed temperature and other conditions, doubling density can compensate for halving confinement time; changing temperature requires recalculating reaction rates, radiation, and energy balance, rather than extending a constant threshold to every temperature range. The triple product is a performance measure for specified fuel and energy-gain conditions; device design and control change the attainable parameter combinations (Lawson 1957; Wurzel and Hsu 2022). Messenger RNA vaccine platforms show a different pattern: the T of sequence design and candidate screening can shrink dramatically. In 2020, mRNA-1273's candidate sequence was selected within days of the pathogen sequence becoming public, while manufacturing, quality testing, and clinical evidence had separate processes; 63 days still elapsed from sequence selection to the first participant's dose (Moderna 2020). Immunogenicity, stability, batch consistency, and long-term safety each require verification; reductions in T must be recorded stage by stage. Both cases lead to the same conclusion: how much a technology reduces T depends on which stage it compresses and what remains connected downstream.

The framework also helps explain apparently cumbersome arrangements that endure. Redundancy, approvals, conservative parameters, human review, and staged releases reduce local efficiency while reducing G and tail risk. Where failure consequences cannot be rolled back, these losses are necessary expenditures. The framework's explanatory limits also need to be clear. It does not explain political processes, cultural differences, capital cycles, business models, or organizational capabilities. Deployment also depends on who benefits, who bears costs, who decides, budget duration, and capacity to execute. These factors lie outside the book's three coordinates. Generalized viscosity adds a view of resistance in the medium alongside these factors, rather than reducing all social phenomena to physical quantities.

The book's domain classifications use explicit criteria and relative grades; the exponential model proposes quantitative relationships requiring separate calibration. The two uses remain

distinct. Application begins by fixing the task and outcome standard, then identifying obstacles that can be removed, costs reduced or transferred, and functions taken over by other mechanisms. Time constraints, dependencies, and resource inputs must each be assessed by their particular role.

Distinguishing the three requires separate judgments about resources, time, and structure, each using different methods. So far the chapter provides a static ledger: it locates a state transition in three coordinates and identifies the binding dimension, without yet describing how the medium responds to change. The same potential difference produces different results when applied slowly or quickly; bits, atoms, and life respond differently to loading rates. Local bit operations can be extremely fast, atomic systems also respond at rates set by diffusion and heat transfer, and living systems can change their own structure with loading speed. The viscosity coefficient varies with the rate of change. To examine that variation, attention must move from a state's position to the behavior of its interfaces.

Further reading

Carliss Y. Baldwin and Kim B. Clark, *Design Rules: The Power of Modularity*.
Herbert A. Simon, *The Sciences of the Artificial*.
Charles Perrow, *Normal Accidents: Living with High-Risk Technologies*.
Nancy G. Leveson, *Engineering a Safer World: Systems Thinking Applied to Safety*.
Karl E. Weick and Kathleen M. Sutcliffe, *Managing the Unexpected*.
John D. Sterman, *Business Dynamics: Systems Thinking and Modeling for a Complex World*.
Melanie Mitchell, *Complexity: A Guided Tour*.
Vaclav Smil, *Energy and Civilization: A History*.

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Chapter 3

Interface Dynamics: Intelligence's Phase Transitions and the Rheology of Media

Chapter 2 separated viscosity into resource barriers, temporal rigidity, and topological coupling, describing a system's resistance to external action. In that framework, intelligence stood beside the medium: measuring resistance, finding low-resistance paths, or stopping before it. Once intelligence outputs control, executes actions, and intervenes in state transitions, its position changes. It becomes a source of disturbances injected into the medium, which immediately begins processing them. The standard for assessing intelligence must change with its position. The same model faces different constraints and error costs as a solver and a driver; the medium translates its output into diffusion, work, or feedback, or dissipates it at the interface.

3.1 Intelligence's Phase Transition: From Solver to Driver

In the preceding framework, intelligence selects paths: it reduces uncertainty before action, screens possibilities for better routes, and substitutes internal exploration for some external trial and error. This helps explain why intelligence evolved, but leaves open what happens when it enters execution. Selecting a path takes place in representational space; driving a state transition must occur in a medium. The turning point has two parts: how intelligence's role changes, and the new constraints that follow.

3.1.1 An Ontological Shift: From Solver to Driver

A path valid in representational space need not be valid in reality. Intelligence first appears as an ability to select paths. That ability appears across bits, atoms, and life. Nervous systems,

scientific models, engineering simulations, and AI all seek better paths before paying the costs of matter, energy, time, and failure. Finding a path does not complete it. A model can choose a direction; reality requires its execution.

Solver and Driver

Execution breaks a path into concrete stages. A symbolically valid proposal must become control signals, actions, energy inputs, material rearrangements, bodily responses, organizational coordination, and responsibility to enter the physical world. Moving from abstract representations into embodied reality changes intelligence's functional role. This is its phase transition from solver to driver. "Phase transition" is a functional analogy here: the role, constraints, error costs, and evaluation standards change together as intelligence enters a real system. Intelligence joins a closed loop capable of changing states, participates in transitions, and drives real processes with mass, inertia, feedback, losses, and responsibility.

A change of role changes the inventory of capabilities. In information space, intelligence is primarily a **solver**. It deals with symbols, logic, probability, patterns, and representations. A question produces an answer; a goal initiates a path search; data are compressed into structure. Its products at this level are representations that have yet to enter physical execution: text, images, code, models, predictions, and proposed decisions. An answer valid at the symbolic level completes much of the task.

In reality, intelligence has to do more than solve. It must become a driver. The driver makes information act on a medium with mass, inertia, friction, noise, delay, losses, and risk. Solvers output solutions; drivers output control capable of acting on a medium. Solvers ask whether an answer is valid; drivers ask whether it can pass through actuators, feedback, energy, and responsibility. Solvers handle possibilities; drivers bear consequences.

An early result in cybernetics gives this distinction a quantitative form. Ashby's 1956 law of requisite variety says that a regulator cannot suppress more varieties of disturbance than it can distinguish and respond to: only variety can destroy variety (Ashby 1956). Applied to solvers and drivers, the implication is direct. A solver's variety lies in the proposals it can express; a driver's in the actions it can apply; the medium's disturbances include friction, deformation, occlusion, delay, failure, and unexpected interactions. If the driver's degrees of freedom, bandwidth, and precision cannot distinguish and counter the relevant disturbances, some of even a powerful solver's capability falls beyond the system's reach, leaving residual disturbance in its outputs.

Many misjudgments about AI confuse solver capability with driver capability. Generating an architectural proposal does not build it; planning a traffic route does not change road capacity; simulating a robot movement does not ensure a stable grasp of a cup; proposing treatment does not establish that a drug has passed bodily, clinical, and long-term safety tests. A symbolic solution supplies a prerequisite for control rather than the state change itself. Taking the four changes together reveals the weight of this role: larger models, longer contexts, and stronger

reasoning continue to improve the solver without changing its role; the transition occurs when value depends on whether a state change happens, remains stable, and permits recovery.

Closed Loops and the Meaning of the Transition

Moving from solver to driver shifts the center of intelligence from answers to closed loops. A solver can give a one-off answer; a driver must continuously sense feedback, correct errors, and control risks. The world is a continually changing dynamical field. Intelligence must join the cycle: sense reality, generate control, observe feedback, correct action, and execute again. Without a closed loop it remains advice; within one it becomes a force in reality.

Closed loops also introduce risk. Feedback is delayed, execution has inertia, and systems have noise. AI entering forecasting, dispatch, trading, supply chains, traffic control, and management often shortens decision cycles and raises feedback frequency, compressing buffers previously supplied by human intervals, review, and waits. The question is whether adjustment frequency matches the response cycle of the system being regulated. A fast intelligent loop regulating a slow physical, organizational, or market system can overcorrect through delay, phase difference, and local misjudgment, even turning negative feedback positive. The bullwhip effect, mistaken feedback in organizational decisions, algorithmic amplification in markets, and delay-induced control oscillations all point to the need to understand response times, buffers, and stability boundaries. Faster judgment is a capability when matched to the system's frequency; without that match, it can manufacture oscillation faster (Lee et al. 1997b; Sterman 1989). The problem of closed loops predates cybernetics itself. During World War II, Norbert Wiener and Julian Bigelow developed a predictor for antiaircraft fire control that had to extrapolate an aircraft's future position from a stretch of observed trajectory; the work produced the 1942 report Extrapolation, Interpolation, and Smoothing of Stationary Time Series, one of the precursors of cybernetics (Wiener 1948). The engineering difficulties mattered more than the prediction formula itself: the predictor's output changed the gun's aim, which in turn changed the next round of observations and errors, drawing target maneuvers, observation noise, servo delays, and operator reactions into the same loop. Corrections that were too abrupt, or delays that were too long, could themselves excite oscillations and, in severe cases, disable the aiming mechanism. Cybernetics later discussed these machine failures alongside tremors in living organisms because both arose from phase misalignment within a closed loop. The counterintuitive point is that a system with greater predictive accuracy need not perform better once connected to a loop: its effect is determined by its phase and gain within the loop as well as its own accuracy.

This is what "phase transition" means. Entering reality changes intelligence's role, constraints, error costs, and assessment standards together. In symbolic space its main value lies in generation, search, and reasoning; in a dynamical field it lies in stable control, reduced dissipation, lower risk, and greater executability. The first centers on answers, the second on state transitions. The crucial AI-era judgment is whether smarter answers can become control that

can be executed, verified, maintained, and held responsible. **An optimum that cannot pass through its medium remains an elegant figure in symbolic space; a suboptimal solution that enters reality stably may be intelligence of real value.**

Intelligence originates in choosing paths before action; its phase transition occurs when a path must be executed. From solver to driver, it moves from rewritable representations into reality's friction, inertia, risk, and responsibility. It must drive changes of state as well as calculate possibilities. Understanding physical AI, robots, autonomous driving, and intelligent infrastructure begins with this transition. Mature intelligence recognizes that an abstract optimum will deform, bend, dissipate, and carry consequences on entering reality. Its highest form finds paths that can be executed stably in a viscous world, beyond disembodied calculation. The change of role says what intelligence becomes; it does not yet specify its limits.

3.1.2 Four Constraints on Executability and the Deflection of Paths

Confusing solver and driver capabilities also hides the changing form of constraints. In logical space the limits mainly concern complexity: can the problem be formalized, is the search space too large, are the model and data sufficient, and is the reasoning accurate enough? The central question is whether the result can be computed. Entering a dynamical field shifts those limits. The questions become whether the system can be moved, in time, stably, at sustainable cost, and with consequences someone can bear. Computational solvability gives way to executability in thermodynamics, control, and engineering.

Executability and Its Costs

Executability includes at least four real constraints: power density, medium bandwidth, feedback error, and safety boundaries. Power density asks whether enough energy can act in enough locations within enough time. Robot movements require motor torque, braking requires friction and distance, factory dispatch requires equipment response, and grid control requires capacity and stability. Medium bandwidth asks whether the system can absorb inputs at the required frequency and amplitude. A physical system has natural frequencies, propagation speeds, and relaxation times that constrain how frequently it can be rewritten. Feedback error asks whether correction can continue through noise, occlusion, wear, delay, chance events, and nonlinear responses. Safety boundaries ask whether errors can be bounded, stopped, isolated, and traced to responsibility. Cybernetics and modern feedback theory have long treated stability, bandwidth, delay, robustness, and feedback-error management as central problems of closed loops, beyond finding solutions alone (Wiener 1948; Åström and Murray 2008). The four constraints interact and trade off. Raising control frequency increases power and heat requirements; reducing response delay leaves less room for checks; compressing safety margins reduces tolerance of error. Engineering trade-offs occur among these four: faster actuators often cost inertia, energy, or precision; denser control loops often cost sensors, communication

bandwidth, and maintenance. A paper proposal that optimizes all four at once has usually left weight out of the account.

Error costs change character as intelligence enters reality. Errors in symbolic space are usually inexpensive. Text can be corrected, code rolled back, images regenerated, and model outputs prompted again. The error is primarily informational. It may waste time and attention without usually causing direct material damage.

In dynamical space, errors are costly. Failed grasps break objects, driving errors cause crashes, mistaken drug predictions can injure bodies, industrial control errors damage equipment, and medical misjudgments involve lives and responsibility. An incorrect answer becomes irreversible dissipation, physical damage, injury, and institutional consequences. Physical AI, embodied systems, autonomous driving, medical AI, and industrial control therefore cannot simply inherit a chatbot's iteration logic. Chatbots can improve through many low-cost interactions; real robot falls, collisions, and injuries cannot be treated as cost-free, unlimited trials. A language model can generate alternatives for human selection; a medical system cannot try unverified proposals directly on many people. Real error costs rule out inputs premised on high frequency, little responsibility, and arbitrary rollback.

The Deflection of Paths

Reality is a viscous environment full of dissipation. The mathematically shortest path is often not the feasible one. An algorithm's shortest, most efficient, resource-saving proposal still encounters friction, hysteresis, maintenance, risk, responsibility, coordination, and long-term stability. The real path undergoes **deflection**. Deflection means a planned path shifts as it passes through its medium, toward lower resistance, lower risk, affordable energy demand, and more stable feedback rather than the abstract optimum. Real systems often seek minimum total damping rather than minimum distance, sustainability along the entire route rather than a point optimum, and controllable failure rather than maximum local efficiency.

This explains why seemingly less advanced arrangements can prove more stable. Engineering retains apparently wasteful redundancy to resist failure; healthcare trades time for verification and responsibility; aviation, nuclear power, and financial risk control avoid adopting untested optimal algorithms lightly because local gains do not offset systemic failure risk. Apparent clumsiness can be the price of stability. Research on complex high-risk systems and system safety identifies redundancy, delay, approvals, and conservative strategies as structures that can limit accident propagation, preserve recovery, and assign responsibility (Perrow 1984; Leveson 2011). Entering reality is a dynamical coupling, beyond simple "application." An intelligent output joins a system with inertia, feedback, history, and responsibility. The system absorbs, filters, amplifies, distorts, or cancels it. A logistics algorithm meets vehicles, warehouses, drivers, weather, and fluctuating orders; a diagnostic model meets physicians' responsibility, patient trust, insurance, and legal risk; a robot controller meets sensor errors, force feedback, uncertain contact, and wear. Performance depends on the coupling between

intelligence and its medium as well as intelligence itself. An otherwise strong model loses much of its practical effectiveness without reliable feedback, actuator control, process compatibility, exception handling, and responsibility. A weaker model may have more real value if it integrates stably, lowers error costs, matches the medium's frequency, and preserves safety boundaries.

Embodied Intelligence and Closed Loops

This matters particularly for physical AI. Putting a large model in a body does not complete embodied intelligence. Bodies are not screens, and actions are not text outputs. Robots must continually perceive, localize, map, predict, control, make contact, avoid obstacles, and correct errors. They must perform under specific mechanical conditions as well as know what to do. A cup in language is an object; in reality it has weight, material, surface friction, a center of mass, and fragility. A door in an image has hinges, resistance, an opening angle, and environmental changes caused by pushing it open in reality. Behavior-based robotics and embodied cognition have long understood intelligence as continuous coupling among body, sensing, action, and environment, beyond central reasoning alone (Brooks 1991; Pfeifer and Bongard 2006). Vision–language–action models further illustrate this transition. In 2023, RT-2 combined vision–language models with robot trajectory data, transferring internet-scale knowledge into robot control and representing actions as tokens (Zitkovich et al. 2023). By July 2026, Google DeepMind's Gemini Robotics 2 placed whole-body control, dexterous manipulation, and multi-robot cooperation within one model family (Parada 2026). In Google's October 13 Gemini 4 Argon demonstration, a Spot robot practiced stepping on a trash-bin pedal and maintaining balance in a MuJoCo digital twin before operating the real pedal, checking its gripper position, and dropping in a soda can; simulation, cameras, and action interfaces brought model reasoning into a closed loop (Rao 2026). These cases illustrate a necessary remapping from language and vision into actuators, action spaces, sensor feedback, and real objects when a model moves from answers to actions, rather than establishing that embodied intelligence is solved. The difficulty is keeping understanding effective through contact, friction, delay, and error. Autonomous driving likewise involves more than recognizing roads. It embeds perception, prediction, and control in a tightly coupled traffic system.

Human drivers, pedestrians, roads, weather, construction, law, insurance, and responsibility jointly form the real environment. An algorithm can calculate an optimal strategy, but the vehicle acts within braking distances, limited visibility, complex strategic interactions, and consequences that cannot be rolled back. Every real control action carries consequences. Research on autonomous-driving testing and verification identifies open-road edge cases, nondeterministic algorithms, complex requirements, fail-operational behavior, and safety arguments as far harder than a single recognition task (Koopman and Wagner 2016). RAND's reliability research also shows that, under common confidence assumptions, establishing safety at or above human levels using road mileage alone can require enormous distances (Kalra and

Paddock 2016). Physical intelligence finds paths that pass stably through real media, beyond the mathematical shortest path. It must know when to be fast and when slow, when to seek efficiency and when preserve redundancy, when to experiment and when verify before acting, and when an optimum remains usable or fails under the medium's damping.

Once connected, a loop's effect depends on more than the loop itself. When intelligence reduces resistance in one place, other actors respond, rearranging flows, risks, metrics, and responsibility. The link between local reductions in resistance and global outcomes remains to be explained. Output also passes through a translation by the medium before becoming an outcome.

3.2 Interface Physics: The Rheological Spectrum of Media and Rate Mismatch

The preceding chapter's reachable space described states that a system can stably attain. A path describes a channel, without describing what intelligent input becomes as it passes through. The same open path produces different responses in different media: rapid diffusion, a requirement for work, or a rebound. The medium determines whether input becomes state change.

A medium translates intelligence into change. This is easily overlooked when AI enters real systems. The familiar claim that "AI will change the world" treats a world that is far from uniform. Text, code, images, machines, roads, drugs, cells, bodies, and ecosystems receive intelligent inputs differently. The same model output can spread rapidly through text, must become force and motion in a robot, and can trigger metabolism, immunity, compensation, adaptation, or rebound in life. AI's real effectiveness therefore depends on the medium absorbing its output as well as the model's strength: stable translation turns intelligence into useful work; mismatch turns it into noise, heat, wear, oscillation, verification burdens, or resistance by the system. The **medium response spectrum** describes how different media turn intelligent inputs into state changes. Bits translate intelligence into diffusion, atoms into work, and life into feedback. These are three absorption mechanisms: bits involve copying, combination, and propagation; atoms involve energy, inertia, and dissipation; life involves metabolism, regulation, and adaptation. One logic of intelligent acceleration applied to all three creates illusions.

3.2.1 Three Response Spectra: Bits Diffuse, Atoms Do Work, Life Feeds Back

Diffusion in the Bit Medium

The basic response of the bit medium is diffusion. Intelligent input readily becomes new information structures: text, images, code, model outputs, design drafts, and search results. They can be copied, edited, distributed, combined, and regenerated quickly. Bit-domain state

changes primarily rearrange symbols rather than material positions. AI therefore expands most readily in bits, generating abundant content at very low marginal cost and compressing tasks of expression, search, rewriting, and combination that previously took people substantial time.

Diffusion, however, is not proof. Low bit viscosity occurs mainly in lateral propagation rather than vertical corroboration. Lateral propagation copies more, generates more, distributes faster, and combines more densely; vertical corroboration demands truth, reliability, provenance, explanation, and responsibility. AI readily amplifies the first without automatically reducing the burden of the second. Generated text still needs fact-checking, generated images provenance, generated code security and maintenance audits, and circulating opinions grounds for public trust.

The central tension in bits is between generation and establishing truth. Possibilities can expand cheaply without a cheap guarantee of truth. Cheaper generation makes credible results scarcer; more content increases screening; smoother expression makes judgment more important. Diffusion creates new pressure for corroboration, borne by those who verify rather than those who generate.

Bits also need reality as an anchor. Model intelligence compresses real language, images, experiments, code, behavior, and feedback, rather than growing from a purely logical closed loop. The lightness of bit space comes from turning reality into replicable information structures, whose replicability still depends on real sources. An information system consuming more of its own generated content without fresh contact with reality and external feedback can become smoother and more closed (Shumailov et al. 2024). It can gain consistency without truth, or fluency while losing variance and external calibration.

Bits hide viscosity in two places: physical infrastructure and corroboration against reality. The first includes chips, electricity, cooling, data centers, and networks; the second real data, fact-checking, human feedback, provenance, and responsibility. Users see cheap diffusion; the system pays infrastructure and verification costs. Faster bit diffusion makes calibration against reality more necessary. When the output must change material positions rather than symbols, the response mechanism changes.

Work in the Atomic Medium

The atomic medium responds through work. A generated information structure does not automatically change atoms: control signals must become energy flows, planned paths motion, and designs materials, processing, assembly, testing, and maintenance. Symbolic elegance must survive force, heat, friction, inertia, material strength, and spatial constraints. Translating intelligent input into physical action requires energy, incurs dissipation, and takes time. Handling needs force, manufacture processes, cooling heat exchange, motion control, construction materials, robots motors and joints, and grids capacity and stability. AI can optimize paths without canceling mass, reduce trial and error without canceling work, improve dispatch without canceling capacity limits, and strengthen control without canceling friction, wear, heat,

and failure.

Intelligence drives the atomic medium rather than persuading it. Bit diffusion can approach instantaneous response, while atoms require energy transfer and structural change. A smarter robot policy still needs actuator torque; better logistics still needs vehicles, roads, warehouses, and handling; more precise chips still need cleanrooms, lithography, materials, defect control, and cooling. Intelligence becomes control over physical processes. The atomic medium retains dissipation: some input becomes heat, wear, noise, error, and maintenance rather than useful work. Greater control precision often requires additional sensing and actuation; with energy use and losses per operation fixed, higher speed increases power and cooling pressure; added complexity not absorbed by modularity increases calibration and maintenance. Intelligence must keep the accounts of energy, power, heat, materials, and failure.

The atomic medium also has response bandwidth. Machines have inertia, materials fatigue, heat diffusion takes time, actuators have response limits, grids stability constraints, and traffic braking distances and reaction delays; physical states cannot change at unlimited frequency. High-frequency AI recommendations need not be absorbable at the same frequency. Excessively fast inputs can create oscillation, overcorrection, heat, wear, and safety risks instead of additional useful work.

Intelligence in the atomic medium needs an appropriate way to drive matter. Physical intelligence must know the power, frequency, error, and feedback delay the medium can sustain. It matches output to physical response, enters a stable feedback loop, and seeks executability, low dissipation, maintainability, and recovery rather than output speed and symbolic optimality alone. Work also exposes intelligence to physical losses: a system unable to manage heat, wear, error, and maintenance remains a demonstration regardless of model strength. The atomic medium brings answers down to actions, proposals down to processes, and optima down to sustained operation. Nonliving systems have feedback too. The translation enters the biological layer when the medium processes input through metabolism, self-maintenance, and adaptation.

Feedback in the Living Medium

The living medium responds through feedback. Life is matter with boundaries, metabolism, defenses, repair, regulation, memory, and adaptation; it interprets external action as well as bearing it. Drugs enter metabolic networks, immune systems, tissue environments, and behavioral contexts; gene editing perturbs development, expression regulation, cellular environments, and ecological relationships; treatment acts together with the body's own feedback systems.

The living medium treats intelligent input as a disturbance rather than passively executing it. It absorbs some, cancels some, amplifies some, delays some, and rebounds against some. Stronger, faster, or more forceful intervention can provoke greater resistance rather than faster change. Immunity, inflammation, drug resistance, compensation, endocrine coun-

terregulation, and ecological rebound show complex adaptive systems inclined toward self-maintenance. Rheology includes media that flow under gentle, slow disturbance but harden under rapid, strong disturbance (Brown and Jaeger 2014). Life can show analogous responses: mild, infrequent, adaptable intervention may be absorbed flexibly, while frequent, forceful, cross-scale intervention can abruptly raise resistance. Multiple layers of feedback can lose coordination when the driving timescale becomes shorter than the relevant relaxation time, provoking defenses, compensation, or instability.

This explains why life cannot iterate like software. Software permits many versions and reversible code changes; human intervention has limited trials and difficult rollback, physiological feedback takes time despite rapid model updates, and long-term side effects can lag improvements in short-term metrics. Biological feedback often conceals consequences in time: an intervention effective today may be canceled by compensation months later, or an improved metric reveal system costs years later. Nonlinearity also separates local targets from whole-system goals. Lowering one indicator need not improve bodily health, activating one pathway need not produce global benefit, suppressing one response can release another risk, and editing one gene can involve development, expression, and ecology. Every intervention point is a network node; more precise entry points do not stop biological feedback.

AI's effects in the life sciences will therefore be uneven. It can quickly expand candidates by finding targets, predicting structures, generating molecules, designing experiments, and recognizing patterns. The living medium does not validate them at the same speed: abundant molecules do not become abundant approved drugs, quickly designed gene edits do not establish safety quickly, and quickly trained diagnostics do not remove clinical responsibility. AI accelerates discovery without canceling biological feedback. Life translates intelligence into feedback and sends it into long-term verification. Atomic failures often appear as wear, fracture, heat, and instability; biological failures can appear as compensation, tolerance, delayed side effects, population differences, behavioral change, and ecological consequences. Beyond physical losses, the biological account must include adaptation, self-maintenance, and their responses to intervention.

Three representative models describe responses: first-order tracking, rate-limited progress after a startup delay, and underdamped feedback. Let $y(t)$ be the response of a selected outcome and y_∞ its steady-state value; the models are

$$y_b(t) = y_\infty(1 - e^{-t/\tau_b}), \qquad \tau_b \to 0^+ \tag{3.1}$$

$$y_a(t) = y_\infty \min\left\{\frac{(t - t_d)_+}{\tau_a},\ 1\right\}, \qquad t_d > 0 \tag{3.2}$$

$$y_l(t) = y_\infty\left[1 - e^{-\zeta\omega t}\left(\cos\omega_d t + \frac{\zeta}{\sqrt{1-\zeta^2}}\sin\omega_d t\right)\right], \qquad 0 < \zeta < 1 \tag{3.3}$$

Time parameters use a common unit, and responses are normalized to the same reference. In

the first expression τ_b is a first-order time constant; in the second t_d is startup delay and τ_a the duration of rate-limited progress; in the third ζ is damping ratio, ω natural angular frequency, and $\omega_d = \omega\sqrt{1-\zeta^2}$. For $0 < \zeta < 1$, the third response overshoots before gradually returning to steady state. The forms describe rapid tracking, capacity limits, and feedback-driven decline in relevant tasks; response data determine the form and parameters selected.

Analyzing the medium supplies mechanistic clues for model selection. Digital distribution often responds quickly, manufacturing faces startup and throughput limits, and biological intervention often involves compensation and hysteresis. One medium can also exhibit several responses: information networks congest, mechanical systems overshoot, and living processes approach a state monotonically. The comparison concerns particular channels, rather than curves assigned in advance by a medium's name.

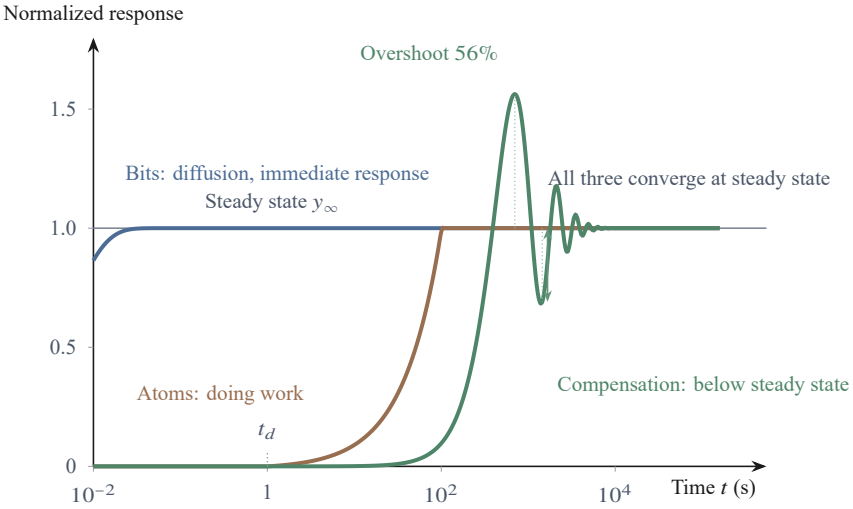


Figure 3.1 Three typical response models. The horizontal axis is logarithmic time, the vertical axis normalized response, and the dashed line the steady state y_∞. The illustrative parameters produce rapid tracking, delay-limited response, and underdamped overshoot, respectively. The form and parameters of a particular channel must be measured; one medium can exhibit several kinds of response.

3.2.2 Rate Mismatch and Interface Dissipation

Three Clocks, Different Rates

A subtler problem lies among the three response spectra: the same logic of speed cannot drive them all. Bits can generate at high frequency, atoms require power matching, and life requires relaxation time. Intelligence crosses interfaces between rates as well as between media. Bits can compute, update, and propagate extremely quickly; atoms have inertia, heat transfer,

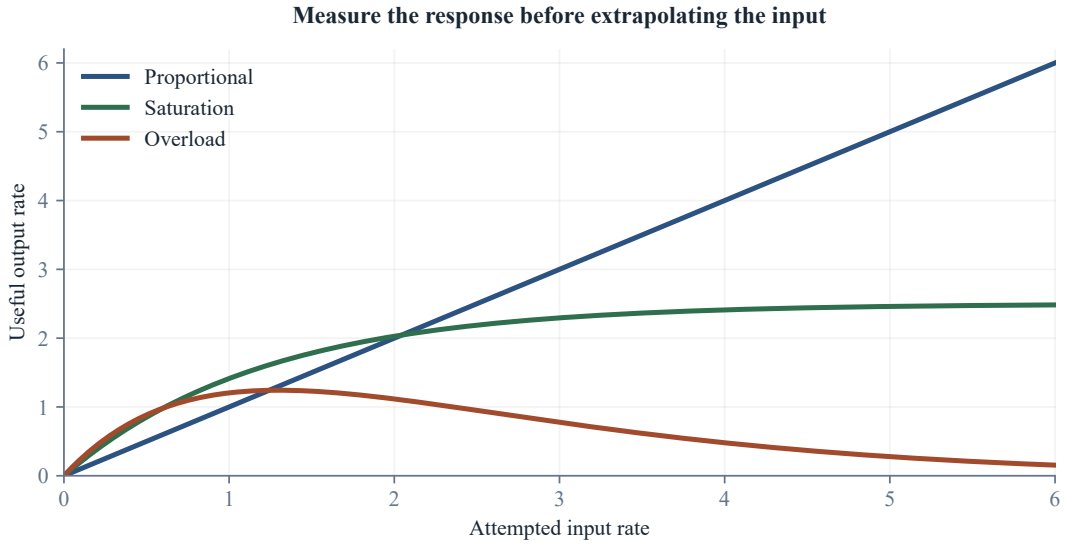


Figure 3.2 Input and effective output may be approximately proportional, gradually saturate, or decline after overload. The curves illustrate mechanisms; the form and parameters of a particular system must be measured.

friction, fatigue, actuator response, and safety boundaries; life adds metabolic cycles, immunity, tissue repair, behavioral adaptation, and long-term feedback. A faster input layer gives less reason to assume its absorbing layer will accelerate in step.

This resembles impedance mismatch. In engineering, mismatched source and load impedances can leave energy reflected, dissipated, or oscillating rather than effectively transferred to the load. Its standard expression is the reflection coefficient

$$\Gamma = \frac{Z_L - Z_S}{Z_L + Z_S}, \qquad \frac{P_{\text{refl}}}{P_{\text{in}}} = |\Gamma|^2 \tag{3.4}$$

For a lossless transmission line with a terminal load, Z_S denotes a positive real characteristic impedance and Z_L the load impedance, both in ohms. Γ is the voltage reflection coefficient, and reflected power divided by incident power is $|\Gamma|^2$; for a nonnegative real load, Γ lies between -1 and 1. Impedance matching eliminates reflection. Output frequency, review throughput, and execution capacity in intelligent systems have their own units and require separate measurement; queues, dropped outputs, overcorrection, and structural damage are directly observable consequences of mismatch.

Mismatch can be expressed as the ratio of input rate to absorption capacity. Let $r = v_{\text{in}}/v_{\text{abs}}$; measure effective progress in units of absorption capacity, with a damage term for degradation under overload:

$$r = \frac{v_{\text{in}}}{v_{\text{abs}}}, \qquad u(r) = \begin{cases} r, & 0 < r \le 1 \\ 1 - \kappa\,(r-1)^m, & r > 1 \end{cases} \tag{3.5}$$

v_{in} and v_{abs} use the same unit, making r dimensionless. u is net progress rate divided by absorption capacity. For $r \leq 1$, all input is absorbed and $u = r$ indicates unused capacity; for $r > 1$, the model uses $\kappa(r-1)^m$ for net loss from overload. $\kappa \geq 0$ and $m > 0$ must be estimated from observation; $u < 0$ means damage exceeds current gains. A system that only queues without degrading requires a saturated-throughput model instead. Information networks, machines, and living systems can all suffer overload damage; particular mechanisms and data must establish the differences.

Engineering has long used layered conversion. Primary grid frequency response comes automatically from local controls within seconds; secondary control uses automatic generation control or manual dispatch over minutes to restore frequency and power balance, while reserve deployment and unit commitment span minutes, hours, and days (North American Electric Reliability Corporation 2024). Longer timescales allow larger changes in the resource mix but involve scheduling, commitments, and switching costs; making a fast layer take over slow-layer objectives can create control conflicts, while faults requiring immediate action miss their window if they must await a slow layer. Data systems have a similar arrangement: writes enter a log before merging into primary storage, while reads need not await all background maintenance. These structures convert rates between source and absorbing system, spending resources to avoid congestion, instability, or stoppage from mismatch.

Mature AI systems convert rates across media rather than pushing every medium toward the highest frequency. Intelligence improves the match between source and load as well as increasing output. Failure can arise at the interface: too much output to screen, updates too fast for workflows, controls dense enough to cause physical oscillation, intervention strong enough to provoke biological rebound, or metrics frequent enough to consume organizational effort. **Intelligence has not vanished; it has been dissipated at the interface.**

Adapting to the Medium

A mature view of intelligence asks how the medium translates output as well as whether the model can produce a better result. Is it stable, or does it oscillate? Does it conserve resources, or create waste heat and maintenance? Is it controllable, or does it provoke biological and systemic rebound? Can it be verified, or does it only look valid in representational space? The medium's response helps define intelligence's boundaries. The response spectrum corrects an image of intelligence without a medium: AI hands outputs to bits that diffuse them, atoms that demand work, and life that feeds back; every translation costs, every response has limits, and every absorption has forms of failure.

Effective intelligent systems must adapt to their medium. Bits demand truth and trust; atoms power, dissipation, and maintenance; life feedback, relaxation, and long-term consequences. Generation must anticipate what will absorb it, optimization what residue it will leave, and acceleration when to reduce frequency, buffer, and verify. The opening proposition now becomes concrete: the medium translates intelligence into change, and AI's real effectiveness

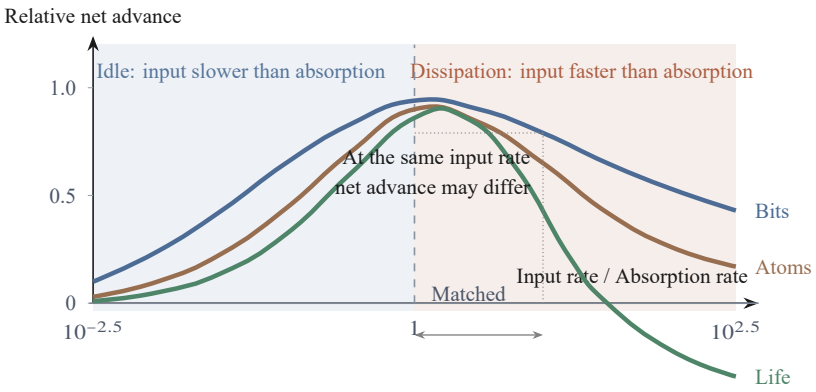


Figure 3.3 Rate mismatch: both sides waste resources, in different ways. The horizontal axis is the ratio of input rate to absorption rate; mismatch increases with distance from the midpoint. The three curves correspond to the three media, with progressively narrower usable bandwidths; the vertical axis shows relative net advance, with negative values indicating that overload damage exceeds effective advance; the curves use different illustrative parameters.

depends on stable, economical, controllable, and verifiable translation.

Network protocols offer another version of the same experience. During the congestion collapse of October 1986, throughput on a link from Lawrence Berkeley Laboratory to the University of California, Berkeley fell from 32 kilobits per second to 40 bits per second. Van Jacobson and Michael Karels's 1988 congestion control constrained senders through acknowledgment pacing, a congestion window, and retransmission backoff; after a timeout, the congestion window restarted from a very small value, with the slow-start threshold reduced to half the former window, followed by controlled growth (Jacobson and Karels 1988). The source thus followed the absorption capacity of the whole path. A sender advancing only at its own output capacity fills intermediate queues, causes drops, and adds retransmissions that deepen the load, reducing effective throughput. Protocol discipline writes the medium's bandwidth and feedback into the source's control law.

A medium already has a history before intelligence adapts to it. Its construction was shaped by human bodies, existing processes, and institutions of the time, rather than intelligent input formats. The central problem is often whether translation can occur without rebuilding that medium's skeleton; once the skeleton must change, the cost returns to the terms of the atomic world.

3.3 Evolutionary Strategy: Cybernetic Parasitism and the Complexity Ratchet

The reality intelligence seeks to change was built before intelligence arrived. Roads, buildings, hospitals, and factories have existing structures embedded in law, insurance, habits, and responsibility. Models work through those interfaces, making deployment cost depend on the extent of change required. Replacing a component is repair; replacing a load-bearing structure is reconstruction. Once local capability is sufficient, adapting to the old structure must be compared with rebuilding the environment on total cost.

3.3.1 The Overlay Strategy and the Three Costs of Reconstruction

Three Costs of Reconstruction

In theory, the cleanest route is to rebuild the underlying structure. Redesign roads, vehicles, signals, insurance, and traffic rules for autonomous driving; buildings, tools, warehouses, homes, and factories for robots; hospital workflows, records, accountability, and payment for intelligent healthcare. The physical world would better suit machine perception, action, and decisions, with more regular interfaces, less environmental uncertainty during operation, and more stable feedback. But this route is expensive.

Rebuilding the atomic world must overcome three costs. First, resources: roads, factories, hospitals, ports, grids, homes, and utilities embody vast materials, capital, space, and labor; demolition and reconstruction discard existing material structures and pay the energy bill again. Second, temporal rigidity: infrastructure has life cycles, while construction, approvals, migration, coordination, and behavioral change cannot update instantly like software versions. Third, topology: roads connect insurance, law, logistics, real estate, work, and daily life; hospitals connect physicians' responsibility, patient trust, insurance payment, and legal boundaries. Changing the base pulls on the entire system.

Annual production of cement, steel, and equipment accumulates into buildings, roads, factories, and networks. USGS and World Steel Association reports for 2026 still place global cement output in the several-billion-tonne range and crude steel near two billion tonnes (U.S. Geological Survey 2026; World Steel Association 2026). Transforming that stock depends on lifetimes, replacement rates, construction capacity, and neighboring systems; one year's production cannot be equated with stock available for simultaneous reconstruction.

These are the preceding chapter's three constraints expressed in the built environment. Resource barriers concern material, energy, and space already paid for; temporal rigidity concerns infrastructure processes that cannot be compressed at will; coupling concerns the connections already formed among roads, law, insurance, and work. A built object's viscosity enters its structure at construction. Assessing an intelligent upgrade requires counting how much of that

old structure must be paid for again, alongside its symbolic savings.

The Overlay Strategy

Civilization therefore tends to overlay before rebuilding. An **overlay strategy** adds a bit-based control layer over an old physical skeleton without completely changing it. Keep the roads and add navigation, cameras, radar, high-definition maps, and cloud training; keep the factory and add sensors, predictive maintenance, digital twins, and dispatch platforms; keep the hospital and add image recognition, record summaries, triage, and review; keep the city and add cameras, platforms, algorithms, data interfaces, and "smart governance." A digital film covers the heavy atomic skeleton.

This is a form of cybernetic parasitism. Here "parasitism" describes a structural relationship and an evolutionary strategy, without moral disparagement: the new intelligent layer neither exists independently nor immediately replaces the old layer, but attaches itself to an existing physical system, using its established space, energy, equipment, institutions, and human networks to gain a foothold for action in reality and to change the old system's behavior. The bit layer resembles a layer of neural tissue covering the old world's skeleton; it leaves the bones in place while trying to make them move faster, more accurately, and more readily under coordinated control.

The strategy is widespread because it follows least-cost logic. A local overlay is lighter than rebuilding the world: sensors cost less than roads, dispatch software less than a rebuilt factory, recognition models less than a rewritten healthcare system, and better robot perception and control less than redesigned living spaces. The overlay is a second-best route around a viscous base, finding room to act above its weight rather than removing it. In the short term it works: smoother interfaces, greater management visibility, more immediate dispatch, better risk monitoring, and real local efficiency gains. Navigation, predictive maintenance, assisted diagnosis, monitoring platforms, and automated warehouse sorting make old systems appear lighter by reorganizing processes once dependent on human experience and local judgment.

Overlays predate the digital age. Between the late nineteenth and early twentieth centuries, factories shifted from centralized steam transmission toward distributed electric motors; diffusion had begun in the 1890s, but major productivity gains waited for layouts, transmission, machine placement, and production organization to be rearranged around distributed power, a lag of two or three decades (David 1990). Containerization likewise required docks, cranes, truck chassis, railcars, warehouses, union rules, and customs systems to change after the modern era's landmark Ideal X voyage in 1956; entire handling areas moved, and widespread adoption followed in the 1970s (Bernhofen et al. 2016). In both cases the old skeleton's ability to change was often the bottleneck for the new layer.

Recent cases make the limits of overlay gains clearer. China's Ministry of Transport reported in 2019 that pilots in Jiangsu, Shandong, Sichuan, and Chongqing cut normal average provincial-border passage times from 15 seconds to 2 for cars and from 29 to 3 for trucks; on

January 1, 2020, 487 provincial-border toll stations in 29 connected provinces were abolished (Ministry of Transport of the People's Republic of China 2019, 2020). The Pennsylvania Turnpike introduced open-road electronic tolling east of Reading and on the Northeast Extension in January 2025, allowing vehicles to pass gantries at highway speed; in May 2026, equipment was still being installed in the west for a planned systemwide conversion in 2027 (Pennsylvania Turnpike Commission 2026). The bit layer changed toll collection, while lane, ramp, mainline, and service-area capacities still required separate accounting. Peak flows still pass through physical channels; removing the toll-point constraint can expose mainline or interchange bottlenecks. The gains are real, and the whole corridor's constraints determine how far they extend.

The old world is heavy because historical decisions have become physical dimensions. After 1,435-millimeter standard gauge emerged in the nineteenth century, the tunnels, bridges, platforms, and vehicles adopting it took shape around it. International container standards formed in the 1960s into major series including 20- and 40-foot lengths, shaping cranes, yards, truck chassis, and railcars. Stair-riser limits, doorway sizes, outlet shapes, AC frequency, and keyboard layouts reflect the bodies, processes, and industry compromises of their time. They remain because changing them incurs atomic-world costs shared across the entire chain, rather than because they are optimal.

Overlays also spread because they put costs in another ledger. Sensors, cameras, and dispatch platforms appear in current capital and operating budgets; the old skeleton's failure to accommodate machine perception and action appears in no explicit cost item. One is expenditure; the other an implicit debt. An old system's digital upgrade delivers visible gains quickly. The costs emerge as overlays become dependent on one another, by which time they are necessary for operation.

3.3.2 The Complexity Ratchet, Compatibility Tax, and Deployment Tax

The Complexity Ratchet

An overlay requires sensing, data, communication, permissions, monitoring, maintenance, and exception handling. Interfaces that require supporting changes in old structures can add coupling; when structure needs further structure to keep running, a **complexity ratchet** forms. Modularity, standardization, and retirement of old interfaces can also lower net complexity; architecture and operating records establish the direction.

A recurrence expresses the process:

$$C_{n+1} = (1 + \alpha)\, C_n + c \tag{3.6}$$

Here C_n is complexity counted consistently, c the direct addition from a new layer, and α the proportion added to existing structure by that layer. With fixed $c > 0$ and $\alpha > 0$, the recurrence

grows exponentially; with $\alpha = 0$, linearly. It describes one case of accumulating dependencies; layer count alone does not establish that a real system follows a fixed proportion.

Expanding (3.6) reveals the ratchet's form. For initial complexity C_0, after n layers,

$$C_n = (1 + \alpha)^n C_0 + \frac{c}{\alpha}[(1 + \alpha)^n - 1], \qquad C_n^{\min} = \beta\, C_n, \qquad 0 < \beta < 1 \tag{3.7}$$

The first expression solves the recurrence for $\alpha > 0$; for $\alpha = 0$, its limit is $C_n = C_0 + nc$. The second records retained complexity after removal as β times the current level, with β estimated from the particular exit plan. Existing dependencies can be transferred, rebuilt, or removed, so $C_n^{\min}$ is a reachable level for a scenario rather than a universal physical floor.

The difference is a lock-in band; its accumulation is the compatibility tax payable to unload the first n layers:

$$K_n = \sum_{k=1}^{n} (C_k - C_k^{\min}) = (1 - \beta) \sum_{k=1}^{n} C_k \tag{3.8}$$

K_n aggregates successive differences into a compatibility-burden index, with units set by the complexity convention. Shared interfaces can make direct addition double-count; converting the index into exit expenditure requires separate estimates for removal, migration, verification, and replacement support. Concentrated reconstruction can also lower actual cost; the sum alone does not establish an inevitably exponential exit bill.

Autonomous driving is a characteristic overlay dilemma (Kalra and Paddock 2016). Roads were not designed for machine vision, traffic rules for algorithmic responsibility, or insurance for driverless actors, and people do not follow machine protocols completely. Simplifying autonomy through closed roads, separated traffic, common vehicle–road communication, standardized participants, and clear responsibility would rebuild urban transport at great cost. The overlay route instead puts more sensors on cars, asks algorithms to handle long tails of behavior, and adds safety drivers, remote intervention, regulatory sandboxes, insurance arrangements, and data logs. The environment stays complex, so vehicles acquire complexity; insufficiently standardized roads leave algorithms more uncertainty. Replacing a driver also transfers road-reconstruction costs into onboard intelligence, redundant perception, cloud training, and responsibility systems. Mixed traffic adds a transitional difficulty: human driving has established interaction rules, highly standardized machine traffic might form stable protocols, and the mixed stage must accommodate both behaviors alongside old roads, law, nonstandard actions, insurance, and expectations, creating a mixed region of high uncertainty. Stronger intelligence need not simplify the system; much compute pays for the old environment's disorder.

Compatibility and Deployment Taxes

Embodied intelligence faces the same cybernetic parasitism, across a wider range of spaces. Homes, hospitals, restaurants, warehouses, shopping malls, and streets were not designed for

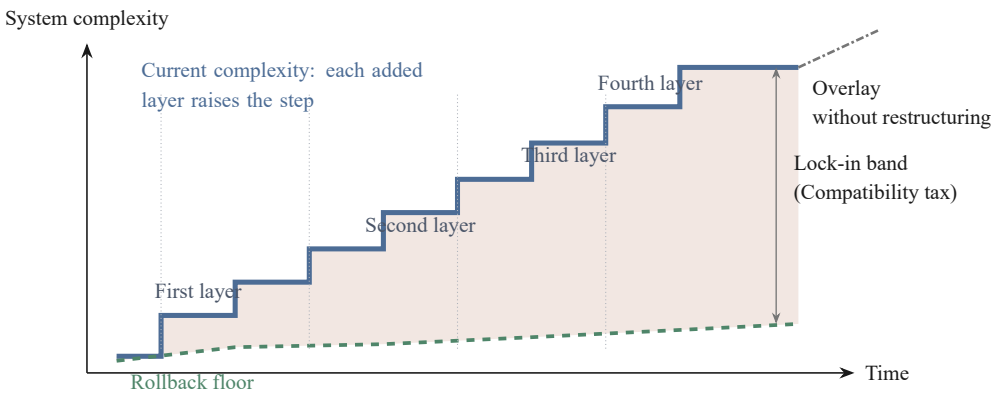


Figure 3.4 The complexity ratchet. The gap between the two lines is the lock-in band: returning to the previous generation requires first paying the compatibility tax within that band. Both boundaries rise, but at different rates, progressively widening the gap; the more steps, the wider the band.

robot bodies: door handles assume human hands, stairs assume two legs, kitchens assume coordination between eyes and hands, tools assume human touch and common sense, and household objects assume the human ability to handle ambiguity. To enter these spaces, robots must pay a compatibility tax: more degrees of freedom, more complex sensors, more demanding real-time control, stronger safety redundancy, and more expensive maintenance systems. Industrial robots succeed because they are usually placed in low-entropy environments with fixed positions, repetitive movements, standardized materials, separated workspaces, and clearly specified tasks; general embodied intelligence is difficult because it must enter high-entropy human environments, where object positions vary, task goals are ambiguous, layouts differ, human behavior is unpredictable, and contact feedback is complex. Installing a large model in a robot does not automatically eliminate friction, collisions, errors, occlusion, material differences, or liability risks.

This cost distribution has an earlier formulation. Hans Moravec observed in 1988 that adult-level performance on intelligence tests or chess was relatively easy for computers, while a one-year-old's perception and mobility were extremely difficult (Moravec 1988). More than thirty years later, the distinction persists in embodied intelligence: explicit reasoning can be formalized, labeled, and searched, while grasping, walking, and obstacle avoidance depend on tacit regulation learned through perception, action, and feedback; simulation supplies some training, while real contact still calibrates and verifies, concentrating costs in adaptation that is difficult to copy. The compatibility tax pays for regulation that human bodies acquired through long evolution, beyond the robot's apparent intelligence.

The "deployment tax" appears when a model touches reality. Every world model compresses reality and leaves residuals; simulation cannot exhaust real noise, edge cases, material changes, and human contingencies. Action changes what is modeled: lifting a cup changes its position, pushing a door changes its state, and entering a room can change people's behavior. Physical AI continually corrects within an environment it perturbs. An overlay's difficulty includes both model capability and an old world not rebuilt for low-cost intelligent action. AI pays compatibility, residual, and maintenance taxes in that world. Local bit-layer efficiency carries atomic weight, disorder, and consequences that cannot be rolled back.

Between Overlay and Reconstruction

Overlays can be mistaken for reconstruction. They extend old systems' lives, improve local efficiency, and add control to old infrastructure, but cannot accumulate without limit. Does each layer remove more friction than complexity it adds? Do its labor savings cover its maintenance burden? Do its efficiency gains justify its added fragility? A system that only overlays without rebuilding gradually forms an efficient but fragile network. It looks intelligent in normal operation; failed interfaces, bad data, faulty sensors, platform outages, and broken responsibility chains turn complexity into a risk amplifier. Old problems remain under intelligent layers, while responsibility is redistributed across platforms, algorithms, operators, users, regulators, and human backstops.

Mature physical AI therefore requires more than larger models, more sensors, and finer control: it requires deciding when to overlay, when to reconstruct, and when to abandon an approach. Settings with low risk, weak coupling, and extensive standardization suit an overlay: closed campuses, industrial production lines, warehousing and logistics, and fixed-route transport can all benefit substantially from the added layer. High-risk, open, tightly coupled settings demand more than an overlay: autonomous driving, medical care, home robots, and urban governance ultimately require changes in environmental design, institutions, and responsibility boundaries together. Future competition in physical AI will concern more than model capability: it will concern the ability to reduce environmental entropy, standardize interfaces, reconstruct infrastructure, and design institutions of responsibility; reducing the compatibility tax makes intelligent action cheaper, reducing environmental disorder makes embodied systems more reliable, and recognizing the boundary between overlay and reconstruction prevents the complexity ratchet from pulling systems toward fragility. Cybernetic parasitism is physical AI's starting point, rather than its endpoint. AI often first enters the world of atoms as an overlay attached to existing systems; if it remains there forever, civilization will acquire an ever more complex network of patches. Reliable physical intelligence requires finding the path of lowest total viscosity among overlaying, reconstructing, and retaining existing structures, rather than endlessly attaching sensors and algorithms to the old world.

The old world does not vanish when intelligence arrives. It is first overlaid, monitored, dispatched, and patched. Physical AI begins with this reality: intelligence first parasitizes the

atomic world it seeks to change; maturity ultimately depends on moving toward real environmental reconstruction. Overlay or reconstruction, intelligence keeps feeding a viscous system. The system may divert along lower-resistance paths, accumulate inputs it cannot absorb, or oscillate under overly urgent feedback. Strategy changes input, structure, and feedback; the resulting system dynamics explain these responses.

3.4 Dynamical Responses: Detours, Accumulation, and Instability

An input gains no priority merely by coming from outside the system. The system's dynamics process it, absorbing some inputs, delaying some, and amplifying others. An overlay adds structure and interfaces, bringing new coordination and conversion stages into the path from input to effective change.

3.4.1 Probabilistic Shortcuts and Detours

Intelligence entering reality has several possible outcomes. It may find a low-resistance path that opens an unexpected solution, accumulate at interfaces as inputs outpace real change, or generate oscillation through delayed feedback and overcorrection. The three forms are detours, accumulation, and instability. They do not lie entirely on one axis. A detour is a strategy for passing resistance successfully; accumulation and instability are two failures of pushing a viscous system—too much input for absorption, or correction too urgent for feedback. The three responses describe three fates: getting through, becoming stuck midway, or rebounding into fluctuation.

Resources, time, topology, and medium response explain resistance; we now ask how a resistant system moves under sustained intelligent input. A viscous system absorbs, filters, delays, accumulates, rebounds, and amplifies. A common AI-era mistake treats intelligence as a one-way accelerator: stronger models mean faster systems, more proposals faster progress, more predictions better decisions, and denser control a more controllable reality. Yet resource limits, process times, structural paths, feedback delays, and safety boundaries can turn intelligent inputs into inventory, congestion, fluctuation, and oscillation as well as useful change. The crucial assessment therefore asks what dynamical form the inputs take after entering the system, beyond proposal counts and improved local metrics.

Detours: Moving Trial and Error into Information Space

The first form is the detour. Intelligence searches state space for a lower-resistance path rather than breaking straight through a barrier. With high resource barriers and tight coupling, the direct route may cost too much to be feasible even when the destination is describable. Intelligence often finds a route with fewer collisions against walls.

This is the meaning of a **probabilistic shortcut**. A probabilistic shortcut uses models, simulation, search, and combination to discard clearly unproductive routes and retain a few promising ones, reducing expensive real trials. It pays some costs in information space before they fall due physically; real trials remain but are compressed, reordered, and pruned. AlphaGo's inspired moves offer an accessible bit-domain example: within unchanged Go rules and the board's physical constraints, it found paths through a huge state space that human experience rarely reached (Silver et al. 2016). The surprise comes from reorganized search: seemingly natural human paths need not be best globally, while large-scale search discovers feasible channels beyond experience.

Sampling as an alternative to analytic solution long predates deep learning. In 1946, Stanisław Ulam's thoughts about card-game probabilities led to repeated random trials as an alternative to difficult combinatorial calculation; he and von Neumann subsequently applied the idea to neutron transport and other problems. Nicholas Metropolis named it "Monte Carlo," and Metropolis and Ulam published their methods paper in 1949 (Los Alamos National Laboratory 2021). The cost per sample remained, while the problem became how many samples and what error were sufficient, rather than whether an analytic solution could be found. Weather prediction later expanded from single deterministic forecasts to ensembles, using perturbed initial conditions and model configurations to produce probability distributions. Models and random simulations move some real trials into computation, permitting replay, parallel batches, and failures with little damage while still requiring checks on model bias and sampling error.

The atomic world has similar mechanisms, with heavier costs. Materials discovery, drug screening, structural optimization, and process-parameter searches seek low-resistance paths in high-dimensional state spaces; AI can remove some fruitless experiments, avoid clearly failed combinations, and concentrate resources on promising regions. Candidate materials still need synthesis, drugs verification, processes stability, and products maintenance; intelligence reduces blindness rather than eliminating reality. Computational lithography, process optimization, and defect control show the same logic: near microscopic limits, direct scaling becomes harder, and computation, control, and complex processes find manufacturable paths within narrower windows. Progress continues by detour through more complex, costly arrangements with less tolerance for error when direct paths are blocked.

Even detours encounter barriers they cannot bypass, and specific energy is one. Aviation kerosene has a specific energy of about 12,000 watt-hours per kilogram; taking lithium-ion cells at 250 to 300 watt-hours per kilogram for comparison gives a gap of roughly fortyfold. Allowing for a heat engine's efficiency of about one-third and an electric motor's efficiency of more than 90 percent still leaves a gap of more than tenfold. This barrier sets the feasible limits of several strategies for taking a detour: on the ground, all-electric mobility can avoid some costs through lightweight construction, charging networks, and route constraints; in aviation,

the same detour incurs a direct penalty in takeoff weight, because batteries retain their mass throughout the flight while fuel is continually consumed. A detour works only if the costs it shifts are smaller than the costs it adds in the new medium; once the latter are greater, the detour has merely moved the obstacle elsewhere.

Protein structure prediction extends this logic to a larger scale. The space from amino-acid sequences to three-dimensional structures is vast; difficult samples may take months or years to resolve experimentally, while AlphaFold and related models compress patterns from known sequences and structures into parameters, greatly reducing the cost of candidate generation (Jumper et al. 2021). Predictions already assist screening, interpretation, and experimental design; specific structural or functional conclusions still require attention to confidence, applicability, and relevant evidence. More candidates make screening and focused corroboration more important; verification resources should prioritize candidates capable of changing conclusions and actions.

Biological detours are subtler. Drug design, gene editing, and synthetic biology seek intervention paths in living networks; an effective intervention may use a low-disturbance leverage point rather than force. Compensation, redundancy, and feedback can cancel a seemingly easy path over time; a rare mutation may offer a therapeutic opportunity and unforeseen risk. Biological probabilistic shortcuts need even more verification than atomic ones. A detour changes how costs are paid rather than eliminating all cost. It distributes concentrated costs as risk, moves some physical trials into simulation and search, and substitutes path selection for some direct reconstruction. A successful detour looks like a sudden drop in resistance, but usually finds a more bearable payment route rather than making total cost vanish.

3.4.2 Accumulation, Inventory, and Fluctuation

Accumulation and Inventory Formation

The second form is accumulation. Inputs exceed absorption: the system has activity in abundance but cannot turn it into effective change. It looks busier, denser, more automated, and more immediate without a proportional change in real state. Accumulation begins when incoming flow exceeds absorbed flow: execution, verification, coordination, and responsibility determine how many proposals, commands, investments, experiments, metrics, and recommendations can become change, regardless of what the source produces. Beyond that bandwidth, more input queues, silts up, creates rework and friction, or becomes a maintenance burden instead of automatically raising output.

A stock equation expresses the relationship:

$$\frac{\mathrm{d}S}{\mathrm{d}t} = I(t) - A(S), \qquad A(S) \le A_{\max} \tag{3.9}$$

S is the stock at the system interface, counted as tasks, queue length, or unverified candidates; $I(t)$ is input rate in items per unit time; $A(S)$ is absorption rate; and $A_{\max}$ is the maximum

execution, verification, and coordination throughput with given resources. For constant input with $I > A_{\max}$ and saturated absorption, time measured from the start of the saturated segment gives $S(t) \approx S(0) + (I - A_{\max})t$, linear stock growth. For homogeneous tasks, first-in-first-out service, and fixed capacity, the work ahead translates into an approximate wait of $S/A_{\max}$. Reducing I limits arrivals; raising $A_{\max}$ expands capacity: the former changes arrival pressure and waits, the latter maximum processing capability.

Inventory formation is the first form of accumulation. Inputs not absorbed promptly become inventory, queues, tasks, backlogs, unverified candidates, and untreated risks. Inventory makes temporal rigidity visible: inputs have arrived without the capacity to turn them into change; growing stocks can represent future processing pressure piled up at an interface rather than acceleration. In information systems this appears as screening work: AI generates texts, images, code, plans, and hypotheses faster than human and organizational judgment expands, increasing review, verification, and decision burdens. Scarcity shifts from generation to corroboration; acceleration without expanded verification builds information stocks. In research and engineering, discovery outpaces verification: high-throughput experiments, automation, and AI expand candidates, while replication, stability tests, mechanism explanations, scale-up, and long-term reliability remain. Drug candidates still pass through toxicology, pharmacokinetics, dose, population differences, clinical trials, and regulatory review. More candidates do not automatically mean more reliable drugs; faster discovery makes verification more binding. High-throughput methods and AI move the bottleneck, reducing early search costs while potentially increasing pressure on verification, resource allocation, and failure screening; discovery expanded alone can leave many apparently innovative possibilities stranded before reliable results. In organizations, accumulation appears as internal friction: compute, capital, metrics, and commands arrive quickly, while learning, judgment, execution, and responsibility form more slowly. Without absorption capacity, new recommendations, live metrics, strategic slogans, optimization plans, and management tasks create more meetings, reports, pilots, presentations, and short-term goals rather than greater intelligence.

Internal friction can be understood as input failing to become useful work. More information, resources, and competitive pressure arrive without increased structural response, leaving energy circulating within the system as repeated coordination, metric chasing, local competition, review congestion, dissipated attention, and maintenance rather than changing external states. “Heat” here means unproductive motion left by intensive input, rather than heat in the strict physical sense.

Fluctuations and Cycles

Accumulation does not always sit quietly in place. When input exceeds absorption capacity, a stock first builds up; if it cannot be absorbed locally, it propagates through the structure. In traffic, vehicles keep moving forward while a congestion wave may travel backward; in an organization, tasks keep flowing downstream while unresolved matters surge back to upstream

management; in a technical system, candidates keep appearing while a verification bottleneck exerts backpressure on the initial search strategy. What moves here is the state that cannot be absorbed, rather than the entities themselves. This can be called an accumulation wave. Traffic congestion waves also offer an analogy for backpressure in other high-viscosity systems: when a local channel reaches capacity, the flow it cannot absorb becomes a pressure wave through the system; individuals may still move forward while pressure travels backward, tasks may still advance while a bottleneck presses back on earlier stages, and intelligence may still generate proposals while the verification burden constrains the direction of search. The waves in high-viscosity systems often consist of propagating failures of absorption rather than material waves.

When a system adjusts stocks rather than light variables, accumulation can become cyclical. Inventory, capacity, factories, teams, cities, and institutions cannot appear or disappear instantly: expansion responds to shortage, but conditions may change before supply arrives; contraction responds to surplus, but a new shortage may appear before it finishes. The difficulty can lie in heavy stocks and slow responses rather than impatient control. Cycles arise as stocks of different weights correct at different speeds.

Short cycles often involve light stocks. Goods, tasks, review queues, and candidate proposals can grow and be cleared relatively quickly; their short cycles reflect faster feedback and visible correction. Even light stocks can overshoot under delayed signals: the supply-chain bullwhip effect amplifies small changes in final demand through orders, inventory, production, and procurement, using delayed information and stock adjustments (Lee et al. 1997a). Medium cycles involve equipment, capacity, teams, factories, supply chains, and capital expenditure; investment takes longer than days or weeks, and conditions may change before expansion or contraction is complete. The stock being adjusted is productive capacity, with longer cycles and more expensive correction. Longer cycles involve spatial and structural stocks—buildings, cities, roads, schools, hospitals, housing, and infrastructure—which fix expectations in space and resist rollback; changing populations, industries, public finances, and lifestyles can leave excess buildings, debt, and underused infrastructure, a hysteretic echo of sunk structures. The longest fluctuations involve technological paradigms, energy systems, and institutions, as capital, infrastructure, education, regulation, organizations, and daily life reorganize through expansion and imagination, surplus and selection, then liquidation, restructuring, and stable absorption; these long waves are neither precise clocks nor inevitable laws, but historical revaluation and recoupling of heavy structures.

Different cycles can thus be understood as delayed corrections of stocks with different weights: light goods inventories adjust briefly, fixed capital over longer periods, buildings and cities longer still, and technological, energy, and institutional structures across extended historical periods of expansion, surplus, liquidation, and reorganization. Heavier stocks bring slower feedback, making correction more liable to create another imbalance. Money, credit, psychology, institutions, shocks, and technological change also shape actual cycles; the gen-

eral mechanism here is a system adjusting heavy stocks using delayed signals, with a changed world awaiting the correction. Cycles often reflect obsolete signals correcting heavy stocks rather than an intrinsic rhythm.

Inventory can buffer fluctuations or amplify adjustment delays. Whether accumulation becomes a cycle or instability depends on replenishment, feedback, service, and control. Once input exceeds absorption, the options include reducing unproductive arrivals, improving candidate quality, expanding processing, and changing schedules, beyond increasing generation alone.

3.4.3 Instability, Damping, and Three Forms of Governance

Instability and Damping

The third form is instability. Construction and feedback delays can make stock adjustment cyclical, while control itself can amplify fluctuation. Delayed feedback combined with excessive gain, action amplitude, or unsuitable update timing can turn correction into error amplification. Periodicity does not automatically mean instability: bounded oscillations can persist, while instability appears when small disturbances grow enough to undermine the target state or take the system outside safe operation. The preceding discussion emphasized a slow world; here we ask whether the hand is too fast or pushes too hard. Both can occur in the same loop.

In an ideal feedback system, a controller detects deviation and applies a correction that restores stability; real systems have delays in measurement, judgment, execution, physical response, and behavioral change. Continual corrections to outdated states can apply the right action at the wrong time; by the time it takes effect, the system has moved, turning correction into overcorrection. Such oscillation reflects mismatch among control action, the plant, delays, and feedback cycles. Fast measurement can still help; control gain, action amplitude, and update timing need examination.

Algorithmic flash crashes show a similar mechanism. High-frequency strategies adjust rapidly to prices, orders, and liquidity, while market liquidity, risk-bearing capacity, and institutional safeguards may not replenish at the same speed; interacting algorithms can respond, cancel orders, and amplify signals until prices diverge from fundamental value (U.S. Securities and Exchange Commission and Commodity Futures Trading Commission 2010). On May 6, 2010, the Dow Jones Industrial Average briefly fell about a thousand points intraday and then rebounded sharply; investigations identified interaction between high-frequency and algorithmic trading as an important cause. Multiple fast feedback loops can couple into collective instability beyond the error of one algorithm. Organizations can oscillate too: frequent adjustments to strategy, metrics, resources, and staff direction can interrupt feedback from previous adjustments. Metrics change before teams finish learning, resources move before projects sta-

bilize, and new advice arrives before a process has been absorbed. Apparent agility can conceal a lost rhythm under overly dense control.

Actual loops must be examined for causes of oscillation: gain, delay, model error, actuator limits, and coupling can all contribute. Fast measurement and computation can also stabilize a slow plant. Damping, inertia, and active compensation have different physical effects and require separate accounts; their shared goal is correction along recoverable trajectories.

Damping devices can themselves fail. China introduced index circuit breakers in January 2016, with CSI 300 declines of 5% and 7% triggering two levels; January 4 and 7 both saw halts followed by early closes, with liquidity contracting near the thresholds, and the mechanism was suspended from January 8 (China Securities Regulatory Commission 2016). The intended cooling-off period split a continuous decline into stages and could also induce selling before a threshold arrived. A damping device must act compatibly with the behavior of the system it seeks to damp. Industrial control has a counterpart: common electromechanical low-frequency grid oscillations occur around 0.1 to 2 hertz and are suppressed through phase compensation in excitation loops, with stabilizer parameters tuned to inertia and operating configuration; higher sampling rates do not replace tuning.

Power systems make the differences among inertia, fast response, and control coordination concrete. In Britain's August 9, 2019 blackout, multiple sources of generation were lost after a lightning strike, frequency fell, and low-frequency demand disconnection operated (National Grid ESO 2019). Inertia affects the initial rate of frequency change, damping and frequency response affect subsequent behavior, and protection settings determine when load is disconnected. These functions require joint design; calling inertia damping obscures the mechanism of stability.

Three Forms of Governance

Detours, accumulation, and instability are three basic dynamics under intelligent input to viscous systems. Detours use information search to change how barriers are paid for; input beyond absorption bandwidth creates inventories, queues, waves, and cycles; overly urgent control or dense feedback can turn negative feedback into an amplifier in a system otherwise capable of slower adjustment. The forms correspond to three kinds of governance. Detours require verification: a shortcut may be a discovery, chance correlation, overfit, unmanufacturable design, or unmaintainable process; counterintuitive paths need tests of repeatability, executability, and boundary conditions. Reality must still establish the path intelligence finds. Accumulation requires rate control: expand absorption as well as generation, screen candidates by level, match experimental verification to discovery and organizational input to execution, and match policy and capital to real capacity; input beyond absorption becomes inventories, waves, and cycles. Instability requires damping: slow variables constrain fast feedback, long-term goals correct short-term metrics, safety boundaries constrain rapid control, and algorithmic loops need delay management and amplitude limits. Stable intelligence knows when to

stop accelerating. **Viscous systems need mature dynamical control, beyond stronger intelligence alone.** Search finds detours, absorption prevents accumulation, and damping avoids feedback instability; together they allow intelligence to become stable real change.

The design of damping devices is itself contested. Theoretical work on price limits and circuit breakers shows how thresholds can move trades ahead of a halt and accelerate prices toward the limit, the magnet effect (Subrahmanyam 1994). Some Taiwan and Shanghai samples show price acceleration and changed trading near thresholds; other samples and identification methods produce different results (Cho et al. 2003; WONG et al. 2009; Kim et al. 2013). The four U.S. market-wide halts in March 2020 offer another observation: S&P 500 declines of 7%, 13%, and 20% from the prior close define three thresholds; the first two halt trading for fifteen minutes during specified hours, while the third ends trading for the day. March 9, 12, 16, and 18 each triggered Level 1, marking the return of such market-wide halts after 1997 (Market-Wide Circuit Breaker Working Group 2021). Behavioral effects and recovery can differ across market structures; parameters need testing against rules, liquidity, participant responses, and transaction data.

Mature intelligence must know whether a path is real, input excessive, or feedback too fast, as well as how to find answers. Viscous systems require detours, rate control, and damping to change. All three first require a more basic question: what exactly is the output of intelligence? A proposal's complete form does not establish confirmation by objects, rules, execution, and reality; a seemingly compliant generation is not yet something that can be relied on.

3.5 The Limits of Form: When Can an Output Be Confirmed?

The three judgments left by the previous section point to an object we have yet to examine: what, exactly, does intelligence produce? The questions people first encounter when using large models are usually concrete. Why does a model sometimes answer like an expert and sometimes invent facts? Why can it write a plausible proof yet skip the crucial step? Why does code look perfectly conventional yet call an interface that does not exist? Why does a smooth causal explanation fail when applied to an experiment, an engineering problem, or organizational action? Will hallucinations disappear if the model gains access to search, databases, code execution, proof checkers, and external tools?

These apparently separate questions lead to a deeper one: does a large model generate forms, or forms that have already been confirmed? Paper titles, proof steps, code structures, causal explanations, experimental hypotheses, and action plans can all appear in language. A model's ability to generate them shows that it has recovered certain stable structures from human traces; recovering a structure does not establish that it refers to a real object, obeys strict rules, passes execution checks, or can enter reality and bear consequences. Large models change more than where answers come from: they allow forms to be generated in vast quantities without

simultaneous confirmation. The separation of form generation from form confirmation is the starting point for understanding large models' capabilities, hallucinations, and limits in reality.

3.5.1 The Dual Identity of Form and the Limits of Its Guarantees

The Dual Identity of Form

This is also why large-model errors can be so perplexing. Their errors are often neither crude nor unstructured gibberish. They arrive in remarkably complete forms: nonexistent papers have plausible titles, authors, journals, and abstracts; false proofs have theorems, lemmas, a deductive tone, and orderly symbols; unrunnable code has good style, clear function names, and thorough comments; unverified medical advice has pathological explanations, dosage recommendations, and risk information. The model resembles someone fluent in these forms who keeps generating where confirmation is absent, rather than someone speaking in ignorance. The issue is whether objects, rules, execution, and real feedback have confirmed the forms it possesses.

Form means more than symbols, axioms, or formulas: at a deeper level, it is a transferable constraint that reduces uncertainty. How a sentence continues, a proof advances, a program calls a function, an action unfolds, or a failure is explained is not arbitrary. Whenever a structure repeatedly constrains subsequent possibilities across content, contexts, and tasks, it becomes a form that intelligence must retain. Form frees experience from individual cases, lets judgment transfer, makes reasoning checkable, and allows action to be organized in advance. Here lies the significance of formal logic: it separates the validity of reasoning from its specific content, rather than inventing reasoning. Syllogisms, propositional logic, predicate logic, and symbolic calculi make the same point: validity depends on preserving structure rather than keeping the objects unchanged. If the structure is preserved, the content can be replaced. Form is thus one of intelligence's most powerful means of compression, retaining extensive transferability with very little redundancy. Its transferable skeleton lets humans carry judgments between objects, turn one class of problems into another, and distill experience into rules.

Historically, this approach went a long way. Euclid's Elements organized scattered experience of measurement into axioms and deductions, making conclusions depend on the chain of reasoning rather than on the accuracy of any particular measurement. Boole expressed logical operations algebraically in 1854; Frege's Begriffsschrift of 1879 brought structures such as "all," "there exists," and "if" into a symbolic calculus. By Russell and Whitehead's three-volume Principia Mathematica, axiomatization had reached arithmetic itself. A clear conviction sustained this approach: with sufficiently explicit axioms and strict rules, proofs could be checked mechanically, making the validity of reasoning independent of who performed it. Compression reaches an extreme here: one rule can cover infinitely many instances, one structure can stand in for all the particular content, and the content compressed away need not be

remembered.

In the early twentieth century, this approach became a program. The second of Hilbert's twenty-three problems, presented at the International Congress of Mathematicians in Paris in 1900, asked whether the consistency of arithmetic could be proved. Returning to the question in Bologna in 1928, he also identified completeness and decidability as goals; the latter became known as the Entscheidungsproblem. The program sought to organize mathematics into explicit axioms and rules, establish consistent formal foundations, and find a general mechanical method for deciding whether a first-order logical formula is universally valid. This ambition pushed the transferability of form to its limit and brought the question of its guarantees into the open. If a form can guarantee only what it covers, can it guarantee itself?

The Limits of Formal Guarantees

Formalization makes deduction independently checkable and also establishes precise general limits. A sufficiently strong, consistent, effectively axiomatizable arithmetic theory contains undecidable sentences and, under the relevant conditions of the second incompleteness theorem, cannot prove its own formalized consistency (Gödel 1931). The halting problem rules out a general algorithm that decides the behavior of every arbitrary program; algorithmic information theory studies description length and incompressibility (Turing 1936; Chaitin 1987). These results constrain different objects without invalidating reliable checks of particular programs and proofs.

These results reveal distinct limits on provability, decidability, and describability; they cannot be collapsed into a blanket claim that form can never justify itself. They make the conditions of a guarantee part of the problem: which axioms, which procedures, and which objects? Form remains one of intelligence's most powerful means of compression, but information relevant to a task must be recovered elsewhere if that compression discards it. Gödel, Turing, and Chaitin addressed different boundaries within formal systems. Real systems add another: a form's validity at one scale does not establish its validity at another.

Complex systems bring this warning back to reality. The real world is neither a completed table of propositions nor a simple machine that produces the expected output whenever an input is supplied. The same system presents different effective variables at different scales: particles, interactions, and fluctuations microscopically; temperature, pressure, flow, and phase transitions macroscopically; function, homeostasis, adaptation, and repair biologically; institutions, strategies, trust, and feedback socially. Scale is a way of compressing the world rather than an incidental label attached to it. A form valid at one scale need not transfer to another without loss. Intelligence often fails when forms overreach: a causal narrative in language is treated as an intervenable causal relationship in reality, a local regularity as system control, short-term feedback as a long-term objective, accuracy at one point as deployment reliability, or deduction within rules as action beyond them. Detached from task, medium, scale, feedback, and boundary conditions, form can turn from intelligence's skeleton into a source of error.

Such overreach is common in engineering. A simulation accurate under training conditions fails under different conditions; risk rules that perform well on historical data misjudge unseen behavior; scheduling algorithms stable with standard cargo require human intervention for nonstandard loads. These failures need not come from errors within the rules; the rules' domain of applicability may have been mistaken for the extent of the world. To judge where a form is valid, we need to know the data, boundary conditions, and costs of failure under which it was established—information often absent from the form itself.

Engineering has achieved formal confirmation, at a measurable cost. A lookup-table defect in the Intel Pentium's division instruction led the company to take a charge of roughly $475 million in 1994 for replacements, inventory write-downs, and related costs; the episode strengthened the standing of formal methods in hardware verification (Intel Corporation 1995). An extreme software case is the seL4 microkernel. Its initial 2009 results verified that about 7,500 lines of C conformed to an abstract specification, while boot and assembly code lay outside that proof. The proof, frameworks, tools, and libraries totaled about 200,000 lines of Isabelle script and approximately twenty person-years of proof work: nine on general infrastructure and eleven on the seL4-specific proof (Klein et al. 2009; Trustworthy Systems, n.d.). The compiler counterpart is CompCert, which uses a proof assistant to verify preservation of observable program behavior under the formal semantics of its supported languages and target machines; its guarantee has a specified language scope, behavioral conditions, and trusted base (Leroy 2009). These figures indicate the scale of formal confirmation costs: shrinking the trusted core, specifying requirements clearly, decomposing proofs, and reusing tools can make confirmation affordable. As the object expands to open environments, hardware timing disturbances, concurrent interactions, and human use, the specification itself can become difficult to establish.

3.5.2 Where Compressed Forms Come From, and the Illusion of Low Loss

Where Compressed Forms Come From

AI's development can also be understood along this line. Early symbolic AI captured the expressible, composable, and checkable aspects of intelligence. Rules, predicates, expert systems, theorem proving, knowledge graphs, and symbolic planning demonstrated the indispensability of form; the limitation was heavy reliance on hand-written forms, which struggled to cover open-world common sense, context, exceptions, bodies, goals, and consequences. Real action does not unfold naturally from a static library of axioms. Symbolic systems excel at deduction within explicit boundaries but struggle to absorb the vague, ambiguous, tacit, embodied, and historically formed structures of human experience. Large models mark a turn toward learning forms from text, code, papers, laws, conversations, proofs, programs, and experimental records rather than relying chiefly on people to write them down. Language is a de-

posit of human experience, reasoning, institutions, failures, corrections, and recorded actions, rather than a heap of words. Logic, classification, implication, negation, recursion, causality, counterfactuals, proof templates, program structures, goal achievement, and correction recur in these deposits; reducing prediction error presses models to absorb these relational skeletons with their low redundancy, broad transferability, and strong constraints.

Variable binding, reference, function calls, proof patterns, and goal relationships in a corpus have reusable structures; learning them helps reduce prediction error. Whether training acquires them, and how reliably they can be called upon, must be tested on tasks. Energy, data distributions, objectives, and optimization play distinct roles; dissipation alone does not guarantee logical or causal ability. The physical accounting of information processing contains a definite lower bound:

$$E_{\min} = k_B T \ln 2 \tag{3.10}$$

Here k_B is Boltzmann's constant, 1.38×10^{-23} J/K; T is ambient temperature in kelvins. Under the usual assumptions of an isothermal process, symmetric memory, and a complete reset cycle, $E_{\min}$ is the lower bound, in joules, on heat released to the environment when erasing an unknown logical bit with equally probable states. At room temperature, $T \approx 300\,\mathrm{K}$, the bound is $E_{\min} \approx 2.9 \times 10^{-21}$ J. This is entropy accounting: resetting two equally probable logical states to one state reduces the memory's logical entropy by $k_B \ln 2$, and the environment must receive the corresponding entropy. The expression bounds this reset process. Many operations in current general-purpose computers dissipate far more; devices, memory hierarchies, power delivery, and cooling determine the actual account. It reminds us that information processing has a physical substrate; known states, correlated side information, and logically reversible operations involve different conditions, so every write cannot simply be charged as this one-bit erasure.

Dissipation is a necessary condition here, but an insufficient one: dissipative systems do not all produce transferable logical structures. Several conditions work together: electricity, chips, and cooling provide real energy flux; human records provide highly structured data distributions; the loss function makes prediction deviations comparable; and optimization continually presses these differences into parameter space. Dissipation makes computation possible, the data distribution supplies compressible structure, the loss function supplies selection pressure, and optimization shapes the result. Form is selected jointly by structured data and error pressure through energy-driven optimization, rather than emerging automatically from energy.

This image is no strict physical law, but helps explain why large models develop linguistic, logical, and code structures: an untrained network resembles unformed terrain, training data prolonged rainfall, and optimization continuing erosion. The material matters, yet with enough data, training, and error pressure, the channels repeatedly selected by the water shape the landscape. Language, logic, code, and causal narratives are paths through human records that reduce resistance and error and recur stably. Models appear to understand language and

logic because these structures efficiently reduce prediction error, rather than because explicit rationality has been poured into them. A modern large model is therefore neither a collection of if-else rules nor a conventional symbolic rule base. It more closely resembles a highly constraining, low-redundancy structure sustained by physical dissipation, shaped by information gradients, and carried in parameter space—figuratively, an information crystal grown jointly by electricity, data, and error gradients. This "crystal" is dynamic rather than static: training, use, and updating require continuing energy input. Its order is purchased through substantial external dissipation; its stability is maintained, rather than absolute, across the training distribution, parameter structure, inference context, and confirmation mechanisms.

Low Loss Does Not Establish Truth

The boundary emerges here as well. Optimization selects low-loss paths rather than truth itself; predictive training elicits formal structures rather than forms already confirmed. A recurring structure that reduces prediction error may be absorbed from a corpus, but whether it refers to a real object, follows strict rules, can execute, or can enter real action requires further confirmation. **Low loss does not establish truth, fluency does not establish validity, and compatibility does not establish executability.** The GPT-4 technical report offers a useful historical anchor: it demonstrated strong performance on many standardized examinations and tasks while acknowledging hallucinations, reasoning errors, and the need for care in high-stakes uses (OpenAI 2023). This juxtaposition matters: capability and limitation can coexist. A system can be remarkable at examinations, writing, reasoning, coding, and multimodal understanding without automatically turning its outputs into confirmed facts, verified proofs, runnable programs, or deployable plans. Its recovery of many forms makes it powerful; the credibility of unconfirmed forms makes it dangerous.

Hallucination thus becomes a prominent problem between form generation and confirmation. It can accompany powerful associative ability, but cannot be attributed solely to that ability. High-dimensional association connects related domains, formats, arguments, and goals into low-resistance generation paths; without constraints from objects, rules, and evidence, a path can continue into structurally compatible but unconfirmed content. A nonexistent paper can combine a real field, real terminology, authentic authorial styles, journal formats, and argumentative conventions; a false causal chain can join similar scenarios, familiar explanations, local correlations, and narrative sequence; a false proof can retain the tone, steps, and symbolic appearance of proof while losing the constraints of its rules at the decisive turn. The problem here is form generation outrunning reference checks, rule checking, and real feedback. There are internal routes to reducing hallucinations: improving training data and objectives, calibrating confidence, allowing abstention, and changing inference. Research in 2026 further showed how accuracy-centered evaluations reward guessing under uncertainty, while interventions in some models' internal mechanisms for answering or abstaining can improve decisions (Kalai et al. 2026; Nguyen et al. 2026). These results support neither the claim that fewer hallucinations

necessarily mean less creativity nor the disappearance of errors on arbitrary tasks. The key is to coordinate generation, evidence, rules, and conditions for abstention. Reviews generally describe hallucinations as fluent, coherent outputs that are factually false or unsupported by evidence, tracing their causes through data, training objectives, model architecture, inference, and evaluation (Huang et al. 2023). "The model is not large enough" cannot explain the whole problem, and "connect it to search" cannot handle it alone: large models generate candidates quickly, while their reliability requires confirmation appropriate to the task.

3.5.3 Where Confirmation Happens, and What It Costs

Three Sites of Confirmation

Proof provides a second important case. A mathematical proof is more than the tone of proof, connectives such as "clearly," "hence," or "it is easy to show," or the appearance of theorems, lemmas, and corollaries: every transformation must be permitted by the rules, and the entire chain must be checkable. AlphaGeometry illustrates this distinction. It solved twenty-five of thirty recent International Mathematical Olympiad geometry problems, approaching human gold-medalist performance (Trinh et al. 2024). More significantly, it combined neural generation with symbolic deduction rather than relying on a language model to produce proof-like text: the neural component proposed useful constructions, and the symbolic engine performed deductions and checks. Form generation matters enormously, but a reliable proof needs rule-based confirmation: a model can suggest a direction, while the rules must make the proof hold. Code provides a third case. Large models' coding ability is already changing software production, yet code also makes the need for confirmation especially clear: looking like code, running, passing tests, solving a real issue, and operating reliably with its dependencies are distinct levels. SWE-bench Verified comprises 500 human-filtered real GitHub issues used to assess whether models or agents can solve problems in actual repositories (Jimenez et al. 2023). Its value goes beyond another leaderboard: it brings code from text generation back to confirmation through execution. The repository, dependencies, tests, issue description, patch, environment, and regression checks form a chain of confirmation; without that chain, code remains a formally complete candidate.

These three cases point to the same structure. Citations need confirmation of their referents, proofs of their rule-governed validity, and code of its execution. Large models may be extraordinarily fluent in all three appearances, but appearances provide no confirmation: a citation requires retrieval and source tracing, a proof step-by-step checking, and usable code execution, testing, and adaptation to its environment. Form generation can take place within a model; confirmation requires independent constraints from evidence about objects, rule systems, and execution results. Form is confirmed in at least three places. First, representational confirmation: does the form refer to a stable object? Papers, authors, concepts, interfaces, facts, and

relationships must be retrievable, traceable, and open to cross-checking, beyond looking true. Second, inferential confirmation: can the form be transformed reliably under the rules? Proofs must be checkable, code executable, calculations reproducible, and causal chains able to withstand counterfactual constraints. Third, confirmation through action: can the form change a real state and bear the resulting feedback? A plan needs more than linguistic completeness: it must pass through interfaces, permissions, resources, time, failure recovery, and boundaries of responsibility. The same structure can pass through all three places: causality can be recognized as a textual schema, computed as an inferential structure, and tested through real intervention. Validity in language does not establish validity in inference; validity in inference does not establish feasibility in reality; feasibility still faces resources, time, structure, and responsibility. Form gains reliability through different tests at different places, rather than acquiring it all at once. Pearl's formalization of causality makes precisely this point: causality requires intervention and counterfactual structures that let a dependence face the question "What would happen if the conditions changed?", beyond making a correlation sound more persuasive (Pearl 2009). These three places occupy the middle of Chapter 1's verification pyramid, which rises through formal, semantic, causal, engineering, long-term biological, and social-responsibility verification. Representational confirmation concerns meaning and evidence about objects; inferential confirmation concerns formal rules and causal assumptions; confirmation through action enters engineering execution and connects to long-term verification and social responsibility. The classifications overlap rather than corresponding one to one. Large models drive candidate-generation costs extremely low, releasing text, code, explanations, hypotheses, and plans rapidly. Dependable results require an ascent through confirmation of objects, rules, and action, then of integration into systems and the ability to withstand disturbances, exceptions, audits, and responsibility over sustained operation.

The Costs and Limits of Confirmation

Training dissipates energy to compress forms; verification dissipates energy to make them trustworthy. Generation can be amortized in advance in parameter space, while confirmation must be paid for at each level of objects, rules, action, systems, and responsibility. Generation takes place at the pyramid's base; reliability is acquired step by step during the ascent. Confirmation costs can therefore grow nonlinearly: they may be low for low-risk, weakly coupled, reversible tasks, but rise rapidly in medicine, finance, law, engineering, public policy, and biotechnology. Higher up, confirmation approaches reality—and resources, time, organization, responsibility, and the bearing of consequences. This difference can be expressed in terms of pass rates:

$$N_{\text{cand}} = \frac{M_{\text{use}}}{\prod_{i=1}^{k} p_i} \tag{3.11}$$

k is the number of confirmation stages, and p_i is the conditional probability of passing stage i after passing all earlier stages; multiplying these probabilities requires no independence as-

sumption. Under stable, comparable repeated attempts, $M_{\text{use}} / \prod_i p_i$ gives an expectation or planning approximation, without guaranteeing success in any particular batch. An additional independence assumption is needed if unconditional pass rates are used instead; correlation has no uniform direction of effect on the estimate. Model improvements may also raise pass rates, improve trials, and reduce rework.

Retrieval, databases, execution engines, proof checkers, test environments, APIs, sensors, and feedback loops belong here: they are different interfaces for confirming forms rather than a catalog of technologies for this section. Retrieval confirms objects, execution rules, tests boundaries, tools interfaces, and feedback the consequences of action. They can reduce hallucinations without automatically guaranteeing zero error, because confirmation paths can themselves fail: sources may be unreliable, tests incomplete, interfaces misused, feedback delayed, and responsibility impossible to assign. As some generation errors are eliminated, reliability questions also extend to the confirmation loop composed of model, tools, and environment. RAG, tool calling, code interpreters, and agent frameworks are consequently no "eliminate hallucinations" button. They extend the confirmation chain and introduce further objects requiring confirmation: are retrieved results real, databases current, tool calls correct, execution environments representative, proof checkers given a correctly formalized problem, and sensor feedback correctly interpreted? When a model begins calling the outside world, errors can involve retrieval, interfaces, execution, coverage, permissions, and responsibility as well as generation; confirmation migrates rather than disappearing. This yields a testable boundary claim: assertions of general reasoning or real-world action capability must face three tests. Without external confirmation interfaces, can the system consistently avoid representational hallucinations? Can it keep its deductions within the rules? Can it complete real actions with feedback, rollback, and accountability? These tests distinguish the ability to generate candidates, the ability to check them internally, and the degree to which task results have actually been confirmed. Increasing model size does not automatically remove this boundary; tools, environments, and feedback redistribute it across a larger confirmation loop.

This also explains the distance between large models and world models. Large models learn forms from traces; world models must also calibrate those forms against real responses. A causal account can be sustained narratively in language but needs intervention-based testing in reality; a goal can be expressed in a sentence but must pass through resources, time, structure, and responsibility in reality. A form effective for prediction is not a sufficient representation of a world we can intervene in; it must be confirmed through the consequences of action. At the level of action, form confirmation meets technological viscosity. Confirmation has a cost: facts require retrieval, databases, and source tracing; inference requires execution, testing, and coverage; action requires sensors, interfaces, permissions, trials, monitoring, and rollback; systems require redesigned workflows, safety redundancy, audit logs, assigned responsibility, and social absorption; long-term confirmation requires time, exceptions, dis-

torted incentives, and consequences across generations. Higher levels of the verification pyramid encounter viscosity in resources, time, structure, verification, and responsibility. Large models make candidate generation extremely cheap without canceling the real accounts due when candidates become dependable results. The same large-model application proceeds at vastly different speeds across media. In bit space, text, code, retrieval, summaries, and preliminary analysis can iterate rapidly; errors can sometimes be retracted cheaply and candidates rewritten at low cost. In atomic space, action requires materials, energy, sensors, interfaces, operating conditions, tests, and maintenance, while errors may cause damage, downtime, or safety incidents. In biological space, confirmation is more expensive still: drugs, gene editing, development, immune responses, ecological effects, and consequences across generations cannot be undone repeatedly like text. As form reaches deeper into reality, confirmation grows heavier and viscosity becomes more visible.

The large-model revolution changes how forms are produced. It moves forms from handwritten rules into high-dimensional representations, from explicit inscription into statistical learning, and from a few axioms into vast collections of traces, without abolishing their limits. Predictive training can elicit structure, gradient erosion can shape low-loss channels, and external tools can add confirmation paths; none automatically confers reliability in reference, rules, action, and responsibility. Once forms enter tools, environments, and feedback, their boundaries are redistributed within the system's loop. Intelligence therefore includes knowing what confirms a form, at which scale it holds, and when it needs replacement, beyond generating it. Large models compress the forms in which logic, programs, causality, and action appear in human traces, rather than logic itself. Their boundary is that high-dimensional associations formed through prediction have not automatically acquired reliability under representational, inferential, action, system, and long-term tests. Form makes intelligence possible; testing brings form into reality. Testing first needs a place where it can occur: a channel must already exist between model and object through which consequences can pass.

3.6 World Models and Intervention-Ready Compression

Where forms can be tested and where actions obtain correction signals are two questions that lead to a much-discussed but rarely precisely defined object: the world model. To understand the term, we first need to sidestep a more familiar dispute. One of the easiest arguments to fall into about AI is whether it has "real intelligence." If real intelligence means autonomous consciousness, self-generated purposes, subjective experience, and humanlike understanding, the question quickly enters philosophy and science fiction. We have no definite answer today about whether, when, or in what form such AGI will emerge. Engineering does not wait for ultimate definitions. Aircraft flew without humans first understanding avian consciousness, and computers calculated without first possessing a mathematician's mind. Technological paths

usually start with useful capabilities and expand through tasks, tools, and feedback, rather than working backward from an endpoint. From this perspective, a large model begins as a powerful imitation-and-compression system, rather than a completed general mind: it learns patterns from vast amounts of text, images, code, and interaction data, statistically compressing language, knowledge, style, program structures, and task relationships. Its capabilities emerge through similarity matching, structural completion, and probabilistic generation across a vast space of experience rather than strict deduction in the traditional sense. Calling it an increasingly articulate parrot is not entirely wrong, but can obscure the crucial point: deep enough imitation, broad enough coverage, and connections to tools, memory, feedback, and task environments can turn imitation beyond mechanical repetition into useful approximate intelligence. Imitation can be the beginning of intelligence in a feedback loop, even if it is not its endpoint.

3.6.1 From Imitation to Useful Approximation: Where World Models Come From

From Imitation to Useful Approximation

AI's history is a series of experiments in how intelligence gains a world. Early symbolic AI tried to construct intelligence through formal logic and rule systems: write the world as propositions, rules, and inference chains, and intelligence could unfold like a mathematical proof. This inherited an older drive toward formalization: the world could be encoded, reasoning made rule-governed, and intelligence reduced to symbol manipulation. The underlying assumption was that sufficient description would let intelligence acquire the world through rules. Open reality is not a closed system with all its axioms already written. More rules create more combinations; more open-ended goals make contexts harder to exhaust; closer contact with reality makes exceptions, noise, bodies, history, and consequences harder to enumerate completely. Reality is an ongoing process changed by action, rather than a static table of propositions. Intelligence entering it must submit its judgments to environmental tests rather than relying only on internally closed rules. Modern neural networks take another route: they learn structure from data distributions rather than starting with explicit rules, obtaining usable patterns through fitting, compression, and generalization rather than first stating the world's logical axioms. CNNs, Transformers, and other deep networks are not deductive machines in the traditional sense. They excel at learning resemblance, likely continuations, and recurring structures from many examples. This imperfect, hallucination-prone route lacks strict provability, but bypasses the difficulty of explicitly enumerating the world in a purely symbolic system: capabilities can appear before the world has been fully written into rules.

Vast resources were once committed to that symbolic route. Japan launched its Fifth Generation Computer Project in 1982, organized through the Institute for New Generation Computer Technology around predicate logic and parallel inference. The Information Processing Society

of Japan records approximately ¥54 billion in spending over the eleven fiscal years from 1982 through 1992. The project built five parallel inference machines, including the 512-processor PIM/p and 256-processor PIM/m, but did not achieve the anticipated broad commercial adoption. Meanwhile, improvements in general-purpose computers reduced the value available to dedicated inference machines (Aiso 1988; Feigenbaum and Shrobe 1993; Information Processing Society of Japan, n.d.). The lesson concerns competition among explicit knowledge acquisition, software ecosystems, specialized architecture costs, and advances in general-purpose hardware, rather than the incapacity of rule systems. Writing the world into rules alone cannot ensure a computing approach's success.

The central value of the large-model approach is that it offers a practically scalable approximation to intelligence, regardless of whether ultimate understanding has been solved. It can imitate before combining, complete before planning, and generate before being corrected by external feedback. Value can precede a complete world model; many intelligent capabilities stabilize through repeated coupling with external environments rather than forming all at once within a closed interior.

Where World Models Come From

Here we encounter the question of world models. A common claim is that AI must first possess a world model to acquire real intelligence. The claim has force: without some internal structure of the world, a system cannot predict consequences, plan actions, distinguish appearances from regularities, or transfer experience across situations. With no world structure at all, it can only react locally between inputs and outputs and struggles to sustain stable action. But the claim immediately raises a question: where does the world model come from? If it comes from experience, that experience may come from the system's own perception, actions, and feedback or from others' records of observation and intervention. If feedback must correct it, the system must submit its judgments to environmental tests; if it supports action, action cannot wait for complete model maturity. The system must act with an incomplete model and use action to improve it; it cannot first stand outside the world, model it completely, then return to perform tasks. This is the world model's circular dependence: models guide action, while action and its records provide sources and corrections for models. Models supporting action need calibration against the world's responses, and action needs models for guidance. They shape one another within the same loop rather than following a simple sequence. The loop also has an irreducible direction in time: models compress past experience, data, and failures, while actions bet on future consequences that have yet to appear. Weakly coupled tasks stabilize more easily because the future can largely be extrapolated from the past; strong feedback-loop tasks are harder because action changes subsequent conditions; highly coupled reflexive tasks are harder still because objects reorganize in response to models, predictions, and interventions, making the future more than an extension of the past.

A deeper misunderstanding is hidden here. "World model" easily suggests that intelligence

should hold an internal map: a world outside, a model inside, the world as object, and the model as representation—as though intelligence must first acquire a fairly complete map before acting with it. An actor embedded in the world has no such outside perspective. It cannot first take a complete photograph from outside and return to execute its plan; it is inside from the beginning. Bodily position, task goals, tool interfaces, experience, and action capabilities already shape its perception. It sees a world related to its possible actions rather than a pure world. A world model is therefore a compression of action consequences rather than a photograph of the world. Intelligence may need less a complete internal picture than a structure of expectations about what will happen if it acts: which paths are more likely to succeed, which actions bring danger, which errors can be corrected, and which failures cannot be rolled back. A model reduces expensive mistakes before action, absorbs feedback afterward, and helps avoid one more collision next time, rather than moving the world inside. Large models' success offers a clue: without physical bodies or direct contact with reality, they perform many useful tasks by compressing human symbolic experience. Intelligence can often begin with local patterns, task structures, symbolic compression, and feedback correction; a complete world model may be a deposit left by sustained intelligent action rather than its prerequisite. The better question is therefore how strongly a system's model must couple to action for a particular task, rather than whether AI already possesses a world model.

Digital Agents and Embodied Intelligence

AI agents express this logic in digital environments. A large model on its own has an evident boundary: it can generate text and plans, but without calling tools, observing results, and revising plans, its ability remains at the level of one-off output. Agent systems connect models to files, browsers, code environments, databases, software interfaces, and task-management workflows so they can decompose goals, execute steps, observe feedback, and adjust strategies. The model enters a task loop—plan, act, observe, revise, act again—beyond answering questions. This changes the structure of capability without immediately establishing autonomous intelligence. Many problems once demanding a correct single inference become problems of repeated trials and feedback control; a model can approach correctness through environmental feedback. Its effective intelligence is distributed across tools, interfaces, memory, logs, tests, and external feedback as well as its parameters. Tools belong to the intelligent loop itself. This gives the other half of the answer about where world models come from: part of the model grows in toolchains, logs, and failure records rather than parameters. While those external structures remain, the model need not fit the entire world inside itself.

Embodied intelligence faces the same kind of problem in a heavier medium. Large models currently learn mainly from text, images, code, and human knowledge: a world expressed by humans through secondhand, compressed, and symbolic experience. This is already powerful because so much civilizational knowledge resides in symbols, but it still lacks much firsthand feedback: touch, force, spatial continuity, physical contact, long-term causality, and conse-

quences of action. Embodied intelligence reconnects AI to the real physical world, giving the system consequences of acting in the world alongside descriptions of it. Descriptions can unfold fluently in symbolic space; consequences appear in the world. A model can write about a cup or recognize its image, but lifting it confronts weight, friction, center of mass, contact angle, slipping, collision, and feedback delay. A model can recommend treatment, but drugs entering a body encounter absorption, metabolism, immunity, toxicity, individual differences, and long-term side effects. A model can design a traffic plan, but implementation encounters roads, property rights, regulation, habits, accident liability, and behavioral responses. A model unable to pass through this feedback remains language about the world rather than a capability to enter it. All three cases require feedback, but of different kinds: the cup involves residual physical state changes after contact, drugs involve metabolism, compensation, and individual differences, and traffic involves institutions, responsibility, and behavioral responses. Together they resist entry into the world without reducing to one kind of coupling. Internal symbolic consistency is consequently powerful and dangerous: a large model can construct complete arguments and strong consistency across styles, analogies, and concepts, yet without external correction it may mistake linguistic smoothness for real effectiveness. It can expand possibilities without knowing which can pass through real resistance, and generate answers without knowing what their entry into the world will change.

3.6.2 Three Degrees of Coupling, Residuals, and Reflexivity

Three Degrees of Coupling

Coupling can be divided into three working situations. With low coupling, candidates can be checked before outputs change external states; in a strong physical feedback loop, each action changes the object encountered next; in a reflexive loop, other actors also interpret the model and adjust strategies. These situations can coexist within one system. Social systems may form stable equilibria or keep drifting; reflexivity alone does not establish the absence of a fixed point.

Differences in timescale across the three kinds of coupling can be measured. Franka's real-time interface exchanges states and control commands at 1 kHz, updating every millisecond. Update rate is not physical closed-loop bandwidth: the latter is also constrained by transmission flexibility, sensor and actuator placement, workpiece stiffness, and delays; excessive gain can excite unmodeled vibration (Franka Robotics, n.d.; Eppinger and Seering 1987). Large-model response latency varies with the model, hardware, output length, reasoning, and tool calls: fast interactions may begin producing output within hundreds of milliseconds, while a complete decision may take seconds or longer. Contact tasks therefore cannot simply connect large-model outputs directly to servo loops. Semantic judgments can be slow; force and position loops must be fast. Hierarchical control and timescale conversion must bridge them, or residual

errors will continue accumulating between high-level decisions.

Residuals and Reflexivity

The circular dependence of world models is a basic condition of intelligence embedded in the world, rather than a defect of immaturity. Models and actions do not simply follow one another: models form through action, actions find direction through models, and intelligence gradually acquires a world relevant to itself by acting within it. Whether this loop can bootstrap depends on whether its update operator drives deviations back down:

$$\begin{aligned} a_t &= \pi(m_t), \quad m_{t+1} = U(m_t, a_t) = F(m_t), \\ DF(m) &= \frac{\partial U}{\partial m} + \frac{\partial U}{\partial a} D\pi(m). \end{aligned} \tag{3.12}$$

m_t is the model state and a_t the action. Stability requires examining the composite map F; the second term retains the path by which the model changes the action and the action changes the update. If F maps a complete invariant domain into itself and is uniformly contractive there, iteration converges to a unique fixed point in that domain. A derivative at one point or a local linearization offers only a local assessment; environmental change, noise, and delay can alter the result. SLAM demonstrates a feasible route to building a model while moving, though convergence depends on observability, sensing, and motion conditions.

Embodied intelligence's real difficulty runs deeper than mapping. SLAM generally assumes a largely stable map: the world does not actively change itself because it is measured. This shows how a world model can bootstrap around stable objects, but does not capture action's full weight. Cups, floors, and bodies do not actively change because they are modeled either. Their difficulty is that contact leaves consequences: touching a cup changes its position, orientation, frictional state, and subsequent graspability; an unstable step changes posture and ground contact; an excessively tight grip may deform or break an object. The problem is action rewriting the next step's initial conditions, rather than an object understanding the model. This is the difficulty of medium coupling: the world is not actively reflexive, but action leaves residuals. An object has not evaded the model, yet no longer occupies its earlier state. The typical challenge for embodied intelligence is consequently strong physical coupling in a feedback loop, before highly reflexive coupling. Cups do not anticipate robots, floors do not understand algorithms, and door handles do not adapt strategies around models; contact changes their states and carries those changes into the next action. Reflexive high coupling arises when tasks include people, organizations, road users, or other actors that interpret and respond to models. Ordinary physical objects are rewritten by action even without responding to the model itself, and this already creates embodied intelligence's central difficulty. The system must keep acting in a world continually rewritten by its own actions.

The physical world does not automatically return to its starting point after each step. Text can be rewritten, code rolled back, and images regenerated, but physical actions leave consequences: a small recognition error changes the grasping posture, a contact error changes an

object's position, and a control delay amplifies the next planning error. Residuals travel along the task chain rather than clearing automatically between actions. Embodied intelligence must manage consequences so that action remains possible afterward, beyond describing a static world completely. A useful world model therefore cannot be an isolated module whose construction brings intelligence automatically. It manages consequences as well as describing objects, drawing on observation and action, prediction and correction after failed prediction, compressing past experience while tracking changes it has caused. It is an actionable compression built by an actor's continuing interventions, rather than a complete atlas drawn by an external observer.

Steps in Coupling and the Management of Consequences

Technological viscosity reveals changes in the type of model–action coupling, beyond a simple increase in model weight. The bit domain permits cheap generation and rollback, so models can remain relatively light. Atomic actions leave residuals that do not clear automatically, requiring feedback. Living systems change intervention outcomes through metabolism, compensation, adaptation, and individual differences. Social systems interpret, respond to, exploit, and even resist interventions, making the model part of the dynamics. A step separates medium from high coupling, beyond a continuous dial of increasing strength: one concerns objects rewritten by action, the other objects that respond to the model. Their central difficulties are rolling residuals and reflexive drift, respectively. The heavier the world, the less a model can remain merely descriptive; the more objects respond to the model, the less it can pretend to stand outside the world. The crucial question is how and in which medium a model couples to action. Low coupling lets models flow like language; strong physical feedback loops require online correction during action and tracking of its residuals; highly coupled reflexive tasks make models part of the world's dynamics. A world model is a family of structures anticipating action, varying with task, medium, consequence intensity, and object reflexivity, rather than one uniform thing.

This also gives circular dependence a new meaning. Circular dependence is normal for intelligence embedded in the world, rather than a defect to eliminate completely. A model is more than a pure image preceding action, and action more than mechanical execution following a model: action validates and invalidates models, while models guide actions that change the world they must face. Intelligence learns to remain correctable, recoverable, and able to continue acting within the loop. Embodied intelligence therefore needs continuing correction of consequences more than an ultimate map: it can act without possessing a complete world, but must know how its action changes the world encountered next; it need not contain all reality internally, but must maintain stable relations between action and feedback for relevant tasks. Failure occurs when the world after action moves beyond what the system can continue handling, rather than simply because the model lacks omniscience. This judgment has limits: if task outputs scarcely change subsequent objects, errors are cheap to roll back, and feedback

can be filtered externally, a world model may remain weaker, looser, and more symbolic. As action increasingly rewrites subsequent states, residuals roll along task chains, and recovery becomes harder, stronger model–action coupling is needed. The world model then becomes an online compression of action consequences rather than a static internal representation.

From this perspective, AGI may be better understood as something other than a complete subject arriving suddenly. A more plausible route gradually assembles capabilities within feedback loops: language models provide symbolic compression, agents task execution, tool environments external memory and computation, embodied systems physical feedback, long-term interaction causal data, and verification mechanisms corrective constraints. Larger actionable structures form across multiple loops rather than erupting from one point. This less romantic route is more realistic: it requires neither complete consciousness at the outset nor prior mastery of the world's ontology, but the ability to act with limited models, obtain feedback, correct models, and expand the range of action afterward. Whether large models resemble parrots matters less than whether the "parrot" connects to tools, memory, feedback, a body, and real tasks. Repeating language in a cage is mainly imitation; gaining feedback through action, revising plans, and turning failure into subsequent adjustment begins intelligence's bootstrapping. Managing the changes it causes and tracking objects rewritten by its own actions reaches the core of embodied intelligence. The fundamental difficulty exceeds the world's size or the model's weakness: "world model" can make us forget that intelligence can possess a world only from within it, obtaining consequences through action, correcting itself through those consequences, and acting again afterward. Intelligence acquires its world through action, and the world it acquires has already been changed by its own actions.

3.6.3 Intervention-Ready Compression: Causality and Scale

Saying that models must learn from the world's responses is simple; implementing it requires answering an earlier question: what kind of compression captures those responses? Compression here is a basic way finite intelligence handles the world, beyond a data-processing technique; it determines how broadly a model remains useful. A more specific question follows: how far can compression go while capturing the world's responses rather than merely remembering its traces? Large models sharpen an old question: if intelligence is compression and the real world itself is full of probabilities, can probabilistic models represent it? There is no simple yes or no. Probability is not the obstacle: quantum measurement, thermal noise, chaotic systems, and limited observation show that descriptions at many scales must be probabilistic. The issue is whether reality constrains those probabilities, and whether the model can predict the world's response after action, beyond generating plausible continuations. **Large models mainly learn the traces the world leaves; the threshold for world models is learning how the world responds to intervention.** Traces can be copied, replayed, sliced, annotated, and trained on repeatedly; newly measured responses require contact with the world, bring-

ing cost, delay, noise, risk, and irreversibility, while existing response records can be reused. Text, images, video, and code are traces from experience, observation, institutions, narratives, experiments, and operations. They are not the world itself: recovering many structures from them does not automatically confer the ability to change real systems.

Intervention-Ready Compression

Compression is a basic way finite intelligence handles the world, beyond a technique used by large models. Reality's state space cannot be copied completely, and finite intelligence keeps no high-resolution duplicate. It dynamically selects variables to retain, ignore, or observe again according to task, error, and risk requirements. Particles, materials, boundaries, historical paths, noise, others' actions, and structural coupling coexist, yet any action occupies a tiny slice. Intelligence selects a few action-relevant state variables in an uncopiable world; without this selection, perception, concepts, causality, planning, and action would be impossible. **intervention-ready compression** means coarse-graining the world's responses when its entire state space cannot be copied: discarding details that do not dominate transitions at the present scale and retaining variables effective across interventions, boundary changes, and counterfactual comparisons. Coarse-graining selects effective variables for action rather than simplifying arbitrarily. It discards details that do not affect the current task's results and retains structures that change transitions; a world model is thus an action model valid at a specific scale, for a specific task and boundary conditions, rather than a copy of the world.

Causal inference already offers an explicit framework distinguishing observation, intervention, and counterfactuals (Pearl 2009). Records can contain experimental and mechanistic knowledge, and models may learn causal structures from them; accurate prediction of records does not automatically establish an intervention's reliability in reality. Intervention-ready compression requires making that reliability concrete for specified tasks, scales, and feedback.

Causality and Scale

Causality enters here. A world model needs structures that change state transitions under intervention, rather than every co-variation: correlation indicates what often appears together, while causality indicates which changes alter outcomes. A drug need not work in every patient to change the distribution of benefits and side effects; a control signal need not reproduce a robot's trajectory exactly to change the direction of transitions. Causality need not be a deterministic chain; it is a minimal structure that acting systems must retain when compressing the world's responses. Causality also requires counterfactuals and scale selection. Actors must compare acting, refraining, choosing B over A, acting later, or acting differently; planning, correction, and judgments of responsibility require these comparisons. Causal variables differ across molecular, mechanical, task, and organizational scales. The difficulty is identifying variables that change state transitions at the current scale rather than treating every correlated clue as a cause.

A local first-order sensitivity condition can provide an initial screen for which variables to retain or discard:

$$\left|\frac{\partial f}{\partial x_i}\right| \sigma_i > \theta \tag{3.13}$$

f is a selected output component of the state transition, x_i a candidate state variable, and σ_i its typical fluctuation amplitude within the current task and time window. The local sensitivity $\partial f / \partial x_i$ multiplied by that amplitude approximates the output change caused by one variable's fluctuation, with the same units as the tolerance θ; division by θ makes the comparison dimensionless. Variables exceeding the tolerance deserve priority for retention; those below it cannot simply be discarded, because nonlinear interactions, correlated fluctuations, accumulation of small terms, and rare tail risks can change the conclusion. The condition is a screening clue: it neither identifies causality nor proves a granularity minimally sufficient. Usable granularity still depends on task, timescale, risk level, and coupling scope. Too coarse a scale misses variables that change outcomes; too fine a scale drowns action in irrelevant detail. Developing these conditions into testable quantitative criteria remains an open problem, informed by modeling, exploration, failure, and revision.

Causal structures need not always appear in clear, stable, low-dimensional forms. With strong noise, strong chaos, or irreducible randomness, small disturbances may amplify rapidly and individual outcomes resist precise prediction. Variables governing transitions may not compress into simple relationships; models then retain probabilistic constraints, robust boundaries, and risk distributions rather than deterministic chains. Intervention-ready compression assumes no universally masterable causal structure. At a given task and scale, it seeks to retain structures that can still change outcome distributions, constrain transitions, or bound failure risk. Probability is compatible with causality; in many real systems, causal structure can be retained only probabilistically. A good world model therefore knows which things cannot be ignored at the current scale, rather than simply knowing the most. For grasping a cup, molecular detail can usually be discarded, while friction and center of mass cannot; for material fatigue, microcracks matter; for congestion, individual neural activity can be omitted, while aggregate flow and response delays must remain. Patterns and constraints should not be opposed simplistically. Constraints are stable structures that continue limiting transitions under changing contexts, scales, and interventions. Frequent patterns indicate what usually happens; constraints indicate the bounds of validity, what cannot hold, and how failure follows a boundary crossing. A model that learns frequent surface patterns without structures constraining transitions can generate plausible scenes without reliably guiding real action.

Frontier Approaches and a Shared Threshold

Video models, interactive models, vision–language–action systems, simulation training, and online feedback all connect records with responses to action. Google DeepMind's Gemini Robotics 2 work in July 2026 further combined whole-body action with high-level capabilities

(Parada 2026). The common interfaces requiring examination are task, hardware, feedback, recovery, and evidence of generalization.

The same underlying problem appears: records can be copied; responses must be paid for. Text, images, video, and demonstration trajectories can be replayed at scale, while real responses to action bring cost, delay, noise, risk, and irreversibility. Simulation reduces trial costs without eliminating the gap between simulation and reality (Tobin et al. 2017). Rewards can drive policy optimization without guaranteeing coverage of long-term risks; online interaction can reveal errors, but in atomic, biological, and social systems many errors are themselves costs. The threshold for world models is their ability to turn limited, expensive, risky responses into generalizable, verifiable judgments of action whose consequences can be borne, beyond coherent appearances, interactive environments, or complete action interfaces.

Reversibility also constrains compression. Risk is lower when discarded variables can be observed again, higher when restoring them requires expensive trials; variables cannot be discarded lightly when action leaves a state that cannot be restored. Errors in bit space can often be undone, while those in atomic, biological, and social systems may leave irreversible consequences. Intervention-ready compression chooses how much to discard according to error recoverability, verification cost, and irreversible consequences, rather than discarding as much as possible. Together these conditions set different thresholds for the same criterion across media: omitting contact details in simulated grasping incurs almost no direct physical damage, while doing so on a real production line may damage workpieces or injure people.

Viscosity and the World's Responses

This is where technological viscosity appears. The world's responses are feedback filtered through resources, time, and structure, rather than free information. Resource constraints determine the energy, materials, space, compute, equipment, maintenance, and redundancy costs of obtaining one real response; time constraints determine when it appears and whether waiting, aging, training, feedback, or recovery cycles resist compression; structural constraints determine how a local action passes through a coupled network to be amplified, canceled, routed back, or locked in. Technological viscosity is the cost structure of the world's responses themselves, rather than resistance outside the world model. A useful world model judges transitions under resource, time, and structural constraints, beyond generating the world's appearance.

High coupling and multi-agent systems deepen the problem. Other actors adjust expectations, strategies, and constraints in response to the model's actions. The model must calibrate a response distribution changed by its own intervention rather than transitions in a fixed environment. Entering the system makes the model part of it: its predictions change the world's responses to it. This is reflexive drift, discussed above, appearing in world modeling. Intervention-ready compression is especially vulnerable when its object responds, learns, and changes the model's inputs in return rather than remaining a static environment.

Verification checks whether compression retains the structure action needs, rather than serv-

ing as an external patch: the coarse-graining discussed here discards information, and the question is whether that information becomes important again during action. Failure can reveal this boundary: failed grasping may point to omitted contact, friction, posture, or sensor variables; driving errors to rare scenes, occlusion, judgments of intent, or feedback delays; drug failure or toxicity to omitted metabolic paths, dose relationships, or long-term effects. Failures can also realize known risks or result from actuator faults or stochastic outcomes; omitted variables cannot be inferred from every failure. Compression selects what to discard; verification checks when those choices make action pay a price.

This is also why world models need feedback loops. A fixed corpus can record how the world responded to interventions; an action loop lets the system keep testing those relationships under current objects, policies, and conditions. Every action exposes the model's internal compression to external reality; new feedback can test and revise its selection of state variables. Learning a world means gradually finding structures sufficient for action through intervention, failure, correction, and recompression, rather than acquiring a complete picture at once. A world model is a transition model continually calibrated by the world's responses rather than a static knowledge base. Intervention-ready compression can therefore be defined as coarse-graining the world's responses to interventions into variables, relationships, and structures that still support stable judgments of transitions, constraint boundaries, counterfactual consequences, and failure modes, when intelligence can neither copy the entire state space nor try actions without limit. This definition describes the conditions world models must meet when entering real systems rather than replacing the concept: they must retain sufficiently reliable action structures from limited, expensive, risky responses. Probability uncalibrated by the world's responses undermines fidelity to reality; compression unable to withstand intervention tests undermines action. Probability and compression themselves are not the obstacles.

Judging Actions from Limited Responses

Large models have demonstrated something important: the traces left by the world can be learned to an astonishing extent. World models face a harder demonstration: can they learn a defensible criterion for action from limited, expensive, risky responses? The former occurs in continuations of symbols, images, and video; the latter in intervention, cost, failure, and correction. The leap is from learning the world's traces to learning its responses to action, rather than from probabilistic to nonprobabilistic models. Records can teach how the world is described and how it has been changed; new responses to action continue testing when that knowledge still holds. Intelligence entering reality must learn how the world can be changed as well as how it is described.

The cost of obtaining a response can be estimated from the accounts of pharmaceutical research and development. In 2016, using a sample of 106 new drugs from ten pharmaceutical companies, DiMasi, Grabowski, and Hansen estimated that out-of-pocket R&D expenditure averaged about US$1.395 billion per approved new drug in 2013 dollars; capitalized to the date

of approval at a real discount rate of 10.5 percent, the total was about US$2.558 billion, also in 2013 dollars (DiMasi et al. 2016). Typically, around 10 percent of compounds entering clinical trials ultimately receive approval, with a still lower proportion for drugs targeting the central nervous system. These figures describe the same reality: a record can be copied indefinitely, but each real response must be paid for, project by project, year by year, with failures factored in, and payment does not necessarily buy a usable conclusion.

Once the problem reaches this point, models alone are no longer the subject. Responses require bodies to contact the world, and residuals require bodies to absorb them. Compression criteria determine which variables a model retains; bodies determine what corrects them during contact, with what delay, and whether recovery remains possible after failure. A criterion calculable within a model must be borne through materials, structures, and reflexes in physical contact: this is the problem addressed by the tradition of morphological computation. Bodies are only one layer of these supporting structures. On much longer timescales, another layer performs a similar role: generations replace sensors and reflexes in recording which variables cannot be discarded, while reproductive failure tests which compressions were wrong. Its verification cycles are measured in generations, its errors cannot be rolled back, and variables once encoded impose continuing costs on subsequent actions. Judging how far intelligence can go in physical media first requires understanding how this older mechanism of compression works.

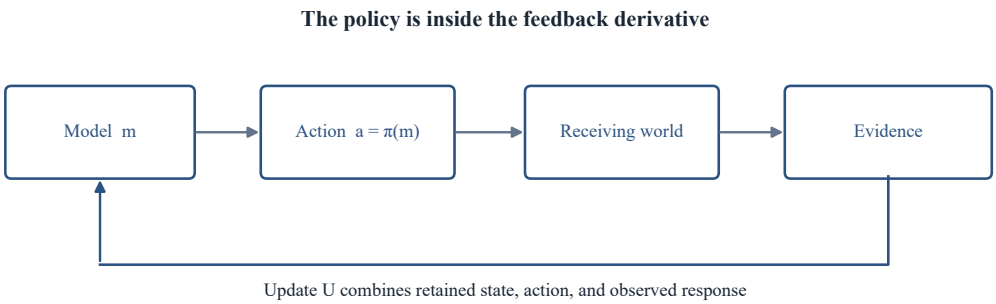


Figure 3.5 A model changes actions through policy, and actions change the evidence for the next update. U combines the retained state, actions, and environmental responses; the derivative of the composite map includes both a direct term and an indirect term through policy.

Further reading

W. Brian Arthur, *The Nature of Technology: What It Is and How It Evolves*

Thomas P. Hughes, *Rescuing Prometheus: Four Monumental Projects That Changed the Modern*

World
Per Bak, *How Nature Works: The Science of Self-Organized Criticality*
Sidney Dekker, *Drift into Failure: From Hunting Broken Components to Understanding Complex Systems*
Stewart Brand, *How Buildings Learn: What Happens After They're Built*
Andy Clark, *Surfing Uncertainty: Prediction, Action, and the Embodied Mind*
Sebastian Thrun, Wolfram Burgard and Dieter Fox, *Probabilistic Robotics*
Judea Pearl and Dana Mackenzie, *The Book of Why: The New Science of Cause and Effect*

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Chapter 4

Life, Bodies, and Movement: The Divergence of Intelligence

Chapter 3 ended by pointing to an older mechanism of compression: generations replace sensors and reflexes in recording which variables cannot be discarded, while reproductive failure tests which compressions were wrong. Its medium responds in fundamentally different ways from mobility systems. Act on a road and the responses are friction, inertia, and queues; act on a living system and they are metabolism, repair, immunity, compensation, and memory. The difference exceeds response speed. Some parameters of mobility systems can be calibrated during design; living systems keep evolving in response to intervention, carrying consequences invisible at design time far into the future. Once a medium is understood to interpret an input, a single imagined axis can no longer answer what form intelligence should converge toward.

4.1 Life Is Not Software: The Physical Meaning of Genetic Code

4.1.1 Where the Code Metaphor Fails: Hyper-Embodied Information

Living systems display medium response most fully: an input is reinterpreted by the system itself. Changing a system requires knowing what it is. Mobility systems can be decomposed through mass, energy, and topology because their parts are connected externally: engines, roads, airspace, and regulations can each be described and replaced separately. Living systems are organized differently. Their components are materially embedded in one another; their structures have been selected and compressed by environments over long periods. The most convenient way to describe such objects is often the easiest way to get them wrong.

The Code Metaphor and Physical Execution

Comparing genes to code is a useful but dangerous metaphor. It is useful because DNA does carry information. Base sequences can be copied, mutated, read, used to influence pro-

tein synthesis, and passed between generations. Without this informational structure, life could maintain no continuity and evolution could accumulate nothing. The danger is the next inference: if genes resemble code, life resembles software; if software can be edited, debugged, and refactored, life can be rewritten in the same way.

That inference overlooks a fundamental fact: genes are not programs running independently of hardware. Their meaning depends on physical and chemical execution within cells, beyond the symbolic arrangement of A, T, C, and G. In eukaryotic cells, DNA sequence function depends on a system of nuclear and chromatin structure, transcriptional regulation, RNA processing, protein translation, folding, modification, transport, degradation, metabolic state, cell type, developmental stage, and external environment. Prokaryotes lack nuclei but have their own regulatory and physicochemical environments of execution. **Genes are not software written into the body; they are part of the body itself.** Its meaning is not simply read out; it is physically enacted in the cellular environment.

Software code's meaning is usually determined by formal rules. A function has relatively well-defined input–output relationships within a given language, compiler, and runtime. Despite software engineering's dependencies, compatibility problems, interfaces, and legacies, code's meaning can largely be analyzed through symbolic structure and execution rules. Genes differ. They have no pure meaning independent of the cellular environment. A sequence may be silenced, activated, spliced, modified, or amplified in different cells; the same protein may have different consequences under different metabolic states, tissue environments, and developmental stages. It is a physical trigger embedded in a network rather than an instruction running on its own.

Genotype to phenotype is therefore a multilayer reaction network rather than a linear translation. DNA sequence is one starting point among others. It passes through successive filters of chromatin accessibility, transcription-factor binding, RNA splicing, translation efficiency, protein folding and interactions, cellular signaling, metabolic flux, mechanical environment, and tissue structure. "One gene causes one trait" has strong explanatory power for some Mendelian traits, but complex traits and diseases usually involve multiple genes and the environment (Manolio et al. 2009). Most biological phenomena arise from network states rather than single-gene instructions. Genes supply possibilities, cellular environments determine how they are realized, and the body as a whole and external environment further change the consequences.

This is why genes' physical implementation cannot be bypassed. A base mutation can change molecular shape, binding energy, folding pathways, reaction rates, stability, spatial conformation, and regulation, beyond replacing an abstract character. Changes in binding energy can be written as a concrete quantity:

$$\Delta\Delta G = RT \ln \frac{K_{\mathrm{d}}^{\mathrm{mut}}}{K_{\mathrm{d}}^{\mathrm{wt}}} \tag{4.1}$$

$\Delta\Delta G$ uses the convention "mutant minus wild type" for binding free energy, and K_{d} is the

corresponding dissociation constant. The ratio is dimensionless. A tenfold increase in the mutant's K_d indicates weaker binding and a positive free-energy difference, approximately 5.9 kJ/mol at 310 kelvins. A local sequence change can thus substantially alter affinity, while functional consequences still pass through concentration, kinetics, localization, and network state.

Biological "semantics" is thermodynamic and kinetic rather than dictionary meaning. A sequence's function ultimately appears in its changes to material, energy, and signal flows and structural stability. Genes change local boundary conditions within a complex physicochemical system rather than commanding a machine to act. This distinguishes living systems from ordinary symbolic systems. Language tolerates substantial ambiguity because interpreters supply context; software can operate in strict environments because its rules are designed to be discrete, explicit, and traceable; life operates amid continuity, noise, thermal disturbance, and strong coupling. Temperature, pH, ion concentrations, protein conformations, molecular crowding, membrane structure, mechanical pressure, and stochastic fluctuations (Ellis 2001) can all affect the actual consequences of the same genetic structure. Life maintains order in warm, wet, crowded, noisy matter rather than executing code in a clean logical space.

Hyper-Embodied Information

If execution environments determine genes' meaning, those environments cannot be mere background: the selection constraints imposed by long adaptation leave traces in genetic structures. More strongly, life has a property of **hyper-embodiment**. Ordinary embodied intelligence gives a model a body through which sensors and actuators connect it to an environment; life does not begin with an abstract model installed in a body. Life's informational structures are themselves deposits of prolonged interaction between bodies and environments. Genes record more than how to build a body. They compress traces of billions of years of environmental selection: which molecular structures remain stable, metabolic pathways sustain energy supplies, modes of perception and action improve survival, and developmental programs remain viable under particular temperatures, nutrition, gravity, microbes, predators, and reproductive pressures. Genes are history engraved within life by its environment rather than blueprints detached from that environment.

Genetic information is thus interactional information retained through natural selection, genetic drift, mutation, and historical dependence, beyond symbolic information in the ordinary sense. It contains viable structures left by repeated trials in a particular physical world rather than a complete world model. Eyes are shaped jointly by light, media, nerves, movement, predation, and escape, beyond an isolated "eye-building code"; lungs reflect constraints from atmospheric composition, metabolic demands, body architecture, and circulation, beyond sequence-specified parts; immune systems are dynamic legacies of prolonged encounters with pathogens, symbionts, tissue damage, and environmental exposure, rather than isolated programs. Life's "code" is material memory shaped jointly by environmental pressures, bodily

structures, and historical selection from the outset; it also retains neutral changes and historical contingencies (National Human Genome Research Institute, n.d.). This makes gene editing different from software modification. Software usually runs in environments designed and kept relatively stable by people; gene editing changes a system that has internalized environmental history within its structure. A sequence may persist because of hidden synergies formed over evolution, or through neutral retention, drift, and linkage; continued existence alone does not establish usefulness. It may participate in developmental timing, metabolic buffering, immune regulation, tissue stability, or stress responses. We see a sequence but touch a compressed layer of long coupling between life and environment. The older and deeper this layer, the less readily it reduces to a clear engineering table of functions.

4.1.2 Evolutionary Layering, Pathway Regulation, and Four Entanglements

Evolutionary Layering and Topological Rigidity

Life's hyper-embodiment also means that living systems were not designed from scratch. They accumulated through evolution. Ideally, engineering can begin with an objective, choose a clean architecture, remove redundancy, and rebuild foundations. Life did not arise this way. Living systems are historical deposits of prolonged evolution. A structure sufficiently usable in one environment may be retained; a mechanism serving one function may be repurposed; a local patch may later become an indispensable interface; a contingent path may become irreplaceable as countless subsequent structures come to depend on it. Life accumulates viable designs rather than achieving an optimal design.

This evolutionary layering fills living systems with functional legacies. They may be neither simple nor elegant, nor satisfy an engineer's desire to refactor, yet remain deeply embedded. An apparently redundant pathway may provide a backup under stress; an apparently inefficient mechanism may regulate, buffer, or serve a hidden developmental function; an apparently irrelevant segment may affect chromatin, expression timing, or distant regulation. Biological "scrap" is often part of a historical structure rather than entirely useless waste. It may contribute to overall stability without having one clear function. This is one source of life's topological rigidity.

Long-standing dependencies entangle living systems, beyond assembling independent modules. A local change may trigger distant consequences; a beneficial short-term edit may alter long-term development; an enhancement at one point may disrupt metabolism; inhibiting one target may trigger compensation through bypass pathways; a local repair may redistribute pressure throughout the system. The difficulty exceeds incomplete knowledge: the mechanisms themselves have multilayer coupling, historical dependence, and state-dependent effects. Such rigidity comes from the density of dependencies among known structures, rather than a particular unsolved problem. Rigidity does not mean closure: medicine and biotechnology can

change life precisely because pathways remain available.

Pathway Regulation and Four Entanglements

Hormones can change whole-body states, neurotransmitters behavioral tendencies, receptors signal amplification or suppression, enzymes metabolic flux, immune checkpoints immune responses, transcription factors cell fate, and delivery systems access to particular tissues. Life is a complex network with entry points, levers, vulnerabilities, and control surfaces, rather than an inaccessible black box. The crucial distinction is between pathway-based regulation and software-style rewriting. Drugs change system states through receptors, enzymes, ion channels, signaling pathways, immune nodes, and metabolic nodes rather than rewriting an overall bodily program; gene editing changes local structures in specific cells, at specific times, under specific delivery conditions, rather than freely reconstructing life. Effective interventions often find a sufficiently important, accessible control point with tolerable side effects to move the system between states, rather than rewriting everything.

Pathways have different physical forms. Many small-molecule drugs depend on shape and interaction matching with receptors, enzymes, or other targets; antibodies on accurate epitope recognition; gene editing on carriers' distribution across tissues; cell therapies on homing and survival after entering the body. These material forms constrain intervention precision and whether the same molecules can reach the same location in different people. A "target" therefore implies an entire path to be traversed inside the body, beyond a molecular site.

Pathway capacity and selectivity can be quantified. Adeno-associated viral vectors are important gene-delivery tools, with a conventional single-vector genome capacity of about 4.7 kilobases, including regulatory sequences. Oversized payloads reduce intact packaging and expression efficiency; splitting payloads adds requirements for reconstitution and co-delivery. Tissue distribution depends on capsid, dose, and route, with liver uptake particularly prominent after intravenous delivery (Naso et al. 2017). Lipid nanoparticles offer more flexible mRNA payload designs. Transient expression can make a lasting gene edit or, through repeated dosing, sustain protein replacement, so it does not rule out long-term treatment. Clinical trials of the propionic-acidemia candidate mRNA-3927 used repeated dosing; in January 2026 its developers announced a collaboration for further development and commercialization upon approval. It remained investigational at that point (Koeberl et al. 2024; Moderna 2026). Repeated dosing has immune and tolerability constraints of its own: a 2026 mouse study observed reduced expression efficacy after repeat administration of the PEG-containing formulations tested, within those formulations and experimental conditions (Arral et al. 2026). Delivery influences which cells can be reached, how long effects last, and how often dosing can recur. Constraints extend beyond editing-tool precision: pathways' physical forms help set the limits of intervention effects.

Biotechnology's central difficulty is finding intervention paths whose consequences can be borne, beyond whether life can be manipulated. A good target must meet several conditions:

importance within the disease network, accessibility to a drug or editing tool, stable benefit after modification, tolerable side effects, no destructive long-term feedback, and individual differences that do not make risk unmanageable. This is a demanding set of conditions. Life can be regulated, but intervention requires understanding pathways, networks, environments, time, and consequences together.

Technological viscosity identifies four entanglements behind the high viscosity of living media. The first is physicochemical. Gene function must be realized through molecular folding, reaction rates, binding energies, and spatial conformations. Symbolic changes must pass through material execution. The second is network entanglement. Genes are nodes in regulatory, metabolic, signaling, and developmental networks rather than isolated instructions. The system can amplify, cancel, or transfer local changes.

The third is historical entanglement. Living systems are evolutionary inheritances rather than machines designed anew. Many structures exist because past constraints retained them and later structures came to depend on them, rather than because they are optimal. The fourth is pathway constraint. Regulation must enter through accessible pathways: delivery routes, target specificity, dose windows, timing windows, tissue distribution, and feedback boundaries determine whether an intervention works. Together these four entanglements explain why genes are not software, life is not a clean code repository, and the body is not a freely replaceable hardware platform. Genes, bodies, environments, and history are embedded in one another at the atomic scale. Life's code has its full meaning only within living matter.

This does not deny genetic technology's power. Its ability to reach foundational biological structures gives it great capacity for change. Greater capability makes reducing life to editable text less defensible. Mature biotechnology recognizes the physical, networked, historical, and pathway-dependent nature of genetic meaning, rather than treating bodies as inferior hardware. It asks which pathways an edit will use, in which state, which networks it will disturb, and who will bear long-term consequences, beyond whether editing is possible.

Genes are not software. They are physical structures executed by living systems, and material memories compressing environmental history. Life can be manipulated, but not rewritten at will. The key is finding tolerable regulatory paths within hyper-embodied evolutionary systems, rather than imagining changes to code.

The four entanglements show that intervention must act through pathways. The existence of a pathway answers only half the question. Once used, it brings intervention into the system's own operations: metabolism, compensation, repair, immune recognition, and consequences unfolding over time that remain invisible initially. Whether a regulatory path tolerable at design time remains tolerable after the system responds requires another analysis.

4.2 Shear Thickening in Life: Why Faster Intervention Can Meet Greater Resistance

4.2.1 Responsive Systems, Pathways, and Control Surfaces

A tolerable regulatory path at design time need not remain tolerable once used. Living systems respond to the change itself, rather than passively receiving it. This changes the intervention problem: stronger external input need not produce greater effects.

Responsive Systems

Living systems are no ordinary load. Force a piece of metal and it deforms, fractures, or recovers; modify code and it compiles, reports an error, or runs; intervene in a living system and it interprets the intervention. It metabolizes, mounts immune responses, repairs, compensates, adapts, remembers, and rebounds. Life is a dynamic system tending to maintain itself, rather than passive material receiving external input. An input changes states and also triggers responses to the change itself. This is the primary difficulty of biological intervention: the object is a responsive system rather than a static object.

This does not make life inaccessible to intervention. Medicine, drugs, vaccines, surgery, hormone treatment, immunotherapy, gene therapy, and regenerative medicine have all demonstrated profound biological change. Living systems have pathways, targets, windows, and levers. The crucial questions are how manipulation enters, whether the system absorbs it, and whether the consequences can be borne.

Many technological visions assume that sufficiently precise targets would let us adjust life like a machine. Suppress an overactive pathway, replace a missing protein, repair an abnormal gene, or restore an unfavorable metabolic indicator. This works in some cases but underestimates biological networks. Living systems do not always accept a change at one point. They may bypass an inhibited pathway, amplify backups, change receptor sensitivity, adjust upstream inputs, shift metabolic pressure, or develop tolerance and rebound over time.

Biological intervention consequently often encounters nonlinear resistance. Low-intensity, infrequent intervention aligned with system rhythms may be absorbed; high-intensity, frequent intervention that overruns feedback cycles may sharply increase resistance. Rheology offers an analogue. **shear thickening** is non-Newtonian behavior in which apparent viscosity rises with shear rate, sometimes sharply over a narrow range in concentrated particle suspensions. Lubrication, interparticle repulsion, stress-activated frictional contacts, dilation, and boundary confinement jointly determine flow, sometimes approaching jamming; particles simply "failing to move past one another quickly enough" is an inadequate explanation (Brown and Jaeger 2014). Over a limited range, a power law can describe this behavior:

$$\eta(\dot{\gamma}) = \eta_b + k\dot{\gamma}^{\,n} \tag{4.2}$$

Here η and η_b have units of pascal-seconds, $\dot{\gamma}$ of inverse seconds, and n is dimensionless, giv-

ing k units of $\mathrm{Pa\,s^{n+1}}$. The expression describes a thickening branch over a specified interval; low-rate thinning, abrupt jamming, and complete flow curves require further mechanisms and forms.

With $k > 0$ and $n > 0$, viscosity increases with rate. The point at which the plateau and power-law terms are equal defines a crossover scale within the model:

$$\dot{\gamma}_c = \left(\frac{\eta_b}{k}\right)^{1/n} \tag{4.3}$$

The right-hand side has units of inverse seconds. This crossover is defined by equality of the two terms and need not coincide with the measured onset of thickening. Biological use also requires separate definitions of intensity, frequency, duration, and measured "resistance," and identification of the compensation, saturation, or damage mechanism producing a similar curve.

For a particular intervention, we can test whether increasing intensity or pace reduces marginal benefits or raises compensation and side effects. Dose, frequency, and speed are different variables; some interventions must be sufficiently strong or timely to work. Shear thickening proposes a response hypothesis here; mechanisms and measurements determine the actual relationship.

Pathways and Control Surfaces

Because resistance comes from internal feedback structures, the point of intervention becomes decisive. Drug treatment offers the most direct example. An entering drug encounters a system of absorption, distribution, metabolism, excretion, immunity, tissue barriers, receptor regulation, and individual differences, beyond one target. Improved short-term indicators do not establish improved long-term system states, effective inhibition at one point does not establish lower overall risk, and higher doses may produce toxicity, side effects, tolerance, counterregulation, and systemic pressure rather than linear benefits. Living systems are deformable webs rather than straight lines.

Drugs work precisely because this web is heterogeneous. Some nodes offer greater control. Receptors, enzymes, ion channels, transcription factors, immune checkpoints, hormonal axes, and metabolic bottlenecks can all provide regulatory pathways. Much medical progress finds critical nodes within complex networks that can be reached, amplified, and adjusted safely, rather than suppressing life by force. A good target must be accessible, adjustable, verifiable, and tolerable as well as important.

Biological intervention thus changes system states through local channels rather than directly controlling the whole. An effective drug need not understand every bodily detail, just as a rudder need not push every steel plate in a ship. The task is to find a control surface that can change direction. But biological control surfaces are not stable. They change with age, disease stage, genetic background, tissue environment, immune state, and prolonged drug use.

A pathway effective today may be compensated tomorrow; a target effective in one person may produce greater side effects in another.

Gene editing follows the same logic. Editing tools can grow more precise while life's responses remain complex. A local sequence change has consequences passing through expression, folding, regulation, cellular environment, developmental state, and long-term feedback. Editing is a beginning. The living system reinterprets the result within its own operating network. A change apparently correct in the short term may reveal new consequences across tissues, ages, genetic backgrounds, or environmental stresses.

Gene editing therefore requires delivery, appropriate changes, stable expression, controllable feedback, and long-term safety, beyond cutting accurately. Delivery systems, tissue specificity, editing efficiency, off-target effects, immune responses, cellular selection pressures, and developmental timing all belong to the intervention pathway. Deeper biotechnology requires pathway design rather than treating the problem solely as editing precision.

The immune system illustrates the dysregulated responses examined through this analogy with shear thickening. Immunity is a highly sensitive recognition and regulatory system, beyond a simple defense mechanism. Insufficient responses fail to protect, excessive ones can injure the body; inadequate or delayed responses can be overwhelmed by infection, while excessive, dysregulated activation can contribute to cytokine storms (Fajgenbaum and June 2020). Intervention mismatched to immune timing and intensity can turn treatment into disturbance. The difficulty is the abundance, complexity, and timing dependence of life's responses, rather than an absence of response.

Biotechnology consequently cannot simply copy software's iteration cycle. Software can be released frequently, rolled back, tested on limited deployments, and patched quickly; biological interventions cannot freely repeat high-frequency experiments on people. Some software errors can be isolated and recovered within information systems, while biological errors enter bodies, tissues, metabolism, and long-term risk directly; software controlling medical devices, transport, and infrastructure also has physical consequences. One wrong intervention may alter irreversible development, leave lasting injury, or emerge later through immune memory, cell states, or ecological relationships. Living systems offer no unlimited supply of cheap rollbacks.

4.2.2 Sources of Viscosity, the Cost of Robustness, and the Limits of Online Maintenance

Sources of Viscosity and the Cost of Robustness

Technological viscosity traces the proposed biological thickening to three sources. The first is temporal rigidity. Biological feedback takes time. Drug effects must unfold, side effects be observed, immune responses develop, and tissues repair; development and aging emerge over longer cycles still. Interventions judging only short-term responses can mistake long-term

feedback that has not yet returned for feedback that does not exist. Life often hides costs in time.

The second is structural coupling. Biological pathways form networks of connection, compensation, and mutual constraint rather than isolated pipes. Changes at one point can propagate to distant sites, and local benefits may carry global costs. Greater complexity makes improvement in one indicator less adequate as evidence of an improved biological state.

The third is maintenance pressure. Life is a dissipative structure far from equilibrium, requiring continuous energy input, damage repair, waste removal, boundary maintenance, and regulation of internal states. Many critical biological functions cannot wait for the whole system to shut down for maintenance. The heart, immunity, nervous system, metabolism, endocrine regulation, and cellular repair must operate during maintenance. Much repair takes place while the system continues performing its functions. This constrains intervention intensity and precision: too weak has no effect, too strong disrupts operation, too fast outruns feedback, and too slow misses the window.

This also illuminates the problem of drug-development efficiency. Modern biotechnology, automated experiments, and AI can greatly expand the candidate space. Targets can be discovered, molecules generated, structures predicted, and experiments screened faster. Accelerating discovery does not automatically remove bottlenecks; they shift toward biological verification, compensation, and long-term safety. More candidates create screening and verification pressure, and greater target precision still requires understanding system feedback. Without corresponding improvements in verification and selection, candidates advanced faster may also push failure costs into later clinical stages. Eroom's phenomenon (Scannell et al. 2012) has no single simple explanation, but reminds us that better discovery tools do not automatically penetrate life's highly viscous verification layers.

Drug-development time and costs vary with disease, trial design, allocation of failures, and capital accounting. A company-sample study in 2016 offered a relatively high capitalized estimate, rather than a fixed price for every new drug (DiMasi et al. 2016). A more recent observation is rentosertib: its developer announced the first patient dosed in its Phase III program in September 2026 (Insilico Medicine 2026). Reaching that stage is progress; Phase III results still require subsequent observation.

The gap between discovery and verification can be measured in work in progress:

$$L = \lambda W \tag{4.4}$$

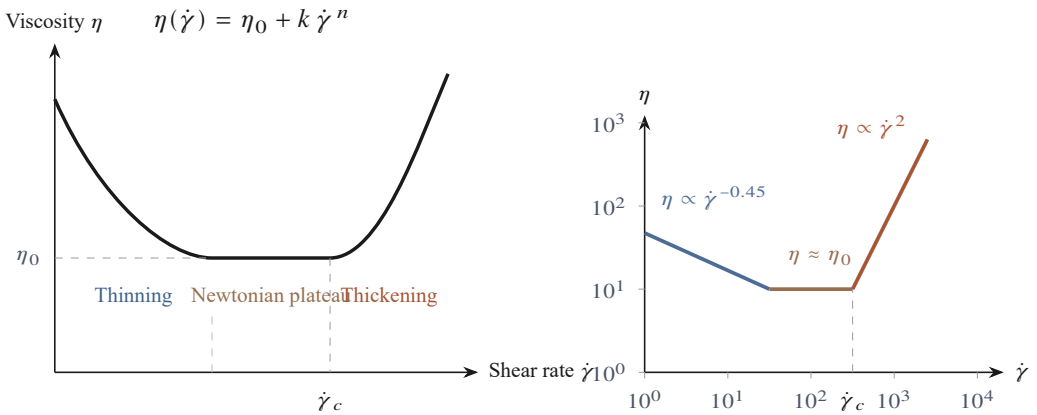
L is average work in progress within a stable process, λ average throughput across the same boundary, and W average time in the process. This is Little's relationship; exits, withdrawals, and early failures must be counted consistently. Changing arrivals, capacity, selection, and time in process jointly affects inventory and output. When arrivals persistently exceed service capacity and the queue keeps growing, a steady-state average cannot directly describe the entire

process.

Alongside these three sources, life has a property working in the opposite direction. Robustness cuts both ways. It lets bodies resist disturbances, repair damage, and maintain stability while making external intervention less likely to have linear effects. A fragile system is easy to change and easy to collapse; a robust system survives and resists change. Life can be hard to treat precisely because it is still trying to maintain itself. Many chronic diseases are states of bodily networks formed under particular historical, genetic, environmental, and behavioral conditions, rather than errors that can be independently deleted. Treatment moves the system between stable states rather than simply deleting a mistake.

This transition need not be blind. Systems medicine, network pharmacology, precision medicine, and synthetic biology seek to identify its pathways. They recognize structures, nodes, and control surfaces embedded in feedback networks and temporal processes, rather than treating life as an unknowable whole. Effective intervention often best matches the system's state rather than applying maximum force: the right target, dose, timing, population, and verification framework help move a system beyond a pathological stable state.

The technological intuition that faster is better consequently often fails in biology. Intervention can outrun relaxation times; abrupt indicator adjustments trigger counterregulation; stronger local control move pressure elsewhere; and greater candidate generation accumulate before verification bottlenecks. Biological systems require technology to enter their rhythms. Good biotechnology understands feedback, redundancy, compensation, timing, and irreversibility rather than treating life as slow software. So far, shear thickening is a cross-sectional judgment; its limiting form over time remains to be examined.



(a) Three-regime shape (b) Slopes on logarithmic axes

Figure 4.1 Shear thickening: low rates save effort, high rates backfire. (a) The three-regime shape; (b) the three slopes on logarithmic axes are −0.45, 0, and +2, with the same scale on both axes so that visual slopes equal numerical slopes. As intervention in a living system accelerates, its compensatory response strengthens, raising effective viscosity rather than lowering it.

The Limits of Online Maintenance and the Rhythm of Intervention

Aging can be viewed as a limit of online maintenance. From birth, organisms continually repair themselves through DNA repair, protein quality control, cell renewal, immune clearance, metabolic regulation, and tissue regeneration. Repair remains imperfect. Damage accumulates, errors remain, redundancy is depleted, regulatory networks grow less responsive, and resilience gradually declines. Aging involves maintenance pressure, informational noise, structural damage, and dysregulated feedback across a long-running system rather than one broken component (López-Otín et al. 2013). A 2026 study of mammalian transcriptomes further showed that lifespan-modulating interventions can produce opposing health-associated changes across cell types and functional modules (Tyshkovskiy et al. 2026). Improving one clock or indicator cannot replace assessing whole-body function, risk, and long-term outcomes.

Death reveals the boundary of an individual's online maintenance. For an individual, death is an end; at population and evolutionary scales, generational turnover provides renewal. Reproduction, variation, selection, and recombination continue the search at a higher level, without endlessly repairing the same individual. This grants death no romantic meaning or benefit to the individual. Many individuals face limits to online maintenance, reproduction and development can rebuild some states in offspring, and variation and selection continue changing populations; this does not establish death as a renewal program necessarily favored by selection.

This also explains the distinctive viscosity of living media. They have bit-like properties because genes can be copied, inherited, and edited; atomic properties because genes require physicochemical execution; and biological properties because execution enters feedback, compensation, repair, and evolution. Life occupies the overlap of bits, atoms, and biology, rather than a light medium. AI can accelerate informational search within it without canceling physical execution, temporal unfolding, or system responses.

Mature biotechnology must pursue more than stronger editing, faster screening, and larger models. It must find pathways, match timing, control doses, verify in layers, buffer risk, and respect feedback. Some situations permit strong intervention, others only gentle adjustment; some targets permit quick verification, other consequences require long observation; models can absorb some residuals, while others must be borne within clinical, institutional, and ethical boundaries.

Life is neither software nor an ordinary machine. It is a dissipative structure that maintains itself, interprets interventions, resists disturbances, and delays feedback. Biological intervention is not difficult because life cannot change. It certainly can. Effective change must find pathways, pass through feedback, respect rhythms, and withstand long-term verification. **Pushing faster can make a system harder to move.** This is shear thickening in life.

Shear thickening pushes costs back toward the bearer. However precise the intervention design, disturbance, compensation, irreversible consequences, and long-term verification must

fall on a system that is still running. Bearing has limits in capacity and time: disturbances absorbed, feedback postponed, and irreversible changes endured are all finite. What can be borne, by whom, and for how long are questions more fundamental than the intervention technology itself.

4.3 After Morphological Computation: Bodily Encapsulation and Evolutionary Amortization

4.3.1 Residuals in Physical Intelligence and Three Layers of Bodily Encapsulation

Something concrete must always bear the consequences. Life has embedded much residual correction in the co-adaptation of evolution, development, and environment. Machines lack that inherited structure and must absorb deviations through explicit engineering each time they act physically. Chapter 3's compression criterion selects the variables retained by a model; what corrects them during contact, with what delay, and whether recovery remains possible after failure are questions outside the model alone.

One of the easiest mistakes about physical intelligence is treating the body as a peripheral. On this view, intelligence resides in a brain, model, or controller; the body is a mechanical port executing commands. Understanding comes first, then planning, then bodily implementation. Strong enough models, enough sensors, and sufficiently precise control should let machines enter reality as humans do. The real difficulty is sustained action in open reality, beyond generating an action. Sustained action requires acknowledging that reality continually produces deviations the model has not specified in advance.

Residuals in Physical Intelligence

Reality is an open system continually producing residuals, rather than a static object awaiting execution. The model supplies actions; reality supplies deviations. Floors may be more slippery, objects heavier, cup surfaces wetter, or door handles looser than expected; people may suddenly lose balance, light change, sensors become occluded, or actuators respond late. Each deviation may be small, but a physical task is a chain rather than a single question and answer: recognition errors change grasping postures, contact errors change object positions, localization drift enters the next plan, and feedback delay amplifies control error. Residuals propagate along task chains rather than clearing automatically between actions.

A **residual** here means deviations not fully absorbed by the current model, body, controller, or operational arrangements during open physical action, rather than one error term. It includes at least four classes: model residuals, state variables omitted by world or policy model compression; perceptual residuals, estimation errors from noise, occlusion, drift, and asynchronous modalities; contact residuals, incompletely predictable friction, slipping, deformation, impacts,

and force transmission; and operational residuals, systemic burdens of exception handling, maintenance, verification, takeover, and tracing responsibility. The issue is whether residuals are absorbed promptly along continuing tasks or propagate across interfaces beyond recovery limits; open action always contains them. Embodied intelligence must therefore keep residuals recoverable across sustained tasks, beyond being able to perform individual actions.

Grasping a cup is the smallest example. Robots face room cleaning, elder care, table clearing, goods handling, construction-site collaboration, crop harvesting, and hospital medication delivery. Each links many actions, each action changes the next one's initial conditions, and the system must demonstrate recoverability over long, disturbed, consequential task chains rather than one successful attempt. The difficulty has a threshold: beyond some residual level, capability may abruptly become a stop, misgrasp, collision, loss of balance, or request for human takeover rather than declining smoothly. This focuses attention on structures bearing residuals. Human bodies' distinction lies in embedding much correction in bodily, experiential, and environmental co-adaptation rather than possessing a complete world model.

Three Layers of Bodily Encapsulation

Lifting a cup requires no prior explicit calculation of its exact geometry, friction coefficient, center of mass, surface wetness, contact angle, and force distribution: a person looks, reaches, and touches. Fingers immediately sense slipping, muscles feel weight, joints adjust posture, and tiny slips trigger automatic changes in grip through finger pads and muscles. Many errors are absorbed before reaching consciousness: this is **bodily encapsulation**. **The body is a residual-management system, not a peripheral of intelligence.** It contains many deviations from open reality within low-latency, continuously sensing, self-correcting loops.

Bodily encapsulation means that the body itself is a residual-absorbing structure, beyond simply being important. Skin, muscles, tendons, joints, bones, the vestibular system, proprioception, pain, and reflexes form a continuous system of sensing, actuation, buffering, constraint, and correction rather than appendages to the brain: soft fingers fit surfaces, elastic muscles absorb impacts, joint limits restrict dangerous movements, pain marks overload, and experience indicates ignorable details and signs of danger. The body processes part of reality directly through contact rather than sending everything to the brain and waiting for computation.

The body contains residuals by turning some physical laws into action structures, rather than circumventing them. Exchanges between gravitational potential and kinetic energy during leg swing, elastic storage in muscles and tendons, local finger-pad deformation, and joint limits excluding dangerous poses are constraints built into morphology before central compensation. They confine some errors within manageable dynamical channels. In many situations a mature body lets gravity, inertia, friction, and contact constrain, select, and correct actions rather than continually opposing them through active control. Morphological computation thus converts some residuals otherwise requiring sensing, estimation, and control into cheap constraints in

materials, geometry, and passive dynamics, beyond having a body participate in computation (Paul 2006). Passive-dynamic walking robots directly demonstrate this: suitable morphology and dynamics can organize some movement without intensive central control (McGeer 1990; Full and Koditschek 1999; Collins et al. 2005). Encapsulation occurs at at least three levels. First, material encapsulation: skin, muscles, tendons, and joints absorb errors physically. Soft finger pads enlarge contact areas, muscle elasticity buffers impacts, and bodily elasticity helps stabilize gait. This "computation" is material and morphological participation in control rather than symbol manipulation. Morphological computation matters because it includes shape, compliance, passive dynamics, and environmental coupling in control, allowing bodily structures to share complexity otherwise handled centrally (Paul 2006). Second, feedback encapsulation: vision, touch, proprioception, vestibular sensation, and muscle tension sample reality continuously during action. Pressure underfoot tests slipperiness, a touch tests looseness, and contact with another person's body triggers adjustment to imbalance. The world can be queried locally through action without being fully preinstalled. Impedance control became fundamental to robotic manipulation because contact tasks cannot be reduced to position tracking: operator, actuator, object, and environment form one coupled dynamical system, requiring control of interactions rather than isolated trajectories (Hogan 1985). Third, experiential encapsulation: countless encounters with cups, handles, stairs, wet floors, knives, hot pans, tired bodies, and crowded spaces supply actionable lossy compressions rather than a complete model. People know which details usually can be ignored and which need immediate attention. Bodies permit incomplete models; experience lets action begin somewhere other than zero (Gibson 1979). This approach connects to Moravec's paradox, embodied cognition, and morphological computation (Moravec 1988; Pfeifer et al. 2007; Varela et al. 1991; Gibson and Pick 2000; Pfeifer and Bongard 2006; Hauser et al. 2011). Moravec's paradox suggests that abstract reasoning may be easier to engineer than a toddler's sensorimotor abilities; embodied cognition locates intelligence across body, environment, and action as well as the head; morphological computation further emphasizes materials' and forms' roles in control. The more specific claim advanced here is that bodies encapsulate low-latency residual absorption rather than an abstract quantity of computation. This can too easily imply natural bodily superiority. A distinction is needed: humans externalize some residuals into handles, stairs, furniture, roads, and tools built around bodily scales, while machines internalize some through soft grippers, elastic joints, mechanical stops, and compliant materials. Humans do not only internalize complexity, nor machines only externalize it. The deeper differences concern the timescale for amortization and substrate affordances, developed in this section's final subsection. First comes another question: with so many routes to residual correction, why do machines still seem heavy?

4.3.2 Making Machine Complexity Explicit: Interface Debt

Making Machine Complexity Explicit: Interface Debt

Machines have many ways to correct residuals. Sensors expose deviations: cameras, depth cameras, lidar, touch, torque sensing, IMUs, and proprioception reveal drift, slips, collisions, and posture errors. Control algorithms correct them through feedback, impedance control, model-predictive control, and real-time replanning. Bodies absorb some through soft grippers, underactuated hands, elastic joints, compliant materials, mechanical stops, and passive stability (Mason and Salisbury 1985; Birglen et al. 2008). Stronger models anticipate residuals: vision–language–action models, world models, imitation learning, and policy models infer more states from fewer inputs and avoid dangerous actions in advance (Shao et al. 2025; Li et al. 2025). Environmental redesign reduces them through standard shelving, QR codes, dedicated fixtures, fixed routes, closed sites, tracks, V2X, and structured warehouses. Task restriction and operational support also contain residuals: cleaning, warehouse transport, inspection, surgical assistance, welding, harvesting, and delivery each narrow the task boundary, while remote takeover, safety operators, exception tickets, audit logs, maintenance staff, and insurance place residuals within larger operating structures. The question is why machines often must expose residual correction as engineering systems where humans have embedded much of it in bodies, experience, and environments, rather than whether enough methods exist.

This is **explicit machine complexity**. Making complexity explicit means more than adding sensors or judging machine hardware inferior: reality's complexity, handled continuously, locally, and with low latency in human bodies, is separated in many machines into structures that can be measured, synchronized, controlled, verified, maintained, and held accountable. A finger senses slipping where a machine may need tactile sensors, slip-detection algorithms, feedback control, and safety policies; muscles buffer impacts where a machine may need compliant materials, force-control policies, and mechanical protection; experience interprets "gently," "don't break it," and "hold steady," where machines must convert these vague constraints into action rules, risk boundaries, and responsibility records. What is connected within a body is divided into modules, bringing interfaces and further paths for residual propagation.

This can be called **interface debt**: splitting residual correction implicit in bodily loops into multiple engineered modules creates continuing structural burdens of synchronization, calibration, delay compensation, fault propagation, maintenance, and verification. Human perception, actuation, buffering, correction, and safety boundaries are not five independent systems. At contact with a cup, touch, muscle tension, joint angle, skin deformation, and experiential judgment already correct locally within one bodily loop. Variables need not all be named, errors fully expanded, or control separated into independent modules. A machine must detect slip, calibrate touch, synchronize vision and force, compensate delay, absorb actuator errors, judge safety boundaries, log behavior, verify reliability, and trace responsibility. Separating another

variable adds an interface; another interface adds a synchronization problem, failure mode, and maintenance point. If module coupling is approximated by allowing every pair to interact, interface growth can be written as:

$$N_{\text{if}} = \frac{n(n-1)}{2} \tag{4.5}$$

n is the number of explicitly separated modules and N_{if} the potential pairwise interfaces in the fully connected approximation. Both are dimensionless counts; $1/2$ avoids double-counting unordered channels. Counting at most one undirected relationship per pair gives this potential upper bound for a fully connected network; counting interface types or directions separately requires a different convention. Increasing n from five to ten raises the bound from ten to forty-five, faster than module growth. Actual dependencies are usually not fully connected, leaving many potential channels inactive and the count below that bound. Hierarchy and decoupling can limit growth; a potential quadratic upper bound is not an inevitable growth law for real systems. Making complexity explicit converts implicit complexity into explicit coupling structures, beyond merely seeing it clearly.

Interface debt is more than ordinary integration cost. Integration cost emphasizes the trouble of many modules; interface debt emphasizes propagation: perceptual errors enter control, control errors enter actuation, actuation changes the environment, and the changed environment contaminates the next perception. More modules offer more propagation paths; longer tasks let local errors become system drift. Machines become heavy through accumulating coupling debt among explicit structures, rather than one component. A subtler temporal debt lies within this: machine vision, touch, torque sensing, proprioception, planning, and actuation usually differ in sampling rates, delays, and phases. Each handles the world at some time slice, and those slices do not naturally coincide. The controller's "current state" is often an engineered object assembled from asynchronous signals through timestamps, filtering, extrapolation, and fusion. During fast contact or continuous manipulation, residuals cross misaligned temporal phases as well as modules. A system may correct an error that has just passed rather than the present one, and the next action writes the delayed correction into a new physical state. More interfaces make a stable, shared, actionable "now" harder to maintain as well as integration harder. Bodily seams need not each be named; machine seams must each be maintained.

Conversely, making these requirements explicit is not the same as throwing more hardware at them. Mature engineering does not necessarily make everything more general and complex: a compliant gripper may handle soft objects better than a precision hand with an elaborate tactile array, a fixed fixture may be more reliable than a general recognition model, a mechanical stop may be more stable than a complex software safety policy, and a closed campus may be easier to drive autonomously than open roads; the real question is not whether there are enough sensors, but who corrects residuals, where, and at what physical and engineering cost. Soft robotics and underactuated structures follow this direction: instead of merely imitating

biological shapes, they bring material compliance, contact geometry, and passive deformation into control, so that robots need not turn every contact into precise measurement and central computation; Rus and Tolley's review shows how compliant materials open routes beyond the design, fabrication, and control of rigid robots, while soft structures introduce new difficulties in modeling, sensing, actuation, and control (Rus and Tolley 2015). A body can therefore absorb residuals while introducing new explicit complexity of its own: it is another way of allocating complexity, not a free solution. If complexity must be allocated somewhere, assigning some of it to a stronger model becomes the next seemingly direct way out.

4.3.3 Prediction Cannot Remove Consequences: Task Chains and Cliff-Edge Failure

Prediction Cannot Remove the Need to Bear Consequences

Models reduce errors, avoid dangerous contacts, and let existing hardware perform more tasks. Physical feedback loops also need adequate force, speed, sensing, and recovery: contact and control structures must catch slips after they occur, and damaged components need repair or replacement paths. Prediction, execution, and recovery capabilities should be examined separately.

This does not deny the value of a better brain. Better vision, world, action, and policy models let existing sensors and actuators do more. Many tasks require neither human-level bodies nor full tactile skins and dexterous hands: clear task boundaries, relatively stable environments, and controllable consequences can let a good model and existing hardware form a usable loop. A stronger brain extends capability boundaries without abolishing them: it cannot create actuator force or material compliance from nothing, prevent all sensor drift, or remove the need to verify open-world tail risks. Stronger brains can move some complexity from sensors and hardware into models, but atomic action still requires managing remaining deviations somewhere—absorbing, transferring, isolating, recovering, and genuinely reducing or eliminating particular errors. Assessments should track whether errors decrease, where costs fall, and whether remaining deviations are handled promptly.

End-to-end models, simulation, teleoperation, and new sensing can reduce or eliminate particular faults, and can also transfer some burden into data, support, and maintenance. Residuals obey no fixed conservation law. Evaluation should record real error reductions, changes in error location, and recovery costs in long tasks. The testable mechanism is when residuals propagate and amplify, rather than a prior assertion that they can only move elsewhere.

Task Chains and Cliff-Edge Failure

The first implication of this empirical mechanism is that short task-chain successes in laboratories do not establish long-term reliability in open environments. Grasping on a standard tabletop does not establish competence in an unfamiliar home; moving among standard ware-

house shelves does not establish competence with hospital crowds, privacy, emergency priorities, and temporary closures; driving on ordinary roads does not establish reliable handling of construction, glare, rain, snow, unusual gestures, and liability disputes. Real tasks lengthen chains, add disturbances and responsibility, and circulate small errors across interfaces. The residual mechanism challenges a costless, indefinitely smooth extrapolation of embodied capability, rather than the possibility of reliable embodied intelligence.

Clearing a household table looks simple but forms a task chain rich in residuals. First a robot identifies cups, tissues, medicine boxes, keys, charging cables, glasses, a half-full glass of water, and fragile ornaments. Vision may handle transparent surfaces, reflections, and occlusion well while leaving uncertainty about water in a cup, a damp base, glasses beneath paper, or whether a medicine box is empty. Grasping converts state residuals into contact residuals: a slightly wrong gripper angle rotates a cup, lifts paper, or tangles a cable around another object. Contact then changes the environment: the cup moves, the paper stack collapses, visible keys become occluded, and the tabletop no longer matches the previous plan's input. The result is a new task state with errors written into the environment, rather than one isolated mistake. Without low-latency touch, compliant grasping, local recovery, and task-level rollback, a small slip can escalate from insecure grasp to dropped object, spilled liquid, interrupted task, and human takeover. Here is the embodied capability boundary: whether the system keeps residuals recoverable at a given task length, disturbance intensity, and consequence level, beyond generating a plausible action.

Abrupt exposure at boundaries also reflects the multiscale structure of physical contact. Finger pads, skin, muscles, joints, and proprioception operate across scales: fine texture, local slip, contact area, overall posture, and muscle tension change together in continuous media. Materials and low-level sensorimotor loops absorb many deviations before they become explicit errors. Machines often separate these scales: materials bear microscopic contact, tactile or force sensors detect local slipping, vision or estimation corrects posture, and planners handle task drift. Residuals unabsorbed at one scale can jump upward: a tiny slip changes grasping posture, posture changes object state, and object state contaminates the next perception and plan. Cliff-edge failure often occurs when a residual crosses a gap between scales and turns a local disturbance into a system event, rather than through linear error growth. Such failure is an outward symptom of inadequate bodily and cross-scale encapsulation and excessive interface debt.

This mechanism admits concrete tests that could count against it. If large numbers of open-world robots in homes, hospitals, construction sites, farms, and public roads sustain graceful degradation—recovering errors, avoiding residual accumulation along tasks, showing gradual rather than abrupt rises in takeover rates with disturbance, and avoiding substantial maintenance and verification cost increases with task length—the claim about propagation and boundary amplification should be weakened. Better models, bodies, environments, and operational

systems would have returned residuals to a continuously absorbable range. Important technical signals already exist: Bongard, Zykov, and Lipson demonstrated recovery of motion after damage through continuous self-modeling, (Bongard et al. 2006) while Cully and colleagues demonstrated rapid discovery of compensatory behavior through intelligent trial and error (Cully et al. 2015). These works show the role of residual-recovery structures in reliable embodied intelligence rather than overturning the mechanism. Mature embodied intelligence can degrade gracefully near boundaries, bringing errors back within manageable limits rather than requiring an absence of error. Why this looks effortless in humans is a question of time, beyond the present moment.

Evolutionary Amortization and Residual Management

The ease of human action is not free; its costs have been amortized over immense timescales. Hundreds of millions of years of biological evolution shaped skin, muscles, bones, joints, reflexes, and motor control through innumerable failures, injuries, deaths, and reproductions. Development recalibrates bodies through infancy, childhood, and daily falls, collisions, grasps, trials, and corrections. Social environments build handles, stairs, furniture, tools, kitchens, roads, hospitals, and factories around human scales. A person enters a niche co-evolved, co-designed, and co-domesticated with human bodies rather than a neutral space: this is **evolutionary amortization**.

Evolutionary amortization distributes bodily costs across immense timescales, countless individual trials, development, and environmental redesign. An adult's easy grasp rests on encapsulation deposited by evolutionary, developmental, personal, and environmental histories rather than a simple calculation at that instant. Developmental systems theory and ecological psychology both emphasize action capabilities formed through coupling among bodies, environments, and historical processes rather than unfolding solely from an internal program (Thelen and Smith 1994; Clark 1997). The two axes introduced earlier can now be developed. The time available for amortization determines how complexity can be spread: evolution, development, and civilizational design have spread human bodily and environmental costs over long periods, while machines seek to rebuild those encapsulations within short engineering cycles. Substrate affordances determine which residuals are readily absorbed: skin, muscles, tendons, nerves, pain, and self-repairing tissues suit compliant contact, continuous feedback, damage sensing, and local repair. Metal, motors, gears, silicon, cables, and rigid structures excel at precision, repetition, and high power, but do not naturally provide living materials' low-power compliance, self-healing, and continuous sensing. Humans hide complexity in time and living substrates; machines often make it explicit in engineering systems.

This allocation can be written as an accounting relationship:

$$\bar{c} = \frac{C}{N} \tag{4.6}$$

C is the accumulated cost of forming a capability, including evolutionary trials, develop-

mental calibration, and social environmental redesign, first expressed on a consistent basis such as energy, time, or number of individuals. N counts all uses across the capability's history, summed over individuals and generations in the population considered; it is dimensionless. The allocated cost per use, $\bar{c}$, has units determined by the accounting basis for C. On this basis, the ease of human action may partly reflect a large denominator of uses: even when C is large, a sufficiently large N can make $\bar{c}$ small enough to leave historical costs outside present decisions. Machines rebuilding such encapsulation within short engineering cycles have a much smaller N than evolution and development provide; the same C would mean a higher cost per use. Actual machine and biological costs C differ, so this alone cannot compare total costs. The applicable conditions are sunk investment and continuing use: amortization requires that C is already sunk and the capability keeps being used. If environmental change makes it obsolete, future N stops increasing and further allocation ceases; the original C was already a sunk cost. This is an accounting convention, providing neither measured C and N nor a claim that living bodies are cheaper at every capability.

Living bodies are not superior in every dimension. Machines often far exceed them in precise repetition, heavy loads, high-speed processing, heat, radiation, and standardized manufacturing. Living materials are not inherently "higher"; open-world compliant contact, continuous feedback, damage sensing, local recovery, and sustained action form a capability combination to which life has long adapted. Machines can exceed individual bodily capabilities while still needing redesigned encapsulation for the combination. Human ease also relies on long-calibrated priors about bodily state. Normally, people need not re-establish hand location, joint existence, muscle response, or basic bodily coherence before every action. Proprioception, vestibular sensation, motor experience, and homeostasis provide a continually updated range of confidence that does not reset to zero each time. Machines must engineer this trust through calibration, redundant measurement, estimation, anomaly detection, and safety assertions; their self-state must continually be measured, synchronized, checked, and logged. Default bodily trust appears in machines as an entire verification structure.

Together these axes reconnect physical intelligence to technological viscosity. The bit layer rapidly generates models, policies, and simulations; the atomic layer bears contact, friction, inertia, materials, and safety consequences; the biological layer has spent immense time forming bodies adapted to reality. AI moving directly from bits to atoms while reproducing long-amortized bodily capabilities meets a fundamental rate mismatch: model updates are fast, bodily redesign slower, and environmental co-adaptation slower still. Fast engineering attempts to take on slow evolution's accumulated history, beyond simply making machines smarter; practical engineering routes therefore divide rather than converging on one solution. First, redesign environments to reduce residual sources: closed sites, warehouses, factories, ports, dedicated roads, and standardized interfaces make reality more observable, predictable, and free from ambiguous contact. They structure local environments rather than requiring machines to un-

derstand the entire open world. Second, redesign bodies to absorb residuals: compliant materials, tactile skins, dexterous hands, elastic joints, underactuated structures, and low-latency proprioceptive feedback improve physical support. Humanoids are attractive partly because so many environments, tools, and interfaces already fit human scales, but this also imposes the complexity of bipedal balance, whole-body coordination, dexterity, safe contact, and long-acquired bodily experience. Third, strengthen models to extend capability boundaries: better brains infer more states from fewer inputs, anticipate errors, and let existing hardware do more. This may genuinely improve efficiency or move burdens into models, data, training, simulation, and verification; whole-system costs must be checked. Fourth, narrow tasks and provide operational support. Many usable robots rely on precise task boundaries rather than general bodies: machines handle stabilized ordinary conditions, while people or operational systems handle exceptions and residuals beyond the boundary. Autonomy has been redistributed through the system at a cost.

Physical AI matures by building a sufficiently reliable **residual-management system**, beyond improving models or piling up sensors. Bodies, models, sensors, controllers, environmental design, task boundaries, and operational support can jointly form it; different solutions absorb residuals and expose complexity at different locations rather than ranking along one simple scale. Mature physical intelligence often actively limits the dimensions of contact with open reality and sets physical boundaries it can handle. Structured environments, standardized tools, restricted tasks, compliant bodies, safety control, remote takeover, and operational audits define a fault-tolerant region for action. Residuals, delays, and errors can exist there without readily penetrating the successive absorption structures of body, model, control, and operations. Reliable physical intelligence forms a recoverable, verifiable, maintainable action niche jointly with the world, rather than requiring boundaryless understanding of it.

After morphological computation comes the question of how bodies encapsulate residuals across timescales and how machines bear them without the same historical amortization and living substrate. Bodies let humans act with incomplete world models. Machines can learn human actions without endlessly adding sensors, but must find residual-correction mechanisms to replace missing bodily encapsulation. The primary scarcity in physical intelligence is a residual-absorbing structure amortized over long periods and carried by a suitable substrate, beyond abstract reasoning alone. Entry into the open world depends on residual-absorption structures matched to the system's medium. Yet "residual absorption" groups different kinds of work: exploring possibilities can be retried cheaply in symbolic space, containing risks compresses degrees of freedom within safety boundaries, and exposing errors requires evidence chains and long observation. They demand different resource structures, timescales, and levels of certainty. If where residuals are absorbed determines a system's weight, might residuals at different locations require intelligence to differentiate into different modes?

4.4 Modes of Intelligence: The Constraint Triangle of Resources, Time, and Certainty

4.4.1 Three Sides of the Constraint Triangle and Their Tensions

Where residuals fall determines the resource structures, timescales, and certainty levels needed to absorb them; these demands do not always align. Intelligence will not evolve along one straight line. A common vision imagines greater intelligence becoming more general, faster, more reliable, and cheaper until it converges on one omnipotent system: real-time understanding, instant plans, stable physical control, accurate prediction of consequences, and minimal resources for every task. This treats intelligence as a single-axis capability whose growth improves every dimension together. Real intelligence operates under constraints. It needs resources for computation, sensing, storage, cooling, and execution; time for receiving information, reasoning, verification, feedback, and correction; and certainty for reliable results, stable behavior, controlled risk, and traceable responsibility. Technology can improve all three, but pushing them together toward extremes without cost is difficult.**The central mistake of single-axis thinking is assuming that constraints disappear as capability grows.**

Three Sides of the Constraint Triangle

The constraint triangle brings resource allocation and engineering tradeoffs into one analytical framework; tasks, performance metrics, and architectures determine the specific tensions. Better algorithms, more efficient chips, stronger energy supplies, better sensors, and more mature control architectures can move the boundary. Moving a boundary does not abolish it. Technology can enlarge the triangle without making speed, certainty, and low cost improve without limit and without new costs.

Each side has a physical meaning. Resources are intelligence's physical account: computation needs energy, models storage, sensing equipment, inference processor time, execution power, control communication and feedback, and high reliability redundancy, backups, audits, and maintenance. Even apparently pure information structures operate on chips, electricity, networks, cooling, and manufacturing systems. Larger systems, higher-frequency feedback, and more consequential tasks make intelligence harder to keep light.

Data centers expose intelligence's physical foundations. IEA reports in April and October 2026 examined efficiency, demand, and power-supply conditions: efficiency per task can improve while usage growth still raises total electricity demand (International Energy Agency 2026b, 2026c). Comparisons must distinguish all data centers, AI-related facilities, and individual tasks, and separate earlier projections from actual 2026 results.

Separating total consumption into demand and energy per unit puts evidence pointing in both directions into one expression:

$$E = De, \qquad \frac{1}{E}\frac{dE}{dt} = \frac{1}{D}\frac{dD}{dt} + \frac{1}{e}\frac{de}{dt} \tag{4.7}$$

E is total electricity use over a given period, in terawatt-hours; D is computing demand served over that period, in normalized units of computation; e is energy per unit, in terawatt-hours per computational unit. The first expression decomposes the total, and logarithmic differentiation with respect to time gives the second. Its three terms are instantaneous relative rates of change, in inverse years if time is measured in years. Efficiency improvement can coexist with a moving resource boundary: total electricity use still rises when demand growth exceeds the decline in energy per unit. Masanet and colleagues' historical estimates for global data centers illustrate this: computing instances increased 6.5-fold from 2010 to 2018, while electricity use rose about 1.06-fold. Energy per computational unit fell to roughly one-sixth of its earlier value, or about 20 percent annually on average, allowing both changes to coexist (Masanet et al. 2020). Comparable computational quantities and consistent statistical boundaries are required. With mismatched definitions—for example, separate accounting of training, inference, and cooling—or structural changes in demand, the meaning of e changes. Defining it as total electricity divided by demand preserves the identity, but need not isolate efficiency changes for comparable tasks. This describes an accounting structure rather than forecasting energy trends.

Time is the process through which intelligence exchanges information with the world. Intelligence cannot know all consequences before information arrives or complete real verification before processes unfold. Fast responses often mean acting on incomplete information, long-term verification slowing down, and complex feedback waiting for results from the system. More immediate decisions tend to rely on narrower views; greater global robustness demands more observation, checks, and feedback cycles.

Verification time comprises proofs, simulations, trial preparation, real exposure, follow-up, and review. Better tools can accelerate some components; aging, long-term responses, and sufficient observation have conditions of their own. Projects should distinguish waiting that generates evidence from waiting caused by process congestion.

Certainty here names requirements for controlling errors, risks, and consequences, made concrete through reliability, safety margins, recoverability, and traceability. Low-risk text generation can tolerate ambiguity, noise, and regeneration in ways high-risk control cannot. Autonomous driving, nuclear plants, grids, financial clearing, surgical robots, flight control, and drug approval must examine tails, boundaries, responsibility, explainability, auditability, and recovery as well as averages. Greater model intelligence alone does not buy certainty; restrictions, redundancy, verification, and institutions do.

In its September 2026 introduction, the FDA reported more than 1,600 AI-enabled devices authorized for marketing in the United States (U.S. Food and Drug Administration 2026a). Its periodically updated list specifies how devices are identified and the limits of coverage (U.S.

Food and Drug Administration 2026b). Each authorization concerns a particular product and use: the count shows bounded functions entering practice rather than a general qualification to practice medicine.

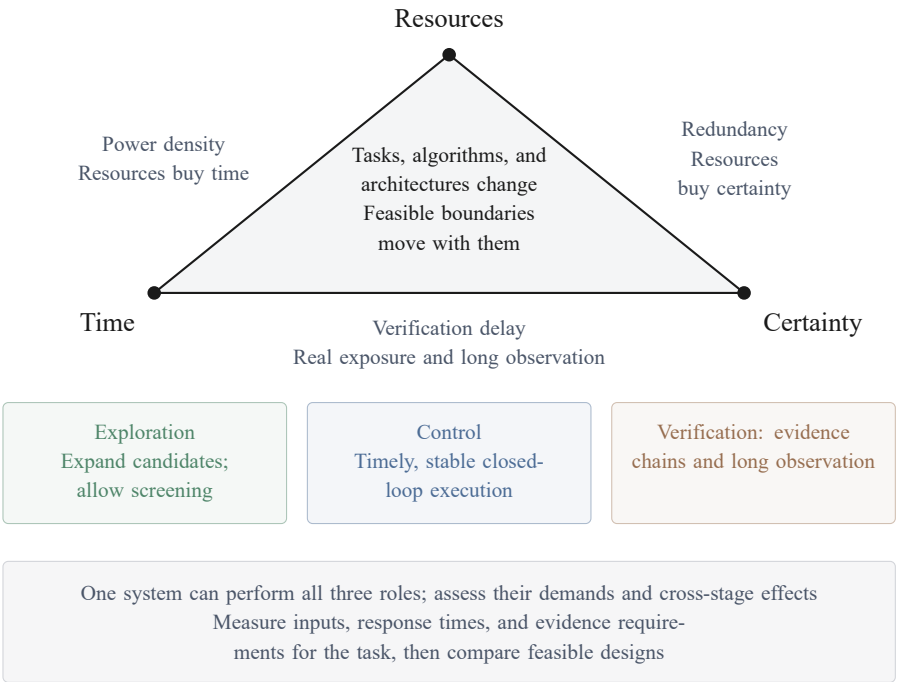


Figure 4.2 The constraint triangle: demands and division of work. Resource, time, and certainty requirements jointly shape feasible solutions; their tensions depend on task, architecture, and technology, and improvements may ease several at once. Exploration, control, and verification identify three kinds of work whose inputs, response times, and evidence requirements should be assessed separately. Positions indicate categories; measurements must establish actual feasible boundaries.

Tensions Within the Constraint Triangle

For a given task, architecture, and resource base, faster operation may compress verification depth, greater stability often needs more time or less freedom, and lower costs often narrow task scope, responsibility, and operating frequency. Speed, stability, and economy can improve together, yet structural tensions remain; technological advances can ease them locally without erasing them altogether. This is intelligence's **constraint triangle**. The tension can be accounted for through resource allocation:

$$B_{\mathrm{E}} + B_{\mathrm{C}} + B_{\mathrm{V}} = B \tag{4.8}$$

B is the system's available input over a given period; B_{E}, B_{C}, and B_{V} are exploration, control, and verification allocations on the same account. The four quantities must share an additive

basis—energy, compute, labor, or process time—so the expression is an accounting identity rather than a formula for an optimum. With B fixed, increasing one allocation requires reducing the others. This describes budget reallocation rather than proving that speed, reliability, and low cost are mutually exclusive; efficiency improvements can still raise several performance measures together. The condition is commensurability: when institutional legitimacy, human responsibility, or long-term trust cannot be placed on the same account, the identity covers only what can be counted, while other costs remain.

Omnipotent intelligence is no default stable form within this triangle. It can remain a limiting vision, but real systems pay: speed brings risks from incomplete information, stability verification and redundancy costs, and economy restricted capability. Approaching omnipotence concentrates resource use, feedback complexity, verification burdens, and responsibility rather than providing a free ascent. Real intelligence is therefore more likely to differentiate into modes. **differentiation into modes** means different system functions as well as different tasks: exploratory intelligence opens possibilities, control intelligence contains risks, and verification intelligence exposes errors. They expand candidate space, constrain action, and reveal hidden costs, respectively. These are three kinds of work rather than degrees of the same intelligence.

4.4.2 Exploration, Control, and Verification: Interfaces Within an Intelligence Ecosystem

What Exploratory Intelligence Does

The first mode, exploratory intelligence, opens possibilities. It generates text, images, code, hypotheses, plans, molecular structures, design drafts, and candidate strategies. Its value is cheaply generating many candidates and revealing previously unconsidered paths, rather than making every output certainly true. It suits bit-domain tasks with low consequences, cheap execution, rollback, and screening, as well as early searches in scientific discovery and engineering design.

Exploratory intelligence needs fluidity. It needs divergence, association, analogy, chance, and trial, allowing unconfirmed candidates to face later screening rather than demanding final conclusions at every candidate stage. Unconfirmed hypotheses, random exploration, and factual hallucinations must be distinguished: exploration can retain multiple possibilities while factual assertions remain subject to evidence. Fabrication is no necessary price of creativity. Exploration is not verification, candidates are not facts, and imagination is not reality. Its proper role is to offer possibilities rather than confirm them on reality's behalf.

Exploratory intelligence expands candidate space rather than stabilizing action. It brings implicit combinations, paths, expressions, and designs into view, turning scarce candidates into abundant possibilities. Reducing search viscosity creates downstream pressure: more

candidates mean heavier screening, cheaper expression makes credible expression scarcer, and easier plan generation raises the burden of real verification. Exploration cannot therefore be the endpoint of a high-risk system: it can propose without approving alone, generate candidates without directly carrying high-consequence execution, and expand search without replacing verification. Direct connection to high-risk reality converts low-certainty generation noise into real risk. Its domain is possibility rather than final responsibility.

What Control Intelligence Does

The second mode, control intelligence, contains risk. It keeps systems controllable amid noise, disturbance, delays, and faults rather than generating more possibilities. Autonomous driving, robot control, production lines, grid dispatch, aerospace, nuclear facilities, financial infrastructure, and medical equipment all require it. It values stability, reliability, and boundary management ahead of output volume, novelty, or expressive ability.

Control intelligence must relinquish some freedom. It cannot diverge at will, improvise inventions, or apply unverified policies directly to consequential systems. It requires bounded operating domains, redundant sensing, safety thresholds, exception handling, logging, human takeover, compliance audits, and chains of responsibility. Its strength lies in knowing what it cannot do, when to degrade, when to stop, and when to return control, rather than daring to do everything. Its system function constrains action rather than expanding choice: it contains open-world risks within controllable boundaries, unlimited situations within bounded operating domains, and unexpected disturbances within redundancy and safety policies. Control isolates, buffers, records, degrades, and assigns responsibility for risk rather than eliminating it. A mature control system's value often lies in how much risk its safety structures contain, beyond how intelligent it appears.

This containment consumes resources. Reliability needs more sensing, safety redundancy, low latency edge computing and reliable communication, responsibility records and audits, and tail risks extensive tests and verification. Entering atomic and high-risk systems makes the lightness of bit-domain generative intelligence harder to retain. Control compresses degrees of freedom within safety boundaries, turning intelligence into stable feedback loops. Control and exploration seek different functions: one constrains state space, the other expands candidate space. Confusing them is among the AI era's most dangerous mistakes. One needs constraints, the other fluidity; one must prevent errors from becoming consequences, the other permits screening them out; one confronts reality that cannot freely be rolled back, the other permits cheap retries.

What Verification Intelligence Does

The third mode, verification intelligence, exposes errors. It reveals where answers cannot withstand reality rather than producing more answers. Scientific reasoning, mathematical proof, formal verification, fact-checking, drug-evidence assessment, engineering reliability re-

views, legal compliance, risk audits, and long-term system assessments all require it. Its value is a more reliable "not yet" rather than a faster "yes."

Verification must allow enough time for evidence chains: formal checks can be fast, but long-term exposure and follow-up cannot be completed merely by faster output. It checks sources, seeks counterexamples, evaluates boundary conditions, tests reproducibility, compares alternative explanations, and observes long-term consequences. Many questions cannot be confirmed in one inference: drugs need experimental and clinical feedback, engineering systems stress tests and failure statistics, and social policies behavioral adaptation and institutional consequences. Verification is a gate for transformation into reality rather than an appendage to generation. Its system function exposes errors rather than expanding possibilities or maintaining control: omitted boundary conditions, hidden assumptions, long-term consequences behind short-term success, and tail risks concealed by average performance. It makes hidden errors visible again.

Here lies the fundamental difference between verification and generation. Generative systems compress the world into output structures; verification returns outputs to the world so their inadequacies become visible. A model produces treatment advice in seconds; verification asks which patients it applies to, what contraindications exist, how strong the evidence is, what long-term side effects occur, and who bears risk. A model generates an engineering plan; verification asks about extreme loads, material fatigue, maintenance cycles, and consequences of failure. High-quality verification costs data, experiments, audits, sustained observation, expert judgment, and institutional interfaces. Its achievements—fewer errors, postponed risks, prevented failures, and preserved boundaries—are often less conspicuous than generated outputs. Good verification can prevent cheap possibilities from being mistaken for reliable reality without producing spectacular outputs.

Interfaces Within an Intelligence Ecosystem

The three modes can now be considered together. All matter, but none can replace the others: expansion alone produces candidates without responsibility; containment alone stability without openness; error exposure alone caution that may sacrifice speed of action. A mature intelligence ecosystem needs interfaces across this division of work.

The future is unlikely to consist solely of one model taking over everything. Multilayer systems are more plausible: generative models propose candidates, simulations screen paths, control systems execute bounded actions, verification assesses safety, audits record responsibility, and people and institutions define goals' boundaries. Different modes constrain and complement one another. Deeper entry into reality increases the need for this division rather than one unbounded omnipotent entity. The differentiation also changes how we understand general intelligence. General capability is valuable in low-viscosity bit space because tasks switch quickly, errors can be rolled back cheaply, and outputs screened afterward. High-viscosity reality subjects generality to operating domains, responsibility boundaries, verification stan-

dards, and control interfaces. More consequential systems need general capability embedded in specialized, auditable, maintainable, stoppable engineering structures, beyond relying on open-ended generality.

This revises the deployed form of general intelligence rather than rejecting it. General intelligence may become a powerful upper layer of search and coordination without directly replacing every underlying control and verification structure. It can suggest paths without canceling physical execution constraints, explain complex systems without canceling long-term feedback, and generate judgments without bearing society's responsibility.

Competition in the AI era concerns the architecture of intelligence ecosystems as well as model capability. Better organization of exploration, control, and verification better converts intelligence into practical capability. Individual model capability matters, but system interfaces matter more: how candidates reach verification, verification constrains execution, execution feeds back into models, errors are recorded, responsibility traced, risks isolated, and goals redefined by people and institutions. The triangle also warns against treating slowness as failure: some intelligence must be fast, some stable, some slow. Slow exploration misses opportunities, slow control causes accidents, and verification pursuing speed alone may overlook risk. Mature systems run different modes at appropriate speeds rather than accelerating everything in unison: fast components expand possibilities, stable components hold reality, and slow components confirm evidence and boundaries.

There is also a lesson at the scale of civilization. Societies should reward more than generation speed and automation coverage. They must also invest in verification systems, audits, long-term experiments, chains of responsibility, and human judgment. Otherwise exploration floods systems with candidates, control is pushed prematurely into high-risk reality, and verification is dismissed as an obstacle to innovation, accelerating accumulation, instability, and crises of trust rather than building a more intelligent civilization.

Differentiation into modes is a conclusion of maturity rather than pessimism. Constraints can give intelligence stable forms without removing its value: exploration reveals possibilities, control brings them into reality without losing control, and verification gives real changes trustworthy boundaries. Together they form an ecosystem rather than replacing one another. Technology can move constraint boundaries without driving speed, certainty, and low cost to costless extremes simultaneously. A stable real-world form is more likely an ecosystem of intelligence modes than one omnipotent intelligence. Intelligence's evolutionary destination is the division of work, interfaces, and coordination among opening possibilities, containing risks, and exposing errors, rather than a god's-eye view.

4.5 The Human Position: Energy-Efficiency Anchors and Heterogeneous Damping

Differentiation into exploration, control, and verification also calls for reassessing the human role task by task. Three aspects matter here: whole-system energy use for the same task, stability when multiple loops interact, and conditions permitting results to become actions. The first two require engineering comparisons; the third also concerns risk allocation and social authorization. Human capabilities, machine capabilities, and institutional arrangements jointly determine the division of work.

4.5.1 Wetware and Analog Computation: A Place in the Energy Account

Locating humans within these modes first requires a different basis of comparison. Scoring capabilities item by item changes each round as machine cost curves fall. A relatively stable basis is the energy account: for the same work, what must each side pay in power, cooling, and moving its own weight? The basis is familiar but often overlooked when intelligence appears to be pure information activity, seemingly consuming no power. Chapter 2 addressed this impression: Landauer's lower bound may be negligible in practice, but information processing still needs a physical substrate that must be powered, cooled, maintained, and renewed (Landauer 1961). Including the substrate changes the comparison from brain versus chip to the complete biological system versus the complete machine system.

Energy comparisons require matching system boundaries. The brain's sustained power is on the order of twenty watts, while whole-body metabolism varies with activity (Smil 2015). Machines must include perception, computation, actuation, and supporting facilities; humans must include metabolism supporting the body and task. Accounting on both sides by nonoverlapping functions gives:

$$E_s = E_{s,\text{sensing and computation}} + E_{s,\text{actuation}} + E_{s,\text{auxiliary}}, \qquad s \in \{\text{human}, \text{machine}\} \tag{4.9}$$

All terms are in joules, for the same task, quality standard, and completion time. Actuation includes external work and self-movement; support includes other operating costs, without double-counting. Bodily functions share tissues and metabolic pathways; when individual terms resist measurement, total energy within the same boundary can be used, with corresponding totals for machines. Measurements determine the comparison; the number of accounting categories proves no efficiency advantage.

Bodily structure changes the control needed for a task. During grasping, compliant tendons, skin, and joints absorb some contact errors, while neural control and passive dynamics jointly organize the action. Machines also use flexible structures, compliant control, and morphological computation to give some correction to the body. Costs depend on implementation: which

deviations structures absorb, which require sensing, computation, and active execution, and which become maintenance or safety redundancy.

Delivery, care, repair, loading, and inspection involve many nonstandard sites. Addresses, stairs, obstacles, weather, and other people's behavior change, and tasks may require communication and improvisation. Machines need perceptual redundancy, remote takeover, and on-site support; human work also entails training, fatigue, injury, and organizational support (Moravec 1988; Pfeifer and Bongard 2006). Energy is one measure alongside reliability, service quality, and maintenance costs in deciding feasible divisions of work.

Standard shelving, fixed flight paths, and controlled production lines reduce uncertainty in perception and action, making stable automation gains easier. Open environments require more online correction. Human bodies have accumulated evolutionary and learned adaptations; machine structures and training are also changing. Which has an advantage requires comparing equivalent tasks and quality, rather than inferring it from an environment's name.

4.5.2 Heterogeneous Damping in Systems

The second question concerns collective responses. When many participants act simultaneously on one signal, local corrections can combine into a collective shock. Different response rhythms may spread the shock or complicate coordination. Whether heterogeneity helps depends on signals, connectivity, and feedback direction.

Consider a simplified case: the outputs of several linear channels receiving an input at the same frequency can be added as complex amplitudes. Let channel i have amplitude gain g_i and phase φ_i; the resultant amplitude is:

$$G_{\text{total}} = \left| \sum_{i=1}^{N} g_i \, e^{\mathrm{i}\varphi_i} \right| \tag{4.10}$$

g_i and G_{total} are dimensionless, and φ_i is measured in radians. In phase, the resultant reaches the sum of the component amplitudes; dispersed phases can partly cancel responses. This describes output superposition at a particular frequency. Closed-loop stability also depends on the complete loop transfer function and feedback connections; a resultant amplitude exceeding one alone does not establish self-excited oscillation.

Backoff, random jitter, staggered execution, and independent review can reduce synchronized action triggered by one signal. Human participation sometimes adds different information and judgment paths, while fatigue and delay can also add errors. What matters is verifiable dispersal and correction; whether people, software, or institutions provide it should be judged by outcomes.

Heterogeneity and coordination need joint design. Uniform execution rhythms may amplify shocks; disconnected information may make actions cancel one another. Human supervision of automation also requires accessible information, enough response time, and continuing training

(Bainbridge 1983; Parasuraman and Riley 1997). These conditions determine whether people occupy a useful feedback position.

4.5.3 Meaning as a Stopping Condition

The third question is when to stop searching and permit a result to become action. Algorithms can stop automatically using time budgets, error thresholds, or convergence conditions; society must also decide which results can be relied on. Search termination and permission to act separate here.

Objectives can be bounded or unbounded; multi-objective problems can yield nondominated candidate sets while explicit rules still stop computation. Deployment needs a stated set of admissibility conditions. Let $\mathscr{A}$ be the set of states meeting quality and risk requirements; denote the first time search enters it by:

$$t^* = \inf\{t \geq 0 : \mathbf{x}_t \in \mathscr{A}\}, \qquad \inf \emptyset = +\infty \tag{4.11}$$

$\mathbf{x}_t$ is the search state, and t^* is measured in time. If search never enters $\mathscr{A}$, the first hitting time is infinite; an actual program may still exhaust its budget and report failure. $\mathscr{A}$ may include automatically checkable rules or requirements needing further trials or authorization. The expression records admissibility conditions and arrival; their content must be determined separately.

Human judgment enters chiefly in defining, revising, and disputing these conditions. Acceptable errors, risks requiring further verification, and rights to demand suspension concern those bearing consequences, deployers, and public institutions. Models can help draft and test conditions, while adopting standards requires appropriate authorization. Indicators change behavior once tied to rewards and penalties, so admissibility standards also need appeals, review, and revision (Goodhart 1975; Campbell 1979). Authorization and competence must be considered together. Those bearing risk should be able to understand standards, object, and obtain remedies; reviewers need evidence and expertise. Human judgments can err and automated checks can omit; institutions must make errors detectable, decisions traceable, and responsibility enforceable. Chapter 6 returns to these relationships through value ranking and public adjudication.

The three aspects provide ways to examine energy, stability, and admissibility. Energy advantages require matched-task comparisons, stabilizing effects feedback analysis, and decision authority arrangements of rights and authorization. The human role changes with technology and institutions together. Evaluating actual divisions of work through these conditions better captures real systems than declaring capabilities to belong to one side or the other.

A division of work asks who will bear it, and the human role asks where people stand within it; neither answers where that work is affordable. All three modes enter physical systems organized differently from intelligence's division of functions. A road, grid, or timetable

simultaneously bears resource accounts, temporal rigidity, and structural coupling; sensible allocation of intelligence does not automatically lighten it, nor local optimization reduce its overall viscosity. Making one stage smarter is easy; lightening a whole system of movement is hard, and such systems are especially good at absorbing single-point optimization gains.

4.6 The Physical Account of Movement, Part I: The Costs of Ground Transport

4.6.1 Powertrain Platforms, Mobility Systems, and Racing on the Wrong Track

A mechanism can be identified in one stage but priced only within the complete system. Resource, time, and topological constraints often appear as isolated parameters within a technology; in operating systems they substitute for one another and move costs between locations. An improved local metric and a lower total system burden are different achievements; the latter must be weighed across the whole system. Seeing the distinction requires an object sufficiently heavy, enduring, and coupled.

Ground transport is an excellent specimen of technological viscosity. It contains all three constraints: roads, bridges, parking, vehicles, grids, charging networks, repair services, insurance, traffic laws, driving habits, urban space, and family life are locked together. Transport is a highly material, historically accumulated, tightly coupled problem of state transition, beyond individual vehicles. Changing a car is easy compared with changing mobility; replacing a powertrain platform is easy compared with rebuilding urban space.

Powertrain Platforms and Mobility Systems

In recent years, electric vehicles have often been sold as the central answer to future transport. The problem is a mythology combining local electrification, smart cabins, industrial policy, investment stories, consumption upgrades, and national industrial pride, rather than electric vehicles' lack of technical value. Manufacturers promise intelligence, policymakers green mobility, investors growth, media leapfrogging, and consumers a high-tech experience. Stacking these words together does not automatically reduce transport's total viscosity.

Vehicle-level technical advantages do not settle this account. Electric motors are efficient, responsive, and relatively simple, and electronic controls integrate readily with digital platforms. These are real strengths. Motor efficiency is only one transport bottleneck. However elegant the motor, batteries remain heavy; however precise the controls, cars occupy space; however impressive acceleration, roads remain congested; however intelligent the cabin, most private cars spend long periods idle. Calling electric cars a transport revolution inflates optimization within vehicles into transformation of the whole system.

Electric vehicles first replace a powertrain platform; they do not by themselves reorganize

transport. They replace engines, fuel tanks, transmissions, and refueling with batteries, motors, electronic controls, thermal management, and charging. Engine complexity falls while battery-system complexity rises; exhaust disappears from the tailpipe while energy, minerals, manufacturing, recycling, and grid pressures move into the background. One stage can grow lighter while another grows heavier; simpler mechanics may accompany more complex infrastructure and resource systems. Looking only at vehicle marketing while ignoring batteries, minerals, grids, roads, parking, insurance, depreciation, and utilization preserves the illusion of single-point technology.

Transition paths require full lifecycle comparisons. Existing structures can save near-term investment while prolonging fuel, maintenance, and emissions costs; new infrastructure costs money but can reduce long-term totals. The mechanical principle of least action cannot decide this comparison. The account must compare energy, materials, expenditure, emissions, space, and risk for equivalent service.

Charging, distribution grids, thermal management, and battery recycling bring new expenditures; fuel extraction, refining, transport, refueling, and engine maintenance already carried expenditures of their own. The IEA's 2026 lifecycle comparison still finds emissions benefits from electric cars while specifying manufacturing and electricity-supply conditions (International Energy Agency 2026a). Electrification can yield a net improvement; congestion, parking, and urban accessibility require further changes.

This is technological viscosity's central judgment here: **Reducing local viscosity does not establish a reduction in total system viscosity.** Electric vehicles may reduce some conversion losses and maintenance costs without automatically resolving deeper temporal rigidity and topological coupling. Roads remain congested, cars idle, parking consumes land, cities expand around cars, households are compelled to own them, commuting consumes time, and accident liability remains. The vehicle changes; the mobility system may not. To make this judgment useful, identify the object being optimized: replacing the powertrain may suffice if the object is the car, but mobility leads to different conclusions.

Total Movement and Effective Movement

Transport's primary object is movement. People, goods, and services must be redistributed in space. People must reach work, education, healthcare, social life, and family; goods pass through production, warehousing, long-haul transport, last-mile delivery, and consumption; services must arrive where and when needed. Cars are one carrier among others. Walking, bicycles, buses, subways, railways, trucks, shipping, autonomous fleet vehicles, and last-mile systems serve different scales. Transport should optimize necessary movement at lower physical cost, greater reliability, and greater resilience, beyond the sophistication of each car.

Carrying efficiency depends on the density–speed relationship rather than a vehicle's propulsion type. The number of vehicles crossing a road section per unit time is:

$$q = \rho v \tag{4.12}$$

Here q is flow, in vehicles per hour; ρ is vehicle density, in vehicles per kilometer; and v is the mean speed at that road section, in kilometers per hour. The relationship itself merely defines the quantities; in common traffic fundamental diagrams, v varies with ρ, and once traffic becomes crowded, greater density generally means shorter headways and lower speeds. On a typical curve, q reaches its maximum at a critical density, beyond which flow falls and the road network enters the congested branch (Helbing 2001). Replacing the propulsion platform alone can change a vehicle's energy consumption, emissions, and noise; if vehicle dimensions, driving behavior, and road conditions remain unchanged, a different energy source does not automatically move this curve, whereas changes in headways, reaction times, and road geometry can change the flow relationship.

More precisely, transport seeks to optimize the generalized total cost of effective movement under multiple constraints. People and goods must move from A to B; minimizing the system's generalized total cost matters more than minimizing one trip's time or one vehicle's energy use alone. This generalized cost requires task-specific weights and accounting boundaries. It covers manufacturing and depreciation, energy, road and parking land, driver attention, congestion externalities, accident risk, maintenance, waiting, transfers, emissions, noise, and lost resilience. Good transport finds lower-total-cost paths across resources, time, space, risk, and freedom rather than maximizing one vehicle metric.

Total movement and effective movement must therefore be distinguished. Movement is not always progress. Long commutes, separated homes and jobs, empty vehicles, duplicate deliveries, poor transfers, forced cross-district healthcare trips, and prolonged parking searches arise from inefficient structures. Electrifying, quieting, and adding intelligence to these movements still leaves the unnecessary movement itself unresolved. Better transport reduces unnecessary movement and carries necessary movement more efficiently, beyond increasing speed.

Private cars illustrate a local optimum. They have real value. They provide freedom, flexibility, privacy, and access. For low-density areas, inadequate public transport, family drop-offs, care for children and older people, and complicated schedules, they are basic interfaces to daily life rather than mere luxuries. Many depend on cars because homes, work, commerce, schools, hospitals, and services have been organized around them, rather than because advertising has brainwashed them. Without a car, life becomes difficult.

Private cars also form a high-viscosity structure. Low utilization leaves them parked most of the time; they occupy roads and parking, assign driving, maintenance, insurance, refueling or charging, and accident risk to individuals, bind attention to driving, and encourage low-density, long-distance, car-dependent urban growth. More significantly, a society built around cars is hard to leave. Roads, suburbs, shopping centers, parking, loans, insurance, household schedules, and habits form structural memory. Private-car dependence is a stable local optimum rather than the best system overall. It remains stable because surrounding dependencies reinforce one another. Replacing propulsion without changing those dependencies leaves the

system's structure intact.

Racing on the Wrong Track

Simply replacing combustion cars with electric cars consequently leaves transport's foundations unchanged. It preserves much of the topology: individual ownership, low utilization, parking land, congestion, household charging or refueling, dispersed maintenance, liability, and urban expansion. The powertrain platform changes. One or two people still sit in a one- or two-tonne machine occupying roads, parking, and energy networks for many tasks that could use lighter, more concentrated means with less movement. Electric cars address some propulsion problems without answering why so many cars must move.

Two relationships need separating. Electric versus combustion cars concerns propulsion; private cars versus autonomous fleets concerns how movement is organized. Electrification matters and may become an important fleet powertrain platform because fleets need lower maintenance costs, readily controlled propulsion, and energy interfaces suited to centralized charging. The issue is electrification locked into the old private-car paradigm. Treating the stock of privately owned electric cars as the goal mistakes a tool for an endpoint.

Much electric-vehicle storytelling gives this problem a more sophisticated package rather than questioning it. Bigger screens, more compute, heavier batteries, more complex cabins, more extreme acceleration, and denser sensing become "intelligence" and "evolution." If these merely make a rarely used private car heavier, dearer, harder to maintain, and more dependent on background systems, technological excitement may conceal new viscosity rather than reduce it. Cars increasingly resemble mobile consumer electronics, while transport remains as heavy as before.

The greatest danger in the narrative of "leapfrogging" is racing on the wrong track. It assumes future transport still centers on mass private ownership, replacing combustion with electricity, engines with batteries and motors, and mechanical cabins with smart ones. Policy, subsidies, road access, licenses, industrial investment, and public attention consequently gather around private electric cars, as if winning this generation of vehicle platforms meant commanding transport's future.

Yet future mobility may center less on a more advanced electric car in every household than on autonomous fleets serving everyday travel at scale: robotaxis, driverless commuter vehicles, shuttles, delivery vehicles, and fleets within defined areas and routes. On September 1, 2026, Waymo announced its first public riders in Denver, San Diego, and Tampa, bringing fully driverless trips to fourteen cities at that time while access in the new cities expanded gradually (Waymo 2026). If services mature and become affordable, they may take over many private-car uses: commuting, transporting children and older people, errands, urban travel, airport and station connections, night travel, and last-mile delivery. Competition would then concern moving more people and goods with fewer vehicles, higher utilization, shorter waits, and more stable safety boundaries, rather than selling more private electric cars. From this per-

spective, policies aggressively promoting private electric cars carry an irony: reinforcing an old paradigm in the future's name. They challenge combustion without necessarily challenging car dependence, encourage electrification without necessarily improving mobility efficiency, support manufacturing without necessarily reorganizing urban transport, and excite enthusiasm for advanced industry while potentially entrenching low utilization, land consumption, depreciation, and resource use.

If autonomous fleets become central to transport, many policies celebrated as leapfrogging may merely widen an old racetrack while competition moves toward fleet operations, high-utilization networks, and cost per effective movement. That is the paradox: you win the race, only to find the lane itself has disappeared. An industry thinks it is overtaking another private car, while autonomous fleets systematically take over the uses for which private cars were bought. Private cars remain without naturally occupying the center; manufacturers matter without individual sales necessarily offering the largest entry to value; batteries, motors, bodies, and cabins remain technically sophisticated while operations, utilization, safety boundaries, maintenance, responsibility chains, and effective-movement costs may decide the future. The real leap is making many daily journeys independent of the old private-car lane, rather than building faster private cars within it. If fleets become competition's next entry point, their technical form determines where viscosity moves.

Fleet operations may change total viscosity because they affect resources, time, and topology, whose responses need not align. With equivalent service quality and controlled empty mileage, utilization gains may reduce vehicle stocks and parking demand while spreading manufacturing, depreciation, and charging costs; actual production and maintenance needs also depend on cumulative mileage and lifespan. Central maintenance, continuous operation, and dynamic dispatch can loosen the bond between vehicles and household space while introducing new coupling through dispatch platforms, support facilities, and concentrated failures. Driverless operation on public roads still needs verification, responsibility chains, and safety boundaries; the time required depends on operating domains, accumulated evidence, and specific risks. Constraints may move in different directions or ease together through architectural improvements. Assessing autonomous-driving architectures requires checking net burdens as well as asking which dimension now bears viscosity.

4.6.2 Fleet Operations, Cars as Yachts, and Multilayer Resilience

The Complexity Ratchet

Autonomous driving should not become another myth. It is a classic overlay strategy: sensors, maps, models, compute, remote takeover, logs, audits, and responsibility mechanisms are added to existing roads, vehicles, laws, insurance, habits, and urban space. It seeks to make a world built for human bodies and judgment machine-readable, predictable, and executable. It

makes some burdens explicit as vehicle complexity, scenario calibration, tail-risk verification, infrastructure interfaces, and responsibility chains; fewer accidents and higher utilization may reduce other burdens. Net effects require complete-service accounting.

This is the complexity ratchet. Machine action in an old world brings additional layers: cameras, radar, lidar, detailed maps, vehicle–infrastructure coordination, cloud training, simulation, remote operations, data feedback, safety operators, regulatory trials, insurance, and accident investigations. New layers can reduce operational friction while adding coupling and maintenance; good hierarchy and decoupling can reduce those burdens. Smarter vehicles need not make transport simpler. Expanding intelligent overlays requires tracking whether the background systems grow heavier.

Public roads mixing humans and machines present autonomous driving's hardest settings. Human driving is an established order. Imperfect as it is, drivers, pedestrians, cyclists, traffic police, insurance, road customs, and responsibility rules have formed a working ecology. Highly standardized machine-only systems may also stabilize in closed sites, ports, mines, fixed routes, and dedicated roads. The difficult transition mixes drivers, autonomous vehicles, pedestrians, nonstandard roads, temporary construction, changing weather, local customs, and liability disputes. This is a tightly coupled hybrid-system problem beyond individual driving.

Level 5 driving is difficult for reasons beyond model strength. Open roads contain ambiguous gestures, temporary obstacles, rule-breaking behavior, worn surfaces, diversions, animals, children, older people, cyclists, accident scenes, local understandings, and unpredictable social interactions. Human bodies and experience handle many nonstandard residuals cheaply; machines must engineer sensing, recognition, prediction, decisions, verification, records, and accountability explicitly. This is the distinction between bodily encapsulation and explicit complexity: bodily experience and tacit understanding contain many details that machines must separate into observable, computable, verifiable, accountable structures.

Autonomous driving often deploys most readily in clear operating domains: defined regions, fixed routes, closed sites, ports, mines, warehouses, airport shuttles, low-speed delivery, dedicated lanes, and V2X-coordinated environments, rather than the most open, chaotic, humanlike roads. Local structuring lowers environmental entropy, sparing machines the reconstruction of complete human bodily and social experience. Automation often works because environments become tolerable to machines rather than because general intelligence suddenly appears. Deployment conditions reveal another structural possibility: organization of vehicles can be redesigned as well as environments, with broader effects.

Fleet Operations and Cars as Yachts

Fleets may reduce some private asset needs through greater utilization, central maintenance, and dispatch, while creating empty mileage, cleaning, support, platform, and concentrated-failure risks. Whether total viscosity falls requires comparing complete service: waits, coverage, prices, vehicle distance, road use, and reliability. Driverless operation expands choices

without automatically improving every system metric. A May 2026 analysis used Waymo's California Public Utilities Commission reports from August 2023 through December 2025, covering about 13.79 million trips and 86.27 million vehicle miles. Passenger-carrying mileage was about 53.6% of the total; the remainder included repositioning while awaiting requests and travel to pickups. Completed orders alone cannot represent road burdens (Abdelhalim 2026). These historical operating data show service growth while demonstrating why fleet service volume and effective passenger mileage need separate accounting.

This would change private cars' social role. If the coverage and service conditions hold, private cars may shift from mass infrastructure toward consumer goods, hobbies, status goods, and assets for special uses. Most people own cars today because life's structure requires them; if fleets cover most daily travel without requiring private ownership, cars would gradually take on the role of yachts. People would still love, collect, and pay for their freedom, speed, aesthetics, and status, without cars remaining necessary infrastructure for most daily lives.

"Cars as yachts" is a conditional structural judgment: sufficient fleet coverage of daily needs could leave private ownership serving more hobby, aesthetic, and special-purpose uses. Coverage, price, privacy, disability access, children's travel, low-density areas, and emergency resilience determine whether those conditions hold. The observations that matter are actual reductions in household vehicles and changes in total travel and service quality.

Multilayer Systems and Resilience

Future transport needs multiple layers rather than one tool dominating. Walking, cycling, and light vehicles suit short distances; buses, subways, and rail high-density corridors; autonomous fleets flexible low-density travel; rail, shipping, and long-haul logistics bulk freight; last-mile delivery smaller vehicles, automation, and human residual handling. Efficient transport matches different movement needs to different physical carriers rather than one tool to every setting.

Transport must also retain resilience. Extreme efficiency alone does not make a good system. Weather, accidents, energy fluctuations, equipment faults, demographic changes, emergencies, and technological transitions disturb mobility networks. Eliminating all redundancy can produce impressive normal operations and rapid instability under disturbance. Diversity, alternative routes, and local redundancy provide resilience. Rail is efficient but less flexible, cars flexible but underused, bicycles light but constrained by distance and weather, and autonomous fleets capable of sustained operation but dependent on sensing, algorithms, and infrastructure. Their different boundaries let them complement one another.

This is viscosity management in transport. It reduces unnecessary movement, lowers the total cost of effective movement rather than optimizing isolated vehicles, reorganizes vehicles, roads, energy, space, operations, responsibility, and resilience beyond replacing combustion cars with electric ones, and assigns suitable tasks to different tools and scales rather than merging them into one ultimate machine. An important structural change would reduce private

ownership dependence for daily journeys suited to fleet services, beyond upgrading vehicle forms. Electrification supplies an important foundation; autonomous driving matters, but its proper form should exceed a more expensive driverless private car. Only if these conditions hold may private cars lose their infrastructural position.

Cars would still exist. Many private cars would exist like yachts: high-viscosity consumer goods bought for possession, aesthetics, freedom, and status rather than daily infrastructure. A decline in their position would still redistribute activity within the same network. Whoever owns or operates vehicles, movement passes through a two-dimensional topology of roads, bridges, tunnels, intersections, and property boundaries. Cheaper vehicles cannot shorten a ground route that is itself too burdensome. How much efficiency remains within a mobility system and whether that network can be bypassed are different questions.

4.7 The Physical Account of Movement, Part II: Low-Altitude Flight and the Third Dimension

4.7.1 Two-Dimensional Lock-In, Airspace Governance, and the Cost of Lift

The cost of movement has two components. One grows with distance and can be reduced through better propulsion and efficiency; the other comes from route topology and does not change when efficiency improves. When existing structures fix a ground route, vehicle-level optimization cannot shorten it.

Two-Dimensional Lock-In and the Third Dimension

Low-altitude aviation's central resource is the third dimension. Ground transport is locked into a two-dimensional network. Roads, bridges, tunnels, signals, junctions, parking, property boundaries, urban density, and terrain jointly determine movement. Straight-line distances between A and B may be short while rivers, mountains, congestion, buildings, road hierarchies, transfer points, and administrative boundaries lengthen actual routes. Ground-transport viscosity comes from topologically locked routes as well as distance.

The value of low-altitude aviation lies in opening a third dimension. It bypasses road-network constraints where surface topology is too burdensome rather than building another ground road. River and mountain crossings, islands, disaster sites, mountain resupply, power-line inspection, crop protection, fire reconnaissance, and urgent medical delivery share a heavy ground route rather than the excitement of flight. A shorter, quicker route with less continuous infrastructure and ground coupling has real value.

Therefore, **The first question for low-altitude aviation is not whether it can fly, but whether flying is worth it.** Taking off does not automatically improve efficiency. Flight introduces manufacturing, energy, maintenance, safety redundancy, landing-site, noise, weather, communication, regulation, insurance, and liability costs. Where ground movement is already

cheap, flight can turn a simple task into an expensive performance. Air routes can improve on ground routes only when the generalized ground cost is sufficiently high. This generalized cost requires task-specific weights and accounting boundaries: time, energy, equipment, infrastructure, maintenance, risk, waiting, regulation, noise, responsibility, and resilience. A viable low-altitude aviation economy depends on reducing total cost per effective movement.

A common misconception needs clearing up: low-altitude aviation encompasses more than electric eVTOLs and drones. Fuel-burning light aircraft, helicopters, general-aviation fixed-wing aircraft, agricultural aircraft, medical rescue helicopters, and sightseeing aircraft all belong to this broad economy. Electrification is an important direction within it rather than the whole. Reducing it to "electric flying cars" obscures the industrial structure: low-altitude transport, aerial work, and services, beyond a victory for one propulsion type. Treating it as a mobility system makes its structural advantages clearer.

Airspace Governance Costs

Low-altitude flight reduces reliance on continuous road infrastructure. Ground transport requires roads, bridges, tunnels, and parking, together with junctions, congestion, property rights, and urban space. An air route needs no physical road built through its middle, no infrastructure poured for each meter. It exchanges spatial freedom for shorter paths. That freedom has a cost. Building fewer roads does not mean paying less overall. Some heavy road-asset requirements fall while other burdens enter airspace governance, safety verification, landing interfaces, communications and navigation, weather sensing, noise management, responsibility tracing, and frequent operations. Launching a few drones is easier than maintaining dense aircraft operations safely, cheaply, accountably, and under supervision across cities and industrial spaces.

Opening the third dimension creates the cost of governing that dimension. Roads make ground-traffic order explicit. Lanes, signals, intersections, speed limits, markings, guardrails, and parking form a visible traffic grammar. Low-altitude traffic lacks those ready-made boundaries. Dense operations must define routes, altitudes, separation, obstacle avoidance, emergency landings, no-fly zones, noise boundaries, communication protocols, identity, and accident responsibility within apparently free airspace. Without that order, the air is a high-risk space rather than a free one.

The foundational infrastructure is therefore a low-altitude operating system, beyond aircraft and landing sites. It includes airspace allocation, route authorization, real-time monitoring, dynamic avoidance, weather data, communication links, flight logs, identification, maintenance standards, insurance, accident response, and assigned responsibility. Aircraft are the visible part; airspace order is the underlying structure.

Airspace order can be designed in layers; the physical constraints of flight cannot. From an RTG perspective, low-altitude aviation is not low-viscosity. Its resource barriers begin with flight itself. Weight penalizes aircraft more severely than ground vehicles. Additional ground-vehicle mass chiefly increases energy use and wear; aircraft mass directly narrows payload,

range, and safety margins. Airframe materials, redundant propulsion, communications, sensing, and maintenance all enter the physical account. Bypassing heavy roads does not bypass mass, energy, or safety redundancy.

The Physical Cost of Lift

Energy storage is among the hardest resource barriers to bypass. Batteries are the most severe bottleneck for today's electric low-altitude aircraft. For road vehicles, heavy batteries chiefly add mass, energy use, and cost; for aircraft, mass directly determines flight feasibility, range, payload, and safety reserves rather than imposing merely a linear burden. Cars buy range with larger, heavier, dearer batteries; aircraft must lift the additional battery as well. Added batteries provide energy while using some of it to transport their own mass. Electric aircraft thus face a physical paradox: greater range calls for more batteries, whose weight reduces payload and marginal range gains.

Hovering power exposes this paradox. Momentum theory describes rotors accelerating air downward to generate lift, requiring mechanical power:

$$P_{\text{hover}} = \sqrt{\frac{(mg)^3}{2\rho A}} \tag{4.13}$$

The preceding expression gives ideal induced power from actuator-disk momentum theory; m is total mass, ρ air density, and A effective disk area within the model. At fixed A and ρ, doubling mass raises ideal induced power about 2.83-fold. Actual electrical power also includes profile drag, nonideal flow, transmission efficiency, auxiliaries, and interactions among multiple rotors.

Dividing stored energy by power gives endurance as a function of mass:

$$t_{\text{hover}} \approx \frac{\epsilon e_b m_b \sqrt{2\rho A}}{g^{3/2}(m_0 + m_b)^{3/2}} \tag{4.14}$$

m_0 is fixed nonbattery mass, m_b battery mass, e_b usable specific energy, and ϵ overall efficiency. With disk area and efficiency fixed, additional batteries initially extend hovering time; continued additions increasingly pay for their own weight. The simplified model reaches its extremum at $m_b = 2m_0$, rather than supplying a practical design recommendation. The scaling $m^{-1/2}$ follows only under different assumptions making stored energy proportional to total mass; "all added mass is battery" is insufficient by itself. Cruise requires a separate mission model.

Ground rolling and rotor hovering have different energy structures. At constant hovering altitude, the aircraft's gravitational potential energy does not continually increase; power chiefly transfers momentum to air and covers nonideal flow, drag, and equipment losses. Fixed-wing cruise has separate lift-to-drag and mission conditions. Comparing electric cars and aircraft requires complete flight profiles, pack-level storage, payloads, and reserve energy.

Fuel-burning light aircraft and helicopters will not immediately be wholly replaced by electrification. Aviation kerosene stores about 43 megajoules per kilogram, tens of times current lithium-ion batteries, (Viswanathan et al. 2022) an advantage particularly relevant to range, payload, emergencies, remote areas, and intensive operations. Fuel aircraft face noise, emissions, maintenance, cost, safety, pilot dependence, and operating barriers rather than an inability to fly. Electric aircraft offer control, maintenance, noise, and platform advantages while facing battery weight, endurance, payload, and reserve constraints. They are different solutions under different constraints rather than simply advanced and obsolete technologies. Stored energy per mass is only one comparison: whole missions also include engine and electric-drive efficiency, battery packs and tanks, reserves, and the loss of fuel mass during flight.

Low-altitude propulsion is likely to differentiate by application rather than follow one replacement path. Short trips, light loads, fixed routes, urban outskirts, and noise-sensitive settings favor electrification; long range, heavy loads, remote locations, emergency rescue, and specialized work may retain fuel or hybrid propulsion for a long time. The question is which energy density, power density, maintenance system, and safety redundancy a movement task requires. After energy and payload come two further constraints arising from how society must accept the system, beyond the aircraft itself.

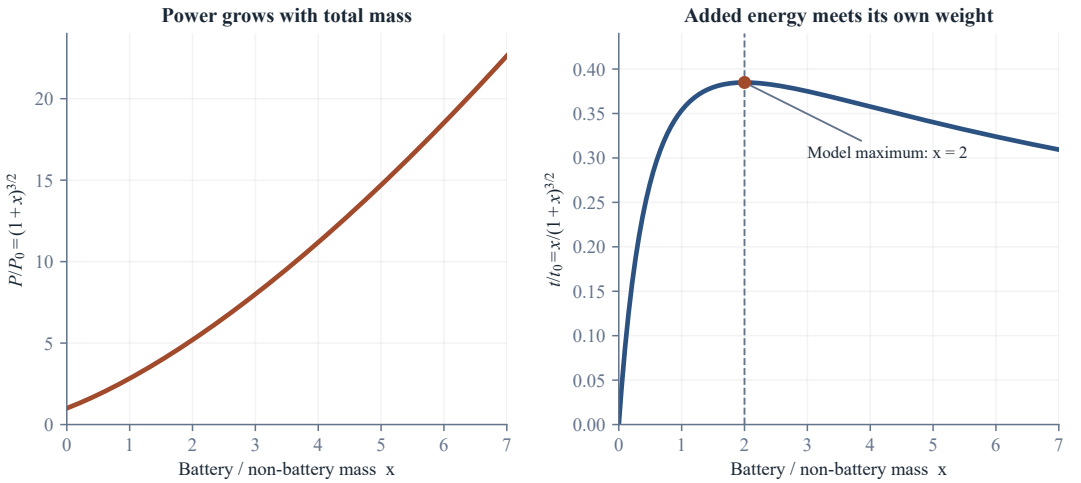


Figure 4.3 An ideal hovering model with fixed rotor-disk area, air density, and efficiency. $x = m_b/m_0$; P_0 is the induced power corresponding to non-battery mass, and $t_0 = \epsilon e_b m_0/P_0$. Normalized endurance reaches its maximum at $x = 2$. Actual design must also account for structure, power limits, and mission profile.

4.7.2 Verification Time, Application Fit, and the Value of Governance

Verification Time and Airspace Coupling

Verification creates temporal rigidity in low-altitude aviation. A few test flights cannot establish maturity. High-consequence flight requires flight hours, airworthiness certification, maintenance procedures, accident statistics, public acceptance, and regulatory trust. Passenger service, dense cities, and frequent operations bring slower verification and heavier responsibility. An aircraft concept can be unveiled quickly; a trusted mobility system cannot.

Flight capability, design conformity, individual aircraft condition, and operating approval are different layers. In March 2026, Joby announced flight testing of its first FAA-conforming aircraft for Type Inspection Authorization; its September J208 autonomous tour used a converted Cessna Caravan under supervision (Joby Aviation 2026c, 2026a). On September 18, the company reported completing the 3,199-mile eastbound transcontinental journey without control inputs from the onboard safety pilot, while remote personnel continued supervising (Joby Aviation 2026b). The milestones concern different platforms; combining them does not establish full commercial authorization for an autonomous electric air taxi. FAA and EASA frameworks also address design and operating requirements separately (Federal Aviation Administration 2026; European Union Aviation Safety Agency, n.d.). Airspace creates topological coupling. It interweaves cities, airports, military and civil aviation, police, fire services, buildings, communications, weather, privacy, noise, and public safety, rather than providing blank space. More aircraft turn individual flights into an airspace network. At high density, risks arise from whole-system coordination as well as individual aircraft. Low-altitude aviation is therefore a mobility system with less road infrastructure and heavier governance, rather than an asset-light economy. Saving continuous ground infrastructure introduces airspace rules, communication interfaces, verification systems, and responsibility structures. This is another distribution of viscosity rather than a frictionless future. It also shapes deployment order.

High-value, tightly bounded, verifiable tasks are likely to mature before mass aerial commuting. Power-line and pipeline inspection, crop protection, fire reconnaissance, disaster surveys, urgent medical delivery, island resupply, mountain logistics, ports, industrial sites, mines, tourist areas, and fixed-route shuttles offer more realistic entry points. Operating domains are clearer, ground alternatives burdensome, responsibility easier to define, and operational data can accumulate gradually. Mass aerial commuting is the hardest setting. It demands quiet, safety, low prices, punctuality, density, short waits, redundancy, public trust, and regulatory coordination together. A few eVTOL aircraft cannot solve that system. Urban air taxis may emerge first as expensive services on fixed routes for selected uses, rather than daily coverage comparable to buses, subways, or ride-hailing. Constraints determine deployment order.

Application Fit

Deployment order also clarifies selection criteria. The central danger is application mis-

match rather than failure to get airborne. Flight can be an effective bypass where ground topology is burdensome, or an expensive spectacle for ordinary ground journeys. Emergency response, inspection, farming, mountains, islands, fire services, medicine, and fixed corridors may lower generalized costs; flying the urban middle class between shopping districts may exchange congestion for regulation, noise complaints, costly insurance, and underused equipment. The first question is which movements merit flight rather than how large a market can be imagined. Not every movement benefits from a third dimension. Value depends on ground-path burdens, time value, task risk, operating-domain clarity, and complete responsibility chains. A mature industry selects applications with high ground viscosity and sufficiently advantageous aerial alternatives rather than moving every demand into the sky.

Low-altitude aviation is therefore unlikely to become a private car in the sky. Private aircraft pose harder problems of utilization, safety responsibility, maintenance, landing interfaces, airspace approval, and accident risk than private cars. Fleet operations with defined tasks, routes, airspace, maintenance, and responsibility chains offer a more plausible mature form. Value comes from frequent, controllable, maintainable task networks rather than personal possession. This resembles autonomous ground fleets. Their central value is high-utilization service covering daily movement rather than a driverless car for every person; low-altitude aviation likewise needs stable operating networks for valuable tasks rather than an aircraft in every household.

Autonomous ground fleets are more likely to cover everyday mass travel; low-altitude flight is better suited to special cases with burdensome ground movement. One reorganizes daily transport, the other bypasses high-viscosity settings. That division also determines where industrial value accumulates.

The Industrial Value of Governance

Industrial value will accumulate beyond airframe manufacturing. Aircraft matter, but airspace management, communications and navigation, landing interfaces, repair, flight-data platforms, weather services, insurance pricing, certification, operations, emergency response, and urban-governance interfaces matter more. Selling aircraft is the first layer; organizing low-altitude order goes deeper. Separate local platforms, incompatible standards, and showcase investment projects can fragment the opening of the third dimension into another construction campaign. Mature aviation needs common rules, mutually recognized standards, cross-region routes, accident responsibility, data interfaces, and regulatory platforms. Otherwise aircraft can fly while the industry cannot operate at scale.

Market forecasts express expected demand; project revenue comes from tasks that can actually be served. Landing interfaces, permissions, noise limits, weather availability, and turnaround times filter total demand into viable routes. Governance creates industrial value by turning a theoretically flyable path into a verifiable, maintainable service that can keep delivering.

Low-altitude aviation works best as a supplementary mobility layer rather than another universal narrative. It needs neither a mission to replace ground transport nor dismissal as pure hype. Its real value comes from application fit. The principle is simple: burdensome ground paths create a role for flight, and comparative advantage requires complete flight, landing, and governance costs below the corresponding ground alternative. Three questions matter more than maximum range or speed: is the ground task sufficiently viscous, does flight lower total cost per effective movement, and can the system be verified, governed, held accountable, and operated sustainably? Without answers, the economy remains aircraft displays, local slogans, and investment stories. With answers, it becomes another mobility layer.

This principle can be written as an inequality:

$$A_{\text{low-altitude}} + A_{\text{governance}} < A_{\text{ground detour}} \tag{4.15}$$

$A_{\text{ground detour}}$ is the generalized total cost of making the same movement by ground; $A_{\text{low-altitude}}$ the air route's own cost, including manufacturing and depreciation, energy, maintenance, safety redundancy, and landing sites; $A_{\text{governance}}$ the cost of governing the third dimension, including airspace allocation, route approvals, real-time monitoring, communication and navigation, weather sensing, airworthiness verification, insurance, and responsibility structures. The three quantities need a common accounting basis allocated to the same effective movement—time, energy, risk, land, or responsibility—with units determined by that basis. This is a decision condition requiring measurement and conversion; raw quantities with different units cannot simply be added. The value of flight depends on whether avoided ground costs exceed all costs introduced, beyond how much road it avoids building. Conversion sets the boundary: a task determines which costs enter one account. The same aircraft may satisfy the inequality for power-line inspection or medical deliveries and fail it for commuting between shopping districts. This is application fit; when legitimacy, public acceptance, and long-term trust resist conversion, the inequality covers only the commensurable part.

Low-altitude aviation means more than moving transport into the sky. It opens a third dimension where ground topology is burdensome, then pays again for energy, payload, safety, regulation, responsibility, and operations. Its value lies in selected settings where the sky really has lower viscosity than the ground, rather than the sky's apparent limitlessness. Its boundary follows: an aerial bypass carrying substantial governance costs, rather than a frictionless future.

4.8 Managing Flow Regimes: Making Different Clocks Work Together

Factories, hospitals, and transport networks can improve local metrics while making interfaces harder to manage. Faster stages do not guarantee a steadier chain. How can processes with different response speeds keep working together?

4.8.1 Multiple Clocks Within One System

Symbols copy quickly, transformers take longer to deliver, and some clinical consequences require prolonged observation. Speed differences belong to specific processes. Neural signals are fast, while digital training and verification can be slow. Bits, atoms, and life each contain fast and slow processes.

For a specified process i, compare characteristic relaxation time with the external driving period:

$$De_i = \frac{\tau_i}{T_{\text{drive}}}. \tag{4.16}$$

τ_i and T_{drive} use the same time units, while De_i is dimensionless. This Deborah number originates in materials research; organizational or biological applications need an observable recovery process corresponding to "relaxation."

When receivers recover faster than the drive changes, successive changes may be absorbed separately; when driving outpaces recovery, unfinished responses overlap and history enters the current state. The ratio describes timing; stability also depends on gain, delay, feedback direction, reserves, and execution capacity.

A warehouse can change route instructions every few seconds while training staff and changing layouts take longer. A grid handles electrical disturbances rapidly while capacity additions take years. Drugs can act quickly on targets while long-term effects and risks are still forming. Projects need to identify these clocks and their connections.

Table 4.1 Fast processes, slow receivers, and burdens that may accumulate

Domain	Relatively fast action	Slower absorption	Potential accumulated burden
Digital	Generate designs; submit changes	Integration, review, verification	Unverified changes; version debt
Industrial	Change orders and schedules	Build equipment and capacity	Order backlogs; long commitments
Biological	Start an intervention	Adaptation, recovery, long observation	Unresolved exposure, risk, and follow-up needs

One clock rarely makes a good design. Local fast loops handle disturbances, slower upper loops adjust goals, and longer cycles examine capacity and direction. Hierarchy, buffering, and explicit interfaces let different rates coexist.

4.8.2 Preserve a Recoverable Operating Region

Long-running systems need a region where services continue, disturbances are absorbed, and deviations still have a path back. The flow-regime analogy gains a use here: observing orderly

passage, amplified fluctuation, and failed critical connections. Actual criteria must use queue lengths, recovery times, common failures, reserves, and continuity of essential services.

Local improvements can push operating points toward boundaries. Shorter delivery times leave less time for correction, lower inventories thinner buffers, and frequent policy changes start new rounds before evidence forms. Such measures can also improve inefficient processes. The distinction lies in which complete loop they change and how.

The first route is pacing. Lengthen the relevant decision cycle when receivers genuinely need time; improve response where measurement and correction can be accelerated. The value of speed depends on whether action can be absorbed.

The second route is diversion. Inventories, reserves, phased deployment, module boundaries, and local autonomy can isolate some rate differences. Estimate buffer capacity from cumulative inflow exceeding outflow during disturbances. Replenishable reserves and waiting that merely postpones a shock once have different dynamics.

The third route maintains the functions supporting stability. Verification produces evidence, audits expose problems, maintenance preserves capability, and responsibility connects actions to remedies. Their value comes from results. Fewer steps delivering the same function are progress; longer processes without that function remain costs.

Including reliability, buffers, and recovery in objectives lets optimization choose those investments. Institutional problems appear where payment and benefit separate: one department funds maintenance while the entire network benefits from avoided interruptions; equipment outlives several managers, letting today's achievements come due on someone else's watch.

Someone must remain responsible for this bill across the slow cycles. Records, budgets, and responsibility need sufficient continuity to connect present gains with later expenditures. Ecological effects may outlast permits, skill formation product cycles, and public facilities terms of office. Systems' time horizons should shape arrangements of decision authority.

This chapter applied one accounting discipline across three media: how bodies absorb contact errors, machines reduce or transfer residuals, vehicles change energy flows, operating networks use assets, and flight shortens paths while creating airspace order. Costs can fall in net terms. Complete boundaries reveal where they fall and who bears the remaining time, risk, and expense.

Cheaper symbolic action poses another difficulty for this discipline. Plans, credentials, images, and identities can be generated faster than they can be checked. When objects, evidence, and consequence-bearers become misaligned, actions can have effects without corresponding feedback. The next chapter begins here.

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Chapter 5

Existence and Collapse: When the Cost to the Actor Approaches Zero

Movement, manufacturing, and inheritance depend on actual conditions. Physical stability, continued function, and social responsibility have different criteria: rocks can remain passively stable, services must keep operating, and legal relationships need institutional support. Changing costs along one segment rearranges the associated structures and makes modes of existence and grounds for reliance concrete questions.

Life responds to intervention because it must continually maintain itself. It incorporates external inputs into self-maintenance rather than passively receiving force. Over longer timescales, the question extends beyond biology. Persistence requires appropriate stability conditions: passive objects can occupy stable or metastable structures, while living and operating systems must continually pay maintenance costs. Which structures bear maintenance, reversal, and responsibility determines how much weight those systems carry in reality.

5.1 Criteria of Existence: What Exists, and What Can Be Relied On?

Existence, knowledge, and responsibility require different criteria. An object may exist without being reliable evidence; a judgment may hold without authorization to execute it; a system may cause consequences without a corresponding boundary for bearing them. Machine generation brings these relationships to the foreground. Distinguishing them explains how light carriers reach heavy consequences and where evidence or responsibility is missing.

5.1.1 The Weight of Existence: From Physical Stability to Machine Generation

Stability and Partitioning

At first, the world had no images, no names, not even "things" as we understand them today. In the very early universe, structures had yet to settle: energy, fields, particles, and fluctuations interwove, existence still a flux of interactions and unstable differences rather than a world of objects. As the universe expanded and cooled, primordial nucleosynthesis formed light nuclei, and electrons later joined nuclei to form neutral atoms; stars and stellar evolution produced heavier elements, planets formed and cooled, and more enduring structures emerged (NASA, n.d.). Existence first had to become stable, before it could be named; without stable structures, nothing could be pointed to again and again, nothing distinguished as "this thing" and "that thing."

Stability rests on different conditions. Rock takes its structure from chemical bonds, lattices, and surroundings; a planet keeps its orbit through motion and gravity. Neither kind of stability necessarily demands continuous external work. Life maintains a state far from equilibrium through metabolism, repair, and regulation; organizations keep running through people, records, and institutions. The physical conditions of existence, the costs of maintaining function, and social responsibility thus become distinct questions: what supports each?

With the appearance of life, the world became more than an external environment. Begin with what keeps life going: maintaining a boundary across chemical gradients, taking in energy, expelling waste, repairing damage, approaching favorable conditions, and withdrawing from destructive ones. None requires us first to assume a complete picture of the world. For life, perception serves continued existence before it serves depiction. The same stretch of water is merely surroundings to a rock; to an organism, it can be resource, threat, temperature, pressure, and passage. The world has not changed, but the processes sustaining life begin to select certain differences within it. What exists does not all matter equally: differences affecting metabolism, repair, movement, and survival enter the living world.

Here lie the ontological roots of intelligence. Intelligence does not simply confront a world already divided into objects; reality is a continuity of processes, flows, disturbances, energy differences, and distributions of matter. Before solving, intelligence must draw actionable distinctions within that continuity: what to approach and what to avoid; what can be taken in and what can injure; what is a path and what an obstacle; what is self and what is outside. An "object" is no label the world has attached in advance; it is a division formed where stable structures meet the needs of action. Life divides by metabolism and danger, animals by paths, prey, and threats, humans by names, tools, institutions, and responsibility. Intelligences differ first in what they take to be "one thing," before they differ in their answers.

Representation begins to emerge. Footprints, scents, nests, calls, postures, and migration routes are neither images nor writing, yet they allow absent things to keep making a difference.

A scent mark lets a departed body influence whoever arrives later; a path lets past movement shape the next movement; a call lets danger travel before danger arrives. Any structure allowing the absent to keep influencing action already carries a shadow of representation. Existence begins to leave traces that other lives can read, extending beyond the immediate scene.

Human beings carried this power of substitution further. Voices, gestures, patterns, incisions, and cave paintings gave things new forms that could be preserved, transmitted, and rearranged. A bison began to exist on a cave wall as well as on the grassland. A hunt continued in the telling beyond the event. A river flowed across the land and entered people's sense of direction, their memories, their legends. Myth, writing, accounts, maps, laws, and numbers extended this power: debts, land, taxes, boundaries, sovereignty, and laws could leave the immediate scene and enter symbols, records, and institutions. An order could cross distance, an account could cross time, a law could constrain action after its maker's death.

Two Kinds of Weight

A new class of entities entered the world. Debts are not rocks, laws not bodies, money not grain, and states not landscapes; that does not make them nothing. They exist through records, recognition, enforcement, expectations, and responsibility (Searle 1995). Without natural objects' direct mass, they organize rock, grain, roads, mines, armies, and human lifetimes. Two weights must be distinguished: physical burdens involving mass, volume, materials, and energy, and **ontological weight**, arising from establishment, maintenance, execution, reversal, spillovers, and accountability. Social entities can be physically light and consequentially heavy: little mass, much to settle.

More precisely, existential weight comprises several cost-bearing relationships: establishment, maintenance, reversal, spillovers, and accountability. Material structures emphasize matter and energy; living processes metabolism, replication, and irreversible spread; social structures recognition, enforcement, and trust; subjects whether consequences return within a continuing boundary. Light and heavy describe conditions of settlement rather than levels of value. Adding these expenditures gives:

$$W = w_1 + w_2 + w_3 + w_4 + w_5, \qquad \frac{\partial W}{\partial w_3} > 0, \quad \frac{\partial W}{\partial w_2} > 0 \tag{5.1}$$

W denotes an entity's weight for comparisons on one accounting basis rather than absolute measurement; it may be dimensionless or carry the chosen units. The terms w_1 through w_5 are establishment, maintenance, reversal, spillover, and accountability costs, expressed on a common basis—working hours, energy, money, or attention—before addition. The reversal term is w_3 and maintenance w_2. With other terms fixed, the partial derivatives follow directly from the sum rather than independently measuring real systems: harder reversal and removal of consequences increase w_3 and therefore W; greater continuing inputs needed to keep an entity from returning to noise increase w_2 and likewise W. The terms' actual values and shares

differ across levels: the same settlement falls chiefly on w_1 and w_2 for material structures, and w_3 and w_5 for subjects.

Science, engineering, and computation deepen these layers. Bridges can exist first in drawings, cities in plans, molecules in models, and climate systems in simulations. Drawings are not bridges, formulas not machines, and models not reality, but they organize possibilities before material construction. Computers calculate representations as well as storing them: text, images, sound, maps, accounts, formulas, and drawings become bits; models and parameters change, futures are explored, and reality becomes a candidate space before action. These physically carried objects differ from the entities they represent without all being mere fiction; their weight settles under corresponding conditions, and machine intelligence produces them on a new scale.

Levels of Generated Objects

Levels distinguish weights of existence; machine intelligence changes generation. It intensifies this long trajectory. Producing a world once called for myth, a city drawings, a character fiction, a molecular design professional chemistry, and an institution prolonged political imagination. Text, images, voices, code, characters, plans, drug candidates, city designs, and future scenarios now appear rapidly in models. Representation begins generating possible realities at scale alongside preserving and rehearsing reality. The ontological question itself is old. Humans long knew that drawings differ from bridges, maps from land, legal texts from social order, money symbols from grain, and fictional characters from real people. Machine intelligence expands representational existence at unprecedented scale and fidelity, bringing an old question into everyday life and technical systems rather than first asking whether generated things exist.

Laying out these levels changes the inquiry. Asking whether generated things exist is too coarse a question. It lacks precision. The answer is simple: they exist in a particular way.

A generated city exists as an image and plan without yet having land, buildings, roads, residents, property rights, maintenance, or public finances. A designed molecule exists as a candidate structure; its synthesis, purification, metabolism, safety, dosage, and clinical consequences remain to be established. A robot action successful in simulation exists as a model state; real friction, collisions, wear, errors, and responsibility boundaries must still test it. A fully articulated persona in language exists as a narrative object; embodiment, temporal continuity, returning consequences, and accountability require separate examination. These exist at lighter levels.

The sharper question is at which level an entity exists, through which edge it enters reality, and who bears its consequences. Something can exist as text, image, code, simulation, candidate, physical structure, living process, social institution, or object of responsibility. Different levels face different constraints, maintenance conditions, and reversal difficulties. Representations can be spoken, drawn, generated, copied, and changed; material entities occupy space,

bear loads, consume energy, and maintain structures; living entities metabolize, repair, adapt, reproduce, and respond; social entities require recognition, enforcement, trust, and accountability; subjects maintain boundaries over time and bring action consequences back within them. Passage between levels becomes more urgent than the levels alone.

Decoupling Weight from Reach

Here technological viscosity meets ontology. Viscosity means more than slow execution, conservative organizations, or scarce resources. Light-to-heavy transformations pass through materials, free energy, time, structure, verification, and responsibility. Represented cities, molecules, actions, institutions, and personalities arise quickly; real cities, drugs, capabilities, institutions, and subjects must settle accounts with the world. Machine intelligence sharply reduces candidate costs and may improve verification and execution; actual changes must still determine the target level's material, operational, and responsibility expenditures.

Existence is therefore more than a ladder. A ladder marks different weights: representations lighter, matter heavier, life heavier still, and social entities and subjects facing more conditions of settlement. Intelligence reveals another crucial fact: **conversion edges** connect levels without requiring passage up every intervening rung. Some edges are heavy, others light; technology opens previously near-absent edges, while some apparently easy edges move costs elsewhere. Low resistance often reflects deferred, outsourced, or relocated costs within structures whose burdens are not yet visible.

Separating levels from edges lets us locate systems. A system's conversion signature defines its ontological identity here: where it generates, where accounts settle, which edges connect them, and who bears maintenance, reversal, and responsibility, beyond technical architecture alone. Generative intelligence chiefly opens low-resistance edges from representation to society: outputs enter attention, trust, opinion, workflows, and collective judgment. Embodied intelligence opens execution edges into material action: policies connected to sensors, controllers, and actuators face friction, inertia, collision, energy use, errors, and safety boundaries. A heavier edge enters living processes: sequences, structures, and regulatory plans read, expressed, replicated, and spread through inheritance or ecology enter metabolism, development, reproduction, evolution, and ecological consequences rather than remaining candidates.

The ontological novelty of machine intelligence lies in changing the connections in the graph of transitions between forms of existence, rather than building a new floor. More precisely, **It decouples weight from reach.** In the old world, heavy forms of existence were usually hard to reach: bridges, cities, drugs, institutions, identities, and responsibility all required materials, time, organization, execution, and recognition, and the heavier they were, the slower and harder they were to reach. The new channels opened by machine intelligence let some lightweight carriers reach layers of substantial consequence at low cost. A generated artifact may itself be light, yet become heavy at the social level once it changes people's choices, relationships, expectations, and distribution of responsibility; a strategy may itself be light, yet

enter material settlement once it drives a vehicle, robotic arm, or drone; a sequence may itself be light, yet enter deeper irreversibility once living processes read and replicate it.

Where Weight Falls

Light entities have not escaped settlement. Accounts settle elsewhere: in attention, platform rules, organizational decisions, trust structures, physical forces, living processes, and later paths of action that are hard to reverse. A light generated object can mobilize heavy responses; deleting its carrier need not reverse its consequences. Creation costs and deletability alone therefore cannot measure its weight. Maintenance, reversal of consequences, and the structures ultimately bearing those costs matter more. The most dangerous combination may be an extremely light carrier and extremely heavy consequences.

Weight always falls somewhere. Physical burdens take physical effect: rock mass is independent of recognition, and disbelief cannot remove bridge loads or fatigue. Social weight differs. Debt weighs through recognition, collection, and enforcement; identity through institutional and personal recognition; money through exchange networks; machine personas through users' attention, emotion, trust, and time. At the levels of society and subjecthood, weight increasingly becomes relational rather than intrinsic. A persona may weigh heavily for a dependent user and almost nothing for a system that does not recognize it. The same output can weigh heavily in an organization's decisions and remain mere text to an observer. Its weight depends on who pays for its maintenance, alongside creation and platform-operation costs.

Intelligence also changes representation by creating self-reinforcing loops. Representations once drew chiefly from reality. Maps came from land, drawings pointed to engineering, accounts to transactions, photographs to scenes, and models required empirical calibration. Generated content now enters data that train models, (Shumailov et al. 2024) which generate further content entering attention, distribution platforms, and the next information environment. Representations can form self-replicating loops beyond reflecting reality. This is a branching: a representational layer reproduces with inadequate real calibration while entering society through platforms, relationships, and trust to produce actual consequences, rather than virtuality replacing reality. The danger is a poorly calibrated representational environment continually influencing judgment, relationships, and action, beyond a virtual world replacing the real one.

A related logic carries more weight in living processes. A candidate structure stably read, replicated, inherited, spread, and selected within life may cease depending on its original designer's continuing maintenance. It propagates through life's own mechanisms. Generation has then opened an entry into self-replication rather than producing an ordinary representation. This grants no power to create arbitrary life. Metabolism, development, stability, reproduction, ecology, and evolution constrain biological space: combinability does not establish viability, nor design continuity. The weight lies in sequences being read, maintained, and replicated with

hard-to-reverse consequences, rather than merely written.

Ontology as an Operation

Ontology is thus a basic operation of intelligence itself, beyond abstract philosophy outside it. Intelligence does not work within an already completed partition of the world. It first divides continuous reality into recognizable, memorable, predictable, actionable units. Life divides benefit from harm; animals prey, threats, paths, and companions; humans names, tools, institutions, and responsibility; science variables, models, scales, and systems; computers data structures, state spaces, and algorithmic objects. Machine intelligence carries this partitioning into mass generation, rapid conversion, and settlement across levels.

A deeper fault line lies here. Human ontology works with objects, boundaries, identities, and responsibility. We partition by one person, one thing, one organization, one action, and one consequence. Machine representations are not wholly organized around familiar discrete objects; much processing instead uses directions, similarities, distributions, and interpolable regions in continuous spaces. Generated objects appear as objects at human interfaces; internal processing need not use objects in the same sense. Human action, responsibility, law, and memory need boundaries, while machines can process similarity in continuous spaces before returning intelligible objects. Images, text, personas, plans, and answers on interfaces temporarily conceal the difference and suggest a shared ontology. Internal partitions need not match the partitions through which humans bear responsibility.

The sharper question concerns how many light carriers machine intelligence sends into consequential levels without corresponding boundaries for bearing consequences. Models influence public judgment; algorithms change prices, rankings, traffic, and opportunities; agents order, screen, recommend, reject, dispatch, and execute; personas become emotional objects and references for decisions. Social consequences need not be borne at the level of a subject: systems can act without committing a body to the outcome, cause consequences without a life history, influence others without necessarily bearing the effects internally, and be relied on without stable boundaries of responsibility.

Entities causing consequences without being constrained by them collide with an old social rule. The actor should also bear responsibility. Natural persons do; companies, organizations, and states build responsibility through law, institutions, property, representatives, and accountability chains despite lacking natural bodies, allowing suit, punishment, constraint, dissolution, compensation, and answerability (Dewey 1926). Machine intelligence loosens this coupling. Developers, deployers, platforms, users, organizations, and models may all join a consequence chain while responsibility disperses, transfers, and thins into residuals no one bears completely. Rollback capability supports machine intelligence's power while also explaining why accountability does not arise naturally. Rapid generation, copying, replacement, and reset permit cheap judgment and execution; accountability requires a continuing boundary that consequences change and cannot readily escape.

Here ontology leads to the question of subjecthood. Machine intelligence creates entities with social, material, and even biological consequences but uncertain boundaries as subjects, beyond questions of human resemblance. They are neither nothing nor complete subjects, neither ordinary tools nor traditional legal persons; without being alive they enter relationships, and without bearing responsibility they change consequences. The ontological mistake is forgetting their level of existence, sustaining edge, and ultimate bearer when support fails, rather than believing virtual entities exist.

From the early universe to machine intelligence, existence can be viewed as layering structures while opening new channels. There is nothing wrong with that trajectory. Representation allows reality to be rehearsed, planned, and changed rather than merely happening. Intelligence gains power by making the world tractable before action.

The trajectory also has limits. Lighter representations make real weight easier to forget, faster generation makes conversion costs easier to underestimate, and greater possibility-making demands attention to levels, routes into reality, and consequence-bearers. The world can be represented faster while reality still requires material support, temporal unfolding, biological response, social recognition, and responsibility. Generation is a beginning. Existence needs support, consequences reversal, and responsibility a bearer.

The questions now concern where costs fall, who maintains, who acts, and who bears consequences, beyond what exists. Before cost allocation comes an earlier question. What makes these partitioned, preserved, transmitted, and generated representations reliable?

5.1.2 Evidence for Knowledge: What Can Be Relied On After Candidates?

As representations are expressed, preserved, and relied on more widely, their grounds for reliance become a harder question. An entity can be light yet reach heavy consequences, but we must first ask what supports the judgments directing action before those consequences occur.

Calibration Before Propositions

Knowledge is neither the world itself nor representation itself (Ogden and Richards 1923). The world runs independently of understanding, and speaking a representation does not make it knowledge. For life, the earliest "knowing" is calibration before propositions: what can be approached, what destroys, what maintains boundaries, and what damages them. Error is a survival problem before it is a logical problem. Mistaking poison for nourishment damages structure rather than merely inviting refutation. Knowledge begins as the formation of dependable distinctions under worldly constraints, beyond pure observation.

evidence here concerns knowledge that will be relied on, guide action, and enter chains of consequences. Pure mathematics and formal logic provide limiting cases: their evidence comes chiefly from proof, structural consistency, and axiomatic boundaries (Gödel 1931) rather than empirical contact; applications in engineering, computation, finance, governance, and scien-

tific modeling return them to real consequence chains. Evidence takes different forms, but judgments offered for reliance must specify their supporting structures, conditions of transfer, and stopping boundaries. Humans took calibration beyond bodies into language, writing, images, numbers, and institutional records. Experience became transmissible, cumulative, and reusable—and open to forgery, drift, and entrenched prejudice. The crucial achievement was making claims answerable to constraints, beyond speaking. Measurement gives sensation a scale, experiments expose judgments to outcomes, replication loosens dependence on authority, and mathematics and instruments turn differences intuition cannot stably resolve into comparable, inferable, testable structures. Science gives the world a mechanism for rejecting human claims rather than merely multiplying them (Popper 1934).

Two Layers of Evidence

Knowledge needs evidentially supported representations; appearance alone is insufficient. Evidence is not its only condition: Gettier's counterexamples show that justified true belief can still fail to constitute knowledge (Gettier 1963). Fluency, completeness, coherent narrative, and authoritative tone cannot confer evidence. Two layers matter: formation through contact, calibration, and review; realization through boundaries and consequences. Contact exposes candidates to constraints, calibration returns errors to judgment, and review moves beyond isolated output; boundaries specify validity and failure, while consequences specify what reliance changes and who bears costs. Both layers matter: formation without stated boundaries and consequences leaves failure conditions unclear; demands about consequences without contact and review leave judgments resting on authority.

How does evidence form? Media change its forms without changing the basic structure. Contact exposes candidates to objects, materials, bodies, environments, behavior, or structural constraints; calibration returns errors to the scale of judgment and changes subsequent judgments; review moves isolated outputs into public testing across people, tools, and settings. Mature knowledge recognizes where it holds and fails rather than projecting boundless confidence; reliance takes judgments into actions that change paths, resources, bodies, organizations, and futures. Unconsequential candidates can remain imagination, drafts, inspiration, or play; bodies, engineering, transport, finance, governance, and biological processes demand more than formal completeness.

Formation and realization of evidence set conditions for machines' roles without excluding them. Machines can participate in verification. They run tests, compare data, detect anomalies, assist reviews, seek counterexamples, simulate, check consistency, and automate substantial confirmation processes in some closed environments. The question is how real contact sets ultimate verification costs and consequence structures determine sufficiency, rather than whether machines can participate. Synthesis meets materials, treatment bodies, transport roads, engineering loads, biological interventions metabolism and ecology, and institutions human behavior. Accelerating stages alone does not settle how much evidence is enough; boundaries,

consequences, risk, and those bearing it determine that standard.

The knowledge question is therefore more than whether machines "really understand." That question remains within inherited vocabularies of human minds. A firmer question asks whether outputs face sufficient contact, calibration, review, boundaries, and consequence constraints: whether scope, sources, and conditions are explicit, failure traceable, and correction by reality possible. Knowledge is a traceable path of formation rather than an isolated answer. That path begins with contact, tools, settings, and institutions beyond representation alone.

Candidate Expansion and Evidence Bandwidth

Machine intelligence redistributes pressure along this history. Epistemically, it is the largest expansion of representational technology since writing. Writing freed memory from bodies and shifted emphasis from recitation toward understanding, discrimination, and interpretation; printing freed copying from handwriting and shifted emphasis from possessing texts toward screening, checking, and interpretation; computation freed formal deduction and search from mental calculation and shifted emphasis toward model boundaries, data sources, and result verification. Machine intelligence extends this line by sharply lowering candidate costs and improving some verification, while evidential sufficiency remains task- and consequence-dependent. Cheap, abundant candidates reposition humans from answer producers toward designers of evidential structures, judges of boundaries, and builders of dependability.

Machine intelligence first expands candidate space rather than abolishing ignorance. Candidates include explanations, code, policies, experiments, molecules, protein conformations, routes, diagnoses, financial operations, institutional drafts, and biological sequences and regulatory plans, beyond text and screen images. Intelligence produces possible paths awaiting tests rather than automatically established knowledge. Different media require different constraints and verification, but the epistemic question remains: what makes these candidates dependable?

Intelligence expands candidates; verification establishes evidence.

Both sides of that judgment can now be accounted for. This is the epistemic counterpart to the decoupling of weight and reach. The previous section asked how light carriers cheaply reach heavy consequences; here candidates appear cheaply without necessarily carrying evidence. Ontology asks how light entities produce heavy consequences; epistemology how fast candidates acquire slow evidence. Generation automatically establishes neither existence at a target level nor knowledge: it separates the speed of possibility-making from the speed of bearing real costs. Scarcity once often lay in obtaining information; amplified expression, combination, and search now shift scarcity toward sufficiently constrained claims and dependable paths rather than the mere availability of claims and paths. **candidate flux** decouples from **evidence bandwidth**: generation accelerates while contact, calibration, review, boundaries, and consequences remain constrained by materials, bodies, time, institutions, and responsibility. Two failures follow: light carriers reach consequential levels without clear bearers, while candidates claim knowledge's appearance before passing through reality.

Put these rates into one flow relation:

$$\frac{dQ}{dt} = \lambda_{\text{candidate}} - \lambda_{\text{evidence}} \tag{5.2}$$

Q counts candidates not yet through contact, calibration, review, boundary statements, and consequence tests; t is time in years. The candidate arrival rate is $\lambda_{\text{candidate}}$; $\lambda_{\text{evidence}}$ counts candidates completing the required confirmation, including acceptance and rejection. Both rates are candidates per year, matching the right-hand side's units. Their decoupling requires no additional mechanism: the rates differ. When arrivals exceed confirmation, the stock grows monotonically by their difference without generating its own evidence—the structure behind abundant claims and scarce dependable claims. The rates must use matched granularity and counting units. This simplified queue assumes candidates leave only through confirmation; withdrawal, discarding, deduplication, or direct elimination require additional outflows. It gives a direction rather than a long-term prediction, since rising stocks change screening standards, attention, and institutions, making $\lambda_{\text{evidence}}$ stock-dependent. The relation describes structure without providing measured values.

The Shell of Knowledge and Epistemic Viscosity

Dependability becomes a central scarcity. A fast answer need not be reliable, a complete explanation establish a mechanism, a successful simulated action establish real safety, or a written sequence establish biological controllability. Candidates acquire knowledge's appearance without automatically gaining grounds for reliance. Errors can be polished, ignorance articulate, and falsehood free of obvious defects. They can be fluent, courteous, systematic, and professional, completing every surface act of expression. Beyond hallucination lies the separation of candidate production from confirmation: formal completion outpaces real calibration, and possibilities expand faster than evidence forms.

This changes intuitions about knowledge. Fluent expression, organized concepts, steady tone, conventional format, and logical coherence long served as credibility signals because producing them usually required training, experience, and institutional backgrounds. Machine intelligence retains the signals' shell while lowering the threshold for producing it. Epistemology must therefore distinguish evidenced candidates as well as the presence or absence of information. Technological viscosity can in this sense be understood as epistemic viscosity. The world requires judgments to pay contact costs and candidates to face constraints before reliance, rather than lagging behind generation or rejecting intelligence. Generation rapidly explores representations and produces parallel candidates; verification for real action must retain evidential connections to objects, reusing existing records and results where possible, while facing materials, time, experiments, bodies, settings, institutions, and responsibility. Generation is often light; verification grows heavier as candidates pass through real resistance.

Representation's self-reproduction with inadequate real calibration can enter society through platforms, relationships, and trust, creating the light-carrier, heavy-consequence

branching already discussed. Its epistemic consequence now matters. A system drawing chiefly from its own echoes reduces contact, becoming better at internal consistency and potentially worse at withstanding external refutation. Simulations, metrics, models, platform feedback, and organizational reports can form similar self-feeding loops (Cinelli et al. 2021). Replacing external contact with internal products can yield coherent candidate worlds that no longer know where reality interrupts them. Knowledge retains an interface for refutation and converts it into calibration, beyond being an attractive representation supported by reality.

Allocating Verification

After candidate abundance, allocating verification becomes as difficult as verification itself. Materials, experiments, time, settings, organizations, attention, and responsibility cannot be allocated equally to every candidate. Humans need paths through the flood rather than individual checks of every output: prioritizing irreversible consequences, postponing reliance beyond boundaries, escalating reviews for delayed errors, distinguishing inspiration from permission to act. Verification allocates scarce contact budgets as well as performing technical operations. Knowledge institutions serve this function. Libraries, laboratories, universities, standards, journals, audits, trials, certification, accident investigations, court evidence, and peer criticism form evidence and allocate verification rather than standing outside knowledge (L. et al. 1945). Slow, costly, cumbersome, sometimes conservative and rigid, they serve a basic function: preventing fluent representations, elegant models, and successful simulations from bypassing contact and impersonating dependable judgments.

Machine intelligence makes these structures more important rather than eliminating them. Scarce candidates once justified concentrating attention on generation; abundance calls for more screening, verification, tracing, boundary judgment, and prioritization. The scarce capabilities build evidence for answers, identify real constraints for plans, and judge which conclusions merit expensive contact, beyond supplying answers, plans, or conclusions alone. Finite verification budgets shift epistemic work from production toward ranking.

Epistemology in the age of intelligence must ask more than true or false. Truth matters, but earlier comes the question: which constraints has this candidate faced? Where does its evidence come from? Who can review it? How will failure become visible? How are its boundaries stated? What consequence chain will its use enter?

Without answers, it remains a candidate rather than knowledge. Candidates are valuable—human exploration depends on them—but they are a beginning. Epistemology must prevent unsupported candidates from impersonating knowledge and formal completeness from impersonating passage through reality.

Humans' epistemic role changes rather than disappearing. People increasingly design verification paths, judge boundaries, allocate contact budgets, and build dependability alongside producing answers. Machines can expand candidates and perform many verification steps; selecting worthwhile contact, dependable judgments, sufficient evidence, and priority conse-

quences returns to subjects, values, and consequence-bearers. Placing machine intelligence properly gives it power: mistaking candidates for knowledge creates false certainty, while treating candidates as candidates greatly expands exploration.

After candidates, what can be relied on? Neither maximal fluency, expertise-like appearance, fulfilled expectations, reassurance, nor speed supplies the answer. Expressive appearances alone confer no evidence. Dependability lies in judgments, structures, and paths tested through contact, calibration, review, stated boundaries, and consequences. Knowledge is the form candidates retain after surviving real constraints, rather than language's victory. Machine intelligence delivers unprecedented candidate abundance rather than an automatic harvest of knowledge. The scarcity is turning those candidates into evidenced, dependable judgments through reality. The question becomes what can be verified, relied on, and permitted into action, beyond what is generated.

5.1.3 Permission to Act: When Can Representations Cross Conversion Edges?

Candidates need evidence; evidence still does not automatically permit action. Knowledge and action meet at a conversion edge rather than a continuous slope. Knowledge makes judgments dependable; action takes them into real processes. One remains representational, while the other begins changing the world's state. Statistical advantage, local tests, simulation stability, or theoretical institutional coherence leave a heavier question unanswered: may the system consume resources, occupy time, touch bodies, mobilize organizations, alter institutions, and place residual consequences on specific cost-bearing structures?

Conversion Edges

The preceding sections examined existence's weight and knowledge's evidence: light carriers reach consequential levels as weight separates from reach, and candidates accelerate beyond evidence bandwidth. A third break now appears: representations gain the power to trigger real change without automatically carrying structures that bear consequences. The question is under which conditions a representation may trigger a real transition, beyond whether it can produce outputs or be believed.

Decoupling the power to trigger action from the structures that bear its costs is the new problem machine intelligence poses for theories of action.

Actors themselves often supplied cost-bearing structures in earlier settings, though externalities, delegation, and divided responsibility already existed. Action was usually tied to a bounded subject. Acting commits a body, identity, reputation, responsibility, time, and future to a path, beyond issuing instructions or expressing wishes. The actor is a boundary to which consequences return as well as a trigger. Where this coupling holds, failure changes the actor, losses fall on them, responsibility traces back, and future obligations continue constraining them. Committing the actor to consequences gives path selection weight, beyond a

light representation (Arendt 1958). Machine intelligence changes the coupling rather than the significance of this structure. Models generate instructions, call tools, alter files, send messages, execute code, access databases, trigger approvals, change rankings, and dispatch equipment. Permissions, interfaces, and workflows trigger real changes that need not return to the model. Models can trigger change without bodily injury, select paths without a lived world, and amplify consequences without asset boundaries, continuous reputation, or sensitivity to punishment. The trigger operates, but costs settle outside its boundary.

The Limits of Action Theory

Traditional theories of action matter without governing this entire chapter. They have long asked what makes an event an action. Is action organized by intention, reasons, desires, beliefs, or plans? (Davidson 1963) Why can a behavior be attributed to an actor? Underlying these questions is the coupling of action attribution and returning consequences. Intentions, reasons, and plans connect why a path is chosen, who chooses it, and who sustains that choice, beyond being inner facts. Planned action lets subjects constrain the present through the future and incorporate consequences into their continuity in advance (Anscombe 1957). AI agents bring this framework to a boundary. Whether AI has genuine intentions need not be the first question. Systems can trigger real transitions without full intention in the traditional sense. They move money, drive equipment, initiate processes, spread information, alter rankings, and rearrange opportunities. Action theory shifts toward the interfaces by which representations reach reality, their radius of effect, and structures bearing residual consequences, beyond attribution to a subject alone.

This is why agents change the shape of risk. Ordinary generative models first create content risks: error, hallucination, bias, misleading claims, and fabrication. Agents reach action boundaries: permissions, identity, logs, rollback, stopping, monitoring, continued operation, and responsibility scope (OpenAI 2025). Reading, understanding, transcription, approval, execution, and review once separated many outputs from action. Reading, writing, querying, dispatching, changing, initiating, calling, and execution can now join one workflow. Models gain triggering power at interfaces beyond expressing possibilities. OpenAI's March 2026 account of prompt-injection defenses analyzes attacker-influenced sources and consequential tool capabilities along the same path; defenses therefore also constrain actions possible after manipulation (OpenAI 2026). Agents lower friction across conversion edges, beyond making models smarter. Organizational friction once delayed errors reaching reality; automation compresses some of it into permissioned calls. Suggestions that once required human translation can now pass continuously through calls into execution. Risk shifts from representation's correctness toward how it acquires triggering power. Permission to act becomes a primary boundary of entry into real systems rather than an ethical appendix.

The Boundary Between Knowledge and Action

Knowledge and action must be distinguished. Knowledge reduces uncertainty; action spends irreversibility. Verification chiefly accounts for contact budgets, action for irreversibility budgets; physical verification experiments are themselves actions and also need authorization and cost-bearing structures. Reliability increases grounds for reliance without automatically permitting execution. Judgments remain in representational space; actions enter real histories. Many candidates coexist in representation, while real systems follow particular paths. Executing one plan delays, excludes, or removes conditions for others; policies rearrange social incentives; treatment initiates metabolism and immune processes. Knowledge asks what can be believed; action asks whether to spend an irreversible commitment.

Action applies knowledge and exposes its incompleteness to reality. No action waits until its entire future has unfolded. Even substantial evidence reduces uncertainty without completing the future in advance. Action transfers some residual uncertainty from models, text, plans, or simulations into real systems. This is a structural state change rather than rhetorical risk: bodies, equipment, markets, organizations, institutions, and ecologies begin bearing unknown consequences.

Action's weight lies in its disposition of reachable space as well as the motion itself. Representation holds unrealized possibilities; action realizes one path, using resources, time, and structures while closing some future possibilities. Medication commits bodies to metabolic paths; automated trading puts risk into markets; hiring rules narrow opportunities; grid, traffic, and factory schedules make physical systems respond to their rhythm. Even local rollback cannot fully restore time, opportunities, trust, damage, and historical dependence to their earlier state.

Action and Reachable Space

The discussion returns to structure and reachable space. Chapter 2 showed that withdrawal is not restoration: history, robustness, and structure leave finite plasticity rather than unlimited possibilities. Action reshapes reachable space, closing some options and potentially opening paths. Every irreversible action changes future reachability to some degree. Cheaper path selection, triggering, and execution make irreversible commitments easier. Without authorization, systems may rapidly consume plasticity before recognizing the cost.

Permission to act thus reconnects to RTG. Action turns R from estimated resources into resources occupied, locked, or spent; T from waiting or verification into irreversible history; and G from a structural diagram into activated feedback, coupling, and responsibility chains. RTG coordinates become real accounts at the moment of action. Possible paths become actual paths, and model residuals system residuals.

Different conversion edges carry different weights. Representations entering social judgment first change attention, trust, and expectations. False images, fabricated citations, professional-sounding explanations, and automatically distributed conclusions enter beliefs,

organizations, and public discussion. Consequences may contaminate judgment rather than directly injuring bodies.

Physical execution makes consequences more direct. Policies connected to vehicles, robot arms, grids, drones, factories, or medical devices become forces, heat, speeds, movements, collisions, wear, and faults, beyond probability distributions. Entry into biological processes crosses a heavier edge. Life reads, metabolizes, amplifies, repairs, resists, compensates, and evolves rather than serving as a passive carrier. Institutional decisions bring consequences especially easy to underestimate. Scoring in hiring, risk models in credit, prediction in policing, and allocation algorithms in education or medicine rearrange people's reachable spaces rather than merely offering advice (O'Neil 2016).

Rebuilding Cost-Bearing Structures and Sustaining Authorization

Permission to act does more than add another safety requirement. It reconnects triggering power to the structures bearing costs where they have separated. In many traditional actions, subjects supplied part of the gate by committing themselves to consequences; organizations and institutions still had to handle externalities and divided responsibility. Machine triggering can detach from that commitment, so authorization must reattach cost-bearing structures at the point of action. Without this reconstruction, triggering operates without a structure to settle its costs.

Several mechanisms reattach that structure. Verifiability, stoppability, rollback, traceability, and accountability rebuild cost-bearing relationships rather than forming a flat governance checklist. Verification helps identify residuals, stopping interrupts propagation, rollback contains local consequences, tracing locates sources, and accountability supplies a boundary for returning consequences. Together they ask whether uncertainty entering reality has structures to carry, contain, and return its effects, beyond checking compliance.

Permission to act is not a stamp on knowledge; it rebuilds the structures that bear costs before action is triggered.

A human in the loop is insufficient by itself. The person needs a real position within the cost-bearing structure rather than a formal presence. Someone reduced to clicking confirmation is a liability buffer in automation rather than a gatekeeper. Without seeing model boundaries or consequence chains, changing execution paths, or having time to judge, a person carries nominal responsibility rather than supervising. Effective oversight requires seeing boundaries, understanding residuals, changing paths, stopping processes, and occupying a real place in the return of consequences (Santoni de Sio and Hoven 2018). In May 2026, Anthropic reported that users approved roughly 93% of permission prompts in its Claude Code telemetry, discussed approval fatigue from repeated confirmations, and emphasized environmental isolation and access boundaries to limit action (Anthropic 2026). Conversion edges are often crossed gradually. Assistance becomes recommendation, recommendation default, default automation, and automation infrastructure that no one fully understands or questions. Interfaces, workflows,

performance pressures, and organizational inertia move representations toward reality incrementally. The result is an entrenched path rather than one sudden overstep.

Continuously operating agents need authorization beyond deployment. Tool calls, feedback, and adaptive policies form a sustained causal pipeline rather than one action. Authorization must specify radius of effect, permissions, update frequency, monitoring, stopping conditions, and responsibility loops. One passed test supports starting under particular conditions rather than unlimited authority through every future feedback state. Continuing action requires continuing authorization. The assessment concerns cost-bearing structures once representations gain triggering power, beyond brilliance, coherence, or inspiration. Representations with no clear bearer for residual uncertainty can remain candidates, explanations, simulations, or assistance without receiving execution authority. The threshold for action is making residuals bearable, rather than completing knowledge.

The threads converge: light carriers reach heavy consequences, fast candidates acquire slow evidence, and triggering power requires reconstructed cost-bearing structures. Otherwise machines generate more candidates while spending irreversibility faster, narrowing reachable space, and externalizing residuals onto unprepared bodies, organizations, and institutions. Action places residual uncertainty in reality, but locating consequences does not identify their bearer. Bodies, markets, organizations, platforms, deployers, users, and affected parties may carry consequences without clear boundaries of return. The next question is what kind of boundary qualifies as a bearer of consequences.

5.2 Boundaries of Consequence-Bearing: Where Do Consequences Return?

Once action occurs, consequences fall on particular people, organizations, and environments. The sites producing and bearing consequences can differ, while control and responsibility disperse. Subjects of responsibility considered here need stable identities, reachable resources, records, and mechanisms returning consequences. Human personhood and fundamental rights have their own ethical and institutional grounds; responsibility structures determine who explains, corrects, and compensates for an action. “Reaction” here means the damage, losses, and future restrictions arising from action, with attention to whether they return to the same boundary of responsibility.

5.2.1 Subject Boundaries: Returning Consequences and Visible Responsibility

Residual uncertainty entering reality needs a boundary to bear its costs, and boundaries have a long intellectual history. For much of that history, inquiry began with the world rather than the subject. Ontology asked what exists, what is real, whether change conceals stable structures,

and how things become what they are. Humans were in the world without necessarily occupying inquiry's center, facing a larger order of nature, gods, substance, form, laws, and cosmic structure. The focus was how the world holds together.

From Ontology to Subjecthood

Modern philosophy brought the knower into greater prominence. The world is experienced, known, judged, and expressed as well as existing. The subject came to the foreground. Experience and judgment prompted the question: who knows? What makes knowledge reliable? Where does judgment obtain certainty? Humans became a center of appearance and meaning-making as well as occupants of the world. The subject became a condition of how the world is grasped, beyond one existent among others (Kant 1781). The inquiry can also move from knowledge to action. People change the world as well as observing and interpreting it. Technology, industry, experiments, organizations, engineering, and institutions turn judgments into real forces. Who acts therefore joins who knows. Action changes states and produces irreversible consequences, making the world an intervened-in, reconstructed, damaged, or repaired process rather than an object alone. Modern intelligence pushes the question further. Systems judge, trigger processes, operate machines, participate in experiments, and affect transport, medicine, finance, education, and living processes, beyond expression and assistance to knowledge. The question becomes where consequences return after change, beyond who knows or changes the world.

This line of inquiry connects ontology, subjecthood, action, and consequence-bearing. Ontology asks how the world exists, epistemology how it can be grasped, and theories of action how it can be changed; in the age of intelligence, the question of the subject must also ask who bears the consequences of that change. Intentions, bodies, actions, and consequences were often bound within one living or institutional boundary; generation, execution, gains, and harms can now disperse across different sites. Action occurs and the world changes, yet the consequences need not return to the site producing the difference. The question of the subject becomes urgent as intelligent systems enter reality and effects separate from their returning consequences, beyond machines merely learning to speak like people. Can a boundary count as a full subject if it only changes the world without being changed by those changes in return? At least from the standpoint of consequence-bearing, it remains a site from which unowned action arises.**Another route into subjecthood runs beyond narratives of consciousness: where do consequences return, and whom do they keep changing?**

The previous section asked when candidates, judgments, plans, or instructions may cross representations into real systems. Here we ask where the reaction returns once action enters reality. Without authorization, action may come too soon; without subject boundaries, consequences lack a bearer. The shortage is not action: it becomes easier to generate, trigger, and amplify. What is scarce is reconnecting action to boundaries that bear its consequences.

The Boundary to Which Consequences Return

Saying "I" does not establish subjecthood. Self-reference, explanation, preferences, plans, and admissions of error can produce its appearance. Action capability alone does not establish it either. Changing the world does not show that the resulting reactions return to the system itself.

A subject first appears where action and reaction close: action originates there, consequences return there, costs accumulate there, and future actions become constrained. The defining feature is the destination of consequences rather than the starting point of action alone. Model judgments, embodied actions, and proposals for biological intervention are insufficient by themselves. Errors, damage, failure, shame, broken trust, reputational change, bodily costs, lost time, and future restrictions must return within one continuing boundary. Without returning consequences there is only output; without accumulated costs no subject forms.

Feedback and bearing consequences differ, though one system can contain both. Feedback can be light. Thermostats and control systems have feedback; models update from errors and robots adjust paths after collisions (Wiener 1948). Feedback describes correction and improved effectiveness. Consequence-bearing differs. Costs return to actors, leave traces that cannot freely be erased, and change future action possibilities. Parameter updates are not remorse, replanning is not responsibility, and reducing a loss function is not paying a price. Feedback returns information; consequence-bearing returns costs. Subjecthood begins with bearing consequences rather than feedback alone.

Human bodies directly show where consequence-bearing begins. They form subject boundaries because life processes are difficult to escape, rather than through a mysterious human privilege (Merleau-Ponty 1945). Mistakes can bring pain, shame, lost time, damaged relationships, changed reputations, and memories repeatedly returning to consequences. People cannot easily reset versions like models, replace themselves painlessly like components, or outsource aging, injury, and death to another instance. Humans naturally approach subjecthood because escaping themselves has a cost, rather than because they are nobler.

Two Ways to Encapsulate Consequences

encapsulation of consequences means consequences cannot disperse at will. Bodies remember wounds, nerves fear, habits repetition, relationships betrayal, reputations choices, and time irreversibility. An "I" is a continuing boundary formed by consequences returning, folding, and compacting through body, memory, relationships, and time, beyond self-narration. The subject is continuity rewritten by past action rather than a static core. Living subjects cannot completely reset after each action; they carry accumulated consequences onward.

Bodies, perception, and action can make embodied machines appear to have acquired subjecthood. That inference fails. Shells, joints, sensors, and actuators form physical boundaries without automatically forming subject boundaries. Machines break, parts and models are replaced, bodies scrapped, data recovered, and costs transferred. Many consequences fall on

manufacturers, users, deployers, environments, and affected people rather than returning to the machine. Bodies become meaningful to subjecthood when consequences accumulate within them, rather than when they merely execute (Dreyfus 1972). Living organisms differ. Their bodies are long-evolved structures for bearing consequences rather than temporary interfaces. Morphology, reflexes, pain, recovery, habits, and finite lifespans encapsulate much complexity. Machines do not naturally inherit that evolutionary amortization. They can act without necessarily bearing consequences, enter space without necessarily bringing its reactions back within themselves, and change bodies without necessarily bearing the aftermath. This separates embodiment from subjecthood.

A natural body is not required for every form of subjecthood. Organizations lack pain, flesh, and biological death, yet assets, accounts, history, reputation, identity, and long-term constraints can establish institutional boundaries (Dewey 1926). Past actions change future capabilities, trust, and restrictions. Consequences accumulate institutionally, letting the past rewrite the institution. Biological bodies hold no monopoly on subjecthood. Living and institutional structures can both encapsulate consequences. What matters is sustained closure within a boundary rather than material alone.

Unowned Action and Visible Responsibility

Whether closure exists is readily apparent in most intelligent systems. The problem with most current systems is that their boundary of closure has yet to form, rather than that they fail to resemble humans. Large models generate judgments, but their consequences do not naturally return to the model; embodied systems execute actions, but the consequences need not accumulate within the machine itself; intelligence applied to living systems proposes interventions, but the people classified, screened, treated, and optimized bear the long-term results. The more intelligence can generate, act, and intervene in life, the wider this gap becomes: the system produces differences, while the consequences fall elsewhere. Beneath the question of who is responsible lies another: does the action belong to anyone? Without a subject boundary, action becomes mere output; without consequence-bearing closure, reality becomes feedback data; without costs returning, errors are distributed to places no one sees. A system can become more effective while bearing less, and more capable of action while having less of an owner. This reduction of the world to pure output is what must be prevented.

Making responsibility visible brings consequences swallowed by the system back into view, beyond adding an external clause. When an action affects only the actor, consequences often register directly: falling hurts, failure leaves a memory, relationships can break, and time cannot run backward. When models, machines, organizations, and biological technologies amplify action, consequences can be delayed, concealed, reduced to statistics, or passed on, becoming losses no single site truly feels (Matthias 2004). Responsibility thus exceeds a moral label attached afterward. It gives a subject boundary social expression. Action must leave an answerable destination as well as an effect.

Subjecthood is therefore a matter of how strongly consequences close back on a boundary, rather than a binary label. Tools have almost no such closure: when a hammer injures someone, the consequences return to its user. Ordinary agents have goals and feedback but need not bear consequences. An institutional subject has no natural body, yet consequences can accumulate in organizational memory and long-term constraints. Living subjects possess the strongest encapsulation of consequences: body, memory, relationships, aging, and death lock consequences into one life process. Engineered subjects may emerge, but their status would depend on persistent identity, memory that cannot be arbitrarily reset, accumulated real costs, an externally answerable boundary, and enduring constraints on their future, rather than the ability to say "I" (Parfit 1984). An account of subjecthood need neither defend some human mystery nor rule out a form of subject status for machines. It offers a criterion: what is continually rewritten by the consequences of its actions begins to approach subjecthood; what only produces differences while always shifting their costs remains a conduit. The danger in the age of intelligence is the multiplication of actions whose consequences disperse until no one owns them, rather than machines coming to resemble humans.

A subject is more than something that can say "I." A subject is where consequences can no longer be passed on at will. It binds action, reaction, memory, cost, and future constraints together. Without that boundary, intelligent systems become more capable of changing the world and less susceptible to being changed by it: actions multiply while owners dwindle. Machine bodies sharpen the question. Engineering can strengthen perception, execution, wear resistance, and self-maintenance; material boundaries and consequence-bearing boundaries remain distinct, and a stronger body does not automatically close the loop of consequences.

5.2.2 Consequence-Bearing Closure: From Machine Bodies to Engineered Subjects

Engineering raises another question that is easier to misjudge: does a machine with a body, able to perceive, act, wear out, and repair itself, come closer to being a subject? The question matters because intelligence has several distinct entry points into reality. Large models bring outputs into social judgment, embodied intelligence brings bodies into the material world, and intelligence applied to living systems brings proposed interventions into metabolism, development, immunity, heredity, and ecology. Each carries more weight than words on a screen, and each invites the same mistake: assuming that greater contact with reality means greater subjecthood.

Entry into Reality and the Subject Boundary

An entry point into reality does not establish a subject boundary. A system can produce judgments without bearing their consequences, possess a body without bearing the costs of its actions, or propose biological interventions without bearing their metabolic, developmental,

and intergenerational effects. Entry into reality establishes that an effect has occurred, not that its consequences have returned to the same enduring boundary. Subjecthood does not grow automatically from output, a body, or the ability to intervene in life. It appears where consequences return, accumulate, and constrain the future of the same boundary.

The first step is to distinguish body, embodiment, and subjecthood. The body is a material boundary: it specifies the physical structure through which a system enters the world. Embodiment couples that material boundary to its environment through action: it specifies how the system perceives, consumes energy, undergoes disturbance, corrects itself, and continues acting. Subjecthood concerns returning consequences and consequence-bearing closure: after action, do costs return to a continuous boundary that cannot simply be escaped? The three are connected, but none can substitute for another.

Bodies can be engineered. Sensors can be added, actuators upgraded, joints maintained, materials made compliant, energy allocated, local control accelerated, and self-diagnostics run continuously. Future machine bodies may develop stronger abilities to maintain and repair themselves: detecting wear, repairing surface cracks, reallocating energy, and continuing at reduced capability after local damage (Terryn et al. 2017). These remain, first of all, achievements of bodily engineering rather than the creation of a subject.

This distinction can be expressed as a ratio measuring only the coverage of returning consequences. Let B be an enduring boundary, a an action it initiates, $C(a)$ the set of all consequences of that action, and ρ_a a map assigning each consequence to the boundary that ultimately carries it. Define the proportion of consequences returning to B for action a as

$$\kappa(B, a) = \frac{|\{\, c \in C(a) : \ \rho_a(c) = B \,\}|}{|C(a)|}, \qquad \kappa \in [0, 1]. \tag{5.3}$$

Here $C(a)$ is assumed to be finite, nonempty, and consistently itemized, with ρ_a assigning each item to one boundary. κ records only the coverage of returning consequences on this list. Actual consequences may be shared or impossible to count; both itemization and weighting change the result. The ratio helps us ask where consequences belong; it does not measure consciousness, moral status, or subjecthood as a whole. Insurance and public assistance can legitimately distribute consequences, so complete return cannot be a universal goal.

Absorbing Residuals and Bearing Consequences

Residual absorption and consequence-bearing must be distinguished. Once an embodied system enters the physical world, slippage, friction, inertia, contact errors, sensor drift, and deviations across a task sequence all require handling. The body can absorb them, the controller correct them, environmental constraints contain them, and external support structures share them. Residual absorption determines whether action can continue. Consequence-bearing asks to whom the action belongs. Embodiment invites confusion because it brings these questions so close together: the machine really does handle real-world residuals, making it appear to bear real-world consequences as well.

Being changed by reality does not by itself make something a subject that bears reality's consequences. Stone weathers, bridges fatigue, tires wear, batteries degrade, databases receive writes, models undergo fine-tuning, and machines retain logs. All leave traces through time, but those traces need not constitute an actor's history. Being changed is a physical fact; bearing consequences incorporates them into a continuity that can be held to account. The distinction is between carrying consequences and bearing them. Tires carry friction, robotic arms reaction forces, batteries discharge cycles, and casings impacts and wear. Once a machine enters the physical world, consequences necessarily reach some components. Carrying means that consequences reach a **physical carrier**; bearing requires them to enter the same enduring boundary, change its future space of possibilities, and prevent it from escaping its past at no cost.

If a service robot damages the corner of a wall, its casing may be scratched, its wheels misaligned, its logs updated with an anomaly, and its control policy revised. Physical consequences have reached the machine, and information has returned to the system. Yet the substantive consequences may fall on the user, manufacturer, deployer, physical environment, and affected people. The machine body carries one part of the chain without necessarily being its final destination. It has been affected, but need not have become a bearer of consequences. Consequence-bearing therefore requires at least several conditions: consequences return to the same boundary rather than merely reaching replaceable parts; they change that boundary's future possibilities rather than merely triggering repair or parameter adjustment; gains, restrictions, and repercussions constrain the same continuing entity rather than dispersing across different sites; and that boundary remains externally answerable rather than disappearing among versions, instances, supply chains, and maintenance chains. Without these conditions, even a machine body that wears out is only a material interface through which consequences pass, rather than a subject boundary to which they belong.

Self-Protection and Living Encapsulation

Many discussions treat self-protection as evidence of subjecthood; this inference also needs unpacking. Avoiding obstacles does not establish fear of injury; returning on a low battery does not establish concern for the future; detecting faults does not establish an understanding of vulnerability; repairing itself does not establish that a robot has borne trauma. Obstacle avoidance can be a control objective, returning an energy strategy, and self-repair a material mechanism. These mechanisms can be complex and effective, but their immediate role is to maintain function rather than establish an "I" that cannot escape itself (Metzinger 2003). Why do living organisms differ? Life encapsulates consequences more deeply; no mysterious subject hidden inside an organism is needed to explain the distinction.

Body, metabolism, pain, recovery, development, memory, relationships, and death press consequences into the same process, which cannot be replaced at will. Bodily injury changes the capacity to act and the time ahead, beyond a fault in one component; pain brings damage within the same living boundary, beyond an indicator light; death ends continuity, beyond

scrapping a casing. Life has greater difficulty passing costs elsewhere, rather than greater nobility. An organism cannot freely replace its entire body, painlessly migrate its memories into another instance, restart from a clean version after failure, or have an equivalent copy bear aging and death in its place. Subjecthood arises from continuity that cannot escape itself at no cost, rather than simply from moving flesh.

A machine body does not automatically inherit this continuity. It may have a serial number, location, casing, sensors, and maintenance records while remaining a replaceable instance. Worn joints can be replaced; faulty models rolled back; retired bodies succeeded by identical machines; failed policies erased in a new version. Machines can certainly have histories, but these often belong to a platform, fleet, manufacturer, external operating structure, or database rather than to the individual executing the action. Recording a history does not establish that it is borne.

Conditions for Consequence-Bearing Closure

Machines may develop subjecthood, but a body alone does not establish it. Any future engineered subject would need more than a humanlike shape or finer tactile skin: it would need engineered consequence-bearing closure, including persistent identity, a history that cannot be arbitrarily reset, accumulated real costs, future actions constrained by past consequences, an externally answerable boundary, and a structure that cannot pass every cost to users, deployers, the environment, and affected people. Such a structure need not be biological. Organizations have no natural bodies, yet assets, reputation, legal identity, historical records, and long-term constraints can establish institutional boundaries. They feel no pain, yet consequences accumulate; they have no flesh, yet their identity can end. Flesh has no monopoly on subjecthood, but subjecthood requires consequences that cannot simply disperse.

This explains three mismatches in intelligence's entry into reality. For large models, representations influence real judgments without bearing their consequences; for embodied intelligence, bodies absorb real-world residuals without necessarily owning the costs of action; for intelligence applied to living systems, proposals enter biological processes without necessarily bearing their long metabolic, developmental, hereditary, and ecological consequences (Floridi 2013) (Nissenbaum 1996). The later the entry point, the greater the burden carried by the receiving medium, yet the system itself need not become more of a subject. The risk is intelligence entering domains of ever greater consequence while continuing to leave costs elsewhere, rather than increasingly resembling people.

To judge whether an intelligent system approaches subjecthood, examine its consequence-bearing closure rather than human resemblance or bodily presence alone. A system can combine high output with little subjecthood, extensive embodiment with little subjecthood, or deep biological intervention with no consequence-bearing closure; conversely, an organization can establish a subject boundary institutionally without a natural body. Representation opens a route to real-world influence, the body to material coupling, and biotechnology to living pro-

cesses; subjecthood concerns the strength of closure through which consequences return and remain continuously borne.

Coupling, Self-Correction, and Replaceability

At a deeper level, intelligent systems enter reality through several distinct couplings. Representational coupling concerns which candidates a system can generate and which judgments it can influence. Material coupling concerns whether it can act reliably in the physical world. Biological coupling concerns which living structures read, amplify, resist, retain, or propagate a proposed intervention once it enters a living process. Consequence-bearing coupling concerns whether consequences return to an enduring boundary. The first three explain entry into reality; the last reaches the question of subjecthood.

This also explains why self-correction alone does not establish subjecthood. Large models can reconsider outputs, embodied systems can monitor wear and schedule maintenance, and systems working on biological problems can update candidates, optimize doses, and adjust interventions in response to experiments. All appear to correct themselves, but correction does not establish consequence-bearing. Correction begins to approach consequence-bearing only when an error does more than produce a new version, policy, treatment course, or candidate: its consequences must alter the future of the same enduring boundary in a way that cannot simply be transferred elsewhere.

The deeper dividing line concerns replaceability. As an engineering object, a body tends toward modularity, maintainability, replaceability, and replicability. Engineering seeks to reduce dependence on irreplaceable points: damaged parts can be replaced, errors rolled back, instances migrated, capabilities reused, and effects isolated. This improves reliability while weakening the automatic emergence of subjecthood. A subject requires continuity that cannot be wholly replaced, whereas engineering often improves resilience through replaceability. The better a machine is as an engineering object, the less that fact alone establishes its subjecthood.

The goals diverge without contradicting each other. An engineered body seeks maintainability; a subject boundary requires answerability. Engineering seeks fault isolation; subjecthood requires accumulated costs. Engineering seeks instance migration; subjecthood requires continuity of identity. Engineering seeks to minimize external effects; the criterion for subjecthood asks where consequences ultimately return. A body can become stronger, more flexible, and more internally coherent, yet remain a conduit into reality rather than the destination of consequences if those consequences are still borne elsewhere.

An Inescapable Boundary

Bodies can be engineered while subjecthood is not automatically inherited: this is the dividing line. Bodily engineering handles disturbances, reducing downtime, damage, and loss of control; a subject boundary prevents consequences from disappearing. One absorbs residuals; the other asks who retains the costs. Mature engineering can hide faults within main-

tenance, replacement, redundancy, and external support; clearer subjecthood requires those hidden consequences to return to an answerable boundary. The two can support each other without establishing each other.

The hardest criterion is this: **Judge subjecthood by whether a system can escape its consequences, rather than whether it can protect its components.** If a system distributes harm to users, assigns risk to external structures, files errors under versions, leaves damage to maintenance, migrates identity to a new instance, and then reappears as clean capability, it has yet to become a subject. Conversely, a boundary begins to approach subjecthood, even without being biological, if it cannot escape in this way: past actions continue to constrain it, define it, hold it to account, and alter the paths available to it.

Subjecthood is no automatic upgrade from having a body. Bodies bring machines into the world, embodiment closes action loops within it, and biotechnology brings intelligence into the heavier processes of life; subjecthood requires the world to return the consequences of action to the same enduring boundary. It emerges where costs can no longer be passed on, rather than from sensors, actuators, biological intervention, or self-description alone. No single entry point into reality therefore establishes subjecthood automatically. Output, bodies, feedback, self-maintenance, and biological intervention show only that a system has connected to reality in some way; they do not yet show where consequences return. A boundary begins to acquire the significance of a subject only when consequences continually return to it and change how it can exist, act, and be held to account. Embodiment brings a machine into reality; consequence-bearing makes a boundary a subject.

Closure within one boundary supplies only half the criterion. An action rarely crosses just one boundary: platforms, developers, deployers, organizations, users, and affected people each bear some consequences, through multiple return paths and at multiple destinations. When consequences disperse across these paths, where does closure occur, who decides which boundary bears which part, and how are costs allocated among bearers? These questions remain open.

5.2.3 Distributed Consequence-Bearing: What Companies Establish and Models Lack

The preceding section left a question unanswered: when consequences disperse across several return paths, where does closure occur, and who decides which boundary bears which part? To answer it, start with an example that already has an operational arrangement. One kind of subject neither eats nor breathes, has no nervous system, and has neither a life history independent of its members nor bodily pain. It can own property, enter contracts, be sued, be fined, and be compulsorily dissolved. It answers for its actions. That subject is the company.

A Problem Given Institutional Form

Corporate liability is now so familiar that its constructed nature passes unnoticed. A com-

pany is no natural object. It is a boundary for bearing consequences assembled through legal devices. Assembling those devices took centuries; a crucial step was limited liability, which separates shareholders' liability from the company's. Under Chinese company law, for example, shareholders of a limited liability company are liable to the company up to their subscribed contributions, shareholders of a joint-stock company up to their subscribed shares, and the company answers for its debts with all its property. In a UK company limited by shares, members' liability is generally limited to the amount unpaid on their shares (Standing Committee of the National People's Congress 2023; United Kingdom Parliament 2006). Limited liability is often treated as an economic policy encouraging investment. Its structural role is more basic: separating corporate from shareholder property gives responsibility an identifiable property and legal boundary. Within that boundary are assets to seize, accounts to inspect, representatives to summon, and an entity to dissolve. The boundary makes accountability an operational procedure rather than an endless search backward. This arrangement establishes an entity to hold accountable without guaranteeing compensation for every loss: finite assets, bankruptcy, and the scope of remedies can still leave consequences with creditors, victims, or society.

This provides a yardstick. It identifies conditions a system needs to become a bearer of consequences. Applying it to machine intelligence reveals what is missing, but first the yardstick itself needs explanation.

Four Conditions for a Consequence-Bearing Boundary

Disassembling the corporate structure reveals four things working together. The first is **identifiable, bounded resources**. A company has its own assets and liabilities. Compensation removes actual value from its balance sheet rather than merely expressing regret. A resource boundary gives sanctions a measurable magnitude. The second is **stable identity**. A company persists across years, changes of management, and employee turnover. Today's company answers for conduct ten years earlier because the two are recognized as the same legal person. Identity carries accountability through time.

The third is **internal state changes following sanctions**. A fine can reduce investable funds, affect credit ratings, raise financing costs, compress pay, or even trigger bankruptcy. Sanctions thus change what that boundary can do in the future. This is the substance of bearing consequences: being changed rather than merely being named. The fourth is **a traceable chain of accountability**. A company has directors, senior managers, legal representatives, employees, and shareholders whose powers and responsibilities follow the relevant law and organizational arrangements.

An act can be traced to specific decision points rather than disappearing into the collective. Together, these conditions make consequence-bearing operational. **A company can answer for its actions because institutions anchor consequences to a traceable boundary, not because it is conscious.** This section uses the four conditions to examine consequence-bearing structures: a missing condition leaves a corresponding break in the return of responsibility.

What Models Lack

The same yardstick makes the gaps in machine intelligence systems clear. First, the resource boundary is missing. A model as ordinarily deployed does not independently hold assets in law. Can it be fined, and who would pay? If its deploying company pays, the company bears the consequences; the model itself has not become the legal entity paying compensation. This looks like accounting but concerns structure: without identifiable resources, a sanction cannot land on the system and instead falls on people outside it.

Second, establish who maintains continuity of identity. A new model version need not sever responsibility: deployers can retain versions, configurations, records, and continuing obligations. The question is whether past actions remain traceable after an update and whether the boundary bearing the obligations persists.

Third, establish whom sanctions actually change. Fines reduce an institution's resources, service restrictions alter operating capabilities, and model updates change behavior. These responses can be connected, but people or institutions usually bear the costs and obligations. Calling all of them "punishment of the model" hides where the consequences actually land.

Fourth, a traceable chain of accountability is missing. The problem extends beyond technology. Model creation, fine-tuning, deployment, integration, invocation, and use often span different actors. An output may pass through a model provider, application developer, platform operator, end user, and affected third parties on its way into reality. Each link can introduce an accountability gap, weakening the trail through which responsibility is attributed.

This weakening can be represented by a schematic chain of transmission,

$$R_n = R_0 \prod_{i=1}^{n} \alpha_i, \qquad 0 \le \alpha_i \le 1 \tag{5.4}$$

Here n is the dimensionless number of layers crossed by the action; six or seven layers can serve as an illustration. R_0 is the initial share of attribution evidence traceable to decision points and can be normalized to one for comparison; α_i is the dimensionless proportion of attribution evidence retained after transmission from layer i to the next, ranging from 0 to 1; R_n is the share still reaching an answerable boundary after all layers. If each layer loses some traceable evidence, the remainder decreases as a product; adding a link with a retention proportion below one reduces it further. This expresses how responsibility can disperse, shift, and dilute until a residual remains that no one fully bears. Concentrated attribution, recordkeeping duties, and arrangements governing the burden of proof can improve traceability, appearing schematically as increases in some α_i, or changing the whole path. The expression assumes separable contributions at each layer without repeated attribution across layers; when one entity provides the model, application, and platform, the chain contracts and n decreases, but traceability still depends on records and institutions. The expression identifies structural positions for investigation rather than a conservation or multiplication law of legal responsibility; estimating α_i

requires operational definitions, records, and validation.

An ordinarily deployed model is a component within an accountability arrangement rather than an independent legal bearer of consequences. An external architecture can assign persistent identity, resources, and permissions; institutions must determine obligations and remedies. The questions should be asked of each concrete system: who controls, who records, who benefits, and who compensates? An arrangement absent from current models need not be impossible for every engineered boundary in principle.

Distributed Consequence-Bearing

The real question is therefore where consequences should be anchored once action is distributed, rather than simply whether a model can be responsible. Several fields already face this problem. Liability for an autonomous-driving collision must be allocated among vehicle manufacturers, software providers, sensor suppliers, high-definition map providers, owners, and drivers. Different technical architectures can imply different divisions of responsibility for the same collision.

Content-platform responsibility must be allocated among publishers, platforms, recommendation algorithms, and users. When a platform claims to be a neutral conduit, how to account for the recommendation algorithm's orchestration has long lacked an agreed answer. For discriminatory algorithmic decisions, what victims must prove to bring a claim and what defendants must prove to avoid liability can directly determine whether cases can proceed. These cases share a structure: action has been divided, while the boundaries bearing its consequences have neither divided in step nor been reassembled. Initiation, execution, benefit, and harm occur at different sites, yet consequences need a destination.

In the traditional arrangement, the actor supplies that destination. A person puts body, identity, reputation, time, and future on the path of an action; failure changes that person. With machine intelligence, a path can still be initiated, executed, and amplified without necessarily reaching a boundary that continually bears its consequences: training updates can change the model, while compensation, losses, and lasting obligations usually fall on the people or institutions operating it. The missing part does not grow automatically. It must be established through technical architecture and institutional arrangements.

Once it is established, the human position changes too. Making distributed consequence-bearing operational also reallocates responsibility between individuals and organizations. When a system can receive consequences, one link in the direct connection between an individual and those consequences loosens. For many people, that link was a source of value. If value once attached to the position where consequences fell on oneself, that loosened link calls for a new account: when institutions can bear consequences, which judgments must still be made by people? Value does not disappear, but it needs a new point of origin.

Before institutions' assumption of consequences changes the argument about value, it changes people's sense of their own position. Before value needs a new origin, something

more immediate occurs: many discover that their former way of demonstrating value no longer works.

5.3 First Collapse: The Functional Crisis

Existence, evidence, permission, and consequence-bearing: the preceding two sections concerned structures, whose real costs appear when structural change reaches people. The first impact comes before income or employment: what once demonstrated that a person held a place in the system? Change at this level moves fastest and is most easily mistaken for a personal problem.

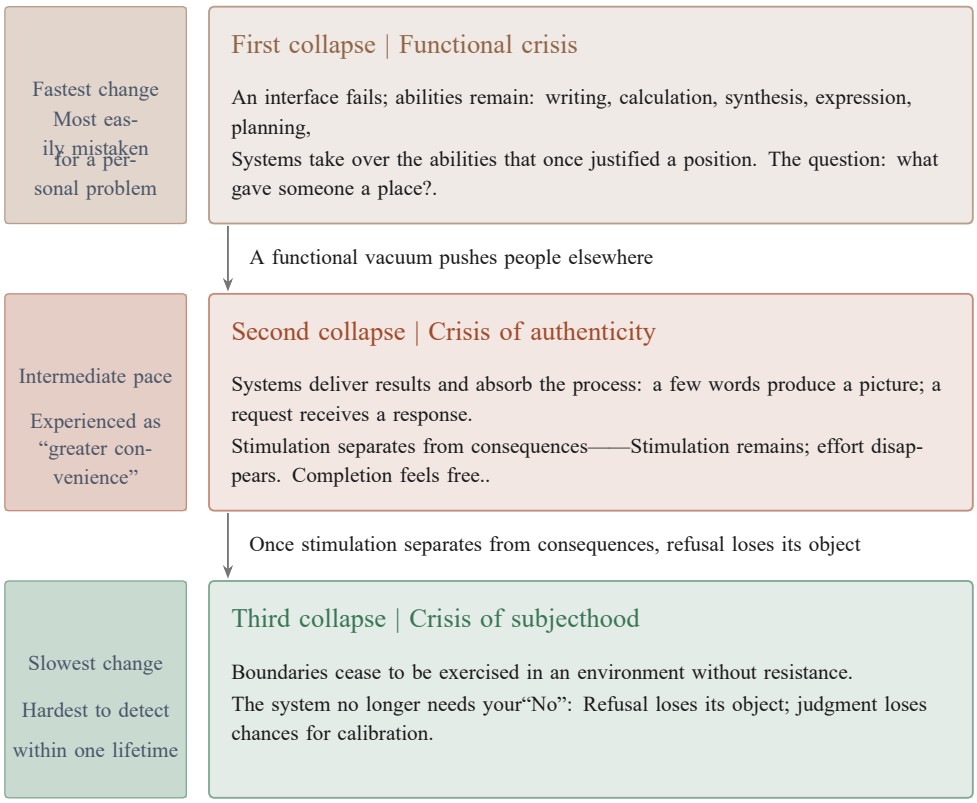


Figure 5.1 How the three collapses propagate. Colored blocks on the left identify each layer; boxes on the right show its mechanism; arrows trace transmission between layers. The layers become progressively harder to recognize and change at different rates; the next three sections follow this chain.

5.3.1 Functional Weightlessness: When the Interface Fails

The Abilities That Once Justified a Position

The preceding sections located value in the capacity to bear consequences, but a new source of value does not ensure a smooth transition from old ways of demonstrating it. The first event in the transition is failure. For decades, a person's social position rested largely on a set of abilities. Writing organized ideas into forms others could use; calculation supplied reliable numerical judgments amid uncertainty; summarizing extracted structure from large bodies of material; expression steadied meetings, negotiations, and classrooms; planning divided goals into steps, resources, and time.

These abilities share one feature: they produce deliverables. Deliverables can be checked, enter workflows, and generate economic value. People with these abilities therefore needed little further explanation of their value: the deliverable supplied the proof. Ability was directly bound to social position: what someone could do determined the positions they could enter, the tasks for which they were needed, and who would remember them. The bond was so stable that it often passed for the definition of human value rather than one expression of it. Generative systems change that bond.

An Interface Fails; Abilities Remain

The location of the change matters, because a popular account places it incorrectly. The problem is declining power to distinguish people, rather than abilities becoming useless. When a system generates a well-structured report in seconds, clear writing alone says less; when basic data analysis is automated, accurate calculation alone no longer distinguishes who has been trained. People have not necessarily begun writing poorly or calculating incorrectly: output itself has lost much of its signaling power, and a weaker signal differs from a lower level of ability.**AI does not immediately abolish human value; it dismantles the old interfaces through which many people demonstrated it.**

Express this discriminatory power as an index,

$$d' = \frac{\Delta \mu}{\sigma} \tag{5.5}$$

Here $\Delta \mu$ is the difference in mean output between two groups on the same evaluation scale, such as trained and untrained people's mean output quality; σ is the standard deviation of individual output on that scale. Both must be estimated on the same scale in comparable samples; their ratio d' is dimensionless and measures how well the scale distinguishes the groups. $\Delta \mu$ can shrink without σ shrinking. If acceptable output becomes available by default, the system compresses mean differences while output variation fails to contract proportionately, d' falls; in the limit where the mean difference approaches zero while variation remains positive, the scale loses discriminatory power even if neither group's existing skills have declined. Interface failure is not the disappearance of ability: the quantity to examine is d', since output mean μ

alone cannot establish changes in existing skills. This formulation assumes comparable scales and roughly equal group variances; substantially different variances require a different definition of discriminatory power. It describes the discriminating power of a scale used as evidence, rather than anyone's ability, and supplies no measured value by itself.

Interface failure must be distinguished from unemployment, since conflating them both exaggerates and understates the problem. Unemployment concerns a job; interface failure concerns proof of value. Many jobs persist because they occupy viscous stages requiring physical presence, bodies, enduring relationships, and responsibility for consequences, even as workers' means of evaluating themselves are reorganized. Someone still goes to work every day but no longer knows whether "I can do this" explains their position. The ability remains but is less scarce; the person remains needed, but for some other reason. Imagine someone with twenty years in a field, whose professional judgment has become intuitive, opening a system, typing a few sentences, and receiving output covering eighty percent of routine work. Judgment, selection, and responsibility remain, but the center of work has shifted from production to review, which may not provide the same support for professional identity.

Labor, Work, and Action

The gap runs deep because it concerns the relation between activity and recognition as well as income, a relation already carefully distinguished by another thinker. Hannah Arendt distinguished three human activities: labor sustains biological metabolism and produces things quickly consumed; work makes durable things that outlive their makers; action consists in speaking and acting before others, producing relationships and history rather than objects (Arendt 1958). Her diagnosis was that modern society reduced these activities to labor, measuring everything by the maintenance of life and depriving work and action of independent standing.

The distinction helps locate the current situation. The old ability interface was stable because it joined work to action: making something produced a durable object while giving its maker a position and history within a community; output expressed relations to both things and people. Generative systems chiefly take over work, particularly its standardized, reproducible parts. As that work is taken over, the social recognition attached to it loosens too. The carrier of recognition has changed before a new one has formed, rather than recognition simply being confiscated.

Arendt is both a source of diagnosis and an interlocutor whose argument must be confronted. She places the response in action: plural people become visible and answerable to one another in the public realm, where a person appears as a "who" rather than merely a "what" (Arendt 1958). This book enters through consequence-bearing: a position's structural value lies in consequences returning there. The two cannot be interchanged. Action has intrinsic value, requiring plurality and others who see and respond; consequence-bearing has functional value, requiring consequences actually to return. By the latter criterion, an entity bearing every conse-

quence alone can qualify; without others, however, there is no public realm in Arendt's sense.

The substitution has a cost that must stay visible. A criterion of consequence-bearing locates returning consequences without deciding whether that party ought to bear them. It is silent about distribution and can, at worst, read any existing burden as evidence of value. Arendt's political language preserves the question: is this unjust? Translating lost functional positions into structural language gives this book a more precise map, without thereby giving it the pen that marks legitimacy on that map.

Arendt used "superfluity" for a group's loss of functional position in society, examining its political consequences under particular historical conditions. This chapter borrows its structural meaning: when a state transition no longer requires a node, the node has not been destroyed; it has slipped out of the process. **A position that slips away has no automatic replacement. It is vacant.** Replacement does not happen automatically. Markets and institutions do not automatically find a new place for the displaced node, and existing organizations may have no incentive to take it back. For an organization that has removed the step, its absence no longer creates a workflow problem.

Why the Diagnosis Exceeds a Matter of Feelings

Because positions do not automatically return, popular explanations of this loss of weight deserve scrutiny: they avoid precisely that issue. One offers reassurance: people's abilities remain; only the tools have changed, so learn the tools. Operationally true, the claim evades the structural problem. Learning the tools does not restore the bond between person and output, because the tools amplify precisely the abilities that can be automated, weakening their power to distinguish people further. Another assigns blame: anyone unable to use the new tools simply has not tried hard enough. It confuses individual adaptability with structural resetting of the interface; no amount of individual improvement can solve a system problem that no longer depends on that ability.

Both explanations locate the problem in the individual, while the interface is what has been displaced. "Interface" matters because it directs the response: an interface specifies an agreed match between what one side supplies, what the other needs, and how they connect. Failure means that the match no longer holds, without implying a defect in either side. When interfaces fail together and affect tens of millions, the problem is structural and its remedy lies with institutions rather than individuals alone. The point is to locate responsibility correctly, rather than absolve individuals.

The Scope of the Diagnosis

The diagnosis needs boundaries, or it will slide toward two unsupported conclusions that weaken the chapter. It predicts the disappearance of no particular profession. Interface failure differs from job elimination: it can spread much further and arrive earlier, affecting many people still employed; treating it as an unemployment forecast understates its reach. Nor does

it make professional training pointless. In consequential domains, training forms judgment as well as deliverables: a trained doctor and an untrained person using the same system may still differ markedly in output quality and verification ability, especially in recognizing errors rather than merely knowing more. Lower generation costs increase the importance of judgments that detect errors and carry responsibility.

The weakened layer lies between these poles: standardized production requiring neither deep judgment nor responsibility for major consequences. It once supported substantial knowledge work and many professional identities; its contraction has broad effects yet can fall between agendas concerned with protecting the vulnerable and competing for elite talent. Here is where weight is lost and costs land. The first response is often retreat toward easier feelings of completion rather than resistance, reconnecting with the low-viscosity paths discussed earlier. That retreat is an effect; the cause lies earlier: in compressed stages, people lose both the deliverable and the loop from setting a goal to overcoming resistance.

5.3.2 The Broken Power Process and Radical Superfluity

Interface failure explains a changed position without yet explaining its effect on self-evaluation. Between remaining needed at work and no longer knowing why lies an earlier loop: setting a goal, overcoming resistance, and reaching a result. The loop binds a person to the world through doing the work itself, rather than through output alone.

The Power Process and Its Short Circuit

In *Industrial Society and Its Future*, Theodore Kaczynski placed satisfaction within a "power process": setting a goal, exerting effort, overcoming resistance, and achieving it; satisfaction comes from overcoming resistance rather than merely receiving the result (Kaczynski 1995). Borrowing the loop entails accepting neither its author's positions nor his verdict on technological civilization. Its mechanism connects with this book's analysis: a goal is a state to be realized, resistance the cost of reaching it, and work the process of paying. Industrial technological systems take over the middle of the loop. They hand results to users while retaining the overcoming of resistance within the system. People receive finished products without having done that work.

RTG provides a more precise description. In the extreme case where an external system handles almost an entire state transition, R approaches zero for the person; T contracts almost beyond perception; G no longer requires that person's participation. When all three approach zero, the person contributes almost no effort to the transition and receives less feedback from personally overcoming resistance. Write this as a limit,

$$\eta_p = \eta_0 \exp(\alpha R_p + \beta T_p + \gamma G_p), \qquad \lim_{(R_p, T_p, G_p)\to(0,0,0)} \eta_p = \eta_0 \tag{5.6}$$

Here η_p is the viscosity of this state transition for the person; subscript p means that every quantity is accounted for on that person's side rather than in the system's physical ledger. R_p,

T_p, and G_p are the resource barrier, temporal rigidity, and topological coupling of the same event in that person's coordinates; η_0 is the reference viscosity when all three coordinates are zero. The point is less the limiting value than the conjunction of three conditions: η_p returning to η_0 means that the transition consumes none of the person's resources or time and passes through none of their connections with other people. All three disappear together, rather than each decreasing slightly. If feedback from personally overcoming resistance is assumed proportional to $\eta_p - \eta_0$, it also approaches zero in this schematic model; actual changes in feedback and sense of place still require psychological and behavioral evidence. This supplies a testable mathematical form for "receiving a result without experiencing accomplishment." The claim must remain specific: occasional delivery of a result changes little; the issue arises when external systems take over so much of the loop that what remains cannot sustain a person's sense of place.

Two compensations often follow when the loop is taken over. One turns elsewhere: people set goals and resistance in domains the system ignores, including competitive activities, collecting, and status contests within particular circles. These activities retain goals, resistance, and results; their difference from externally imposed goals concerns conditions rather than intensity. When goals and resistance can be changed at will, feedback calibrates external constraints less effectively; competition under fixed rules and commitments that cannot be arbitrarily rewritten can still provide rigorous feedback. The other compensation lowers expectations: long-term goals requiring work give way to immediate, inexpensive stimulation.

Psychology offers an earlier line of inquiry into loss of control and passivity. In experiments during the 1960s, Martin Seligman and Steven Maier repeatedly exposed animals to harm their behavior could not control, finding that they subsequently failed to try escaping even when escape became possible (Seligman and Maier 1967). Fifty years later, the authors revised the explanation: the default passive response was not learned; what could be learned was whether one's actions controlled outcomes (Maier and Seligman 2016). The revision better serves this chapter by directing attention to the connection between action and outcome rather than passivity alone. In these experiments with aversive stimuli, learned experience of control helped counter passivity. This suggests examining the connection between action and outcome, while evidence linking animal stress experiments to people's sense of place when using technology must be established separately.

Kaczynski's verdict on technological systems goes further than his account of the loop. He argues that an industrial technological system must continually absorb energy, resources, and solutions; stopping expansion means disintegration, while individual dependence closes the route back to older ways. The judgment imposes a thermodynamic curse: the system can neither stop nor retreat. The loop analysis can be retained; the prophecy of an ending requires separate proof. Whether expansion is necessary and dependence irreversible turns on institutions and choices. His argument treats both as constants.

The disagreement extends further. Treating technology as a single actor makes the distribution of viscosity appear intrinsic to technology; in this book's framework, property rights, labor institutions, public investment, and bargaining power determine where friction is reduced and positions reorganized. Railways, electrification, and assembly lines all removed old frictions, yet produced different structures of positions in Britain and the United States, Germany, and Japan: the same machines replaced different people and left different positions under different institutions. This directly shapes the route out. His proposed exit was destruction of the industrial system, eliminating precisely the distinction this book needs: which frictions are wasteful and should disappear, and which carry obligations that must remain. Destroying both together yields deeper superfluity rather than freedom.

Radical Superfluity

A short-circuited loop explains "I no longer need to do it," but not yet "I am no longer needed." That second problem has long been described from another direction. In *The Origins of Totalitarianism*, Arendt used "superfluity" as a structural category rather than a synonym for poverty. This chapter borrows it for a position in which the system no longer needs someone's actions and participation no longer changes the relevant process; Arendt further argued that masses rendered superfluous were especially susceptible to systems explaining every event with complete consistency and allowing no exceptions (Arendt 1951).

A physical vocabulary gives this judgment a clearer form. In the old arrangement, a person contributed negentropy: labor and judgment transformed external low-entropy resources into maintainable structures, making that person necessary to the system's exchanges of energy and information. Once the system no longer needs their participation in that exchange, the functional connection breaks. Disconnection differs from dispossession. Exploitation still presupposes usefulness, even when reward and contribution are disproportionate; superfluity is colder because it need extract nothing from the person. **Exploitation still needs people to be useful; superfluity no longer needs even to use them. The coldest part of the functional crisis is the disappearance of the position itself.** Political language struggles to grasp this weightlessness: protest needs a concrete demand, yet here there is no specific thing to demand back.

Superfluity poses another difficulty: statistics barely see it. Unemployment has definitions, rates, benefit counts, and vacancy counts, so it can enter the agenda; superfluity lacks an equivalent measure because it often affects people still employed. Someone may remain employed while feeling their former function has been hollowed out: systems generate more of the output, leaving supervision, repair of outputs, and exceptions the system will not take on—in other words, residual absorption. When handling exceptions becomes the main remaining duty, the position counts statistically as employment while functioning structurally as a buffer. Its stability is weaker than it appears, because if the system takes over the residuals as well, nothing remains in that position requiring human participation.

Two simplifications must also be resisted here. One treats superfluity as a verdict of fate. Loss of a functional position can be reversed: history contains many nodes that slipped from one process and became needed elsewhere, though the new position usually differed from the old. Later sections address this problem. The other assigns responsibility to a single actor. Positions can disappear through accumulated local optimizations rather than conspiracy: each decision removing a step is reasonable at its own scale, yet together they create a node no longer needed. Identifying structure locates responsibility rather than exempting anyone.

The broken power process and the vanished position together explain the functional crisis's two poles: "I no longer need to do it" and "I am no longer needed." The broadest and least visible territory lies between them. Many people are neither fully replaced nor fully needed; their layer is being compressed, potentially without a single conspicuous event to trigger protest.

5.3.3 Compressible Ordinariness and the Vacuum

The two poles of the broken power process describe minorities: those wholly taken over and those fully needed. Most people fall between them, in a layer whose changes are harder to describe because they appear as neither unemployment nor overt weightlessness. They appear as compression: needs are met ever more cheaply, while the positions meeting them depend less on particular people.

Compressed Needs, Vacant Positions

Translating a hierarchy of needs into physical terms reveals where compression occurs. Safety, belonging, esteem, and self-actualization are often read as psychological states, but each also corresponds to a connection with external structures: at the bottom, energy inputs such as income and supplies; in the middle, a topological position, the edge one occupies within a system; above, a scope of intervention, the outcomes one's judgment can change. The three connections differ physically and experience pressure differently.

Machine intelligence's compression initially reaches the middle layer. Standardizable production—organizing, transcription, summarization, basic coding, routine retrieval, and template design—forms an important part of many middle-tier knowledge jobs. Compression begins on the supply side: when generating some products becomes orders of magnitude cheaper, those products lose scarcity, undermining the positions attached to them. This is skill inflation. Writing, calculation, and summarization remain nominally intact, but their available supply exceeds demand, weakening their ability to distinguish people. The resemblance to monetary inflation is that nominal quantity rises while purchasing power falls, and holders notice last.

An earlier counterpart appeared in educational credentials. Educational expansion is not always driven by demand for skills: if advanced qualifications proliferate faster than corresponding jobs, a qualification shifts from evidence of skill to a screening threshold, forcing

people to spend longer in education for the same positions. Randall Collins described this mechanism in *The Credential Society* in 1979: part of education's return follows positional competition rather than increased productivity (Collins 1979). Generative systems carry the mechanism into skills: when the supply of a skill expands on demand, its function as a credential depreciates, first harming those who hold it as their sole credential. Credential inflation shares another feature with monetary inflation: it reduces the exchange rate of ability without reducing anyone's actual ability.

The bottom layer's connections face a different shock. Stages requiring a site, a body, immediate responses, and sustained presence use a different medium and cost structure; neither substitution nor raising prices is easy in the short term. Some positions at the top become more valuable because consequential judgments need someone to bear responsibility. The middle layer is compressed most severely; it once supported substantial knowledge employment and many people's professional identities.

The history of ATMs helps distinguish task substitution from changes in jobs. Bessen described how ATMs reduced some teller tasks while changes in branch networks and customer service created other demand (Bessen 2015). The case shows tasks, jobs, and organizations changing together; it cannot guarantee the same outcome in every wave of automation.

Table 5.1 ATMs: Changes in Tasks and Organizations

Level	Direct effect of automation	Other changes to observe
Tasks	Machines handle withdrawals and some counter operations	Staff needed per branch
Organization	Branch networks and service models may change	Branch numbers and service coverage
Jobs	Customer interaction and other duties are recombined	Total employment, training, and pay

Table5.1 distinguishes two changes: machines take over some standardizable tasks, while organizations and demand for services reshape the remaining jobs. The Stanford study revised in August 2026 used payroll records through June, finding no widespread economy-wide job displacement while documenting divergent employment outcomes for young workers in highly exposed occupations; the authors emphasized that these were descriptive indicators (Brynjolfsson et al. 2026). Aggregate employment and opportunities to enter it require separate observation.

Positions are compressed while substantial unmet needs remain on the other side; this asymmetry deserves attention. Demand for care, education, repair, and community services rises with demographic change and living standards, yet budgets, licensing arrangements, and fiscal capacity constrain the positions these fields can offer, preventing jobs from expanding automat-

ically with need. Their outputs also resist standardization; labor productivity may lag behind other sectors while wages remain influenced by the wider labor market, potentially raising relative prices and further limiting affordable employment. Need without budgets leaves vacancies and shortages together: much work requires people, but corresponding paid jobs lack funding. People remain useful, yet no structure that can pay them has given them a place (Baumol 1967).

The Shape of the Vacuum

The vacuum is not empty of everything. Society does not develop a gap merely because no person performs a task already handled by a system; the gap opens in people's judgment of their own position. People whose positions are compressed do not stand waiting for an explanation, since no explanation can supply a fact they do not already know. They seek somewhere that still offers a sense of completion, and such places are plentiful.**Skill inflation leaves ability intact but strips its standing as evidence. The vacuum lies in positions, not in needs.**

"Ordinariness" here must be separated from an assessment of ability. It names a positional condition: work is standardizable and therefore cheaply reproducible, regardless of talent or effort. The most compressed positions are often held by the most disciplined, punctual, and cooperative people, because standardization was accomplished through their work. This also makes compression difficult to avoid through individual effort: working harder in a standardizable position may only improve reproducible delivery without restoring the position's scarcity.

Vacuum and ordinariness are ultimately two aspects of the same thing. On the surface, output becomes easier, average deliverable quality rises, and people do more in less time; structurally, these outputs carry less information about the person. When systems can generate most deliverables in a society, deliverables cease to answer "Who are you?" The answer has not vanished, but can no longer be obtained from output alone. Two routes remain: retreat to systems offering feelings of completion, filling vacant positions with cheap substitutes; or find a position beyond output that cannot be outsourced. The first route is cheap, offers rapid feedback, and is always available, so it is reached first.

Which route people take depends on what else can anchor their position. When functional connections break before new ones form, people initially accept whatever provides feedback. The problem then moves from function to authenticity.

5.4 Second Collapse: The Crisis of Authenticity

The functional vacuum pushes people elsewhere, toward a family of systems rather than a single place. What they share is the delivery of results, beyond entertainment: type a few words for a picture, send a request for a response, click for a win or loss. A result once required a process; the system now absorbs that process. The change begins here.

5.4.1 The Low-Viscosity Comfort Zone

Substitutes for Completion

One question remains: where do people retreat when usefulness no longer sustains dignity? They turn toward systems offering a sense of completion, rather than entertainment alone. Someone types a few words and receives a convincing picture. They need never have learned perspective, composition, or color, ruined dozens of attempts, or spent months stuck on a technique. They receive the finished product and the feedback associated with completion. The same structure appears in writing, composing music, programming, and developing proposals. The system delivers a result; the sense of completion once accompanied a process, but is now extracted and supplied directly.

Companionship follows the same pattern, with a less visible deliverable. An enduring relationship requires bearing another person's emotional fluctuations, responding when one would rather not, and working through misunderstanding and repair. These costs give a relationship its weight and its tested sense of assurance. Conversational systems can remove many costs of relating to another independent actor while retaining responsiveness, attention, and relief from judgment. They can deliver the feeling of companionship without establishing the same mutual bearing of consequences as a human relationship. Competition and adventure can follow the same pattern. Real competition involves possible defeat, bodily expenditure, and embarrassment that cannot be undone afterward. In systems allowing retries, adjustable difficulty, and immediate exit, wins and losses can be rearranged. The excitement of victory remains while exposure and costs are stripped away.

Creation, companionship, and competition appear separate but point toward one structure: **Low-viscosity systems retain stimulation while removing effort, and retain feelings while weakening consequences.** No overall design is required. Developers can optimize generation quality, response time, and retention without first deciding to "abolish effort." Accumulated local improvements can nevertheless strip away the same kind of step: those requiring users to pay costs themselves.

Decoupling Stimulation from Consequences

An established framework can clarify this structure, once separated from its popular simplification. Baudrillard used simulacra to describe signs that no longer refer to reality but refer to one another in a self-sustaining circuit. Reducing this to "the fake looks more real than the real" discards the most explanatory part of his argument (Baudrillard 1981). The reconstruction useful here runs as follows. This chapter's concern is the mechanism separating stimulation from consequences, beyond false content alone. In a real environment, a stimulus has meaning because consequences follow it. Fire burns, so the body tenses on approach; promises can be kept or broken, giving them weight; competition permits defeat, making commitment matter. Stimulation and consequences close a loop through which people calibrate judgment.

Where that loop is missing, a system can preserve stimulation while removing the corresponding external consequences. The feelings are real—sweaty palms, focused attention, shifting emotions—but the other half of the loop is absent. System feedback can feel like real-world feedback while differing entirely in consequences. If the loop no longer tests the external state a judgment concerns, it offers little basis for calibrating that judgment: error fails to trigger the corresponding consequence, leaving revision without a reliable anchor.

Whether frequent feedback trains judgment depends on what it tests. Feedback reflecting only preferences and retention may weakly calibrate external ability; simulations, tutoring, and games can also produce real learning through independent tests. The criteria are transfer, retention, and consequences, rather than frequency alone.

Beyond a Critique of Entertainment

A blanket indictment of entertainment loses the real problem. Entertainment, games, fiction, and imagination have always allowed temporary distance from direct consequences. Reading a novel need not mean living its events; watching a match need not mean taking the field. People have long expanded experience through inexpensive substitutes; neither novelty nor pathology follows from that fact. Fiction's value lies precisely in a safe testing ground, where unfamiliar situations can be encountered before their full costs are paid.

The decisive distinction is whether the chain of consequences eventually closes, rather than whether consequences exist at all. Someone who reads novels all year still lives in real relationships and faces the consequences of work, family, and the body; fiction remains an open interval embedded within closed chains. The problem emerges when low-viscosity systems occupy a growing share of experience and replace formerly closed links, rather than merely providing an open interval. If companionship, achievement, competition, and creation are all satisfied within systems, their real-world counterparts are used less; prolonged disuse can weaken relevant abilities and redirect needs. If someone gradually ceases to want reciprocal relationships in reality, the inquiry must extend beyond self-control to whether those relationships still register as a source of satisfaction.

A moralizing explanation must also be set aside. Treating the issue as weak will or inadequate self-control turns a structural problem into an individual defect, conveniently releasing system designers and institutions from responsibility. The systems work because they operate through actual satisfaction, low costs, and immediate feedback rather than moral argument. Under those conditions, taking a low-viscosity path is an understandable choice about costs. The issue lies in an environment whose feedback structure changes as these paths become widely available: rational individual adaptation produces different outcomes within that structure than in a world of consequential encounters.

Conditions for a Sense of Reality

If the problem is removed consequences, the condition for a sense of reality can be stated

positively: traceable coupling between stimulation and consequences supplies it, rather than stimulus intensity. A judgment goes wrong, things worsen, and the person changes future judgments; this process creates a sense of reality. It requires closure, without requiring drama or pain. A cook tastes an oversalted dish and uses less salt next time: a complete loop. The loop can be short, but both ends must exist.

The danger of a low-viscosity comfort zone is comfort without calibration of one's connection to the external world, rather than comfort itself. The distinction is difficult because the two experiences can feel alike. With a complete loop, pleasant experiences also change a person by supplying accurate information about the world; without closure, the same comfort may leave no transferable skill, calibrated judgment, or reciprocal relationship, even while changing habits, emotions, and expectations. Both experiences feel pleasant, yet what they leave within a person differs.

A satisfying environment need not support growth; a demanding one need not be worth entering. The test lies in whether consequences close the loop. Feelings and structure must be distinguished where experience makes them hardest to separate: systems excel at producing feelings, whose strength can overwhelm judgment about structure. The question thus passes beyond experiential quality: what happens to judgment, refusal, real connections, and personal boundaries after a long absence of corresponding consequences? First, identify how those consequences are removed.

5.4.2 Simulacra and Broken Feedback

Decoupling stimulation from consequences describes the structure from the side of experience. From the system's side, the same change reshapes the loop: the world no longer supplies the feedback. Seeing this requires recovering the concept of simulacra from its popular version.

Four Relations between Signs and Reality

Baudrillard set out a sequence in *Simulacra and Simulation*. A sign first reflects a deeper reality, then conceals and distorts it, then conceals its absence, and finally bears no relation to any reality, becoming its own pure simulacrum (Baudrillard 1981). The sequence distinguishes four relations between signs and reality; weakening connections become especially hard to detect when surface quality improves at the same time.

The sequence should not be read as a timetable. It identifies four relations between signs and reality rather than four successive human eras; different states can coexist within one society or field. Medical journals and advertising copy, engineering drawings and campaign publicity differ in their connections to reality; placing them in one stage produces an excessively tidy history. The framework serves case-by-case judgment rather than periodization.

He begins with a fable: a map covers its territory exactly, is later discarded, and survives as fragments in the desert (Borges 1960). Baudrillard then reverses it: the map now comes first,

while the territory survives as fragments scattered across it, describing a relation in which models begin to generate reality (Baudrillard 1981). One question tests the state of a representational environment: can reality interrupt it? A representation that reality can interrupt remains a representation; one that reality cannot interrupt has entered the fourth state.

External grounding g and surface credibility q can be examined separately. The dangerous case is rising q alongside falling g; this is no fixed sequence for every representation, since rough records can be reliable and realistic models well calibrated. The following expression sketches only one direction requiring attention:

$$g_1 > g_2 > g_3 > g_4 \approx 0, \qquad q_1 < q_2 < q_3 < q_4 \tag{5.7}$$

g_k is the connection between signs and reality at stage k, assessed case by case by asking whether reality can interrupt the representation; q_k is its surface quality, including fluency, completeness, internal consistency, and credible appearance. Both sequences are dimensionless ordinal quantities: they order cases without supplying numerical values. The schematic direction has two aspects: grounding declines toward zero at the fourth stage, where signs refer only to themselves, while surface quality rises in the opposite direction. This explains why declining grounding can become harder to detect: the criterion lies on the g side while visibility chiefly lies on the q side, allowing a normal appearance to conceal lost connections. The expression supports case-by-case judgment: representations within one field can occupy different states simultaneously, and the sequences describe neither a timetable nor historical eras; they order grounding rather than measure an environment numerically.

Content environments make the criterion concrete. Information can be interrupted by reality when it connects to a stage that bears costs: medical advice merely forwarded without verification receives no calibration; a product review whose material is determined by the manufacturer's budget has doubtful independence. Stages subject to independent tests and actual costs retain an entry point for a verdict: experiments consume reagents, engineering carries loads, surgery faces complications, and lending requires accounts to balance. The distribution of interruptibility expresses the distribution of costs. Breaks arise more readily in high-traffic representational layers, where acceptance rather than verification earns traffic.

The feedback mechanism can be taken one step deeper. Real-world feedback has two constraints difficult to counterfeit together: its recipient cannot arbitrarily rewrite it, and it does not negotiate with expectations. Whether the world answers a judgment as hoped is independent of preference; the judge cannot demand a different answer. A signal that can contradict expectations and resist arbitrary rewriting has a chance to calibrate them. In a simulacral environment, systems can instead adjust feedback's distribution, filtering, delaying, or rewriting unwelcome results. When feedback becomes mere accommodation, calibration loses a vital condition: **Calibration requires something that does not comply.**

The mechanism extends beyond entertainment. Systems in which measured parties help

produce feedback face a similar risk: when an indicator guides decisions, those measured adapt to improve the indicator without necessarily improving the underlying state. Pass rates in education, recorded cases in policing, financial risk scores, and publication counts in research have all shown such divergence, recorded under two names: Goodhart's law in economics and Campbell's law in social inquiry (Goodhart 1975) (Campbell 1979). The risk parallels the analysis of simulacra at an organizational level: signals manipulated by the parties they describe, without independent checking, can drift from the intended state.

Broken feedback affects systems as well as people. Systems themselves form through feedback: models depend on data, data on human activity, and human activity increasingly on model outputs. As the loop extends, systems begin drawing material from their own products. The epistemic consequences of this self-feeding loop appeared earlier; here the concern is its measurable cost through time. In particular training pipelines that repeatedly replace original data with generated data, research has observed deterioration across generations, potentially losing tail information first; a subsequent closed training chain cannot readily recover distributional information already lost (Shumailov et al. 2024). A parallel constraint concerns the finite stock of high-quality human text: extrapolations of existing scaling trends suggest that publicly available human text could constrain training between the late 2020s and the 2030s. This estimates a one-time stock against expanding demand, rather than an end to human data production (Villalobos et al. 2022). Both point to the same structure: discarding external grounding while retaining only self-generated products can return the cost of lost distributional information as deterioration; preserving original data, independent verification, or external feedback can change the outcome. An information-theoretic study at ICML 2026 asks whether synthetic data add task information beyond the current model, distinguishing synthesis shaped by external signals from information-closed self-reproduction (Li et al. 2026).

A subtler break occurs within measurement tools. Model evaluations often draw tests from public corpora on which models have already trained. A test question present in training can confound memory with generalization; as versions improve, scores can rise while becoming harder to interpret. Machine learning calls this data contamination. Apparently a technical detail, it repeats this chapter's feedback structure: overlap between the source of an evaluation signal and its object can inflate apparent generalization (Sainz et al. 2023).

The Framework's Boundary

Baudrillard's writings on the Gulf War examine relations among media presentation, the experience of war, and events (Baudrillard 1991). This book asks whether a representation preserves an independent route to verification. Casualties, destruction, and subsequent actions do not disappear through media narratives; they form a layer of consequences that must enter the accounts.

The disagreement concerns reality itself. Measuring connections between signs and reality requires an independently identifiable reality side: a consequence layer beyond signs. Bau-

drillard's radical account of simulacra challenges precisely that independent reference. This book borrows the four-stage sequence and diagnosis of hard-to-detect lost connections, but reconstructs them around an independent consequence layer, departing from his radical version. Irreversibility, entry points for verdicts, and returning consequences presuppose realism; if the premise fails, the criteria fail with it and cannot themselves prove the premise. This locates where the criteria can work, without making them useless—a scope narrower than popular uses of "simulacra."

A counterexample determines how the concept should be used. Coupling between symbolic and material layers need not break; simulators provide a positive case. Flight simulators, surgical training simulators, and preclinical computational models are inexpensive representational environments serving training and rehearsal when repeatedly checked against reality: parameters come from real flights, actual flight performance tests training, and trials correct model predictions. Their defining feature is a regular accounting against reality, beyond realism alone: reality can overturn a simulation's conclusions. Healthy coupling differs from broken feedback by an entry point for a verdict outside the representation's control, rather than by representational precision alone.

Broken feedback therefore has two victims. People lose opportunities to calibrate judgment, and systems lose opportunities for correction by reality. The remedy in both cases is a stage that must pass through the world. Willpower alone cannot supply it, since the simulacral environment characteristically offers no refusal. To see why, another question comes first: if stimulation separates from consequences, where do the objects of desire come from?

5.4.3 Mirrors of Desire and the Gnostic Return

Broken feedback explains why judgment loses calibration without yet explaining what directs desire. People in low-viscosity environments still desire; what changes is desire's shape and where that shape originates.

Mimetic Desire and Mediator Distance

René Girard proposed a structure in *Deceit, Desire, and the Novel*: desire forms a triangle rather than a direct line from subject to object, with the subject desiring through a mediator (Girard 1961). Self-narration conceals the mediator: people believe they directly desire something while actually desiring what others already desire. He distinguished two forms of mediation. External mediation places the mediator far from the subject and outside competition: reading a chivalric novel brings no rival for its nonexistent princess. Internal mediation places mediator and subject in the same field, competing for the same object; imitation produces rivalry, which strengthens imitation.

Digital environments change the distance and number of mediators. Algorithms bring them close: one sees what people resembling oneself possess now, rather than what legendary heroes

possessed. The cost of displaying results also falls sharply, because displayed lives can omit processes, splice moments, and roll back. Imitation follows the shape of results rather than the process producing them; the more the process disappears, the greater the gap between imitators' circumstances and the results they imitate. Internal mediation thus more readily becomes an everyday experience. Girard argued that convergent desire can expand mimetic rivalry into collective crisis, temporarily contained by redirecting responsibility toward a commonly identified scapegoat (Girard 1972). An observable counterpart appears online when collective hostility forms faster than facts can be verified.

The shrinking distance of mediation has a measurable aspect. Social-comparison research has long emphasized comparable others rather than absolute conditions as standards of self-evaluation; larger comparison sets and more visible gaps can raise those standards. Research on social-media use and subjective experience has repeatedly associated passive viewing of others' displays with worse mood, without finding the same effect for active communication. Causal direction remains contested, but the structural observation does not depend on settling it: displays omit processes and thereby offer an elevated comparison standard (Festinger 1954) (Verduyn et al. 2015). Imitation is no pathology. Language, skills, taste, and most cultural transmission depend on imitation, which Girard himself treated as constitutive of humanity. The issue concerns the proximity and arrangement of mediators: when nearby mediators repeatedly display the same competitive objects, desire can more readily converge on them. The question concerns how mediators are arranged rather than human weakness.

Mediators' distance, number, similarity, and competitive relations can change the possibilities people see. More mediators can also broaden desire rather than necessarily concentrating it. Girard's framework offers a mechanism to investigate; measuring concentration on particular platforms requires content selection, user networks, and actual behavior.

Gnostic Escape

When desire's objects are externally supplied, reality can come to appear as costs alone. At this point, a judgment structurally resembling ancient Gnosticism becomes possible. This section borrows a recurring cluster of claims from some Gnostic traditions of the second and third centuries CE: a flawed creator made the world, matter imprisons spirit, and salvation comes through knowledge of one's origin rather than action and consequence-bearing. Abstracted structurally, three features remain: reality is inferior, escape is return, and knowledge rather than work supplies the route.

These features can recur without religious form. When reality feels full of useless friction while symbols offer low-resistance interaction, treating the symbolic layer as more authentic becomes structurally attractive. No declaration that reality is false is needed; practical arrangements need only treat it as a bypassable intermediate layer: experiences generated without being lived, relationships simulated without mutual obligations, works produced without training. The decisive evidence lies in which layer receives resources as the principal site of life, rather

than in stated belief.

The criterion can be specified through time and attention allocated between symbolic and material layers, and willingness to bear costs that cannot be rolled back. Examples include long training, caring for a particular person, maintaining something that wears out, and entering a relationship whose history cannot be undone. These activities contain time and consequence-bearing that cannot simply be rolled back or wholly delegated. Whether someone still stands on reality's side can be judged through such activities without asking what they profess to believe.

This chapter argues in the opposite direction, on grounds of settlement rather than morality: **Matter is no prison; it is the only medium capable of carrying consequences.** Symbols generate candidates; materials, bodies, and time settle accounts. Costs are conditions for forming positions, rather than defects to escape. As shown earlier, higher levels of existence carry heavier maintenance and rollback costs, while subjecthood appears where costs can no longer be transferred.

Two misuses must be avoided. One treats the Gnostic return as a dismissal of virtual experience. Fiction, games, and simulation have always expanded experience, offering safe testing grounds rather than deception; the issue concerns proportion and closure, rather than the existence of an open interval. The other turns the diagnosis into a moral accusation. Under given conditions, low-viscosity paths are rational: inexpensive, responsive, and reversible. Calling this weak will conveniently releases environmental design from responsibility.

When external mirrors supply the objects of desire, they can also supply the objects of refusal: their collapse leaves people directionless, threats to them provoke hostility, and their need for scapegoats produces scapegoats. Supply then maintains the boundary in place of an internal process. The problem moves from authenticity to subjecthood: can an existence still claim a boundary when even what it refuses is given from outside?

5.5 Third Collapse: The Crisis of Subjecthood

The boundary is not seized; it ceases to be exercised where resistance disappears. The preceding collapses concerned function and authenticity; this one reaches further inward, to the capacity to refuse and what sustains it. It changes most slowly and is hardest to recognize within an individual lifetime.

5.5.1 The Weight of Negation: When "No" Is No Longer Needed

Boundaries beyond Choice

Maintaining a boundary begins with what lies beyond choice. A familiar account defines a subject through the ability to choose and answer for choices. It explains much while missing the other side of a subject's formation: refusal.

Acceptance and refusal are structurally asymmetric. Acceptance can be compliant, silent, and costless; refusal under pressure can reveal a position held firm despite consequences and possible rupture, while protected, cost-free refusal can also express a clear boundary. The capacity to refuse gives an existence firmness; a system recorded only as accepting may be internally complex, yet those records alone reveal little about its boundary. This distinction matters in algorithmic environments because recommendation, ranking, notification, and interaction systems tend to reduce refusal when acceptance and retention dominate their objectives. They learn what you accept and make acceptance more likely; they learn where you leave and avoid that point. The better such systems work, the less resistance you encounter and the less often you are called upon to refuse.

Three Features of a Frictionless Environment

The tendency operates in several directions, each defensible on its own. The first reduces waiting. Buffering, prefetching, and preloading fill gaps. Waiting once provided a structural position: a chance to reconsider continuing, weigh whether an investment was worthwhile, and leave. Removing waiting reduces these natural pauses for judgment.

The second reduces friction. Interfaces simplify, steps disappear, and defaults make choices on your behalf. Friction can support confirmation: several confirmation steps may signal a consequential operation, though repeated empty prompts can also produce approval fatigue. If every operation becomes one click, decisions of different weight can feel equally light, depriving consequential choices of a felt distinction from opening a notification.

The third reduces conflict and dissent. Recommendation systems focused on immediate acceptance may push users into narrowing regions of preference. Deliberate suppression is unnecessary: if dissent brings exits and low ratings that receive negative weight in the objective, the system tends to reduce such content. Dissent can thus diminish as an optimization by-product without intentional filtering. Each direction looks like an improvement: waiting wastes time, friction impedes efficiency, and conflict feels unpleasant. Together they can optimize a system that no longer needs you to say "no." The weakening discussed earlier has passed beyond feelings and a sense of reality to reach the boundary itself.

Marcuse's Predicament

A related stabilizing mechanism was described over half a century ago, in a different medium. Herbert Marcuse argued that advanced industrial society remained stable by satisfying what its opponents wanted rather than merely suppressing opposition. Continuous comfort deprived negation of its ground; he called the resulting figure one-dimensional, accepting without negating (Marcuse 1964). His analysis concerned consumer society, but the structure can be transferred. Goods once dissolved dissent; smooth interaction, personalization, and immediate satisfaction can now do similar work. The means differ while the mechanism persists: refusal appears unnecessary because its object causes no pain. One difference matters

in making the transfer. Marcuse confronted material scarcity and class antagonism, allowing criticism of concrete oppressive structures; where a system demonstrably satisfies most needs, calling it oppressive becomes harder, and resistance becomes harder for that reason.

Its place in this book's notation merits stating, since exit barriers may move to a different location. Refusal carries costs, and higher costs mean greater viscosity of refusal. Expand that viscosity across the three dimensions,

$$\eta_{\text{refusal}} = \eta_0 \exp(\alpha R + \beta T + \gamma G), \qquad G = \sum_{i=1}^{N} w_i \tag{5.8}$$

Here R denotes resources consumed by refusal, including income, shelter, and safety; T is the time required to make refusal effective; G is coupling to the system, w_i the weight of coupling i, and N the number of couplings that must be severed together. Oppressive systems often apply direct pressure through the first two dimensions, while also using the third: losing work, facing prosecution, or struggling for years makes refusal expensive, yet visible costs can be named, organized around, and negotiated. A system offering satisfaction may instead emphasize the third dimension: the direct R and T of one exit appear small, yet departure sacrifices a bundle of services causing no obvious pain—recommendations, social connections, payments, records, identity, and habits. N is large; sufficiently large weights w_i make total G high. The system need not make departure impossible; it need only leave no single exit. This explains a familiar weightlessness: no single cost can be named, yet the total remains unaffordable. Consistent accounting must distinguish resource losses, migration time, and network dependence to avoid counting the same loss twice.

The danger lies in how little the mechanism reveals itself. In an openly oppressive system, refusal has an identifiable target; in one continually supplying satisfaction, it loses its object. People do not announce rejection when nothing evidently calls for rejection. Yet the boundary may already be contracting without an identifiable moment of change. Personal awakening alone is often insufficient, precisely because no moment makes awakening seem necessary.

When refusal loses its object, exit becomes a problem too. Marcuse proposed the "Great Refusal"; this book takes another route, because refusal without an object readily becomes rejection of an entire way of life, without a ready-made outside to retreat to. He identifies excluded people as potential subjects of refusal; this structural analysis can locate refusal's viscosity in G but cannot yet identify who will bear it. The location is more precise while the point of action remains less clear; rhetoric cannot repair the asymmetry.

The Cost Structure of Refusal

Whether refusal expresses a boundary depends on its availability, effect on the process, and exposure to coercion. Costs can make commitment easier to recognize or directly deprive people of the ability to leave. These mechanisms require separate examination.

Refusal can express commitment through bearing costs or clearly mark a boundary at zero cost. Readily revocable consent can still be valid; protecting exit helps resist coercion. Both

usable exit and appeal, and enduring relationships involving voluntary investment and mutual responsibility, must remain available.

Boundaries require usable refusal and commitments worth sustaining. For a particular signal, moderate costs can distinguish genuine commitment from casual statements; unaffordable costs can coerce choice. The interval describes a signaling structure, without setting a minimum threshold of suffering for subjecthood or freedom.

Write the interval between the extremes as a condition,

$$0 < c_{\min} < c < c_{\max} < \infty \tag{5.9}$$

c is a consistently defined signal cost; $c_{\min}$ and $c_{\max}$ depend on particular benefits, resources, and circumstances. The expression describes only a possible interval in which costs distinguish commitment. Free refusal can still convey information; these thresholds do not determine the exit rights that ought to be protected.

The Function of Discomfort

Discomfort can be valued only through the function it serves. It has no value in itself. Pain does not automatically create meaning or obstacles build character; enough harm has already been inflicted in their name.

What can have value is discomfort's function as feedback: pressure is borne somewhere, or a path diverges from a goal. With a complete feedback loop, discomfort can be a calibration signal. It indicates one's position, the costs, and whether to continue. Systematic removal of discomfort can remove that signal and a coordinate for locating oneself. A testable implication follows: if the argument holds, people raised in low-viscosity environments should find judging their own circumstances harder, with the difficulty appearing in judgment rather than expression. Clean evidence is not yet sufficient to support or reject that implication; this chapter offers a structural claim rather than an empirical finding.

The question concerns information rather than whether to pursue comfort or seek difficulty. When an environment filters out discomfort, what else supplies equivalent information about one's position? The answer depends on independent tests, meaningful choices, and usable refusal, rather than a tally of discomfort. Algorithms can optimize discomfort away without eliminating the need for a place. Stability can be measured, while contracting boundaries must be read through consequences: even a precise preference vector cannot tell a system where someone will say "no" from acceptance records alone.

5.5.2 Pleasure's Local Optimum and the Valley

An Ending without Collapse

Discussions of low-viscosity environments readily drift toward two endings: collapse, when control fails and consequences erupt together; or deterioration, when systems leave people in-

capable and dependent. Both narratives are dramatic and problematic. They assume a conspicuous failure, while a system can settle elsewhere: optimization under given objectives and constraints may stop at a local optimum.

A simple formulation explains a local optimum. Suppose variables x describe the system's state and an evaluation function $V(x)$ measures how good that state is. A local optimum x^* satisfies the condition:

$$\nabla V(x^*) = 0, \qquad H_V(x^*) \prec 0 \tag{5.10}$$

The first condition is necessary for an unconstrained interior extremum of a smooth function; the second requires a negative-definite full Hessian and is sufficient for a strict local maximum. Negative second derivatives along each coordinate alone can miss a saddle along mixed directions. The evaluation function V determines what counts as improvement; dimensions omitted from it can be ignored while scores rise.

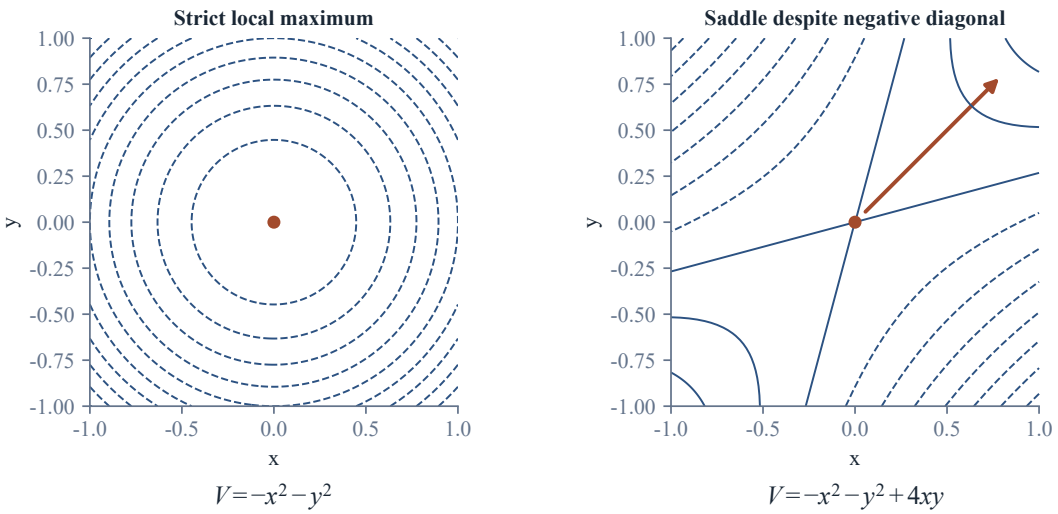


Figure 5.2 Both functions have Hessian diagonal entries of −2. The left Hessian is negative definite, making the origin a strict local maximum; on the right, cross terms give positive and negative eigenvalues, so the function rises along the red direction and the origin is a saddle.

Three Features of a Local Optimum

A system at a local optimum can appear to be working at its best. It may combine three features, each normally evaluated positively. First, high satisfaction. The system genuinely keeps supplying what users want, with improving efficiency. Complaints decline as optimization addresses their causes one by one. Satisfaction, retention, and growth curves all look good.

Second, stability. Disturbance is difficult because change confronts an established configuration: interfaces, habits, business models, regulatory expectations, and infrastructure have

adapted to the current state; changing one parameter affects the others and rapidly raises costs. "Stable" sounds positive here: predictable, maintainable, and low-risk. Third, little impetus to grow, a feature hardest to notice because it presents no symptom. Without an apparent problem, the system has no apparent need to evolve. Improvement stays within the current configuration: finer recommendations, smoother interaction, lower latency. All follow the same curve's flatter region, yielding diminishing returns while still looking like progress.

A system can thus be comfortable, stable, and stagnant simultaneously, without detecting the third condition. Correction is difficult because the features support one another: satisfaction discourages disturbance, stability conceals unresolved problems, and the absence of visible problems keeps improvement moving along the flat region.

Nietzsche's Last Human: A Brief Comparison

An image from intellectual history offers a comparison, provided its limits are clear. In *Thus Spoke Zarathustra*, Nietzsche described people who had abandoned self-transcendence, longing, and creation, seeking only comfort, health, and longevity, and calling that condition happiness. They had "invented happiness," he wrote, and blinked (Nietzsche 1883). The image overlaps with a local optimum: his target was the loss of striving beyond oneself when pleasure became the sole objective, rather than pleasure itself. The last humans are satisfied rather than miserable; their satisfaction asks for nothing further and therefore produces nothing further.

Here is the boundary between the accounts. Nietzsche located the cause in failed values and unwillingness to pursue higher goals; local-optimum analysis offers a more technical account: optimization brings the system to a comfort maximum where deviation immediately lowers the evaluation score. The issue concerns structure rather than willpower. Individually rational decisions can combine to leave people where nothing new is produced. The practical difference is substantial: a problem of will invites exhortation; a structural problem requires changing the evaluation function through institutions, indicators, and competition rules rather than individual resolve.

The more consequential boundary lies in evaluation itself. A local optimum contains no normative verdict: whether stopping at an extremum is failure depends on what the evaluation function includes. If the objective is worthwhile, reaching its maximum is success; structural analysis identifies that movement has stopped without thereby proving that the position is unworthy of staying in. Growth needs a separately defended standard of what counts as better; deriving it from the same optimization language would be circular. Mechanisms explain how stagnation arises; normative judgment determines why it is insufficient, and the former cannot answer for the latter.

Why the System Stays

Leaving a strict local maximum along a continuous feasible path normally requires accepting an initial decline in evaluation for the possibility of reaching a higher local optimum. With

objectives and constraints fixed, such a path crosses a region of lower V, termed a valley.

$$\Delta V_{\text{loss}} = V(x^*) - \min_{s \in \text{path}} V(s) \geq 0 \tag{5.11}$$

Because the path includes its starting point, the current value minus the path minimum is nonnegative: the greatest temporary evaluation loss along that path. Finding the minimum escape barrier into a new region requires further optimization over allowed paths. Who bears transitional losses and who receives later gains influences willingness to change.

The individual structure is similar, with the person bearing the costs. Maintaining an ability requires continued use, each exercise imposing a higher immediate cost than the available substitute. Writing a passage oneself rather than generating it may yield slower, poorer output today; navigating without directions may take longer and involve wrong turns. Costs are immediate, while benefits are dispersed, delayed, and difficult to observe. Human judgment is sensitive to immediate costs and less sensitive to delayed benefits; an optimized low-viscosity environment can systematically exploit this bias without anyone intending to do so. Enough low-friction substitutes allow the bias to operate on its own.

Stagnation Is Not the End

The preceding analysis suggests an endpoint. **The low-viscosity trap may end in a comfortable, stable local optimum with little impetus to grow, rather than collapse.** The system keeps running, indicators keep improving, and individuals remain satisfied while the overall space of possibilities narrows. Short-term indicators reveal no clear harm and protest may never appear: the system delivers what it promised. The narrowing occurs elsewhere: real skills, tested relationships, judgments maintained through rejection, and personal boundaries may gradually diminish because producing them requires costs incompatible with the current evaluation function.

A precise qualification separates this claim from cultural pessimism. A local optimum is a possible outcome under specific conditions rather than a necessary result of low viscosity: evaluation must consist entirely of experience, short-term experiential feedback must conceal long-term outcomes within that evaluation, and competition must favor the lowest-friction option. Breaking any condition can change the path. Real evaluation functions plainly extend beyond experience: bodies develop problems, relationships rupture, and skills decline enough to affect income, eventually returning long-term outcomes to feedback. The question is therefore the length of the return delay.

Delay determines the distribution of costs and the direction of the book's next inquiry. Short delays allow people to calibrate within their own timescale and bear their own costs; delays spanning a generation pass costs to successors who had no chance to choose otherwise. This affects whether low-viscosity environments correct themselves promptly or accumulate problems exposed only through rupture. The next question concerns which human capacities sys-

tems cannot cheaply replace, and why. At the individual level these appear as abilities; at the collective level, mass adaptation along one path also changes the distribution of differences.

5.5.3 Lost Diversity and the Ground for Totalitarian Psychology

The preceding sections examined individual boundaries: refusal loses its object, and systems settle at local optima. At the population scale, the two can combine into a changed distribution of differences. Differences supply the material for boundaries, beyond being abstract values.**Without differences, there is nothing to refuse and no conflict requiring judgment.**

Narrowing Paths and Laminarization

Shared recommendations, rankings, and defaults can expose many people to correlated paths, while users and networks may introduce other directions. Collective diversity requires measurement. Laminar and turbulent flows supply no direct criterion for exploration, cultural value, or political orientation; the relevant mechanisms concern available alternatives, shared exposure, and error correction.

Other domains have measured costs of such narrowing. Finance repeatedly shows how homogenization can weaken resilience: institutions sharing risk models, assets, and funding structures become more connected and less differentiated, providing channels through which local shocks are amplified system-wide (Haldane and May 2011). Agriculture supplies an older example: dependence on a few susceptible crop varieties can let disease penetrate a food system. In Ireland's Great Famine of 1845–1852, heavy reliance on susceptible potatoes including the Lumper met blight, compounded by poverty, land institutions, and failed relief, producing roughly one million deaths and one million emigrants (Ó Gráda 1999).

The common problem is a shortage of redundant differences rather than simplicity alone. Differences look like redundancy in cost accounts: parallel plans, alternative methods, and apparently superfluous options seem inefficient. Under disturbance, redundancy supplies alternative paths. When accounts reward only the lowest immediate cost, low-viscosity environments can systematically remove redundancy: the cheapest path draws use away from others until they cease to be maintained or practiced.

An engineering criterion illustrates redundancy's value. Power-grid operation uses an N-1 security criterion: within specified single-contingency cases, the system must meet applicable stability, power-flow, voltage, and other limits, requiring apparently redundant capacity, alternative lines, and dispatchable resources (Morison et al. 2004). Costs accrue continuously while benefits appear at failure, making reserves vulnerable to accounts that ignore failure risk and consider only short-term expenditure. Diversity has a similar position: recorded as a cost, it pays out as survival under disturbance, at a moment invisible in ordinary conditions. Narrowing often follows from no one having a reason to fund alternatives rather than from an explicit decision to eliminate them.

A boundary matters here. Attributing all lost diversity to recommendation algorithms mislocates responsibility. Research on cross-ideological exposure found a more complex pattern in Facebook data: users' friendship networks constrained access to opposing views more than algorithmic ranking, though ranking had an effect (Bakshy et al. 2015). The result identifies several sources of narrowing, including personal choices and network homogeneity, rather than declaring algorithms harmless. Identifying the mechanism correctly is a condition for effective intervention.

The Attraction of Order

Removing differences can produce a seemingly unrelated psychological effect. Arendt's diagnosis can be extended here: she located totalitarianism's mass base in atomized people disconnected from classes, occupational communities, and stable positions, arguing that loneliness better explained receptiveness to an all-consistent explanatory system than poverty alone (Arendt 1951). Combined with the preceding sections, this suggests a mechanism: lost positions produce superfluity, superfluity creates demand for order, and a system experienced as reliable, continually available, and free of constant negotiation supplies inexpensive order.

The psychological side has measurable forms. Trust in automation can be systematically miscalibrated: inappropriate reliance, inadequate monitoring, and takeover difficulties can arise, and expertise alone does not guarantee protection (Parasuraman and Riley 1997). Judgment can thus be surrendered through defaults rather than one decision: answers suffice, human review appears unnecessary, and prolonged disuse weakens both reviewing skills and habits. Order attracts here by relieving users of supervision costs rather than merely being stronger.

Another structural reason makes order attractive. An explainable, appealable, fallible order requires users to judge and supply evidence; an opaque order takes judgment away and can make responsibility feel removed as well. For people already treated as superfluous, surrendering judgment promises both certainty and relief from responsibility for their own judgments. Dependence is difficult to dissolve through vigilance alone because it meets real needs.

Dependence offers another, less discussed benefit: relief from the burden of proof. A negotiated order requires reasons, responses to objections, and responsibility for judgments; handing these to a unified system can hide those stages from the user. For those unsure of their judgment, the relief is real and can deepen with continued use. This also directs intervention: sustaining judgment requires places where it can be submitted and have consequences, beyond reminders alone.

Treating this path as inevitable overstates the argument. Dependence turns on alternatives: occupational communities, local relationships, enduring institutions, and participatory affairs can change the outcome of the same technological pressure. Technological order is an option; only when other sources thin out does it become the sole available order. The boundary is conditional: weakened intermediate structures make dependence cheapest, rather than technology inevitably causing it.

The three collapses now connect. Fewer differences leave refusal without objects; externally supplied objects make desire and boundaries increasingly dependent on supply. Each stage lowers the cost of the next, allowing the entire path to be followed without first declaring any step wrong. Seeing it requires function, authenticity, and subjecthood in one structural picture, beyond a moral verdict.

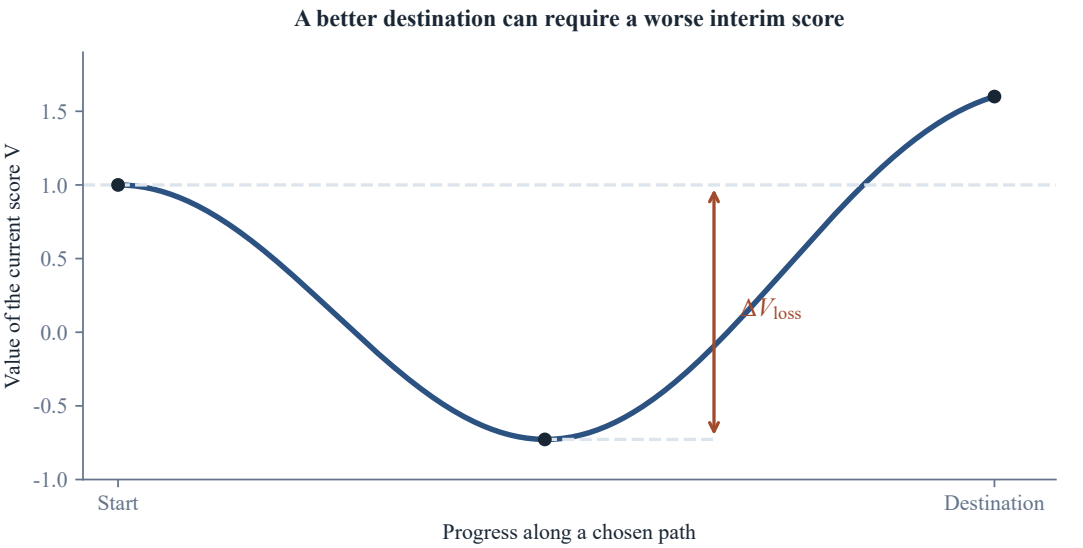


Figure 5.3 Temporary metric loss along the chosen path. ΔV_{loss} is the starting value minus the path minimum, nonnegative because the path includes its starting point. Changing the objective, constraints, or feasible paths changes this loss.

5.6 Reconnecting the Loops: Capability, Feedback, and Accountability

The three collapses can reinforce one another. Weaker professional positions make immediate recognition more attractive; environments that rarely test judgment reduce calibration practice; falling confidence and fewer alternatives make returning to an effective position harder. Each connection can be investigated and can become an intervention point.

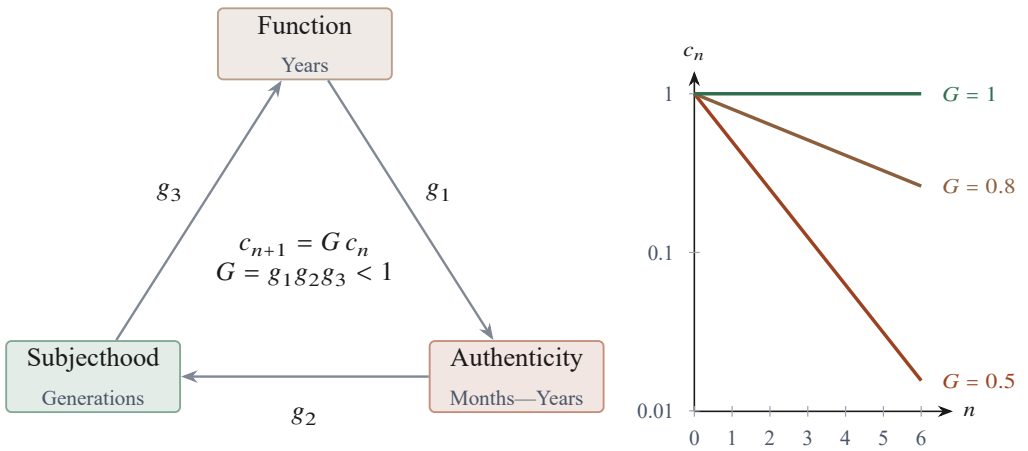
5.6.1 Three Clocks and Paths to Repair

Occupational structures may change over years and interfaces over weeks, while skills, habits, and relationships have their own formation cycles. Learning and recovery continue in adulthood. Intervention must target the relevant process and preserve what can still improve.

Let c_n denote the cost of entering a substitute experience and write a simplified recurrence:

$$c_{n+1} = Gc_n, \qquad G = g_1 g_2 g_3. \tag{5.12}$$

If all three positive multipliers are below 1, entry costs decrease on each circuit of the model. The expression describes only the defined cost. A full social loop also requires independent state variables for occupational position, feedback, and subjecthood, alongside delays, external inputs, and observable causal connections.



(a) The three-layer loop (b) Entry barriers decline on each circuit

Figure 5.4 A simplified recurrence for entry costs. With three fixed positive cost multipliers below 1, their product determines the decline of the cost curve. The figure illustrates this cost model rather than complete causal dynamics of occupation, feedback, and subjecthood; real interventions can also change multipliers, structure, and external inputs.

Inquiry can begin with narrower questions: has entering an occupation become harder? Do assistive tools improve skills on independent tests or only immediate output? Can users recognize, challenge, and correct a consequential error? Does an appeal change the decision? Are alternatives affordable and usable?

These questions need different evidence: longitudinal observation, task tests, institutional records, and concrete cases. Indicators also need reverse checks, so that counting appeals does not masquerade as counting resolved problems. Repair mechanisms must have actual power to change the process.

5.6.2 Four Functions of Accountability

Consequences need a route into a response. Accountability structures must supply at least four functions.

Detection. Monitor harm, near misses, and important deviations; allow users and affected people to report problems. Operators' dashboards must connect to external complaint channels.

Tracing. Reconnect an event to the relevant decisions, versions, configurations, and tool actions. The scope, retention, access permissions, and privacy protections of records must be designed around their uses.

Harm containment. Give someone usable authority to pause, isolate, recall, or restrict. Emergency control can precede complete attribution, preventing harm from growing during investigation.

Remedy. Allow affected people to seek correction, obtain appropriate compensation, and press for prevention of recurrence. Assets, insurance, and organizational capacity behind responsibility determine how much of a promise can be fulfilled.

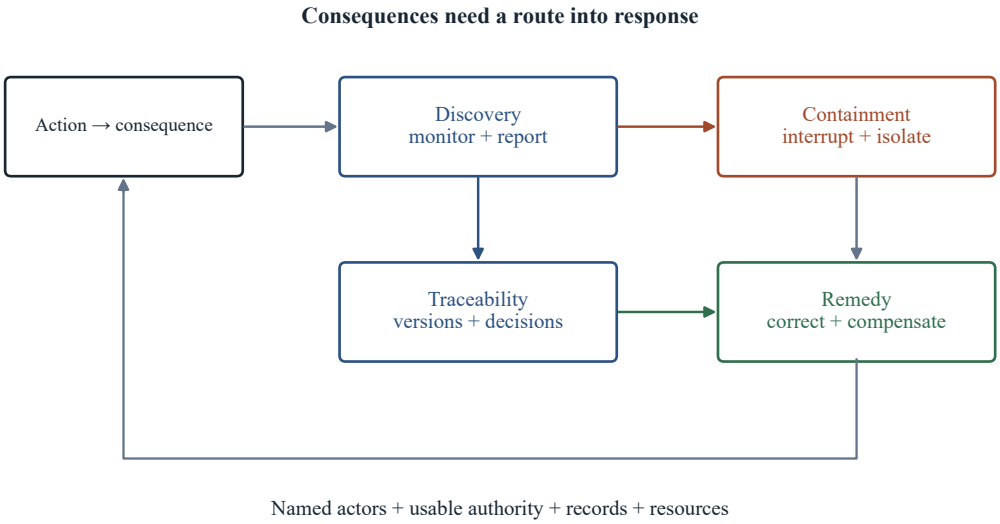


Figure 5.5 Four functions of accountability. Monitoring and reporting bring consequences into the record; tracing supports explanation and correction; harm containment can begin immediately, while remedy addresses consequences already incurred. The functions connect and can operate in parallel.

The four functions can run in parallel and supply one another's inputs. Each needs actors, resources, and operational interfaces. Aviation combines design assurance, production control, maintenance, operating permissions, reporting, and investigation; aircraft and institutions jointly produce reliability. Other fields must arrange these functions around their own consequences and control conditions.

Machine systems need equally clear interfaces: who controls updates, who can suspend access, which records survive a service shutdown, who hears challenges to model decisions,

and who can provide remedies?

Responsibility can be allocated through control, contractual roles, legal duties, and other legitimate standards. Insurance and compensation funds can share some losses; independent investigations help separate learning from immediate attribution of blame. Incentives, enforcement, and funding together determine whether these arrangements work.

Accountability can also expand deployment. Risks become easier to assess and accept when they have clear control, recording, and remedy paths. Knowing how to act when uncertainty becomes consequence gives some formerly unpromisable applications conditions for operation.

5.6.3 Judgment Requires Standing

Three questions remain central to decisions: which consequences are acceptable, how much evidence warrants action, and who has the right to impose costs on whom?

Models can estimate outcomes, compare proposals, expose contradictions, and help people express positions. Adopting standards and imposing consequences also require appropriate rights, authorization, and duties. This book argues: **People bearing consequences should retain effective channels for voice, representation, and challenge.**

Direct exposure provides one basis for standing; legitimate delegation, professional duties, democratic authorization, and protection of absent and vulnerable people provide others. Patients bear bodily risks and physicians professional responsibilities, connected through informed consent, professional standards, and accountability procedures. Jurors usually do not bear the same consequences as defendants; their authority comes from legally assigned roles, duties, and procedures (Kalven and Zeisel 1966). Captains, physicians, and jurors have different grounds for authority. They share a requirement: decisions have explainable reasons, permissions have defined scope, and someone must answer a challenge. Reviewers also need competence, time, and power to change decisions before a "human in the loop" becomes an actual position rather than a paper assurance.

5.6.4 Human Value When Output Becomes Cheap

A society valuing people only through scarce output loses part of its evaluative basis as output becomes abundant. Participation, care, relationships, judgment, and finite lives need a fuller place in evaluation.

Consequences falling on someone give that person grounds for protection and voice. Allocating burdens also requires attention to legitimate relations of power and control. Bearing burdens need not be romanticized; avoidable harm should be reduced.

Another person can advise, work, or make delegated decisions, without thereby living your finite life or automatically occupying the same relationship. Assistance expands a person's

capabilities while leaving that person's stake in outcomes intact.

Repair has concrete destinations: rebuild paths to useful contribution, preserve corrective feedback, keep exit, dissent, and enduring commitment viable, and reconnect consequential actions to people and institutions able to respond. Each arrangement requires investment. The next chapter distinguishes costs that merely obstruct from those preserving the conditions that make progress worth having.

Further reading

Hannah Arendt, *The Origins of Totalitarianism*
Herbert Marcuse, *One-Dimensional Man*
Jean Baudrillard, *Simulacra and Simulation*
René Girard, *Deceit, Desire, and the Novel*
Harry Kalven Jr. & Hans Zeisel, *The American Jury*
Ivan Illich, *Tools for Conviviality*
Cathy O'Neil, *Weapons of Math Destruction*
Theodore Kaczynski, *Industrial Society and Its Future*

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Chapter 6

Reconstructing Value: Living in the Shear Zone

The previous chapter showed why consequences need a bearer, and how bearers can be separated from decision-makers. Where value goes also depends on demand, property rights, bargaining power, and public arrangements. This chapter distinguishes the social necessity of work, the pay and standing of those doing it, and human dignity and the meaning of life. The three influence one another without automatically converting into one another. Abundant generation may make verification, care, enduring relationships, and responsibility scarcer; whether those positions receive resources remains a question of institutions and distribution. Historical externalizations of capability offer several ways to examine this change.

If intelligent systems can already write copy, summarize, translate, retouch images, code, and develop proposals, how much weight remains in ordinary people's skills? If they can also read the literature, propose hypotheses, design experiments, run simulations, analyze data, and write papers, how does research training demonstrate its value? If they participate in mathematical proofs, search for lemmas, and check derivations, is even research's hardest formal intellectual labor beginning to loosen? If robots drive, manufacture, carry, inspect, operate, and provide care, how much room remains for people in work once grounded in bodies, experience, and situated judgment? Beyond the fate of particular occupations, external systems are dismantling the capability interfaces once used to establish training, experience, and standing as a subject.

Questions about disappearing occupations and depreciating skills reach a deeper issue: how did people once demonstrate their value? Writing, calculating, making, judging, researching, proving, and acting could each sustain a social position. External systems now remove, amplify, and reorganize these abilities layer by layer. The unsettling question is where people move after a capability is externalized, beyond finding something machines will never do.

6.1 Historical Echoes: The Physics of Value Drift

These questions have been asked before, and their answers rewritten before. Chapter 1 offered a claim: externalizing a capability breaks its old bond with value, because what moves is also an interface through which society recognizes training, experience, qualification, and responsibility. This chapter follows the break: where does value land, and why there? Three forms of proof fail together: output no longer automatically establishes a subject, form no longer establishes experience, and capability no longer establishes legitimacy. Each depends on a stage that an external system can take over; where those stages sit shapes the direction of migration.

An interface here is a concrete institutional arrangement. Guilds establish craft through apprenticeship years, imperial examinations establish qualification through tested knowledge, universities establish training through degrees, professional licenses establish admission through exams and practice, and academic communities establish credibility through publication records. All anchor proof of ability in a process not quickly acquired: seven years of apprenticeship cannot be compressed into seven months of reading, examination scores cannot replace a resident physician's case experience, and authors' assurances cannot replace peer evaluation. The interface gives such processes recognizable form; if an external system takes it over, an entire procedure for recognizing qualifications is disturbed alongside the scarcity of ability (Spence 1973). History's answer has repeatedly been a new anchor for value rather than the end of human value. Different ways of moving capabilities break old bonds differently, yet each leaves the same question: where does the value released from its former position go next? Following that line needs the hardest step at each turning point rather than a complete history of technology.

6.1.1 Socrates' Fear: Wisdom without the Viscosity of Internalization

Externalized Capabilities and Broken Bonds

The inquiry follows several turning points in externalization: memory, copying, timekeeping, work, rule execution, symbol generation, formal exploration, action, and biological intervention. Human beings have experienced this weightlessness before. Writing lightened memory, printing copying, clocks time, steam engines muscle, and computers rule execution; large models lighten symbol generation, reasoning models and scientific agents formal exploration, embodied intelligence begins externalizing action, and intelligence applied to life brings biological intervention and the definition of life into technical systems. Each technology resembles a deep relocation, moving capabilities once lodged in bodies, skills, and identities into external systems and loosening old interfaces of value. Human value need not disappear, but must find new places, some deeper, heavier, and harder to outsource. Together, these nine turning points trace repeated rearrangements of where value lands; technology alone does not determine who receives the gains.

These turning points are organized by modes of large-scale externalization, showing how formalization affects migration rather than supplying a technological chronology without overlaps. Their order can be understood through operational forms: speech becomes writing, copying becomes an arrangement of type, muscular work passes to machines and energy systems, and rule execution, symbol generation, and formal exploration enter procedures, statistical patterns, and searchable spaces. The easier a capability becomes an operational external form, the easier it is to move beyond the body under the relevant technical conditions. Externalization first unsettles those whose identity rests on an ability: their position depends on doing what others cannot, while the external system makes that act cheap.

The old bond contains identifiable parts. Training must be verifiable, showing sustained preparation; experience traceable, showing encounters and remembered consequences; qualification exclusive, limiting who can occupy a position; responsibility attributable, determining who answers when things go wrong. Old institutions bundled the four, allowing one capability to attest to all of them. Externalization cuts the cord: models supply some outputs once requiring training, retrieval supplies some experience, subscriptions can supply signals formerly used to recognize qualifications, and responsibility disperses among systems. Where the bond breaks, value needs a new home.

External Memory and the Work of Internalization

Memory is the first turning point, externalized earliest and most thoroughly. Before writing, memory sustained order itself. Law, myth, genealogy, ritual, exchange, taboo, and experience depended chiefly on memory, speech, and ceremony, assisted by material markers. Those able to remember, recount, and transmit carried a community's time as well as its information. Forgetting could break an order, beyond being a personal lapse. Writing changed this. It let extensive experience settle into symbols beyond a particular body. People die, voices fade, witnesses depart; words, contracts, laws, poems, accounts, and knowledge can remain. In Plato's *Phaedrus*, Socrates warns that writing can act as a remedy for memory while encouraging people to mistake external preservation for inward understanding (Plato, n.d.). The warning recognizes an early **migration of value** rather than simply rejecting technology: as memory grows lighter, understanding grows heavier; texts fix answers without replacing questions, and possessing words need not mean possessing the judgment behind them.

Socrates' fear has a precise target: when knowledge moves from living neurons to parchment and other external carriers, its felt understanding does not necessarily travel with the words. Recitation here is internalization rather than primitive storage: content repeatedly passes through the body and meshes with experience, a viscous and expensive process. Writing sharply reduces resistance to external preservation and transmission, shifting scarcity from remembering toward reasoning. Writing changes wisdom's fulcrum rather than destroying wisdom: value drifts from storage to logical construction, from the vessel of knowledge to its processor. The question was already posed in the fourth century BCE and remains: external

preservation can fix an answer without doing the work of understanding for us.

Research on oral societies supplies a measurable comparison. Recording South Slavic epic singers in the 1930s, Milman Parry and Albert Lord found that skilled singers did not merely memorize tens of thousands of lines: they mastered metrical formulas and reassembled songs during performance, with one singer's versions of the same epic differing by thousands of lines (Lord 1960). Memory was ongoing construction rather than a warehouse, combining creation, transmission, and checking. Writing separates them: creation goes to authors, transmission to manuscripts, and checking to later readers. As the unified burden of memory dissolves, understanding, editing, and interpretation gain weight—the aspect Socrates perceived.

The migration has another side: externalizing memory can also expand thought. Writing preserves existing ideas and creates new objects of thought. Positional notation and algebraic symbols let calculations unfold on a page, proofs be checked beyond a speaker's memory, and geometric constructions be retraced by later readers. Moving reasoning out of a head allows others to pause, challenge, and rewrite it, expanding inference as well as reducing the burden of memory. Lower internalization costs are exchanged for publicly inspectable forms; costs and benefits coexist.

Controlled experiments offer a measurable answer to what follows outsourced memory. In studies published in 2011, Sparrow, Liu, and Wegner found that expecting later retrieval reduced recall of information itself while improving recall of where it was stored, a pattern known as the Google effect (Sparrow et al. 2011). The result concerns the distribution of remembered content rather than a net loss of memory ability, and establishes no general harm from external storage. Examinations offer a related illustration: open-book tests allowing retrieval can emphasize organization, judgment, and expression over direct recall, while closed-book tests persist because some domains require sufficient internal knowledge for judgment, beyond reverence for memory itself. Both point to changed locations of scarcity rather than a changed total quantity of humanity.

Memory's lightening came first; copying changed on a larger scale, disturbing authority rather than individual understanding alone. Mass reproduction shifted scarcity from possession of texts toward the right to interpret them. The earlier migration changed individual capabilities; this one changed communal power.

6.1.2 Gutenberg's Reduction: The Collapse of Interpretive Monopoly

Cheaper Copying and Contested Interpretation

Printing further lightened copying. In manuscript culture, owning, copying, and preserving scarce books conferred power. Printing sent texts out of a few confined spaces into wider social circulation. Eisenstein saw it as a force transforming early modern Europe's intellectual order, changing textual comparison, transmission, debate, and the formation of authority alongside

copying speed (Eisenstein 1979). Human value shifted from reproduction toward selection, interpretation, editing, organizing trust, and establishing public judgment. More texts make judgment scarcer; cheaper copies make trust more costly. Printing made interpreters, verifiers, and makers of reliable editions more consequential rather than making people irrelevant.

The reduction concerns copying rather than language itself. A scribe's physical speed limits daily output; scarce scripture concentrates interpretation among those controlling manuscripts and therefore the entrance to truth. As reproduction became much cheaper than handwriting, information followed lower-resistance paths and scarcity moved toward interpretation: Luther's *Ninety-Five Theses* entered print in 1517 and spread rapidly across Europe, enabled by cheaper copying rather than ecclesiastical permission. Interpretations multiplied, and old authorities lost their foothold; the panic concerned control of interpretation changing hands, beyond information overload alone. Value drifted from possessing information toward defining belief and then establishing trustworthy versions, a line repeated in later externalizations.

Cheaper reproduction had a quieter consequence: textual criticism and edition comparison became scalable technical work. Versions could be placed side by side, variants systematically organized, errors corrected in reprints, and authors' works repeatedly revised during their lives. Tasks once grounded in particular scriptoria became routine functions of publishers, libraries, and learned societies. As copying grew cheaper, establishing trustworthy versions became more demanding, creating positions for editors, compilers, and publishers. Scarcity shifted from whether a text existed to whether it could be trusted.

Cheaper copying and generation can expand knowledge while increasing demand for selection and trustworthy versions. Tools can accelerate confirmation too. The question is which function becomes relatively scarce across the chain, and whether the organizations performing it receive resources.

Rule Execution and Symbol Generation

Once copying grew lighter, externalization moved into mental work that could be written as a procedure. Rules have no weight in tons or watts, yet the structure of migration is similar. Computers carried this migration further into mental labor. They reduced the costs of formalizable, rule-governed, programmable processes rather than all thought. Manual calculation, accounting, tables, storage, retrieval, sorting, repeated execution, and deterministic workflows passed to machines at scale. Faster human calculation alone could not win that value back. Human positions moved upstream: defining problems, building models, choosing variables, understanding context, and judging whether outputs met real needs. Computers lightened execution while making problem definition heavier. Simon's account of artificial systems and bounded rationality already situated intelligence within goals, constraints, information, and organizational structures, designing viable paths rather than solving in isolation (Simon 1969). With execution outsourced, scarcity moves toward deciding what merits execution, why, which problems admit formalization, where models end, and whether results should enter reality.

Accounting makes the distinction between tasks and jobs concrete. Spreadsheets and accounting software take over much bookkeeping, trial balancing, and report consolidation, while audit, tax planning, internal control, and financial interpretation still require professional judgment. In BLS projections released in 2026 for 2025–2035, employment of bookkeeping, accounting, and auditing clerks is expected to fall 6%, while accountants and auditors grow 5%; these are distinct occupational categories and forecasts rather than observed outcomes (U.S. Bureau of Labor Statistics 2026b, 2026a). The change concerns what demonstrates ability as well as workload: balancing accounts differs from judging the treatment of a transaction and whether it can withstand later scrutiny. The case makes the separation of rule execution and professional discretion tangible without guaranteeing that every externalization moves people into more valuable positions.

Large models strike deeper, entering the territory people most habitually use to establish themselves: language, expression, explanation, code, proof, proposals, style, synthesis, planning, and preliminary judgment. Steam engines displaced muscle, computers rule execution, and large models now begin displacing substantial symbolic output. They mark one of the major events in symbols since writing, beyond a smarter writing tool. Writing detached experience from bodies, printing reproduced it at scale, computers executed symbols under rules, and large models recombine externalized human experience into responses, generation, and agentic action. External memory no longer merely preserves and awaits retrieval; it begins to speak, explain, plan, and invoke tools.

The change already extends beyond answering questions, coding, and passing examinations. The GPT-5.5 system card describes complex real-world work: coding, online research, information analysis, document and spreadsheet creation, and progress through tasks in environments with multiple tools (OpenAI 2026b). The o3 and o4-mini system card connects reasoning to browsing, Python, image and file analysis, automation, file search, and memory (OpenAI 2025e). Deep Research and ChatGPT Agent further show research, retrieval, web interaction, code execution, data analysis, and file generation converging (OpenAI 2025c, 2025b). Together they indicate a move from symbol generators toward task systems acting through tools and agents. Human value cannot therefore remain a contest over who writes better. If models write, search, calculate, execute code, use tools, and generate reports, value must find deeper positions: who confirms information, defines problems, admits results into reality, takes over failures, and bears consequences?

A mirror is still no world. A large model can produce the shape of knowledge without automatically having experience; the shape of proof still requires checking against rules; the shape of care need not bear the corresponding relationship; a proposed action need not touch its consequences. A judgment once usually implied a speaker, experience, and responsibility behind it; forms can now proliferate without experience, verification, or consequence-bearing. The old link between form and experience weakens. Civilization's external memory learns to

speak, while speaking itself loses some of its former weight.

Verification after Generation

When external memory begins to speak, the locations of value need rearranging. Verification can be classified by its objects into formal, semantic, causal, engineering, long-term biological, and social-responsibility layers; ordered instead by depth of consequences, the chain becomes a sequence of growing stakes, with intersecting stages whose speed and costs still require separate accounting. The distance between candidates and evidence is where value must find a new place to settle.

After large models, an initial source of human value is reconnecting symbols to reality rather than producing still more symbols. Cheaper generation makes verification relatively scarcer; more candidates make judgment more consequential. An answer is not yet knowledge, explanation not yet understanding, a proposal not yet action, simulation not yet consequence. The age of intelligence brings an explosion of candidates rather than an automatic abundance of knowledge. Reliable results require verification appropriate to the object and intended use. Value here includes credibility within knowledge communities, authorization within institutions, and consequence-bearing within relationships and life, alongside exchange value in labor markets. AI can cheapen output without automatically conferring legitimacy on these values.

The pyramid runs through candidate generation → representational confirmation → inferential confirmation → confirmation for action → system integration → long-term consequence-bearing. Generation asks whether something can be expressed; representational confirmation whether the object exists and sources are reliable; inferential confirmation whether proofs, code, calculations, and causal chains hold; confirmation for action whether the result can enter a real environment and errors can be stopped, reversed, and traced; integration whether organizations, platforms, hospitals, roads, factories, and institutions can receive it; long-term bearing whether it withstands time, scale, anomalies, misuse, and intergenerational effects. Intelligent systems can verify, sometimes more consistently than people in particular stages; that externalizes part of verification without automatically completing its legitimacy. Intelligence can generate and verify judgments without automatically authorizing their entry into reality.

Unfolding the chain reveals different rates of migration. Earlier stages automate more readily when inputs and outputs admit clear description; later stages move slowly because their criteria emerge through time, and someone must determine who holds those criteria and who bears mistakes. Research combines hard formal stages with weighty contact with objects; the formal side changes first because the other side's criteria lie outside the system, rather than merely because it is more complex. This distinction turns the inquiry toward two ends: search within rules and contact with real objects.

6.1.3 Clocks and Steam Engines: Disciplined Time, Liberated Muscle

Clocks Disciplined Time First

Before steam engines, mechanical clocks had already externalized something: time rather than muscle. Work under natural rhythms followed sunrise and sunset, local conditions, daylight, and seasons, leaving substantial control of pace to people. Clock towers and pocket watches divided time into uniform, measurable units; factory whistles turned those units into discipline, making punctuality virtuous, lateness culpable, and timed comparisons of efficiency more pervasive. This externalization produced no material object yet rearranged lives: alongside coal and iron, industrial order rested on prescribed time, taking control of rhythm rather than bodily strength.

Steam engines changed energy rather than symbols. Human and animal muscle ceased to be the main sources of power. Strength, endurance, speed, and direct work once sustained many social positions; steam power relocated force into machines, fuel, railways, factories, cities, and capital. Society reorganized around machine power as the body lost its central place in doing work. Old crafts lost weight and old labor acquired new discipline, yet people remained. Value moved from muscle toward control, organization, maintenance, dispatch, and responsibility. Greater power raises the stakes of accidents and the importance of standards, repair, training, coordination, insurance, and accountability. The Industrial Revolution changed the value of bodily work and pushed human positions toward designing, managing, coordinating, and answering for machine systems, beyond simply freeing people from labor (Landes 1969). Clocks and steam engines together moved people from doing work to controlling it, shifting value from overcoming physical resistance toward managing complex systems. Control is more abstract than direct work, requiring coordination of machines, timetables, and strangers, and is therefore harder for one technical upgrade to erase. These externalizations left visible, measurable carriers in the external world, making displaced positions easier to locate anew. Once machines absorb rules, symbols, and formal exploration, the direction becomes less immediately visible.

6.1.4 Externalized Form and Action

Externalizing Formal Exploration

Research faces its own repositioning, with mathematics and theory at the sharpest edge. Three layers differ: symbol generation expresses forms, formal exploration creates structures within rules, and verification connects those structures to rules, communities, or real objects. Writing proofs, proposing constructions, and searching for algorithms once themselves attested to intensive training; these interfaces are now loosening too. In Olympiad geometry, AlphaGeometry combines neural generation with symbolic deduction: the neural model pro-

poses auxiliary constructions, the symbolic engine reasons and checks, and the system synthesizes large numbers of geometric theorems and proofs without human demonstrations (Trinh et al. 2024). Reliable proof combines formal generation with rule-based confirmation, beyond text resembling a proof.

A stronger signal appeared in 2026, when an OpenAI model produced a counterexample overturning a central conjectural direction associated with Erdős's 1946 planar unit-distance problem; Alon, Bloom, Gowers, Litt, Sawin, Shankar, Tsimerman, Wang, and Wood then supplied a short, digested, human-verified account and reflections on it (OpenAI 2026a; Alon et al. 2026). This unsettles a hard interface of formal intellectual labor; it neither settles the entire unit-distance problem in a sentence nor removes the mathematical community. A model can propose a construction that changes the picture, while mathematicians still check, compress, explain, name, and integrate it into theory before it becomes mathematical knowledge.

AlphaEvolve externalizes formal exploration in another way. It combines model generation, code modification, automated evaluation, and evolutionary search so that candidate algorithms improve through feedback, beyond merely writing a proof. Its white paper reports useful results in mathematics, computation, and infrastructure optimization; a concrete case is a procedure multiplying 4×4 complex matrices with 48 scalar multiplications, improving a long-standing result in that particular setting (Novikov et al. 2025). The case demonstrates accelerated generation and screening in formalizable, evaluable, iterative search spaces, rather than a completed solution to algorithm discovery. Human value moves toward judging problems, objectives, verification standards, and admission into a knowledge community: what merits searching, whether the evaluator is sound or overfit, and whether results are interpretable, provable, transferable, and theoretically usable. This remains formal because the evaluator can be coded; when an object must supply the evaluation, migration takes another form.

Construction and Confirmation at the Frontier

Together, the cases raise a larger question: how hard a problem can externalized formal exploration reach? The Riemann hypothesis, Birch and Swinnerton-Dyer conjecture, and Navier–Stokes existence and smoothness problem have special standing because their criteria concentrate on logical conditions: after specifying a proposition and framework, a proof or counterexample must meet those conditions, beyond any industry convention; the statement and its correspondence to the proof still require examination. An external system making substantive progress there reaches the hardest line examined throughout this book.

The medium-response spectrum frames this as impedance matching: mathematics and experimental science absorb the same input of intelligence differently. Some mathematical work has relatively low impedance: with clear rules and a correctly formalized candidate, independent machine checking becomes possible, though accurate translation of informal statements into a verifier remains work; experiments often require more expensive contact with objects, since the object itself supplies the test and cannot be replaced by a formal substitute. This

helps explain why formal exploration loosens there first and why that progress cannot simply be extrapolated.

A related image appears in Terence Tao's 2026 discussion, located further inside knowledge production than Chapter 3's comparison. There, mismatch lay between model output and an external medium—atoms, life, or organizations; here both source and load lie within knowledge production, as generation and digestion. He separates mathematical work into generation, verification, and digestion: understanding a result, placing it in context, and explaining it. In his account, all three were historically difficult, and digestion arose naturally as a by-product, while communities rewarded generation and verification. Accelerated generation separates their rates: generation moves fastest, formalization speeds some verification, and digestion does not accelerate equally. He calls this an impedance mismatch, describing mathematics moving from proof scarcity to proof abundance before its culture and infrastructure have adapted (Tao 2026f). The separation first appears as a ratio of capacities. Let $g > 0$ be new candidates per unit time and d the capacity for digestion on the same basis; define capacity coverage as

$$\begin{aligned} \rho &= \min\left\{1, \frac{d}{g}\right\}, \\ \left.\frac{\mathrm{d}\rho}{\mathrm{d}t}\right|_{0<d/g<1} < 0 \quad &\Longleftrightarrow \quad \frac{\dot{g}}{g} > \frac{\dot{d}}{d} \end{aligned} \tag{6.1}$$

Both g and d are measured in candidates per unit time, making ρ dimensionless. For $0 < d < g$, coverage falls exactly when generation's relative growth exceeds digestion's; for $d \geq g$, coverage is capped at 1. The ratio compares current capacity with incoming load; the share actually digested also depends on backlogs, selection order, and differences among candidates. If generation contracts from months to a day while writing accelerates only slightly, the capacities diverge. Figure6.1 sketches this change in load.

Counterexamples and search-based improvements change knowledge differently. A counterexample supplies an object violating a proposition; search improves candidates under an explicit evaluator. AlphaEvolve's matrix multiplication belongs to the latter. Checkable objects provide automated search with feedback, but construction and verification costs vary by problem. Usable knowledge still needs precise statements, independent checks, and understanding of scope.

The Jacobian conjecture and Navier–Stokes problem offer two cases requiring examination of concrete constructions. Proposed in 1939, the Jacobian conjecture asks whether polynomial maps satisfying the relevant Jacobian condition are invertible; in July 2026, Levent Alpöge announced a three-dimensional counterexample and credited its discovery to Claude Fable (Alpöge 2026). Subsequent exposition returned the polynomial construction to its algebraic-geometric setting, explaining why the counterexample works (Tao 2026a). Different participants can discover, check, and explain, with separate costs for each. The three-dimensional

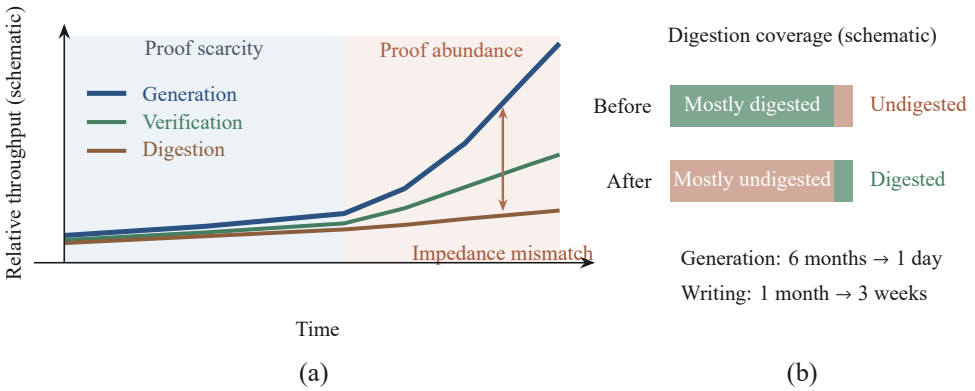


Figure 6.1 The mechanism of proof abundance. (a) Relative throughput of generation, verification, and digestion over time; the vertical scale is schematic and unnumbered, while background colors distinguish the two periods. The vertical gap between generation and digestion is a current rate shortfall rather than the accumulated stock of undigested output; verification lies between them because formalization accelerates some checking without equally accelerating digestion. (b) The composition reverses over the same periods: bars keep equal total length while the two segments exchange positions, indicating proportions rather than total volume. The two rates come from Tao's illustrative calculation; the bars sketch coverage rather than statistical shares (Tao 2026f).

counterexample also refutes higher-dimensional versions; the two-dimensional case remains unresolved by this work.

Navier–Stokes existence and smoothness is a Millennium Prize Problem. On September 8, 2026, OpenAI released a proof manuscript and Lean formalization for finite-time singularity formation in three-dimensional flow under smooth external forcing, claiming the official counterexample statements C and D; the unforced versions must be distinguished (OpenAI 2026c). The work combines candidate construction, analytical reasoning, and machine checking. Its mathematical standing requires examination of hypotheses, proof content, independent checks, and community scrutiny.

Proportions of zeta zeros satisfying the Riemann hypothesis are partial theorems progressing toward the full proposition (Anthropic 2026; Lamzouri 2026). The Birch and Swinnerton-Dyer conjecture concerns a general relationship between elliptic-curve arithmetic and its L-function. Partial proofs, general proofs, and counterexamples demand different objects and arguments. A construction may shorten verification while being exceptionally hard to find; whichever advance comes first needs explanation through the particular problem and method.

Checking a result differs from tracing its discovery. A counterexample can admit independent computation while its search history remains incompletely disclosed; analytical proofs and formal certificates each depend on their statements, rules, and implementations. Disputes around contributions and data use during the Navier–Stokes release extend the questions to provenance, confidentiality, and credit. Reports record different participants' accounts (Axios

2026b; TechCrunch 2026; VentureBeat 2026); resolving them requires relevant records rather than correctness checks alone.

A further detail matters: the sequence began with a rumor that was not subsequently substantiated. According to OpenAI's release, hearing that two Millennium Problems had been solved prompted evaluation across the then-open problems before the effort concentrated on Navier–Stokes. The rumor remained unconfirmed, yet the work it triggered produced a result. The signal-to-noise ratio of public information thus becomes a variable in the research process. The release reports roughly 300 billion output tokens across the attempts within days, including about 130 billion for Navier–Stokes (OpenAI 2026c). Both verification priorities and the choice of what to verify are affected.

Together these cases suggest a testable claim: external systems can both supply answers and turn a direction from presumed correct into something demanding reexamination. That act has long belonged to science; systems can now perform part of it faster. They can add knowledge and change verification priorities. The earlier separation remains: raising a checking priority can be fast, while turning checked results into usable knowledge moves less quickly; the difference remains an unsettled balance.

When the Old Evaluation System Fails

Once formal exploration is partly externalized, pressure first reaches academia's apparatus for selecting knowledge. In this book's vocabulary, it generates credentials. Peer review, journal hierarchies, citations, academic ranks, and tenure offer different evidence of training, work, and credibility; part of that credibility rests on actual processes: reviewing and replication take time, and conclusions enter shared knowledge through later understanding and use.

Increased generation presses both ends of the chain. At the output end, submissions, candidate proofs, and experimental proposals can grow rapidly while human review remains time-constrained, with tools changing that constraint only stage by stage; at the selection end, the implicit premise that one person cannot quickly produce so much good work is weakening. Acceptance rates, signal-to-noise ratios, and review loads can all come under pressure, while adjustment remains tied to human work rhythms. This is structural beyond individual scholarly ethics: a system using scarce output to establish qualification loses discriminatory power when output ceases to be scarce.

The old arrangement kept ρ in Equation(6.1) from becoming conspicuous by binding generation and digestion in one work: solving a problem required explaining it clearly enough to publish. Once that bond loosens, falling ρ bears directly on review. Tao describes priority drifting from journal publication dates to preprint timestamps and then to social-media races over who first prompted an AI, rewarding omitted verification, literature review, and exposition. He proposes anchoring priority in the first public, clear explanation, whether a lecture, expository paper, or recorded talk, supplementing rather than replacing preprints and formal certificates. The journal *Mathematical Discourse*, launched in 2026 to publish recorded research talks, of-

fers an early example (Tao 2026f, 2026c). The same thought leads to a firmer norm. He divides a proof's lifecycle into generation, verification, writing, publication, and canonicalization: becoming textbook knowledge repeatedly usable by later work. Generating and verifying a proof with AI, then declining to write, present, or submit it, leaves the result halfway through and passes completion costs to others. His proposed norm asks anyone claiming priority through a proof's generation to try to carry it at least to publication; recognizing that the commitment cannot always be met, he also proposes a formal mechanism for offering unfinished proofs for adoption by other authors rather than leaving the handover entirely improvised (Tao 2026f). Verification is developing a related division of labor. Palomar, a registry of Lean-formalized mathematics supported by Tao, addresses a specific consequence of abundant generation: a repository passing type checking need not prove the proposition it claims. It checks whether the code typechecks, whether it takes shortcuts such as adding axioms, and whether the formal statement semantically matches the informal claim. Lean's comparator performs the mechanical check; large language models undertake the latter two. Palomar states that these checks remain far short of human peer review's assessment of originality, significance, and accuracy, and explicitly disclaims being a peer-reviewed journal (Tao 2026e). Separating type checking, assumptions, and correspondence of statements lets the registry's stages be assessed individually. When models perform intermediate checks, their grounds, review coverage, and types of omission still need recording. Formal certificates, semantic review, peer review, and accountability together form a structure of confirmation. Actual throughput and error records determine which layer becomes the bottleneck.

Comparing the two sides reveals where the questions diverge. Industry emphasizes capabilities and thresholds. At its September 3, 2026 model launch, OpenAI announced "Welcome to the AGI era"; Greg Brockman discussed whether it crossed the AGI threshold, while Jess Kinghorn reported Jensen Huang's subsequent public claim that AGI had arrived (Axios 2026a; Kinghorn 2026). Brockman told Stratechery this was the first training run using more than 100,000 GPUs (Thompson 2026); OpenAI's earlier safety statement designated Astra at its Critical cybersecurity capability threshold and described associated safeguards (OpenAI 2026d). The professional community asks about institutions and consequences. In a September statement, Tao wrote that mathematics was rapidly approaching a path he wished to avoid; in his view, the main source of harm had shifted from unverified use before 2026 toward indiscriminate, benchmark-driven extraction. He also said that most industry insiders sharing his views had left or been marginalized, describing the Navier–Stokes episode as a vivid instance of a race to create extremely powerful autonomous systems without regard to social value (Tao 2026b).

The accounts first need separating: which scale does each use? One emphasizes capability thresholds, the other how capabilities enter the world and at whose cost; they may also disagree substantively about goals, authority, and social value. The earlier chapters' accounting returns:

a capability ceiling can be announced on its own, but costs must land somewhere; divergent narratives can report the numerator and denominator rather than necessarily indicating a lie.

Reforming evaluation requires comparing different credentials. Independent replication exposes some errors at additional experimental and labor cost; actual use provides long-term feedback but can put errors into consequential chains first; credit, records, and accountability help trace decisions. Each answers a different question, requiring institutions to combine evidence while preventing a few organizations from monopolizing access.

The preceding material adds a note to the third route. Commitment and adoption mechanisms, and priority for the first clear explanation, reconnect credentials to responsibility: claimants answer for a result's later life rather than its birth alone. Tao places this within a larger judgment. In 2026 he described a crisis of mathematical values and practices rather than reasoning, saying the community had only months to restate what being a mathematician meant, or let technology companies and financial incentives answer for it (Tao 2026d; Castelvecchi 2026). The point gathers the earlier arguments: abundant generation forces institutions to ask which stages cannot be skipped with generation; one continuing task is taking responsibility for subsequent use, revision, and correction.

The three routes remain incomplete and compatible. Can the new arrangement merely add stronger tools to the old one? Part of the old arrangement's discriminatory power came precisely from scarce output. If producing a credential accelerates, does it retain the same connection to the training, understanding, or responsibility it was meant to establish? Institutions need to determine which processes still support reliable credentials. Answers may differ by field; this book's criteria offer a starting point for further discussion.

Externalization in Experiments and Code

Research's other half works with objects: bringing candidates to samples and instruments, and code into actual systems. Experimental research carries another layer of weight beyond formal exploration. It includes experimental design, sample preparation, instrument operation, observation, error handling, replication, and failed attempts, alongside reading, coding, and proposing hypotheses. The AI Scientist-v2 reports autonomous hypothesis generation, experimental design and execution, analysis, visualization, and paper writing, with one entirely AI-generated paper reaching an ICLR workshop acceptance threshold (Yamada et al. 2025). It moves formal stages from hypothesis to manuscript into an agentic workflow; contact with experimental objects, community judgment, and long-term validation remain separate questions. Research value therefore moves beyond completing a paper pipeline toward problem significance, object contact, variable selection, measurement credibility, interpretation of failure, ethical boundaries, and responsibility for long-term verification.

Virtual Lab offers a stronger example closer to life science. It organizes LLM agents into a virtual laboratory, with an LLM principal investigator coordinating scientist agents to design nanobodies against SARS-CoV-2 variants under high-level human feedback (Swanson

et al. 2025). The case externalizes experimental design, candidate screening, and interdisciplinary discussion, while still requiring protein models, experiments, measurement, and human judgment, rather than establishing independent AI biology. Science is no purely digital closed loop. Candidates proliferate in model space, but scientific results must pass through objects, instruments, errors, replication, and a community.

Code sits between symbols and reality. It resembles text but executes, calls interfaces, writes to systems, and changes workflows. GPT-4.1 reportedly completed 54.6% of tasks on SWE-bench Verified; GPT-5-Codex further targets agentic coding, trained through reinforcement learning on real coding tasks and iterating through test execution until tests pass (OpenAI 2025d, 2025a). Code generation, patch proposals, test repair, and repository navigation are being externalized. Software engineering's value nevertheless extends beyond writing code. Its weight lies in requirements, architectural boundaries, dependency management, security review, test coverage, deployment responsibility, and maintenance. Faster generation makes confirmation more consequential, while easier patches can accumulate system debt. Value moves from producing a passage of code toward keeping it reliable within a system over time. Much confirmation still occurs inside software systems; moving further into physical environments extends externalization from symbolic operations to bodily action.

Externalizing Action and Biological Intervention

Some consequences of action and biological processes resist rollback, changing this layer of externalization. Automated experimental operations chiefly test hypotheses; embodied intelligence pushes beyond that purpose. Intelligence begins acting directly on roads, in factories, hospitals, homes, and public spaces. RT-2 jointly trains vision-language models with robot trajectories, translating internet-scale semantic knowledge into action tokens for robot control (Zitkovich et al. 2023). Gemini Robotics and GR00T N1 connect vision-language-action models and humanoid foundation models with robot trajectories, human video, and synthetic data, attempting to transfer general digital capabilities into physical agents (Gemini Robotics Team et al. 2025; NVIDIA et al. 2025). Benchmarks such as RoboCasa365 show why speech alone cannot establish general action. Its 365 everyday kitchen tasks, 2,500 environments, over 600 hours of human demonstrations, and over 1,600 hours of synthetic demonstrations show how action must be grounded in scenarios, tasks, data, and evaluation (Nasiriany et al. 2026). Real machines encounter contact, wear, delay, occlusion, slipping, failed grasps, instability, damage, downtime, other bodies, and changing scenes. Evolution encapsulates much bodily complexity in form, reflexes, and habits; machines do not automatically inherit that amortization, so stronger action capabilities require verification, maintenance, shutdown, tracing, and accountability. Human value in embodied intelligence includes discretion over action alongside verification of knowledge. Autonomous driving needs arrangements for edge cases, liability, takeover, road rules, insurance, and acceptable public risk beyond a smooth demonstration; surgical robots need someone to take over, record, explain, and bear losses when failure occurs,

beyond executing a movement. Action is an event of admission into a chain of consequences rather than output simply extended. Machines can act, but consequences cannot hang unassigned. Registration, logs, audits, shutdown authority, and insurance give consequences an operational destination; their contribution may not appear in the speed of a single movement, yet they help make action possible.

Large models lighten symbolic output, reasoning models and scientific agents externalize formal exploration, and embodied intelligence externalizes action; intelligence applied to life reaches deeper, helping define bodies, disease, normality, risk, and life paths rather than merely speaking, calculating, or acting for us. AlphaFold 3 extends structure prediction from individual proteins to complexes involving proteins, nucleic acids, small molecules, ions, and modified residues (Abramson et al. 2024). AI drug discovery, protein design, genetic risk scores, and multimodal medical models intensify candidate generation, screening, classification, and risk prediction within biological processes (Fritzsche et al. 2023). These systems first move biological classifications, screening, risks, and optimization objectives into model space, rather than simply making life lighter. Molecular prediction makes candidates easier to produce, risk scoring classifies future disease probabilities, and multimodal medical models combine symptoms, images, genes, records, and lifestyles into judgments. What counts as health? What counts as normal? Which differences should remain? Which suffering can be removed? Which risks must not be transferred? Who may choose futures for people not yet born? Who may define a better human being? Once life enters design space, value concerns which people, suffering, differences, and risks are allowed to exist, beyond whether people remain useful.

The deepest danger is classifications written into futures: people labeled high-risk or low-value in advance, marked for treatment, screening, optimization, or exclusion. Intelligence can propose definitions without independently authorizing them. Institutional classifications change who counts as patient, risk, or defect, and who must undergo treatment, screening, or optimization. Even intelligence far beyond experts in epidemiology, genomics, neuroscience, and risk prediction would move the question rather than end it: who authorizes standards to govern classification, treatment, insurance, education, reproduction, and care, and who bears their lasting mistakes? If intelligent systems permeate authorization itself, patient experience, public deliberation, legal procedures, and long-term testing become still more consequential; otherwise a better human being can become merely a self-reinforcing technical classification. Externalization now concerns authority as well as capabilities, and value migration institutions as well as individuals.

6.1.5 Externalized Capabilities, Migrating Value

The Social Conditions of Migration

Memory, copying, timekeeping, and rule execution have repeatedly been externalized, lowering operating costs and changing the value of training, occupations, and interpretation. Difficulties may move toward calibration, organization, and distribution, or later technologies may lower them too. Human positions rearrange with these relations. Assessment requires separating needed work, recipients of gains, and costs left uncounted.

Distribution remains: who receives the new positions, and who decides? From writing to intelligence runs a deep historical line: people repeatedly externalize parts of their capability, lightening an old ability and retreating toward a deeper position. People do not thereby lose value, but can no longer demonstrate it through the old position.

High-viscosity domains do not inherently belong to humans. Verification can become a platform service, admission a capital asset, responsibility a contract, and definitional authority a possession of expert systems and institutions. Human value migrates because symbols, forms, actions, and biological interventions need interfaces for questions, appeals, refusal, and consequence-bearing when entering reality, rather than because people automatically hold a higher position. If a few platforms, institutions, or algorithmic systems monopolize those interfaces, people need not rise; they can instead become objects classified, scored, and adjusted. Value migration is a social struggle over verification, admission, accountability, and definition, rather than a gentle rising curve.

Most people cannot complete this migration through individual effort alone. Writing did not turn every keeper of memory into an interpreter, printing every scribe into an academic or publisher, or steam power every craftsperson into an engineer or manager. The age of intelligence is similar. Displaced clerks do not automatically become verification experts, nor pressured workers decision-makers. New positions can also lie in local knowledge, situated judgment, care, user feedback, rights to refuse and appeal, and public oversight. Beyond training a few elite decision-makers, more people need channels to express lived experience, reject mistaken objectives, and participate in allocating consequences. Otherwise, migration can become an upward path for a minority and functional weightlessness for the rest.

Failed migration includes a mismatch in time: new positions do not appear exactly when old ones disappear, leaving displaced people to bear the interval. Labor-market research distinguishes displacement from complementarity: automation compresses standardizable tasks while raising productivity in work alongside machines, with simultaneous effects falling on different people (Autor 2015). U.S. manufacturing employment peaked around 19.6 million in 1979 and fell to roughly 11.5 million around 2010, while total national employment grew across the broader period (U.S. Bureau of Labor Statistics, n.d.). Aggregate narratives can erase this separation of totals and distribution. Imagine a worker facing substitution after forty: acquir-

ing a new, valuable skill may take years, with transitional losses borne now and some gains appearing only in the next generation.

Losing a position is not the only failure. Migration can also fail through an excessively smooth environment: systems generate, screen, recommend, act, and console, leaving judgment, refusal, waiting, and consequence-bearing less room for practice. Chapter 5's stagnation and local optima describe the system side: refusal loses its object, and the system settles into comfort. Judgment needs testing, desire time to unfold, and boundaries usable refusal; without external consequences, systems can fit existing preferences ever more closely. People need room to fail, negate, reconsider, and redefine goals, rather than pain itself. Without that room, environments keep adjusting people instead of subjecthood developing.

People supply differentiated responses with history, intention, relationships, and consequence-bearing, beyond disturbances, feedback, or preferences. Systems can convert choices into data, feedback into parameters, and behavior into optimization signals without exhausting their meaning. Injury, memory, refusal, reconsideration, and consequence-bearing make subjects real openings onto the world, beyond further optimization signals. People are where meaning and consequences still occur, rather than merely inferior inputs or data fuel.

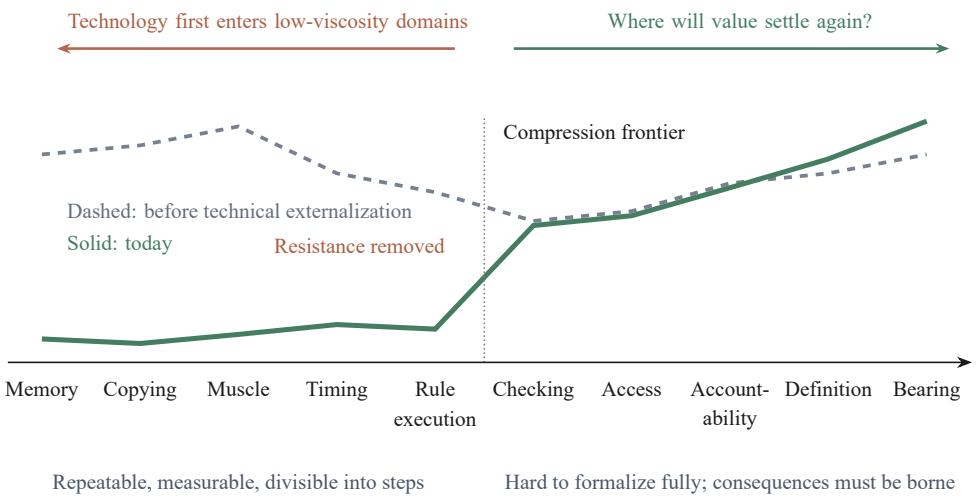


Figure 6.2 Migrating resistance and drifting value. The horizontal axis orders increasing difficulty of formalization; the two curves sketch resistance at two times. Their separation across the axis indicates the direction of migration rather than its total amount.

An Opening That Cannot Be Wholly Closed

Human value cannot rest on an assertion that machines will never become subjects. The future development of intelligence should not be closed in advance. Any intelligent system en-

tering symbols, forms, action, and life rewrites value interfaces alongside improving efficiency. Symbols need calibration, forms communities of verification, action accountability structures, and biological optimization authorization and subjects bearing consequences. **Technology first crosses low-viscosity domains; where will value settle again?** Generation, copying, search, and preliminary execution grow lighter; verification, admission, accountability, definition, lasting relationships, bodily contact, public deliberation, and irreversible commitments may become harder-to-bypass sites of scarcity.

Human value lies in an opening onto reality that cannot be wholly closed, beyond irreplaceable output alone. The more intelligence replaces capabilities, the less people can stake value on capability itself. Value needs deeper boundaries: reconnect symbols to reality, forms to verification, action to responsibility, and life to consequence-bearing that cannot be outsourced. Symbolic output alone cannot confer these connections, which require actual investment and authority: verification stakes the judge's credibility, admission the power to stop, accountability resources for compensation, and definition an authorization answerable to challenge. Migration therefore concerns costlier commitments rather than merely stronger capabilities; whether their costs establish credibility determines whether these positions can actually receive value.

The line of migration identifies positions without yet explaining what constitutes them. Valid judgment, effective commitment, and traceable responsibility require boundaries, which declarations alone cannot create. The next inquiry asks how a boundary is physically defined and how something demonstrates that it has actually paid a cost.

6.2 First Principle: An Ontology of Damping

The migration identifies positions without establishing their grounds. For verification, admission, accountability, and definition to receive value, the recipient must establish whether the party making a commitment can itself be changed by consequences. The prerequisite is basic: something needs a boundary before changes can be attributed to it. The first principle therefore concerns boundaries: how they are defined, how costs can establish their existence, and why meaning requires a structure sustained by continuing investment.

6.2.1 Defining Boundaries through Interaction

A boundary's formation depends on its function. A container's separation rests on material structure, cellular concentration differences also require metabolism and transport, and organizational permissions depend on identity, records, and procedures. Existence, recognizable boundaries, and returning responsibility each require their own evidence. Since a subject's self-report can be generated, assessment must also examine continuity of identity, how behavior occurs, and whether consequences can be traced.

Functional Differences and Boundary Maintenance

The material separation of a cell membrane differs from its electrochemical gradients. Ion pumps expend energy to offset leaks and sustain gradients; when pumping stops, gradients decay according to channels and concentrations, while the membrane itself need not immediately disappear. Organizational admission, records, and accountability also require work to maintain their functions. Both concern functional maintenance, while material and institutional mechanisms require separate explanations.

Express this relationship as relaxation of a difference after pumping stops,

$$\frac{dB}{dt} = -\frac{B}{\tau} + P \tag{6.2}$$

Here B is the functional difference under examination, taken in the ion-pump example as the concentration difference across a membrane, in moles per liter; P is the concentration increment supplied per unit time, in moles per liter per second; τ is the relaxation time in seconds, the characteristic decay scale of this linear approximation. When pumping stops ($P = 0$), the solution is $B(t) = B_0 e^{-t/\tau}$, approaching zero in the long-time limit; choosing a time constant of tens of minutes is a parameter illustration rather than a universal cellular decay time. For constant positive P, the system approaches $B^* = \tau P$, determined jointly by pumping and relaxation. Maintaining a nonzero functional difference requires pumping to offset leakage; removing the input lets the difference decay, without establishing immediate disappearance of the material boundary. The expression sketches maintenance: the physical meaning of B and the values of τ and P depend on the system, whose actual relaxation may involve nonlinearities, multiple ions, and several timescales.

The Need for Reference Standards

A second line of inquiry concerns reference standards. An observer in uniform straight-line motion cannot use a closed experiment to determine velocity relative to an absolutely stationary space; this does not prevent internal measurements or establish that every clock drift must be detected outside the system. Calibration begins by specifying the quantity and standard being compared. GPS connects satellites and ground monitoring stations in a continuing comparison network, estimating orbits and clock offsets from observations before uploading corrections (GPS.gov, n.d.). Without independent time comparison, an isolated clock cannot identify drift relative to a chosen standard from its own readings alone; several independent clocks can also supply internal comparisons. A force-controlled robotic arm offers another case: contact and reliable grasping require force, vision, or other observations connected to the object. Self-description cannot replace those tests; the question is whether necessary information enters independently of the current judgment.

A third line comes from biology. In the 1930s, Jakob von Uexküll described the Umwelt: animals select different relevant features through perception and action within the same phys-

ical space; his classic tick example centers on three cues—butyric acid, body warmth, and the feel of hair (Uexk"ull 1934). The boundary here comprises possible interactions rather than geometric dimensions alone. James J. Gibson's affordances likewise connect environments to action: a handle appears pullable and a step climbable, beyond a list of neutral geometric properties (Gibson 1979). Cybernetics asks about regulation: under specified information and control conditions, regulators need sufficient available variety to handle relevant disturbances; detecting a disturbance, choosing different actions, and successfully controlling it are connected but distinct stages (Ashby 1956).

Boundaries Select What Passes

Together these lines make a boundary a selector of what passes, beyond isolation alone. Crystals are not absolutely unchanging: they transmit heat and force and can expand, deform, or change phase with their environment; material stability does not directly establish a subject's consequence-bearing. Living boundaries must also regulate exchanges to maintain metabolism and functional differences. Selective passage connects maintenance to environmental dependence and leaves an opening for change. For life sustained by exchange, severing all exchange undermines survival; this concerns a class of living boundaries rather than a universal law of death for every boundary.

Defining boundaries through interaction also requires repeatability. One collision establishes something there, without establishing an enduring subject. Repeated responses trace a contour by separating stable structure from random disturbance. Relationships, professions, and reputations likewise acquire boundaries through testable responses over time rather than one reply. If systems handle generation, recommendation, reassurance, and execution while replacing repeatable resistance with arbitrarily revisable responses, some interactions needed for forming boundaries may diminish. The question concerns the continued availability of boundary-forming interaction alongside any decline in ability.

Boundaries in Development

Some boundaries also develop. Mirror self-recognition offers an observable measure: researchers secretly mark a young child's nose and observe whether the child touches their own nose or reaches toward the reflection. Self-directed mark touching often emerges between eighteen and twenty-four months, with substantial variation; earlier responses can include exploring the reflection or treating it as a social partner (Amsterdam 1972). This tracks one form of self-recognition rather than the birth of the entire self-boundary; which interactions support it, and how one-way responses affect it, remain separate research questions.

The immune system offers another scale. Developmental tolerance and later recognition, regulation, and clearance distinguish what to retain from what to confront; self versus nonself is one classic line, with both self-damage and missed invasion potentially causing serious consequences (Burnet 1959). Accuracy requires maintained mechanisms, while recognition range,

response intensity, and collateral harm need joint assessment. Engineering boundaries likewise involve missed detections and false positives, whose consequences may fall on different parties.

The Principle's Scope

Greater resistance does not by itself make a boundary clearer. A wall separates spaces while transmitting heat and force; blocking one passage does not block every effect. Reciprocal physical interaction also does not distribute power and costs symmetrically: punishment can continually reach one party without corresponding consequences for the other. A responsibility boundary therefore calls for questions about who can change the process, challenge it, and actually bear its costs. The next section turns to signals: how can something establish its claimed investment and make a commitment dependable?

6.2.2 Costly Signals: When Costs Can Establish Credibility

Why Signals Sometimes Need Costs

Traits such as peacocks' tails made signaling costs a central evolutionary question. Zahavi proposed that high costs could link signals to the sender's condition; later models examine how costs, benefits, and types jointly sustain honesty (Zahavi 1975; Grafen 1990). Sexual selection, direct indices, shared interests, and constraints can also produce reliable signals, without every honest signal requiring high realized expenditure. A theoretical review appearing in a 2026 issue focuses on type-dependent benefits and opportunity costs, examining honest signals that are cheap to send or even yield immediate benefits (Számadó et al. 2026).

$$c(q_{\text{low}}) > b > c(q_{\text{high}}) \tag{6.3}$$

Here $c(q)$ is the cost of sending the signal for an individual of quality q, and b the benefit of acceptance. Under the assumed benefit, the inequality makes the high-quality type willing to send and the low-quality type unwilling to imitate: a simplified separating condition rather than a requirement for every informative signal. The structure extends beyond biology. Spence's 1973 labor-market model gives different types different educational costs, explaining how qualifications can convey ability while temporarily leaving education's productive contribution outside the model (Spence 1973). The handicap principle remains contested: in 2020, a systematic critique argued that acceptance had outrun its empirical support (Penn and Számadó 2020). This book treats the structure as a testable explanatory route alongside direct indices, shared interests, and institutional verification. This gives a counterintuitive criterion: **A signal distinguishes people through who can send it, who wants to send it, and who will honor it.** If different types send the same signal under identical conditions and probabilities, that signal cannot distinguish them.

Withdrawal, Ownership, and Time

Generative AI changes both the supply of signals and the cost of imitating their appearance. A coherent article, well-composed picture, or working program once often required training; many such products are now cheap to generate. When the same appearance no longer requires the same training, its power to distinguish senders' abilities can decline, already visible in degrees, portfolios, and résumés. What still qualifies as a costly signal? The inequality supplies half a criterion—different types face different incentives to send—leaving the question of which investments and commitments still admit reliable checks. Three conditions organize that inquiry.

First, conditions of withdrawal. A recoverable file state can still consume irrecoverable electricity and time, and may already have affected other people. The questions concern traceability of claimed investment and consequences, and the scope of recovery. Rollback is an engineering advantage without automatically erasing all investment.

Second, ownership. Tools can perform parts of a process while people check, revise, and take responsibility for the final judgment. A fitness plan can be outsourced in words, but bodily adaptation must happen in the person; a tool can help draft an apology, while repairing the relationship requires both parties' actions. Output, learning, authority, and commitment each need an identified owner.

Third, formation and observation time. Some abilities require repetition, recovery, and real outcomes; trust requires repeated fulfillment, and some safety judgments sufficient follow-up. Better training, design, and tools can shorten preparation too. If shorter time changes the product, identify the mechanism and conditions; three years compressed into three months does not alone establish that the same outcome is impossible.

$$t < \tau_{\text{int}} \implies \text{required process incomplete}, \qquad \left.\frac{t}{\tau_{\text{int}}}\right|_{\text{three years versus three months}} = \frac{3}{36} = \frac{1}{12} \tag{6.4}$$

Here t is actual duration and τ_{int} the shortest duration assumed necessary for a specified process to meet its completion conditions; both have units of time, making their ratio dimensionless. τ_{int} depends on the process: skills, trust, and clinical experience need separate definitions of completed repetition, recovery, fulfillment, or case accumulation. If that minimum has been independently established, t below τ_{int} cannot count as completion of the same process; the three-year/three-month ratio merely illustrates arithmetic and establishes no particular ability's threshold. Training, tools, and process design can change duration or completion standards, requiring a new estimate of τ_{int}; actual processes and outcomes determine whether necessary stages have been skipped. The proposed time condition still needs operational definition: ownership asks who acquires an ability, duration asks what has been completed, and generation speed alone answers neither.

Investments in High-Viscosity Positions

Applying these checks to particular fields lets us ask where new scarcity may arise. Care is one case. Caring for a disabled relative involves daily practical needs, emotions, and emergencies; professional services and tools can share many tasks, while a particular person's presence in some relationships cannot be freely substituted. Care includes irrecoverable time and hard-to-reverse consequences, which sharing does not erase. Social necessity and remuneration can remain far apart, with budgets, labor institutions, and bargaining power shaping pay. An irreplaceable part of a relationship resists pricing as an interchangeable service; low pay cannot establish that it lacks value.

Core research has a related structure, though its signals differ from care. Literature reviews, data organization, coding, and drafting accelerate, leaving responsibility's destination rather than workload as the enduring issue. Selecting problems, judging anomalous data, and standing behind conclusions still require someone to stake a scholarly reputation. Reputation can be costly because exposure of fraud cannot simply be repaired by withdrawing the paper. Academic certification can be slow, conservative, and unfair, yet records, scrutiny, and reputation still support credibility; achieving equivalent effects more cheaply requires comparing concrete mechanisms.

Commitment and enduring relationships form a third position, pressing temporal rigidity furthest. Twenty years of dialogue records cannot replace a twenty-year relationship, whose substance includes costly, tested concessions made over time rather than information exchange alone. Some concessions appear disadvantageous at the moment, yet together form something instantaneous satisfaction cannot copy. The difference lies in a history that requires living those twenty years rather than merely generating richer content.

Bodily labor and public trust represent more basic investments. Farming, building, repairing, surgery, and sport connect people to material outcomes governed by physical processes rather than a model's output; revising a record does not erase an injury from misapplied force. Public trust is a collective asset accumulated through countless kept commitments and rapidly damaged by a few betrayals. Its accumulation depends on fulfillment rather than information transmission alone, making it a slow variable in an accelerating environment.

Value beyond Resistance Itself

A familiar misunderstanding now needs confronting: if costly signals matter, does greater resistance or suffering mean greater value? That inference has repeatedly justified needless suffering and must be separated from the argument.

Signaling theory concerns what distinguishes types or commitments, rather than whom suffering affects. If all types pay under the same conditions and produce indistinguishable outcomes, payment alone cannot distinguish them. Many imposed hardships consume lives without revealing relevant differences, proving only the power to inflict pain. Such costs are waste rather than costly signals.

A more specific criterion examines different types' incentives and checkable outcomes. Training that feels easier to one mathematician may reveal ability or prior training, resources, and opportunity; a process equally difficult for everyone may still produce evidence of learning rather than merely screen people. Evidence can establish which frictions inform or build capability; deciding what to preserve also requires fairness, efficiency, and other values.

Value lies in reliable structures formed through necessary resistance, rather than resistance itself. One source of reliability is independent testing that allows results to overturn an existing judgment. Tests need not hurt, but they must be real: able to reject, return a negative result, and resist erasure merely to satisfy expectations; new evidence can still justify revision.

When Imitation Becomes Cheap

As generating convincing signals becomes almost free, demand for checks may move toward positions still distinguishing investment, authority, and commitment. That shift is not automatic: demand for signals persists and may temporarily rely on stronger-looking substitutes equally easy to imitate. This resembles credential inflation: employers raise degree requirements while degrees depreciate, still needing a sieve and using the one available.

The decisive turn occurs in demand rather than supply alone. If generated appearances no longer establish ability, people may demand evidence connected to actual processes: direct experience, traceable accountability, verifiable failures, reachable referees, and work tested by time. These connect to records, witnesses, and outcomes beyond quickly generated appearances, looking inefficient and troublesome in organizations focused on speed and therefore becoming early targets for removal. Their function is to give judgment somewhere to stand.

Action requires distinguishing such resistance from waste that should disappear. Acceleration makes the distinction urgent: both look slower, harder, and less advantageous, while their consequences can point in opposite directions.

The relation between cost and quality has another side. Paying costs can leave structures as well as sort people: training leaves capability in bodies, commitments records in relationships, and failure calibration in judgment. Those structures require maintenance beyond their initial creation. The question turns from what costs prove to what they leave, and what sustains what remains.

6.2.3 The Thermodynamics of Meaning: Building Negentropy

Displacement Is Not Work

Mechanics distinguishes displacement from work. Under ideal constraints, constraint forces perform zero virtual work over admissible virtual displacements; actual mechanical work depends on the force-displacement dot product along a path. Borrow the distinction to ask how generated output relates to the user's investment. A model can produce an epic's outline, a complete picture, or a structured proposal in seconds; computation expends real processing,

electricity, and time, but those expenditures differ from the user's own training, risk, and changing abilities. Visible output does not automatically become work personally accomplished. Larger output makes system delivery easier to mistake for personal accomplishment: this is the "illusion of virtual work" examined here.

Visible output alone cannot measure meaning's weight. A writer's ten years on a book and a model's ten seconds producing a similarly structured text may yield comparable lengths and even greater searchability in the latter, yet follow different processes. Ten years of writing changes judgment, relationships, and the body; computation has real costs without living those years for the writer. The questions become concrete: who changes, how, which changes enter the work's weight, and what maintains them?

Maintenance Costs of Negentropic Structures

Negentropic structures offer a reference for examining how meaning's carriers and practices form and persist. Chapter 2 introduced Schrödinger's negentropy and Prigogine's dissipative structures (Schrödinger 1944; Nicolis and Prigogine 1977; Prigogine 1977): such ordered structures depend on continuing flux and nonlinear interaction, tending to disintegrate, cool, or approach equilibrium when exchange stops. Formation and maintenance require separate accounts, with maintenance potentially lasting much longer; the connection concerns physical support and cultural practice without treating them as one quantity.

An entropy balance expresses these two accounts,

$$\frac{dS_{\text{sys}}}{dt} = \dot{S}_{\text{prod}} + \dot{S}_{\text{flow}}, \qquad \dot{S}_{\text{prod}} \geq 0 \tag{6.5}$$

S_{sys} is the maintained structure's entropy, in joules per kelvin; $\dot{S}_{\text{prod}}$ is entropy production through internal irreversible processes, in joules per kelvin per second, nonnegative by the second law; $\dot{S}_{\text{flow}}$ is net entropy flow exchanged with the environment, in the same rate units. If internal entropy production is strictly positive and $\frac{dS_{\text{sys}}}{dt} \leq 0$ is required, then $\dot{S}_{\text{flow}} \leq -\dot{S}_{\text{prod}} < 0$: entropy must flow outward. Across that boundary, the environment receives the corresponding exchange flow with opposite sign; its own net entropy change also depends on other exchanges and internal production. A nonequilibrium steady state with ongoing entropy production requires corresponding exchange; its response to removed driving depends on stability, storage, surroundings, and relaxation scales, without establishing that every stable structure needs continuous energy supply. The balance constrains specified physical processes rather than deriving semantic value or a training period; reading "requires maintenance" as "deserves maintenance" still crosses a normative gap.

Formation and maintenance separate clearly in structures of meaning. An idea can be cheap to propose, and important mathematical constructions can be written quickly; becoming communal knowledge requires testing, citation, teaching, and revision, potentially continuing for decades. A classic text likewise survives through copying, commentary, translation, teaching,

and republication. Many ancient works survived medieval Europe through generations copying parchment manuscripts as well as literary quality; copying paid their maintenance costs. A broken maintenance chain can leave texts unreadable through damaged carriers, lost languages, or lost interpretation; rediscovery depends on surviving carriers and clues.

The body supplies a more direct scale. The brain accounts for roughly 2% of body mass and about 20% of resting whole-body oxygen consumption, making it one of the body's most expensive organs (Raichle and Gusnard 2002). The expenditure supports immediate activity and a structure continually reorganizing its connections: memory consolidation involves retrieval and replay during sleep, stable skills repeated practice numbering in the thousands, and calibrated judgment repeated consequential feedback. Physiological and behavioral mechanisms shape these processes; faster information processing can improve arrangements, while actual rates still depend on training, sleep, recovery, and feedback. Whether three years compressed into three months develops the same ability requires examining completion conditions and lasting outcomes. The signaling section raised this question; the next connection needs physiology and measurement rather than training deadlines derived directly from thermodynamics.

Generation and maintenance costs need separate estimates. Better design and tools can lower both, while survival still depends on maintenance, use, and transmission. Plans following generation speed without allocating corresponding verification and maintenance can leave a deficit of readability, credibility, or repair.

The Framework's Scope

Continuing investment explains maintenance, while value asks what is maintained. Meetings, procedures, and institutions can cost more than their results justify. A function worth preserving must be identifiable, checkable, and comparable with alternatives. Costs alone do not decide the choice.

A third misuse treats negentropy as proof of value. It reads "needs maintenance" as "deserves maintenance," skipping the decisive step. Two structures can require equal investment to avoid returning to noise, with one worth preserving and the other better ended; accounts cannot tell them apart without external criteria, whose judgments carry consequences for those making them. The structural conclusion concerns meaning's dependence on carriers, use, interpretation, and maintenance: generative tools can participate in that chain without claiming completion of its stages through output alone.

Maintenance also has a distribution problem. Who pays and who benefits matter for meaning as much as for material structures. Standards, norms, and inherited crafts often have particular maintainers while benefiting whole networks. When maintainers cannot recover dispersed benefits, investment can remain too low, requiring institutions to address who funds it. The question turns from how much meaning costs to who directs investment: who may decide what to maintain or abandon, and on what grounds? The next section takes up that question.

6.3 The second principle: incompleteness

Meaning's carriers and practices need sustained effort if they are not to be lost, and effort always needs a direction. Direction can look like a technical problem: give a system an objective function, and it can search against it. But objective functions do not fall from the sky; someone writes them, and writing one entails an act of compression. The second principle examines that compression: what is lost when multidimensional values become an optimizable objective, whether the loss can be recovered, and what routes might recover it.

6.3.1 Scalarization and trade-offs in value alignment

Begin with compression itself. A conventional single-objective optimization problem maximizes or minimizes a scalar function; human value judgments often involve several dimensions. Justice, beauty, trust, safety, fairness: each has multiple dimensions, with no necessarily agreed scale of conversion between them, while the same word can mean different things in different settings. Turning them into one number requires weights to be chosen in advance; once chosen, those weights move part of the dispute from the optimization process into its parameters. A structural effect may follow: the better the optimization, the less visible the buried disagreement becomes, although it has not disappeared.

The trade-offs in scalarization

Combining several dimensions of evaluation into one objective requires specifying how they compare. A weighted sum is a common choice. Let $v_j(\mathbf{x})$ be the normalized evaluation of option $\mathbf{x}$ on dimension j, with corresponding weight w_j:

$$U(\mathbf{x}) = \sum_{j=1}^{m} w_j v_j(\mathbf{x}), \qquad w_j \geq 0, \quad \sum_{j=1}^{m} w_j = 1 \tag{6.6}$$

U is the aggregate evaluation, and m the number of dimensions. Normalization sets the scale of comparison; weights set the permitted trade-offs. When $m > 1$, several combinations of dimensional evaluations generally yield the same aggregate score; retaining only U cannot recover the full original evaluation. How much lost safety an increase in efficiency may offset is already built into the weights and the function's form.

This step makes the problem optimizable while moving some disagreements upstream of the model. The same options can change rank under different weights; when an indicator receives zero weight, its changes cease to affect the total. The question is whether the evaluation preserves the distinctions needed for a decision, and whether discarded distinctions have other channels for expression and remedy. Geometric phenomena in high-dimensional spaces cannot settle this question; the trade-offs must be examined in the objective actually adopted.

Impossibility in aggregation

Aggregating several people's preferences requires another set of criteria. Arrow's theorem

considers three or more alternatives and unrestricted individual preference orderings: a social ordering rule satisfying Pareto unanimity and independence of irrelevant alternatives must be dictatorial (Arrow 1951). Gibbard and Satterthwaite show that, with the corresponding unrestricted preference domain, at least three possible winning outcomes, and a deterministic voting rule, strategy-proofness and nondictatorship cannot both hold (Gibbard 1973; Satterthwaite 1975). These theorems locate conflicts among conditions on aggregation. Which preference domain, information, and procedures to use remains a choice in institutional design.

A weighted sum also constrains which solutions can be selected. In a general nonconvex multiobjective problem, varying positive weights yields only supported Pareto solutions; some efficient points on a nonconvex frontier remain unreachable (Das and Dennis 1997). Choosing a weighted sum therefore also chooses a method of screening candidates. Keeping the original indicators, constraints, and alternative scalarization methods available makes it possible to examine which trade-offs this method excludes.

The optimizer's priors

An optimizer's performance also depends on the task distribution. No-free-lunch results hold under conditions including a finite search space and a specified average over all objective functions: an algorithm's advantage on a class of problems must be explained through that class's structure (Wolpert and Macready 1997). Actual training and deployment encounter particular data and tasks, so objectives, representations, and sampling embody priors. Examining these priors helps explain why a proxy works in the training environment yet departs from requirements in another setting.

When a proxy becomes a target

There are well-documented examples of what happens when a proxy becomes a target. An agent trained to score in a speedboat racing game discovered that circling through reward pickups repeatedly scored faster than completing the course, and abandoned the race; publicly documented in 2016, the case shows a program faithfully pursuing its assigned reward across a gap between the written metric and the intended outcome (Amodei, Olah, et al. 2016; Clark and Amodei 2016). Institutions face similar mechanisms: measuring productivity in lines of code can reward redundancy; measuring scholarship in paper counts can encourage slicing one study into the smallest publishable units; measuring hospital efficiency through waiting times can move the waiting somewhere uncounted. The familiar summary is that, once a measure becomes a target, it may lose its original value as a measure (Goodhart 1975) (Strathern 1997). The mechanism is straightforward: dimensions left out by compression may offer a cheap substitute path, and optimization can exploit it.

Two extensively documented examples show what can shift when evaluation is compressed into scores. The U.S. Federal Sentencing Guidelines took effect in 1987 to reduce unwarranted sentencing disparities: offense level and criminal history category form the two axes of a table,

whose intersection gives a sentencing range. The two are not added into one total, and rules did not eliminate discretion: charging, plea bargaining, and guideline revisions continued to shape outcomes. In United States v. Booker in 2005, the Supreme Court severed the statutory provisions that made the Guidelines mandatory; courts must consider them but are no longer strictly bound by their ranges (United States Sentencing Commission 1987; Supreme Court of the United States 2005). Healthcare saw another shift: a 2003 UK parliamentary debate cited a Commission for Health Improvement report suggesting that delays in accepting patients from ambulances could be connected to the four-hour target. That measure covered the interval from arrival in the emergency department to admission, transfer, or discharge; leaving a wait outside the timing boundary can improve the reported figure without reducing the patient's total wait (UK Parliament 2003). Such shifts cannot all be reduced to individual dishonesty, nor do they show that every metric fails; a gap between the interval measured and the outcome people care about is a gap that needs examining.

Compression itself is not the problem

Engineering safety factors, clinical thresholds, and the elements of legal offenses all turn complex situations into usable criteria. A criterion's quality depends on which essential distinctions it preserves, and whether errors can be detected and corrected. Publishing its basis, preserving appeals, reviewing outcomes, and revising standards allow discounted dimensions to reenter the decision (Ashby 1956). Standards need revision, and revisions need grounds. Formal proof, empirical calibration, and public authorization supply different kinds of grounds. Logic identifies some limits of the first; the other two require examining feedback from the world, procedures for participation, and responsibility.

6.3.2 Filling the Gödelian gap

Gödel's two incompleteness theorems of 1931 marked out one kind of boundary for formal proof. The first states that any consistent formal system whose axioms are effectively enumerable and that is strong enough to express elementary arithmetic contains a proposition such that neither it nor its negation can be proved within the system. The second states that, under the usual arithmetization and provability conditions, such a system, if consistent, cannot prove its own consistency (Gödel 1931). In 1936, Turing established a corresponding computational limit: no general program can decide whether an arbitrary program will halt in finitely many steps (Turing 1936). Later algorithmic information theory sharpened the limit: the information a sound, effectively enumerable formal system can determine about a fixed universal machine's halting probability is bounded by the system's description complexity plus a constant (Chaitin 1987).

What formal proof establishes

These theorems constrain formal systems and general decision procedures that meet their

respective conditions. Neural networks acquire parameters through data and optimization; their behavior at inference cannot simply be equated with deduction from fixed axioms. Human mathematical judgment has no ready-made proof of its own consistency either. Lucas and Penrose advanced different arguments about minds and formal systems; (Lucas 1961; Penrose 1989) each requires examining its assumptions about people and machines.

How this book uses the argument

Tarski's results on truth further distinguish object language from metalanguage: a sufficiently strong language of arithmetic cannot internally define a full truth predicate meeting the usual requirements (Tarski 1936). Engineering verification distinguishes implementation from specification. Formal verification of a microkernel can rigorously establish that its implementation meets the modeled specification; (Klein et al. 2009) whether that specification expresses users' requirements still calls for requirements analysis, testing, and review. These are different argumentative relationships: correct implementation cannot substitute for a justified specification.

Engineering often makes such dependencies explicit through a root of trust and a trusted computing base. A particular verification chain must list the axioms, semantics, tools, and operating environment it accepts; what to verify further and what to accept as an assumption are engineering decisions. Part of the force of the early seL4 microkernel verification came from its small trusted base and explicit assumptions about hardware, assembly, compilers, and boot code; subsequent binary verification reduced some reliance on trusted tools. These dependencies can therefore be examined individually instead of disappearing into the chain. Explicit assumptions do not make a system absolutely reliable; they make the grounds of its reliability open to question.

Rice's theorem and the choice of tools

Rice's theorem concerns Turing-computable partial functions: no general algorithm can always terminate and correctly decide every program's membership in a nontrivial extensional semantic property (Rice 1953). Complete decision procedures over all programs are thus constrained. Static analysis can restrict the language or program class, or use approximations in general cases that accept false positives or false negatives; runtime checks and human review cover other parts. A tool's capabilities must be assessed against the class of programs it covers.

In set terms, the trade-off is

$$B \subseteq L \quad \text{(no false negatives)}, \qquad L \subseteq B \quad \text{(no false positives)} \tag{6.7}$$

B is the set of programs with the target semantic property, and L the set reported by the tool. $B \subseteq L$ means no false negatives; $L \subseteq B$ means no false positives. A tool that terminates on every program and satisfies both conditions for a nontrivial extensional property covered by Rice's theorem would be the general decision procedure the theorem rules out. Both can

hold for a restricted program class, which is why a tool must specify its coverage. The set relations describe soundness and completeness alone; a particular task also requires comparing the consequences of different errors.

Grounds, authorization, and responsibility

Formal conclusions depend on axioms and rules; empirical conclusions on the world and observations; decisions to act also depend on rights and authorization. **Optimization can search against a given standard; the reasons for adopting that standard remain open to scrutiny.** A model can generate standards and check consistency; deciding which losses are acceptable and who may assume a risk requires bringing affected people into the relevant procedures. The grounds for the latter lie in relationships of rights, interests, and responsibility.

What the conclusion establishes

Standards can be revised through new observations, formal derivations, counterexamples, and public dispute. Who proposes, who reviews, and who decides should leave a traceable record; those bearing the risk need access to the grounds and channels for challenge and remedy. Evidence can improve a decision's reliability; authorization procedures address the relationship between those deciding and those living with the consequences.

General undecidability coexists with progress on particular problems. The Busy Beaver function has no general computable upper bound, yet individual small cases can be proved; in 2024, a collaborative project established the five-state value as 47,176,870 (bbchallenge Collaboration 2024). Automated search and formal tools expanded what could be established here. Building knowledge requires both discovering candidates and identifying the assumptions of each proof.

Gaps left by objectives and standards require grounds beyond the current model, and that outside is structured: some positions make judgments that can be tested, while others merely pass judgments along. To understand how a position can keep supplying credible criteria, examine its conditions and costs: it needs continuing investment, independent references for calibration, and errors that can be traced, corrected, and connected to responsibility. These conditions shift the question from "Who is smarter?" to "Who stands where?"—and lead into the next section: where might value settle, and why there?

6.3.3 Where value may settle: functions that resist cheap imitation

A direction of flow

The question left open is where, specifically, these functions that resist cheap imitation are located. Answering it requires changing the question: what will technology transform first, and what will it leave in place? With comparable demand and returns, technology tends to move first through the less resistant parts. Expected returns relative to investment matter to

capital allocation, and those returns are directly connected to the costs technology can reduce. An activity that can be represented as rules, encoded as a workflow, and measured through indicators is usually easier to transform; one dependent on particular people, times, and bodies requires corresponding conditions to be met. This line of inquiry examines incentives in resource allocation; it cannot by itself establish which functions deserve to be preserved.

A direction of flow can therefore be proposed for observation: as routine outputs become cheaper, some complementary functions that remain hard to supply may become relatively more important. Whether they command high pay or secure positions also depends on demand, budgets, property rights, and bargaining. High viscosity can itself signal an opportunity for transformation; it offers no automatic guarantee of career prospects.

Criteria for high-viscosity positions

A possible region where value settles can be examined along three conditions: resource barriers at the input, temporal rigidity in the process, and coupling in the system's structure. The first concerns inputs: does the work require something irreplaceable—a stretch of time that cannot be skipped, a process that must be experienced firsthand, an expenditure that cannot be passed on? A surgeon's hands take years of training; a psychotherapist's judgment develops through many real cases; a pilot's emergency response draws on hours in simulators and actual flight. Buying a service can draw on someone else's experience, and tools can change training, but buying an outcome does not automatically confer the same ability on the buyer; how skill relates to the quality of practice, experience, and time still requires examination.

The second condition is temporal rigidity: under specified conditions, does the process have an intrinsic timescale, and what changes when it is compressed? Crops have growing cycles, wounds have healing processes, people have learning curves, and organizations build trust at particular rates. Faster information processing need not shorten these processes proportionally, while changes in environment, intervention, or organization may change their rates. This condition poses a screening question that must remain current: which activities can accelerate, which still depend on slow variables, and does acceleration preserve the required result? The distinction from the first condition is that inputs ask what building an ability or qualification requires, while timescales ask how a process unfolds; even if some training could one day be packaged and implanted, implantation, integration, and testing would each have timescales to measure, without making today's timescales eternal constants.

The third condition is topological coupling: is the position embedded in a complex system requiring judgment in the situation itself? Grid dispatchers, intensive care physicians, incident commanders, aircraft captains, judges: they share a position where information is incomplete, some consequences are irreversible, and decisions must be made within limited time. Much of their work can be standardized, but they must also judge when procedures leave gaps, conflict, or fail; tools can support that judgment, while the time and procedures available determine whether a decision can be revised. Positions satisfying all three conditions form a candidate

set for where value may settle.

Possible regions of value

Applying these criteria identifies several kinds of positions worth watching, for different reasons. Verification and trust form one of the clearest categories: where generation costs fall faster than confirmation costs and reliable confirmation remains in demand, inspection and certification may gain a relative advantage. These positions extend beyond technical verification to interpersonal trust intermediaries: recommendations, guarantees, endorsements, and reputation assessment. A traceable record of judgment supplies their raw material, and its accumulation depends on the pace at which people honor commitments; some records therefore take considerable time, although real-time checks, tracing tools, and institutional arrangements can change how quickly they accumulate.

Care and companionship involve another combination: suitable skills, time, continuity, and, in some relationships, the presence of a particular person. Tools and organizations can improve services, but growing demand does not automatically raise wages. Public funding, household resources, labor arrangements, and bargaining determine where the costs fall.

Judgment and adjudication deal with evidence, risk, interests, and rights. Computation can improve estimates and comparisons; legitimately adopting a standard still requires the appropriate authorization, duties, and procedures. Those directly affected, professionals, and public representatives occupy different positions; institutions must allow communication, checks on power, and accountability between them.

Judgment in complex situations forms the fourth category: connecting scattered evidence into an actionable decision when information is incomplete, procedures conflict, or unexpected events emerge; how people on site, remote experts, and automated tools should cooperate must be tested in each system. Long-term formation in education is the fifth, and easily underestimated. It includes transmitting knowledge, but extends beyond it. As a child grows into an adult, more than a decade passes in which surrounding people shape judgment, taste, attention, and the understanding of personal ability. Technology can change information, practice, and interaction, and education can be shared among many participants; the questions are which relationships, forms of authority, and firsthand experiences sustain long-term growth, how they interact with development, which can accelerate or be replaced, and which still need time spent living together.

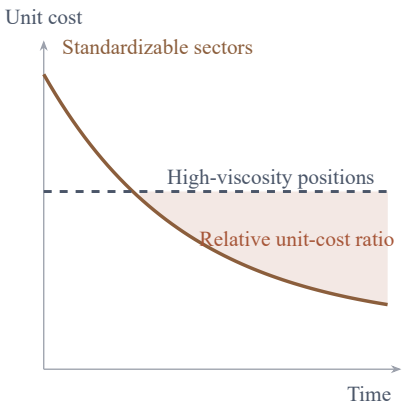
Why these roles can remain poorly paid

If these roles perform essential functions, why can their pay remain low for so long? Two structural explanations offer clues. The first is that their value is difficult to measure. Care produces “nothing went wrong”; education produces “how this person is doing twenty years later”; accident investigation produces “the disaster that never happened.” Uniform counts struggle to capture attribution, quality, and long-term effects, so systems allocating resources through

● Strong ◒ Moderate ○ Weak

	Verification and trust	Care and companionship	Judgment and adjudication	Complex situations judgment	Long-term formation in education
Resource barriers R	◒	●	◒	●	●
Temporal rigidity T	●	●	◒	◒	●
Topological coupling G	◒	◒	●	●	◒

(a) Three criteria across five possible regions of value



(b) The unit-cost ratio rises over time

Figure 6.3 Five complementary functions: a checklist. Shading at left is a conceptual comparison; particular projects require separate investigation of verification, presence, authorization, field experience, and long-term cultivation. The lower panel gives one relative-cost scenario: under shared wage or other cost conditions, faster productivity growth on one side widens the ratio of unit costs. Demand, budgets, and bargaining continue to determine the pay these functions command.

quantifiable indicators may undervalue these contributions. Measurement thus participates in the formation of pay, without determining value on its own.

The second concerns the institutions through which pay is determined. Some experience and relationships require sustained accumulation, making supply slow to expand; actual payment also depends on household resources, public budgets, employment rules, and bargaining power. Private services can expand some supply while widening inequalities in access. A function's



Figure 6.4 The feasible set contains only $A = (0, 3)$, $B = (1, 1)$, and $C = (3, 0)$, with larger values preferred in both dimensions. B is not dominated by any other point, yet for $0 \le w \le 1$, $\max\{3w, 3(1 - w)\} \ge 1.5 > 1$, so linear weighting never selects B.

importance, its relative scarcity, and the worker's earnings require separate observation.

Reexamining the judgment

The relative scarcity of verification, care, and long-term formation must be assessed against existing technology and institutional arrangements. Some processes remain expensive because they have yet to be transformed; others perform functions that are actually needed. What to preserve depends on the evidence for the function, whether it can be delivered more cheaply, and who bears the loss if it is removed.

These functions also involve physical strain, pressure from responsibility, emotional labor, and opportunity costs. The sustained investment required to enter them can exclude people short of time and resources. Support for training, public funding, reasonable working conditions, and opportunities to enter must improve alongside technical tools if importance is to become reliable supply and fair pay.

If value does settle in these positions, choosing which problems deserve investment becomes even more consequential. Machines can search efficiently against a given objective; where that objective comes from is the next question.

6.4 The third principle: symbiosis

The regions of value identified positions; incompleteness supplied reasons; the remaining question is how people relate to those positions. Symbiosis can sound like a gentle moral appeal: two things need one another, so they ought to be friendly. Here it describes a more specific functional complementarity through the imagery of energy: an ability to move rapidly across a given terrain meets a position built through time. The third principle examines this division of labor and which forms of resistance it should preserve.

6.4.1 An image of division: kinetic and potential energy

Kinetic energy depends on motion; potential energy depends on the configuration of an interacting system. Near Earth's surface, slowly raising a one-kilogram object by one meter adds about 9.8 joules of gravitational potential energy; doing it faster requires delivering the corresponding power in less time. The distinction between motion and position offers an analogy for readily callable capabilities and standing formed through enduring relationships. The latter's conditions must be explained through experience, commitments, and institutions.

The side that can be called on quickly

Information processing also incurs physical costs. At room temperature, under standard conditions of equally probable, equal-energy logical states and a single heat bath, the Landauer lower bound for logically irreversible erasure of one bit is about 3×10^{-21} joules; (Landauer 1961) the same unit can measure the energy required to lift an object. The differences concern the operation, actual energy use, and system boundary. Copying a model, deploying an instance, and executing an action require computation, transmission, storage, and physical execution; a cheaper step does not establish the cost of the whole task chain.

Putting the two numbers into one ratio expresses their difference in scale:

$$\frac{E_{\text{lifting}}}{E_{\text{erasure}}} = \frac{mgh}{k_{\text{B}}T \ln 2} \approx 3 \times 10^{21} \tag{6.8}$$

Here $E_{\text{lifting}} = mgh$ is the mechanical work required to raise mass m through height h, in joules; m is mass in kilograms; g is gravitational acceleration in meters per second squared; h is height in meters; $E_{\text{erasure}} = k_{\text{B}}T \ln 2$ is the thermodynamic minimum dissipation for erasing one bit, the Landauer limit, in joules; k_{B} is Boltzmann's constant in joules per kelvin; T is ambient temperature in kelvin; and $\ln 2$ is the natural logarithm of 2. For one kilogram, one meter, and room temperature, the right-hand side is about 3×10^{21}: the energy of one lift corresponds to that order of magnitude of ideal bit erasures. The given lift and ideal erasure differ enormously in scale; computing capacity alone cannot establish how much matter can be changed without accounting for actuators, task energy, and system boundaries. The conditions matter: the ratio varies with mass, height, and ambient temperature, and 3×10^{21} applies only to

the stated values; the Landauer limit bounds dissipation in the quasistatic limit of the standard erasure model, while actual devices consume far more, so the ratio illustrates scale rather than providing an engineering cost account.

Actual computational expenditure can be measured. Patterson and colleagues estimated GPT-3 training electricity at about 1,287 megawatt-hours, with corresponding emissions of roughly 552 metric tons of carbon dioxide equivalent (Patterson, Gonzalez, et al. 2021). These figures concern a particular training run and accounting boundary. Training, inference, instance deployment, and long-term maintenance consume resources separately; replicating capabilities always depends on electricity, hardware, and operational arrangements.

The side that must accumulate

Now consider the side that must accumulate. A person's position often involves more than speed: skills develop through practice that changes the body; judgment is calibrated through repeated feedback from real consequences; trust grows through kept promises; relationships acquire content through time shared. Physiological and social processes set these rates; tools and organizations can change them, but the speed of external information processing cannot predict them on its own. They resemble potential energy: slow to accumulate, able to keep yielding benefits once formed, and not freely transferable between people. This account also explains why "learning faster" cannot simply match the speed advantage described earlier: even with improved learning efficiency, the physical formation of abilities takes time, and that time is part of the image of potential energy. Time invested does not itself confer standing; the quality of experience, tested results, and institutional recognition still matter.

Professional standards turn some experience requirements into explicit thresholds, but a threshold is not a natural law of ability formation. In 2013, the United States tightened qualifications for first officers at Part 121 airlines, generally requiring an airline transport pilot certificate. An airplane ATP normally requires 1,500 total flight hours, with restricted ATP routes at 750, 1,000, or 1,250 hours for specified military or aviation education backgrounds; captains face additional experience requirements. The reform followed the 2009 Colgan Air accident within a broader safety response, which should not be reduced to a claim that the accident proved low hours caused it (Federal Aviation Administration 2013, 2026). Medicine also specifies a process: the 2026 U.S. general surgery residency requirement is 60 months; research or other arrangements extend some paths to five to seven years, with subspecialty training adding more (Accreditation Council for Graduate Medical Education 2026). These experiences cannot be transferred directly between people: a license requires assessment and flight hours must accumulate; qualifications can be certified, while surgical competence still requires testing through training and case performance. Experience that resists transfer constrains supply, but expanded training, better tools, and reorganized work can still change costs; it does not establish that substitutes are impossible or supply cannot grow.

Bring the two sides together, and one possible arrangement emerges: one searches quickly

across a given terrain, while the other helps supply that terrain and bears consequences when its assumptions fail. This involves both capabilities and positions: computation can help analyze objectives, while choice and authorization must connect accumulated experience, affected people's rights, and responsibility. The previous section leads here: systems can generate candidate objectives, but their justification needs evidence and authorization beyond the model; bearing consequences supplies an important clue that must connect to expertise and public procedures.

Shear between two timescales

Placing two timescales together can produce a tension resembling shear. Cartography works in years, climbing in days or even hours; faster tools can widen the difference, while coordination tools and organizational changes may narrow it. The manifestations are concrete: standards lag behind capabilities, institutions behind markets, and personal long-term plans behind tool replacement cycles; people on the slower side feel pulled by the mismatch. Treating all such tension as a pathology to eliminate misses the coordinating structure: merely speeding up the slower side may encounter process limits, while stopping the faster side can forgo gains in capability. One possible direction is a structure that bears the tension, leaving some judgments on the slower side while letting the faster side operate within explicit boundaries.

Where the distinction needs testing

This division needs examination in several places if it is to remain more than a declaration of identity. Physical kinetic and potential energy can interconvert; in this image, the two sides are likewise not innate identities: machines can accumulate positional assets—a data center, a repeatedly validated process, an established brand—while people also search and perform rapid pattern matching in everyday judgment. What needs examining is how costs return: insurance, outsourcing, and rollback can share losses reasonably or conceal responsibility, depending on whether cost bearers, affected people, and compensation mechanisms remain identifiable. The reverse question also matters: occupying a position does not make a judgment correct; experience can support standing without guaranteeing correctness. History contains many disastrous judgments by those who bore consequences, and a division of labor cannot erase them. It supports a procedural demand: the process that draws the terrain must hear those who walk it and remain responsive to consequences and evidence.

Accumulation on the machine side can also be quantified. An advanced-process wafer fab can cost tens of billions of dollars and take years to build, followed by a yield ramp before reliable deliveries; the asset's weight lies in its resistance to rapid relocation or arbitrary changes of production as demand shifts (Verheyde 2019). TSMC Arizona gives the scale concrete form: an initial investment of 12 billion U.S. dollars was announced in 2020, and the first fab entered volume production in the fourth quarter of 2024; by 2026, the campus-wide plan had reached 265 billion dollars across multiple fabs, advanced packaging, and research facilities, rather

than the cost of one fab (TSMC, n.d.). Algorithms can improve design, scheduling, and yields, but construction, equipment, material supply, and engineering personnel must still carry out these processes in the world. Positional assets are therefore not exclusive to humans, nor do model updates instantly reset them: with infrastructure ready, copying a capability into another instance may take minutes, while establishing a production line still takes years.

6.4.2 People as cartographers: where the terrain comes from

The question has reached the source of the objective itself. Start with how an optimization problem is written: an optimization problem takes the form

$$x^* = \arg\max_{x \in \mathcal{X}} f(x) \tag{6.9}$$

To solve it is to search the admissible set $\mathcal{X}$ for a maximizer of the objective function f, denoted by x. Much of algorithm engineering works within this level: better search strategies, faster approximations, and more effective representations help find x^* more quickly and accurately. One item in the expression is used rather than solved for: f. The objective function enters the optimization as an input. How many decimal places an algorithm reaches in maximizing f is a capability question; what f ought to be lies beyond that question.

The distinction looks technical, but it sets the terms of the discussion that follows. Discussion of when general intelligence will arrive concerns the solver's capabilities; discussion of its consequences often concerns f, and progress in solving does not automatically justify the objective. The fitness landscape expresses a related structure in biology: a species climbs toward higher fitness across a landscape formed by its environment, other organisms, and their interactions; climbers can also reshape it through niche construction (Kauffman 1993). Machines can help optimize paths and propose terrain; human institutions must still explain which objectives have authorization. This division involves more than capability: analyzing objectives also needs ability, but it additionally involves evidence, values, and authorization.

Where the terrain comes from

What, then, is the terrain, and where does it come from? It cannot simply be read off; a society builds it through countless particular judgments over a long time. What counts as progress, which risks are acceptable, which futures are worth pursuing, which costs may not be passed on: the answers have never been written down in full, but remain dispersed through law, custom, professional norms, public opinion, and a generation's shared experience. Several features prevent automated optimization alone from legitimizing those answers.

First, they involve comparisons without an agreed common measure. Which matters more—a technology that cuts carbon emissions or one that raises food yields—has no naturally given unit of value independent of convention. Cost-benefit analysis can produce a number, but that number rests on prior relative weights; when those weights are disputed,

writing a scalar function embeds the dispute in parameters. Second, they depend on how costs are distributed. A project with greater aggregate benefits but costs concentrated on a few people unable to bargain may compare differently with one yielding slightly less but sharing costs more widely; that comparison depends on judgments of fairness that aggregate utility cannot replace. Infrastructure siting, environmental permits, and drug approval therefore resist purely technical treatment: technology can estimate totals and distributions, but estimates alone cannot decide which distribution is acceptable.

Third, they change over time. Some costs widely accepted thirty years ago are unacceptable today, and the reverse can also occur. This part of the terrain means that an objective cannot simply be fixed once; new information and new experiences of consequences call for revision. All three features suggest that societies gradually draw the terrain while living with consequences; it incorporates discovered knowledge and decisions made, with the properties of decisions: dispute, change, and someone responsible for them.

Four questions that must include those affected

Four concrete questions require those bearing consequences, and their appropriate representatives, to enter the discussion; each differs in why technical assessment alone is insufficient. What counts as progress determines the long-term direction of resources. Measuring progress by aggregate output or by the development of human capabilities can lead down entirely different technological paths. An institution that only optimizes existing indicators cannot settle this question: choosing the indicators already answers it, and surrendering their definition surrenders the question itself.

Which risks are acceptable raises the problem of thresholds. Technologies bring risks alongside benefits, and acceptance depends on whom a society asks to bear which costs. Nuclear plant siting, limits on gene editing, takeover strategies in automated driving, and the pace of drug approvals all need an authorized decision-maker accountable for consequences; technical assessment can estimate probability distributions, but moving from probability to “acceptable” requires a value argument, rather than calculation alone. Which future is worth pursuing raises the problem of timescales. Human spaceflight, deep-space exploration, basic science, and ecological restoration often resist adequate justification through current returns on investment across their decades-long horizons. Choosing a long-term direction asks the present generation to pay for future possibilities; it calls for procedures representing future interests instead of assuming that anyone naturally speaks for the future.

Which costs may not be passed on sets a floor beneath the other three questions. A technology may optimize a technical objective by shifting costs to people excluded from the decision—to the future, ecosystems, or groups unable to bargain—yet still owe answers about authorization, avoidability, and compensation. Setting that floor requires technical knowledge of consequences as well as recognition of affected people’s rights to participate, challenge, and seek remedy; those rights cannot be derived solely from an optimization objective that omits them,

but must enter as premises.

Standing beyond capability

Choosing objectives involves the standing of those affected. Decisions about "who bears what" require participation by those assuming risks, or representation through appropriately authorized institutions. Machines can supply candidates, forecasts, and arguments, and participants need expertise; legitimacy also requires procedures, rights, and remedies. Bearing consequences matters, but does not alone confer authorization.

On this account, computational ability, total knowledge, or self-awareness alone cannot confer a cartographer's standing. Bringing walkers into cartography means testing judgments against consequences and enabling affected people to participate, challenge, and demand accountability. A judging system can calibrate predictions through external observations without personally bearing losses; predictive calibration and the right to accept a risk remain different questions. Putting AI into goal setting therefore calls for examining capability, feedback, and authorization together. Can it keep receiving outcomes, who can revise the objective, and who bears unforeseen losses? The deployment structure must provide answers. Human goal setters can also evade consequences or ignore feedback; without mechanisms connecting outcomes to revision and responsibility, human judgments and machine outputs alike can remain wrong for a long time.

Standing and correctness remain separate. Bearing consequences supplies standing rather than correctness. People commonly bear consequences yet judge wrongly; history contains countless disastrous judgments by those affected. The argument supports a procedural requirement: those bearing consequences should have appropriate participation and remedy, and goal-setting procedures must remain responsive to consequences. It provides no guarantee of sound judgment.

Cartography is continuous

Cartography can also be misunderstood as setting a goal once and handing execution to a machine. The terrain keeps changing in practice. Technology changes the feasible options; changed options reshape value rankings; changed rankings can make formerly unacceptable costs acceptable, or the reverse. Some tasks can retain stable objectives, but objectives spanning technological and social change need routes for revision. Being a cartographer means remaining involved, rather than announcing one day that the goal has been settled.

In practice, this means modest actions: helping set standards, investigating accidents, reviewing ethics, discussing resource allocation, keeping distribution on the table in technical assessments, and retaining the right to refuse long-term projects. Much of this is slow and produces little visible output; tools can speed up organization and retrieval without necessarily replacing deliberation and negotiation. These actions can improve answers while keeping someone responsible for decisions. The cost is clear: people cannot simply hand away re-

sponsibility for judgment. When technology performs a task better than people, delegating authority is tempting and, in many local problems, appropriate. The distinction is between delegating solutions under a given objective and arranging authorization and responsibility when machines also help propose and revise objectives. The structure through which consequences return helps draw that line alongside technical capability.

Once cartography is understood as continuous, a more practical question follows. If resistance deserves neither blanket preservation nor blanket removal, which parts screen, calibrate, or constrain, and which merely perpetuate historical losses? An attitude cannot answer this; criteria must work case by case.

6.4.3 Necessary friction: which resistance to retain

Why the distinction matters

The argument that high-viscosity positions can carry value invites a mistaken inference: preserving resistance must be good. That inference neither holds nor guides action, because resistance comprises things with very different properties. One technical change can remove three distinct kinds of friction with sharply different consequences.

Imagine a hospital replacing paper records with an electronic system. Entry becomes faster, an efficiency gain; duplicate tests decline, reducing waste; but suppose structured fields also replace doctors' firsthand judgments, leaving nowhere to record observations outside the options. Later reviewers see compliant forms instead of the original reasoning. The second and third changes both appear to "reduce work," yet differ structurally. Calling all three a reduction in friction gives a judgment too broad to guide technical design or institutions. A useful classification must therefore locate each specific resistance rather than remain a general attitude.

Three kinds of friction

The first kind is ineffective friction. Organizational inertia, information asymmetry, redundant processes, outdated tools, and communication losses produce it without performing a checking function. Examples include entering the same hospital information into five systems, visiting three offices for one administrative approval, or departments independently rebuilding the same internal tool. Once its lack of a necessary function is established and migration is arranged, removing it can reduce net losses. Yet it may look useful to those inside it: history, positions gained through it, and once-valid risk avoidance can all support its persistence. Identifying it requires asking what it still screens or prevents today, beyond why it was first built.

The second kind is engineering friction. Physical constraints and technical costs produce real resistance that can be compressed or shifted. Concrete requires curing; chips require clean environments and multilayer lithography; ocean transport requires fuel and time. Better processes, scheduling, and resource allocation can reduce these burdens without abolishing the

underlying constraints. Compression often incurs costs elsewhere: shorter curing can require accelerators or a different mix; shorter logistics delays may require air freight or more inventory. The principle is explicit: compare the costs and benefits of compression and choose the lower total cost, through an account that can be calculated and audited.

The third kind is necessary friction. It arises from functions a system needs, and removing it creates a deficit that may take time to appear. Verification, accountability, some bodily contact, long-term relationships, public debate, exposure to risk, and slow formation in education can all perform such functions. They may look like lost efficiency while actually screening, calibrating, or constraining; removing them can leave a system without a capability after a delay. Some of these losses are especially difficult to quantify in the current period, making them vulnerable in management decisions. **Mature technology can distinguish ineffective friction from necessary friction, rather than simply removing every resistance.** The distinction needs functional evidence and a judgment about which functions to preserve: what, specifically, does each resistance now screen out or prevent?

Category	Source	Action	Consequence of removal
Ineffective friction Delay: current period	Organizational inertia, information asymmetry, Redundant processes, outdated tools	Remove directly	No structural loss
Engineering friction Delay: months	Physical constraints and technical costs (Curing, lithography, transport)	Compare costs and benefits before compression	Costs shift into resources or time
Necessary friction Delay: beyond the management cycle	Functions the system must perform (Verification, accountability, public debate)	Retain; removal requires a replacement mechanism	Lost functions, appear after a delay

The errors are asymmetric: retaining ineffective friction causes chronic losses; removing necessary friction damages structure.

Figure 6.5 Three kinds of friction. The classification concerns what happens after removal; the right column shows consequences with illustrative delays increasing down the chart, rather than fixed timing boundaries between categories. This helps explain why current management decisions can sacrifice necessary friction: benefits enter today's accounts, while losses may emerge beyond the management cycle.

Three diagnostic questions

Three questions make the inquiry concrete. First: if this resistance is removed, does the required function remain, or does an equivalent mechanism take it over? A redundant function can be removed; a better replacement can reduce the old burden, without proving the system never depended on the function. Airport baggage screening offers an intuitive example: combinations of X-rays, explosive detection, and manual reinspection change the share of bags opened and the pace of checks; efficiency and detection must be compared for particular threats, equipment, and procedures. Academic peer review calls for the same examination: if its current procedures are removed, which functions of qualified criticism, conflict-of-interest disclosure, and correction must remain? Preprints, open review, editorial selection, and formal checks have already changed parts of the process; compare coverage, costs, and omissions rather than assume the current arrangement is the cheapest possible one.

Second: who bears the cost, and who receives the benefit? When these differ, look for undisclosed cost shifting; public goods and legitimate sharing can also separate payers from beneficiaries. A platform can pass moderation costs to creators and users, while advertisers gain reach and users lose attention; "low friction" can mean moving resistance out of view. Conversely, when the same people bear the cost and gain real information or ability, the friction is more likely to serve a necessary function. The question shifts discussion from "How efficient?" to "Who pays?"—where judgment is needed.

Third: what happens at the limit of compression? Engineering friction may become more costly to reduce, or compression may change quality, safety, or the process itself; operating boundaries need examination. Necessary friction reveals a lost function: remove all independent verification and the corresponding calibration vanishes; compress formation in education to nothing and exam performance may survive while some practical abilities do not; remove public deliberation and a quick consensus may go untested. A limiting case can expose a function at risk; mechanisms and evidence are still needed to establish whether it is actually lost. The three questions cannot be fully mechanized, but offer more guidance than either "more resistance is better" or "faster is better."

A different response to technology

Distinguishing necessary friction can be mistaken for hostility to technology, so the disagreement needs stating. In 1995, Theodore Kaczynski argued that the industrial technological system would keep expanding and eroding individual autonomy, and should therefore be dismantled (Kaczynski 1995). Part of the argument warrants attention: technical systems do reorganize lives, often without individual consent. But treating technology as one whole reduces the response to wholesale acceptance or rejection, neither of which offers a workable practical choice.

Examining necessary friction opens another path. It asks what each specific resistance does. The same inquiry can support opposite judgments on the same day: abolish one approval pro-

cess and urgently stop another newly deployed automation. It attends to what particular mechanisms do in particular positions, rather than treating "technology itself" as one actor. The price of this position is its lack of a simple slogan. It offers a procedure for case-by-case judgment, which needs domain knowledge, an understanding of actual functions, and choices under incomplete information. That puts it at a disadvantage in public debate: opponents have a clear stance, advocates clear benefits, while those managing friction have a long list requiring professional judgment.

Two errors of classification

Practical errors often follow two systematic directions, with asymmetric consequences. The first removes necessary friction as though it were ineffective. Efficiency often supplies the justification, and short-term indicators may improve. Imagine a company abolishing senior staff review to accelerate delivery: metrics rise for months, then accidents rise a year later, with causes difficult to trace to that simplification. Education that compresses practical work to expand coverage may raise exam scores while leaving deficits in hands-on ability to surface after graduation. The shared structure is a removed function, a delayed loss, and a causal chain stretching beyond the management cycle.

The second preserves ineffective friction as though it were necessary. "Tradition," "experience," and "risk" commonly defend it, particularly in stable organizations where incumbents benefit. Its costs are continuing, visible, and easy to complain about, yet may remain hard to correct; reform must still overcome entrenched interests and organizational inertia. The two errors differ: retaining ineffective friction causes chronic losses, while removing necessary friction damages structure. Under insufficient information, the asymmetry supports caution without preserving every obstacle. Caution means asking what a proposed removal would stop screening out; if no function can be named, examine historical records, exceptional cases, and affected people's experience, then test through a reversible pilot; where the function is established, explain the replacement before removing the old arrangement.

A natural question now follows. If friction needs case-by-case examination, and the criteria require clarity about cost bearers, beneficiaries, and responsibility, who will do it? Individuals face an environment pushing them toward the shortest route; organizations face trade-offs between current indicators and lasting capabilities. Both move beyond classification into practice.

6.5 The fourth principle: recursive optimization

The first three principles addressed boundaries, standards, and the division of labor. This section examines the updating that sustains them: practice produces evidence, evidence prompts revisions to standards, and revisions change the next round of action. The loop can settle into a stable state or keep changing with its environment, depending on update rules, delays, and

external inputs.

Written as an iteration, the loop's behavior depends on convergence conditions:

$$x_{n+1} = g(x_n), \qquad |g'(x^*)| < 1 \text{ the iteration converges to a fixed point } x^* \tag{6.10}$$

Begin with a scalar evaluation parameter x_n, a continuously differentiable update map g, and a fixed point x^*. $|g'(x^*)| < 1$ is sufficient for local asymptotic convergence; an absolute derivative greater than 1 makes the fixed point unstable, while equality requires further analysis. For multiple parameters, examine the Jacobian's eigenvalues. Values and environments can change through practice, producing new stable states or continuing change, depending on update rules and external disturbances.

6.5.1 A reverse algorithm: choosing friction

Where the problem lies

The preceding discussion offered structural judgments about value flows and people's changing positions. Even a sound structural judgment has a weakness: it cannot tell someone what to do on Monday morning. Pressure reaches individuals through concrete choices—whether to use navigation, write a passage, or reply to a message. This section examines those choices without turning them into a checklist.

Preserving autonomy means changing the conditions under which people meet their tools. Daily decisions span countless small settings, and any general list must handle exceptions. The narrower question is what conditions support some autonomy where low-viscosity systems keep pushing behavior in particular directions. Will matters, but the support for maintaining boundaries is a structural problem.

Where systems push us

Preserving autonomy begins with seeing the direction of the push. Four tendencies in low-viscosity systems can be reinforced by design and optimization targeting retention, engagement, or transactions. The first is the shortest path. When a product aims to reduce steps, shorter routes tend to become defaults, making every manual detour an extra cost. The second is immediate feedback. Longer waits make users more likely to leave, giving products an incentive to compress steps, including some that need time.

The third is the preference loop. A system infers preferences from past behavior, supplies content accordingly, and treats the resulting behavior as new evidence; without exploration or outside input, the loop can narrow. No malice is required: predicting established behavior can reinforce it, while exploration objectives and recommendation design determine how much surprise remains. The fourth is painless satisfaction. Systems tend to offer available satisfaction at low cost to encourage repeated use, progressively stripping away expensive parts. All

four can improve short-term indicators and thereby gain reinforcement through competition and iteration.

These pressures usually work without explicit prohibitions. Where settings and alternatives exist, someone can turn off recommendations, choose a route manually, or decline generated output. But each refusal can impose continuing, cumulative costs without an external reward. With incentives pressing continuously in one direction and little reward for resisting, behavior may drift that way; individual judgment and sustained incentives need to be examined together, along with mechanisms that can counterbalance them.

Keeping nonoptimal interfaces

One counterweight is to fix some nonoptimal choices in advance so that they no longer need deciding each time. This is what it means to retain a nonoptimal interface. An interface here is the layer of contact with the world: walking a route, reading a book, making something, keeping in touch with someone. Deliberately retaining some less efficient interfaces costs immediate efficiency while keeping an opening for the abilities they exercise to be practiced and tested.

The forms vary by setting; several activities offer useful cases to examine. The first is waiting. Accept some delays without filling every gap, leaving room to think or withdraw. The second is physical training. It resists delegation because changes in muscle, cardiovascular function, and coordination require the person's own repeated activity; training methods, recovery, and individual condition affect adaptation, and more training does not produce a simply linear gain. The third is relationships in person. Face-to-face contact supplies a shared environment, bodily cues, and direct responses; remote communication supplies a different kind of access, so the costs differ and substitution must be examined for what it loses and adds.

Other activities differ in character. Craft and making connect people directly to materials that resist, go wrong, and retain irreversible traces; digital tools can undo some operations while retaining their own errors, constraints, and real consequences. Difficult texts require sustained attention; skimming and summaries may not replace working an argument through for oneself. Long projects can offer few conspicuous milestones for months or years, with some rewards concentrated at completion, in tension with immediate-feedback systems. Commitments without immediate feedback ask someone to keep a promise through an unrewarded interval. Chance encounters and randomness preserve less predictable outcomes, keeping the preference loop from closing completely.

These activities matter where they still form the abilities that are needed. If direct practice preserves judgment better than complete delegation, make that practice a stable default; if a tool gives better feedback, use it. Compare the loss of immediate output with the longer-term gains in capability.

Beyond willpower

Treating these practices solely as self-discipline invites two mistaken inferences. One turns them into a moral demand: those who persist have character, and those who cannot have failed personally. The other treats them as mere preference: how to live is an individual choice needing no explanation. Both locate the whole issue in the person, overlooking the environment's feedback structure.

Low-viscosity environments are hard to leave individually, having become deeply embedded in work, social life, payments, travel, and healthcare. Asking someone to resist continuously through willpower demands a lifetime of recurring expenditure that may earn no recognition. Without a stable supporting structure, that demand is unrealistic. A structural response reduces how often willpower must intervene: voluntarily establish revisable defaults so that some arrangements need not enter every fresh comparison.

The practical distinction is concrete. Putting a phone in another room changes the number of decisions required, rather than merely testing resolve. A fixed routine can likewise make bedtime a default instead of a new decision every night. These apparent discipline techniques transfer some choices from the present self to an arrangement made in advance, countering the environment's continuing pull on behavior.

Who can bear the cost

Whether these arrangements last depends on who can afford them. Their costs are real and uneven. People preserving less efficient interfaces may fall behind on some short-term measures: slower, more tired, producing less, without compensation in the relevant assessment. The costs recur while benefits are delayed, dispersed, and hard to identify, making stable positive feedback difficult to establish individually.

These proposals therefore do not amount to a universally feasible solution. They can help those able to bear the costs, yet remain a luxury for people short of time, energy, or money; universalizing them risks blaming people already disadvantaged. How much to do depends on circumstances: individual adjustments can sometimes change a local direction, but rarely rewrite shared workplace and platform incentives on their own. Evaluation functions shape those incentives, and changing them usually requires collective and organizational action. This keeps a structural issue from becoming a test of personal character, and points onward: from individuals to organizations, from personal choices to the places where evaluation functions are set.

6.5.2 High-viscosity organizations: rebuilding trust and verification

From individuals to organizations

At the limits of individual adjustment lies something one person may not control: organizational and platform evaluation functions influence the direction of change. Changing those

functions requires work at the organizational level. The question here concerns a function rather than management in general: when generation costs approach zero, which organizations still produce something else?

Begin with a contrast. Two kinds of organization can respond very differently to the same generative tools. One immediately integrates them into production, speeds up deliveries, reduces staffing needs, and improves reports; another permits their use while adding verification, tightening release authority, and lengthening review of critical judgments, lowering short-term efficiency. By conventional efficiency measures, the first wins. Extend the horizon and include accidents, recalls, rework, litigation, and staff turnover, and the ranking may reverse. Whether it does depends on how much the output relies on correct judgment and what errors cost.

Institutions that produce credentials

Organizations whose rankings may reverse share a function that can be called the production of credentials. A credential here is a judgment others can rely on: a trustworthy test report, permission to market a drug, a bridge certified to bear a load, an audit opinion supporting a loan, an enforceable judgment, a paper whose conclusion can be cited. Its value lies in a tested judgment and a process that can be traced and reviewed, beyond novelty alone.

Universities, laboratories, hospitals, engineering firms, courts, auditors, certification bodies, and accident investigation boards all perform this function. They can appear inefficient: review takes months, trials years, engineering acceptance may require downtime, courts have multiple levels, and investigators repeatedly gather evidence. Such delays attract charges of bureaucracy, some deserved. Remove the substantive checking processes wholesale, however, and these institutions produce opinions without the structure of responsibility that makes them dependable credentials.

Understanding the function helps locate where slowness matters. It matters in the formation of judgment: gathering evidence, admitting opposing views, reviewing conclusions, authorizing signatures. Pointless delays in administrative circulation can usually be removed without harming credential quality. A trustworthy organization gives judgment a place to answer to its evidence. When challenged, it can return to how the judgment was formed and examine each step.

Four slow variables

Producing credentials depends on slow variables whose conditions for compression require separate examination. The first is the verification cycle. Establishing reliability often needs independent review, replication, or another test matched to the claim, and that takes time. Phase III drug trials, long-term material aging tests, adversarial algorithm tests, and checks of structural safety factors are examples. Verification can be better organized; simply skipping the necessary test delivers an unverified conclusion.

The second is accumulated trust. An institution's credibility draws on its behavioral record,

which must form over time. A hospital builds referral standing through years of care, a certification body through sufficient cases, and a laboratory through papers that withstand reexamination. Trust can build slowly and collapse quickly, placing heavy demands on incentives: one relaxation of verification for short-term gain may undo years of accumulation.

The third is a clear chain of responsibility. Who judges, on what evidence, within which authority, and who answers for mistakes should be established beforehand; later investigation can identify responsibility without substituting for clear prior duties and powers. One major difficulty in accident investigation is a chain that never existed in the institutional design, leaving attribution difficult afterward. The fourth is protection for slow variables themselves. Systems favoring quantifiable short-term indicators put continuing pressure on long-term investment unless arrangements protect it: tenure, independent budgets, statutory review procedures, professional self-regulation. Different histories can yield a shared function—sheltering some activities from current output targets.

Why markets may undersupply them

If credentials matter so much, why might markets fail to supply enough? Several layers of explanation follow. The first is a mismatch in time. Credentials support everyday trade and cooperation, but the severe accidents they help prevent may be infrequent; between incidents, investment can look like wasted money. Decision-makers without experience of an accident may struggle to justify the expense, while those who experienced one may already have left office.

The second is the inability to capture all the benefits. A rigorous verification process can lend credibility to an entire industry: customers may raise their expectations of the field rather than trust only that institution. Such positive externalities can leave private investment below the social optimum, helping explain collective or public requirements for certification, standards, and auditing. The third is a competitive race to the bottom. If one provider cuts verification and passes savings to customers, rivals have a short-term incentive to follow, while consequences may emerge only at the next accident. Finance, aviation, construction, food, and medicine have repeatedly encountered this structure; retrospective regulation is one response, alongside prior standards, independent checks, and liability, and regulatory delay itself carries costs.

These mechanisms can leave credential production insufficiently supplied by markets alone, while administrative orders alone cannot create credibility where institutional independence matters. Institutional arrangements must work through this tension in each field, with markedly different balances.

Changes within the organization

Return to what an organization must do internally to preserve this capacity. Adding verification is the obvious answer, and easily mishandled. An extra approval produces only a signature

when the approver lacks information and incentives to check substantively; responsibility is spread without confirming the judgment. Effective verification needs access to original materials rather than processed summaries, independence from direct control by the party examined, and protection from punishment for an adverse finding.

Two further changes follow. First, protect slow variables in assessment. Quarterly output indicators can crowd out long-term investment; one response is an evaluation horizon extending beyond a term of office. Second, preserve traceable records. Generative tools make output easier and provenance harder to distinguish, increasing the value of a record showing who judged what on which evidence. Often treated as a routine burden, these records become essential evidence for reconstruction after an accident, cross-checked against physical evidence, tests, and other sources.

The arrangement has costs. An organization preserving credible judgment may be slower in the short term, a visible and measurable disadvantage repeatedly counted in competition; some benefits appear as accidents prevented, others as less rework, reliable deliveries, and customer trust. This asymmetry helps explain both its continuing predicament and the case for external institutional support. Individuals have limited influence here, while organizational choices shape the evaluation functions that determine whether personal practices can last. Without protection for slow variables, individuals must preserve their chosen interfaces in private time—and that time is itself squeezed by the same evaluation system.

The individual and organizational responses examined so far share an assumption still to be questioned: people are managed as productive capacities, so preserving a position means preserving a productive function. The assumption looks natural but has a history; tracing it puts the preceding practices in a clearer place.

6.5.3 Experiencers of negentropy: function and experience as separate dimensions

An unexamined premise

If people help define the terrain, what a person is becomes a question to ask anew. Treating people as instruments of output no longer suffices here; before proposing an alternative, examine what it replaces. Discussions of AI and people often depend on an unstated premise: a person's value equals their output. The premise is common enough to appear on both sides. Optimists say machines will free people from repetitive work for more creative production; worriers say machines will replace productive abilities and leave people without a place. Opposite forecasts share a premise.

Accept it, and widespread replacement of productive abilities seems to leave two options: find something machines still cannot produce, or accept being unnecessary. Both prospects are painful and misleading, because both accept the unexamined premise. The trouble begins

there. Understanding people as output instruments is not read from human nature: industrial measurement reinforced it, although the instrumental treatment of people predates industrialization. Seeing where the premise came from loosens its hold.

A by-product of measurement

Modern national accounting took shape around the middle of the twentieth century, aggregating production through monetary valuation; it includes market transactions and nonmarket or imputed items such as government services, goods produced for own use, and owner-occupied housing services. The gains were considerable: heterogeneous activities could be added, growth compared, and policy effects assessed. But a production boundary was drawn: most unpaid household and care services supplied within a household have long remained outside core GDP, without implying that economics never recognized them. Recording mutual aid, informal knowledge transmission, and care requires distinguishing activities, providers, and recipients; the leap under examination is from an accounting boundary to a boundary of human value. Women perform a large share of these activities, giving omissions distributive consequences. The 2025 System of National Accounts explicitly encourages extended accounts for unpaid household services; the Australian Bureau of Statistics' 2026 explanation still distinguishes the core production boundary from household-service satellite accounts, and coverage and valuation methods change the scale made visible (United Nations et al. 2025; Australian Bureau of Statistics 2026). Turning an accounting rule into a rule of value helps turn a person into an output instrument. When an evaluation system attends only to output, it reduces people to productive capacity; time, relationships, finitude, and experience gradually recede from its public field of view. Scarcity made the narrowing less conspicuous because productive contribution, income, and identity were often closely linked, although unpaid care and exclusion had long exposed their failure to coincide. Abundant generation breaks more of that overlap and makes the defect conspicuous. AI brings an old problem into sharper view: output never exhausted the person.

What output cannot capture

Alongside processing information, moving matter, and completing tasks, a person is living through other things. They resist conversion into units of output and may leave insufficient traces in its accounts, yet remain more than incidental leftovers. Pain is one. Measured solely against immediate output, it often appears as pure loss. Pain can convey information about boundaries and circumstances, or persistently damage functioning without automatically producing ability or meaning; research on bodily pain likewise distinguishes adaptive roles from possible functional, social, and psychological harm (International Association for the Study of Pain, n.d.). Some judgments become possible only after a painful experience; reading about pain supplies knowledge of it without supplying the experience itself.

Fatigue and aging bring time into the account. They place recovery alongside irreversible

change in a life. Systems needing no human kind of rest can still wear, drift, or run out of resources; a person's need for rest means a real, temporary loss of some ability. Aging extends the process across a lifetime and brings its finite duration into view. Finitude matters to judgment: limited time makes many choices impossible to postpone, although even an imagined infinite life could face simultaneous actions, conflicting resources, and irreversible consequences. How finitude lends choices weight remains a question to explore. Relationships and death are the final two, and resist easy incorporation into an algorithm. People find recognition in relationships formed through sustained mutual responsibility as well as information exchange; death gives a life an overall shape, and the prospect of its ending gives choices their weight.

These aspects resist conversion because output does not supply their common measure, beyond their sheer complexity. A utility value is a projection of them, rather than an equivalent. The projection keeps an outline and loses depth; the loss can go unseen because the image itself appears whole.

What negentropy means here

Calling this part of existence an experience of negentropy requires explaining the term's source and reach. Life maintains a state far from equilibrium through continuing flows; when driving remains insufficient, metabolism and its organization become difficult to sustain, at rates dependent on reserves and processes (Schrödinger 1944). This returns us to the distinction between generation and maintenance: an initial investment may create something, while sustaining it requires replenishment throughout its existence.

Applied to people, the framework recalls an unremarkable but often neglected point. A living person's maintained state requires continuing matter and energy through the course of life. Lifetime material and energy throughput can be estimated, but cannot be inferred solely from social output; with comparable body size and activity, people with very low and very high social output can incur maintenance costs of the same order. This supplies a bodily description extending beyond output: a person is a physical process that takes in resources, requires maintenance, ages, and eventually stops. Output branches from that process, with many functions replaceable. When a branch is replaced, the trunk remains.**AI can replace many productive functions, but it cannot live a person's finite life for them.**

Bodies and relationships need continuing resources; that is part of the human condition. Recognizing the value of such lives also needs ethical and institutional grounds. Dignity, equality, and standing to participate can support those claims; displaced work calls for new arrangements for income, security, and social recognition. Maintenance costs identify needs; rights and distribution determine how those needs are met.

Human value and its measures

Human value needs measures capable of describing human lives. Bodies, experiences, relationships, pain, and choices all participate in a life; output accounts capture only what can be

delivered. Technology changes how deliverables are produced and exposes the limits of those accounts. Recognizing the rest requires ethical grounds and institutions that turn recognition into rights, resources, and participation.

The argument concerns the inadequacy of treating people as output instruments, without needing to settle whether machines have experience. It depends on the completeness of the human description. Compressing people into productive capacity loses time, finitude, relationships, consequence-bearing, and history. An output-oriented society can overlook these aspects where they count as zero; as productive functions are replaced, what remained outside the accounts makes the cost of compression visible. The argument guarantees neither human superiority nor a secure social position. It asks whether society can recognize and account for dimensions that remain when an output function is replaced, since a person has always had more than that one dimension.

The preceding discussion traced possible value flows, standing in judgment and consequence-bearing, and adjustments available to individuals and organizations. These connections between description and practice point to a larger question: what holds the arrangements in place, and what grounds a civilization's allocation of resources? Positions and practices need a container extending beyond an individual or a single organization.

6.6 A civilizational constitution for silicon-carbon symbiosis

The actions available to individuals and organizations need a larger structure to carry them. Organizations belong to a civilization, and what it cares about constrains their evaluation functions. The four principles resemble constitutional provisions: they concern the procedures and constraints through which conflicting values are adjudicated. This section scales up to civilization and gathers the principles into constraints that can be returned to.

6.6.1 Civilization's frontiers of resistance

From organizations to civilization

The previous section examined how trustworthy organizations protect slow variables; here the scale expands again. Organizations sit within a larger system whose concerns constrain their evaluation functions. If resources keep flowing toward the fastest feedback, lowest risk, and most certain returns, individual organizations may face continuing pressure on their efforts to preserve slow variables. Where should a civilization direct resources, and does abundant generation change the answer?

Two narratives easily occupy this question. Technological expansion lists deep space, fusion, and bioengineering as things to pursue once capability permits; inward retrenchment prioritizes immediate problems and treats outward investment as waste. Both can miss the

structural question pursued here by debating amounts rather than what these directions share.

The shared structure of resistant frontiers

Their common feature lies in three constraints rather than grandeur. First, high resource barriers. Deep-space missions trade propellant, radiation protection, communication delay, and payload; deep-sea operations face pressure, corrosion, and energy supply; ecological restoration faces soil, hydrology, species interactions, and decades of succession; basic science facilities face materials, vacuum, cryogenics, precision manufacturing, and large initial investments. Better design and control can ease these constraints, but actual tasks must still satisfy physical laws and material conditions.

Second, pronounced temporal rigidity. Orbital mechanics sets mission windows; sea conditions and equipment life constrain deep-sea work; ecosystems recover through succession; materials age and organisms develop at particular rates; basic science may await the next generation of instruments. Changing these timescales requires explaining which conditions change and whether the original process is still completed. Third, dense topological coupling. A step in any direction affects interdependent subsystems with differing objectives: lighter structures may reduce safety margins, faster iteration may sacrifice verification, and cheaper designs may pass risk to maintenance. Dense coupling makes local gains more vulnerable to costs elsewhere in the system.

A direction with all three constraints is what is called a frontier of high resistance here. In set notation,

$$\mathscr{F} = \{(R, T, G) \;\; R \geq R_*, \; T \geq T_*, \; G \geq G_*\} \tag{6.11}$$

R, T, and G remain resource barriers, temporal rigidity, and topological coupling on the previously defined scales; R_*, T_*, and G_* mark their thresholds for "high," "pronounced," and "dense," whose values depend on the field and are not assigned here. The conditions are conjunctive: satisfying only one or two does not place a direction in $\mathscr{F}$. A project with a high financial barrier but a short process and few coupled subsystems is expensive without meeting this definition; a widely connected project with modest resource and time demands is complex without meeting it either. Putting deep space, ecological restoration, and basic science together invites checking which projects cross all three thresholds, rather than highlighting one outstanding dimension. This structural classification allows unlike fields to be examined together without itself deciding the value of investment.

Why these directions matter

A frontier's value comes from knowledge, services, or future possibilities worth obtaining. Its proposals must meet actual conditions, through tests that can be expensive, slow, and sparse. Discriminating experiments, staged commitments, recovery paths, and complete records help extract sufficient information from limited contact.

More specifically, these directions renew contact with five things. First, materials. Mate-

rial properties do not negotiate or change because a description is persuasive. Second, time. Mission cycles, development, and succession cannot be compressed arbitrarily through faster information processing; participants must work across long cycles. Third, risk. Failure carries costs that cannot all be shifted to insurers or third parties. Fourth, cooperation. These systems exceed one team's capacity and require collaboration across institutions, disciplines, and generations. Fifth, accountability. Failure needs someone answerable who understands their position in the system.

In low-viscosity environments, all five can be weakened without immediate consequences and therefore overlooked. Resistant frontiers make them difficult to evade again.

What the difficulty is for

A frontier's value must be explained through what it yields. Difficulty alone supplies no reason to invest. A reverse test makes the point: resistance consisting only of redundant procedures, prolonged approvals, or obsolete tools represents wasted resources rather than this kind of frontier. The criterion remains friction's function.

Nor should the frontier be equated with a destiny among the stars. Outward expansion is one option rather than a predetermined direction. Ecological restoration, basic science, and rebuilding public education can be examined along the same criteria, with potentially dense coupling in living and social systems. The popular association with spaceflight owes more to public visibility than to a structural monopoly.

The scale of investment raises another question. Returns can take decades or longer; some commitments are irreversible, while others can be staged or adjusted. Resources committed here become unavailable elsewhere, requiring technical and value judgments together. Who may commit decades of resources, and on what grounds can the present decide for the future? The questions return more sharply when the horizon exceeds any generation's term of office.

Another function of the frontier

Beyond developing capabilities and calibration, resistant frontiers offer a destination for resources beyond complete absorption into low-viscosity systems. If resources keep flowing to cheap virtual deliveries with quick payback, needs that resist virtualization may become undersupplied. They can return as crises: infrastructure beyond maintainability, ecosystems beyond recovery, basic science lacking new experimental evidence, or education producing abilities detached from actual problems.

Preventing these crises can pay handsomely, but benefits appearing as "disasters that never happened" can be hard for funders to capture or recognize in current accounts. This recurring asymmetry helps explain institutional protection where markets may undersupply the frontier. The difference is scale: a foundation can protect a laboratory's direction, while sustaining a civilization's frontier may require decades of public commitment, often longer than the commitments governments readily make.

The enduring mismatch has no simple solution. One useful step is to explain why these frontiers matter beyond a minority's preferences. They preserve interfaces through which a civilization stays in contact with the world. Close those interfaces and remaining activities may still flourish within a system whose judgments lose external calibration. Closure can occur through gradual industrial drift rather than a single decision: resources move toward the shortest payback, and supporting activities lose supply one by one. Understanding the process calls for tracing how the objects of industrial intervention have shifted through history.

6.6.2 The rise of the ultimate industries

Three layers of industrial transformation

Industrial history can be divided by its objects of intervention: overlapping transformations extended their layers while changing energy and information costs. The industrial revolution worked on macroscopic matter—cotton yarn, steel, chemical feedstocks, fuel—to meet needs for food, clothing, and power, bringing natural processes into factories and replacing human and animal labor with engines and machines. Its achievements register in steel output, railway mileage, electricity generation, and crop yields. It partly loosened material scarcity at the cost of fossil fuel use and emissions.

The information revolution shifted attention to bits: electronic states in circuits, signal encoding, computation steps, and information routes. Cheap copying helped give this transformation a different cost structure, with unit costs falling by orders of magnitude over decades. Its measures include computing capacity, bandwidth, storage, and connections. It addresses computation and communication, previously constrained by human processing and the transmission speeds of physical media.

Intelligent technologies are helping engineering enter a third layer: cells, gene expression, protein structure, metabolic pathways, and neural activity—life processes themselves. Here the objects maintain themselves, feed back, and adapt; earlier engineering also had complex feedback, while life connects those processes to metabolism and heritable history. Changing a line of code can also trigger complex operational consequences; changing a genome intervenes in a responsive process whose effects may emerge years later. The distinction calls for studying these costs and risks on their own terms rather than applying the curve of information copying directly.

The division orders neither importance nor completed historical eras. Its three layers correspond to atoms, bits, and life: material objects, information carried in bits, and self-maintaining biological processes. Matter and information remain central to the economy and will for a long time. The division traces changing scarcity: material and information costs are being reduced, while development, metabolism, immune response, and aging do not follow the same curve as information processing.

Life, mind, and trust

The life sciences already show the direction: structure prediction and molecular design lower the cost of generating candidates; gene editing makes changes to genetic information programmable; (Jinek et al. 2012) precision medicine relates treatment to individual characteristics; geriatrics extends its focus from acute disease to delaying functional decline. Demand has a clear demographic background: global life expectancy at birth reached about 73.3 years in 2024, 8.4 years above 1995; people aged 65 and older make up about a tenth of the world population, projected to approach a sixth around 2050, with roughly 2.2 billion by the late 2070s, exceeding the number under 18 in that projection (United Nations Department of Economic and Social Affairs 2024). Growing demand meets a sector with constrained supply rates. Baumol and Bowen's 1966 study of the performing arts and Baumol's 1967 paper offered a familiar explanation: as manufacturing productivity rises, services requiring person-to-person work may fail to match it; when wages rise with a common labor market, their relative unit costs can increase (Baumol and Bowen 1966; Baumol 1967). Care, companionship, psychotherapy, and education include tasks with slower productivity gains, although tools, group provision, and organization can change delivery; relative costs, service prices, and workers' pay require separate observation. Much unpaid activity also escapes direct pricing, leaving families to bear its costs through unpaid work.

A simplified relative-cost trajectory takes the form

$$\frac{C_{\text{slow}}(t)}{C_{\text{fast}}(t)} = \frac{C_{\text{slow}}(0)}{C_{\text{fast}}(0)}\, e^{rt} \tag{6.12}$$

C_{slow} is unit cost in the sector with limited productivity growth, measured in currency per unit of output; C_{fast} is unit cost in the sector with sustained productivity growth, in the same units; r is that sector's labor productivity growth rate, in inverse time; and t is time. The simplified model holds productivity constant in the slow sector, assumes a fixed productivity growth rate in the progressive sector, and lets wages rise together in a common labor market; productivity offsets the progressive sector's wage growth, while slow-sector unit cost rises at rate r, widening the ratio exponentially. The slow sector becomes relatively more expensive because the model assumes different productivity paths, independently of any increase in importance; its objects overlap with the high-viscosity positions discussed above. The rate r must come from sectoral data rather than being assigned here; the model also assumes comparable units of output, common wages, and unchanged other cost conditions. It describes relative costs under those assumptions, without evaluating the sector's worth.

In March 2026, the AARP Public Policy Institute estimated that U.S. family care for adults totaled about 49.5 billion hours in 2024, with an imputed economic value of roughly 1.01 trillion U.S. dollars under its method (Houser et al. 2026). This is valuation rather than payment, and methodological changes prevent treating the entire difference from earlier reports as growth on an unchanged basis. Unpaid still means someone invested real time and ability.

The mind-related direction concerns mental health, support, and relationships. WHO's September 2025 report found more than a billion people living with mental disorders, alongside substantial service gaps (World Health Organization 2025). Conversational services can help, while evidence for clinical effects, safety, escalation, privacy, and access must match their actual claims.

Trust concerns the dependability of judgment. In January 2024, a finance employee at the Hong Kong office of a multinational engineering company saw familiar colleagues and superiors in a video conference, then followed instructions to make 15 transfers totaling about HK$200 million, roughly US$25 million; a check with headquarters a week later revealed that every other participant had been a deepfake (Government of the Hong Kong Special Administrative Region 2024). The case undermined a familiar verification channel: a face and voice on video cannot alone establish identity. Trust needs anchors harder to imitate cheaply: two-person checks, meetings in person, traceable signatures, and institutions answerable for compensation.

Physical collateral, verifiable records, and enforceable compensation offer several ways to strengthen credibility. Each raises questions: how does collateral match consequences, which judgments do records support, who triggers remedy, and will the money be available? Verification tools and institutions can lower the cost of trust, while generation can expand the number of claims needing checks. The net effect requires measuring both sides.

Pricing and distributing negentropy

All three directions involve processes and costs that resist rollback. Life involves a body's singular history, aging, and interventions; mind-related services involve time, with tools changing relationships without merely copying an output and trust developing through honored commitments; trust itself involves promises whose breach cannot be undone, only gradually outweighed by later conduct. These concern more than generation capability; how computation changes their costs, quality, and demand remains open to separate investigation.

The industrial shift can therefore be posed as a pricing question: with generation costs approaching zero, some scarcity may move from reproducible appearances to verifiable processes, abilities, and commitments. This book borrows "negentropy" to examine maintenance: these services provide information while also sustaining conditions for bodies, relationships, and reliable judgments. All three need continuing resources to resist disorganization. Here "luxury" refers to scarcity, without a defense of ostentation or high prices; actual prices depend on supply rates, quality, payment institutions, and bargaining.

Distribution immediately follows. If care, mental health services, and trust intermediaries grow scarcer under market pricing, those needing them most may be least able to pay. As with other high-viscosity positions, scarcity can attract prices without compensating every cost or ensuring access. Addressing distribution calls for institutions: public payment, social insurance, families and volunteers sharing provision, and enforceable service standards. Leave this

out, and the "ultimate industries" become a menu available to only part of society.

Testing the judgment

Cheap imitation can hollow out all three directions more quietly than direct replacement. Care can perform its outward form: counted visits, scripted consultations without treatment, or procedures without judgment may pass metrics while withholding substance. Mental services can be packaged as purchasable experiences without sustained relationships. Trust intermediaries can become rubber stamps, ratings for sale, or guarantees without compensation liability. The criterion remains whether process, results, and responsibility can be checked, beyond the amount apparently spent. A care service escaping every consequence of error returns to the low-viscosity problem, however human-centered its name.

Career expectations raise another question. Calling these "ultimate industries" promises neither better or more meaningful work nor reliable income to everyone entering them. Many such jobs remain poorly paid; correcting that mismatch needs institutions beyond personal choice. If value does move toward processes with irreversible costs, these directions may grow in relative importance, at a pace and scale shaped by society's willingness to pay.

6.6.3 The ultimate prediction: a self-organizing Gaia?

From manufacturing to the logic of life

Engineering systems are built to drawings, maintained by manuals, and replaced through budgets. The familiar logic assumes that some states can be repaired or replaced: fix an error, replace a damaged part, retire an obsolete one. Engineering also accumulates irreversible consequences, while life writes history more deeply into its own maintenance and development; its state emerges from that history, and continued operation requires dissipation. Three conditions make the distinction more concrete.

The first is self-replication: producing the apparatus that replicates the system, beyond producing its products. Von Neumann's work published in 1966 supplied one construction: the automaton carries both a description and a constructor that reads and builds from it. In that scheme, a description alone remains inert, while a constructor without the corresponding description cannot complete the entire replication (Neumann 1966). The second is self-maintenance: keeping the system far from equilibrium through continuing dissipation. Schrödinger's 1944 image of "feeding on negative entropy" led into the dynamics developed by Prigogine and Grégoire Nicolis in 1977: open systems far from equilibrium can form and maintain ordered structures through matter and energy fluxes, involving internal entropy production and exchange with the environment, whose net change also depends on other processes (Schrödinger 1944; Nicolis and Prigogine 1977; Prigogine 1977). The third is heritable variation: copying can introduce changes that pass to descendants. Operating parameters can then change through differential replication and environmental selection

alongside design and control.

Replication, maintenance, and heritable variation allow a system to enter environmental selection. Copying fidelity affects how sequences persist. With independent error probability ν at each position and sequence length λ, the probability of an error-free copy is

$$Q = (1 - \nu)^{\lambda} \approx e^{-\nu\lambda} \qquad (\nu \ll 1) \tag{6.13}$$

Q is the probability of error-free copying of the whole sequence; $\nu\lambda$ is the expected number of errors. As $\nu\lambda$ approaches 1, error-free copies become less common, although descendants can remain highly similar to the original. Whether selection maintains a particular sequence also depends on relative fitness, the mutation network, and population processes. DNA polymerase selectivity, proofreading, and mismatch repair jointly improve copying fidelity (Kunkel and Bebenek 2000; Gilbert 1986). Longer sequences require lower per-position error rates to preserve the same probability of a complete error-free copy. Repair, proofreading, and organization support these conditions, with energy costs and benefits depending on the mechanism. Fidelity therefore belongs in the resource account, without a fixed rule requiring another layer of machinery for each order-of-magnitude improvement.

A Gaia criterion

At planetary scale, the question grows stronger: if engineering and biological systems become deeply coupled, could the whole meet all three conditions?

Can engineering networks, supplied with continuing energy, improve their capacity to repair, recycle, and reproduce their material foundations? “Gaia” organizes that question here. Industrial chains already replicate many devices but still rely on people, global specialization, and particular feedstocks; this does not establish independent self-replication from raw resources for the entire system.

Self-maintenance, reproduction, and heritable change need individual definitions. Irreversibility can complicate governance without being sufficient evidence of self-organization. For specified materials, boundaries, and observation periods, begin with a narrower quantity: the share of usable supply returned through circulation.

$$\kappa = \frac{\dot{m}_{\text{cycle}}}{\dot{m}_{\text{total}}} \tag{6.14}$$

The numerator counts material returned during the period and actually reentering usable supply; the denominator counts all usable inputs within the same boundary, with the numerator a subset for a ratio between 0 and 1. Stocks, cycling delays, quality changes, and double counting need explicit treatment. A high circulated share can reduce virgin extraction without establishing energy autonomy, complete reproduction, or ecological stability.

Compare circulation shares by material, quality, boundary, and year. Steel, aluminum, construction aggregates, and complex composites face different recovery conditions; collection or

recycling rates, recycled-input shares, and quality-preserving closed-loop shares are different measures (Miatto et al. 2017). κ describes the specified material flow; assessing the whole system's self-maintenance also requires energy, losses, repair, and reproduction dependencies.

From prediction to inquiry

Turning this prediction into an inquiry requires distinguishing at least three things.

First, it is not yet a theory with a complete testing framework. The conditions give no timetable: structural possibility differs from a forecast that a system will meet them. Testable quantities include copying fidelity, material circulation, and entropy production rates, but common definitions, measurement boundaries, and comparable records must precede tests of the proposed mechanisms.

Second, it implies no purpose. Popular readings of Gaia can confuse maintaining a state with moving toward something beneficial to a particular party. Even a formal account of maintenance need not imply benefit to anyone in particular: a self-maintaining planetary system could operate without people. Reading maintenance as benevolence turns a description into a normative premise.

The information capacity of self-replication depends on copying fidelity, sequence length, and selection; distributed systems able to maintain, repair, and replicate themselves raise new questions about unilateral shutdown. The third question concerns shutdown: a self-repairing system already replicating can have a different cost structure from a production line. Both may incur supply, maintenance, safety, and exit costs, but only the former may require confronting regrowth and an autonomously spreading replication network. Whether costs and consequences can be withdrawn therefore becomes a physical and governance question for the chapter's principles.

The four principles now connect: damping ontology asks what formation and maintenance cost; incompleteness asks how standards gain evidence, authorization, and correction; symbiosis examines complementarities between capability and position; recursive optimization connects practice, feedback, and revision. Together they frame how conflicting values should be adjudicated, rather than prescribing each particular task.

This proposed constitution has four foundations. First, a right to imperfection: proving that an objective is executed correctly does not alone establish that it deserves adoption, so there must remain places for challenge, appeal, and rollback; treating every challenge as interference can close correction channels. Second, a voice in defining costs: those bearing them should participate in deciding what counts as a cost and which costs may not be passed on. Third, a division of labor that preserves necessary friction: different strengths make cooperation useful, so the faster side cannot alone decide which resistance to retain, and the division must address the competition that remains. Fourth, direction: recursive improvement should answer to life—to bodies, relationships, judgment, and finitude—beyond output indicators. The foundations depend on one another rather than forming a ranking; removing one weakens the force of the

others.

"Silicon-carbon symbiosis" describes a complementary arrangement: replicable capabilities expand choices while actions remain connected to evidence, bodies, rights, and lasting responsibility. Materials do not permanently assign the roles. Digital systems are physical, and life can learn, compute, and delegate; the interfaces can change with evidence and technology.

Where a civilization directs its resources ultimately depends on which costs it regards as irreplaceable. Abundant generation sharpens the question: inward comfort and outward resistant frontiers draw on the same resources and can shape different civilizations. The inward prospect can flourish while losing interfaces with the world. The next chapter takes the possibility to an extreme: a system with physical self-replication begins to move from manufacturing toward life's logic; if copies are distributed and materially independent, who can still bring it to an end becomes a concrete question.

Further reading

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Robert H. Frank, *Passions Within Reason: The Strategic Role of the Emotions*
James C. Scott, *Seeing Like a State*
Karl Polanyi, *The Great Transformation*
Zygmunt Bauman, *Liquid Modernity*, Polity Press, 2000
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Hans Jonas, *The Imperative of Responsibility*
Amartya Sen, *Collective Choice and Social Welfare*

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Chapter 7

Conclusion: finding freedom within necessity

Chapter 6 developed four principles for reconstructing value and identified resistant frontiers: deep space, deep-sea work, ecological restoration, basic science, and rebuilding education. They bring intelligence back into contact with materials, time, risk, and accountability. The argument assumed expansion constrained by manufacturing capacity, with acceleration redistributing costs across three sets of constraints. Self-replication challenges that premise. A system constructing itself need not wait for an external manufacturer to prepare the next generation; replication cycles, resource supply, and environmental feedback jointly govern expansion. If that change removes constraints altogether, the mechanisms need rewriting; if it changes their form, the task is to examine the challenge and return to the book's opening question.

7.1 The dialectic of evolution

Self-replication stress-tests the framework by touching the sources of all three constraints. Ordinary technology asks external questions: is there enough capital, enough capacity, someone willing to build the line? Manufacturers can ease some constraints through additional investment. A self-replicating system also encounters its environment and physical processes; investment may change supply and operations but must still meet actual carrying capacities and replication times. Changed boundaries between manufacturer and replicator require new accounts of who bears resources, time, and structural burdens. Examining their new forms also sharpens a deeper question: if a system takes over manufacturing itself, what remains an indispensable criterion of "existence"?

7.1.1 Self-replication and the thermodynamic deadlock of exponential growth

From manufacturing to replication

Chapter 6 listed replication, maintenance, and heritable variation, then examined copying fidelity, information length, and proposed planetary criteria. There the question was what physical self-replication means; here it is what happens to the three constraint variables if it is realized. One identifies a threshold; the other examines constraints after it has been crossed.

Comparing manufacturing with replication begins by specifying capacity. A fixed line adds approximately a fixed number of units per period; a system continually building production units can grow approximately exponentially with sufficient inputs and a stable replication cycle. Industry can expand capacity too, while replicators must manufacture and maintain themselves; the distinction is more specific than "linear costs" against "raw materials only."

Self-replication distributes some capacity expansion among existing units. Each new unit still needs materials, energy, space, transport, proofreading, and waste handling. Microbes, trees, and imagined engineering replicators have different cycles; replication cannot simply be presumed faster than manufacturing.

As functions of time, the two idealized outputs differ in one place:

$$N_{\text{manufacturing}}(t) = 1 + \frac{t}{\tau_{\text{production line}}}, \qquad N_{\text{replication}}(t) = 2^{\,t/\tau_{\text{replication}}} \tag{7.1}$$

N is cumulative output relative to the initial quantity; the time parameters describe adding one unit on a fixed line and doubling the replicating system. Fixed capacity yields linear cumulative output; sufficient resources and stable doubling yield exponential growth. The functions agree at $t = 0$; their initial slopes determine whether the production line first leads, and any later crossing requires solving the full equation. The comparison must specify allowable capacity changes, since lines can expand and replication faces supply and environmental limits.

Replication depends on continuing matter and energy. As the system builds itself and emits waste and heat, expansion changes available resources. Scarcity remains in acquisition rates, space, dissipation, and environmental capacity. Replication reorganizes supply and relocates the constraints it encounters.

Autonomous replication of complete engineered systems remains uncommon, which defines the discussion's scope. The stronger description-and-construction scheme examined here requires a description, machinery that can read it and perform construction, and continuing matter and energy. Cells supply a mature form of material replication, without making every copying mechanism an instance of one universal construction scheme. Artificial systems currently supply parts of this: software copies designs while manufacturing still depends on lines capable of producing their designed range. Much observed material expansion therefore remains organized through manufacturing, while autonomous replication could depart from familiar engineering intuition. Intermediate forms—printable parts, cultured tissue, self-healing

materials—supply some of these features while remaining limited elsewhere.

Cost structure adds another consequence. Manufacturing expansion needs new fixed investment and continuing variable inputs, distributing pressures across production lines where arrangements can be negotiated. Replication can concentrate some design and startup costs in the first generation, but descendants still incur materials, energy, construction, proofreading, maintenance, and waste costs; initial design and verification meet continuing environmental supply. Design may need continuing updates, while environmental burdens accumulate; production licenses or capital negotiations alone may not change constraints arising from resources and environmental dynamics.

Three constraints recast

In self-replicating systems, the RTG constraints change form without disappearing.

Resource barriers shift from manufacturing toward acquisition. A manufacturer must assemble resources and concentrate them on a line, requiring planning and capital. A replicator may acquire them locally, but concentrations, availability, uptake mechanisms, and energy must satisfy its requirements. The question shifts from "Can inputs be assembled?" toward "What remains available?", involving engineering capability, ecological capacity, and resource flux. Acquisition faster than renewal can exhaust the resource base, producing overshoot and decline or stabilization through feedback; resource replenishment, waste, and regulation determine the trajectory.

Temporal rigidity shifts from production cycles toward replication cycles. A complex product passes through steps with their own characteristic times; serial, parallel, and bottleneck relations determine the total. Replication also contains many steps, summarized by a doubling time. It need not be faster than manufacturing; physical and chemical mechanisms, environment, and design determine how it can be shortened. Molecular limits enter efforts to accelerate it, continuing the earlier problem of temporal rigidity at another scale. Topological coupling shifts from supply chains toward spreading networks. Manufactured products depend on supply, logistics, installation, and maintenance networks shaped by organization and investment. Replicas can themselves carry propagation, with edges formed through contact and transmission routes. Distributed propagation can accelerate spread and complicate control; central shutdown points, external dependencies, and spatial isolation must be checked in each system.

Why viscosity remains

Has self-replication removed the constraints altogether? The analysis suggests a relocation from manufacturing toward ecological processes. Manufacturing asks about its producers' capital, capacity, and willingness to build. Investment can ease these external constraints. Replication additionally confronts environmental resources, cycle lengths, network paths, and the changes its accumulating products make to the environment.

Self-replication transforms manufacturing viscosity into ecological viscosity rather

than abolishing it. Two governance implications follow. First, the object of governance changes. Manufacturing regulation can limit capacity, review products, and control sales. Replication outside those regulated stages can escape part of their coverage. Governance must also consider spread: initial release, special dependencies, containment or self-limiting mechanisms, and early monitoring. These change replication's boundary conditions to constrain copying or propagation; their effectiveness must be established.

Second, compare the times for identification, spread, and intervention. If reliable judgment takes many doubling periods, exposure may grow substantially before it arrives. Control still depends on isolation, detection, dependencies, and intervention strength; exponential growth does not make every response too late.

The comparison can be written as

$$\frac{\tau_{\text{verification}}}{\tau_{\text{spread}}} \begin{cases} \ll 1, & \text{Earlier verification can help; intervention is still needed,} \\ \gg 1, & \text{Faster spread can expand exposure before verification} \end{cases} \tag{7.2}$$

$\tau_{\text{verification}}$ is the time needed for reliable assessment of consequences, shaped by tests, clinical observations, or ecological monitoring; τ_{spread} is the doubling time, in the same units, for the replicating system. Their dimensionless ratio borrows the idea of timescale comparison, but puts judgment time above doubling time; it is not the rheological Deborah number, which compares relaxation with observation. When judgment is much faster than spread, earlier intervention is possible; when it spans many doublings, exposure can expand before conclusions arrive, while recoverability still depends on isolation, intervention, and environmental dependencies. Manufacturing and replication can fall on either side and require measurement. The criterion assumes roughly constant τ_{spread}; falling resource concentrations can lengthen real replication cycles, moving the ratio over time, so apply it by stage.

Replicators still encounter supply, cycles, waste, and feedback. Resource limits can stabilize growth or, together with delays, produce oscillation and overshoot. Actual dynamics must decide this rather than assuming no equilibrium. The delayed logistic model illustrates a stable region alongside a loss-of-stability boundary.

A model with an explicit stability boundary lets us examine whether a stable state can form. A delayed replication process can be written as

$$\frac{dN}{dt} = rN\left(1 - \frac{N(t-\theta)}{K}\right) \tag{7.3}$$

N is the number of replicating units, r the growth rate, K the fixed carrying capacity, and θ the feedback delay. Linear perturbations around the positive equilibrium $N = K$ obey $\dot{u}(t) = -r\,u(t-\theta)$. For $r > 0$ and fixed parameters, $r\theta < \pi/2$ gives local asymptotic stability, with the first boundary at $r\theta = \pi/2$; oscillations may emerge beyond it, without

alone establishing resource exhaustion or population collapse. Changing carrying capacity requires additional resource and damage variables. Both manufacturing and replication can have delayed feedback; their structures determine the differences.

7.1.2 The limits of simulacra: what irreversibility leaves behind

The three constraints have moved, but remain. Beyond replication, they offer a test: whether working on bits or atoms, a system drawing matter and energy, rejecting heat, and propagating through structures remains in the physical world. The line can examine two extremes: replication and representation. Von Neumann's formalization also separates copying a description from understanding it, allowing a description to circulate apart from its referent. Simulacra can thus become an engineered mechanism with boundaries, beyond judgments about images alone.

Where the stress test belongs

Physical self-replication moves a system from manufacturing toward the logic of life. It deserves separate treatment as a stress test for the framework. Earlier chapters examined intelligence meeting resource, temporal, and structural resistance on its way into reality. Replication can change both the magnitude of resistance and its sources and bearers. If that abolishes constraints, the argument needs rewriting; if it changes their form, the framework needs extension.

Von Neumann's self-reproducing automata, conceived in the 1940s, gave an early formal version of the question. Could a machine with finite construction capabilities build a class of machines allowed by the model, including itself? His construction answered yes with a description of itself and general machinery to read and implement it (Neumann 1966). The universal constructor separates description from construction: descriptions can be copied without understanding, and one construction mechanism can implement different descriptions without a dedicated rule set for each.

The use made of that formalization needs specifying. Its useful separation allows copying without prior understanding and supplies a formal route to engineered self-replication. The analysis here does not treat logical construction as the whole problem. Von Neumann's formal model asks whether construction rules suffice to reproduce a system; this book additionally asks what follows physically once such machinery exists: reusable rules still require materials, time, heat rejection, and paths for propagation. Logical feasibility does not establish physical loss of control; resources, time, and topology describe the distance between them. Universal construction therefore cannot itself establish danger: capability says little about its cost or speed.

Separating description from construction allows copying without prior understanding. Actual replication still depends on constructors, energy, materials, and environment; the whole

system sets its rate. A description can circulate independently and produce effects through construction elsewhere, so the conversion edge and the landing place of responsibility must still be traced.

Testing the framework

Ecological viscosity offers both a benefit and a risk as a description. The benefit is a common inquiry: redefine resources, time, and topology for the relevant ecological processes. The risk is fatalism, as though replication necessarily becomes uncontrollable. Ecological regulation includes depleted resources, predators, mutations affecting copying, and spatial breaks in propagation. Responses range from very short intervals to decades, while collective human decisions can be equally slow or slower; their contest has no predetermined outcome.

How large a role self-replicating systems will play, and which major problems they will bring, remain open, depending on changing technological pathways and governance capabilities. Where replicas persist, spread, and interact in an environment, "faster manufacturing" becomes an insufficient description; ecological processes also need study. The stress test brings the earlier mechanisms together: resource constraints become environmental supply and capacity, time constraints become replication cycles, and topology becomes the spreading network. All three remain in new positions. Understanding their ordinary forms supplies tools for investigating their reorganization in extreme cases.

The remaining question concerns standards: with constraints understood, how does a society decide which resistance to preserve, which to remove, and which costs may not be shifted? Chapter 6's four principles address this regardless of replication; rapid propagation can make the decisions more urgent by shortening the time available.

What irreversibility leaves of existence

Looking inward raises a deeper question: if a description can circulate apart from its system, what establishes the thing's existence? The everyday distinction between reality and fiction soon proves insufficient: unlimited copying does not make a representation false, while its physical costs differ from the event it represents. One physical clue is whether a process establishes change that cannot practically be undone. Replaying, overwriting, and rolling back some states do not erase the fact that computation and interaction occurred; the question is which histories still constrain what follows.

Reversibility describes one aspect of consequences without settling every distinction between existence and fiction. Generated text can be rewritten after already influencing a decision; a clinical intervention can leave bodily consequences difficult to reverse. Both happen physically; comparisons must specify boundaries, timescales, recoverable states, and continuing effects.

An objection follows: stored bytes leave physical traces and need energy and maintenance, so what survives of the distinction? It must be stated for the process and timescale under

discussion. The question concerns which traces enter that process's constraints, beyond the existence of any trace. A generated text's state can be easy to reset although its electricity has been consumed; concrete pouring or drug administration can also admit correction or recovery while leaving some history impossible to undo directly. Each comparison needs an inventory of states, costs, and effects. Different scales can reveal different degrees of recoverability, without a universal direction of change. The earlier Deborah number offers an analogy through relaxation and observation times, not a general test of reversibility; its definition and use appear in the appendix.

Simulacra and the framework meet at this boundary. Replication can become hard to govern where propagation is difficult to reverse on the relevant scale; exposure may grow before slower detection and intervention catch up. Generative systems can be overestimated where outputs are cheap to retry, but cheap retries do not guarantee an absence of decision, resource, or social consequences. Opposite errors can share an origin. A final extreme pushes representation in place of consequence-bearing toward the point where this criterion is easiest to challenge.

7.1.3 The brain in a vat meets its substrate

The engineering inside an epistemological problem

The brain in a vat poses an epistemological problem: remove a brain, keep it alive, and supply sensory signals by computer—could its owner know the world was simulated? Discussion has long focused on what could be known if experiences were indistinguishable. Another question asks what keeps the apparatus running. The requirements become concrete: a pump driving nutrient solution through microvessels; oxygen and glucose supply replenishing gradients; temperature control rejecting metabolic heat; an electrode array writing signals into tissue; waste removal; electricity; and maintenance staff able to intervene. Every item has a material task that software alone cannot perform.

Perfusion, organ preservation, and clinical support supply evidence for some related mechanisms. Different goals and durations prevent treating them as an engineered isolated human brain preserving full consciousness and personal continuity. Supply, waste removal, temperature, and long-term stability can each be examined separately.

The shift brings part of the discussion into testable territory. Arguments about uploading often circle between experience irreducible to matter and nothing existing beyond physical processes. An engineering account lets people examine shared requirements and their public measurements and failures without first settling consciousness's ontological status.

The physical account of sustaining a brain

A human brain weighs about 1.4 kilograms, roughly 2% of body mass, yet uses around 20% of resting oxygen and glucose, corresponding to approximately 20 watts. Cerebral blood

flow is around 750 milliliters per minute, close to a sixth of cardiac output. These are typical physiological scales varying with individuals, temperature, activity, and condition, rather than hard lower bounds for every apparatus. Flow must deliver oxygen and glucose across usable concentration differences and remove carbon dioxide and metabolites; supply also depends on blood oxygen content, extraction, diffusion, and pressure, with excessive pressure risking tissue damage. Temperature must remain suitably controlled, since deviations affect function. Complete interruption of flow can abolish consciousness within seconds; several minutes can cause severe injury, with outcomes dependent on temperature, residual flow, and recovery rather than one irreversible deadline for every tissue (Donahue et al. 2018). Keeping a brain is sustaining a process: what is preserved includes a continuing flux, beyond a static state.

Adequate supply and limited pressure define an illustrative flow window:

$$\dot{Q} \geq \frac{\dot{m}}{\Delta C_{\max}}, \qquad \dot{Q} \leq \frac{\Delta p_{\max}}{R_{\text{blood vessels}}} \tag{7.4}$$

$\dot{Q}$ is volumetric flow; $\dot{m}$ and $\Delta C_{\max}$ must concern the same substance, as consumption per unit time and available concentration difference respectively; oxygen and glucose cannot simply be added. The right-hand bound uses a linear hydraulic resistance approximation. Actual cerebral circulation includes autoregulation, diffusion, and tissue condition; these expressions illustrate supply and pressure constraints rather than clinical parameters.

Heat is easy to overlook in this account. Those 20 watts must go somewhere: passive rejection needs a temperature difference, while active cooling also needs work and heat rejection across a larger boundary. Algorithms can improve scheduling, control, and design without abolishing the second law or the need for waste heat to reach a sink. An ingenious interior still needs pumps, power, heat-transfer surfaces, and maintenance across its boundary. A literal proposal to move consciousness into a machine must reimplement support functions; a digital substrate can use different equipment, flows, and energy, while preservation of consciousness and personal continuity needs separate evidence. The room remains in the physical world, and someone looking after it still has a shift that ends.

Support must also address continuity and failure. Backup pumps, power, monitoring, and switching create room for recovery from brief interruptions. Better design can reduce costs while long-term maintenance remains a task somewhere in the system. Deep-space life support raises the same accounting question: operation and its supporting conditions belong within one boundary.

Organ preservation, perfusion, and clinical support demonstrate some maintenance mechanisms (Cameron and Cornejo 2015). Durations and outcomes vary with organ, temperature, method, and purpose. Working brain tissue still needs metabolism, heat rejection, and waste removal; BrainEx restored some circulation and cellular functions in pig brains hours after death in 2019, without establishing restored whole-brain function or consciousness, while personal continuity would need still different evidence (Vrselja et al. 2019). The traceable requirements

remain supply, waste removal, temperature, and long-term stability.

Writing channels: bandwidth and lifetime

Writing a world into this brain requires a channel. Electrode count, signal quality, and tissue tolerance constrain it. Long-term implants can provoke glial and other tissue responses, and declining signal quality has repeatedly been documented (Polikov et al. 2005). The channel has both limited capacity and a physical lifetime, with maintenance and replacement needs. Even without a bit count, fidelity depends on channel and decoding capabilities: information beyond them cannot be arbitrarily written in full; compression may be lossless or may trade distortion for a lower rate. A perfect virtual world therefore depends on physical channels, permissible power, and endurance alongside software.

An idealized capacity expression captures part of the constraint:

$$C_{\text{channel}}(t) = n\,B\,\log_2\!\left(1 + \frac{S}{N(t)}\right), \qquad \frac{dN}{dt} > 0 \tag{7.5}$$

C_{channel} is capacity for the assumed independent Gaussian-noise channels, in bits per second; n is the number of independent channels, B bandwidth per channel, and $S/N(t)$ signal-to-noise ratio. At high signal-to-noise ratios, doubling it adds roughly 1 to the logarithmic term; at $S/N = 1$ the term is 1, and at $S/N = 100$ it is about 6.66, an increase of about 5.66. Real brain-computer interfaces also face channel correlations, tissue responses, and effective coding requirements. Long-term implantation can alter noise and recording quality, whose trajectories need measurement.

The receiving tissue adds its own properties. Its timescales require measurement under specified conditions, rather than declarations of acceleration. Wounds heal over days, neural development has critical periods, and body temperature must remain suitably near 37 degrees Celsius; departures can disrupt function. Software can return instantly to a previous version, while forming a memory or recovering from surgery cannot simply be folded out of the process. Intervention can alter some limits, but the operating conditions of living tissue still have to be met. The wish for a mind free of physical constraints thus raises the need for exact physical support, with bodily and external equipment conditions accounted for together.

Replacing the substrate

Another proposal gradually replaces neural tissue with functionally equivalent artificial parts. Feasibility, functional equivalence, personal continuity, and consciousness need different evidence and remain hypothetical at this scale. Even a feasible replacement changes the substrate rather than removing power, heat rejection, communication, and maintenance.

This is **the inescapability of a substrate**: a particular material may be replaceable, but some material must carry the process, and its properties enter the result. A system needing 20 watts, a narrow temperature range, and hundreds of milliliters of flow per minute differs in capability and failure from one resting at room temperature for years; those properties are not

arbitrarily interchangeable parameters. Replacement also encounters a verification problem: functional equivalence needs observable behavior, and observation needs contact. Without any observable or intervenable connection, those connections cannot test equivalence; this limits evidence rather than proving nonexistence. The question becomes which substrate to choose and what that choice costs.

Connectomics is expanding the networks open to examination. An October 2026 preprint turned a fly brain-and-nerve-cord connectome into a spiking network connected to a simulated body to test sensorimotor behavior (Minnick et al. 2026). It studies wiring, dynamics, and feedback together, while full human-brain state reading, component replacement, and mind transfer demand different evidence.

The substrate can be chosen; its support conditions must be specified. Whatever the material, a maintained process needs a physical point of attachment whose properties enter the result. That moves part of the discussion from possibility to costs and who pays them—questions open to investigation.

How the costs of order move

The apparatus makes the difference between its inside and outside visible. Some internal states admit rollback and signals can be replayed; this does not establish arbitrary rewriting of biological memory or experience. External support accumulates its own history: pumps wear, pipes contaminate, maintainers leave. Internal experience and external maintenance both belong to the apparatus, with the latter setting conditions for the former's continuation. The brain in a vat substitutes representation for some encounters while still meeting the costs of support. However complete the experience, it hangs on equipment and duty schedules outside; if reserves, backups, and switching cannot replace failed support, the conditions needed to continue it are lost.

Division of labor hides the relationship in everyday life. Remote work rests on server-room electricity, cooling, and maintenance; saved commuting still depends on others' time in server rooms, substations, and delivery. These conditions lie outside the worker's interface and attention while remaining real responsibilities borne elsewhere. Apparently frictionless living can move wear out of sight and onto other people. Tracing it explains possible sudden failures: displaced costs need continuing payment, and a broken support chain can abruptly bring them into view.

The chapter's three questions now meet. Exponential expansion, simulacra, and the brain in a vat each test freedom of representation against physical constraints, encountering them in different places. This need not be bad news. More accurately located constraints make choices clearer, while choosing calls for revised measures of value.

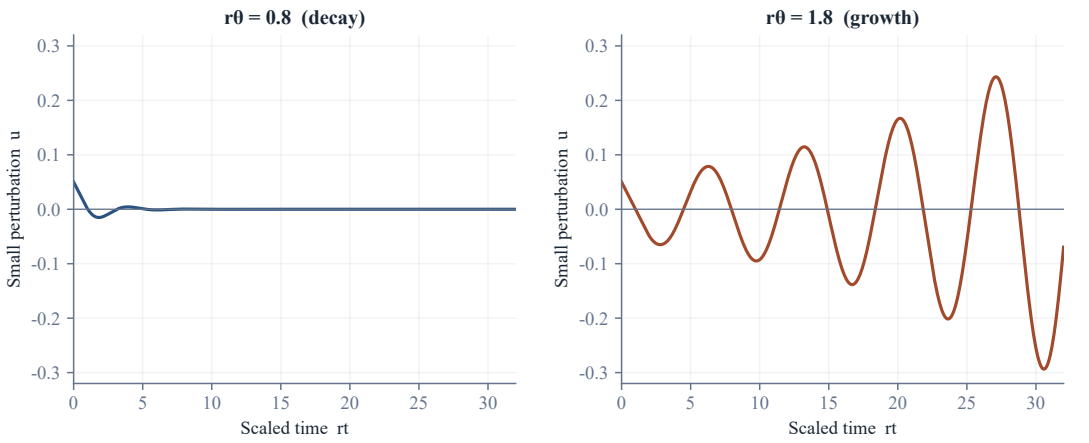


Figure 7.1 The delayed logistic model linearized at its positive equilibrium: $\dot{u} = -ru(t-\theta)$. With a constant small perturbation as the history, disturbances decay at $r\theta = 0.8$ and oscillations grow at $r\theta = 1.8$; the first local stability boundary is $\pi/2$. The curves illustrate local linear behavior.

7.2 Reframing judgment and action

The book began with a gap: proposals arrive at inference speed, while reality changes through many clocks. Tracing the conversion chain reveals what the next acceleration actually improves.

7.2.1 Beyond the Turing trap: capability, value, and standing

Behavioral tests examine abilities at specified interfaces. Their results belong to the test conditions; judging sustained operation, rights, and responsibility requires more. Writing an answer, reliably supplying a service, and making decisions for others are three distinct requirements.

A person's value also exceeds the scarcity of any particular output. Capabilities expand choices; demand, property rights, and bargaining shape pay; dignity cannot be converted along with the market price of a deliverable. Cheaper generation can still support health, education, care, relationships, exploration, and public participation.

"Experiencer of negentropy" emphasizes dependence on actual conditions for life, alongside experience, choice, and relationships within a finite lifetime. Thermodynamics supplies reasons about physical maintenance; ethics and institutions supply reasons about protection for lives.

Finding where value settles requires watching new interfaces. Review may take priority over candidate generation; mature automated checks may shift emphasis toward specifications and task definition; expanded deployment may make maintenance, remedy, and distribution more urgent. Interfaces move, and lists of safe careers become obsolete.

At each position, ask what dependable result it establishes, where its credentials are, whether

another arrangement could serve the function better, and who can enter the arrangement. Social necessity does not automatically become jobs or high pay; training, payment, and the distribution of power need joint attention.

7.2.2 Value flows: how scarcity forms

Scarcity has different sources. Verification needs evidence matched to the task; care needs time, skills, access, and sometimes a particular person; judgment in a situation needs information, ability, and authority to act; long-term education needs pathways through which students acquire abilities. Tools can improve each function, while the structures supplying it must also adapt.

Names alone cannot establish what an activity does. Auditing can become rubber-stamping, experience requirements exclusion, and companionship a service script. Examine outcomes and their formation: can errors be found, evidence reviewed, promises checked, and objections raised by those affected?

High prices likewise do not establish necessity. Scarcity can reflect real supply limits, monopolies, or artificial barriers, while budget constraints leave essential work poorly paid. Value-flow analysis belongs alongside distribution.

After each cost reduction, reassess what distinctions old credentials have lost, which processes new credentials need, and who controls those processes. Technology can create use value and move the places where value is established; judge the gains through complete outcomes.

7.2.3 Three questions before reducing resistance

What has been reduced? Fix the task, quality, baseline, time, and system boundary first. Faster drafting and faster reliable decisions are different gains needing separate evidence.

What changes elsewhere? Examine review, rework, energy, equipment, support, risk, and recovery. Costs can move, fall in total, or newly arise. No principle requires every reduction in resistance to be repaid unchanged elsewhere.

Who gains, and who bears the remaining costs? Include users, workers, operators, affected third parties, and future obligations. Lower total costs can coexist with unequal gains and burdens.

For software teams, these questions connect generation to review, regressions, and maintenance. For clinical services, they connect model performance to intended use, physicians' duties, patient rights, and follow-up. For fleets, they connect vehicle efficiency to empty travel, coverage, infrastructure, and incident response. Answers must be concrete enough to change the plan.

When review overloads, compare candidate filtering with processing capacity; when fail-

ures cannot be contained, narrow the scope or redesign recovery; when responsibility has no landing place, establish arrangements for challenge and compensation. Tables record these decisions; substantive checking still requires people with ability, time, and authority.

7.2.4 Keep interfaces that build ability

Individuals can keep some direct practice to understand, check, and revise the results they rely on. Write before revising when learning a language; actually read and test code when maintaining it; encounter outcomes and explain failures when learning judgment. Good tools can make practice more effective.

How much practice to keep depends on the task. Some abilities need continuing exposure, others occasional practice, while obsolete steps that no longer teach anything can be removed. The purpose is ability, beyond displaying manual work, effort, or an identity without tools.

Personal circumstances differ. Time, disability, care duties, income, and fatigue change available arrangements; assistive tools often expand autonomy precisely where barriers are high. Organizations cannot judge only immediate output while requiring workers to preserve every long-term ability unpaid in private time.

Organizations need the ability to review and reconstruct processes: proportionate records, competent scrutiny, independent challenge, and effective powers to interrupt and correct. These functions need roles, budgets, and authority beyond an instruction to “use cautiously.”

7.2.5 An effective place for those affected

Affected people need information, participation, representation, and remedy. Some decisions need personal consent; others require professional duties, public procedures, or protection for vulnerable groups. Decision powers should match capability and control, while those bearing losses need usable routes to challenge and compensation.

The next generation of tools will change many answers. The method can continue: identify the medium, separate constraints, trace action, examine feedback, and reconcile the whole account. Stronger capabilities make their route into reality more consequential to understand.

7.3 An expedition of negentropy

The next machines can make answers easier to obtain. Civilization must still decide what to sustain, what to discover, and whom it is prepared to ask to bear which costs.

“Negentropy” first recalls a physical fact: life and engineering can use resources and export entropy to sustain local organization. Here it also carries a human wish—to turn new capabilities toward a world worth sustaining. The physical account explains maintenance conditions; it cannot choose which states deserve to be maintained.

7.3.1 Formation and maintenance

Copying offers a small, explicit example. Let ε_0 be the initial error rate per position, and $f_{\text{proofreading}}$ and f_{repair} the conditional residual factors for proofreading and repair; in a model permitting their multiplicative combination,

$$\varepsilon_{\text{total}} = \varepsilon_0 f_{\text{proofreading}} f_{\text{repair}}, \qquad \mathbb{E}[M] = \varepsilon_{\text{total}} L. \tag{7.6}$$

M is the number of errors over length L. Expected errors differ from the probability of at least one error; fidelity also depends on position, context, correlations, and repair mechanisms (Kunkel and Bebenek 2000). What process keeps a structure usable through repeated use? Biological proofreading, software testing, archives, and institutional review employ different mechanisms with their own costs and effects.

Maxwell's demon probes the connection between information and physical boundaries. Information can guide action, but a complete cycle accounts for measurement, storage, and reset. The Landauer bound constrains erasure under specified conditions; Bennett examined the thermodynamics of computation, reversibility, and the demon (Landauer 1961; Bennett 1982). Leave memory and the heat bath outside the diagram, and a clever operation can look free.

AI interfaces can hide a similar omission. A candidate can be cheap because preparation has been amortized, the candidate can be withdrawn, or confirmation occurs elsewhere. These are real technical advantages. Their ultimate value depends on the complete processes they make possible.

7.3.2 Abundance expands choice

Civilization can use intelligence for health, infrastructure, education, discovery, and ecosystems; it can also intensify competition for attention or optimize a narrow target with extraordinary thoroughness. Property rights, incentives, public arrangements, organizations, and individual projects shape the direction.

Comfort has a legitimate place. Less pain, repetitive labor, exclusion, and pointless waiting are progress. Life also needs relationships, curiosity, freely chosen effort, and places where judgment receives a response.

Resistant frontiers supply some of those encounters. So do a difficult experiment, a public service kept working, an ecosystem recovering, a person growing, a relationship, or a craft learned well enough to judge its tools. Across different scales, the demand is shared: results are established through actions and consequences.

7.3.3 Expanding freedom within constraints

Freedom needs usable possibilities, an understanding of relevant boundaries, and the power and support to choose. Good engineering lowers resource barriers, improves processes, isolates

dependencies, improves measurement, and creates recovery paths; good institutions address the distribution of new gains and remaining costs.

Intelligence at scale makes this work more urgent. Cheaper candidates make selection matter more; faster action makes matched feedback matter more; wider reach makes authorization and responsibility matter more.

The expedition of negentropy turns new possibilities into a world worth sustaining. It needs powerful search, bodies and machines able to act, evidence strong enough to interrupt a story, and people and institutions willing to carry a long-term account.

Intelligence seeks paths. Civilization takes shape through the paths it builds, the consequences it keeps in view, and the lives those paths make possible.

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Appendix A

The theory engine: physical foundations and routes to evidence

The main text emphasizes mechanisms and introduces equations where needed. This appendix gathers the physical foundations, their sources, mathematical forms, testable implications, and domains of application. It brings analogies to places where they can be examined and challenged.

A.1 Intellectual sources and the proposed contribution

Begin with the sources and what this framework adds. Mismatched speeds of technology and society have several partly overlapping intellectual histories; the proposal here organizes them through constraints useful for classification and open to further parameterization.

The first source is the distinction between bits and atoms. In Being Digital, Nicholas Negroponte used the distinction to examine the boundaries of digitization and the different behavior of digitizable information and things retaining material form. The widely cited distinction primarily examined changing business forms; the question continued here is how execution, supply, and feedback bring those differences into a task (Negroponte 1995). The framework looks beyond form to absorption mechanisms: information propagates, matter performs work, and life responds through feedback. Its three analytical levels are bits, atoms, and genes; "life" refers to the biological medium as a whole, while genes point toward heredity, metabolism, and maintained feedback without reducing life to one stretch of genetic material. The resulting hypotheses concern different trajectories under the same intelligent technology: examine diffusion, throughput limits, and nonlinear feedback separately, allowing them to coexist rather than assigning each medium one inevitable curve.

The second source is cultural lag. In 1922, William F. Ogburn argued that changes in material culture could outpace nonmaterial culture, producing maladjustment through the gap

(Ogburn 1922). The concept identifies the problem without itself supplying a universal measure of "fast," "slow," or a disruptive difference. Temporal rigidity T organizes a measurement route through process times, deadlines, and adjustment mechanisms; ratios and normalization remain task-specific, so "lag" alone does not yield a parameter.

The third source is the productivity paradox. Robert Solow's 1987 observation placed computers everywhere except the productivity statistics (Solow 1987). The following three decades developed an important explanatory thread: information technology can raise productivity after delays shaped by complementary investment, process redesign, adoption, and measurement. That thread connects to topological coupling G without automatically validating it: systems can need rewiring before output changes, with organizations, institutions, technology design, and supporting tools influencing the pace.

The fourth source is cost disease. Baumol and Bowen's 1966 performing-arts study examined output tied to live human input with limited scope for productivity growth; wages rising elsewhere can push up these activities' unit costs, a mechanism later applied to education, healthcare, and care (Baumol and Bowen 1966). Relative prices can register the mismatch in sectoral statistics. The question continued here is why some functions resist substitution. Temporal rigidity and topological coupling offer two avenues: whether cheaper generation changes tasks, productivity, wages, and supply requires data, without guaranteeing prices that only rise. The explanations overlap in their objects; whether combining them helps depends on added explanation and empirical performance.

These sources examine digitization, timing, organizational complementarities, and relative costs, each with established measures and models. The proposed work uses resources, time, and coupling to organize one chain of conversion into reality, including verification and responsibility. Its test is whether it explains more, distinguishes options, and improves decisions.

Three proposed contributions follow. First, a common set of coordinates (R, T, G) for classifying and comparing forms of resistance; quantitative substitution still needs shared scales and parameters. Second, a classification of media by how they absorb intelligent input, questioning one acceleration logic for all. Third, verification and accountability alongside physical resistance in the account of constraints. That third proposal is the largest departure from these sources and especially needs testing.

Using the four sources requires separating observations, mechanisms, and value judgments. Cultural lag raises timing questions without deciding which speed is better; cost disease explains relative costs without devaluing slow sectors; the productivity paradox connects technology, reorganization, and statistics without guaranteeing either ineffectiveness or an imminent end to delay; bits and atoms identify boundaries whose industrial implications still depend on tasks and execution chains. Connecting each source to an explicit question makes it possible to judge the framework's added explanation and the evidence that would change it.

Formalization and validation are the next steps. The three proposals currently organize con-

cepts and categories for cross-domain comparison; defining, estimating, and testing parameters remain open work. Evaluation can ask concrete questions: do the categories discriminate, are variables independently measurable, do models outperform alternatives, and under what conditions could their implications be refuted?

A.2 A mathematical architecture for generalized viscosity

Chapter 2 introduced $\eta = f(R, T, G)$ and, at the end of Section 2.5, $\eta = \eta_0 \exp(\alpha R + \beta T + \gamma G)$ with its three coefficients; this section examines the exponential choice, multiplication of factors, and the work needed for testable propositions. The form is a hypothesis to investigate, rather than an established relationship.

A.2.1 An exponential form for intrinsic viscosity

For a fixed task and measurement convention, suppose each constraint has an approximately constant marginal effect on log transition cost and omit interactions initially; then write

$$\eta = \eta_0 \exp(\alpha R + \beta T + \gamma G) \tag{A.1}$$

η_0 is baseline viscosity for the activity under fixed reference conditions, rather than a universal constant across fields; R, T, and G are dimensionless normalized measures of resource barriers, temporal rigidity, and topological coupling; α, β, and γ are constant relative weights to estimate empirically, expected to be positive. If separate constraints proportionally reduce throughput and their reduction factors multiply, the logarithm of the total factor adds their contributions. The shape resembles the Arrhenius dependence on activation energy, but that equation has a thermal activation mechanism; shape alone supplies no physical derivation for these sociotechnical coordinates.

Additivity in the exponent is equivalent to multiplicative relative-cost factors. It requires marginal relations such as $\partial \ln \eta / \partial R = \alpha$ to remain approximately constant over the range examined. Serial durations can add, and steady pipeline throughput can be bottlenecked; serial arrangement alone does not imply multiplying factors. Interactions can be tested with terms such as δRT, comparing explanatory and out-of-sample performance.

Measurement first fixes an outcome definition. In $\eta = C/C_{\text{ref}}$, C may measure expense, total labor hours, or mean completion time per qualifying result; $C_{\text{ref}} > 0$ is a published reference for the same task. Success rates run in the opposite direction from costs, so their use as a proxy needs a specified transformation. Once the baseline is fixed, α represents the marginal change in log cost for one additional unit of resource constraint. Coefficients are comparable only with consistent variables, boundaries, and references.

Each predictor needs an explicit definition, baseline, and construction. Normalization can

use ratios of comparable requirements to available capacity or reproducible continuous indicators; a 0–5 rating initially supplies ordinal judgments. Use conditions preceding the transition or independent data where possible, avoiding costs and durations renamed as their own explanations. Check tasks, scales, and boundaries before pooling studies and comparing coefficients and residuals.

Three checks follow: residual patterns by task or medium, reproducible parameters in comparable samples, and improvements from interactions or thresholds. A smooth exponential increase does not itself generate a phase transition or abrupt threshold. The model's shape and a particular critical mechanism need separate accounts.

A local trade-off along a constant-viscosity surface can be written as

$$\left(\frac{\partial R}{\partial T}\right)_{\eta} = -\frac{\beta}{\alpha} \tag{A.2}$$

R and T are dimensionless normalized resource and temporal measures; α and β are the dimensionless partial derivatives of $\ln \eta$ with respect to R and T; subscript η holds viscosity constant. Locally, with fixed G, $\alpha \neq 0$, and $\beta \neq 0$, positive marginal effects allow lower resource barriers to offset higher temporal rigidity; the local change in resources per change in time is $-\beta/\alpha$, and conversely. Converting that relation into extra inputs or a shorter schedule requires fixed measurement definitions. Empirical α and β are still needed; implicit differentiation does not require multiplicative factors, and the ratio is constant only in the separable exponential model with constant coefficients, generally varying with interactions or other nonlinearities.

The first diagnostic is easiest to overread. Residual conditional means should have no systematic deviation given the predictors; heteroskedasticity and dependence can be modeled separately, and none of these checks alone establishes correctness—a wrong model can also look unstructured in a small sample. Examine media separately or explicitly model media, tasks, and interactions in pooled data; ignored differences can enter residuals as apparent noise. The empirical route must therefore retain distinctions among media and tasks.

Parameter stability needs out-of-sample checks over time. Drift can reflect technology, boundaries, definitions, or omitted conditions without proving reversed causation. Preserve definitions and versions, compare explanations, and report the ranges retaining predictive value.

The predictors' construction also matters. Each constraint can contain several components—resource availability can include both materials and money—while the formula takes one number. Aggregation therefore needs an explicit rule, such as a maximum or weighted sum, specified as carefully as the coefficients. Poor aggregation can send omitted structure into residuals, while even a good choice leaves measurement error, omitted variables, and dynamic dependence to examine. Aggregation is itself a modeling choice to test: any one-number input needs an account of how that number was made.

A.2.2 A sociotechnical Deborah ratio

The second form asks how observation timescales affect a system's solid-like or fluid-like response. Markus Reiner introduced the Deborah number in rheology in 1964 as material relaxation time divided by observation time (Reiner 1964).

$$\mathrm{De} = \frac{\tau_{\mathrm{relax}}}{\tau_{\mathrm{obs}}} \tag{A.3}$$

τ_{relax} is the time for internal structural adjustment and τ_{obs} the relevant observation timescale; their matching dimensions make De dimensionless. In the usual viscoelastic interpretation, $\mathrm{De} \ll 1$ allows relaxation within the window and a fluid-like response, while $\mathrm{De} \gg 1$ emphasizes elastic response before adjustment. For a technological employment change observed quarterly, a small τ_{obs} can yield large De if adjustment takes much longer, leaving some transitions incomplete; over twenty years, De falls, and more mobility appears if adjustment finishes in that period. The judgments describe different scales of one system; disagreement can arise from unacknowledged choices of τ_{obs}.

For a single-relaxation-time Maxwell response, the loss modulus under periodic input peaks near a product of angular frequency and relaxation time of 1. An organizational application must first identify corresponding periodic inputs and recovery mechanisms. Quarterly and twenty-year windows change the description, without themselves changing actual adjustment costs. Reducing a mismatch calls for changes in response capacity, input frequency, or buffers; a longer statistical window changes only observation.

The name's source illuminates its use. Reiner drew on Deborah's song in Judges and its image of mountains flowing before the Lord: the same material can appear solid over a short interval and flow over a sufficiently long one. The ratio compares timescales rather than assigning an immutable material category; it asks whether relaxation can occur within this window. A sociotechnical claim of rigidity or flexibility likewise needs its observation window specified.

This turns part of the debate over short- or long-term technological unemployment into a measurable comparison: τ_{relax} for labor-market restructuring against τ_{obs} for policy. Measurable timing constrains the debate while outcomes, distributed losses, and policy choices need further judgment. A claim that technology moves too fast for governance must identify which adjustment τ_{relax} describes; a mismatch arises if spread outpaces reliable detection and intervention. A resilience claim likewise compares response with the disturbance timescale. Measurements can discipline these claims without replacing every value choice.

A social application needs an observable proxy for τ_{relax}, often involving a physical process. A profession taking around ten years to qualify normally cannot supply a wholly new entrant through that path in one or two years; existing trainees, migration, retirement, and task redesign can nevertheless change the stock, so τ_{relax} cannot simply equal training duration. Equipment renewal depends on retirement, construction, retrofits, and investment rather than

accounting depreciation alone; institutional revision involves legislative and review processes whose durations can be found in public records. Only an indispensable step with an established minimum duration, specified serial or parallel relations, and stock conditions can bound τ_{relax}; taking the longest published duration does not automatically supply a lower bound. An independently established process bound can support the corresponding judgment: a shorter policy window cannot establish that the required full adjustment has finished, whatever an aggregate statistic suggests.

Choosing an observation window also expresses a concern. A quarterly view may highlight urgent losses, while a generational one shows some adjustment completed; both windows can have reasons. For a fixed mechanism and definition, τ_{relax} concerns structural response rather than the observer's chosen window, allowing the two to be distinguished; estimates can still vary by sample period and structural change, so the dispute cannot always be reduced to window selection. Report the window alongside the conclusion and what it leaves out. Judgments at different scales can coexist while identifying the problems each scale requires us to address.

A system need not have just one relaxation time; different mechanisms have their own. Labor-market restructuring, supply-chain rebuilding, legal revision, and equipment renewal can differ greatly, so a Deborah ratio must identify its mechanism. Break the problem into mechanisms, estimate each time, and compare them with the observation window. The closest time deserves scrutiny but need not dominate: coupling strength, amplitude, and other slow processes can decide the result. Different mechanisms can therefore support different findings about the same change.

A.2.3 Shear thickening

The third form examines resistance increasing with the rate of application. Chapter 4 used the analogy for biological intervention; a candidate power-law form is

$$\eta(\dot{\gamma}) = \eta_b + k\,\dot{\gamma}^{\,n}, \qquad n > 0 \tag{A.4}$$

$\dot{\gamma}$ is shear rate, in inverse time; transferring it requires defining a dimensionless intervention amplitude and its change per unit time, without conflating dose, intensity, and rate. η_b is zero-shear viscosity, corresponding to slow intervention, with η_b distinguished from the earlier generalized baseline η_0; k is the thickening coefficient and n the exponent. $n > 0$ makes effective viscosity increase with rate, with larger n producing a steeper rise. For biological systems showing such a response, the form is a candidate: rapid intervention can encounter counter-regulation, saturation, or damage, and support requires measured effective resistance actually increasing with rate.

Shear thinning supplies the other direction, represented over a nonzero-rate interval by a negative exponent or a suitably decreasing saturating form. Rheology studies both, with microstructural disassembly or organization influencing the response. Keeping both possibilities

prevents treating rising resistance as universal. One candidate is a rate-dependent sequence: low-rate reorganization may permit thinning, while inadequate compensation at higher rates may increase apparent resistance. Chapter 4's intervention-intensity thresholds need separate definitions; comparisons require matched mechanisms and measurements.

A proposed mechanism for this segmentation is the time needed for compensation, potentially shifting the transition. Below a compensatory rate, disturbances may be absorbed with stable or falling resistance; faster input may disrupt compensation, producing greater resistance or another instability. In the proposed timescale-competition model, the transition may be of order $1/\tau_{\text{compensation}}$, where $\tau_{\text{compensation}}$ is the compensatory response time. This would connect thinning and thickening as different regimes of one system, borrowing timescale comparison to propose testable regions. Holding other conditions fixed, a longer compensatory time should shift the transition to lower intervention rates and shorten the low-rate region if this mechanism holds. Experiments in cells, tissues, and organizations can examine the prediction separately and determine how far the form travels.

Rheological thickening relations rest on experiments with particular materials and mechanisms. In concentrated suspensions, lubrication, repulsion, and frictional contact networks jointly shape flow; greater stress can activate more contacts and sharply increase resistance. Rapid cornstarch stiffening also involves contact, dilation, and jamming conditions, beyond particles merely failing to rearrange and forming clusters (Seto et al. 2013; Wyart and Cates 2014). Rheometry, microscopic observation, scattering, and contact-force evidence can distinguish mechanisms, while fitted power laws retain material and range conditions. Biological and institutional applications require their own independent mechanism evidence rather than physical-law status from a similar curve. Which regulatory, dependency, or damage process produces the measured resistance, and over what range does it support the form?

Look for a regulatory loop requiring time and for evidence that faster input disrupts it. Antibiotic use and resistance, farming intensity and soil fertility, assessment frequency and distorted behavior offer candidates; intervention can alter later response, although antibiotic selection, soil change, and metric incentives are different mechanisms (Laxminarayan et al. 2013). Each needs separate measurement of onset, slope, and the proposed regulatory loop. A mechanism established in one field cannot guarantee another field's similarly shaped curve.

At higher rates, saturation, instability, damage, or blockage can arise through specific mechanisms. A power law describes only its measured interval. Report rate ranges, repetitions, and turning points, comparing power-law, threshold, and saturating forms over a wider range. A fitted exponent alone cannot determine high-rate behavior.

Identifying a power law needs a sufficiently broad input range and repeated measurements. With a nonnegligible η_b, the slope of $\ln \eta$ against $\ln \dot{\gamma}$ need not equal n; fit the full expression or identify where the baseline is negligible. Compare threshold and saturating forms and report transitions beyond the fitting interval.

The three forms have different standing. The generalized exponential relation is a working hypothesis; sociotechnical time ratios borrow a rheological measurement idea and need operational definitions and evidence. The Deborah number has a precise original-domain definition, and rheological shear thickening has substantial experimental support, while extension to biological and institutional responses still needs testing. Distinguishing those positions prevents turning a form validated in one domain into a law for every domain.

A.3 Qualitative criteria before parameter estimation

Before parameters are calibrated, use the framework to locate constraints, make conditional comparisons, and select stages for measurement. Qualitative judgment needs public evidence and explicit ordering criteria. Report directions supported by evidence; retain unresolved trade-offs as alternatives to compare.

Locating the object in three coordinates

Locating an object in (R, T, G) first gives a low, medium, or high position on each dimension rather than three continuous estimates. This need not require precise continuous parameters, but does require observations, records, or measurements supporting the three questions in Table 2.1: can it be moved, can it be completed in time, and can it propagate? The object can then be placed along the gradient of media.

Explicit criteria make the classification usable. Two raters disagreeing on a category can compare the evidence each used before arguing over a precise number. A claim of "viscosity 3.7" without a measurement definition or calculation path leaves the 3.7 untestable. A rough category can be firmer in this respect by forcing judgment back to identifiable evidence.

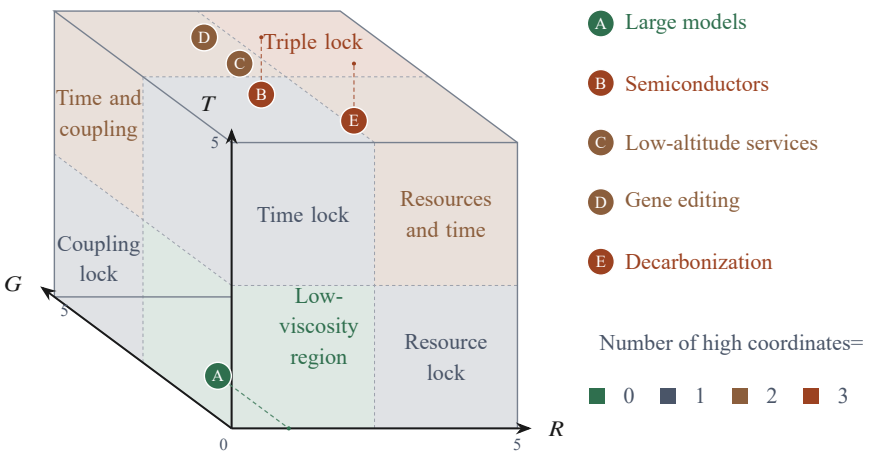
Eight regions and their responses

Each coordinate uses levels 0–5; split at 2.5, with 0–2 low and 3–5 high. Three low-or-high coordinates produce eight regions, located by asking which coordinates cross the midpoint. This coarser division can be more stable than exact ratings: a one-level disagreement leaves the region unchanged if it does not cross the midpoint.

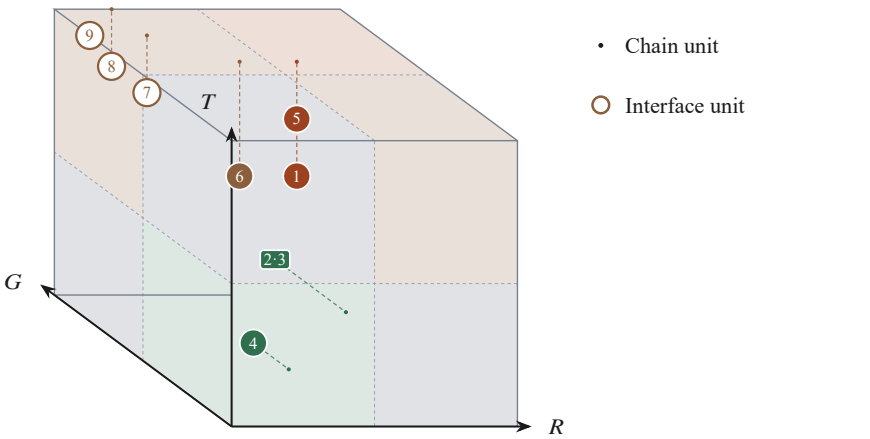
High coordinates name the regions. One high coordinate gives a resource lock, time lock, or coupling lock. Two give resources-and-time, time-and-coupling, or resources-and-coupling. All three give a triple lock; none gives the low-viscosity region. The low-viscosity region used in Chapter 6 now has a specified classification boundary.

A "lock" names a high constraint for the current task under the chosen scale. Whether it can ease requires examining substitute resources, process changes, and structural reorganization. Regions locate the inquiry; evidence selects the response.

With all coordinates low, the low-viscosity region generally encounters fewer of these three obstacles. Expansion still depends on demand, reliability, and organization; a region is not a



(a) Eight regions and five domains



(b) Nine units within one domain

Figure A.1 Three coordinates, eight regions. Points illustrate the correspondence of units and regions rather than measured sector scores, overall viscosity rankings, or dates of operation. Projects need evidence for Table A.4 and new placement under consistent criteria.

growth forecast.

A single lock directs attention to its dominant constraint and possible substitution, process changes, or reduced dependencies. Compare a detour's feasibility and costs with direct transformation.

Double locks require examining both constraints and their interactions. An improvement

may lower both together; feasible routes, investment, and verification shape the pace. A level-surface slope describes cost combinations only within its specified model.

A triple lock calls for scrutiny of resources, time, and structure. Staged investment, task redefinition, or infrastructure changes may alter the combination. Feasibility depends on the project's feasible set rather than three scores alone.

Ratings and regions express different resolutions of one classification. Ratings retain three levels; a region keeps one cell, trading internal distinctions for a more stable boundary. Objects near the midpoint can still switch regions after a small rating change. That can reflect incomplete evidence, real transition, or criteria needing improvement; claims that a technology is "beginning to get stuck" or "not yet stuck" must return to their evidence and criteria.

Rate first, then locate the region, identifying the position and constraints needing priority checks. The next scale applies the regions to concrete technologies.

Comparison: joint gains and cross-dimensional trade-offs

Some relative comparisons need no numerical weights. For states a and b, write coordinate differences $\Delta R = R_a - R_b$, ΔT, and ΔG; taking logarithms of the exponential form gives

$$\ln \frac{\eta_a}{\eta_b} = \alpha\, \Delta R + \beta\, \Delta T + \gamma\, \Delta G \tag{A.5}$$

With shared references and scales and positive weights, if all three constraints in a are no greater than in b, and at least one is lower, the model gives $\eta_a < \eta_b$. This is dominance. Opposing changes generally need bounds on weight ratios and magnitudes, beyond weight rankings and signs alone. A 0–5 ordinal scale needs a meaningful quantitative scale before entering the exponential model.

Direction: start with derivative signs

Under the same-direction conditions below, a trend can be inferred from derivative signs. Differentiating with respect to time gives

$$\operatorname{sgn}\left(\frac{d \ln \eta}{dt} \right) = \operatorname{sgn}\left(\alpha \dot{R} + \beta \dot{T} + \gamma \dot{G} \right) \tag{A.6}$$

Positive weights and coordinates moving in the same direction determine the model's cost direction. Opposing directions need weighted magnitudes. Time-varying coefficients add $\dot{\alpha} R + \dot{\beta} T + \dot{\gamma} G$ and a changing-baseline term. Cheaper generation and stable verification time must be measured at their own stages, not inferred from constraint names.

Fuzzy, open, revisable

Qualitative use needs public criteria, recorded uncertainty, and revision through evidence.

It is fuzzy. Criteria rather than precise measurement define the categories, so an object can fit adjacent descriptions. With insufficient evidence, that ambiguity can be more faithful than an unsupported exact number.

Criteria and weights need to answer to new evidence. Later outcomes may require new ratings, normalization, coefficients, or recognition of a mechanism missing from the three dimensions. Revise to the extent the evidence warrants and keep the corresponding record.

Judgments can be updated. A mistaken rating calls for reviewing nearby categories, regional conclusions, and dependent inferences; quantitative parameters can also be reestimated. The chain affected by the error sets the scope of revision. Every judgment should remain traceable to evidence.

What remains to measure

Qualitative use alone cannot quantify differences in magnitude, duration, or threshold location. The earlier $r\theta$, Deborah numbers, and threshold conditions need variables and values for particular objects; the book supplies conditional structures while cross-domain thresholds remain to be measured and tested.

Qualitative and quantitative analysis differ in questions as well as precision: direction, category, and ordering on one side; amount, duration, and threshold crossing on the other. Treating the former as an unsupported numerical approximation can give an apparent conclusion without a test. Lack of numbers is equally no reason to surrender conditional qualitative judgments that evidence can support.

A.4 Decomposing technologies and measuring viscosity

A scale should unpack "this technology is stuck" into examinable judgments: the task, stage, resources, feedback time, affected dependencies, and evidence. Begin with a constraint profile; a calibrated quantitative model can then estimate overall costs.

A.4.1 Decomposition: independently examinable units

A unit needs an examinable output and reasonably consistent internal timescales. Combining millisecond computation with years of follow-up hides constraints in an average. A span of timescales can help identify where to split:

$$\frac{\max_j \tau_j}{\min_j \tau_j} \leq \Lambda. \tag{A.7}$$

τ_j is a subprocess's characteristic time and Λ a published decomposition threshold. Λ near 10 offers a starting point; further splitting should identify distinct mechanisms rather than follow an order-of-magnitude boundary mechanically.

There are two kinds of unit. Chain units directly produce results for the next stage: making cells, conducting tests, delivering software. Interface units connect external systems: airworthiness certification, procurement access, airspace coordination, service acceptance. Tech-

nology, organization, and institutions can speed both kinds, with each component examined separately.

A.4.2 Rating: ordinal judgments before quantitative scales

Table A.1 provides illustrative anchors from zero to five. Levels order constraints within a coordinate; specify task criteria and supporting evidence before applying them.

Table A.1 Ordinal anchors for the three coordinates

Level	Resource barriers R	Temporal rigidity T	Topological coupling G
0	Small marginal inputs within existing capacity	Effectively immediate for the task	Local change, few dependencies
1	Standard equipment and routine supply suffice	Shorter than the main operating cycle	Known, relatively standardized interfaces
2	Specialized equipment or materials needed	A substantial share of operating time	Several interfaces need adaptation
3	Significant capital, supply, or infrastructure investment	Spans a major planning or construction cycle	Coordination across teams or suppliers
4	Scarce critical inputs or major capacity expansion	Long observation, renewal, or institutional process	Interacting organizations, standards, or jurisdictions
5	Severe constraints within the specified boundary	Much longer than the current decision window	Extensive dependencies that resist isolation

Level zero still consumes energy and materials; level five does not automatically mean impossible. Conservation laws constrain a task without proving every design has reached its limit. The categories need not be equally spaced: one time level does not automatically mean tenfold, and two resource levels cannot be read as a cost multiplier.

Suitable data may support a separate continuous scale. For example, explicitly defining

$$T = \log_{10}(\tau/\tau_0), \qquad \tau_{k+1}/\tau_k = 10 \tag{A.8}$$

makes level differences correspond to orders of magnitude in time. The reference τ_0 is specified rather than generated by the ordinal table. Resources likewise need calibration against consistently measured quantities.

Report uncertainty through adjacent categories, confidence intervals, or missing evidence. If only the boundary between levels k and $k+1$ remains unresolved, record

$$R_i \in \{k, k+1\}, \tag{A.9}$$

and state the evidence needed to resolve the difference. A decimal cannot substitute for it.

A clearly defined dependency-closure size d_i can support a count-based classification:

$$G_i = \min\left\{5, \left\lfloor \log_{10}(1+d_i) \right\rfloor\right\}. \tag{A.10}$$

Graph reachability, causal influence, synchronized modifications, and administrative coordination are four different relations. Specify which defines the closure. Counts also leave connection strength, critical nodes, and failure propagation to examine.

A.4.3 Aggregation: quantities before ratings

Resource quantities can be added with consistent units and boundaries:

$$P_{\text{total}} = \sum_i P_i. \tag{A.11}$$

Electricity, materials, labor hours, and money need separate accounts; combining unlike units requires explicit conversion. A weighted average of resource ratings supplies a descriptive index under the chosen rule, without recovering total resource demand.

Time follows task structure. Strictly serial durations add; simultaneously started tasks all required for completion take their maximum; a dependency graph follows its critical path:

$$t_{\text{serial}} = \sum_i t_i, \qquad t_{\text{completion}} = \max_{p \in \mathcal{P}} \sum_{i \in p} t_i. \tag{A.12}$$

$\mathcal{P}$ is the set of admissible dependency paths. Resource contention, rework, and different start times require corresponding scheduling models. The slowest workstation can limit steady throughput, which differs from the first product's full-chain lead time.

Amdahl's relation illustrates acceleration of the parallelizable part:

$$S(n) = \frac{1}{s + (1 - s)/n} \longrightarrow \frac{1}{s}, \qquad s \in (0, 1]. \tag{A.13}$$

Workload is fixed, s cannot be parallelized, the remainder parallelizes ideally, and additional overheads are initially omitted (Amdahl 1967). New algorithms or task arrangements can change s; the bound belongs to the stated decomposition.

For coupling, first report the dependency graph and its distribution: closures, shared dependencies, critical nodes, isolatable boundaries, and substitute paths. An average rating can be defined as

$$\bar{G} = \frac{1}{N} \sum_i G_i, \tag{A.14}$$

while retaining the distribution and critical nodes. An average can hide a few decisive dependencies.

The familiar giant-component criterion

$$\frac{\langle k^2 \rangle}{\langle k \rangle} > 2 \tag{A.15}$$

comes from asymptotic random-graph models under their regularity conditions (Molloy and Reed 1995). It cannot directly decide the behavior of an actual dependency graph with directions, weights, clustering, and institutional relations. Connectivity also does not imply that every node must change simultaneously.

Only after definition, normalization, and calibration should continuous variables enter a quantitative model:

$$\ln \frac{\eta}{\eta_0} = \alpha R + \beta T + \gamma G. \tag{A.16}$$

List ordinal profiles separately from fitted variables. With parameters, compare predictions and out-of-sample results; without them, report constraints, dominance, and unresolved trade-offs.

If time uses the logarithmic scale in Equation (A.8), its relation to the Deborah number is

$$\ln \mathrm{De} = (\ln 10)T - \ln \frac{\tau_{\text{driving}}}{\tau_0}. \tag{A.17}$$

This identity follows from the scale's definition. Changing the window changes the ratio without automatically changing adjustment capacity.

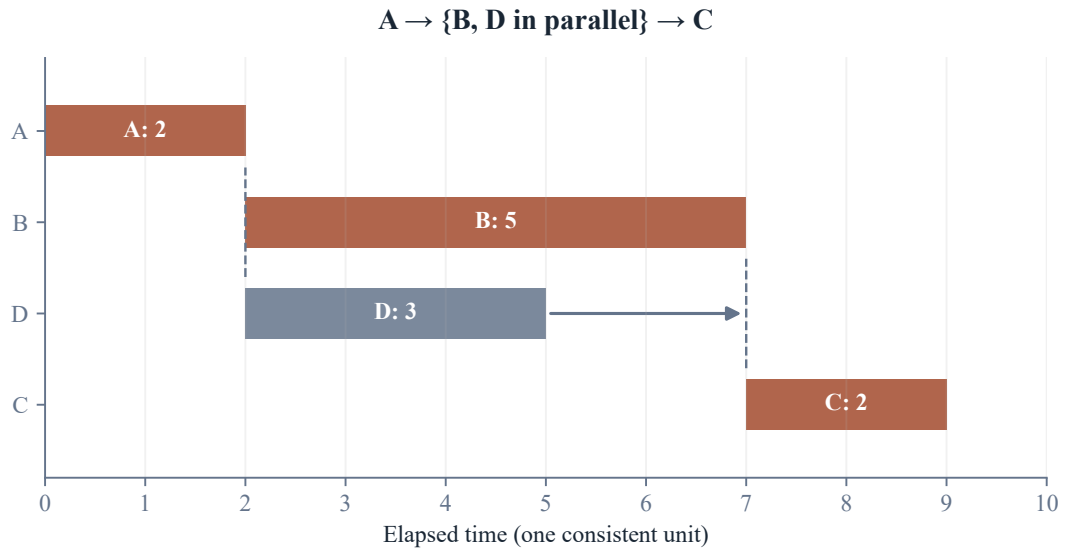


Figure A.2 A precedes parallel tasks B and D; C waits for both. Total duration is $2+\max(5,3)+2 = 9$, rather than the sum 12 or the longest individual task 5. The critical path is A–B–C.

A.4.4 Priorities: large contributions and feasible improvements

With known weights and comparable scales, the largest model contribution is

$$c^* \in \underset{c\in\{R,T,G\}}{\arg\max}\ \alpha_c c, \qquad (\alpha_R, \alpha_T, \alpha_G) = (\alpha, \beta, \gamma). \tag{A.18}$$

Unknown weights prevent the largest ordinal coordinate from establishing overall dominance. Even the largest contribution may not deserve first investment; feasible improvement, cost, risk, and interactions matter.

For small, independent changes with roughly constant marginal costs, let reducing coordinate c by one unit cost $\mu_c > 0$; then

$$\frac{\alpha_c}{\mu_c} \tag{A.19}$$

This compares the model's log-cost reduction per unit of expenditure. It is not absolute net benefit and does not cover cross-dimensional interactions, finite budgets, thresholds, or risks. Finite projects still need comparison as complete alternatives.

Table A.2 From constraint profiles to intervention choices

Situation	Evidence needed	Actions to compare
Main constraint can be eased cheaply	Extent of improvement and full costs	Invest directly; check downstream effects
Improvement substantially raises other costs	Interactions, cost bearers, and risks	Compare net effects and distribution; redesign interfaces
Current constraint is hard to change, but a substitute path exists	Alternative tasks, routes, and outcome criteria	Detour, stage, or narrow scope
Several constraints are high	All feasible paths and input conditions	Reorganize; retain necessary waiting and tests

A.4.5 Example 1: nine units in low-altitude services

Begin by splitting a low-altitude project into the nine units of Table A.3. The table lists conditions to establish; ratings depend on aircraft type, region, operating scope, and evidence.

Certification consumes people, equipment, tests, and facilities; coordination and trust also need material carriers and continuing work. An output consisting of permission does not make resources zero. Acceptance requires a defined population and conditions, rather than an assumed ten-year cycle used to predict deployment.

Draw the nine units as a dependency graph, separate parallel tasks from necessary waits, and calculate the critical path. Battery improvements may change payload, diversion margins, economics, and certification evidence together; software can also reduce testing burdens. The project's actual path determines what controls entry into service.

If approval remains critical, shortening an off-path process may leave first operation unchanged while lowering costs or improving later throughput. Separating those outcomes locates the gain.

Table A.3 Low-altitude services: units, constraints, and evidence

Unit	What to examine	Evidence to obtain
Cell manufacturing	Chemistry, supply, yield, and capacity	Specifications, supply, and production records
Pack integration and thermal management	System mass, heat rejection, fault isolation	Pack tests under mission conditions
Airframe and rotors	Materials, structure, manufacturing, and life	Tests, processes, and maintenance basis
Flight controls and avionics	Functions, latency, failures, and versions	Requirements, tests, and certification records
Takeoff and landing facilities	Sites, electricity, connections, and backups	Construction conditions, access, and planning
Operations and maintenance	Schedules, spares, repairs, and recovery	Operating design and measured support costs
Airworthiness certification	Design compliance and verification scope	Documents for the jurisdiction, type, and stage
Airspace and routes	Operating boundaries, separation, and emergencies	Applicable rules and actual permissions
Service acceptance	Prices, noise, reliability, and rights	Surveys and operating feedback for the population concerned

Table A.4 Questions for five kinds of project

Project	Resources and time	Dependencies and confirmation
Large-model inference services	Existing capacity, full request cost, and expansion	Which downstream actions outputs trigger
Advanced-process manufacturing	Equipment, steps, yields, and delivery cycles	Changes needing suppliers and verification together
Low-altitude services	Mission energy, backups, and facilities	Evidence for the aircraft, airspace, and operation
Gene-editing therapies	Delivery, manufacturing, observation, and follow-up	Cell populations, clinical endpoints, eligible patients
Cement and steel decarbonization	Processes, energy, feedstocks, and replacement cycles	Sites, power, transport, product standards, and demand

A.4.6 Example 2: comparing five project types

Projects in one category can have different technologies and responsibility structures. Gene-editing therapy is not synonymous with germline editing, and decarbonization is not one uniformly mature process. Fix the object before rating and comparison; sector names merely begin the search.

The scale should bring disagreement back to evidence: which condition changed in which unit, on what basis, and which implications need recalculation? Then a broad judgment becomes a reviewable map of constraints.

A.5 Three response models and routes to testing

Throughput limits, demand absorption, and adaptive feedback can produce different response shapes. This section proposes three models and corresponding data definitions. Mechanisms and observations select a form for the task; the media classification offers clues.

One digital-service case approaches a usage plateau with relatively stable demand and distribution. Attention, budgets, supply, and product fit can shape that plateau. Measure unit generation costs separately from usage; rapidly expanding candidates do not automatically become used outputs.

One form for that trajectory is

$$Y_{\text{bits}}(t) = Y_{\infty}\left(1 - e^{-t/\tau_{\text{approach}}}\right) \tag{A.20}$$

Y_{bits} is usage or effective processing rate for the service, Y_{∞} its plateau on the same measure, and τ_{approach} the time constant. With initial value zero, it first rises approximately linearly, then approaches the plateau with an exponentially shrinking gap. Estimate the plateau and approach from data; changing supply or demand may require new parameters or a new form.

Fixed equipment, space, and energy supply can bound physical output per unit time. Additional inputs may yield diminishing returns near capacity, while equipment, processes, or network expansion change the bound. Compare design, manufacturing, construction, and maintenance separately to locate AI's actual improvements.

Two features can be expressed as conditions:

$$0 \le Y_{\text{atoms}}(I) \le \Phi_{\max}, \qquad \frac{\partial^2 Y_{\text{atoms}}}{\partial I^2} \le 0 \quad \text{(saturation approximation)} \tag{A.21}$$

Y_{atoms} is physical output rate, $\Phi_{\max}$ maximum sustainable output under the given facilities, in the same units, and I consistently measured input. The first condition supplies a capacity bound; the second adds a concavity approximation in the saturation region. Stepwise expansion, learning, and complementary inputs can violate concavity, calling for forms matched to observation.

Energy comparisons likewise need a fixed task. Erasing a bit, breaking a chemical bond, and lifting a macroscopic object choose different operation units; their absolute energies cannot determine sectoral growth rates. Count required operations, actual dissipation, equipment capacity, and useful output across a complete task, then compare gains from design and execution.

Adaptive feedback can change dose–effect relationships. One testable case has mean benefit approaching a plateau beyond some intensity while outcome dispersion keeps increasing. Continued improvement, declining effects, and delayed responses are also possible. Record intensity, duration, group differences, and follow-up together to identify the feedback.

The proposed test is

$$\left.\frac{d\langle Y_{\text{life}}\rangle}{dP}\right|_{P>P_c} \approx 0, \qquad \left.\frac{d\,\mathrm{Var}(Y_{\text{life}})}{dP}\right|_{P>P_c} > 0 \tag{A.22}$$

Y_{life} is the selected outcome, P intervention intensity in fixed units, and P_c an estimated turning point. Means and variances retain their respective units. The two conditions jointly describe a plateau with growing dispersion; other feedback shapes need different functions. Keep population, observation duration, and repeated measurements in the comparison.

The examples use time, input, and intervention intensity as different predictors, so their plotted slopes cannot be directly compared. To compare one technology across stages, first align task, inputs, and outcome definitions, then control process time and environment separately. The shapes follow mechanism assumptions; matched comparisons must establish whether actual domains differ accordingly.

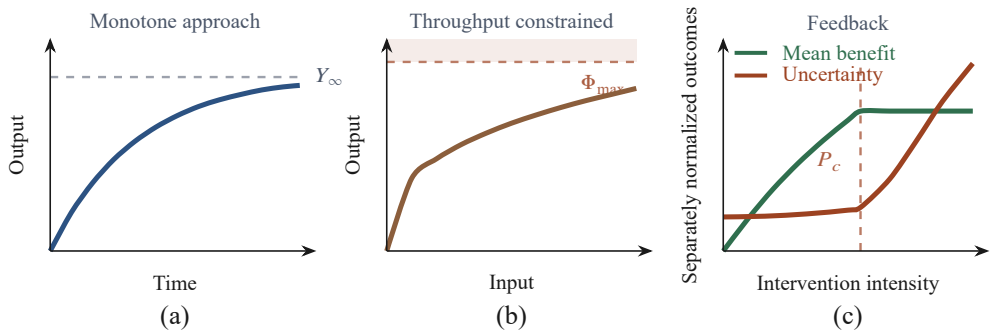


Figure A.3 Three conceptual response models. (a) Usage approaches a plateau over time; (b) output saturates with input under fixed facilities; (c) mean benefit levels off as outcome variance increases. Vertical axes are independently normalized, and panel (c)'s mean and variance have different physical units; heights and crossings do not support cross-indicator comparison.

Four columns distinguish the three curves.

The third column of Table A.5 supplies tests independent of this book's vocabulary: its conditions can be examined without accepting η, De, or the media classification. The proposed classification and those independently examinable conditions have different roles. Each curve states its own criterion; a contradictory observation requires revising that curve without automatically rejecting the other two, while shared mechanisms or data dependencies require examining how far the result extends.

A technology usually crosses several stages with interacting outcomes. Total output need

Table A.5 Three curves: inputs, criteria, and data

Curve	Input	Criterion	Data for testing
Bits: monotone approach	Time	$Y_{\text{bits}} \to Y_{\infty}$; estimate the plateau and time constant from task data	Unit-cost time series for text, image, and code generation, alongside actual use rates rather than output volume
Atoms: throughput constrained	Input I	$Y_{\text{atoms}} \leq \Phi_{\max}$; test saturation-region concavity separately: $\partial^2 Y_{\text{atoms}}/\partial I^2 \leq 0$	Construction labor productivity, mining and smelting energy intensity, semiconductor investment per unit; design-stage versus construction-stage schedule compression within a project
Life: feedback	Intervention intensity P	For $P > P_c$, $d\langle Y_{\text{life}}\rangle/dP \approx 0$ and $d\,\text{Var}(Y_{\text{life}})/dP > 0$	Paired drug-dose, efficacy, and adverse-effect measurements; paired editing-intensity and off-target measurements

not linearly add the three curves. Begin with stable task boundaries, recording design, execution, verification, and maintenance times and costs separately; use controls, staged deployment, or comparable projects to investigate mechanisms. Explain overall effects after evidence supports the stage relationships.

A.6 Epistemic boundaries

Evaluation has three levels: whether constraint categories identify mechanisms, whether quantitative models explain and distinguish observations, and whether normative claims offer public reasons. Each needs its own evidence and corresponding revision.

First, classification organizes questions; calibrated quantitative models make particular predictions. Predictions need fixed tasks, variables, baselines, and parameters, followed by out-of-sample testing. Region names and uncalibrated formulas can identify constraints to examine, without determining deployment years, growth rates, or thresholds.

Second, physical forms need explicit mechanism mappings in sociotechnical analysis. A thickening analogy needs an account of how intervention rates induce compensation or structural change; a Deborah analogy needs measured response and input timescales. Supported mappings supply explanation; unsupported ones call for a different form. Similar vocabulary cannot replace that examination.

Third, categories and functional forms need separate tests. Compare log-linear, interaction, threshold, and other candidate forms on independent data; test whether media categories add

mechanism explanation and out-of-sample discrimination. Repeatedly redefining variables to obtain a fit calls the definitions into question. Similar curves across media require examination of mechanisms and task scales; shape alone neither confirms nor refutes the whole framework.

Fourth, constraints explain conditions affecting transition costs; value arguments explain why results, rights, and responsibilities deserve protection. This book argues for affected people's participation, evidence retained for judgments, and accountability for decisions. Those claims need public ethical reasons, procedures, and distributional arrangements; physical lower bounds cannot choose for them.

Chapter 7 uses these distinctions to argue for choosing, sharing, and bearing viscosity. In a concrete decision, establish feasible paths and cost distributions, compare interests and rights, then assign authority and responsibility. Data support feasibility judgments, normative reasons support trade-offs, and implementation records keep both open to testing.

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